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**Cisco 200-301 Exam Question & Answers**  
**Cisco Certified Network Associate Exam Exam**

<https://www.validexamdumps.com>

# Product Questions: 1238

## Version: 81.2

Topic 1, Exam Pool A

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### Question: 1

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Which command entered on a switch configured with Rapid PVST\* listens and learns for a specific time period?

- A. switch(config)#spanning-tree vlan 1 max-age 6
- B. switch(config)#spanning-tree vlan 1 hello-time 10
- C. switch(config)#spanning-tree vlan 1 priority 4096
- D. switch(config)#spanning-tree vlan 1 forward-time 20

---

**Answer: D**

---

Explanation:

Forward time : Determines how long each of the listening and learning states last before the port begins forwarding.

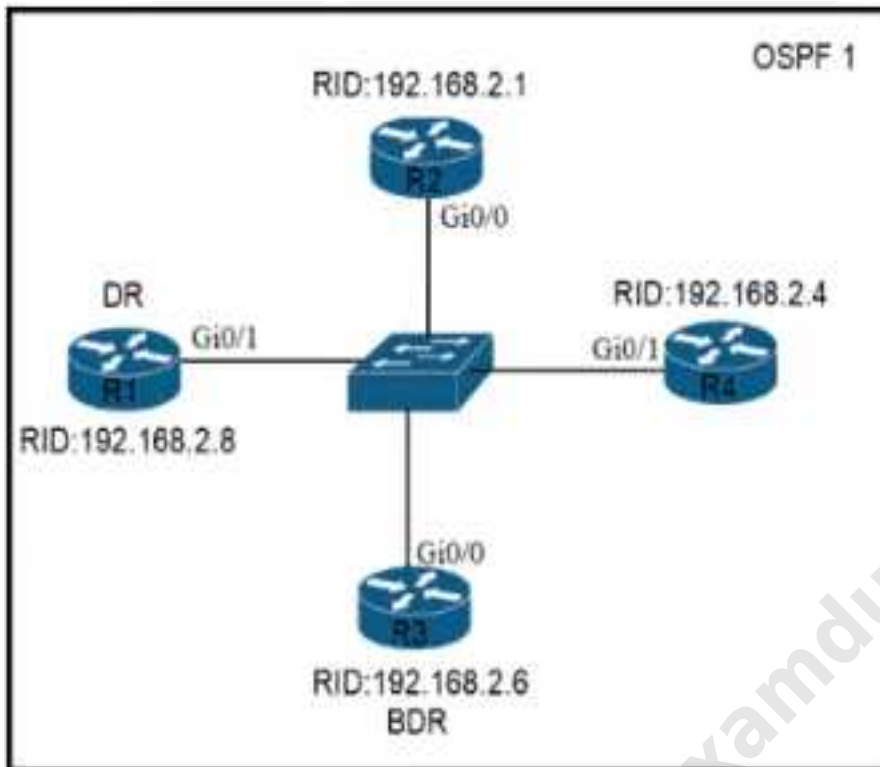
Switch(config)# [ no ] spanning-tree vlan vlan\_ID forward-time forward\_time

Configures the forward time of a VLAN. The forward\_time value can be from 4 to 30 seconds.

<https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst4500/12-2/15-02SG/configuration/guide/config/spantree.html#56177>

**Question: 2**

Refer to the exhibit.



Refer to the exhibit. All routers in the network are configured R2 must be the DR. After the engineer connected the devices, R1 was elected as the DR. Which command sequence must be configure on R2 to Be elected as the DR in the network?

- ☐ R2(config)#interface gi0/0  
R2(config-if)#ip ospf priority 1
- ☐ R2(config)#interface gi0/0  
R2(config-if)#ip ospf priority 100
- ☐ R2(config)#router ospf 1  
R2(config-router)#router-id 10.100.100.100
- ☐ R2(config)#router ospf 1  
R2(config-router)#router-id 192.168.2.7

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: B**

---

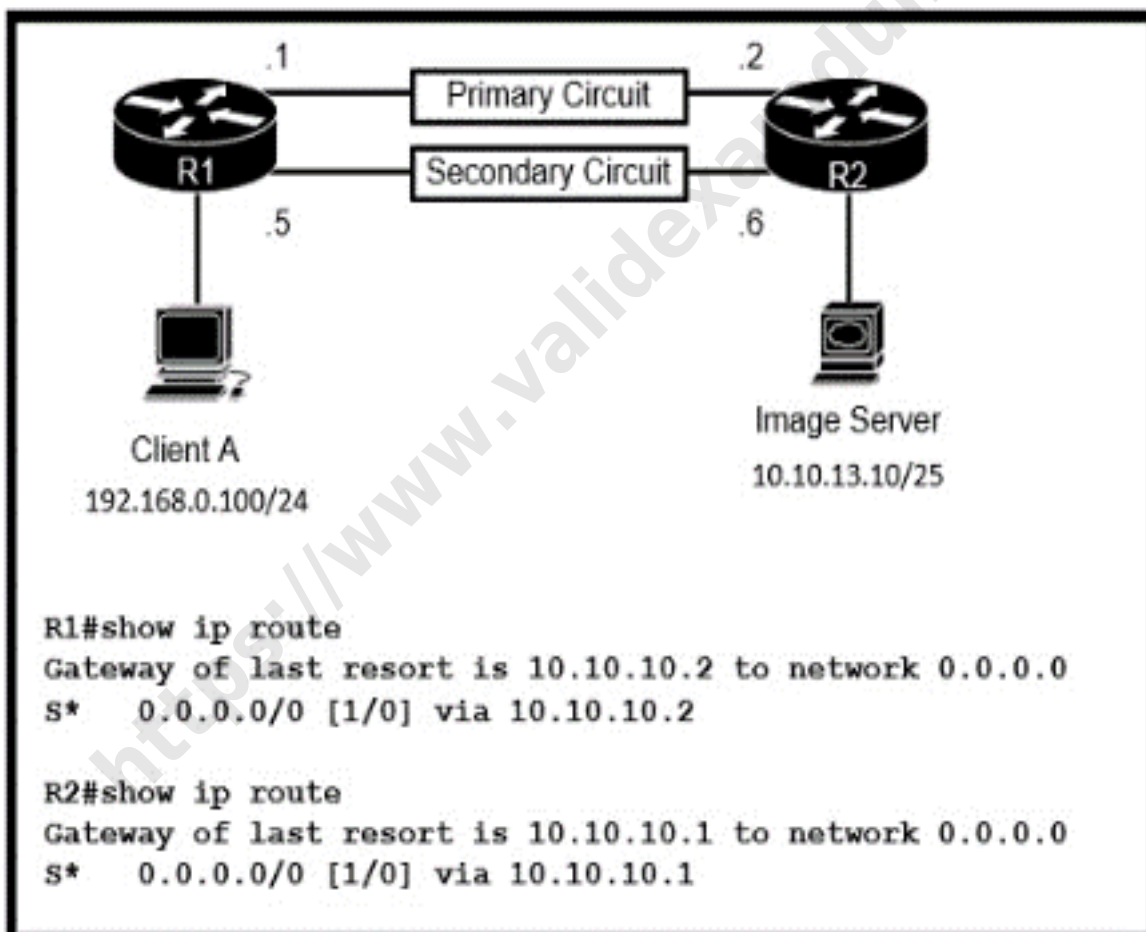
Explanation:

---

**Question: 3**

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Refer to the exhibit.



Refer to the exhibit Routers R1 and R2 have been configured with their respective LAN interfaces The two circuits are operational and reachable across WAN Which command set establishes failover redundancy if the primary circuit goes down?



- ☐ R1(config)#ip route 10.10.13.10 255.255.255.255 10.10.10.2  
R2(config)#ip route 192.168.0.100 255.255.255.255 10.10.10.1
- ☒ R1(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.6 2  
R2(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.5 2
- ☐ R1(config)#ip route 10.10.13.10 255.255.255.255 10.10.10.6  
R2(config)#ip route 192.168.0.100 255.255.255.255 10.10.10.5
- ☐ R1(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.6  
R2(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.5

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: B**

---

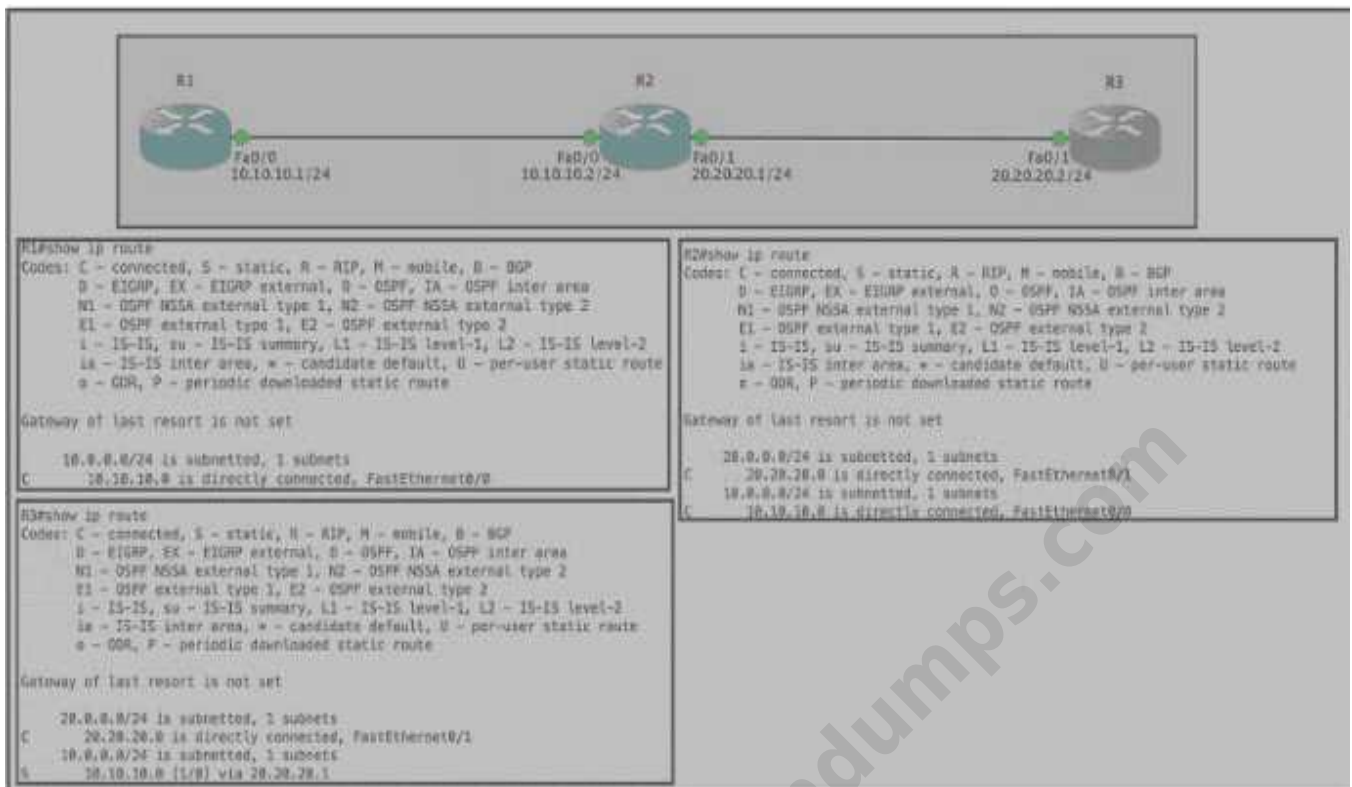
Explanation:

---

**Question: 4**

---

Refer to the exhibit.



Refer to the exhibit Router R1 Fa0/0 is unable to ping router R3 Fa0/1. Which action must be taken in router R1 to help resolve the configuration issue?

- A. set the default network as 20.20.20.0/24
- B. set the default gateway as 20.20.20.2
- C. configure a static route with Fa0/1 as the egress interface to reach the 20.20.20.0/24 network
- D. configure a static route with 10.10.10.2 as the next hop to reach the 20.20.20.0/24 network

**Answer: D**

Explanation:

### Question: 5

What is a benefit of using a Cisco Wireless LAN Controller?

- A. Central AP management requires more complex configurations

- B. Unique SSIDs cannot use the same authentication method
- C. It supports autonomous and lightweight APs
- D. It eliminates the need to configure each access point individually

---

**Answer: D**

---

Explanation:

---

**Question: 6**

---

Which network allows devices to communicate without the need to access the Internet?

- A. 1729.0.0/16
- B. 172.28.0.0/16
- C. 192.0.0.0/8
- D. 209.165.201.0/24

---

**Answer: B**

---

Explanation:

The private ranges of each class of IPv4 are listed below:

Class A private IP address ranges from 10.0.0.0 to 10.255.255.255 Class B private IP address ranges from 172.16.0.0 to 172.31.255.255 Class C private IP address ranges from 192.168.0.0 to 192.168.255.255 Only the network 172.28.0.0/16 belongs to the private IP address (of class B).

---

**Question: 7**

---

When configuring a WLAN with WPA2 PSK in the Cisco Wireless LAN Controller GUI, which two formats are available to select? (Choose two)

- A. ASCII
- B. base64
- C. binary
- D. decimal
- E. hexadecimal

---

**Answer: A,E**

---

Explanation:

Reference: [https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-4/configuration/guides/consolidated/b\\_cg74\\_CONSOLIDATED/b\\_cg74\\_CONSOLIDATED\\_chapter\\_01010001.html](https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-4/configuration/guides/consolidated/b_cg74_CONSOLIDATED/b_cg74_CONSOLIDATED_chapter_01010001.html)

---

**Question: 8**

---

DRAG DROP

Drag drop the descriptions from the left onto the correct configuration-management technologies on the right.

|                                                             |         |  |
|-------------------------------------------------------------|---------|--|
| fundamental configuration elements are stored in a manifest | Ansible |  |
| uses TCP port 10002 for configuration push jobs             |         |  |
| uses Ruby for fundamental configuration elements            | Chef    |  |
| uses SSH for remote device communication                    |         |  |
| uses TCP 8140 for communication                             | Puppet  |  |
| uses YAML for fundamental configuration elements            |         |  |

Explanation:

**Answer:**

|                                                             |
|-------------------------------------------------------------|
| Ansible                                                     |
| uses YAML for fundamental configuration elements            |
| uses SSH for remote device communication                    |
| Chef                                                        |
| uses TCP port 10002 for configuration push jobs             |
| uses Ruby for fundamental configuration elements            |
| Puppet                                                      |
| uses TCP 8140 for communication                             |
| fundamental configuration elements are stored in a manifest |

The focus of Ansible is to be streamlined and fast, and to require no node agent installation.

Thus, Ansible performs all functions over SSH. Ansible is built on Python, in contrast to the Ruby foundation of Puppet and Chef.

TCP port 10002 is the command port. It may be configured in the Chef Push Jobs configuration file .

This port allows Chef Push Jobs clients to communicate with the Chef Push Jobs server.

Puppet is an open-source configuration management solution, which is built with Ruby and offers custom Domain Specific Language (DSL) and Embedded Ruby (ERB) templates to create custom Puppet language files, offering a declarative-paradigm programming approach.

A Puppet piece of code is called a manifest, and is a file with .pp extension.

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**Question: 9**

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An organization has decided to start using cloud-provided services. Which cloud service allows the organization to install its own operating system on a virtual machine?

- A. platform-as-a-service
- B. software-as-a-service
- C. network-as-a-service
- D. infrastructure-as-a-service

---

**Answer: B**

---

Explanation:

Below are the 3 cloud supporting services cloud providers provide to customer:

+ SaaS (Software as a Service): SaaS uses the web to deliver applications that are managed by a thirdparty vendor and whose interface is accessed on the clients' side. Most SaaS applications can be run directly from a web browser without any downloads or installations required, although some require plugins.

+ PaaS (Platform as a Service): are used for applications, and other development, while providing cloud components to software. What developers gain with PaaS is a framework they can build upon to develop or customize applications. PaaS makes the development, testing, and deployment of applications quick, simple, and cost-effective. With this technology, enterprise operations, or a thirdparty provider, can manage OSes, virtualization, servers, storage, networking, and the PaaS software itself. Developers, however, manage the applications.

+ IaaS (Infrastructure as a Service): self-service models for accessing, monitoring, and managing remote datacenter infrastructures, such as compute (virtualized or bare metal), storage, networking, and networking services (e.g. firewalls). Instead of having to purchase hardware outright, users can purchase IaaS based on consumption, similar to electricity or other utility billing.

In general, IaaS provides hardware so that an organization can install their own operating system.

---

**Question: 10**

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## DRAG DROP

Drag and drop the descriptions of file-transfer protocols from the left onto the correct protocols on the right.

The screenshot shows a drag-and-drop interface. On the left, there are six light blue boxes with the following descriptions: "provides reliability when loading an IOS image upon boot up", "does not require user authentication", "uses port 69", "uses ports 20 and 21", "uses TCP", and "uses UDP". On the right, there are two yellow boxes labeled "FTP" and "TFTP", each containing three empty slots for dragging descriptions. A watermark "https://www.validexamdumps.com" is visible across the image.

---

**Answer:**

---

Explanation:

The diagram shows the correct mapping of descriptions to the protocols. The "FTP" box contains three descriptions: "provides reliability when loading an IOS image upon boot up", "uses ports 20 and 21", and "uses TCP". The "TFTP" box contains three descriptions: "does not require user authentication", "uses port 69", and "uses UDP". A watermark "https://www.validexamdumps.com" is visible across the image.



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**Question: 11**

---

Refer to exhibit.

```
Router(config)#interface GigabitEthernet 1/0/1
Router(config-if)#ip address 192.168.16.143 255.255.255.240
Bad mask /28 for address 192.168.16.143
```

Which statement explains the configuration error message that is received?

- A. It is a broadcast IP address
- B. The router does not support /28 mask.
- C. It belongs to a private IP address range.
- D. IT is a network IP address.

---

**Answer: A**

---

Explanation:

---

**Question: 12**

---

Which attribute does a router use to select the best path when two or more different routes to the same destination exist from two different routing protocols.

- A. dual algorithm
- B. metric
- C. administrative distance
- D. hop count

---

**Answer: C**

---

Explanation:

Administrative distance is the feature used by routers to select the best path when there are two or more different routes to the same destination from different routing protocols. Administrative distance defines the reliability of a routing protocol.

---

**Question: 13**

---

Which command prevents passwords from being stored in the configuration as plain text on a router or switch?

- A. enable secret
- B. service password-encryption
- C. username Cisco password encrypt
- D. enable password

---

**Answer: B**

---

Explanation:

---

**Question: 14**

---

A frame that enters a switch fails the Frame Check Sequence. Which two interface counters are incremented? (Choose two)

- A. runs
- B. giants
- C. frame
- D. CRC
- E. input errors

---

**Answer: D,E**

---

Explanation:

Whenever the physical transmission has problems, the receiving device might receive a frame whose bits have changed values. These frames do not pass the error detection logic as implemented in the FCS field in the Ethernet trailer. The receiving device discards the frame and counts it as some kind of input error.

Cisco switches list this error as a CRC error. Cyclic redundancy check (CRC) is a term related to how the FCS math detects an error.

The “input errors” includes runts, giants, no buffer, CRC, frame, overrun, and ignored counts.

The output below show the interface counters with the “show interface s0/0/0” command:

```
Router#show interface s0/0/0
Serial0/0/0 is up, line protocol is up
  Hardware is M4T
  Description: Link to R2
  Internet address is 10.1.1.1/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  --output omitted--
  5 minute output rate 0 bits/sec, 0 packets/sec
    268 packets input, 24889 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    251 packets output, 23498 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 output buffer failures, 0 output buffers swapped out
    0 carrier transitions      DCD=up  DSR=up  DTR=up  RTS=up  CTS=up
```

---

**Question: 15**

---

DRAG DROP

Drag and drop the WLAN components from the left onto the correct descriptions on the right.

|                         |                                                                         |
|-------------------------|-------------------------------------------------------------------------|
| access point            | device that manages access points                                       |
| virtual interface       | device that provides Wi-Fi devices with a connection to a wired network |
| dynamic interface       | used for out of band management of a WLC                                |
| service port            | used to support mobility management of the WLC                          |
| wireless LAN controller | applied to the WLAN for wireless client communication                   |

**Answer:**

Explanation:

|                         |
|-------------------------|
| wireless LAN controller |
| access point            |
| service port            |
| virtual interface       |
| dynamic interface       |

### Question: 16

Which command enables a router to become a DHCP client?

- A. ip address dhcp
- B. ip helper-address
- C. ip dhcp pool
- D. ip dhcp client

---

**Answer: A**

---

Explanation:

Reference: [https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/ipaddr\\_dhcp/configuration/12-4/dhcp-12-4-book/config-dhcp-client.html](https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/ipaddr_dhcp/configuration/12-4/dhcp-12-4-book/config-dhcp-client.html)

If we want to get an IP address from the DHCP server on a Cisco device, we can use the command “ip address dhcp”.

Note: The command “ip helper-address” enables a router to become a DHCP Relay Agent.

---

**Question: 17**

---

Which two encoding methods are supported by REST APIs? (Choose two)

- A. YAML
- B. JSON
- C. EBCDIC
- D. SGML
- E. XML

---

**Answer: B,E**

---

Explanation:

[https://www.cisco.com/c/en/us/td/docs/switches/datacenter/aci/apic/sw/2-x/rest\\_cfg/2\\_1\\_x/b\\_Cisco\\_APIC\\_REST\\_API\\_Configuration\\_Guide/b\\_Cisco\\_APIC\\_REST\\_API\\_Configuration\\_Guide\\_chapter\\_01.html](https://www.cisco.com/c/en/us/td/docs/switches/datacenter/aci/apic/sw/2-x/rest_cfg/2_1_x/b_Cisco_APIC_REST_API_Configuration_Guide/b_Cisco_APIC_REST_API_Configuration_Guide_chapter_01.html)

Reference:

[https://www.cisco.com/c/en/us/td/docs/switches/datacenter/nexus1000/sw/5\\_x/rest\\_api\\_config/b\\_Cisco\\_N1KV\\_VMware\\_REST\\_API\\_Config\\_5x/b\\_Cisco\\_N1KV\\_VMware\\_REST\\_API\\_Config\\_5x\\_chapter\\_010.pdf](https://www.cisco.com/c/en/us/td/docs/switches/datacenter/nexus1000/sw/5_x/rest_api_config/b_Cisco_N1KV_VMware_REST_API_Config_5x/b_Cisco_N1KV_VMware_REST_API_Config_5x_chapter_010.pdf)

The Application Policy Infrastructure Controller (APIC) REST API is a programmatic interface that uses REST architecture. The API accepts and returns HTTP (not enabled by default) or HTTPS messages that contain JavaScript Object Notation (JSON) or Extensible Markup Language (XML) documents.

---

**Question: 18**

---

Two switches are connected and using Cisco Dynamic Trunking Protocol SW1 is set to Dynamic Desirable

What is the result of this configuration?

- A. The link is in a down state.
- B. The link is in an error disables state
- C. The link is becomes an access port.
- D. The link becomes a trunk port.

---

**Answer: D**

---

Explanation:

---

**Question: 19**

---

When configuring IPv6 on an interface, which two IPv6 multicast groups are joined? (Choose two)

- A. 2000::/3
- B. 2002::5
- C. FC00::/7
- D. FF02::1
- E. FF02::2

---

**Answer: D,E**

---

Explanation:

Reference:

<https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/ipv6/configuration/xr-3s/ipv6-xr-36s-book/ipv6-multicast.html>

When an interface is configured with IPv6 address, it automatically joins the all nodes (FF02::1) and solicited-node (FF02::1:FFxx:xxxx) multicast groups. The all-node group is used to communicate with all interfaces on the local link, and the solicited-nodes multicast group is required for link-layer address resolution. Routers also join a third multicast group, the all-routers group (FF02::2).

---

**Question: 20**

---

Which MAC address is recognized as a VRRP virtual address?

- A. 0000.5E00.010a

- B. 0005.3711.0975
- C. 0000.0C07.AC99
- D. 0007.C070/AB01

---

**Answer: A**

---

Explanation:

With VRRP, the virtual router's MAC address is 0000.5E00.01xx , in which xx is the VRRP group.

---

**Question: 21**

---

in Which way does a spine and-leaf architecture allow for scalability in a network when additional access ports are required?

- A. A spine switch and a leaf switch can be added with redundant connections between them
- B. A spine switch can be added with at least 40 GB uplinks
- C. A leaf switch can be added with a single connection to a core spine switch.
- D. A leaf switch can be added with connections to every spine switch

---

**Answer: D**

---

Explanation:

Spine-leaf architecture is typically deployed as two layers: spines (such as an aggregation layer), and leaves (such as an access layer). Spine-leaf topologies provide high-bandwidth, low-latency, nonblocking server-to-server connectivity.

Leaf (aggregation) switches are what provide devices access to the fabric (the network of spine and leaf switches) and are typically deployed at the top of the rack. Generally, devices connect to the leaf switches.



Devices can include servers, Layer 4-7 services (firewalls and load balancers), and WAN or Internet routers. Leaf switches do not connect to other leaf switches. In spine-and-leaf architecture, every leaf should connect to every spine in a full mesh.

Spine (aggregation) switches are used to connect to all leaf switches and are typically deployed at the end or middle of the row. Spine switches do not connect to other spine switches.

---

**Question: 22**

---

Which type of wireless encryption is used for WPA2 in preshared key mode?

- A. TKIP with RC4
- B. RC4
- C. AES-128
- D. AES-256

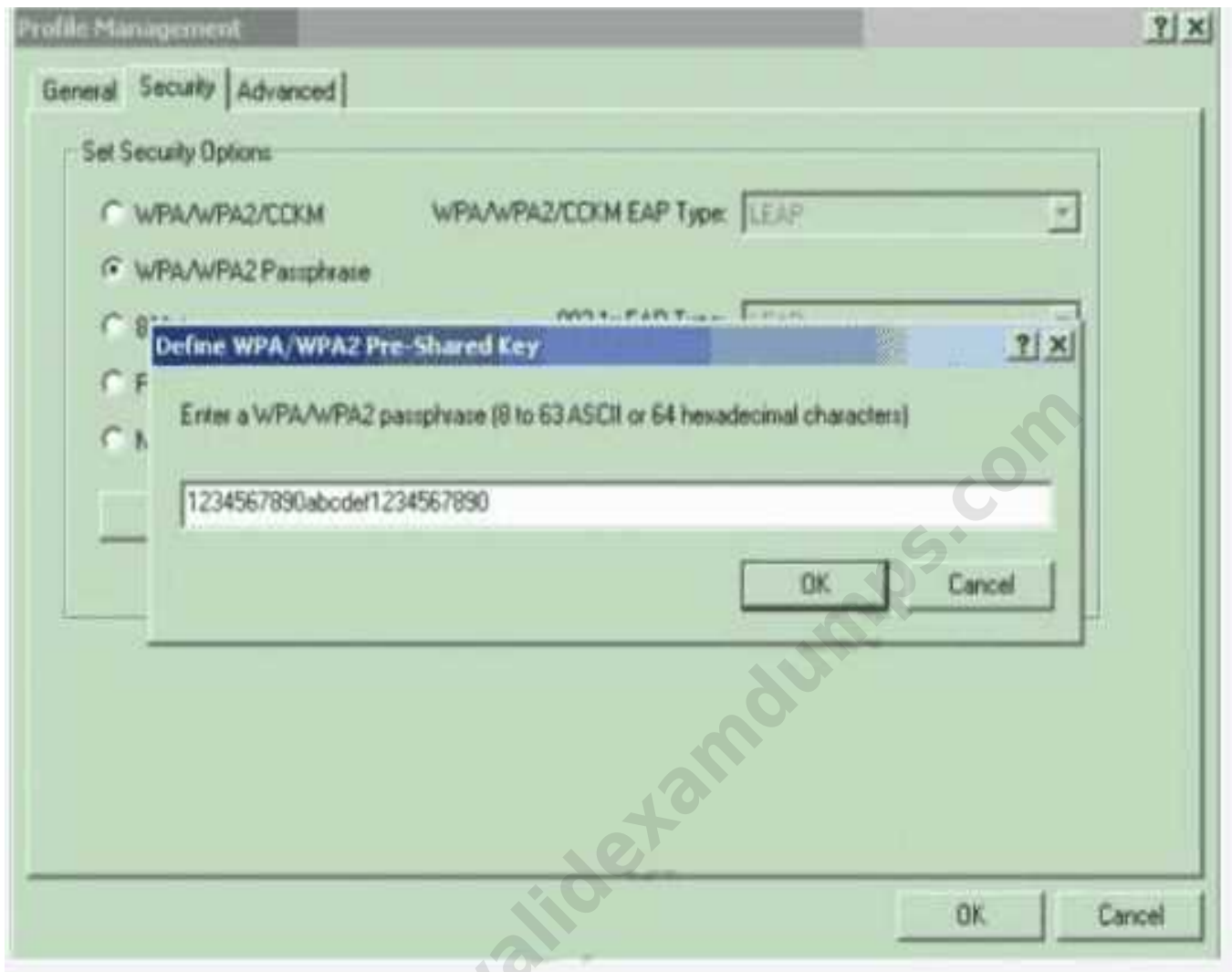
---

**Answer: D**

---

Explanation:

We can see in this picture we have to type 64 hexadecimal characters (256 bit) for the WPA2 passphrase so we can deduce the encryption is AES-256, not AES-128.



<https://www.cisco.com/c/en/us/support/docs/wireless-mobility/wireless-lan-wlan/67134-wpa2-config.html>

---

**Question: 23**

---

Which two actions are performed by the Weighted Random Early Detection mechanism? (Choose two)

- A. It drops lower-priority packets before it drops higher-priority packets
- B. It can identify different flows with a high level of granularity
- C. It guarantees the delivery of high-priority packets

- D. It can mitigate congestion by preventing the queue from filling up
- E. it supports protocol discovery

---

**Answer: A,D**

---

Explanation:

Weighted Random Early Detection (WRED) is just a congestion avoidance mechanism. WRED drops packets selectively based on IP precedence. Edge routers assign IP precedences to packets as they enter the network. When a packet arrives, the following events occur:

1. The average queue size is calculated.
2. If the average is less than the minimum queue threshold, the arriving packet is queued.
3. If the average is between the minimum queue threshold for that type of traffic and the maximum threshold for the interface, the packet is either dropped or queued, depending on the packet drop probability for that type of traffic.
4. If the average queue size is greater than the maximum threshold, the packet is dropped. WRED reduces the chances of tail drop (when the queue is full, the packet is dropped) by selectively dropping packets when the output interface begins to show signs of congestion (thus it can mitigate congestion by preventing the queue from filling up). By dropping some packets early rather than waiting until the queue is full, WRED avoids dropping large numbers of packets at once and minimizes the chances of global synchronization. Thus, WRED allows the transmission line to be used fully at all times.

WRED generally drops packets selectively based on IP precedence. Packets with a higher IP precedence are less likely to be dropped than packets with a lower precedence. Thus, the higher the priority of a packet, the higher the probability that the packet will be delivered

---

**Question: 24**

---

When a floating static route is configured, which action ensures that the backup route is used when the primary route fails?

- A. The floating static route must have a higher administrative distance than the primary route so it is used as a backup

- B. The administrative distance must be higher on the primary route so that the backup route becomes secondary.
- C. The floating static route must have a lower administrative distance than the primary route so it is used as a backup
- D. The default-information originate command must be configured for the route to be installed into the routing table

---

**Answer: A**

---

Explanation:

---

**Question: 25**

---

Refer to the exhibit.

```
Atlanta#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Atlanta(config)#aaa new-model
Atlanta(config)#aaa authentication login default local
Atlanta(config)#line vty 0 4
Atlanta(config-line)#login authentication default
Atlanta(config-line)#exit
Atlanta(config)#username ciscoadmin password adminadmin123
Atlanta(config)#username ciscoadmin privilege 15
Atlanta(config)#enable password cisco123
Atlanta(config)#enable secret testing1234
Atlanta(config)#end
```

Which password must an engineer use to enter the enable mode?

- A. adminadmin123
- B. default
- C. testing 1234
- D. cisco123

---

**Answer: C**

---

Explanation:

If neither the enable password command nor the enable secret command is configured, and if there is a line password configured for the console, the console line password serves as the enable password for all VTY sessions -> The "enable secret" will be used first if available, then "enable password" and line password.

---

**Question: 26**

---

How do TCP and UDP differ in the way that they establish a connection between two endpoints?

- A. TCP uses synchronization packets, and UDP uses acknowledgment packets.
- B. UDP uses SYN, SYN ACK and FIN bits in the frame header while TCP uses SYN, SYN ACK and ACK bits
- C. UDP provides reliable message transfer and TCP is a connectionless protocol
- D. TCP uses the three-way handshake and UDP does not guarantee message delivery

---

**Answer: D**

---

Explanation:

---

**Question: 27**

---

Which mode allows access points to be managed by Cisco Wireless LAN Controllers?

- A. autonomous
- B. lightweight
- C. bridge
- D. mobility express

---

**Answer: B**

---

Explanation:

<https://www.cisco.com/c/en/us/support/docs/wireless/aironet-1200-series/70278-lap-faq.html>

A Lightweight Access Point (LAP) is an AP that is designed to be connected to a wireless LAN (WLAN) controller (WLC). APs are “lightweight,” which means that they cannot act independently of a wireless LAN controller (WLC). The WLC manages the AP configurations and firmware. The APs are “zero touch” deployed, and individual configuration of APs is not necessary.

---

**Question: 28**

---

Which QoS Profile is selected in the GUI when configuring a voice over WLAN deployment?

- A. Bronze
- B. Platinum
- C. Silver
- D. Gold

---

**Answer: B**

---

Explanation:

Reference: <https://www.cisco.com/c/en/us/support/docs/wireless-mobility/wireless-lan-wlan/81831-qos-wlc-lap.html>

Cisco Unified Wireless Network solution WLANs support four levels of QoS: Platinum/Voice, Gold/Video, Silver/Best Effort (default), and Bronze/Background.

---

**Question: 29**

---

If a notice-level messaging is sent to a syslog server, which event has occurred?

- A. A network device has restarted
- B. An ARP inspection has failed
- C. A routing instance has flapped
- D. A debug operation is running

---

**Answer: C**

---

Explanation:

Usually no action is required when a route flaps so it generates the notification syslog level message (level 5).

---

**Question: 30**

---

What are two southbound APIs? (Choose two )

- A. OpenFlow
- B. NETCONF
- C. Thrift
- D. CORBA
- E. DSC

---

**Answer: A,B**

---

Explanation:

OpenFlow is a well-known southbound API. OpenFlow defines the way the SDN Controller should interact with the forwarding plane to make adjustments to the network, so it can better adapt to

changing business requirements.

The Network Configuration Protocol (NetConf) uses Extensible Markup Language (XML) to install, manipulate and delete configuration to network devices.

---

**Question: 31**

---

An email user has been lured into clicking a link in an email sent by their company's security organization. The webpage that opens reports that it was safe but the link could have contained malicious code. Which type of security program is in place?

- A. Physical access control
- B. Social engineering attack
- C. brute force attack
- D. user awareness

---

**Answer: D**

---

Explanation:

This is a training program which simulates an attack, not a real attack (as it says "The webpage that opens reports that it was safe") so we believed it should be called a "user awareness" program.

Therefore the best answer here should be "user awareness". This is the definition of

"User awareness" from CCNA 200- 301 Official Cert Guide Book:

"User awareness: All users should be made aware of the need for data confidentiality to protect corporate information, as well as their own credentials and personal information. They should also be made aware of potential threats, schemes to mislead, and proper procedures to report security incidents." Note: Physical access control means infrastructure locations, such as network closets and data centers, should remain securely locked.

---

**Question: 32**

---



An engineer must configure a /30 subnet between two routers. Which usable IP address and subnet mask combination meets this criteria?

```
interface e0/0  
description to HQ-A371:19452  
ip address 209.165.201.2 255.255.255.252
```

```
interface e0/0  
description to HQ-A371:19452  
ip address 10.2.1.3 255.255.255.252
```

```
interface e0/0  
description to HQ-A371:19452  
ip address 172.16.1.4 255.255.255.248
```

```
interface e0/0  
description to HQ-A371:19452  
ip address 192.168.1.1 255.255.255.248
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: A**

---

Explanation:

---

### Question: 33

---

What is the default behavior of a Layer 2 switch when a frame with an unknown destination MAC address is received?

- A. The Layer 2 switch drops the received frame
- B. The Layer 2 switch floods packets to all ports except the receiving port in the given VLAN.

- C. The Layer 2 switch sends a copy of a packet to CPU for destination MAC address learning.
- D. The Layer 2 switch forwards the packet and adds the destination MAC address to its MAC address table

---

**Answer: B**

---

Explanation:

If the destination MAC address is not in the CAM table (unknown destination MAC address), the switch sends the frame out all other ports that are in the same VLAN as the received frame. This is called flooding. It does not flood the frame out the same port on which the frame was received.

---

**Question: 34**

---

Refer to the exhibit.

```
R2#show ip nat translations
Pro Inside global      Inside local    Outside local   Outside global
tcp 172.23.104.3:43268  10.4.4.4:43268  172.23.103.10:23 172.23.103.10:23
tcp 172.23.104.4:45507  10.4.4.5:45507  172.23.103.10:80 172.23.103.10:80
```

An engineer configured NAT translations and has verified that the configuration is correct.

Which IP address is the source IP?

- A. 10.4.4.4
- B. 10.4.4.5
- C. 172.23.103.10
- D. 172.23.104.4

---

**Answer: D**

---

Explanation:

NAT is used to send a packet to the outside network, using a public IP address to make it routable. The NAT logic is "inside-to-outside" FIRST and "outside-to-inside" THEN. This way, configuring NAT means "choosing a public IP address" for any outbound packet" IN THE FIRST PLACE, where "public IP address" translates to "inside global address". Among the given answers, the only inside global address is 172.123.104.4.

---

**Question: 35**

---

Which feature on the Cisco Wireless LAN Controller when enabled restricts management access from specific networks?

- A. CPU ACL
- B. TACACS
- C. Flex ACL
- D. RADIUS

---

**Answer: A**

---

Explanation:

Reference: <https://www.cisco.com/c/en/us/support/docs/wireless-mobility/wlan-security/71978-acl-wlc.html>

---

**Question: 36**

---

Which command automatically generates an IPv6 address from a specified IPv6 prefix and MAC address of an interface?

- A. ipv6 address dhcp
- B. ipv6 address 2001:DB8:5:112::/64 eui-64

- C. ipv6 address autoconfig
- D. ipv6 address 2001:DB8:5:112::2/64 link-local

---

**Answer: C**

---

Explanation:

The "ipv6 address autoconfig" command causes the device to perform IPv6 stateless address autoconfiguration to discover prefixes on the link and then to add the EUI-64 based addresses to the interface.

Addresses are configured depending on the prefixes received in Router Advertisement (RA) messages.

The device will listen for RA messages which are transmitted periodically from the router (DHCP Server).

This RA message allows a host to create a global IPv6 address from:

- + Its interface identifier (EUI-64 address)
- + Link Prefix (obtained via RA)

Note: Global address is the combination of Link Prefix and EUI-64 address

---

**Question: 37**

---

An engineer is asked to protect unused ports that are configured in the default VLAN on a switch.

Which two steps will fulfill the request? (Choose two)

- A. Configure the ports in an EtherChannel.
- B. Administratively shut down the ports
- C. Configure the port type as access and place in VLAN 99

- D. Configure the ports as trunk ports
- E. Enable the Cisco Discovery Protocol

---

**Answer: B,C**

---

Explanation:

---

**Question: 38**

---

Which output displays a JSON data representation?

<https://www.validexamdumps.com>

A. {  
 "response": {  
 "taskId": {},  
 "url": "string"  
 },  
 "version": "string"  
}

B. {  
 "response" - {  
 "taskId" - {},  
 "url" - "string"  
 },  
 "version" - "string"  
}

C. {  
 "response": {  
 "taskId": {},  
 "url": "string"  
 },  
 "version": "string"  
}

D. {  
 "response". {  
 "taskId". {},  
 "url". "string"  
 },  
 "version". "string"  
}

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: C**

---

Explanation:

JSON data is written as name/value pairs.

A name/value pair consists of a field name (in double quotes), followed by a colon, followed by a value:

"name": "Mark"

JSON can use arrays. Array values must be of type string, number, object, array, boolean or null.

For example:

```
{  
  "name": "John",  
  "age": 30,  
  "cars": [ "Ford", "BMW", "Fiat" ]  
}
```

JSON can have empty object like "taskId": {}

---

**Question: 39**

---

Which command is used to specify the delay time in seconds for LLDP to initialize on any interface?

- A. lldp timer

- B. lldp holdtime
- C. lldp reinit
- D. lldp tlv-select

---

**Answer: C**

---

Explanation:

Reference: [https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst2960/software/release/12-2\\_37\\_ey/configuration/guide/scg/swlldp.pdf](https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst2960/software/release/12-2_37_ey/configuration/guide/scg/swlldp.pdf)

+ lldp holdtime seconds: Specify the amount of time a receiving device should hold the information from your device before discarding it  
+ lldp reinit delay: Specify the delay time in seconds for LLDP to initialize on an interface  
+ lldp timer rate: Set the sending frequency of LLDP updates in seconds

---

**Question: 40**

---

A network engineer must back up 20 network router configurations globally within a customer environment. Which protocol allows the engineer to perform this function using the Cisco IOS MIB?

- A. CDP
- B. SNMP
- C. SMTP
- D. ARP

---

**Answer: B**

---

Explanation:

SNMP is an application-layer protocol that provides a message format for communication between SNMP managers and agents. SNMP provides a standardized framework and a common language



used for the monitoring and management of devices in a network.

The SNMP framework has three parts:

- + An SNMP manager
- + An SNMP agent
- + A Management Information Base (MIB)

The Management Information Base (MIB) is a virtual information storage area for network management information, which consists of collections of managed objects.

With SNMP, the network administrator can send commands to multiple routers to do the backup

---

**Question: 41**

---

DRAG DROP

Drag and drop the threat-mitigation techniques from the left onto the types of threat or attack they mitigate on the right.

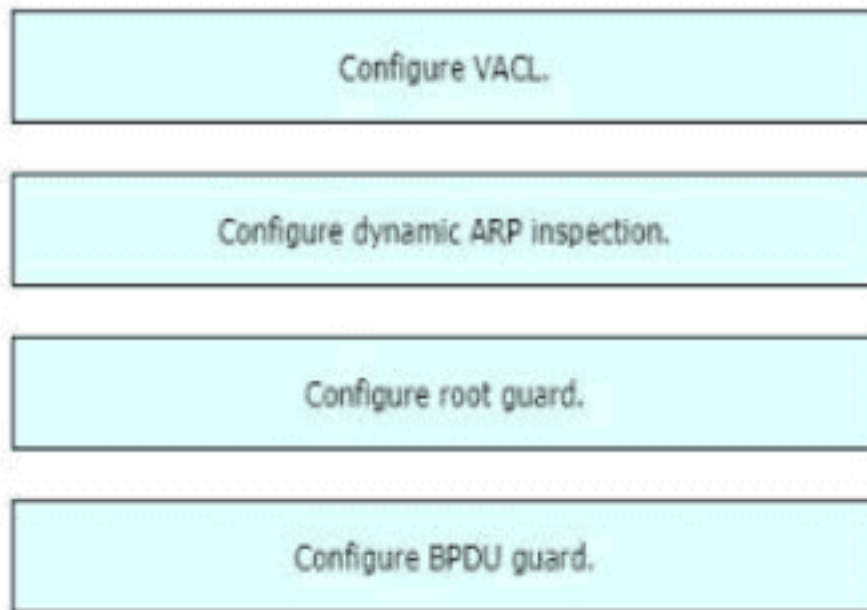
|                                   |                                               |
|-----------------------------------|-----------------------------------------------|
| Configure BPDU guard.             | 802.1q double tagging                         |
| Configure dynamic ARP inspection. | ARP spoofing                                  |
| Configure root guard.             | unwanted superior BPDUs                       |
| Configure VACL.                   | unwanted BPDUs on PortFast-enabled interfaces |

---

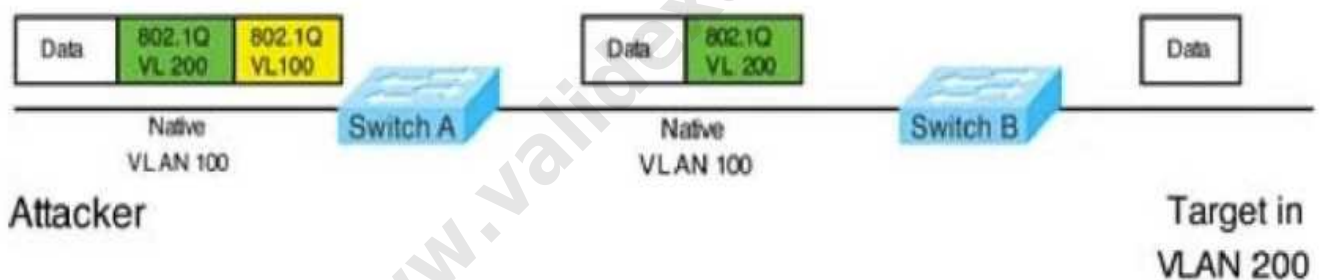
**Answer:**

---

Explanation:



Double-Tagging attack:



In this attack, the attacking computer generates frames with two 802.1Q tags. The first tag matches the native VLAN of the trunk port (VLAN 10 in this case), and the second matches the VLAN of a host it wants to attack (VLAN 20).

When the packet from the attacker reaches Switch A, Switch A only sees the first VLAN 10 and it matches with its native VLAN 10 so this VLAN tag is removed. Switch A forwards the frame out all links with the same native VLAN 10. Switch B receives the frame with a tag of VLAN 20 so it removes this tag and forwards out to the Victim computer.

Note: This attack only works if the trunk (between two switches) has the same native VLAN as the attacker.

To mitigate this type of attack, you can use VLAN access control lists (VACLs, which applies to all traffic within a VLAN. We can use VACL to drop attacker traffic to specific victims/servers) or implement Private VLANs.

ARP attack (like ARP poisoning/spoofing) is a type of attack in which a malicious actor sends falsified ARP messages over a local area network as ARP allows a gratuitous reply from a host even if an ARP

request was not received. This results in the linking of an attacker's MAC address with the IP address of a legitimate computer or server on the network. This is an attack based on ARP which is at Layer 2.

Dynamic ARP inspection (DAI) is a security feature that validates ARP packets in a network which can be used to mitigate this type of attack.

---

**Question: 42**

---

DRAG DROP

Drag and drop the network protocols from the left onto the correct transport services on the right.

SMTP

SNMP

TFTP

VoIP

SSH

FTP

Connection Oriented

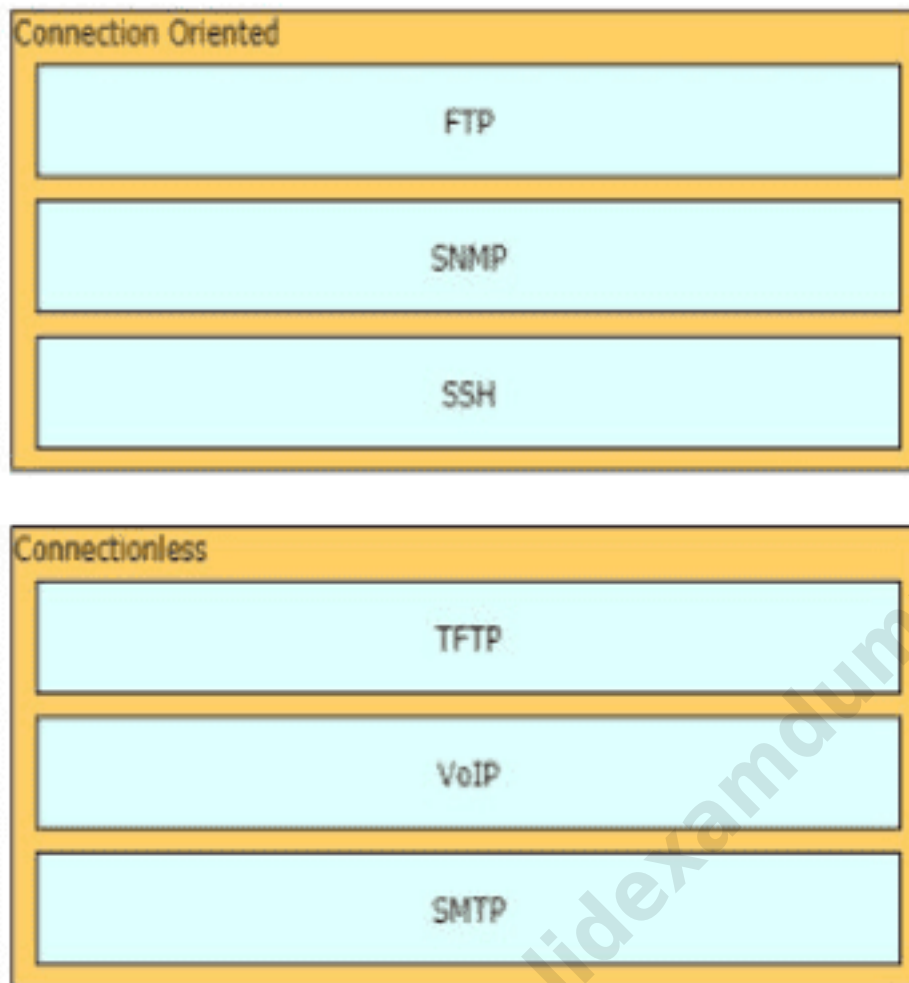
Connectionless

---

**Answer:**

---

Explanation:



---

**Question: 43**

---

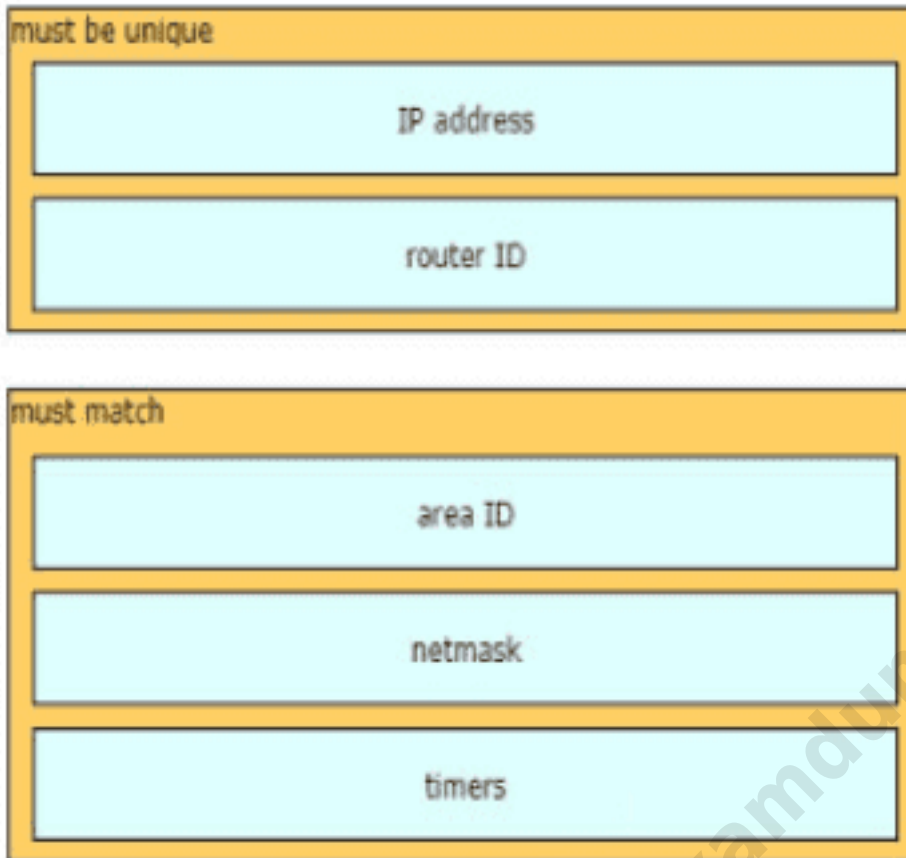
DRAG DROP

A network engineer is configuring an OSPFv2 neighbor adjacency. Drag and drop the parameters from the left onto their required categories on the right. Not all parameters are used.

|                 |                |
|-----------------|----------------|
| netmask         | must be unique |
| OSPF process ID |                |
| router ID       | must match     |
| IP address      |                |
| area ID         |                |
| timers          |                |

Explanation:

**Answer:**

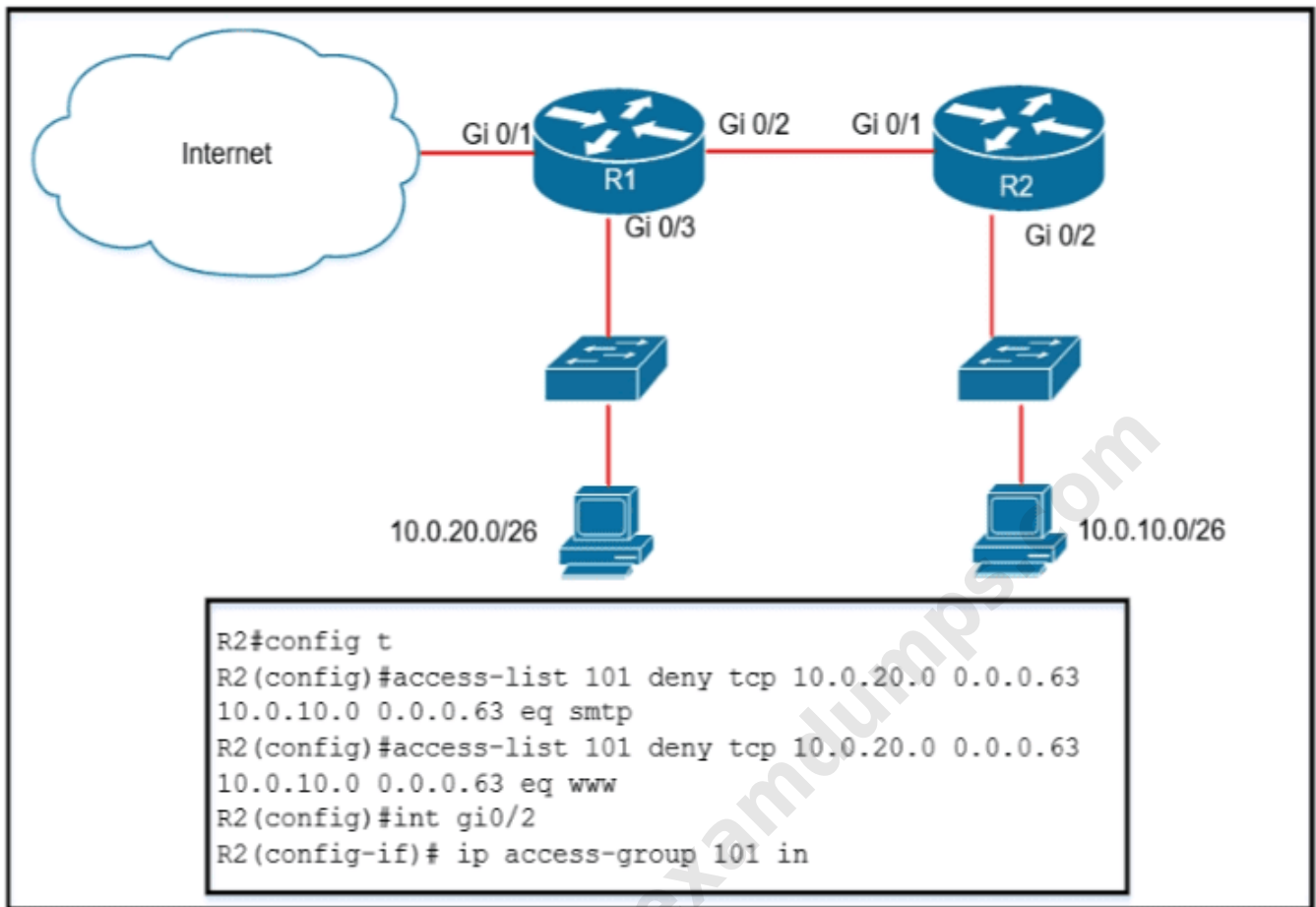


---

**Question: 44**

---

Refer to the exhibit.



An extended ACL has been configured and applied to router R2. The configuration failed to work as intended. Which two

changes stop outbound traffic on TCP ports 25 and 80 to 10.0.20.0/26 from the 10.0.10.0/26 subnet while still allowing all other traffic? (Choose

two )

- A. Add a "permit ip any any" statement to the beginning of ACL 101 for allowed traffic.
- B. Add a "permit ip any any" statement at the end of ACL 101 for allowed traffic.
- C. The source and destination IPs must be swapped in ACL 101.
- D. The ACL must be configured on the Gi0/2 interface inbound on R1.
- E. The ACL must be moved to the Gi0/1 interface outbound on R2.

---

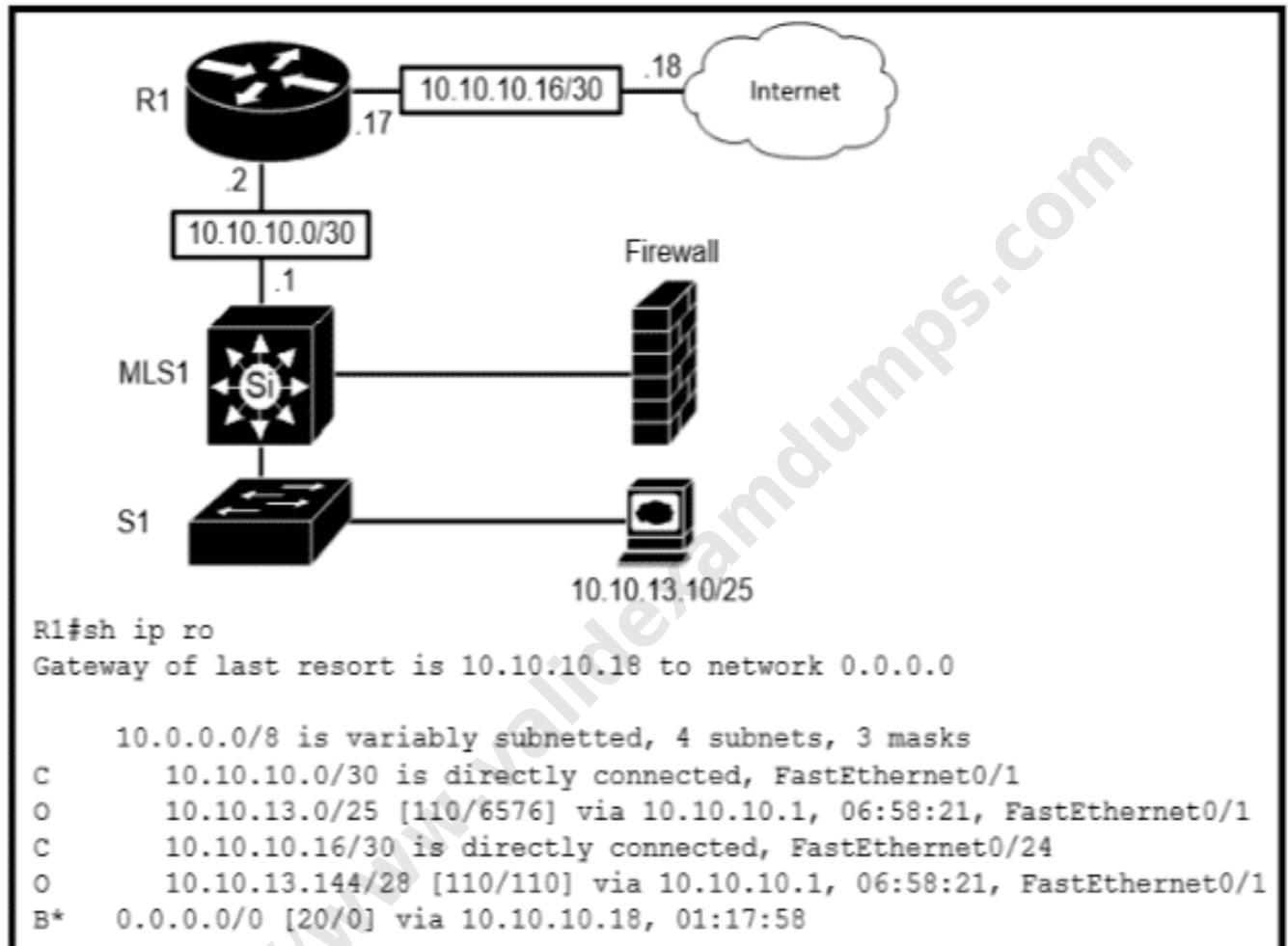
**Answer: B,C**

---

Explanation:

**Question: 45**

Refer to the exhibit.



Which type of route does R1 use to reach host 10.10.13.10/32?

- A. floating static route
- B. host route
- C. default route
- D. network route



---

**Answer: D**

---

Explanation:

From the output, we see R1 will use the entry "O 10.10.13.0/25 [110/4576] via 10.10.10.1, ..." to reach host 10.10.13.10. This is a network route.

Note: "B\* 0.0.0.0/0 ..." is a default route.

---

**Question: 46**

---

Which mode must be used to configure EtherChannel between two switches without using a negotiation protocol?

- A. on
- B. auto
- C. active
- D. desirable

---

**Answer: A**

---

Explanation:

The Static Persistence (or "on" mode) bundles the links unconditionally and no negotiation protocol is used. In this mode, neither PAGP nor LACP packets are sent or received.

---

**Question: 47**

---

Which IPv6 address block sends packets to a group address rather than a single address?

- A. 2000::/3

- B. FC00::/7
- C. FE80::/10
- D. FF00::/8

---

**Answer: D**

---

Explanation:

FF00::/8 is used for IPv6 multicast and this is the IPv6 type of address the question wants to ask.

FE80::/10 range is used for link-local addresses. Link-local addresses only used for communications within the local subnetwork (automatic address configuration, neighbor discovery, router discovery, and by many routing protocols). It is only valid on the current subnet.

It is usually created dynamically using a link-local prefix of FE80::/10 and a 64-bit interface identifier (based on 48-bit MAC address).

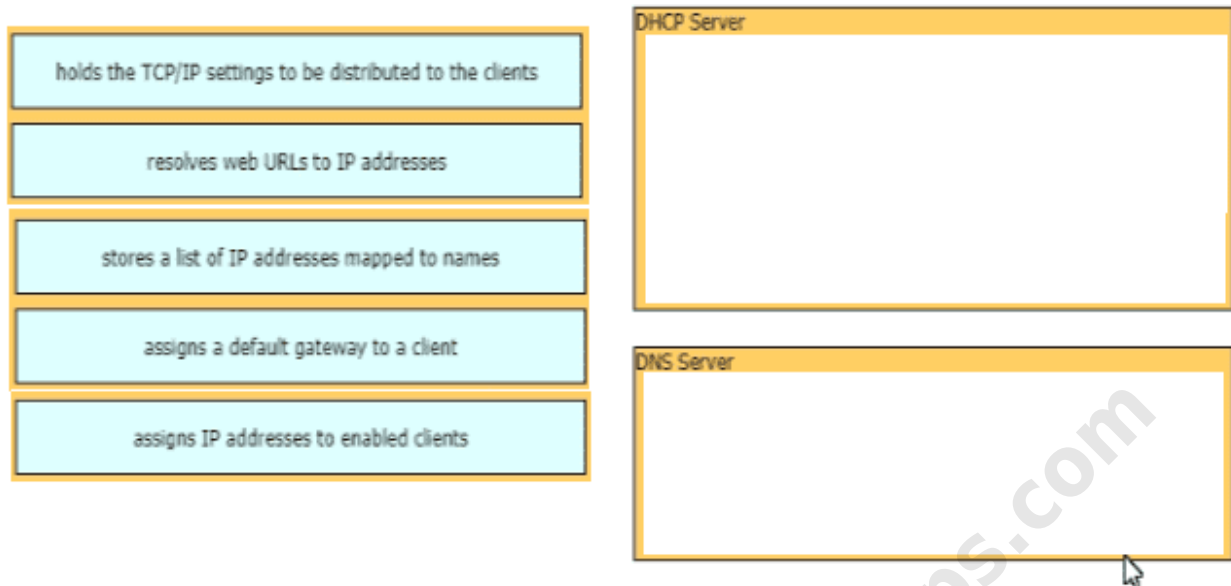
---

**Question: 48**

---

DRAG DROP

Drag and drop the functions from the left onto the correct network components on the right

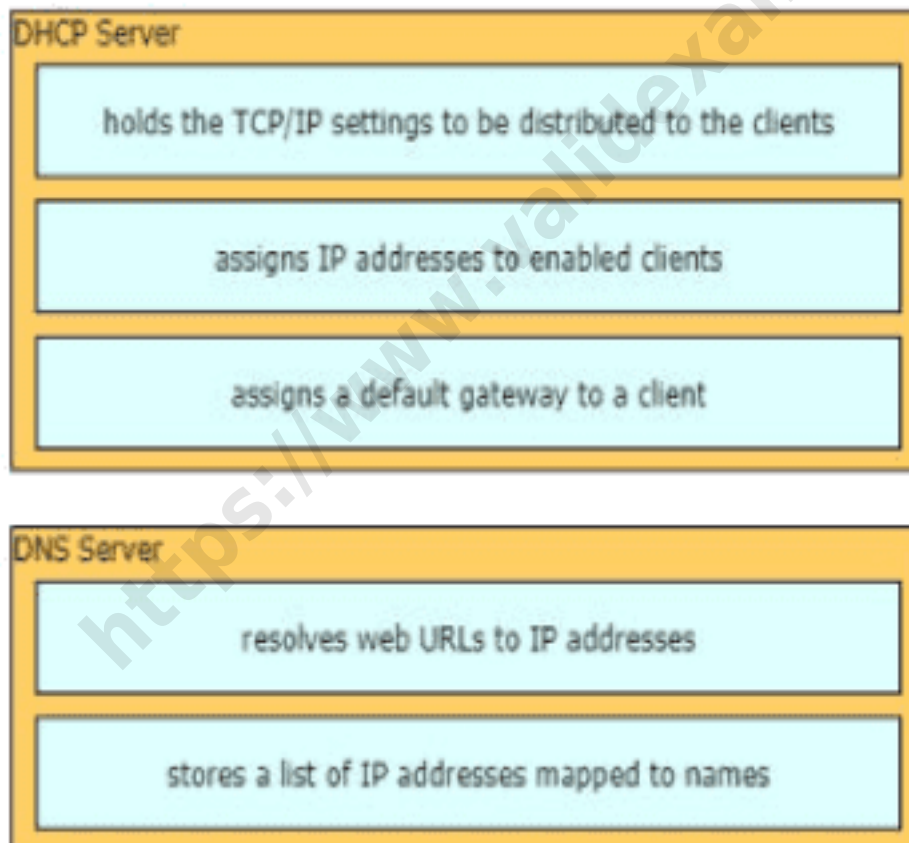


---

**Answer:**

---

Explanation:



---

**Question: 49**

---

Which two capacities of Cisco DNA Center make it more extensible as compared to traditional campus device management? (Choose two)

- A. adapters that support all families of Cisco IOS software
- B. SDKs that support interaction with third-party network equipment
- C. customized versions for small, medium, and large enterprises
- D. REST APIs that allow for external applications to interact natively with Cisco DNA Center
- E. modular design that is upgradable as needed

---

**Answer: B,D**

---

Explanation:

Cisco DNA Center offers 360-degree extensibility through four distinct types of platform capabilities:

+ Intent-based APIs leverage the controller and enable business and IT applications to deliver intent to the network and to reap network analytics and insights for IT and business innovation.

+ Process adapters, built on integration APIs, allow integration with other IT and network systems to streamline IT operations and processes.

+ Domain adapters, built on integration APIs, allow integration with other infrastructure domains such as data center, WAN, and security to deliver a consistent intent-based infrastructure across the entire IT environment.

+ SDKs allow management to be extended to third-party vendor's network devices to offer support for diverse environments.

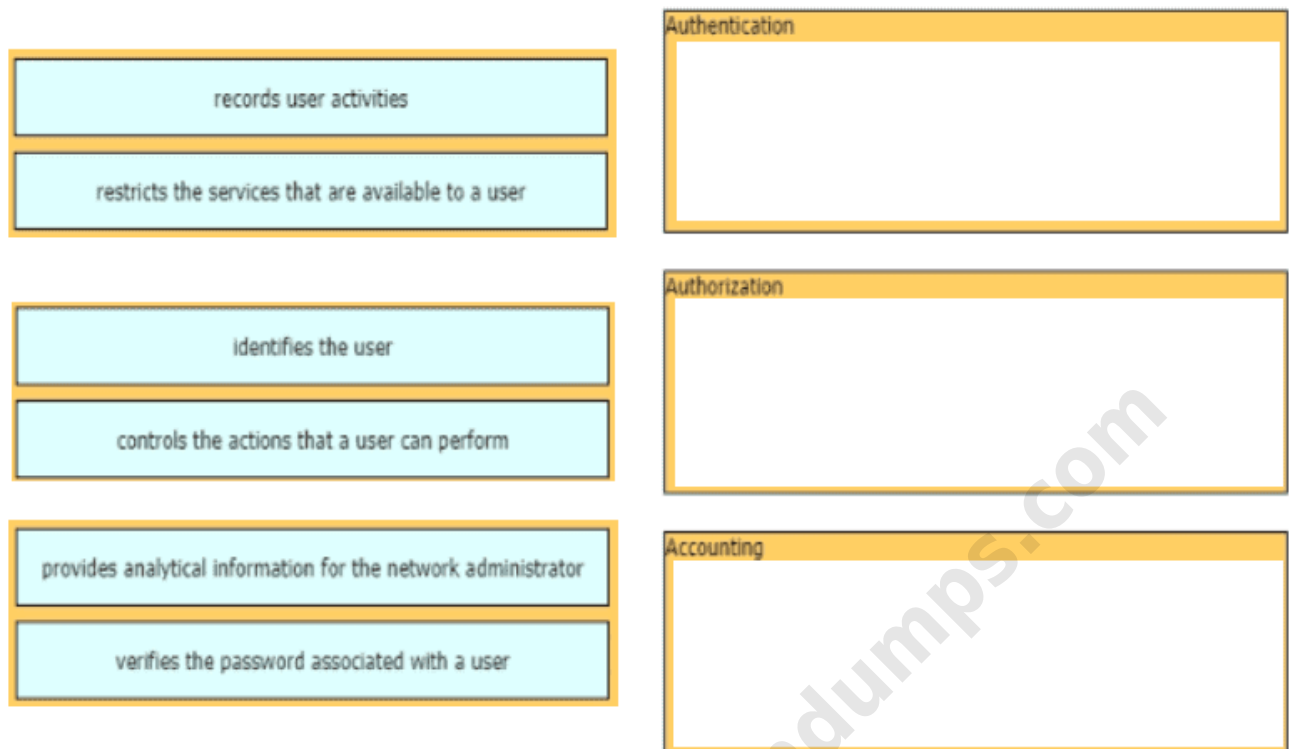
---

### Question: 50

---

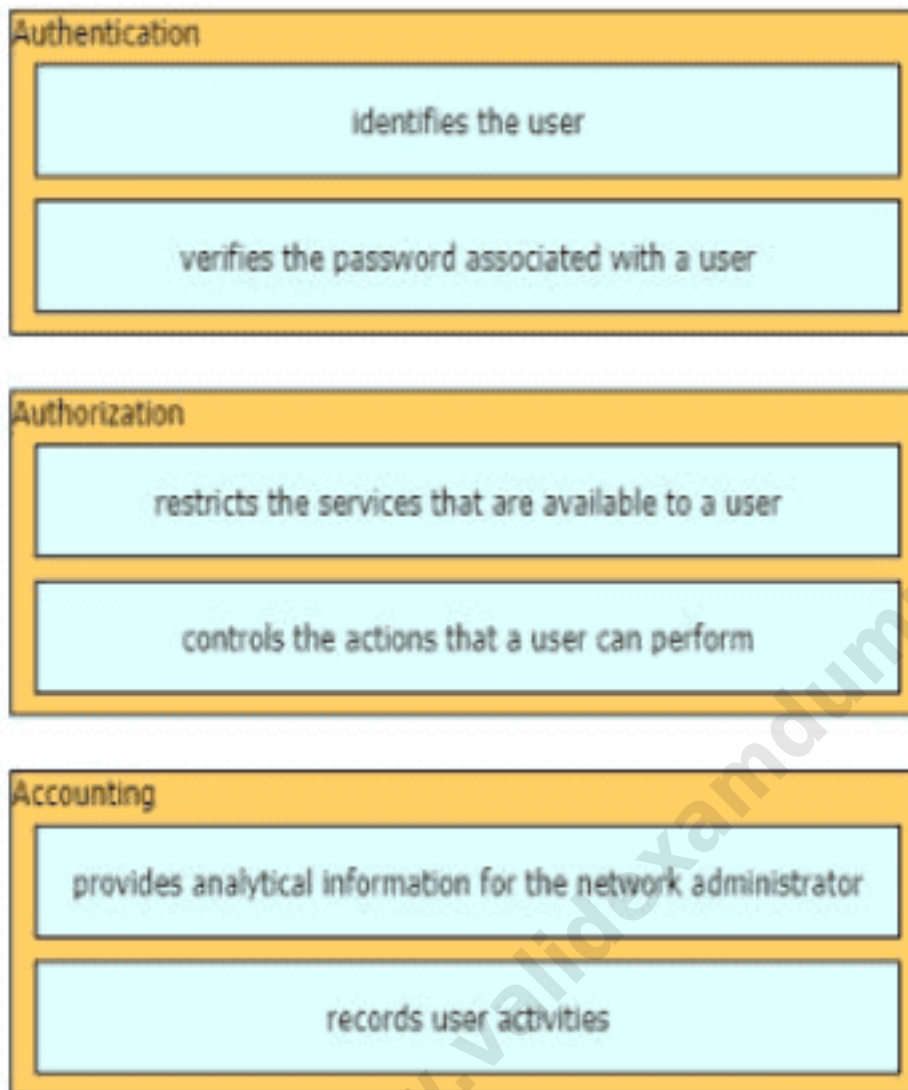
DRAG DROP

Drag and drop the AAA functions from the left onto the correct AAA services on the right



**Answer:**

Explanation:



---

**Question: 51**

---

What is the primary effect of the spanning-tree portfast command?

- A. it enables BPDU messages
- B. It minimizes spanning-tree convergence time
- C. It immediately puts the port into the forwarding state when the switch is reloaded
- D. It immediately enables the port in the listening state

---

**Answer: B**

---

Explanation:

Reference: [https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst3560/software/release/12-2\\_55\\_se/configuration/guide/3560\\_scg/swstpopt.html](https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst3560/software/release/12-2_55_se/configuration/guide/3560_scg/swstpopt.html)

---

**Question: 52**

---

DRAG DROP

Drag and drop the IPv4 network subnets from the left onto the correct usable host ranges on the right

|                   |                                 |
|-------------------|---------------------------------|
| 172.28.228.144/18 | 172.28.228.1 - 172.28.229.254   |
| 172.28.228.144/21 | 172.28.224.1 - 172.28.231.254   |
| 172.28.228.144/23 | 172.28.228.129 - 172.28.228.254 |
| 172.28.228.144/25 | 172.28.228.145 - 172.28.228.150 |
| 172.28.228.144/29 | 172.28.192.1 - 172.28.255.254   |

---

**Answer:**

---

Explanation:

172.28.228.144/23

172.28.228.144/21

172.28.228.144/25

172.28.228.144/29

172.28.228.144/18

---

**Question: 53**

---

Refer to the exhibit.



**Router#**

**Capability Codes:** R - Router, T - Trans Bridge, B - Source Route Bridge  
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,  
D - Remote, C - CVTA, M - Two-port Mac Relay

| Device ID | Local Intrfce | Holdtme | Capability | Platform | Port ID   |
|-----------|---------------|---------|------------|----------|-----------|
| 10.1.1.2  | Gig 37/3      | 176     | R I        | CPT 600  | Gig 36/41 |
| 10.1.1.2  | Gig 37/1      | 174     | R I        | CPT 600  | Gig 36/43 |
| 10.1.1.2  | Gig 36/41     | 134     | R I        | CPT 600  | Gig 37/3  |
| 10.1.1.2  | Gig 36/43     | 134     | R I        | CPT 600  | Gig 37/1  |
| 10.1.1.2  | Ten 3/2       | 132     | R I        | CPT 600  | Ten 4/2   |
| 10.1.1.2  | Ten 4/2       | 174     | R I        | CPT 600  | Ten 3/2   |

Which command provides this output?

- A. show ip route
- B. show ip interface
- C. show interface
- D. show cdp neighbor

**Answer: D**

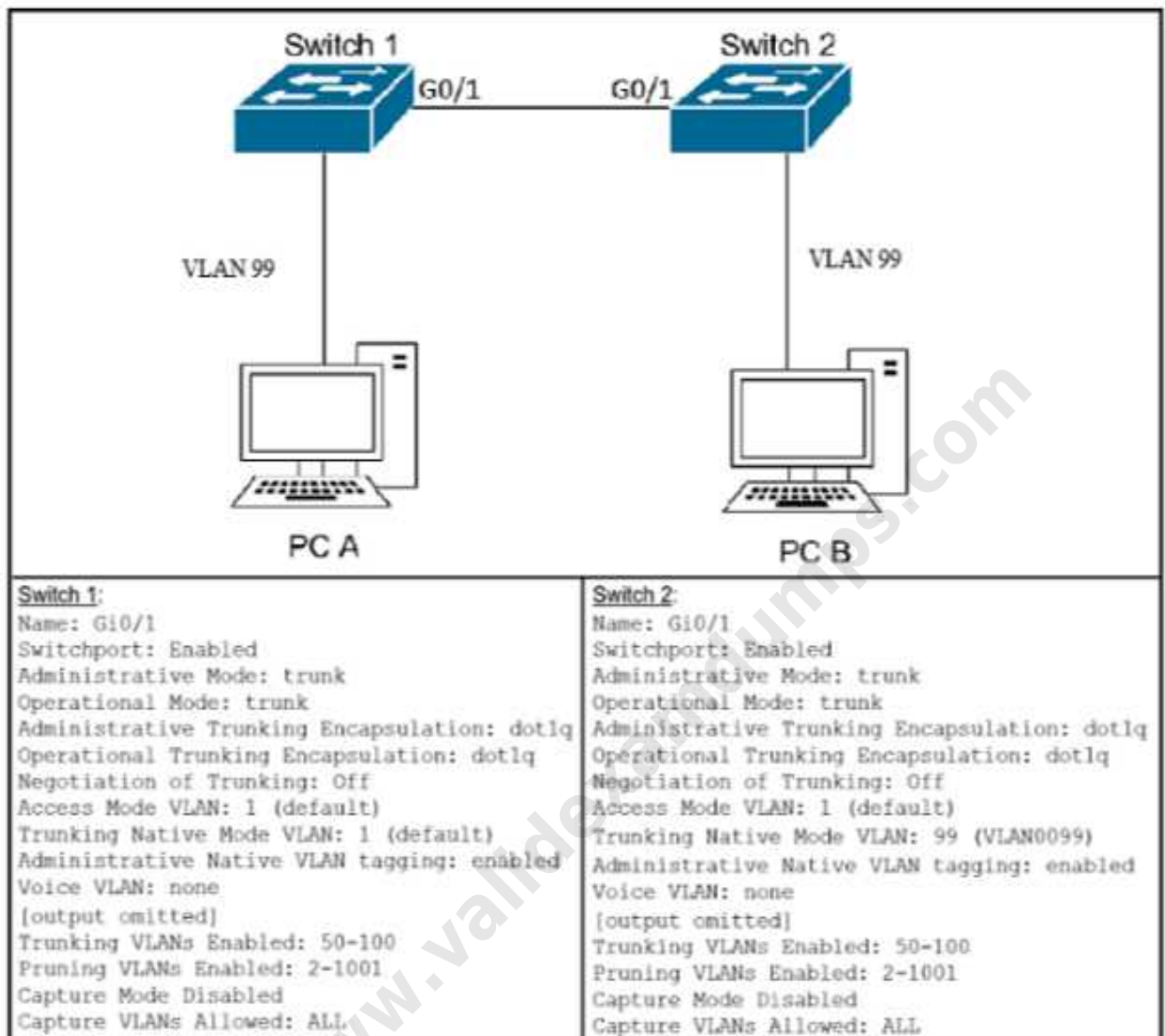
Explanation:

---

**Question: 54**

---

Refer to the Exhibit.



After the switch configuration the ping test fails between PC A and PC B Based on the output for switch 1. which error must be corrected?

- A. There is a native VLAN mismatch
- B. Access mode is configured on the switch ports.
- C. The PCs are in the incorrect VLAN
- D. All VLANs are not enabled on the trunk

**Answer: A**

Explanation:

From the output we see the native VLAN of Switch1 on Gi0/1 interface is VLAN 1 while that of Switch2 is VLAN 99 so there would be a native VLAN mismatch.

---

**Question: 55**

---

Which API is used in controller-based architectures to interact with edge devices?

- A. overlay
- B. northbound
- C. underlay
- D. southbound

---

**Answer: D**

---

Explanation:

---

**Question: 56**

---

Which statement identifies the functionality of virtual machines?

- A. Virtualized servers run most efficiently when they are physically connected to a switch that is separate from the hypervisor
- B. The hypervisor can virtualize physical components including CPU, memory, and storage
- C. Each hypervisor can support a single virtual machine and a single software switch
- D. The hypervisor communicates on Layer 3 without the need for additional resources

---

**Answer: B**

---

Explanation:

---

**Question: 57**

---

Which type of address is the public IP address of a NAT device?

- A. outside global
- B. outside local
- C. inside global
- D. inside local
- E. outside public
- F. inside public

---

**Answer: C**

---

Explanation:

NAT use four types of addresses:

\* Inside local address – The IP address assigned to a host on the inside network. The address is usually not an IP address assigned by the Internet Network Information Center (InterNIC) or service provider.

This address is likely to be an RFC 1918 private address.

\* Inside global address – A legitimate IP address assigned by the InterNIC or service provider that represents one or more inside local IP addresses to the outside world.

\* Outside local address – The IP address of an outside host as it is known to the hosts on the inside network.

\* Outside global address – The IP address assigned to a host on the outside network. The owner of the host assigns this address.

---

**Question: 58**

---

Which option about JSON is true?

- A. uses predefined tags or angle brackets ( ) to delimit markup text
- B. used to describe structured data that includes arrays
- C. used for storing information
- D. similar to HTML, it is more verbose than XML

---

**Answer: B**

---

Explanation:

JSON data is written as name/value pairs.

A name/value pair consists of a field name (in double quotes), followed by a colon, followed by a value:

"name": "Mark"

JSON can use arrays. Array values must be of type string, number, object, array, boolean or null..

For example:

```
{  
  "name": "John",  
  "age": 30,  
  "cars": [ "Ford", "BMW", "Fiat" ]  
}
```

---

### Question: 59

---

How do TCP and UDP differ in the way they provide reliability for delivery of packets?

- A. TCP is a connectionless protocol that does not provide reliable delivery of data, UDP is a connection-oriented protocol that uses sequencing to provide reliable delivery.
- B. TCP does not guarantee delivery or error checking to ensure that there is no corruption of data UDP provides message acknowledgement and retransmits data if lost.
- C. TCP provides flow control to avoid overwhelming a receiver by sending too many packets at once, UDP sends packets to the receiver in a continuous stream without checking for sequencing
- D. TCP uses windowing to deliver packets reliably; UDP provides reliable message transfer between hosts by establishing a three-way handshake

---

**Answer: C**

---

Explanation:

---

**Question: 60**

---

Which two command sequences must you configure on switch to establish a Layer 3 EtherChannel with an open-standard protocol? (Choose two )

- A. interface GigabitEthernet0/0/1channel-group 10 mode on
- B. interface GigabitEthernet0/0/1channel-group 10 mode active
- C. interface GigabitEthernet0/0/1channel-group 10 mode auto
- D. interface port-channel 10switchportswitchport mode trunk
- E. interface port-channel 10no switchportip address 172.16.0.1.255.255.255.0

---

**Answer: B,E**

---

Explanation:

---

**Question: 61**

---

What is an advantage of Cisco DNA Center versus traditional campus device management?

- A. It supports numerous extensibility options including cross-domain adapters and third-party SDKs.

- B. It supports high availability for management functions when operating in cluster mode.
- C. It enables easy autodiscovery of network elements in a brownfield deployment.
- D. It is designed primarily to provide network assurance.

---

**Answer: A**

---

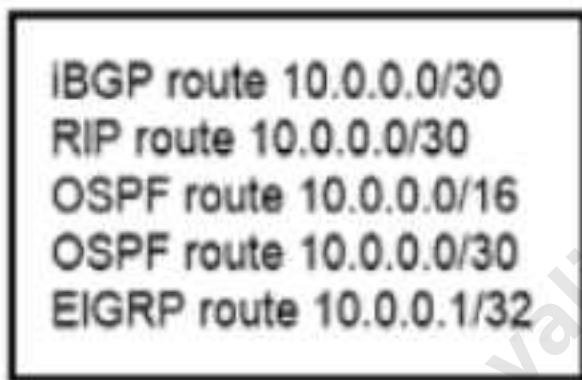
Explanation:

---

**Question: 62**

---

Refer to the exhibit.



```
iBGP route 10.0.0.0/30
RIP route 10.0.0.0/30
OSPF route 10.0.0.0/16
OSPF route 10.0.0.0/30
EIGRP route 10.0.0.1/32
```

A router reserved these five routes from different routing information sources.

Which two routes does the router install in its routing table? (Choose two)

- A. RIP route 10.0.0.0/30
- B. iBGP route 10.0.0.0/30
- C. OSPF route 10.0.0.0/30
- D. EIGRP route 10.0.0.1/32
- E. OSPF route 10.0.0.0/16

---

**Answer: C,D**

---

Explanation:

---

**Question: 63**

---

By default, how Does EIGRP determine the metric of a route for the routing table?

- A. it uses the bandwidth and delay values of the path to calculate the route metric
- B. it uses a default metric of 10 for all routes that are learned by the router
- C. it uses a reference Bandwidth and the actual bandwidth of the connected link to calculate the route metric
- D. it counts the number of hops between the receiving and destination routers and uses that value as the metric

---

**Answer: A**

---

Explanation:

---

**Question: 64**

---

What is a difference between local AP mode and FlexConnect AP mode?

- A. Local AP mode creates two CAPWAP tunnels per AP to the WLC
- B. FlexConnect AP mode fails to function if the AP loses connectivity with the WLC
- C. FlexConnect AP mode bridges the traffic from the AP to the WLC when local switching is configured
- D. Local AP mode causes the AP to behave as if it were an autonomous AP

---

**Answer: A**

---

Explanation:



---

**Question: 65**

---

Router R1 must send all traffic without a matching routing-table entry to 192.168.1.1. Which configuration accomplishes this task?

- ☐ R1#Config t  
R1(config)#ip routing  
R1(config)#ip route default-route 192.168.1.1
- ☐ R1#Config t  
R1(config)#ip routing  
R1(config)#ip route 192.168.1.1 0.0.0.0 0.0.0.0
- ☐ R1#Config t  
R1(config)#ip routing  
R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.1.1
- ☐ R1#Config t  
R1(config)#ip routing  
R1(config)#ip default-gateway 192.168.1.1

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: C**

---

Explanation:

---

**Question: 66**

---

Which function does the range of private IPv4 addresses perform?

- A. allows multiple companies to each use the same addresses without conflicts
- B. provides a direct connection for hosts from outside of the enterprise network
- C. ensures that NAT is not required to reach the internet with private range addressing
- D. enables secure communications to the internet for all external hosts

---

**Answer: A**

---

Explanation:

---

**Question: 67**

---

What event has occurred if a router sends a notice level message to a syslog server?

- A. A TCP connection has been torn down
- B. An ICMP connection has been built
- C. An interface line has changed status
- D. A certificate has expired.

---

**Answer: C**

---

Explanation:

---

**Question: 68**

---

Refer to the exhibit.

```
SW1#show spanning-tree vlan 30

VLAN0030
Spanning tree enabled protocol rstp
Root ID      Priority          32798
             Address          0025.63e9.c800
             Cost            19
             Port            1 (FastEthernet 2/1)
             Hello Time      2 sec
             Max Age         30 sec
             Forward Delay   20 sec

[Output suppressed]
```

What two conclusions should be made about this configuration? (Choose two )

- A. The designated port is FastEthernet 2/1
- B. This is a root bridge
- C. The spanning-tree mode is Rapid PVST+
- D. The spanning-tree mode is PVST+
- E. The root port is FastEthernet 2/1

---

**Answer: C,E**

---

Explanation:

---

### Question: 69

---

What are two fundamentals of virtualization? (choose two)

- A. The environment must be configured with one hypervisor that serves solely as a network manager to monitor SNMP traffic
- B. It allows logical network devices to move traffic between virtual machines and the rest of the physical network
- C. It allows multiple operating systems and applications to run independently on one physical server.
- D. It allows a physical router to directly connect NICs from each virtual machine into the network
- E. It requires that some servers, virtual machines and network gear reside on the Internet

---

**Answer: B,C**

---

Explanation:

---

**Question: 70**

---

What is the difference regarding reliability and communication type between TCP and UDP?

- A. TCP is reliable and is a connection-oriented protocol UDP is not reliable and is a connectionless protocol
- B. TCP is not reliable and is a connection-oriented protocol; UDP is reliable and is a connectionless protocol
- C. TCP is not reliable and is a connectionless protocol; UDP is reliable and is a connection-oriented protocol
- D. TCP is reliable and is a connectionless protocol; UDP is not reliable and is a connection-oriented protocol

---

**Answer: A**

---

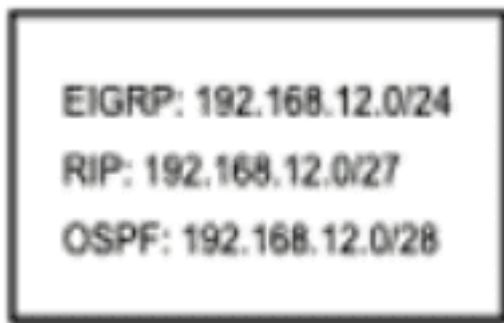
Explanation:

---

**Question: 71**

---

Refer to the exhibit.



How does the router manage traffic to 192.168.12.16?

- A. It selects the RIP route because it has the longest prefix inclusive of the destination address.
- B. It chooses the OSPF route because it has the longest prefix inclusive of the destination address.
- C. it load-balances traffic between all three routes
- D. It chooses the EIGRP route because it has the lowest administrative distance

---

**Answer: A**

---

Explanation:

---

### Question: 72

---

How does Cisco DNA Center gather data from the network?

- A. Network devices use different services like SNMP, syslog, and streaming telemetry to send data to the controller
- B. Devices establish an IPsec tunnel to exchange data with the controller
- C. Devices use the call-home protocol to periodically send data to the controller.
- D. The Cisco CU Analyzer tool gathers data from each licensed network device and streams it to the controller.

---

**Answer: A**

---

Explanation:

---

**Question: 73**

---

DRAG DROP

Drag and drop the attack-mitigation techniques from the left onto the Types of attack that they mitigate on the right.

|                                                     |                                           |
|-----------------------------------------------------|-------------------------------------------|
| configure 802.1x authentication                     | 802.1q double-tagging VLAN-hopping attack |
| configure DHCP snooping                             | MAC flooding attack                       |
| configure the native VLAN with a nondefault VLAN ID | man-in-the-middle spoofing attack         |
| disable DTP                                         | switch-spoofing VLAN-hopping attack       |

---

**Answer:**

---

Explanation:

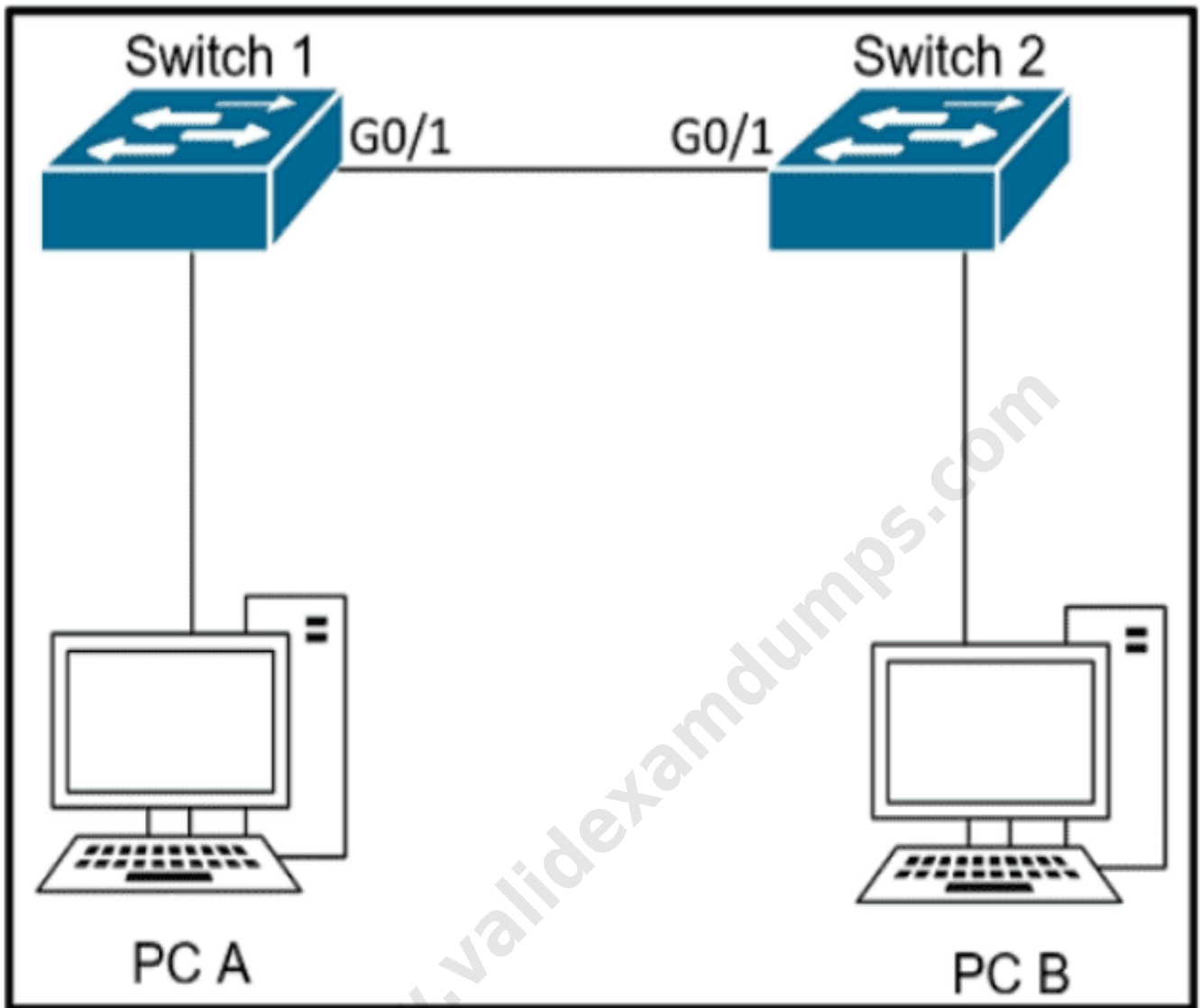
|                                           |                                                     |
|-------------------------------------------|-----------------------------------------------------|
| 802.1q double-tagging VLAN-hopping attack | configure the native VLAN with a nondefault VLAN ID |
| MAC flooding attack                       | configure 802.1x authentication                     |
| man-in-the-middle spoofing attack         | configure DHCP snooping                             |
| switch-spoofing VLAN-hopping attack       | disable DTP                                         |

---

**Question: 74**

---

Refer to the exhibit.



The network administrator wants VLAN 67 traffic to be untagged between Switch 1 and Switch 2 while all other VLANs are to remain tagged.

Which command accomplishes this task?

- A. switchport access vlan 67
- B. switchport trunk allowed vlan 67
- C. switchport private-vlan association host 67
- D. switchport trunk native vlan 67

---

**Answer: D**

---

Explanation:

---

**Question: 75**

---

What are two roles of the Dynamic Host Configuration Protocol (DHCP)? (Choose two)

- A. The DHCP server offers the ability to exclude specific IP addresses from a pool of IP addresses
- B. The DHCP client can request up to four DNS server addresses
- C. The DHCP server assigns IP addresses without requiring the client to renew them
- D. The DHCP server leases client IP addresses dynamically.
- E. The DHCP client maintains a pool of IP addresses it can assign.

---

**Answer: A,D**

---

Explanation:

---

**Question: 76**

---

Which two minimum parameters must be configured on an active interface to enable OSPFv2 to operate? (Choose two)

- A. OSPF area
- B. OSPF MD5 authentication key
- C. IPv6 address
- D. OSPF process ID
- E. OSPF stub flag



---

**Answer: A,D**

---

Explanation:

---

**Question: 77**

---

Which two outcomes are predictable behaviors for HSRP? (Choose two.)

- A. The two routers synchronize configurations to provide consistent packet forwarding
- B. The two routers negotiate one router as the active router and the other as the standby router
- C. Each router has a different IP address, both routers act as the default gateway on the LAN, and traffic is load-balanced between them
- D. The two routers share a virtual IP address that is used as the default gateway for devices on the LAN
- E. The two routers share the same interface IP address and default gateway traffic is load-balanced between them

---

**Answer: B,D**

---

Explanation:

---

**Question: 78**

---

Several new coverage cells are required to improve the Wi-Fi network of an organization. Which two standard designs are recommended? (choose two.)

- A. 5GHz provides increased network capacity with up to 23 nonoverlapping channels.
- B. For maximum throughput, the WLC is configured to dynamically set adjacent access points to the same channel.
- C. 5GHz channel selection requires an autonomous access point.
- D. Adjacent cells with overlapping channels use a repeater access point.

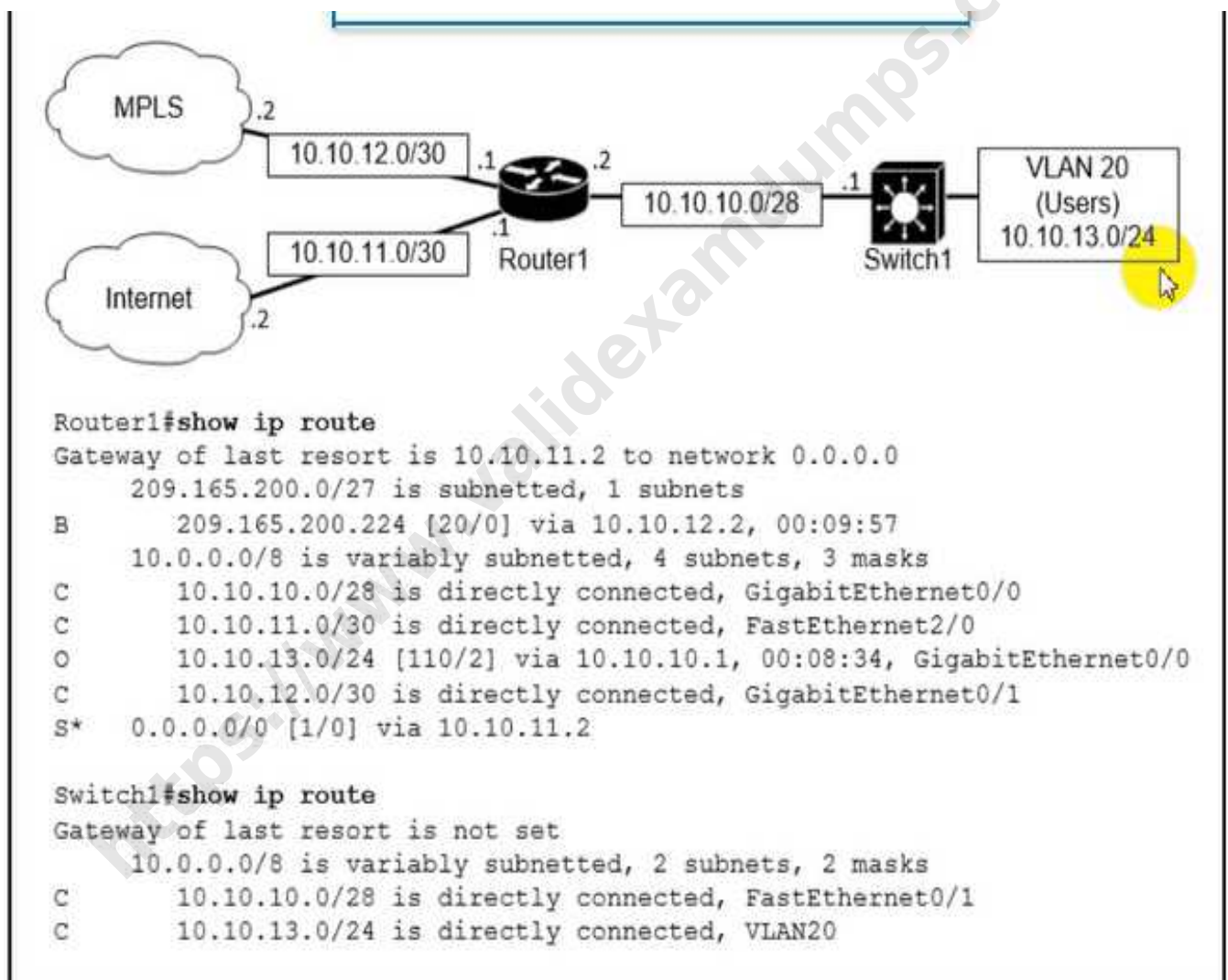
E. Cells that overlap one another are configured to use nonoverlapping channels.

**Answer: B,E**

Explanation:

### Question: 79

Refer to the exhibit.



which path is used by the router for internet traffic ?

A. 209.165.200.0/27

B. 10.10.10.0/28

C. 0.0.0.0/0

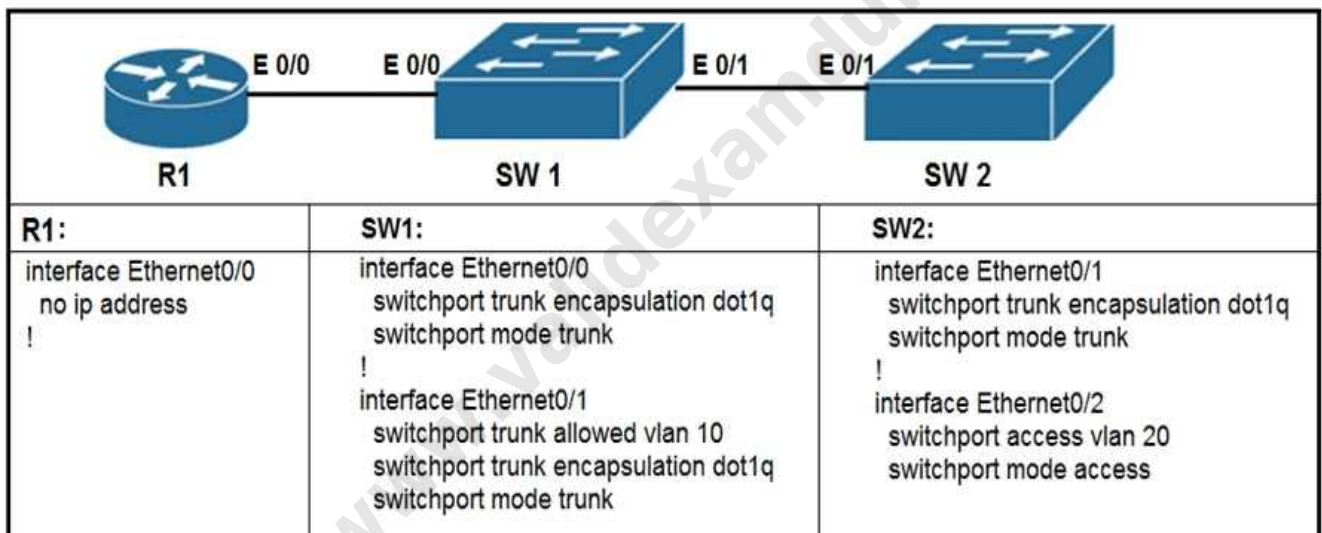
D. 10.10.13.0/24

**Answer: C**

Explanation:

**Question: 80**

Refer to the exhibit.



What commands are needed to add a subinterface to Ethernet0/0 on R1 to allow for VLAN 20, with IP address 10.20.20.1/24?

- A. R1(config)#interface ethernet0/0R1(config)#encapsulation dot1q 20R1(config)#ip address 10.20.20.1 255.255.255.0
- B. R1(config)#interface ethernet0/0.20R1(config)#encapsulation dot1q 20R1(config)#ip address 10.20.20.1 255.255.255.0
- C. R1(config)#interface ethernet0/0.20R1(config)#ip address 10.20.20.1 255.255.255.0
- D. R1(config)#interface ethernet0/0R1(config)#ip address 10.20.20.1 255.255.255.0

---

**Answer: B**

---

Explanation:

---

**Question: 81**

---

which purpose does a northbound API serve in a controller-based networking architecture?

- A. communicates between the controller and the physical network hardware
- B. reports device errors to a controller
- C. generates statistics for network hardware and traffic
- D. facilitates communication between the controller and the applications

---

**Answer: D**

---

Explanation:

---

**Question: 82**

---

Refer to the exhibit.

```
ip arp inspection vlan 2-10
interface fastethernet 0/1
    ip arp inspection trust
```

If the network environment is operating normally, which type of device must be connected to interface FastEthernet 0/1?

- A. DHCP client
- B. access point
- C. router
- D. PC

---

**Answer: C**

---

Explanation:

---

**Question: 83**

---

What is the primary purpose of a First Hop Redundancy Protocol?

- A. It allows directly connected neighbors to share configuration information.
- B. It allows a router to use bridge priorities to create multiple loop-free paths to a single destination.
- C. It reduces routing failures by allowing Layer 3 load balancing between OSPF neighbors that have the same link metric.
- D. It reduces routing failures by allowing more than one router to represent itself, as the default gateway of a network.

---

**Answer: D**

---

Explanation:

---

**Question: 84**

---

What occurs to frames during the process of frame flooding?

- A. Frames are sent to every port on the switch in the same VLAN except from the originating port
- B. Frames are sent to every port on the switch that has a matching entry in the MAC address table.
- C. Frames are sent to all ports, including those that are assigned to other VLANs.

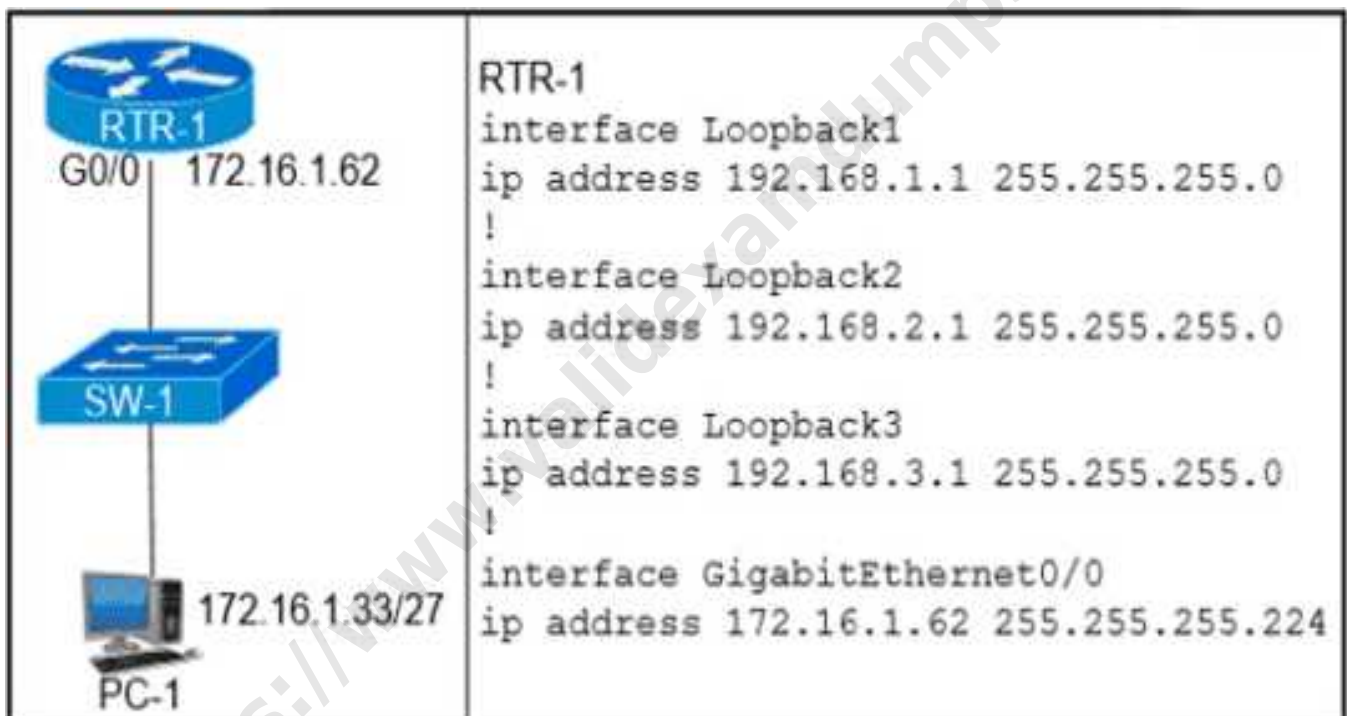
D. Frames are sent to every port on the switch in the same VLAN.

**Answer: A**

Explanation:

### Question: 85

Refer to the exhibit.



Which configuration on RTR-1 denies SSH access from PC-1 to any RTR-1 interface and allows all other traffic?

- A. access-list 100 deny tcp host 172.16.1.33 any eq 22 access-list 100 permit ip any any interface GigabitEthernet0/0 ip access-group 100 in
- B. access-list 100 deny tcp host 172.16.1.33 any eq 22 access-list 100 permit ip any any line vty 0 15 ip access-group 100 in
- C. access-list 100 deny tcp host 172.16.1.33 any eq 23 access-list 100 permit ip any any interface

GigabitEthernet0/0 ip access-group 100 in

D. access-list 100 deny tcp host 172.16.1.33 any eq 23 access-list 100 permit ip any anyline vty 0 15 ip access-group 100 in

---

**Answer: B**

---

Explanation:

---

**Question: 86**

---

In which two ways does a password manager reduce the chance of a hacker stealing a users password? (Choose two.)

- A. It automatically provides a second authentication factor that is unknown to the original user.
- B. It uses an internal firewall to protect the password repository from unauthorized access.
- C. It protects against keystroke logging on a compromised device or web site.
- D. It stores the password repository on the local workstation with built-in antivirus and anti-malware functionality
- E. It encourages users to create stronger passwords.

---

**Answer: C,E**

---

Explanation:

---

**Question: 87**

---

Which technology is used to improve web traffic performance by proxy caching?

- A. WSA
- B. Firepower

- C. ASA
- D. FireSIGHT

---

**Answer: A**

---

Explanation:

---

**Question: 88**

---

Which type of attack can be mitigated by dynamic ARP inspection?

- A. worm
- B. malware
- C. DDoS
- D. man-in-the-middle

---

**Answer: D**

---

Explanation:

---

**Question: 89**

---

What are two roles of Domain Name Services (DNS)? (Choose Two)

- A. builds a flat structure of DNS names for more efficient IP operations
- B. encrypts network Traffic as it travels across a WAN by default
- C. improves security by protecting IP addresses under Fully Qualified Domain Names (FQDNs)
- D. enables applications to identify resources by name instead of IP address
- E. allows a single host name to be shared across more than one IP address



---

**Answer: D,E**

---

Explanation:

---

**Question: 90**

---

How do TCP and UDP differ in the way they guarantee packet delivery?

- A. TCP uses checksum, acknowledgement, and retransmissions, and UDP uses checksums only.
- B. TCP uses two-dimensional parity checks, checksums, and cyclic redundancy checks and UDP uses retransmissions only.
- C. TCP uses checksum, parity checks, and retransmissions, and UDP uses acknowledgements only.
- D. TCP uses retransmissions, acknowledgement and parity checks and UDP uses cyclic redundancy checks only.

---

**Answer: A**

---

Explanation:

---

**Question: 91**

---

Refer to the exhibit.

```
R1# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default
       U - per-user static route, o - ODR
Gateway of last resort is not set
C    1.0.0.0/8 is directly connected, Loopback0
O    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O    10.0.1.3/32 [110/100] via 10.0.1.3, 00:39:08, Serial0
C    10.0.1.0/24 is directly connected, Serial0
O    10.0.1.5/32 [110/5] via 10.0.1.50, 00:39:08, Serial0
O    10.0.1.4/32 [110/10] via 10.0.1.4, 00:39:08, Serial0
```

What is the next hop address for traffic that is destined to host 10.0.1.5?

- A. 10.0.1.3
- B. 10.0.1.50
- C. 10.0.1.4
- D. Loopback D

---

**Answer: B**

---

Explanation:

---

**Question: 92**

---

What are two benefits of controller-based networking compared to traditional networking?

- A. controller-based increases network bandwidth usage, while traditional lightens the load on the network.
- B. controller-based inflates software costs, while traditional decreases individual licensing costs
- C. Controller-based reduces network configuration complexity, while traditional increases the potential for errors
- D. Controller-based provides centralization of key IT functions. While traditional requires distributes management function
- E. controller-based allows for fewer network failure, while traditional increases failure rates.

---

**Answer: C,D**

---

Explanation:

Cisco DNA Center Device Management

3. Monitor the cloud for software update

5. Uses CLI templates to apply a consistent configuration to multiple devices at an individual location

6. Uses NetFlow to analyse potential security threats throughout the network and take appropriate action on that traffic

Traditional device management

2. Manages device configuration on a per-device basis

4. Security is managed near the perimeter of the network with firewalls, VPNs, and IPS

Implements changes via an SSH terminal

---

**Question: 93**

---

What mechanism carries multicast traffic between remote sites and supports encryption?

- A. ISATAP
- B. GRE over iPsec
- C. iPsec over ISATAP
- D. GRE

---

**Answer: B**

---

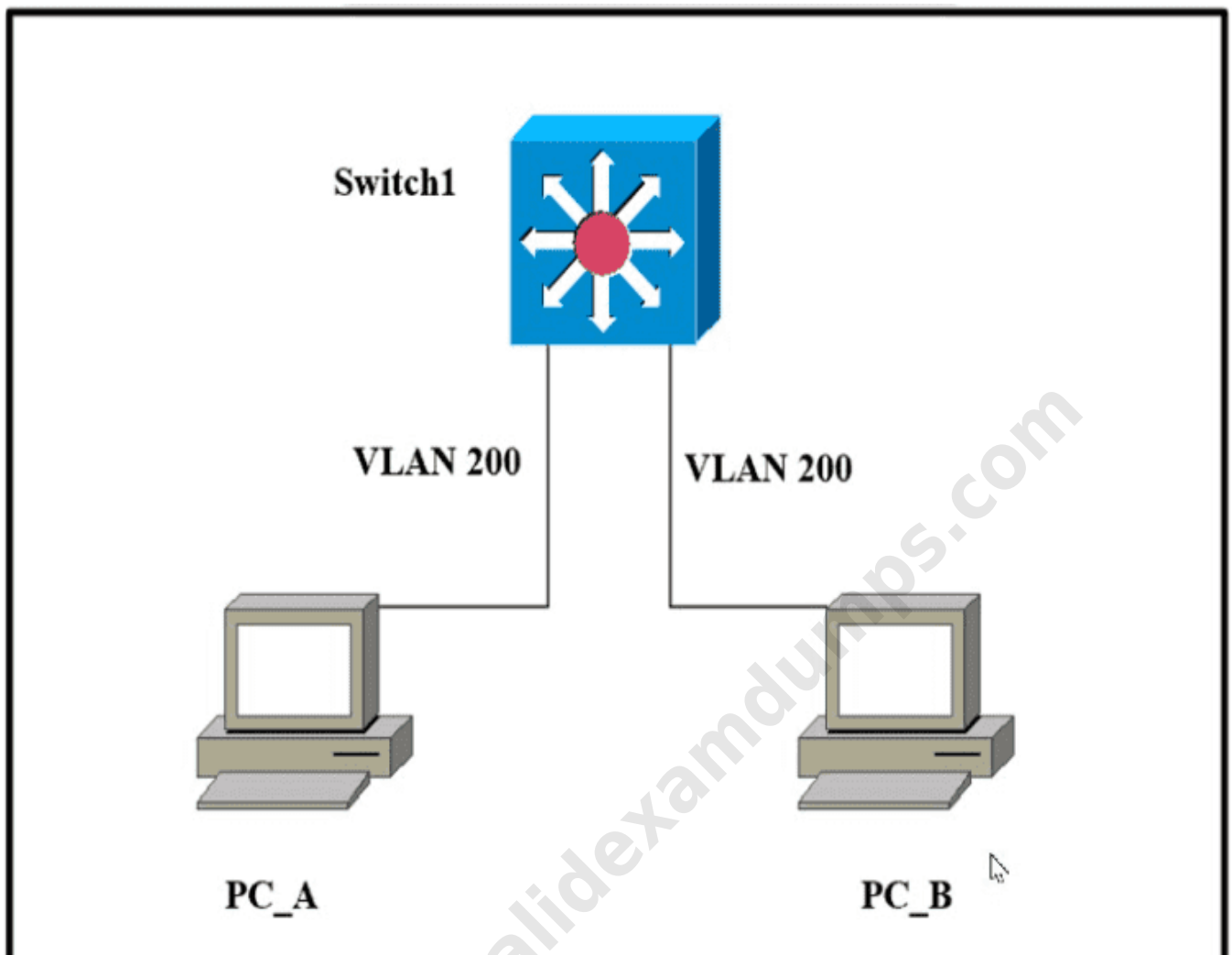
Explanation:

---

**Question: 94**

---

Refer to the exhibit.



Which outcome is expected when PC\_A sends data to PC\_B?

- A. The switch rewrites the source and destination MAC addresses with its own.
- B. The source MAC address is changed.
- C. The source and destination MAC addresses remain the same.
- D. The destination MAC address is replaced with ffff.ffff.ffff.

---

**Answer: C**

---

Explanation:

---

**Question: 95**

---

How will Link Aggregation be Implemented on a Cisco Wireless LAN Controller?

- A. One functional physical port is needed to pass client traffic.
- B. The EthernetChannel must be configured in "mode active".
- C. When enabled, the WLC bandwidth drops to 500 Mbps.
- D. To pass client traffic, two or more ports must be configured.

---

**Answer: A**

---

Explanation:

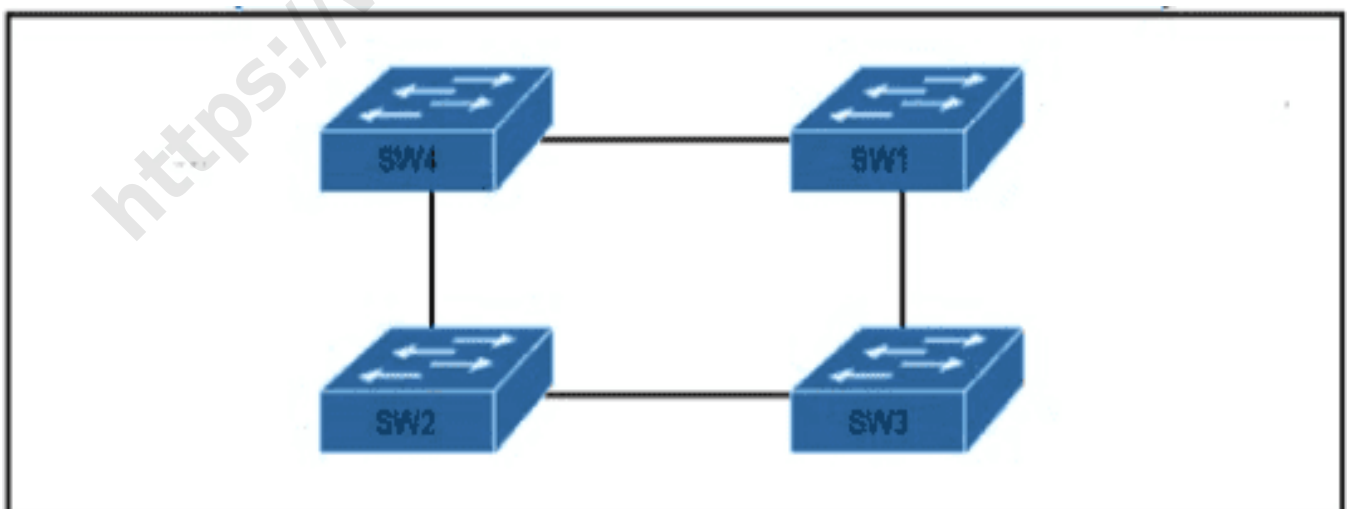
[https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-5/configuration-guide/b\\_cg75/b\\_cg75\\_chapter\\_0100010.html](https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-5/configuration-guide/b_cg75/b_cg75_chapter_0100010.html)

---

**Question: 96**

---

Refer to the exhibit.



Which switch in this configuration will be elected as the root bridge?

SW1: 0C:E0:38:00:94:04  
SW2: 0C:0E:15:22:05:97  
SW3: 0C:0E:15:1A:3C:9D  
SW4: 0C:E0:18:A1:B3:19

- A. SW1
- B. SW2
- C. SW3
- D. SW4

---

**Answer: C**

---

Explanation:

---

**Question: 97**

---

Which device performs stateful inspection of traffic?

- A. firewall
- B. switch
- C. access point
- D. wireless controller

---

**Answer: A**

---

Explanation:

---

**Question: 98**

---

Which configuration ensures that the switch is always the root for VLAN 750?

- A. Switch(config)#spanning-tree vlan 750 priority 38003685
- B. Switch(config)#spanning-tree vlan 750 root primary
- C. Switch(config)#spanning-tree vlan 750 priority 614440
- D. Switch(config)#spanning-tree vlan 750 priority 0

---

**Answer: D**

---

Explanation:

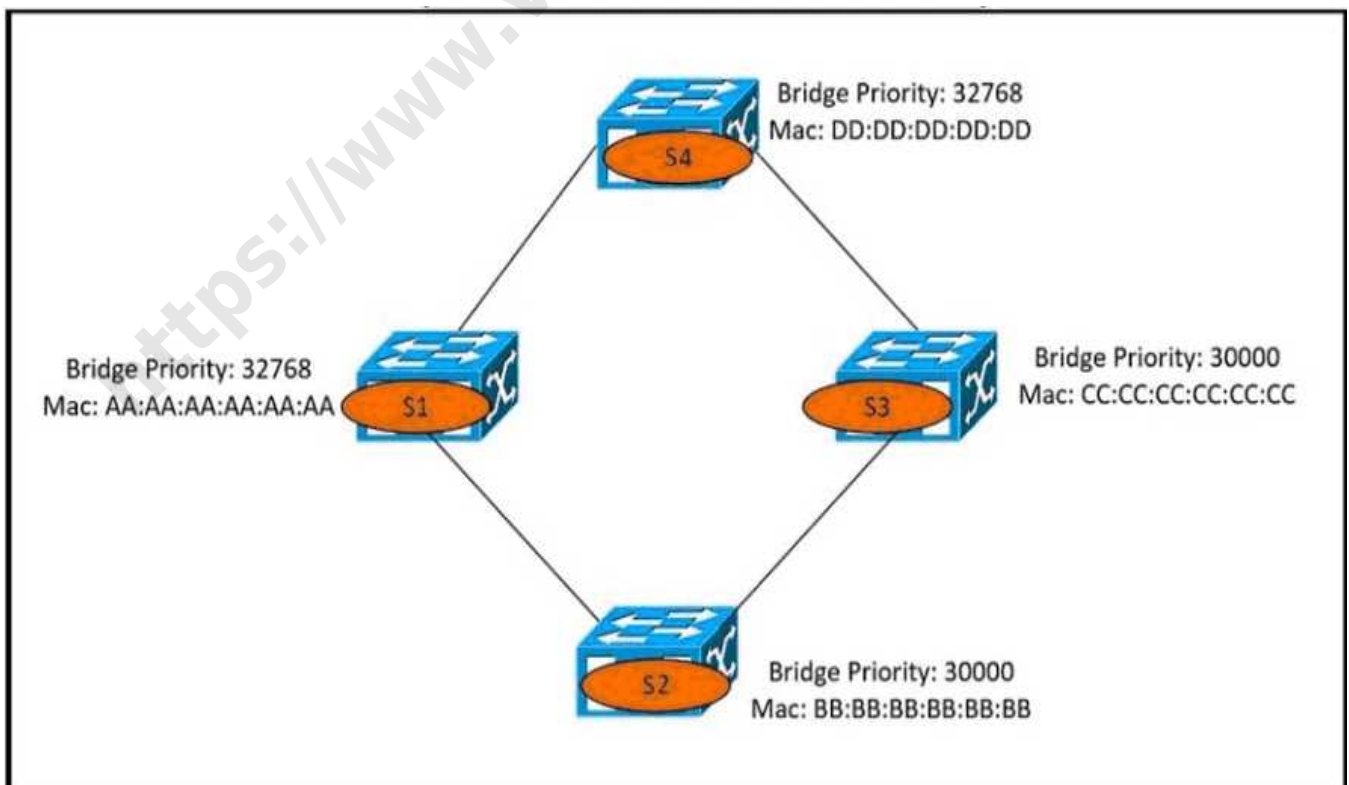
Although the spanning-tree vlan 10 root primary command will ensure a switch will have a bridge priority value lower than other bridges introduced to the network, the spanning-tree vlan 10 priority 0 command ensures the bridge priority takes precedence over all other priorities.

---

**Question: 99**

---

Refer to the exhibit.



Which switch becomes the root bridge?

- A. S1
- B. S2
- C. S3
- D. S4

---

**Answer: B**

---

Explanation:

---

**Question: 100**

---

What protocol allows an engineer to back up 20 network router configurations globally while using the copy function?

- A. SMTP
- B. SNMP
- C. TCP
- D. FTP

---

**Answer: B**

---

Explanation:

---

**Question: 101**

---

What software defined architecture plane assists network devices with making packet-forwarding decisions by providing Layer 2 reachability and Layer 3 routing information?



- A. data plane
- B. control plane
- C. policy plane
- D. management plane

---

**Answer: B**

---

Explanation:

---

**Question: 102**

---

Which WAN access technology is preferred for a small office / home office architecture?

- A. broadband cable access
- B. frame-relay packet switching
- C. dedicated point-to-point leased line
- D. Integrated Services Digital Network switching.

---

**Answer: A**

---

Explanation:

---

**Question: 103**

---

Which two WAN architecture options help a business scalability and reliability for the network?  
(Choose two)

- A. asynchronous routing
- B. single-homed branches

- C. dual-homed branches
- D. static routing
- E. dynamic routing

---

**Answer: A,C**

---

Explanation:

---

**Question: 104**

---

What criteria is used first during the root port selection process?

- A. local port ID
- B. lowest path cost to the root bridge
- C. lowest neighbor's bridge ID
- D. lowest neighbor's port ID

---

**Answer: B**

---

Explanation:

---

**Question: 105**

---

What criteria is used first during the root port selection process?

- A. local port ID
- B. lowest path cost to the root bridge
- C. lowest neighbor's bridge ID
- D. lowest neighbor's port ID

---

**Answer: B**

---

Explanation:

---

**Question: 106**

---

What is a function of Wireless LAN Controller?

- A. register with a single access point that controls traffic between wired and wireless endpoints.
- B. use SSIDs to distinguish between wireless clients.
- C. send LWAPP packets to access points.
- D. monitor activity on wireless and wired LANs

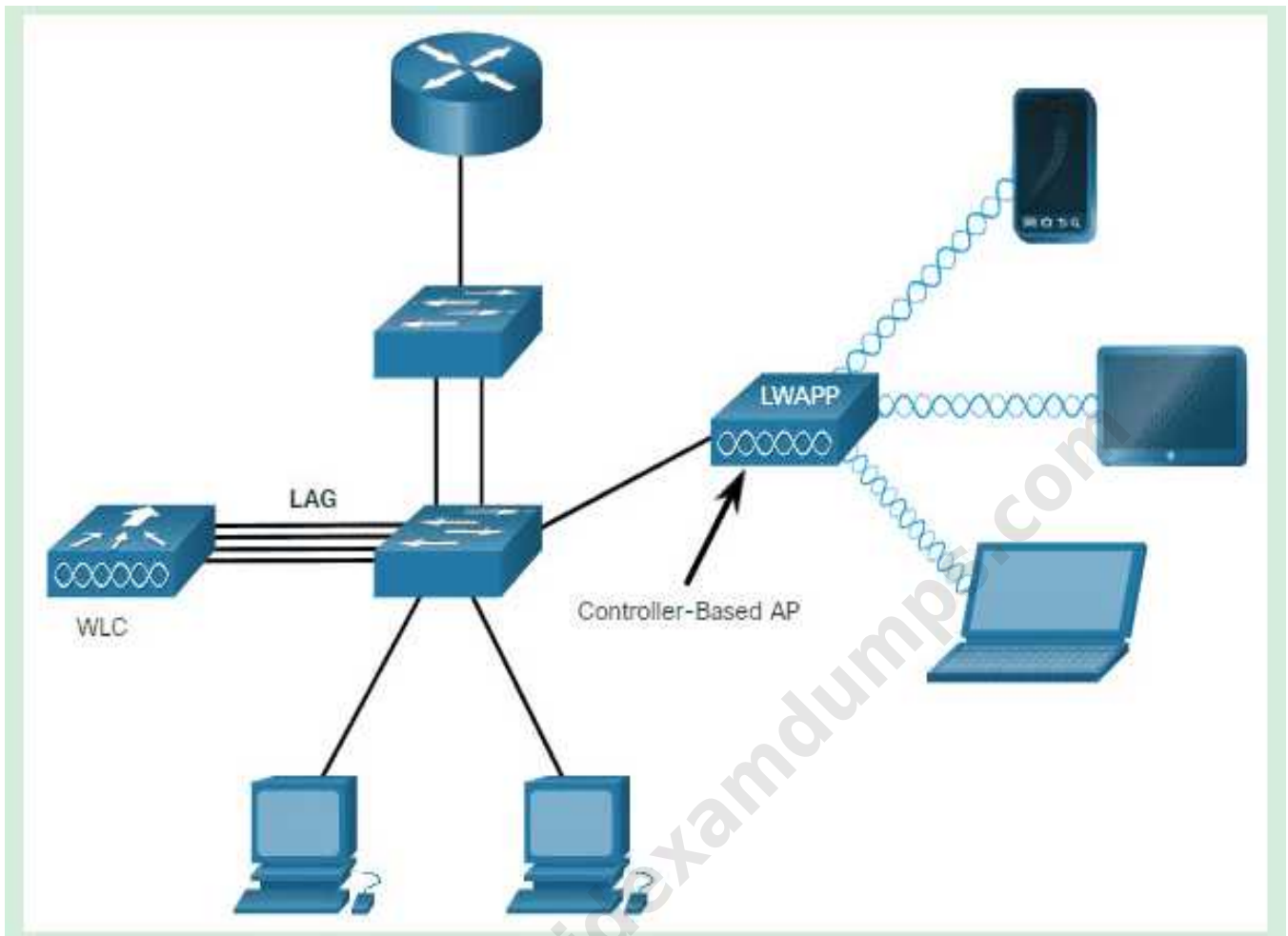
---

**Answer: C**

---

Explanation:

Lightweight APs (LAPs) is devices require no initial configuration. LAPs use the Lightweight Access Point Protocol (LWAPP) to communicate with a WLAN controller (WLC), as shown in the below figure. Controller-based APs are useful in situations where many APs are required in the network. As more APs are added, each AP is automatically configured and managed by the WLC.



---

**Question: 107**

---

Which type of information resides on a DHCP server?

- A. a list of the available IP addresses in a pool
- B. a list of public IP addresses and their corresponding names
- C. usernames and passwords for the end users in a domain
- D. a list of statically assigned MAC addresses

---

**Answer: A**

---

Explanation:

---

**Question: 108**

---

A manager asks a network engineer to advise which cloud service models are used so employees do not have to waste their time installing, managing, and updating software which is only used occasionally Which cloud service model does the engineer recommend?

- A. infrastructure-as-a-service
- B. platform-as-a-service
- C. business process as service to support different types of service
- D. software-as-a-service

---

**Answer: D**

---

Explanation:

---

**Question: 109**

---

Refer to the exhibit.

```
cisco_ospf_vrf {"R1 default":  
  ensure => 'present',  
  auto_cost => '100',  
}
```

Which type of configuration is represented in the output?

- A. Ansible
- B. JSON
- C. Chef

D. Puppet

---

**Answer: D**

---

Explanation:

---

**Question: 110**

---

What are two functions of a Layer 2 switch? (Choose two)

- A. acts as a central point for association and authentication servers
- B. selects the best route between networks on a WAN
- C. moves packets within a VLAN
- D. moves packets between different VLANs
- E. makes forwarding decisions based on the MAC address of a packet

---

**Answer: A,E**

---

Explanation:

---

**Question: 111**

---

Which spanning-tree enhancement avoids the learning and listening states and immediately places ports in the forwarding state?

- A. BPDUfilter
- B. PortFast
- C. Backbonefast
- D. BPDUguard

---

**Answer: B**

---

Explanation:

PortFast

Spanning Tree Portfast causes layer 2 switch interfaces to enter forwarding state immediately, bypassing the listening and learning states. It should be used on ports connected directly to end hosts like servers or workstations. Note: If portfast isn't enabled, DHCP timeouts can occur while STP converges, causing more problems.

<https://skminhaj.wordpress.com/2015/03/04/spanning-tree-stp-rstp-mst-enhancements/>

---

**Question: 112**

---

DRAG DROP

Drag the IPv6 DNS record types from the left onto the description on the right.

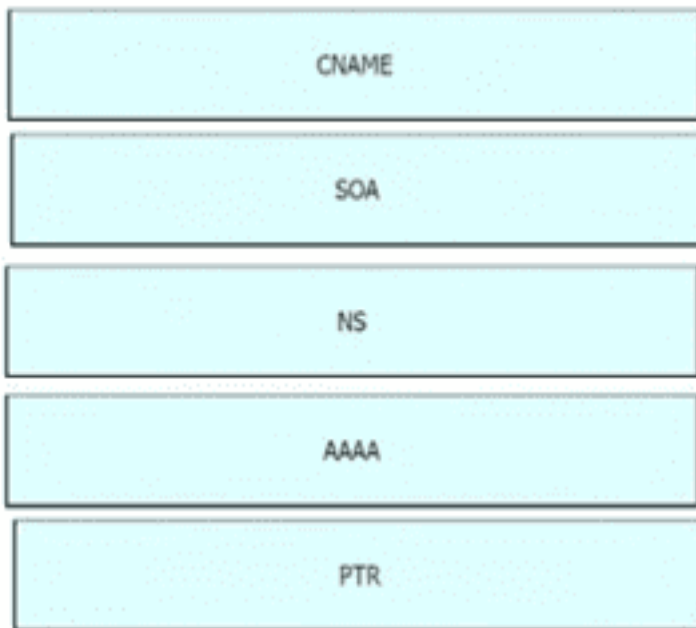
|       |                                                         |
|-------|---------------------------------------------------------|
| AAAA  | aliases one name to another                             |
| CNAME | associates the domain serial number with its owner      |
| NS    | correlates a domain with its authoritative name servers |
| PTR   | correlates a host name with an IP address               |
| SOA   | supports reverse name lookups                           |

---

**Answer:**

---

Explanation:



[https://ns1.com/resources/dns-types-records-servers-and-queries#:~:text=Address%20Mapping%20record%20\(A%20Record,a%20hostname%20to%20another%20hostname.](https://ns1.com/resources/dns-types-records-servers-and-queries#:~:text=Address%20Mapping%20record%20(A%20Record,a%20hostname%20to%20another%20hostname.)

---

**Question: 113**

---

What is a recommended approach to avoid co-channel congestion while installing access points that use the 2.4 GHz frequency?

- A. different nonoverlapping channels
- B. different overlapping channels
- C. one overlapping channel
- D. one nonoverlapping channel

---

**Answer: A**

---



Explanation:

---

**Question: 114**

---

Which function is performed by the collapsed core layer in a two-tier architecture?

- A. enforcing routing policies
- B. marking interesting traffic for data polices
- C. attaching users to the edge of the network
- D. applying security policies

---

**Answer: A**

---

Explanation:

---

**Question: 115**

---

What are two functions of a server on a network? (Choose two)

- A. achieves redundancy by exclusively using virtual server clustering
- B. runs applications that send and retrieve data for workstations that make requests
- C. handles requests from multiple workstations at the same time
- D. runs the same operating system in order to communicate with other servers
- E. housed solely in a data center that is dedicated to a single client

---

**Answer: B,C**

---

Explanation:

---

**Question: 116**

---

In software defined architectures, which plane is distributed and responsible for traffic forwarding?

- A. management plane
- B. control plane
- C. policy plane
- D. data plane

---

**Answer: D**

---

Explanation:

---

**Question: 117**

---

When using Rapid PVST+, which command guarantees the switch is always the root bridge for VLAN 200?

- A. spanning -tree vlan 200 priority 614440
- B. spanning -tree vlan 200 priority 38572422
- C. spanning -tree vlan 200 priority 0
- D. spanning -tree vlan 200 root primary

---

**Answer: C**

---

Explanation:

---

**Question: 118**

---

DRAG DROP

Drag and drop the SNMP components from the left onto the descriptions on the right.

|              |                                                                         |
|--------------|-------------------------------------------------------------------------|
| MIB          | collection of variables that can be monitored                           |
| SNMP agent   | unsolicited message                                                     |
| SNMP manager | responds to status requests and requests for information about a device |
| SNMP trap    | resides on an NMS                                                       |

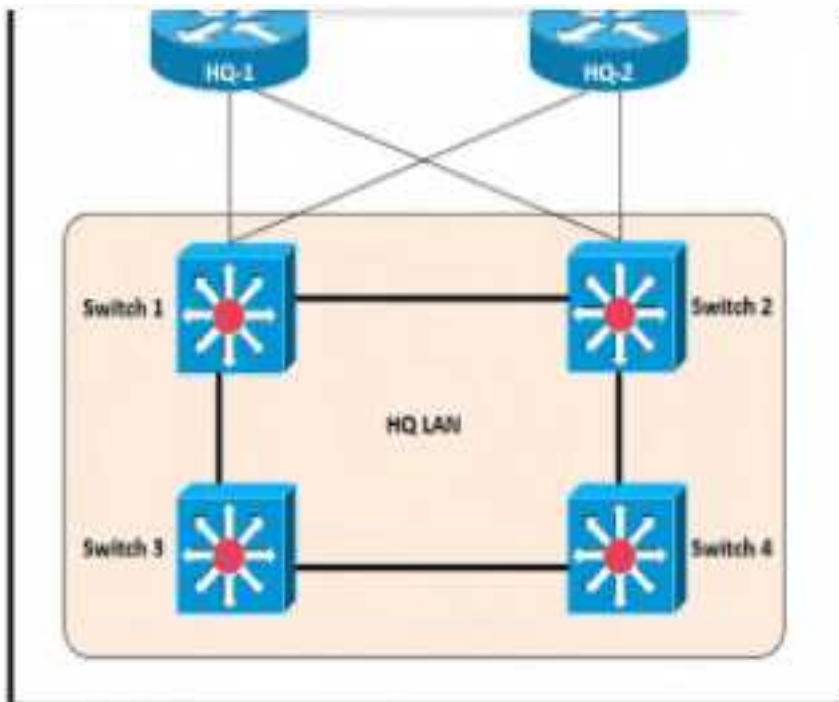
**Answer:**

Explanation:

|              |
|--------------|
| MIB          |
| SNMP manager |
| SNMP trap    |
| SNMP agent   |

**Question: 119**

Refer to the exhibit.



After the election process what is the root bridge in the HQ LAN?

Switch 1: 0C:E0:38:58:15:77  
Switch 2: 0C:0E:15:22:1A:61  
Switch 3: 0C:0E:15:1D:3C:9A  
Switch 4: 0C:E0:19:A1:4D:16

- A. Switch 1
- B. Switch 2
- C. Switch 3
- D. Switch 4

---

**Answer: C**

---

Explanation:

The root bridge is determined by the lowest bridge ID, which consists of the priority value and the MAC address. Because the priority values of all of the switches are not available, the MAC address is used to determine the root bridge. Because S3 has the lowest MAC address, S3 becomes the root bridge.

---

**Question: 120**

---

Which CRUD operation modifies an existing table or view?

- A. read
- B. create
- C. replace
- D. update

---

**Answer: D**

---

Explanation:

---

**Question: 121**

---

An engineer must configure Interswitch VLAN communication between a Cisco switch and a third-party switch. Which action should be taken?

- A. configure IEEE 802.1p
- B. configure IEEE 802.1q
- C. configure ISL
- D. configure DSCP

---

**Answer: B**

---

Explanation:

---

**Question: 122**

---

What is a function of a remote access VPN?

- A. used cryptographic tunneling to protect the privacy of data for multiple users simultaneously
- B. used exclusively when a user is connected to a company's internal network
- C. establishes a secure tunnel between two branch sites
- D. allows the users to access company internal network resources through a secure tunnel

---

**Answer: D**

---

Explanation:

---

**Question: 123**

---

What is a DHCP client?

- A. a workstation that requests a domain name associated with its IP address
- B. a host that is configured to request an IP address automatically
- C. a server that dynamically assigns IP addresses to hosts.
- D. a router that statically assigns IP addresses to hosts.

---

**Answer: B**

---

Explanation:

---

**Question: 124**

---

Which two functions are performed by the core layer in a three-tier architecture? (Choose two)

- A. Provide uninterrupted forwarding service.
- B. Police traffic that is sent to the edge of the network.

- C. Provide direct connectivity for end user devices.
- D. Ensure timely data transfer between layers.
- E. Inspect packets for malicious activity.

---

**Answer: A,D**

---

Explanation:

Cisco is very clear about the purpose of this layer. Its only role is to forward traffic, the fastest it can. Here you don't apply any policy, as you must try to reduce the load of the core so it can focus on routing.

<https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Campus/campover.html#wp708831>

---

**Question: 125**

---

Refer to the exhibit.

```
interface GigabitEthernet0/1
ip address 192.168.1.2 255.255.255.0
ip access-group 2699 in
!
access-list 2699 deny icmp any 10.10.1.0 0.0.0.255 echo
access-list 2699 deny ip any 10.20.1.0 0.0.0.255
access-list 2699 permit ip any 10.10.1.0 0.0.0.255
access-list 2699 permit tcp any 10.20.1.0 0.0.0.127 eq 22
```

A network administrator must permit SSH access to remotely manage routers in a network. The operations team resides on the 10.20.1.0/25 network. Which command will accomplish this task?

- A. access-list 2699 permit udp 10.20.1.0 0.0.0.255
- B. no access-list 2699 deny tcp any 10.20.1.0 0.0.0.127 eq 22
- C. access-list 2699 permit tcp any 10.20.1.0 0.0.0.255 eq 22
- D. no access-list 2699 deny ip any 10.20.1.0 0.0.0.255

---

**Answer: D**

---

Explanation:

Note : Already a statement is there in last to allow SSH Traffic for network 10.20.1.0 0.0.0.127, but Second statement says deny ip any 10.20.1.0 0.0.0.255, so how it will work once it is denied. So the right answer is remove the --- no access-list 2699 deny ip any 10.20.1.0 0.0.0.255.

---

**Question: 126**

---

What is a practice that protects a network from VLAN hopping attacks?

- A. Enable dynamic ARP inspection
- B. Configure an ACL to prevent traffic from changing VLANs
- C. Change native VLAN to an unused VLAN ID
- D. Implement port security on internet-facing VLANs

---

**Answer: C**

---

Explanation:

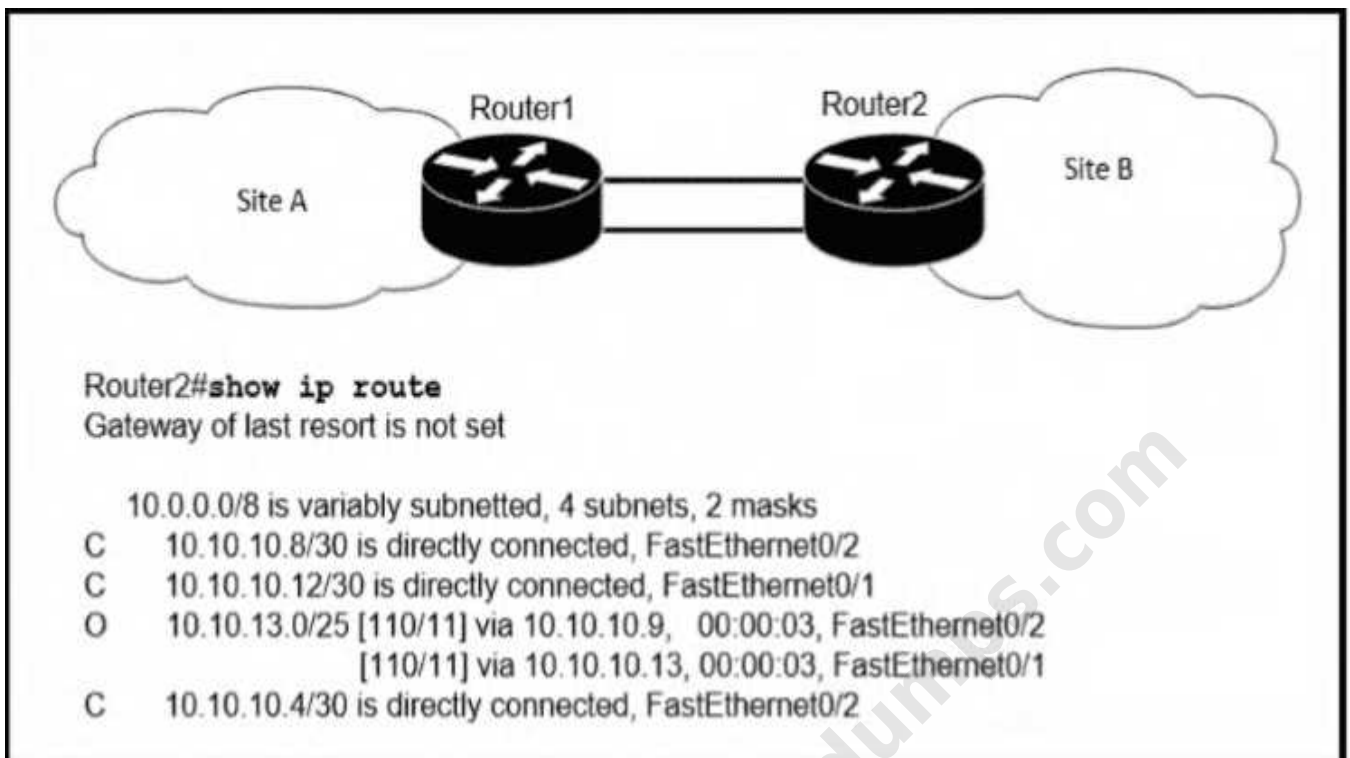
---

**Question: 127**

---

Refer to the exhibit.





If OSPF is running on this network, how does Router2 handle traffic from Site B to 10.10.13.128/25 at Site A?

- A. It load-balances traffic out of Fa0/1 and Fa0/2.
- B. It is unreachable and discards the traffic.
- C. It sends packets out of interface Fa0/2.
- D. It sends packets out of interface Fa0/1.

**Answer: B**

Explanation:

### Question: 128

What is the purpose of traffic shaping?

- A. to mitigate delays over slow links

- B. to provide fair queuing for buffered flows
- C. to limit the bandwidth that a flow can use to
- D. be a marking mechanism that identifies different flows

---

**Answer: B**

---

Explanation:

Traffic shaping retains excess packets in a queue and then schedules the excess for later transmission over increments of time.

---

**Question: 129**

---

Where does the configuration reside when a helper address is configured to support DHCP?

- A. on the router closest to the server
- B. on the router closest to the client
- C. on every router along the path
- D. on the switch trunk interface

---

**Answer: B**

---

Explanation:

---

**Question: 130**

---

What facilitates a Telnet connection between devices by entering the device name?

- A. SNMP

- B. DNS lookup
- C. syslog
- D. NTP

---

**Answer: B**

---

Explanation:

---

**Question: 131**

---

When implementing a router as a DHCP server, which two features must be configured'? (Choose two)

- A. relay agent information
- B. database agent
- C. address pool
- D. smart-relay
- E. manual bindings

---

**Answer: C,E**

---

Explanation:

---

**Question: 132**

---

DRAG DROP

Drag and drop the QoS congestion management terms from the left onto the description on the right.

|      |                                                                                       |
|------|---------------------------------------------------------------------------------------|
| CBWQ | places packets into one of four priority-based queues                                 |
| CQ   | provides guaranteed bandwidth to a specified class of traffic                         |
| FIFO | provides minimum guaranteed bandwidth to one or more flows                            |
| PQ   | services a specified number of bytes in one queue before continuing to the next queue |
| WFQ  | uses store-and-forward queueing                                                       |

Answer:

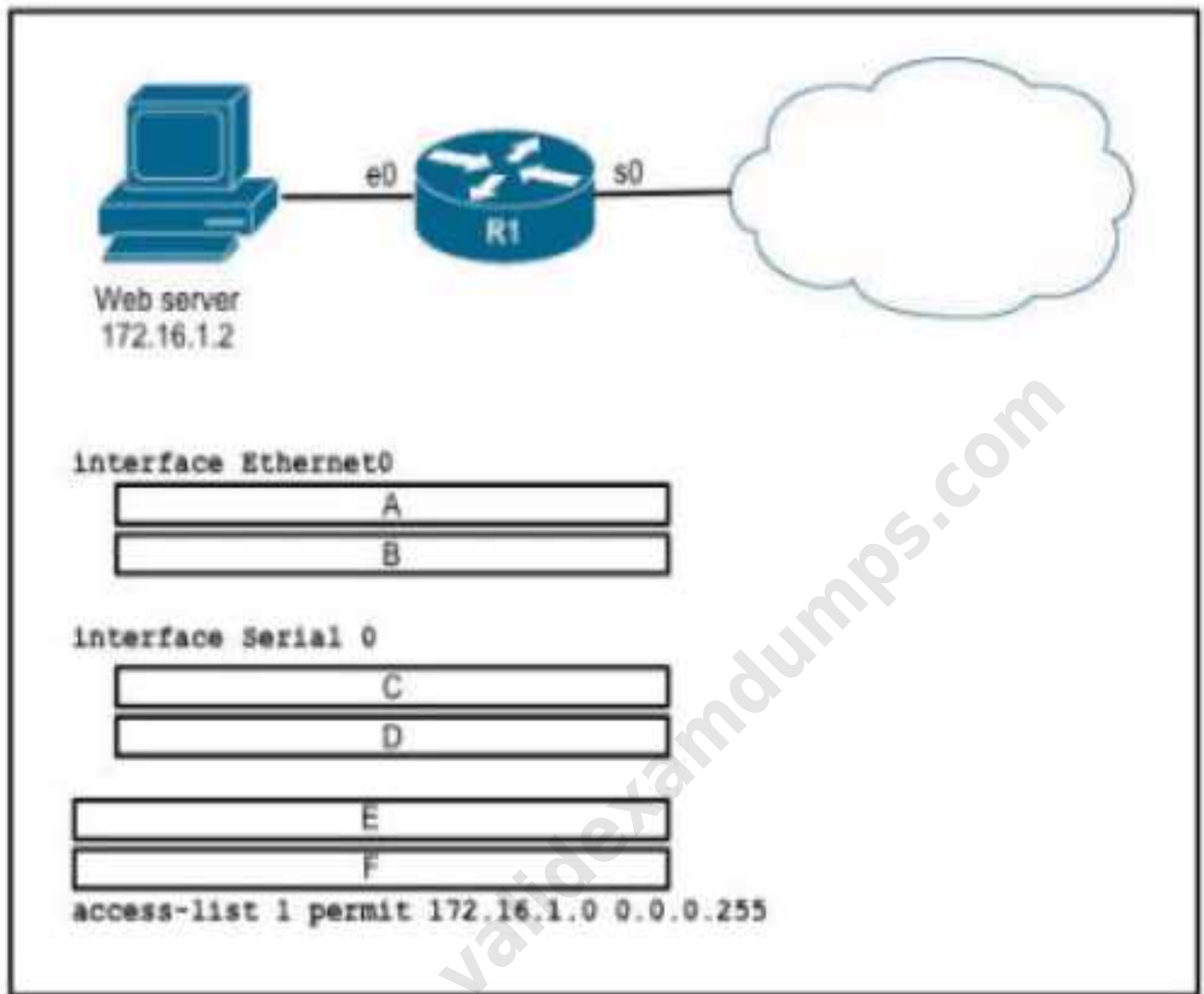
Explanation:

|      |                                                                                       |
|------|---------------------------------------------------------------------------------------|
| CBWQ | places packets into one of four priority-based queues                                 |
| CQ   | provides guaranteed bandwidth to a specified class of traffic                         |
| FIFO | provides minimum guaranteed bandwidth to one or more flows                            |
| PQ   | services a specified number of bytes in one queue before continuing to the next queue |
| WFQ  | uses store-and-forward queueing                                                       |

### Question: 133

DRAG DROP

Refer to the exhibit.



An engineer is configuring the router to provide static NAT for the webserver. Drag and drop the configuration commands from the left onto the letters that correspond to its position in the configuration on the right.

|                                                                         |            |
|-------------------------------------------------------------------------|------------|
| ip address 172.16.1.1 255.255.255.0                                     | position A |
| ip address 45.83.2.214 255.255.255.240                                  | position B |
| ip nat inside                                                           | position C |
| ip nat inside source list 1 interface s0 overload                       | position D |
| ip nat inside source static tcp 172.16.1.2 80 45.83.2.214 80 extendable | position E |
| ip nat outside                                                          | position F |

---

**Answer:**

---

Explanation:



ip address 172.16.1.1 255.255.255.0

ip nat inside

ip address 45.83.2.214 255.255.255.240

ip nat outside

ip nat inside source static tcp 172.16.1.2 80 45.83.2.214 80  
extendable

ip nat inside source list 1 interface s0 overload

---

**Question: 134**

---

DRAG DROP

Drag and drop the DHCP snooping terms from the left onto the descriptions on the right.

|                           |                                                                            |
|---------------------------|----------------------------------------------------------------------------|
| DHCP server               | list of hosts on the network that are unknown to the administrative domain |
| snooping binding database | network component that propagates IP addresses to hosts on the network     |
| spurious DHCP server      | internal device under the control of the network administrator             |
| trusted                   | unknown DHCP server within an administrative domain                        |
| untrusted                 | default state of all interfaces                                            |

---

**Answer:**

---

Explanation:

|                           |
|---------------------------|
| snooping binding database |
| spurious DHCP server      |
| trusted                   |
| DHCP server               |
| untrusted                 |



---

**Question: 135**

---

What is a role of wireless controllers in an enterprise network?

- A. centralize the management of access points in an enterprise network
- B. support standalone or controller-based architectures
- C. serve as the first line of defense in an enterprise network
- D. provide secure user logins to devices on the network.

---

**Answer: A**

---

Explanation:

---

**Question: 136**

---

How do servers connect to the network in a virtual environment?

- A. wireless to an access point that is physically connected to the network
- B. a cable connected to a physical switch on the network
- C. a virtual switch that links to an access point that is physically connected to the network
- D. a software switch on a hypervisor that is physically connected to the network

---

**Answer: D**

---

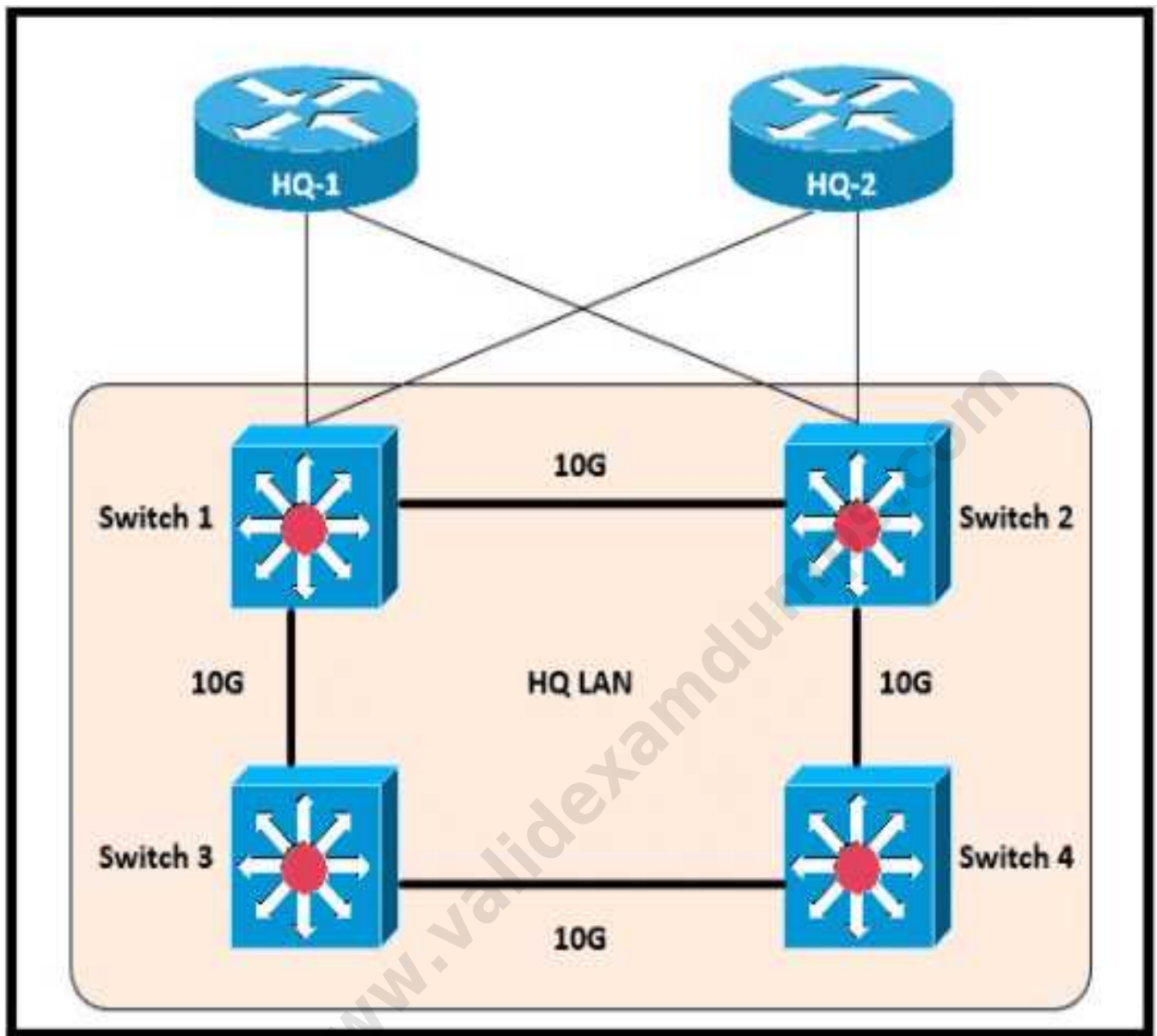
Explanation:

---

**Question: 137**

---

Refer to the exhibit.



Which switch becomes the root of the spanning tree for VLAN 110?

Switch 1  
VLAN 110 - 32778 0018.184e.3c00  
Switch 2  
VLAN 110 - 24586 001a.e3ff.a680  
Switch 3  
VLAN 110 - 28682 0022.55cf.cc00  
Switch 4  
VLAN 110 - 64000 0e38.7363.657f

A. Switch 1

B. Switch 2

C. Switch 3

D. Switch 4

---

**Answer: B**

---

Explanation:

---

**Question: 138**

---

Which device tracks the state of active connections in order to make a decision to forward a packet through?

A. wireless access point

B. firewall

C. wireless LAN controller

D. router

---

**Answer: B**

---

Explanation:

Stateful inspection, also known as dynamic packet filtering, is a firewall technology that monitors the state of active connections and uses this information to determine which network packets to allow through the firewall.

---

**Question: 139**

---

DRAG DROP

Drag and drop the characteristics of network architectures from the left onto the type of architecture on the right.

|                                                             |                                                       |
|-------------------------------------------------------------|-------------------------------------------------------|
| single device handles the core and the distribution layer   | Collapsed Core<br><div></div> <div></div> <div></div> |
| enhances network availability                               |                                                       |
| more cost-effective than other options                      |                                                       |
| most appropriate for small network designs                  | Three-Tier<br><div></div> <div></div>                 |
| separate devices handle the core and the distribution layer |                                                       |

**Answer:**

Explanation:

|                                                           |
|-----------------------------------------------------------|
| Collapsed Core                                            |
| single device handles the core and the distribution layer |
| more cost-effective than other options                    |
| most appropriate for small network designs                |

|                                                             |
|-------------------------------------------------------------|
| Three-Tier                                                  |
| enhances network availability                               |
| separate devices handle the core and the distribution layer |

---

**Question: 140**

---

How does a switch process a frame received on Fa0/1 with the destination MAC address of 0e38.7363.657b when the table is missing the address?

- A. It drops the frame immediately.
- B. It forwards the frame back out of interface Fa0/1.
- C. It floods the frame to all interfaces except Fa0/1.
- D. It holds the frame until the MAC address timer expires and then drops the frame.

---

**Answer: C**

---

Explanation:

---

**Question: 141**

---

DRAG DROP

Drag and drop the SNMP manager and agent identifier commands from the left onto the functions on the right

|                     |                                                   |
|---------------------|---------------------------------------------------|
| show snmp chassis   | displays information about the SNMP recipient     |
| show snmp community | displays the IP address of the remote SNMP device |
| show snmp engineID  | displays the SNMP security model in use           |
| show snmp group     | displays the SNMP access string                   |
| show snmp host      | displays the SNMP server serial number            |

---

**Answer:**

---

Explanation:

`show snmp host``show snmp engineID``show snmp group``show snmp community``show snmp chassis`

---

**Question: 142**

---

A network administrator must enable DHCP services between two sites. What must be configured for the router to pass DHCPDISCOVER messages on to the server?

- A. a DHCP Relay Agent
- B. DHCP Binding
- C. a DHCP Pool
- D. DHCP Snooping

---

**Answer: A**

---

Explanation:

---

**Question: 143**

---

What is recommended for the wireless infrastructure design of an organization?

- A. group access points together to increase throughput on a given channel
- B. configure the first three access points are configured to use Channels 1, 6, and 11
- C. include a least two access points on nonoverlapping channels to support load balancing
- D. assign physically adjacent access points to the same Wi-Fi channel

---

**Answer: B**

---

Explanation:

---

**Question: 144**

---

Refer to the exhibit.

```
switch(config)#interface gigabitEthernet 1/11  
switch(config-if)#switchport mode access  
switch(config-if)#spanning-tree portfast  
switch(config-if)#spanning-tree bpduguard enable
```

What is the result if Gig1/11 receives an STP BPDU?

- A. The port transitions to STP blocking
- B. The port transitions to the root port
- C. The port immediately transitions to STP forwarding.
- D. The port goes into error-disable state

---

**Answer: D**

---

Explanation:

---

**Question: 145**

---

What does a switch use to build its MAC address table?

- A. VTP
- B. DTP
- C. egress traffic
- D. ingress traffic

---

**Answer: D**

---

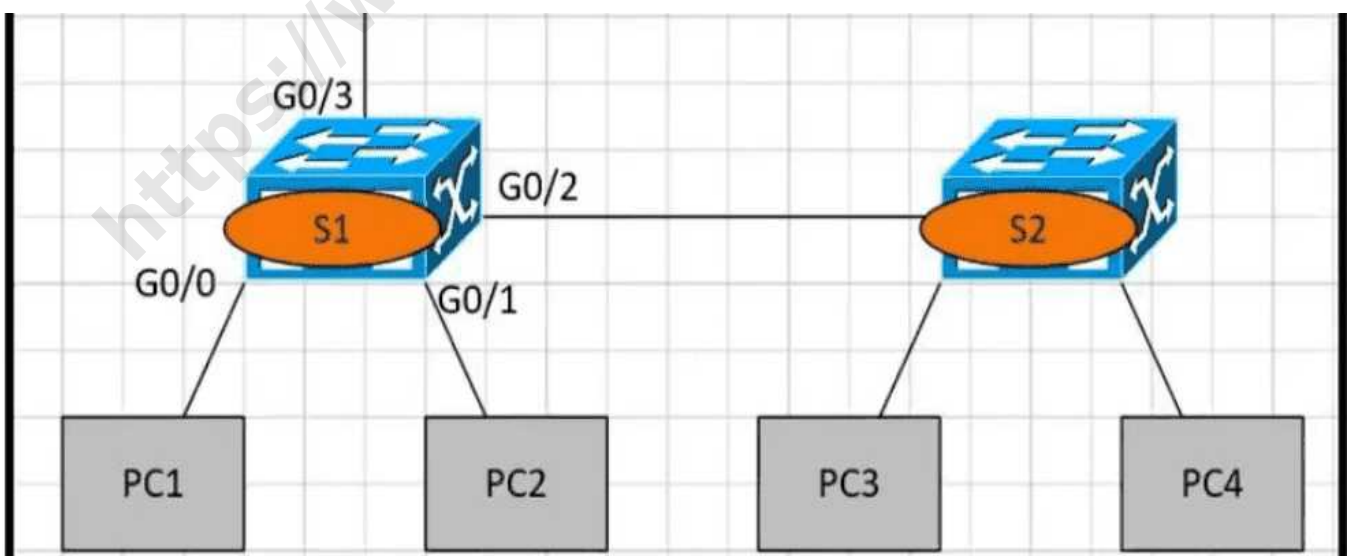
Explanation:

---

**Question: 146**

---

Refer to the exhibit.



PC1 is trying to ping PC3 for the first time and sends out an ARP to S1 Which action is taken by S1?



- A. It forwards it out G0/3 only
- B. It is flooded out every port except G0/0.
- C. It drops the frame.
- D. It forwards it out interface G0/2 only.

---

**Answer: B**

---

Explanation:

---

**Question: 147**

---

What does a router do when configured with the default DNS lookup settings, and a URL is entered on the CLI?

- A. initiates a ping request to the URL
- B. prompts the user to specify the desired IP address
- C. continuously attempts to resolve the URL until the command is cancelled
- D. sends a broadcast message in an attempt to resolve the URL

---

**Answer: D**

---

Explanation:

---

**Question: 148**

---

Which two WAN architecture options help a business improve scalability and reliability for the network? (Choose two.)

- A. asynchronous routing
- B. single-homed branches

- C. dual-homed branches
- D. static routing
- E. dynamic routing

---

**Answer: A,C**

---

Explanation:

---

**Question: 149**

---

Which type of security program is violated when a group of employees enters a building using the ID badge of only one person?

- A. intrusion detection
- B. user awareness
- C. physical access control
- D. network authorization

---

**Answer: C**

---

Explanation:

---

**Question: 150**

---

Which device controls the forwarding of authentication requests for users when connecting to the network using a lightweight access point?

- A. TACACS server
- B. wireless access point
- C. RADIUS server
- D. wireless LAN controller

---

**Answer: B**

---

Explanation:

---

**Question: 151**

---

What is a benefit of VRRP?

- A. It provides traffic load balancing to destinations that are more than two hops from the source.
- B. It provides the default gateway redundancy on a LAN using two or more routers.
- C. It allows neighbors to share routing table information between each other.
- D. It prevents loops in a Layer 2 LAN by forwarding all traffic to a root bridge, which then makes the final forwarding decision.

---

**Answer: B**

---

Explanation:

---

**Question: 152**

---

Aside from discarding, which two states does the switch port transition through while using RSTP (802.1w)? (Choose two)

- A. listening
- B. blocking
- C. forwarding
- D. learning
- E. speaking

---

**Answer: C,D**

---

Explanation:

---

**Question: 153**

---

Which protocol does an IPv4 host use to obtain a dynamically assigned IP address?

- A. ARP
- B. DHCP
- C. CDP
- D. DNS

---

**Answer: B**

---

Explanation:

<https://www.geeksforgeeks.org/how-dhcp-server-dynamically-assigns-ip-address-to-a-host/#:~:text=DHCP%20is%20an%20abbreviation%20for,subnet%20mask%20and%20gateway%20address.>

---

**Question: 154**

---

Which CRUD operation corresponds to the HTTP GET method?

- A. read
- B. update
- C. create
- D. delete

---

**Answer: A**

---

Explanation:

GET: This method retrieves the information identified by the request URI. In the context of the RESTful web services, this method is used to retrieve resources. This is the method used for read operations (the R in CRUD).

<https://hub.packtpub.com/crud-operations-rest/>

---

**Question: 155**

---

In which situation is private IPv4 addressing appropriate for a new subnet on the network of an organization?

- A. There is limited unique address space, and traffic on the new subnet will stay local within the organization.
- B. The network has multiple endpoint listeners, and it is desired to limit the number of broadcasts.
- C. Traffic on the subnet must traverse a site-to-site VPN to an outside organization.
- D. The ISP requires the new subnet to be advertised to the internet for web services.

---

**Answer: A**

---

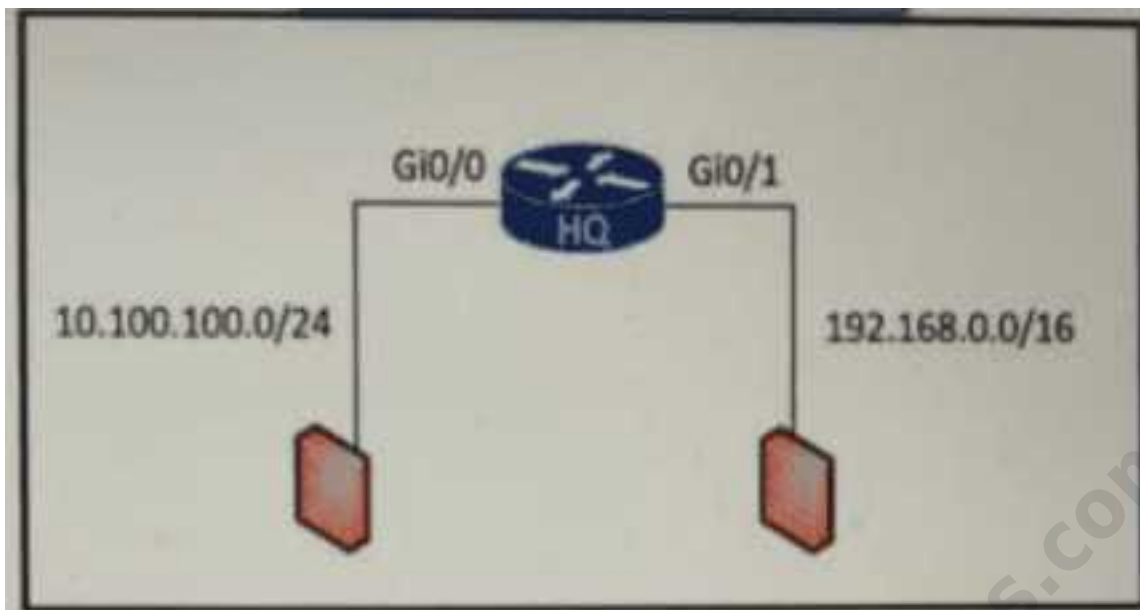
Explanation:

---

**Question: 156**

---

Refer to the exhibit.



An access list is required to permit traffic from any host on interface G0/0 and deny traffic from interface G/0/1. Which access list must be applied?

- ☐ ip access-list standard 99  
permit 10.100.100.0 0.0.0.255  
deny 192.168.0.0 0.0.255.255
- ☐ ip access-list standard 99  
permit 10.100.100.0 0.0.0.255  
deny 192.168.0.0 0.255.255.255
- ☐ ip access-list standard 199  
permit 10.100.100.0 0.0.0.255  
deny 192.168.0.0 0.255.255.255
- ☐ ip access-list standard 199  
permit 10.100.100.0 0.0.0.255  
deny 192.168.0.0 0.0.255.255

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: A**

---

Explanation:

---

**Question: 157**

---

What is the maximum bandwidth of a T1 point-to-point connection?

- A. 1.544 Mbps
- B. 2.048 Mbps
- C. 34.368 Mbps
- D. 43.7 Mbps

---

**Answer: A**

---

Explanation:

[https://www.bsimplify.com/what-is-point-to-point-t1/#:~:text=A%20Point%20to%20Point%20T1,data%20speeds%20\(1.54Mbps\).](https://www.bsimplify.com/what-is-point-to-point-t1/#:~:text=A%20Point%20to%20Point%20T1,data%20speeds%20(1.54Mbps).)

Point to Point T1

A Point to Point T1 service is a private data connection securely connecting two or more locations with T1 data speeds (1.54Mbps).

---

**Question: 158**

---

What is a DNS lookup operation?

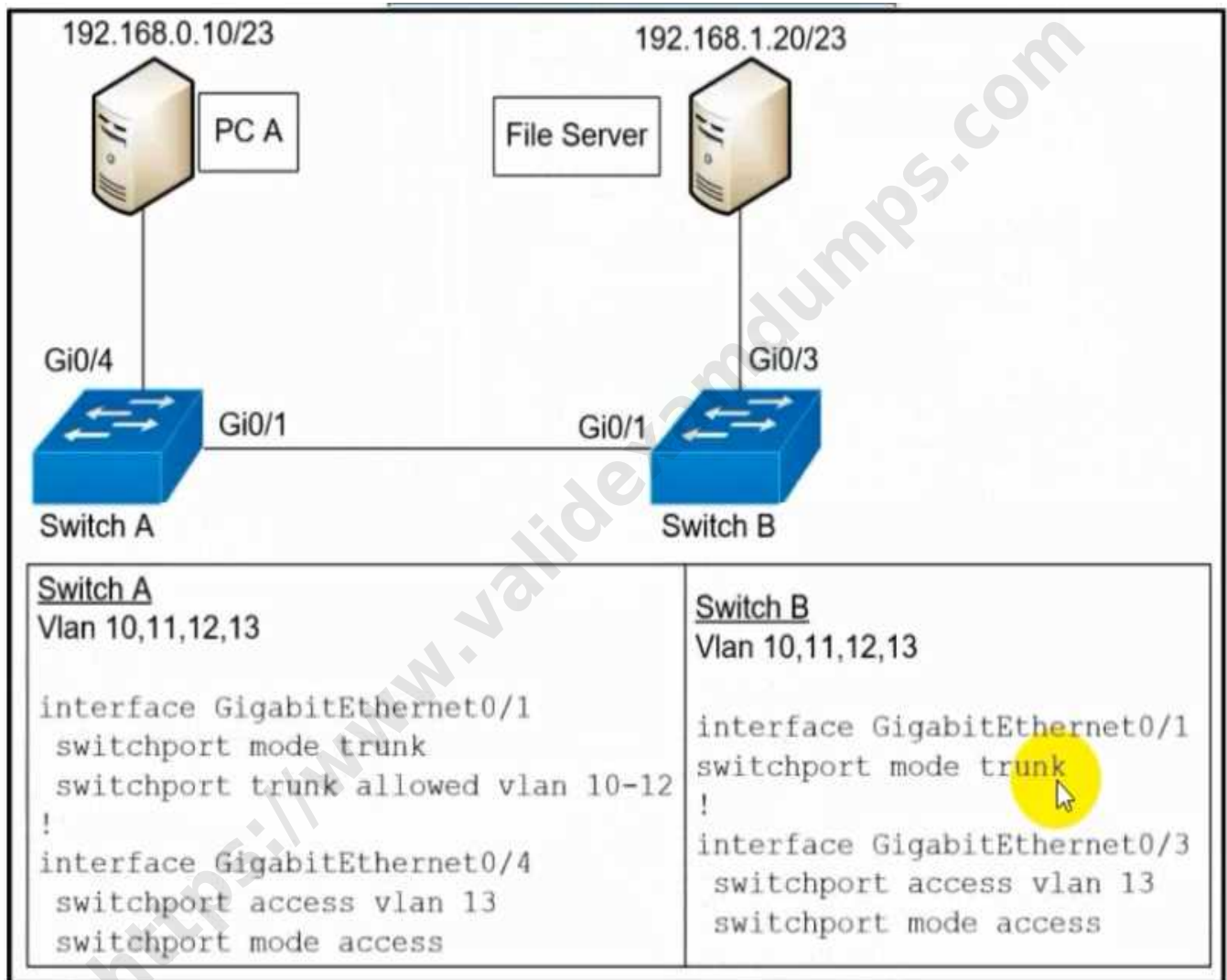
- A. DNS server pings the destination to verify that it is available
- B. serves requests over destination port 53
- C. DNS server forwards the client to an alternate IP address when the primary IP is down
- D. responds to a request for IP address to domain name resolution to the DNS server

**Answer: D**

Explanation:

**Question: 159**

Refer to the exhibit.



A network engineer must configured communication between PC A and the File Server. To prevent interruption for any other communications, which command must be configured?

- A. Switch trunk allowed vlan 12
- B. Switchport trunk allowed vlan none
- C. Switchport trunk allowed vlan add 13
- D. Switchport trunk allowed vlan remove 10-11



---

**Answer: C**

---

Explanation:

---

**Question: 160**

---

Which implementation provides the strongest encryption combination for the wireless environment?

- A. WPA2 + AES
- B. WPA + AES
- C. WEP
- D. WPA + TKIP

---

**Answer: A**

---

Explanation:

---

**Question: 161**

---

What is a characteristic of a SOHO network?

- A. connects each switch to every other switch in the network
- B. enables multiple users to share a single broadband connection
- C. provides high throughput access for 1000 or more users
- D. includes at least three tiers of devices to provide load balancing and redundancy

---

**Answer: B**

---

Explanation:

---

**Question: 162**

---

Refer to the exhibit.

```
import ncclient

with ncclient.manager.connect(host='192.168.1.1', port=830, username='root',
                             password='teset123!', allow_agent=False) as m:
    print(m.get_config('running').data_xml)
```

After running the code in the exhibit, which step reduces the amount of data that the NETCONF server returns to the NETCONF client, to only the interface's configuration?

- A. Use the lxml library to parse the data returned by the NETCONF server for the interface's configuration.
- B. Create an XML filter as a string and pass it to get\_config() method as an argument.
- C. Create a JSON filter as a string and pass it to the get\_config() method as an argument.
- D. Use the JSON library to parse the data returned by the NETCONF server for the interface's configuration.

---

**Answer: D**

---

Explanation:

---

**Question: 163**

---

Which resource is able to be shared among virtual machines deployed on the same physical server?

- A. disk
- B. applications
- C. VM configuration file
- D. operating system

---

**Answer: C**

---

Explanation:

---

**Question: 164**

---

Which WAN topology provides a combination of simplicity quality, and availability?

- A. partial mesh
- B. full mesh
- C. point-to-point
- D. hub-and-spoke

---

**Answer: C**

---

Explanation:

---

**Question: 165**

---

Which command on a port enters the forwarding state immediately when a PC is connected to it?

- A. switch(config)#spanning-tree portfast default
- B. switch(config)#spanning-tree portfast bpduguard default
- C. switch(config-if)#spanning-tree portfast trunk
- D. switch(config-if)#no spanning-tree portfast

---

**Answer: C**

---

Explanation:

---

**Question: 166**

---

What are two functions of an SDN controller? (Choose two)

- A. Layer 2 forwarding
- B. coordinating VTNs
- C. tracking hosts
- D. managing the topology
- E. protecting against DDoS attacks

---

**Answer: B,D**

---

Explanation:

---

**Question: 167**

---

What is a network appliance that checks the state of a packet to determine whether the packet is legitimate?

- A. Layer 2 switch
- B. load balancer
- C. firewall
- D. LAN controller

---

**Answer: C**

---

Explanation:

---

**Question: 168**

---

When DHCP is configured on a router, which command must be entered so the default gateway is automatically distributed?

- A. default-router
- B. default-gateway
- C. ip helper-address
- D. dns-server

---

**Answer: A**

---

Explanation:

---

**Question: 169**

---

What is an appropriate use for private IPv4 addressing?

- A. on the public-facing interface of a firewall
- B. to allow hosts inside to communicate in both directions with hosts outside the organization
- C. on internal hosts that stream data solely to external resources
- D. on hosts that communicates only with other internal hosts

---

**Answer: D**

---

Explanation:

---

**Question: 170**

---

How is the native VLAN secured in a network?

- A. separate from other VLANs within the administrative domain
- B. give it a value in the private VLAN range
- C. assign it as VLAN 1
- D. configure it as a different VLAN ID on each end of the link

---

**Answer: A**

---

Explanation:

---

**Question: 171**

---

What is the purpose of a southbound API in a control based networking architecture?

- A. Facilitates communication between the controller and the applications
- B. Facilitates communication between the controller and the networking hardware
- C. allows application developers to interact with the network
- D. integrates a controller with other automation and orchestration tools.

---

**Answer: B**

---

Explanation:

<https://www.ciscopress.com/articles/article.asp?p=2995354&seqNum=2#:~:text=The%20Southbound%20Interface,communicate%20to%20the%20networking%20devices.&text=The%20overall%20goal%20is%20network,from%20being%20only%20a%20protocol.>

The Southbound Interface

In a controller-based network architecture, the controller needs to communicate to the networking devices.

---

**Question: 172**

---

What causes a port to be placed in the err-disabled state?

- A. latency
- B. port security violation
- C. shutdown command issued on the port
- D. nothing plugged into the port

---

**Answer: B**

---

Explanation:

This mode is the default violation mode; when in this mode, the switch will automatically force the switchport into an error disabled (err-disable) state when a violation occurs. While in this state, the switchport forwards no traffic. The switchport can be brought out of this error disabled state by issuing the errdisable recovery cause CLI command or by disabling and reenabling the switchport.

---

**Question: 173**

---

Which switch technology establishes a network connection immediately when it is plugged in?

- A. PortFast
- B. BPDU guard
- C. UplinkFast
- D. BackboneFast

---

**Answer: A**

---

Explanation:

PortFast is useful to connect hosts and switches to a switch. Access layer switches are more frequently “plugged in” and “plugged out” than distribution or core layer switches. Also, this feature’s target is just to minimize STP convergence time.

---

**Question: 174**

---

Which security program element involves installing badge readers on data-center doors to allow workers to enter and exit based on their job roles?

- A. role-based access control
- B. biometrics
- C. multifactor authentication
- D. physical access control

---

**Answer: D**

---

Explanation:

---

**Question: 175**

---

Which network action occurs within the data plane?

- A. compare the destination IP address to the IP routing table.
- B. run routing protocols (OSPF, EIGRP, RIP, BGP)
- C. make a configuration change from an incoming NETCONF RPC
- D. reply to an incoming ICMP echo request

---

**Answer: A**

---



Explanation:

---

**Question: 176**

---

What is a DHCP client?

- A. a host that is configured to request an IP address automatically
- B. a server that dynamically assigns IP addresses to hosts
- C. a workstation that requests a domain name associated with its IP address
- D. a router that statically assigns IP addresses to hosts

---

**Answer: A**

---

Explanation:

---

**Question: 177**

---

An engineer needs to add an old switch back into a network. To prevent the switch from corrupting the VLAN database which action must be taken?

- A. Add the switch in the VTP domain with a lower revision number
- B. Add the switch with DTP set to dynamic desirable
- C. Add the switch in the VTP domain with a higher revision number
- D. Add the switch with DTP set to desirable

---

**Answer: A**

---

Explanation:

---

**Question: 178**

---

What is a similarity between OM3 and OM4 fiber optic cable?

- A. Both have a 50 micron core diameter
- B. Both have a 9 micron core diameter
- C. Both have a 62.5 micron core diameter
- D. Both have a 100 micron core diameter

---

**Answer: A**

---

Explanation:

---

**Question: 179**

---

What is the benefit of using FHRP?

- A. reduced management overhead on network routers
- B. balancing traffic across multiple gateways in proportion to their loads
- C. higher degree of availability
- D. reduced ARP traffic on the network

---

**Answer: C**

---

Explanation:

---

**Question: 180**

---

Which technology allows for multiple operating systems to be run on a single host computer?

- A. virtual routing and forwarding
- B. network port ID visualization
- C. virtual device contexts
- D. server visualization

---

**Answer: D**

---

Explanation:

---

**Question: 181**

---

What occurs when overlapping Wi-Fi channels are implemented?

- A. The wireless network becomes vulnerable to unauthorized access.
- B. Wireless devices are unable to distinguish between different SSIDs
- C. Users experience poor wireless network performance.
- D. Network communications are open to eavesdropping.

---

**Answer: C**

---

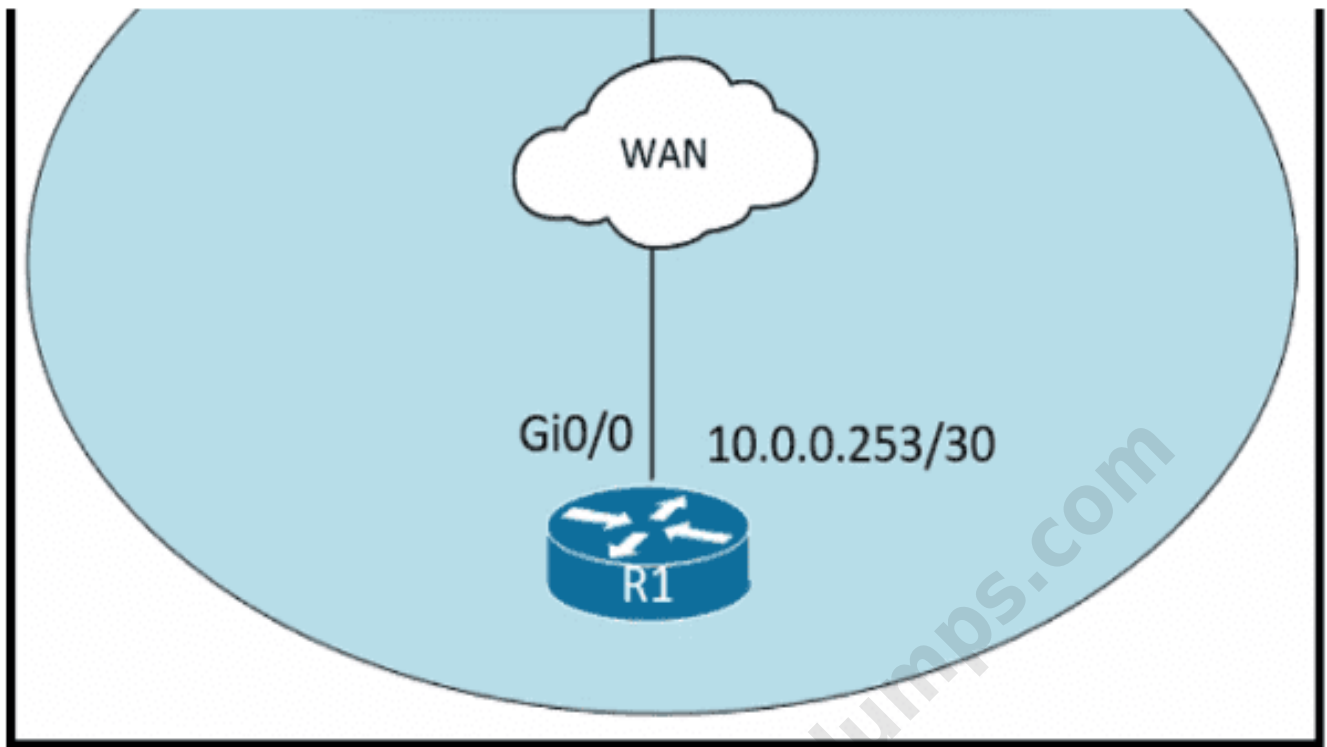
Explanation:

---

**Question: 182**

---

Refer to the exhibit.



An administrator must turn off the Cisco Discovery Protocol on the port configured with address last usable address in the 10.0.0.0/30 subnet. Which command set meets the requirement?

- A. interface gi0/1no cdp enable
- B. interface gi0/1clear cdp table
- C. interface gi0/0no cdp advertise-v2
- D. interface gi0/0no cdp run

---

**Answer: D**

---

Explanation:

---

**Question: 183**

---

Which 802.11 management frame type is sent when a client roams between access points on the same SSID?

- A. Reassociation Request

- B. Probe Request
- C. Authentication Request
- D. Association Request

---

**Answer: A**

---

Explanation:

---

**Question: 184**

---

What are two improvements provided by automation for network management in an SDN environment? (Choose two)

- A. Data collection and analysis tools establish a baseline for the network
- B. Artificial intelligence identifies and prevents potential design failures.
- C. Machine learning minimizes the overall error rate when automating troubleshooting processes
- D. New devices are onboarded with minimal effort
- E. Proprietary Cisco APIs leverage multiple network management tools.

---

**Answer: B,E**

---

Explanation:

---

**Question: 185**

---

An engineer must configure the IPv6 address 2001:0db8:0000:0000:0700:0003:400F:572B on the serial0/0 interface of the HQ router and wants to compress it for easier configuration. Which command must be issued on the router interface?

- A. ipv6 address 2001:db8::700:3:400F:572B

- B. ipv6 address 2001:db8:0::700:3:4F:572B
- C. ipv6 address 2001:Odb8::7:3:4F:572B
- D. ipv6 address 2001::db8:0000::700:3:400F:572B

---

**Answer: A**

---

Explanation:

---

**Question: 186**

---

What describes the operation of virtual machines?

- A. Virtual machines are responsible for managing and allocating host hardware resources
- B. In a virtual machine environment, physical servers must run one operating system at a time.
- C. Virtual machines are the physical hardware that support a virtual environment.
- D. Virtual machines are operating system instances that are decoupled from server hardware

---

**Answer: B**

---

Explanation:

---

**Question: 187**

---

Which WLC port connects to a switch to pass normal access-point traffic?

- A. redundancy
- B. console
- C. distribution system
- D. service

---

**Answer: C**

---

Explanation:

---

**Question: 188**

---

An engineering team asks an implementer to configure syslog for warning conditions and error conditions. Which command does the implementer configure to achieve the desired result?

- A. logging trap 5
- B. logging trap 2
- C. logging trap 4
- D. logging trap 3

---

**Answer: C**

---

Explanation:

---

**Question: 189**

---

DRAG DROP

Drag and drop the 802.11 wireless standards from the left onto the matching statements on the right

|          |                                                                                              |
|----------|----------------------------------------------------------------------------------------------|
| 802.11a  | Operates in the 2.4 GHz and 5 GHz bands.                                                     |
| 802.11ac | Operates in the 5 GHz band only and supports a maximum data rate of 54 Mbps.                 |
| 802.11b  | Operates in the 2.4 GHz band only and supports a maximum data rate that can exceed 100 Mbps. |
| 802.11g  | Supports a maximum data rate of 11 Mbps.                                                     |
| 802.11n  | Operates in the 5 GHz band only and supports a maximum data rate of 54 Mbps.                 |

**Answer:**

Explanation:

|          |
|----------|
| 802.11n  |
| 802.11g  |
| 802.11ac |
| 802.11b  |
| 802.11a  |

### Question: 190

Which two protocols are supported on service-port interfaces? (Choose two.)



- A. RADIUS
- B. TACACS+
- C. SCP
- D. Telnet
- E. SSH

---

**Answer: D,E**

---

Explanation:

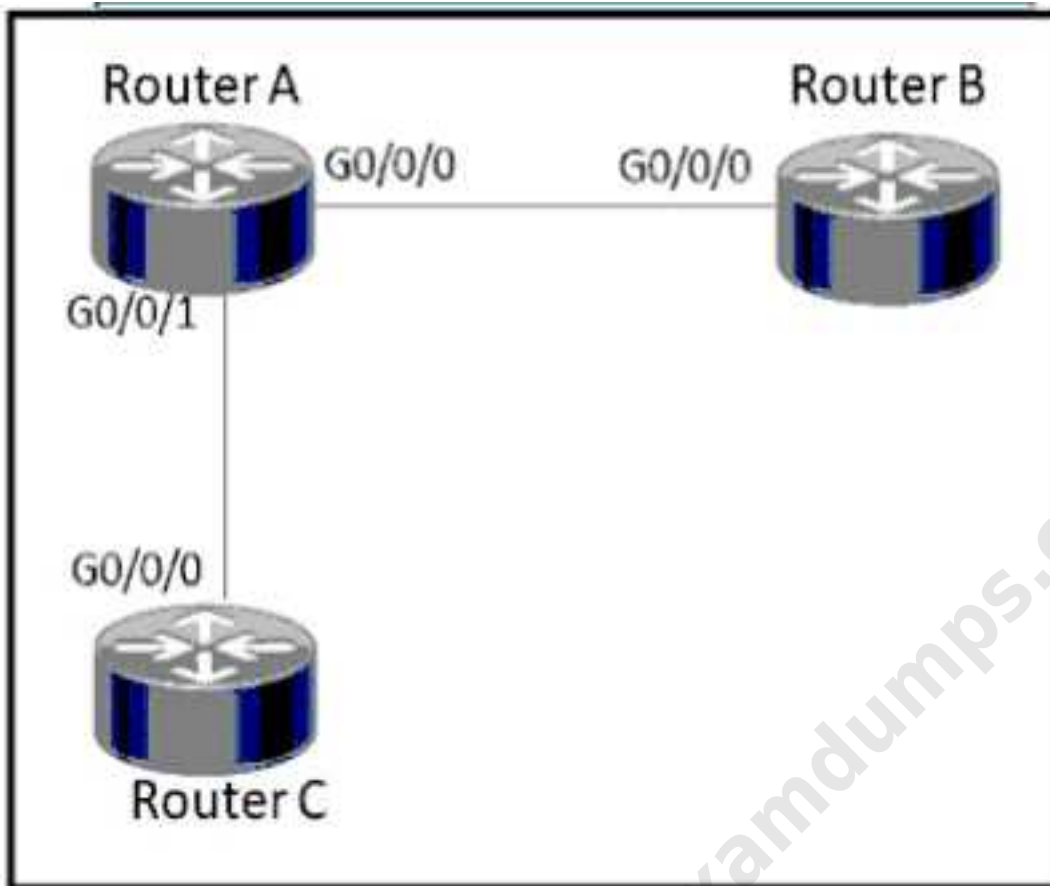
[https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-5/configuration-guide/b\\_cg75/b\\_cg75\\_chapter\\_011110.html](https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-5/configuration-guide/b_cg75/b_cg75_chapter_011110.html)

---

**Question: 191**

---

Refer to the exhibit.



How must router A be configured so that it only sends Cisco Discovery Protocol Information to router C?

☒ #config t  
Router A (config)#cdp run  
Router A (config)#interface gi0/0/0  
Router A (config-if)#no cdp enable

☒ #config t  
Router A (config)#cdp run  
Router A (config)#interface gi0/0/0  
Router A (config-if)#cdp enable

☒ #config t  
Router A (config)#cdp run  
Router A (config)#interface gi0/0/1  
Router A (config-if)#cdp enable

☐ #config t  
Router A (config)#no cdp run  
Router A (config)#interface gi0/0/1  
Router A (config-if)#cdp enable

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: D**

---

Explanation:

---

**Question: 192**

---

Which global command encrypt all passwords in the running configuration?

A. password-encrypt

B. enable password-encryption

- C. enable secret
- D. service password-encryption

---

**Answer: B**

---

Explanation:

---

**Question: 193**

---

What is the function of a hub-and-spoke WAN topology?

- A. allows access restrictions to be implemented between subscriber sites.
- B. provides direct connections between subscribers
- C. supports Layer 2 VPNs
- D. supports application optimization

---

**Answer: B**

---

Explanation:

---

**Question: 194**

---

What uses HTTP messages to transfer data to applications residing on different hosts?

- A. OpenFlow
- B. OpenStack
- C. OpFlex
- D. REST

---

**Answer: D**

---

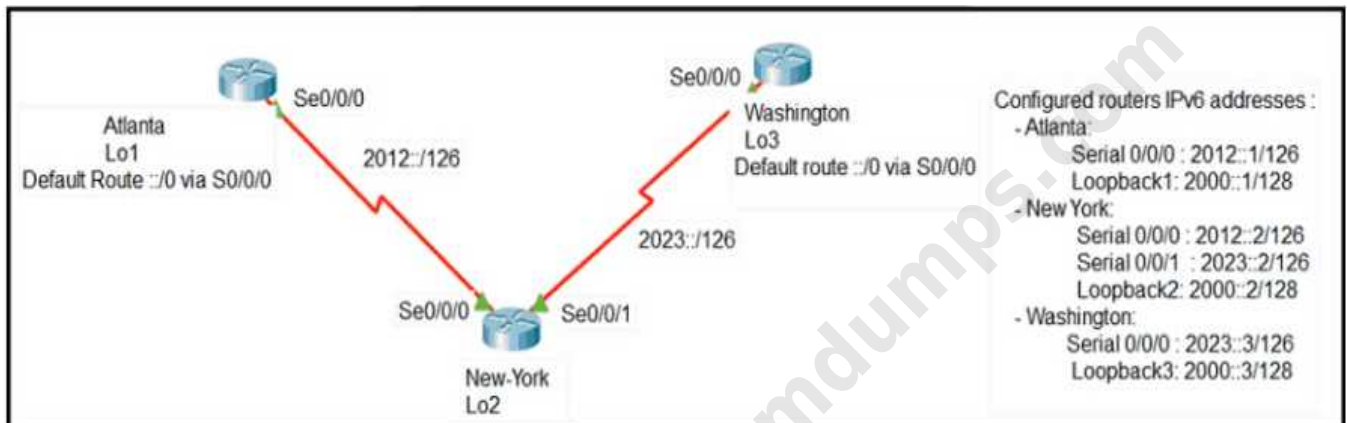
Explanation:

---

**Question: 195**

---

Refer to Exhibit.



The loopback1 interface of the Atlanta router must reach the loopback3 interface of the Washington router. Which two static host routes must be configured on the NEW York router? (Choose two)

- A. ipv6 route 2000::1/128 2012::1
- B. ipv6 route 2000::3/128 2023::3
- C. ipv6 route 2000::3/128 s0/0/0
- D. ipv6 route 2000::1/128 2012::2
- E. ipv6 route 2000::1/128 s0/0/1

---

**Answer: A,B**

---

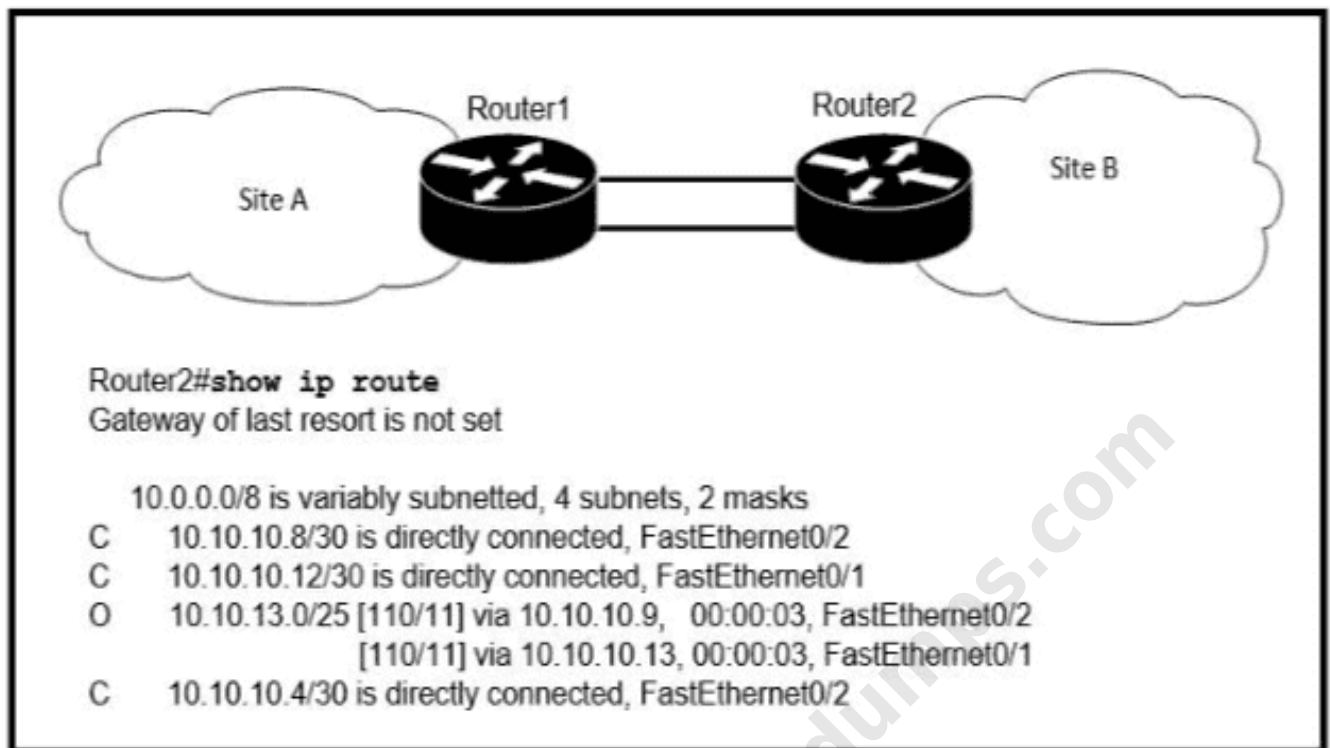
Explanation:

---

**Question: 196**

---

Refer to the exhibit.



If OSPF is running on this network, how does Router 2 handle traffic from Site B to 10.10.13/25 at Site A?

- A. It sends packets out of interface Fa0/2 only.
- B. It sends packets out of interface Fa0/1 only.
- C. It cannot send packets to 10.10.13.128/25
- D. It load-balances traffic out of Fa0/1 and Fa0/2

---

**Answer: C**

---

Explanation:

Router2 does not have an entry for the subnet 10.10.13.128/25. It only has an entry for 10.10.13.0/25, which ranges from 10.10.13.0 to 10.10.13.127.

<https://study-ccna.com/administrative-distance-metric/>

---

**Question: 197**

---

How does HSRP provide first hop redundancy?

- A. It load-balances traffic by assigning the same metric value to more than one route to the same destination in the IP routing table.
- B. It load-balances Layer 2 traffic along the path by flooding traffic out all interfaces configured with the same VLAN.
- C. It forwards multiple packets to the same destination over different routed links in the data path
- D. It uses a shared virtual MAC and a virtual IP address to a group of routers that serve as the default gateway for hosts on a LAN

---

**Answer: D**

---

Explanation:

[https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/ipapp\\_fhrp/configuration/xr-16/fhrp-xr-16-book/fhrp-hsrp-mgo.html](https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/ipapp_fhrp/configuration/xr-16/fhrp-xr-16-book/fhrp-hsrp-mgo.html)

---

**Question: 198**

---

What are two characteristics of the distribution layer in a three-tier network architecture? (Choose two.)

- A. serves as the network aggregation point
- B. provides a boundary between Layer 2 and Layer 3 communications
- C. designed to meet continuous, redundant uptime requirements
- D. is the backbone for the network topology
- E. physical connection point for a LAN printer

---

**Answer: B,C**

---

Explanation:

---

**Question: 199**

---

What is the purpose of using First Hop Redundancy Protocol in a specific subnet?

- A. Filter traffic based on destination IP addressing
- B. Sends the default route to the hosts on a network
- C. ensures a loop-free physical topology
- D. forwards multicast hello messages between routers

---

**Answer: D**

---

Explanation:

FHRP is layer 3 protocol whose purpose is to protect the default gateway by offering redundancy of the gateway in a subnet. This is achieved by allowing two or more routers to provide a backup for the first-hop IP router address. If a failure of an active router occurs, the backup router will take over the address. The routers negotiate their roles (Active/Standby) with each other by multicast hello messages to share the VIP (virtual IP address) between the FHRP routers. The terms Active/Standby vary between the different types of FHRP. The active router will act as the default gateway and the standby router acts as a backup the active router.

---

### Question: 200

---

Which access layer threat-mitigation technique provides security based on identity?

- A. Dynamic ARP Inspection
- B. using a non-default native VLAN
- C. 802.1x
- D. DHCP snooping

---

**Answer: C**

---

Explanation:



---

**Question: 201**

---

What must be considered when using 802.11a?

- A. It is compatible with 802.11a- and 802.11b-compliant wireless devices
- B. It is used in place of 802.11b/g when many nonoverlapping channels are required
- C. It is susceptible to interference from 2.4 GHz devices such as microwave ovens.
- D. It is chosen over 802.11b/g when a lower-cost solution is necessary

---

**Answer: A**

---

Explanation:

---

**Question: 202**

---

When a site-to-site VPN is configured, which IPsec mode provides encapsulation and encryption of the entire original IP packet?

- A. IPsec tunnel mode with AH
- B. IPsec transport mode with AH
- C. IPsec tunnel mode with ESP
- D. IPsec transport mode with ESP

---

**Answer: C**

---

Explanation:

“Encapsulating Security Payload

...

Unlike Authentication Header (AH), ESP in transport mode does not provide integrity and authentication for the entire IP packet. However, in Tunnel Mode, where the entire original IP packet is encapsulated with a new packet header added, ESP protection is afforded to the whole inner IP packet (including the inner header) while the outer header (including any outer IPv4 options or IPv6

extension headers) remains unprotected.

---

**Question: 203**

---

What does physical access control regulate?

- A. access to specific networks based on business function
- B. access to servers to prevent malicious activity
- C. access to computer networks and file systems
- D. access to networking equipment and facilities

---

**Answer: D**

---

Explanation:

---

**Question: 204**

---

On workstations running Microsoft Windows, which protocol provides the default gateway for the device?

- A. DHCP
- B. STP
- C. SNMP
- D. DNS

---

**Answer: A**

---

Explanation:

---

**Question: 205**

---

How are VLAN hopping attacks mitigated?

- A. enable dynamic ARP inspection
- B. manually implement trunk ports and disable DTP
- C. activate all ports and place in the default VLAN
- D. configure extended VLANs

---

**Answer: B**

---

Explanation:

---

**Question: 206**

---

What is the role of a firewall in an enterprise network?

- A. Forwards packets based on stateless packet inspection
- B. Processes unauthorized packets and allows passage to less secure segments of the network
- C. determines which packets are allowed to cross from unsecured to secured networks
- D. explicitly denies all packets from entering an administrative domain

---

**Answer: C**

---

Explanation:

---

**Question: 207**

---

What is a function of the Cisco DNA Center Overall Health Dashboard?

- A. It provides a summary of the top 10 global issues.
- B. It provides detailed activity logging for the 10 devices and users on the network.
- C. It summarizes the operational status of each wireless device on the network.
- D. It summarizes daily and weekly CPU usage for servers and workstations in the network.

---

**Answer: A**

---

Explanation:

---

**Question: 208**

---

DRAG DROP

Drag and drop the DNS lookup components from the left onto the functions on the right.

|                     |                                                                                  |
|---------------------|----------------------------------------------------------------------------------|
| cache               | local database of address mappings that improves name-resolution performance     |
| DNS                 | service that maps hostnames to IP addresses                                      |
| domain              | disables DNS services on a Cisco device                                          |
| name resolver       | in response to client requests, queries a name server for IP address information |
| no ip domain-lookup | component of a URL that indicates the location or organization type.             |

---

**Answer:**

---

Explanation:

|                     |                                                                                  |
|---------------------|----------------------------------------------------------------------------------|
| cache               | local database of address mappings that improves name-resolution performance     |
| DNS                 | service that maps hostnames to IP addresses                                      |
| domain              | disables DNS services on a Cisco device                                          |
| name resolver       | in response to client requests, queries a name server for IP address information |
| no ip domain-lookup | component of a URL that indicates the location or organization type.             |

---

**Question: 209**

---

After installing a new Cisco ISE server, which task must the engineer perform on the Cisco WLC to connect wireless clients on a specific VLAN based on their credentials?

- A. Enable the allow AAA Override
- B. Enable the Even: Driven RRM.
- C. Disable the LAG Mode or Next Reboot.
- D. Enable the Authorized MIC APs against auth-list or AAA.

---

**Answer: A**

---

Explanation:

---

**Question: 210**

---

A network analyst is tasked with configured the date and time on a router using EXEC mode. The date must be set to 12:00am. Which command should be used?

- A. Clock timezone
- B. Clock summer-time-recurring
- C. Clock summer-time date
- D. Clock set

---

**Answer: D**

---

Explanation:

---

**Question: 211**

---

A network administrator is asked to configure VLANs 2, 3 and 4 for a new implementation. Some ports must be assigned to the new VLANs with unused remaining. Which action should be taken for

the unused ports?

- A. configure port in the native VLAN
- B. configure ports in a black hole VLAN
- C. configure in a nondefault native VLAN
- D. configure ports as access ports

---

**Answer: B**

---

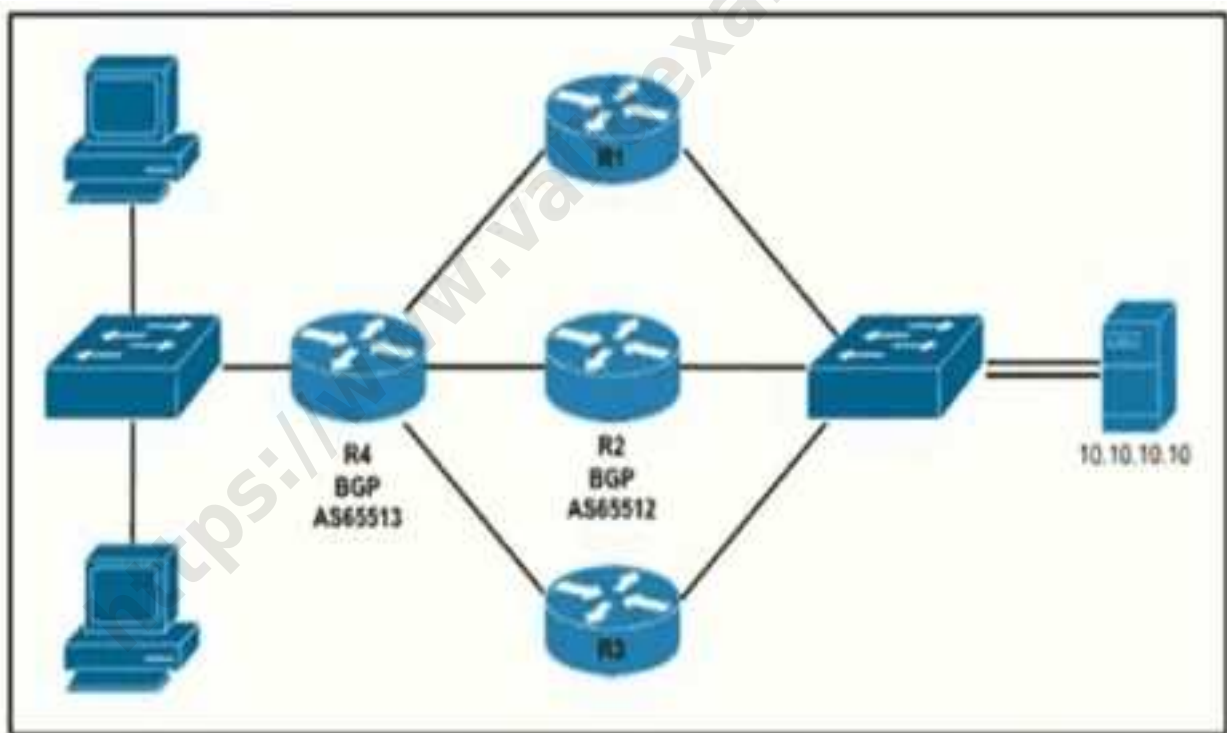
Explanation:

---

**Question: 212**

---

Refer to the exhibit.



Router R4 is dynamically learning the path to the server. If R4 is connected to R1 via OSPF Area 20, to R2 via BGP, and to R3 via EIGRP 777, which path is installed in the routing table of R4?

- A. the path through R1, because the OSPF administrative distance is 110
- B. the path through R2, because the IBGP administrative distance is 200

- C. the path through R2 because the EBGp administrative distance is 20
- D. the path through R3. because the EIGRP administrative distance is lower than OSPF and BGP

---

**Answer: C**

---

Explanation:

---

**Question: 213**

---

Why was the RFC 1918 address space defined?

- A. conserve public IPv4 addressing
- B. preserve public IPv6 address space
- C. reduce instances of overlapping IP addresses
- D. support the NAT protocol

---

**Answer: A**

---

Explanation:

---

**Question: 214**

---

Which HTTP status code is returned after a successful REST API request?

- A. 200
- B. 301
- C. 404
- D. 500

---

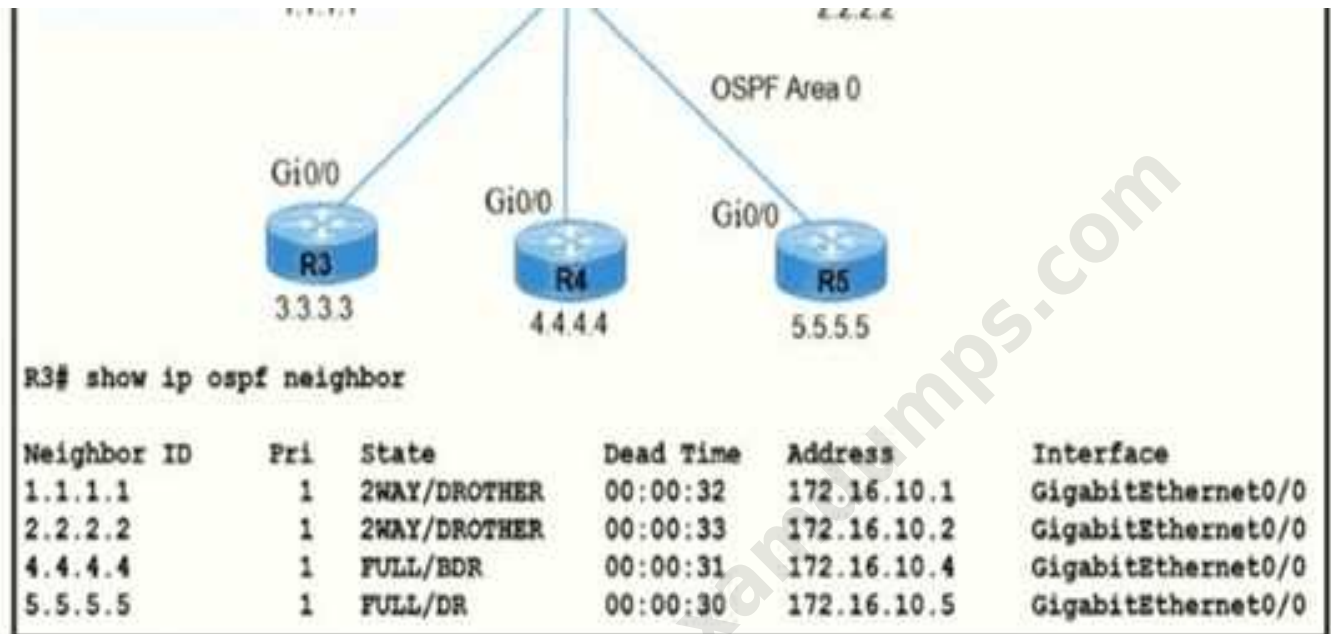
**Answer: A**

---

Explanation:

**Question: 215**

Refer to the exhibit.



R5 is the current DR on the network, and R4 is the BDR. Their interfaces are flapping, so a network engineer wants the OSPF network to elect a different DR and BDR. Which set of configurations must the engineer implement?

A)

```
R4(config)#interface gi0/0
R4(config-if)#ip ospf priority 20

R5(config)#interface gi0/0
R5(config-if)#ip ospf priority 10
```

B)

```
R2(config)#interface gi0/0
R2(config-if)#ip ospf priority 259

R3(config)#interface gi0/0
R3(config-if)#ip ospf priority 255
```

C)



```
R5(config)#interface gi0/0
R5(config-if)#ip ospf priority 120

R4(config)#interface gi0/0
R4(config-if)#ip ospf priority 110
```

D)

```
R3(config)#interface gi0/0
R3(config-if)#ip ospf priority 255

R2(config)#interface gi0/0
R2(config-if)#ip ospf priority 240
```

- A. Option
- B. Option
- C. Option
- D. Option

---

**Answer: D**

---

Explanation:

---

### Question: 216

---

What are network endpoints?

- A. act as routers to connect a user to the service provider network
- B. a threat to the network if they are compromised
- C. support inter-VLAN connectivity
- D. enforce policies for campus-wide traffic going to the internet

---

**Answer: B**

---

Explanation:

---

### Question: 217

---

Which two components are needed to create an Ansible script that configures a VLAN on a switch?  
(Choose two.)

- A. cookbook
- B. task
- C. playbook
- D. model
- E. recipe

---

**Answer: C,D**

---

Explanation:

---

**Question: 218**

---

DRAG DROP

Drag and drop the statement about networking from the left into the Corresponding networking types on the right. Not all statements are used.

This type deploys a consistent configuration across multiple devices.

A distributed control plane is needed.

This type requires a distributed management plane.

Southbound APIs are used to apply configurations.

Northbound APIs interact with end devices.

Controller-Based Networking

Traditional Networking

---

**Answer:**

---

Explanation:

2, 4

1, 3

---

**Question: 219**

---

Which two events occur automatically when a device is added to Cisco DNA Center? (Choose two. )

- A. The device is assigned to the Global site.
- B. The device is placed into the Unmanaged state.
- C. The device is placed into the Provisioned state.
- D. The device is placed into the Managed state.
- E. The device is assigned to the Local site.

---

**Answer: A,B**

---

Explanation:

---

**Question: 220**

---

Which virtual MAC address is used by VRRP group 1?

- A. 0050.0c05.ad81
- B. 0007.c061.bc01
- C. 0000.5E00.0101
- D. 0500.3976.6401

---

**Answer: C**

---

Explanation:

The virtual router MAC address associated with a virtual router is an

IEEE 802 MAC Address in the following format:

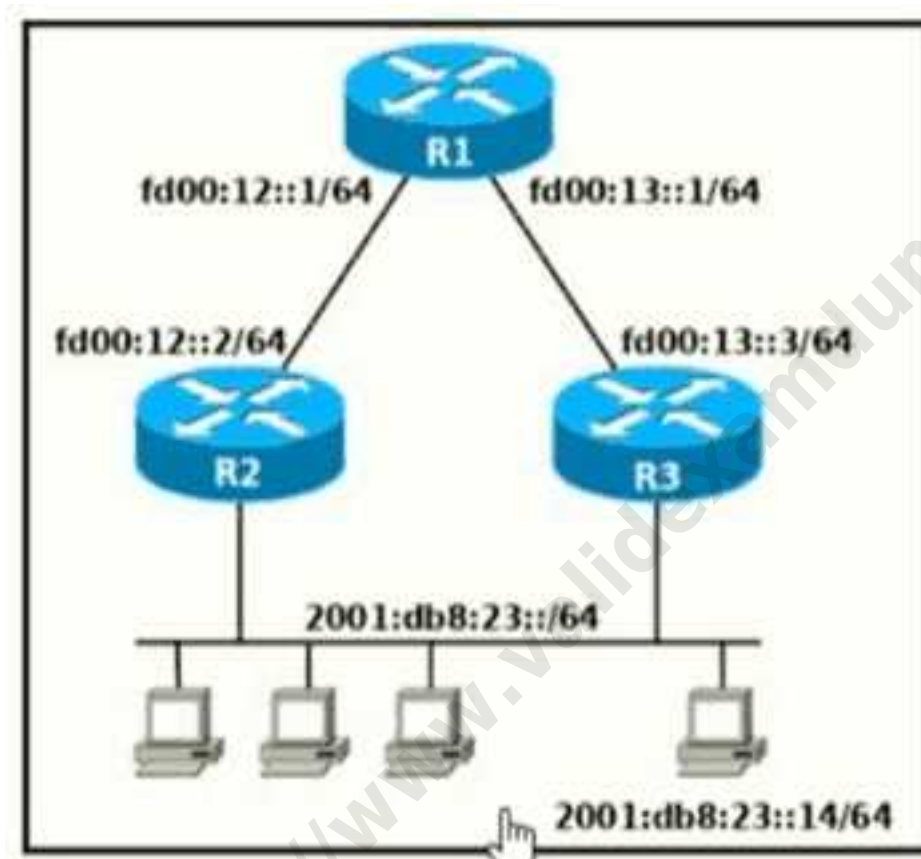
00-00-5E-00-01-{VRID} (in hex in internet standard bit-order)

---

**Question: 221**

---

Refer to the exhibit.



Which two commands, when configured on router R1, fulfill these requirements? (Choose two.)

Packets towards the entire network 2001:db8:2::/64 must be forwarded through router R2.

Packets toward host 2001:db8:23::14 preferably must be forwarded through R3.

- A. Ipv6 route 2001:db8:23::/128 fd00:12::2
- B. Ipv6 route 2001:db8:23::14/128 fd00:13::3
- C. Ipv6 route 2001:db8:23::14/64 fd00:12::2
- D. Ipv6 route 2001:db8:23::/64 fd00:12::2
- E. Ipv6 route 2001:db8:23::14/64 fd00:12::2 200

---

**Answer: D,E**

---

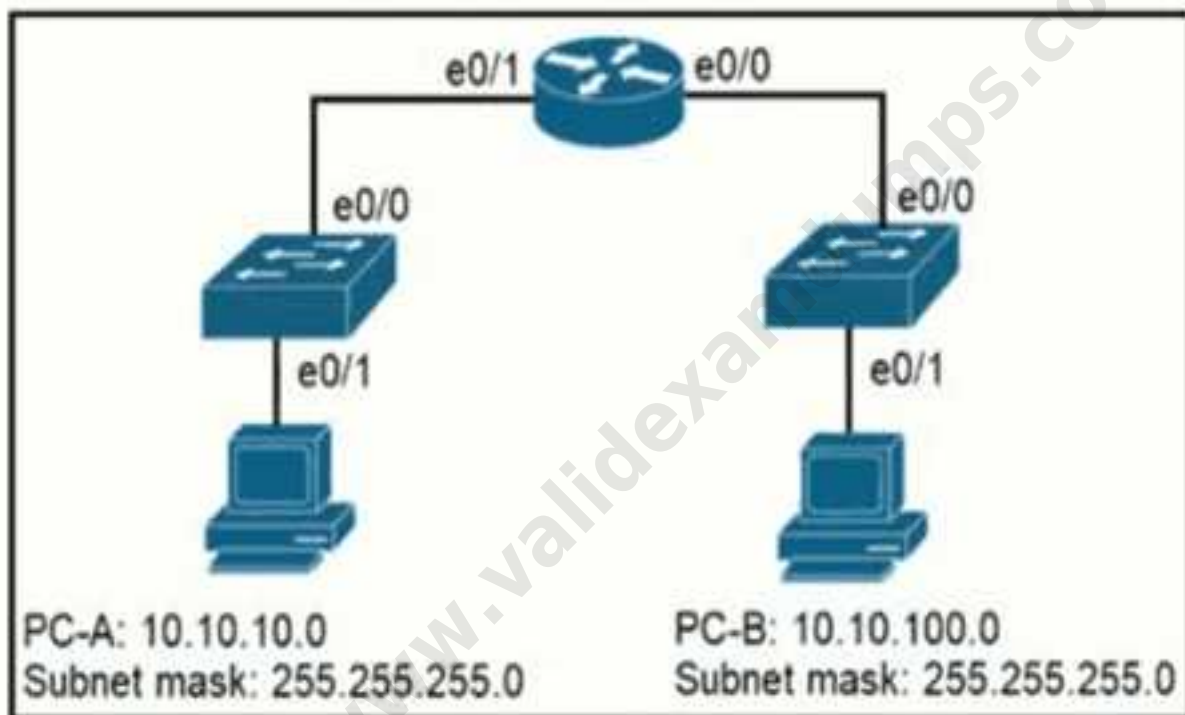
Explanation:

---

**Question: 222**

---

Refer to the exhibit.



When PC-A sends traffic to PC-B, which network component is in charge of receiving the packet from PC-A verifying the IP addresses, and forwarding the packet to PC-B?

- A. Layer 2 switch
- B. Router
- C. Load balancer
- D. firewall

---

**Answer: B**

---

Explanation:

PC--A and PC-B are not in the same network. Switches send traffic in layer 2 and within the same VLA while routers route traffic to different subnet and at layer 3.

---

**Question: 223**

---

In software-defined architecture, which plane handles switching for traffic through a Cisco router?

- A. Control
- B. Management
- C. Data
- D. application

---

**Answer: C**

---

Explanation:

Data plane—Handles all the data traffic. The basic functionality of a Cisco NX-OS device is to forward packets from one interface to another. The packets that are not meant for the switch itself are called the transit packets. These packets are handled by the data plane

---

**Question: 224**

---

Which level of severity must be set to get informational syslogs?

- A. alert
- B. critical
- C. notice
- D. debug

---

**Answer: D**

---

Explanation:

---

**Question: 225**

---

When a switch receives a frame for a known destination MAC address, how is the frame handed?

- A. sent to the port identified for the known MAC address
- B. broadcast to all ports
- C. forwarded to the first available port
- D. flooded to all ports except the one from which it originated

---

**Answer: A**

---

Explanation:

---

**Question: 226**

---

How does QoS optimize voice traffic?

- A. reducing bandwidth usage
- B. by reducing packet loss
- C. by differentiating voice and video traffic
- D. by increasing jitter

---

**Answer: C**

---

Explanation:

---

**Question: 227**

---

What is the function of a controller in controller-based networking?

- A. It serves as the centralized management point of an SDN architecture.
- B. It centralizes the data plane for the network.
- C. It is the card on a core router that maintains all routing decisions for a campus.
- D. It is a pair of core routers that maintain all routing decisions for a campus

---

**Answer: A**

---

Explanation:

---

**Question: 228**

---

What are two similarities between UTP Cat 5e and Cat 6a cabling? (Choose two.)

- A. Both operate at a frequency of 500 MHz.
- B. Both support runs of up to 55 meters.
- C. Both support runs of up to 100 meters.
- D. Both support speeds of at least 1 Gigabit.
- E. Both support speeds up to 10 Gigabit.

---

**Answer: C,D**

---

Explanation:

---

**Question: 229**

---

What is a characteristic of cloud-based network topology?

- A. wireless connections provide the sole access method to services
- B. onsite network services are provided with physical Layer 2 and Layer 3 components



- C. services are provided by a public, private, or hybrid deployment
- D. physical workstations are configured to share resources

---

**Answer: A**

---

Explanation:

---

**Question: 230**

---

What is the difference in data transmission delivery and reliability between TCP and UDP?

- A. TCP transmits data at a higher rate and ensures packet delivery. UDP retransmits lost data to ensure applications receive the data on the remote end.
- B. UDP sets up a connection between both devices before transmitting data. TCP uses the three-way handshake to transmit data with a reliable connection.
- C. UDP is used for multicast and broadcast communication. TCP is used for unicast communication and transmits data at a higher rate with error checking.
- D. TCP requires the connection to be established before transmitting data. UDP transmits data at a higher rate without ensuring packet delivery.

---

**Answer: D**

---

Explanation:

---

**Question: 231**

---

How are the switches in a spine-and-leaf topology interconnected?

- A. Each leaf switch is connected to one of the spine switches.
- B. Each leaf switch is connected to two spine switches, making a loop.
- C. Each leaf switch is connected to each spine switch.

D. Each leaf switch is connected to a central leaf switch, then uplinked to a core spine switch.

---

**Answer: C**

---

Explanation:

---

**Question: 232**

---

DRAG DROP

Drag and drop the IPv6 address type characteristics from the left to the right.

|                                                                  |                      |
|------------------------------------------------------------------|----------------------|
| attached to a single subnet                                      | Link-Local Address   |
| addresses with prefix FC00::/7                                   |                      |
| configured only once per interface                               |                      |
| addressing for exclusive use internally without Internet routing | Unique Local Address |
|                                                                  |                      |

---

**Answer:**

---

Explanation:

**Link-Local Address**

addresses with prefix FC00::/7

addressing for exclusive use internally without Internet routing

**Unique Local Address**

configured only once per interface

attached to a single subnet

---

**Question: 233**

---

Refer to the exhibit.

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
Gateway of last resort is 192.168.30.10 to network 0.0.0.0
 192.168.30.0/29 is subnetted, 2 subnets
C      192.168.30.0 is directly connected, FastEthernet0/0
C      192.168.30.8 is directly connected, Serial0/0.1
 192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
O IA   192.168.10.32/28 [110/193] via 192.168.30.10, 00:18:49, Serial0/0.1
O IA   192.168.10.0/27 [110/192] via 192.168.30.10, 00:18:49, Serial0/0.1
 192.168.20.0/30 is subnetted, 1 subnets
O IA   192.168.20.0 [110/128] via 192.168.30.10, 00:18:49, Serial0/0.1
 192.168.50.0/32 is subnetted, 1 subnets
C      192.168.50.1 is directly connected, Loopback0
O*IA 0.0.0.0/0 [110/84] via 192.168.30.10, 00:10:36, Serial0/0.1
```

What is the metric of the route to the 192.168.10.33/28 subnet?

- A. 84
- B. 110
- C. 128
- D. 192
- E. 193

---

**Answer: E**

---

Explanation:

---

### Question: 234

---

DRAG DROP

An engineer is configuring an encrypted password for the enable command on a router where the

local user database has already been configured Drag and drop the configuration commands from the left into the correct sequence on the right Not all commands are used

|                             |        |
|-----------------------------|--------|
| configure terminal          | first  |
| enable                      | second |
| enable secret \$hfl@4fs     | third  |
| exit                        | fourth |
| line vty 0 4                |        |
| service password-encryption |        |

**Answer:**

Explanation:

|                         |
|-------------------------|
| enable                  |
| configure terminal      |
| enable secret \$hfl@4fs |
| line vty 0 4            |

---

**Question: 235**

---

Which action does the router take as it forwards a packet through the network?

- A. The router replaces the original source and destination MAC addresses with the sending router MAC address as the source and neighbor MAC address as the destination
- B. The router encapsulates the original packet and then includes a tag that identifies the source router MAC address and transmits it transparently to the destination
- C. The router encapsulates the source and destination IP addresses with the sending router IP address as the source and the neighbor IP address as the destination
- D. The router replaces the source and destination labels with the sending router interface label as a source and the next hop router label as a destination

---

**Answer: A**

---

Explanation:

---

**Question: 236**

---

DRAG DROP

Drag and drop the functions of DHCP from the left onto any of the positions on the right Not all functions are used

|                                                                          |   |
|--------------------------------------------------------------------------|---|
| provides local control for network segments using a client-server scheme | 1 |
| reduces the administrative burden for onboarding end users               | 2 |
| associates hostnames to IP addresses                                     | 3 |
| maintains an address pool                                                | 4 |
| assigns IP addresses to local hosts for a configurable lease time        |   |
| offers domain name server configuration                                  |   |
| uses authoritative servers for record keeping                            |   |

---

**Answer:**

---

Explanation:

|                                                                          |
|--------------------------------------------------------------------------|
| maintains an address pool                                                |
| provides local control for network segments using a client-server scheme |
| reduces the administrative burden for onboarding end users               |
| assigns IP addresses to local hosts for a configurable lease time        |

---

**Question: 237**

---

In QoS, which prioritization method is appropriate for interactive voice and video?

- A. expedited forwarding
- B. traffic policing
- C. round-robin scheduling
- D. low-latency queuing

---

**Answer: D**

---

Explanation:

Topic 2, Exam Pool B

---

**Question: 238**

---

Which two values or settings must be entered when configuring a new WLAN in the Cisco Wireless LAN Controller GUI? (Choose two)

- A. management interface settings
- B. QoS settings
- C. Ip address of one or more access points
- D. SSID
- E. Profile name

---

**Answer: D,E**

---

Explanation:

---

**Question: 239**

---



Which communication interaction takes place when a southbound API is used?

- A. between the SDN controller and PCs on the network
- B. between the SON controller and switches and routers on the network
- C. between the SON controller and services and applications on the network
- D. between network applications and switches and routers on the network

---

**Answer: B**

---

Explanation:

---

**Question: 240**

---

What prevents a workstation from receiving a DHCP address?

- A. DTP
- B. STP
- C. VTP
- D. 802.10

---

**Answer: B**

---

Explanation:

---

**Question: 241**

---

An engineer must establish a trunk link between two switches. The neighboring switch is set to trunk or desirable mode. What action should be taken?

- A. configure switchport nonegotiate
- B. configure switchport mode dynamic desirable

- C. configure switchport mode dynamic auto
- D. configure switchport trunk dynamic desirable

---

**Answer: C**

---

Explanation:

---

**Question: 242**

---

If a switch port receives a new frame while it is actively transmitting a previous frame, how does it process the frames?

- A. The new frame is delivered first, the previous frame is dropped, and a retransmission request is sent.
- B. The previous frame is delivered, the new frame is dropped, and a retransmission request is sent.
- C. The new frame is placed in a queue for transmission after the previous frame.
- D. The two frames are processed and delivered at the same time.

---

**Answer: B**

---

Explanation:

---

**Question: 243**

---

A wireless administrator has configured a WLAN; however, the clients need access to a less congested 5-GHz network for their voice quality. What action must be taken to meet the requirement?

- A. enable AAA override
- B. enable RX-SOP
- C. enable DTIM
- D. enable Band Select

---

**Answer: D**

---

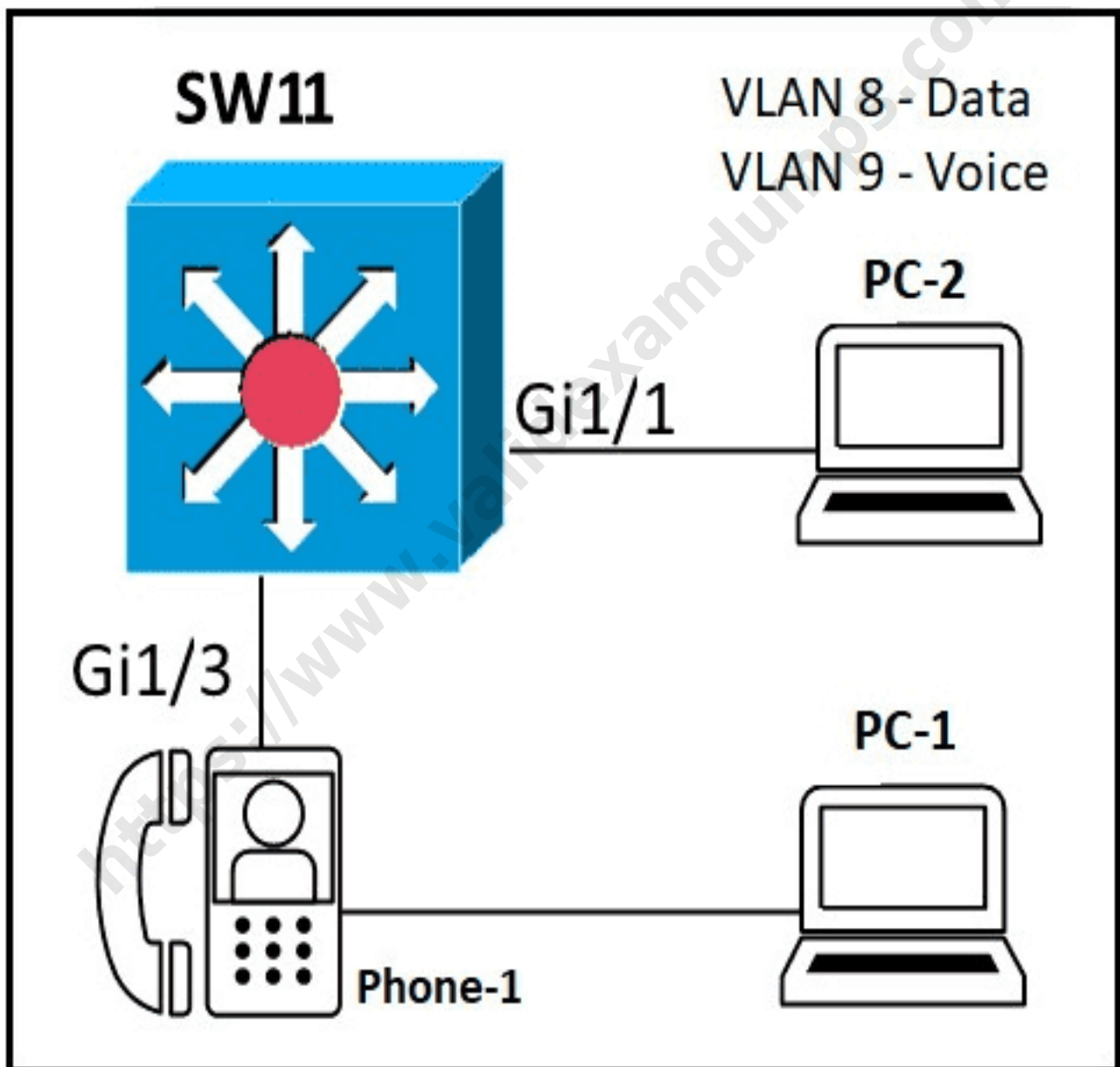
Explanation:

---

**Question: 244**

---

Refer to the exhibit.



An administrator must configure interfaces Gi1/1 and Gi1/3 on switch SW11. PC-1 and PC-2 must be placed in the Data VLAN, and Phone-1 must be placed in the Voice VLAN. Which configuration meets these requirements?

- interface gigabitethernet1/1  
switchport mode access  
switchport access vlan 8  
!  
interface gigabitethernet1/3  
switchport mode access  
switchport voice vlan 8  
switchport access vlan 9
- interface gigabitethernet1/1  
switchport mode access  
switchport access vlan 9  
!  
interface gigabitethernet1/3  
switchport mode trunk  
switchport trunk vlan 8  
switchport trunk vlan 9
- interface gigabitethernet1/1  
switchport mode access  
switchport access vlan 8  
!  
interface gigabitethernet1/3  
switchport mode access  
switchport access vlan 8  
switchport voice vlan 9
- interface gigabitethernet1/1  
switchport mode access  
switchport access vlan 8  
!  
interface gigabitethernet1/3  
switchport mode trunk  
switchport trunk vlan 8  
switchport voice vlan 9

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: C**

---

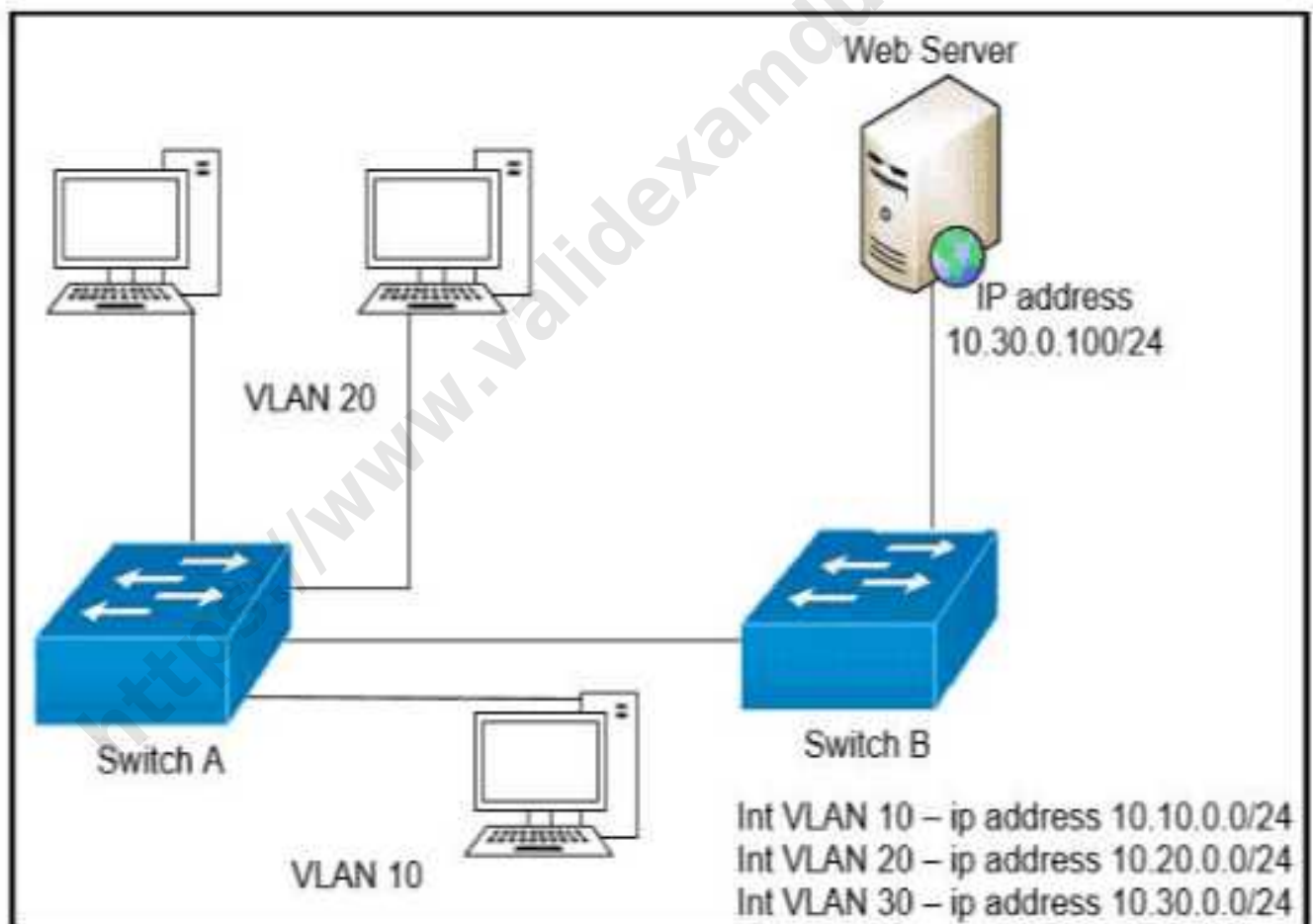
Explanation:

---

**Question: 245**

---

Refer to the exhibit.



A network engineer must block access for all computers on VLAN 20 to the web server via HTTP. All other computers must be able to access the web server. Which configuration, when applied to switch A, accomplishes this task?

```
config t
ip access-list extended wwwblock
deny tcp any host 10.30.0.100 eq 80
int vlan 10
ip access-group wwwblock in
```

```
config t
ip access-list extended wwwblock
deny tcp any host 10.30.0.100 eq 80
permit ip any any
int vlan 20
ip access-group wwwblock in
```

```
config t
ip access-list extended wwwblock
permit ip any any
deny tcp any host 10.30.0.100 eq 80
int vlan 30
ip access-group wwwblock in
```

```
config t
ip access-list extended wwwblock
permit ip any any
deny tcp any host 10.30.0.100 eq 80
int vlan 20
ip access-group wwwblock in
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: B**

---

Explanation:

---

**Question: 246**

---

An engineer must configure a WLAN using the strongest encryption type for WPA2- PSK. Which cipher fulfills the configuration requirement?

- A. WEP
- B. RC4
- C. AES
- D. TKIP

---

**Answer: C**

---

Explanation:

Many routers provide WPA2-PSK (TKIP), WPA2-PSK (AES), and WPA2-PSK (TKIP/AES) as options. TKIP is actually an older encryption protocol introduced with WPA to replace the very-insecure WEP encryption at the time. TKIP is actually quite similar to WEP encryption. TKIP is no longer considered secure, and is now deprecated. In other words, you shouldn't be using it.

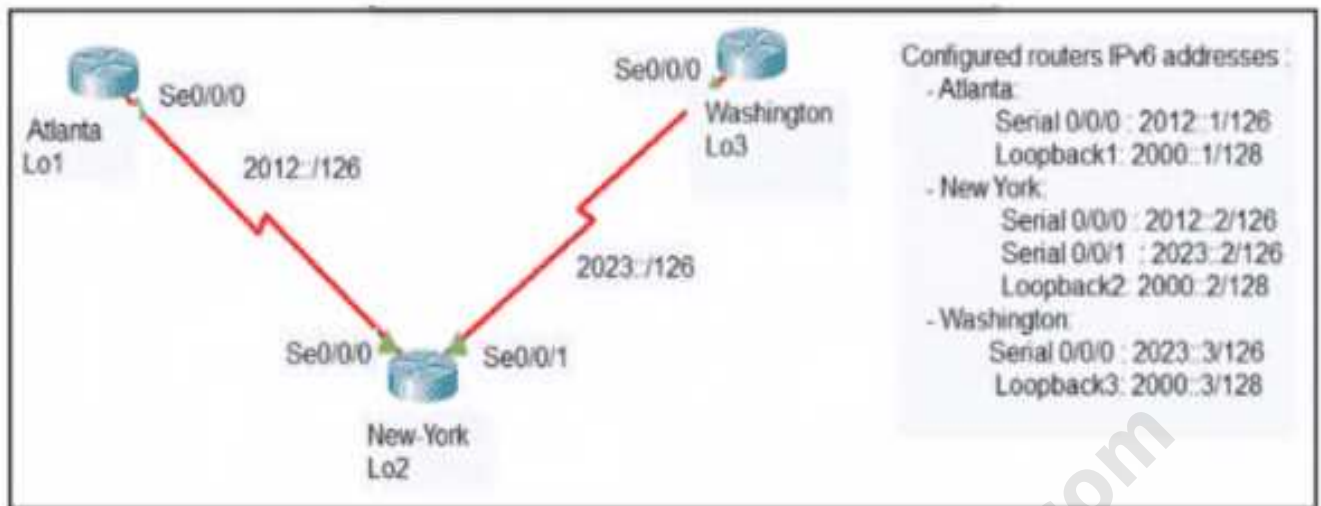
AES is a more secure encryption protocol introduced with WPA2 and it is currently the strongest encryption type for WPA2-PSK.

---

**Question: 247**

---

Refer to the exhibit.



An engineer configured the New York router with static routes that point to the Atlanta and Washington sites. When command must be configured on the Atlanta and Washington routers so that both sites are able to reach the loopback2 interface on the New York router?

- A. ipv6 route ::/0 Serial 0/0/1
- B. ipv6 route 0/0 Serial 0/0/0
- C. ipv6 route ::/0 Serial 0/0/0
- D. ip route 0.0.0.0.0.0.0.0 Serial 0/0/0
- E. ipv6 route ::/0 2000::2

---

**Answer: C**

---

Explanation:

---

**Question: 248**

---

DRAG DROP

Drag and drop the Cisco Wireless LAN Controller security settings from the left onto the correct security mechanism categories on the right.



|             |                                        |
|-------------|----------------------------------------|
| web policy  | Layer 2 Security Mechanisms            |
| Passthrough |                                        |
| WPA+WPA2    | Layer 3 Security Mechanisms (for WLAN) |
| 802.1X      |                                        |

**Answer:**

Explanation:

|                             |
|-----------------------------|
| Layer 2 Security Mechanisms |
| 802.1X                      |
| WPA+WPA2                    |

|                                        |
|----------------------------------------|
| Layer 3 Security Mechanisms (for WLAN) |
| web policy                             |
| Passthrough                            |

### Question: 249

Which unified access point mode continues to serve wireless clients after losing connectivity to the Cisco Wireless LAN Controller?

- A. sniffer
- B. mesh
- C. flexconnect
- D. local

---

**Answer: C**

---

Explanation:

Reference: [https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-5/config-guide/b\\_cg85/flexconnect.html](https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-5/config-guide/b_cg85/flexconnect.html)

---

**Question: 250**

---

What is a difference between RADIUS and TACACS+?

- A. RADIUS is most appropriate for dial authentication, but TACACS+ can be used for multiple types of authentication
- B. TACACS+ encrypts only password information and RADIUS encrypts the entire payload
- C. TACACS+ separates authentication and authorization, and RADIUS merges them
- D. RADIUS logs all commands that are entered by the administrator, but TACACS+ logs only start, stop, and interim commands

---

**Answer: C**

---

Explanation:

---

**Question: 251**

---

What Is a syslog facility?

- A. Host that is configured for the system to send log messages
- B. password that authenticates a Network Management System to receive log messages
- C. group of log messages associated with the configured severity level
- D. set of values that represent the processes that can generate a log message

---

**Answer: C**

---

Explanation:

Cisco Community – Difference between logging level and logging facility

Post by ahmednaas

“The logging facility command basically tells the syslog server where to put the log message. You configure the syslog server with something like:

```
local7.debug /var/adm/local7.log
```

Now, when you use the “logging facility local7” on your device, all messages with severity “debug” or greater should be saved in /var/adm/local7.log.”

Example: on a switch, any process (CDP, SNMP, etc.) can generate a log message. On a syslog server, the logging facility is the place where all received messages with the same priority level are stored.

---

**Question: 252**

---

What are two characteristics of a public cloud Implementation? (Choose two.)

- A. It is owned and maintained by one party, but it is shared among multiple organizations.
- B. It enables an organization to fully customize how it deploys network resources.
- C. It provides services that are accessed over the Internet.
- D. It is a data center on the public Internet that maintains cloud services for only one company.
- E. It supports network resources from a centralized third-party provider and privately-owned virtual resources

---

**Answer: C,E**

---

Explanation:

Private cloud is cloud infrastructure operated solely for a single organization, whether managed internally or by a third party, and hosted either internally or externally.

Most public-cloud providers offer direct-connection services that allow customers to securely link their legacy data centers to their cloud-resident applications.

---

**Question: 253**

---

Refer to the exhibit.

```
R1# show ip route
D    192.168.10.0/24 [90/2679326] via 192.168.1.1
R    192.168.10.0/27 [120/3] via 192.168.1.2
O    192.168.10.0/23 [110/2] via 192.168.1.3
i L1 192.168.10.0/13 [115/30] via 192.168.1.4
```

How does router R1 handle traffic to 192.168.10.16?

- A. It selects the IS-IS route because it has the shortest prefix inclusive of the destination address.
- B. It selects the EIGRP route because it has the lowest administrative distance.
- C. It selects the OSPF route because it has the lowest cost.
- D. It selects the RIP route because it has the longest prefix inclusive of the destination address.

---

**Answer: D**

---

Explanation:

---

**Question: 254**

---

What role does a hypervisor provide for each virtual machine in server virtualization?

- A. infrastructure-as-a-service.
- B. Software-as-a-service
- C. control and distribution of physical resources
- D. services as a hardware controller.

---

**Answer: C**

---

Explanation:

The hypervisor creates and manages virtual machines on a host computer and allocates physical system resources to them.

---

**Question: 255**

---

What is the function of a server?

- A. It transmits packets between hosts in the same broadcast domain.
- B. It provides shared applications to end users.
- C. It routes traffic between Layer 3 devices.
- D. It Creates security zones between trusted and untrusted networks

---

**Answer: B**

---

Explanation:

---

**Question: 256**

---

Refer to the exhibit.

```
ip arp inspection vlan 5-10
interface fastethernet 0/1
  switchport mode access
  switchport access vlan 5
```

What is the effect of this configuration?

- A. All ARP packets are dropped by the switch
- B. Egress traffic is passed only if the destination is a DHCP server.
- C. All ingress and egress traffic is dropped because the interface is untrusted
- D. The switch discard all ingress ARP traffic with invalid MAC-to-IP address bindings.

---

**Answer: D**

---

Explanation:

---

**Question: 257**

---

What is a characteristic of private IPv4 addressing?

- A. traverse the Internet when an outbound ACL is applied
- B. issued by IANA in conjunction with an autonomous system number
- C. composed of up to 65.536 available addresses
- D. used without tracking or registration

---

**Answer: D**

---

Explanation:

---

**Question: 258**

---

Refer to the exhibit.

<https://www.validexamdumps.com>



Which configuration issue is preventing the OSPF neighbor relationship from being established between the two routers?



- A. R2 is using the passive-interface default command
- B. R1 has an incorrect network command for interface Gi1/0
- C. R2 should have its network command in area 1
- D. R1 interface Gi1/0 has a larger MTU size

---

**Answer: D**

---

Explanation:

---

**Question: 259**

---

Refer to the exhibit.

```
R1# show ip route | begin gateway
Gateway of last resort is 209.165.200.246 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 209.165.200.246, Serial0/1/0
    is directly connected, Serial0/1/0
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
S   172.16.3.0/24 [1/0] via 207.165.200.250, Serial0/0/0
O   172.16.3.0/28 [110/84437] via 207.165.200.254, 00:00:28, Serial0/0/1
    207.165.200.0/24 is variably subnetted, 6 subnets, 2 masks
C   207.165.200.244/30 is directly connected, Serial0/1/0
L   207.165.200.245/32 is directly connected, Serial0/1/0
C   207.165.200.248/30 is directly connected, Serial0/0/0
L   207.165.200.249/32 is directly connected, Serial0/0/0
C   207.165.200.252/30 is directly connected, Serial0/0/1
L   207.165.200.253/32 is directly connected, Serial0/0/1
```

A packet is being sent across router R1 to host 172.163.3.14. To which destination does the router send the packet?

- A. 207.165.200.246 via Serial0/1/0
- B. 207.165.200.254 via Serial0/0/1
- C. 207.165.200.254 via Serial0/0/0

D. 207.165.200.250 via Serial/0/0/0

---

**Answer: A**

---

Explanation:

---

**Question: 260**

---

DRAG DROP

Drag and drop to the characteristics of networking from the left onto the correct networking types on the right.

|                                |                                                                           |
|--------------------------------|---------------------------------------------------------------------------|
| focused on network             | <b>Controller-Based Networking</b><br><div></div> <div></div> <div></div> |
| focused on devices             |                                                                           |
| user input is a configuration  |                                                                           |
| user input is a policy         | <b>Traditional Networking</b><br><div></div> <div></div> <div></div>      |
| uses white list security model |                                                                           |
| uses black list security model |                                                                           |

---

**Answer:**

---

Explanation:

|                             |                                |
|-----------------------------|--------------------------------|
| Controller-Based Networking | focused on network             |
|                             | uses white list security model |
|                             | user input is a policy         |
| Traditional Networking      | focused on devices             |
|                             | uses black list security model |
|                             | user input is a configuration  |

---

**Question: 261**

---

A network administrator needs to aggregate 4 ports into a single logical link which must negotiate layer 2 connectivity to ports on another switch. What must be configured when using active mode on both sides of the connection?

- A. 802.1q trunks
- B. Cisco vPC
- C. LLDP
- D. LACP

---

**Answer: D**

---

Explanation:

---

**Question: 262**

---

When a WPA2-PSK WLAN is configured in the wireless LAN Controller, what is the minimum number of characters that in ASCII format?

- A. 6
- B. 8
- C. 12
- D. 18

---

**Answer: B**

---

Explanation:

---

**Question: 263**

---

What are two differences between optical-fiber cabling and copper cabling? (Choose two)

- A. Light is transmitted through the core of the fiber
- B. A BNC connector is used for fiber connections
- C. The glass core component is encased in a cladding
- D. Fiber connects to physical interfaces using RJ-45 connections
- E. The data can pass through the cladding

---

**Answer: A,C**

---

Explanation:

---

**Question: 264**

---

Refer to the exhibit.

```
R1# sh ip ospf int gig0/0
Gig0/0 is up, line protocol is up
  Internet Address 10.201.24.8/28, Area 1, Attached via Network Statement
  Process ID 100, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1
  Topology-MTID      Cost      Disabled      Shutdown      Topology Name
        0             1         no           no           Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.1.1, Interface address 10.201.24.8
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:07

R2#sh ip ospf int gig0/0
gig0/0 is up, line protocol is up
  Internet Address 10.201.24.1/28, Area 1
  Process ID 100, Router ID 172.16.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 172.16.1.1, Interface address 10.201.24.1
  No backup designated router on this network
  Timer intervals configured, Hello 20, Dead 80, Wait 80, Retransmit 5
```

What action establishes the OSPF neighbor relationship without forming an adjacency?

- A. modify hello interval
- B. modify process ID
- C. modify priority
- D. modify network type

---

**Answer: A**

---

Explanation:

---

### Question: 265

---

How does WPA3 improve security?

- A. It uses SAE for authentication.
- B. It uses a 4-way handshake for authentication.

- C. It uses RC4 for encryption.
- D. It uses TKIP for encryption.

---

**Answer: A**

---

Explanation:

---

**Question: 266**

---

A device detects two stations transmitting frames at the same time. This condition occurs after the first 64 bytes of the frame is received interface counter increments?

- A. collision
- B. CRC
- C. runt
- D. late collision

---

**Answer: D**

---

Explanation:

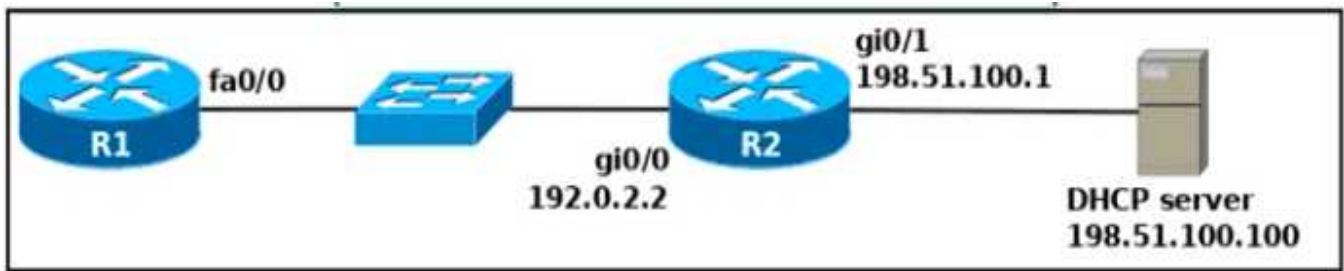
<https://www.cisco.com/c/en/us/support/docs/interfaces-modules/port-adapters/12768-eth-collisions.html>

---

**Question: 267**

---

Refer to the exhibit.



An engineer deploys a topology in which R1 obtains its IP configuration from DHCP. If

the switch and DHCP server configurations are complete and correct. Which two sets of commands must be configured on R1 and R2 to complete the task? (Choose two)

- A. R1(config)# interface fa0/0R1(config-if)# ip helper-address 198.51.100.100
- B. R2(config)# interface gi0/0R2(config-if)# ip helper-address 198.51.100.100
- C. R1(config)# interface fa0/0R1(config-if)# ip address dhcpR1(config-if)# no shutdown
- D. R2(config)# interface gi0/0R2(config-if)# ip address dhcp
- E. R1(config)# interface fa0/0R1(config-if)# ip helper-address 192.0.2.2

---

**Answer: B,C**

---

Explanation:

---

### Question: 268

---

When OSPF learns multiple paths to a network, how does it select a route?

- A. It multiple the active K value by 256 to calculate the route with the lowest metric.
- B. For each existing interface, it adds the metric from the source router to the destination to calculate the route with the lowest bandwidth.
- C. It divides a reference bandwidth of 100 Mbps by the actual bandwidth of the existing interface to calculate the router with the lowest cost.
- D. It count the number of hops between the source router and the destination to determine the

router with the lowest metric

---

**Answer: C**

---

Explanation:

---

**Question: 269**

---

A user configured OSPF in a single area between two routers A serial interface connecting R1 and R2 is running encapsulation PPP By default which OSPF network type is seen on this interface when the user types show ip ospf interface on R1 or R2?

- A. port-to-multipoint
- B. broadcast
- C. point-to-point
- D. nonbroadcast

---

**Answer: C**

---

Explanation:

The default OSPF network type for HDLC and PPP on Serial link is point-to-point (while the default OSPF network type for Ethernet link is Broadcast).

---

**Question: 270**

---

How do AAA operations compare regarding user identification, user services and access control?

- A. Authorization provides access control and authentication tracks user services
- B. Authentication identifies users and accounting tracks user services



- C. Accounting tracks user services, and authentication provides access control
- D. Authorization identifies users and authentication provides access control

---

**Answer: B**

---

Explanation:

---

**Question: 271**

---

An engineer requires a scratch interface to actively attempt to establish a trunk link with a neighbor switch. What command must be configured?

- A. switchport mode trunk
- B. switchport mode dynamic desirable
- C. switchport mode dynamic auto
- D. switchport nonegotiate

---

**Answer: C**

---

Explanation:

---

**Question: 272**

---

Refer to the exhibit.

```
R1# show ip route | begin gateway
Gateway of last resort is 209.165.200.246 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 209.165.200.246, Serial0/1/0
    is directly connected, Serial0/1/0
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
S    172.16.3.0/24 [1/0] via 209.165.200.250, Serial0/0/0
O    172.16.3.0/28 [110/1] via 209.165.200.254, 00:00:28, Serial0/0/1
    209.165.200.0/24 is variably subnetted, 6 subnets, 2 masks
C    209.165.200.244/30 is directly connected, Serial0/1/0
L    209.165.200.245/32 is directly connected, Serial0/1/0
C    209.165.200.248/30 is directly connected, Serial0/0/0
L    209.165.200.249/32 is directly connected, Serial0/0/0
C    209.165.200.252/30 is directly connected, Serial0/0/1
L    209.165.200.253/32 is directly connected, Serial0/0/1
```

A packet is being sent across router R1 to host 172.16.0.14. What is the destination route for the packet?

- A. 209.165.200.254 via Serial0/0/1
- B. 209.165.200.254 via Serial0/0/0
- C. 209.165.200.246 via Serial0/1/0
- D. 209.165.200.250 via Serial0/0/0

---

**Answer: A**

---

Explanation:

---

### Question: 273

---

With REST API, which standard HTTP header tells a server which media type is expected by the client?

- A. Accept-Encoding: gzip, deflate
- B. Accept-Patch: text/example; charset=utf-8

C. Content-Type: application/json; charset=utf-8

D. Accept: application/json

---

**Answer: D**

---

Explanation:

Accept header is a way for a client to specify the media type of the response content it is expecting and Content-type is a way to specify the media type of request being sent from the client to the server.

<http://www.java-allandsundry.com/2012/08/accept-header-vs-content-type-header.html#:~:text=Accept%20and%20Content%2Dtype%20are,the%20client%20to%20the%20server>

---

**Question: 274**

---

Which JSON data type is an unordered set of attribute- value pairs?

A. array

B. string

C. object

D. Boolean

---

**Answer: C**

---

Explanation:

---

**Question: 275**

---

What is the expected outcome when an EUI-64 address is generated?

A. The seventh bit of the original MAC address of the interface is inverted

- B. The interface ID is configured as a random 64-bit value
- C. The characters FE80 are inserted at the beginning of the MAC address of the interface
- D. The MAC address of the interface is used as the interface ID without modification

---

**Answer: A**

---

Explanation:

---

**Question: 276**

---

Which protocol prompts the Wireless LAN Controller to generate its own local web administration SSL certificate for GUI access?

- A. HTTPS
- B. RADIUS
- C. TACACS+
- D. HTTP

---

**Answer: A**

---

Explanation:

You can protect communication with the GUI by enabling HTTPS. HTTPS protects HTTP browser sessions by using the Secure Sockets Layer (SSL) protocol. When you enable HTTPS, the controller generates its own local web administration SSL certificate and automatically applies it to the GUI. You also have the option of downloading an externally generated certificate.

Reference: [https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-0/configuration-guide/b\\_cg80/b\\_cg80\\_chapter\\_011.html](https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-0/configuration-guide/b_cg80/b_cg80_chapter_011.html)

---

**Question: 277**

---

The SW1 interface g0/1 is in the down/down state. Which two configurations are valid reasons for

the interface conditions?(choose two)

- A. There is a duplex mismatch
- B. There is a speed mismatch
- C. There is a protocol mismatch
- D. The interface is shut down
- E. The interface is error-disabled

---

**Answer: B,E**

---

Explanation:

---

**Question: 278**

---

Which network plane is centralized and manages routing decisions?

- A. policy plane
- B. management plane
- C. control plane
- D. data plane

---

**Answer: C**

---

Explanation:

---

**Question: 279**

---

Which action must be taken to assign a global unicast IPv6 address on an interface that is derived from the MAC address of that interface?

- A. configure a stateful DHCPv6 server on the network
- B. enable SLAAC on an interface

- C. disable the EUI-64 bit process
- D. explicitly assign a link-local address

---

**Answer: A**

---

Explanation:

---

**Question: 280**

---

Refer to the exhibit.

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route

Gateway of last resort is 209.165.202.131 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 209.165.202.131
      209.165.200.0/27 is subnetted, 1 subnets
S      209.165.200.224 [254/0] via 209.165.202.129
      209.165.201.0/27 is subnetted, 1 subnets
S      209.165.201.0 [1/0] via 209.165.202.130
```

Which command configures a floating static route to provide a backup to the primary link?

- A. ip route 0.0.0.0 0.0.0.0 209.165.202.131
- B. ip route 209.165.201.0 255.255.255.224 209.165.202.130
- C. ip route 0.0.0.0 0.0.0.0 209.165.200.224
- D. ip route 209.165.200.224 255.255.255.224 209.165.202.129 254

---

**Answer: D**

---

Explanation:

---

**Question: 281**

---

Which two QoS tools provides congestion management? ( Choose two )

- A. CAR
- B. CBWFQ
- C. PQ
- D. PBR
- E. FRTS

---

**Answer: B,C**

---

Explanation:

Type of queuing methods are available:

- First-In-First-Out (FIFO)
- Priority Queuing (PQ)
- Custom Queuing (CQ)
- Weighted Fair Queuing (WFQ)
- Class-Based Weighted Fair Queuing (CBWFQ)
- Low-Latency Queuing (LLQ)

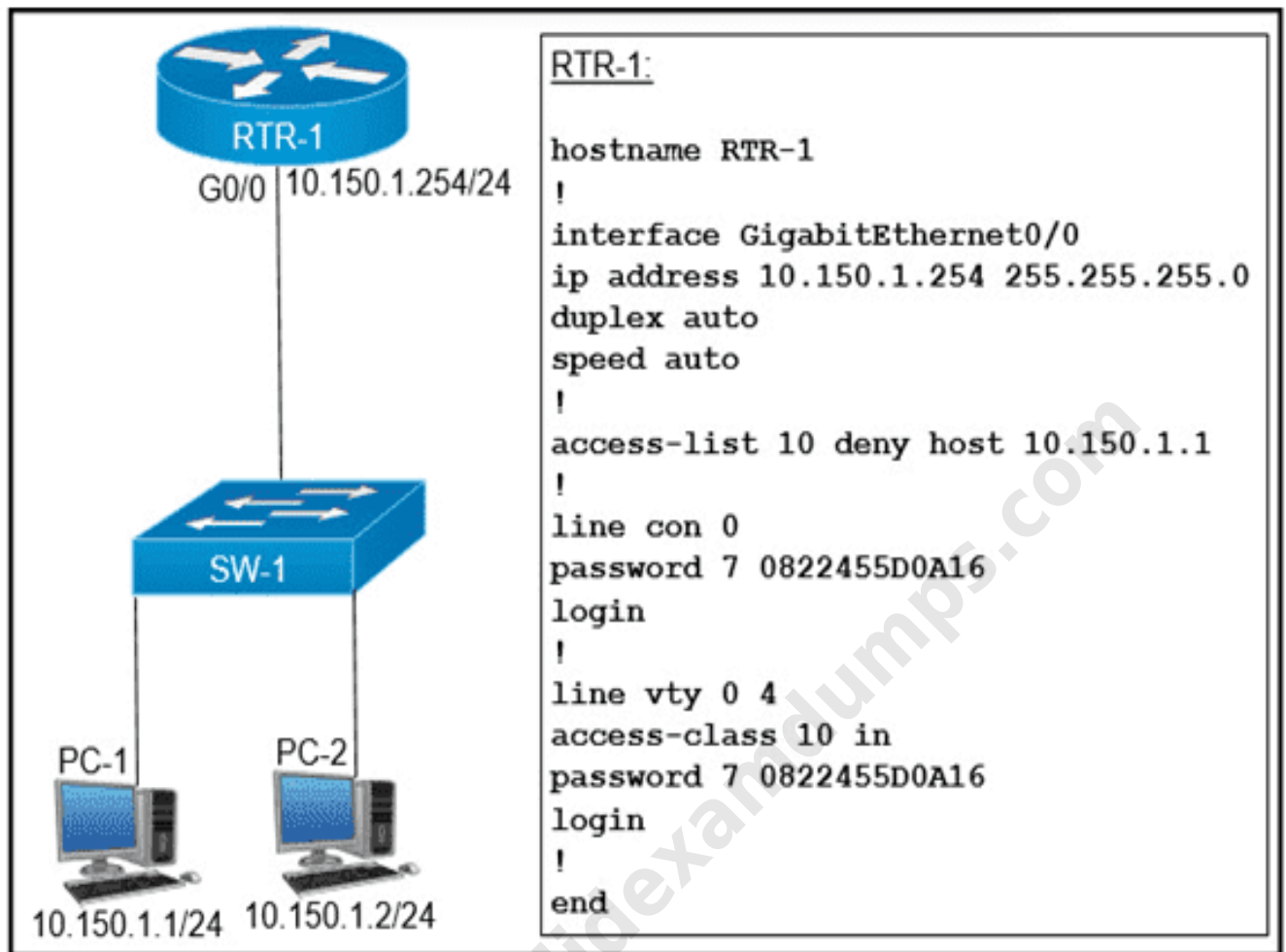
<https://www.orbit-computer-solutions.com/qos-congestion-management-tools/>

---

**Question: 282**

---

Refer to the exhibit.



An access list is created to deny Telnet access from host PC-1 to RTR-1 and allow access from all other hosts. A Telnet attempt from PC-2 gives this message: "% Connection refused by remote host". Without allowing Telnet access from PC-1, which action must be taken to permit the traffic?

- A. Add the access-list 10 permit any command to the configuration
- B. Remove the access-class 10 in command from line vty 0.4.
- C. Add the ip access-group 10 out command to interface g0/0.
- D. Remove the password command from line vty 0 4.

---

**Answer: A**

---

Explanation:

---

**Question: 283**

---



An engineer must configure an OSPF neighbor relationship between router R1 and R3. The authentication configuration has been configured and the connecting interfaces are in the same 192.168.1.0/30 subnet. What are the next two steps to complete the configuration? (Choose two.)

- A. configure the hello and dead timers to match on both sides
- B. configure the same process ID for the router OSPF process
- C. configure the same router ID on both routing processes
- D. Configure the interfaces as OSPF active on both sides.
- E. configure both interfaces with the same area ID

---

**Answer: A,E**

---

Explanation:

---

**Question: 284**

---

Which type of traffic is sent with pure IPsec?

- A. broadcast packets from a switch that is attempting to locate a MAC address at one of several remote sites
- B. multicast traffic from a server at one site to hosts at another location
- C. spanning-tree updates between switches that are at two different sites
- D. unicast messages from a host at a remote site to a server at headquarters

---

**Answer: D**

---

Explanation:

“The original poster makes a correct observation that EIGRP does not work in a pure IPSEC environment. IPSEC was designed to process unicast traffic.

---

**Question: 285**

---

R1 has learned route 10.10.10.0/24 via numerous routing protocols. Which route is installed?

- A. route with the lowest cost
- B. route with the next hop that has the highest IP
- C. route with the shortest prefix length
- D. route with the lowest administrative distance

---

**Answer: D**

---

Explanation:

---

**Question: 286**

---

Which configuration management mechanism uses TCP port 22 by default when communicating with managed nodes?

- A. Ansible
- B. Python
- C. Puppet
- D. Chef

---

**Answer: A**

---

Explanation:

---

**Question: 287**

---

Which 802.11 frame type is indicated by a probe response after a client sends a probe request?

- A. action
- B. management

C. control

D. data

---

**Answer: B**

---

Explanation:

---

**Question: 288**

---

Which two must be met before SSH can operate normally on a Cisco IOS switch? (Choose two)

- A. The switch must be running a k9 (crypto) IOS image
- B. The ip domain-name command must be configured on the switch
- C. IP routing must be enabled on the switch
- D. A console password must be configured on the switch
- E. Telnet must be disabled on the switch

---

**Answer: A,B**

---

Explanation:

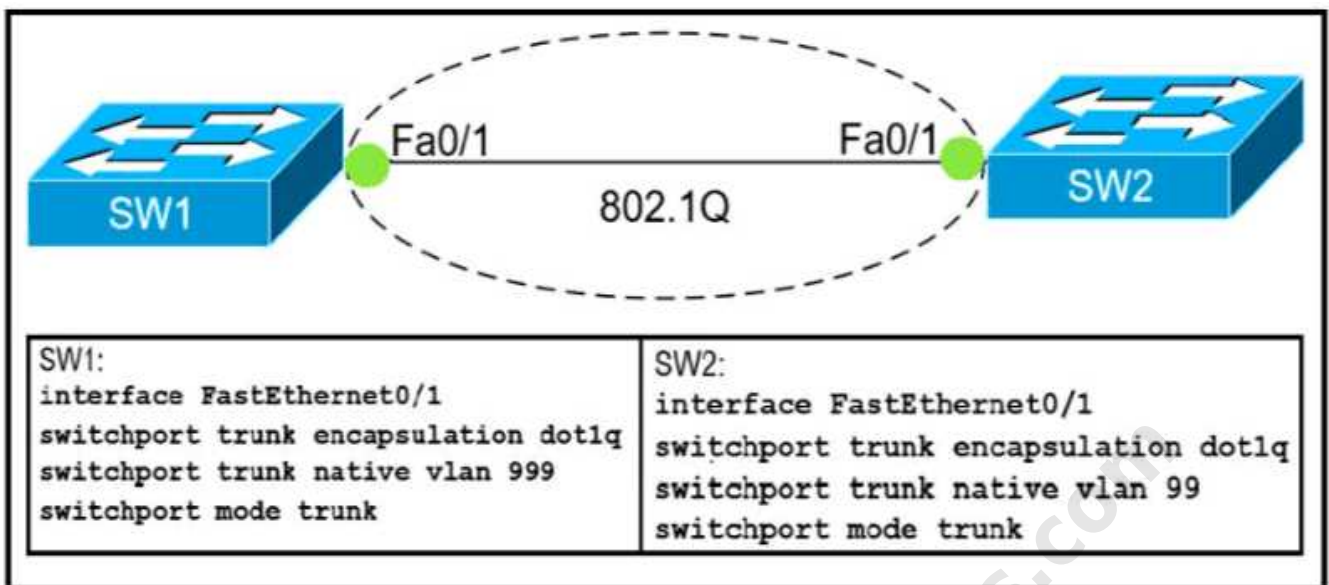
Reference: <https://www.cisco.com/c/en/us/support/docs/security-vpn/secure-shell-ssh/4145-ssh.html>

---

**Question: 289**

---

Refer to Exhibit.



Which action do the switches take on the trunk link?

- A. The trunk does not form and the ports go into an err-disabled status.
- B. The trunk forms but the mismatched native VLANs are merged into a single broadcast domain.
- C. The trunk does not form, but VLAN 99 and VLAN 999 are allowed to traverse the link.
- D. The trunk forms but VLAN 99 and VLAN 999 are in a shutdown state.

**Answer: B**

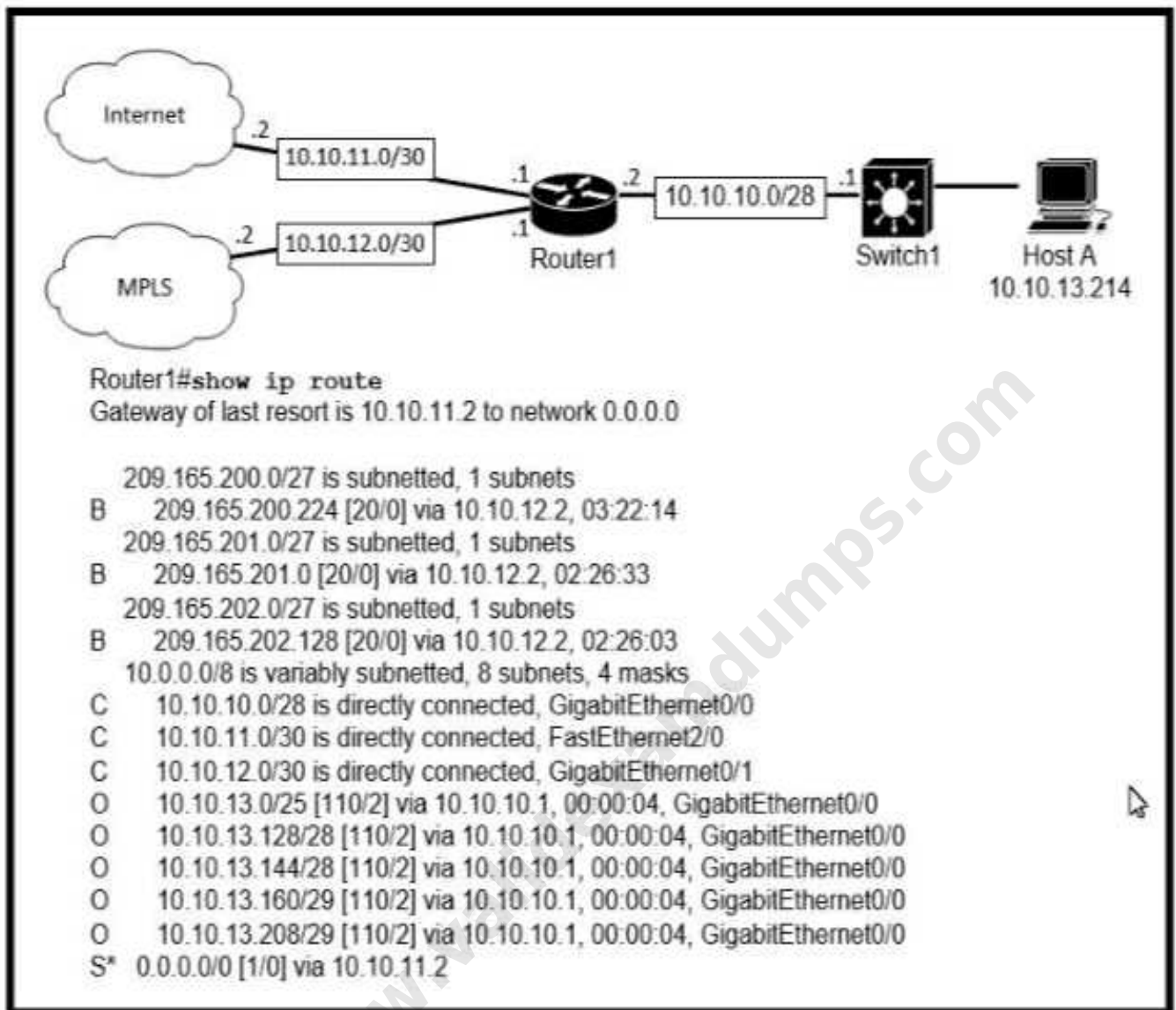
Explanation:

The trunk still forms with mismatched native VLANs and the traffic can actually flow between mismatched switches. But it is absolutely necessary that the native VLANs on both ends of a trunk link match; otherwise a native VLAN mismatch occurs, causing the two VLANs to effectively merge.

For example with the above configuration, SW1 would send untagged frames for VLAN 999. SW2 receives them but would think they are for VLAN 99 so we can say these two VLANs are merged.

### Question: 290

Refer to the exhibit.



Which prefix does Router 1 use for traffic to Host A?

- A. 10.10.10.0/28
- B. 10.10.13.0/25
- C. 10.10.13.144/28
- D. 10.10.13.208/29

**Answer: D**

Explanation:

Host A address fall within the address range. However, if more than one route to the same subnet exist (router will use the longest stick match, which match more specific route to the subnet). If there

are route 10.10.13.192/26 and 10.10.13.208/29, the router will forward the packet to /29 rather than /28.

---

**Question: 291**

---

Refer to the exhibit.

<https://www.validexamdumps.com>

```
R1#config t
R1(config)# interface gil/1
R1(config-if)# ip address 192.168.0.1 255.255.255.0

R1(config)# router bgp 65000
R1(config-router)# neighbor 192.168.0.2 remote-as 65001
R1(config-router)# network 10.1.1.0 mask 255.255.255.0

R1(config)# router ospf 1
R1(config)# router-id 1.1.1.1
R1(config)# network 192.168.0.1 0.0.0.0 area 0
R1(config)# network 10.1.1.0 0.0.0.255 area 0

R1(config)# router eigrp 1
R1(config)# eigrp router-id 1.1.1.1
R1(config)# network 10.1.1.0 0.0.0.255
R1(config)# network 192.168.0.1 0.0.0.0

R2#config t
R2(config)# interface gil/1
R2(config-if)# ip address 192.168.0.2 255.255.255.0

R2#config t
R2(config)# router bgp 65001
R2(config-router)# neighbor 192.168.0.1 remote-as 65000

R2(config)# router ospf 1
R2(config)# router-id 2.2.2.2
R2(config)# network 192.168.1.2 0.0.0.0 area 0

R2(config)# router eigrp 1
R2(config)# eigrp router-id 1.1.1.1
R2(config)# network 192.168.0.1 0.0.0.0

R2(config)# ip route 10.1.1.0 255.255.255.0 192.168.0.1
```

Router R2 is configured with multiple routes to reach network 10.1.1.0/24 from router R1. What protocol is chosen by router R2 to reach the destination network 10.1.1.0/24?

A. eBGP

- B. static
- C. OSPF
- D. EIGRP

---

**Answer: B**

---

Explanation:

---

**Question: 292**

---

Refer to the exhibit.

```
R2#show ip route  
C    192.168.1.0/26 is directly connected, FastEthernet0/1
```

Which two prefixes are included in this routing table entry? (Choose two.)

- A. 192.168.1.17
- B. 192.168.1.61
- C. 192.168.1.64
- D. 192.168.1.127
- E. 192.168.1.254

---

**Answer: B,C**

---

Explanation:

---

**Question: 293**

---

which IPv6 address block forwards packets to a multicast address rather than a unicast address?



- A. 2000::/3
- B. FC00::/7
- C. FE80::/10
- D. FF00::/12

---

**Answer: D**

---

Explanation:

---

**Question: 294**

---

What are two recommendations for protecting network ports from being exploited when located in an office space outside of an IT closer? (Choose two.)

- A. enable the PortFast feature on ports
- B. implement port-based authentication
- C. configure static ARP entries
- D. configure ports to a fixed speed
- E. shut down unused ports

---

**Answer: B,E**

---

Explanation:

---

**Question: 295**

---

Refer to the exhibit.

```
10.0.0.0/24 is subsetting, 1 subnets
C    10.0.0.0 is directly connected, FastEthernet0/1
C    172.160.0/16 is directly connected, FastEthernet0/0
D    192.168.0.0/24 [90/30720] via 172.16.0.2, 00:00:03, FastEthernet0/0
```

Which route type does the routing protocol Code D represent in the output?

- A. internal BGP route
- B. /24 route of a locally configured IP
- C. statically assigned route
- D. route learned through EIGRP

---

**Answer: D**

---

Explanation:

---

**Question: 296**

---

Which type of IPv6 address is publicly routable in the same way as IPv4 public address?

- A. global unicast
- B. link-local
- C. unique local
- D. multicast

---

**Answer: A**

---

Explanation:

---

**Question: 297**

---

Refer to the exhibit.

```

Switch1#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:           1
Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SD)          LACP        Fa0/2 (I) Fa0/1 (I)

Switch1#show run
Building configuration...
interface Port-channel1
!
interface FastEthernet0/1
channel-group 1 mode passive
!
interface FastEthernet0/2
channel-group 1 mode passive

Switch2#show run
Building configuration...
interface Port-channel1
!
interface FastEthernet0/1
channel-group 1 mode passive
!
interface FastEthernet0/2
channel-group 1 mode passive

```

Which change to the configuration on Switch?

allows the two switches to establish an EtherChannel?

- A. Change the protocol to EtherChannel mode on.
- B. Change the LACP mode to active
- C. Change the LACP mode to desirable
- D. Change the protocol to PAqP and use auto mode

---

**Answer: B**

---

Explanation:

---

**Question: 298**

---

What Is the path for traffic sent from one user workstation to another workstation on a separate

switch In a three-tier architecture model?

- A. access - core - distribution - access
- B. access - distribution - distribution - access
- C. access - core - access
- D. access - distribution - core - distribution - access

---

**Answer: D**

---

Explanation:

---

**Question: 299**

---

Refer to the exhibit.

```
R1#show ip route
#output suppressed
```

Gateway of last resort is 192.168.14.4 to network 0.0.0.0

```
C    172.16.1.128/25 is directly connected, GigabitEthernet1/1/0
C    192.168.12.0/24 is directly connected, FastEthernet0/0
C    192.168.13.0/24 is directly connected, FastEthernet0/1
C    192.168.14.0/24 is directly connected, FastEthernet1/0
C    172.16.16.1 is directly connected, Loopback1
    192.168.10.0/24 is variably subnetted, 3 subnets, 3 masks
O    192.168.10.0.24 [110/2] via 192.168.14.4, 00:02:01, FastEthernet1/0
O    192.168.10.32/27 [110/11] via 192.168.13.3, 00:00:52, FastEthernet0/1
O    192.168.0.0/16 [110/2] via 192.168.15.5, 00:05:01, FastEthernet1/1
D    192.168.10.1/32 [90/52778] via 192.168.12.2, 00:03:44, FastEthernet0/0
O*E2  0.0.0.0/0 [110/1] via 192.168.14.4, 00:00:10, FastEthernet1/0
```

If R1 receives a packet destined to 172.161.1, to which IP address does it send the packet?

- A. 192.168.12.2
- B. 192.168.13.3
- C. 192.168.14.4
- D. 192.168.15.5

---

**Answer: C**

---

Explanation:

---

**Question: 300**

---

A packet is destined for 10.10.1.22. Which static route does the router choose to forward the packet?

- A. ip route 10.10.1.0 255.255.255.240 10.10.255.1
- B. ip route 10.10.1.16 255.255.255.252 10.10.255.1
- C. ip route 10.10.1.20 255.255.255.252 10.10.255.1
- D. ip route 10.10.1.20 255.255.255.254 10.10.255.1

---

**Answer: C**

---

Explanation:

---

**Question: 301**

---

An office has 8 floors with approximately 30-40 users per floor. What command must be configured on the router Switched Virtual Interface to use address space efficiently?

- A. ip address 192.168.0.0 255.255.0.0
- B. ip address 192.168.0.0 255.255.254.0

C. ip address 192.168.0.0 255.255.255.128

D. ip address 192.168.0.0 255.255.255.224

---

**Answer: A**

---

Explanation:

---

**Question: 302**

---

DRAG DROP

Drag the descriptions of IP protocol transmissions from the left onto the IP traffic types on the right.

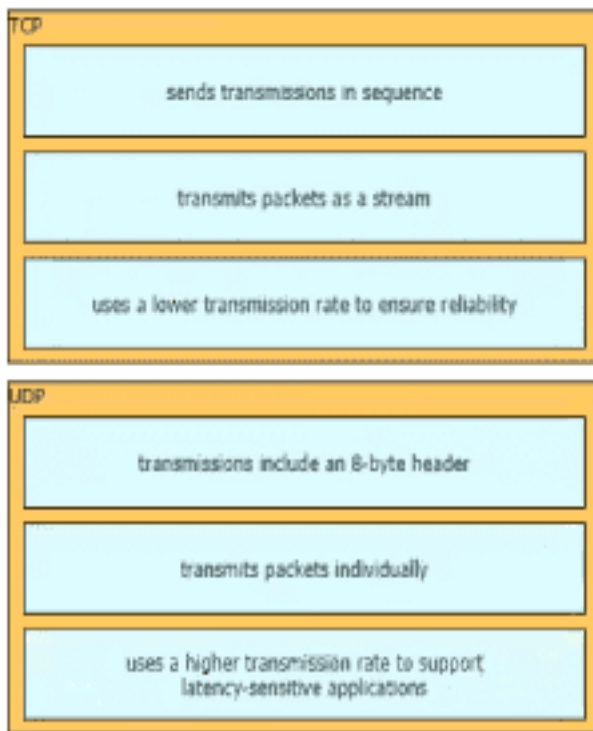
|                                                                           |     |
|---------------------------------------------------------------------------|-----|
| sends transmissions in sequence                                           | TCP |
| transmissions include an 8-byte header                                    |     |
| transmits packets as a stream                                             |     |
| transmits packets individually                                            |     |
| uses a higher transmission rate to support latency-sensitive applications | UDP |
| uses a lower transmission rate to ensure reliability                      |     |
|                                                                           |     |

---

**Answer:**

---

Explanation:



---

**Question: 303**

---

Which command must be entered when a device is configured as an NTP server?

- A. ntp authenticate
- B. ntp server
- C. ntp peer
- D. ntp master

---

**Answer: D**

---

Explanation:

To configure a Cisco device as an Authoritative NTP Server, use the ntp master [stratum] command.

To configure a Cisco device as a NTP client, use the command ntp server <IP address>. For example:

Router(config)#ntp server 192.168.1.1. This command will instruct the router to query 192.168.1.1

for the time.

---

**Question: 304**

---

Which command must be entered to configure a DHCP relay?

- A. ip helper-address
- B. ip address dhcp
- C. ip dhcp pool
- D. ip dhcp relay

---

**Answer: A**

---

Explanation:

---

**Question: 305**

---

Which technology must be implemented to configure network device monitoring with the highest security?

- A. IP SLA
- B. syslog
- C. NetFlow
- D. SNMPv3

---

**Answer: C**

---

Explanation:

---

**Question: 306**

---



Refer to the exhibit.

```
Switch(config)#hostname R1
R1(config)#interface FastEthernet0/1
R1(config-if)#no switchport
R1(config-if)#ip address 10.100.20.42 255.255.255.0
R1(config-if)#line vty 0 4
R1(config-line)#login
```

An engineer booted a new switch and applied this configuration via the console port. Which additional configuration must be applied to allow administrators to authenticate directly to enable privilege mode via Telnet using a local username and password?

- ☐ R1(config)#username admin privilege 15 secret p@ss1234  
R1(config-if)#line vty 0 4  
R1(config-line)#login local
- ☐ R1(config)#username admin secret p@ss1234  
R1(config-if)#line vty 0 4  
R1(config-line)#login local  
R1(config)#enable secret p@ss1234
- ☐ R1(config)#username admin  
R1(config-if)#line vty 0 4  
R1(config-line)#password p@ss1234  
R1(config-line)#transport input telnet
- ☐ R1(config)#username admin  
R1(config-if)#line vty 0 4  
R1(config-line)#password p@ss1234

- A. Option A
- B. Option B
- C. Option C

D. Option D

---

**Answer: A**

---

Explanation:

---

**Question: 307**

---

DRAG DROP

Refer to the exhibit.

```
C:\>ipconfig/all

Windows IP Configuration

Host Name . . . . . : Inspiron15
Primary Dns Suffix . . . . . :
Node Type . . . . . : Mixed
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Wireless LAN adapter Local Area Connection* 12:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 1A-76-3F-7C-57-DF
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Dell Wireless 1703 802.11b/g/n (2.4GHz)
Physical Address. . . . . : B8-76-3F-7C-57-DF
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::e09f:9839:6e86:f755%12(Preferred)
. . . . . : 192.168.1.20(Preferred)
. . . . . : 255.255.255.0
. . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 263747135
DHCPv6 Client DUID. . . . . : 00-01-00-01-18-E6-32-43-B8-76-3F-7C-57-DF
. . . . . : 192.168.1.15
. . . . . : 192.168.1.16
NetBIOS over Tcpip. . . . . : Enabled
```

An engineer is required to verify that the network parameters are valid for the users wireless LAN connectivity on a /24 subnet. Drag and drop the values from the left onto the network parameters on the right. Not all values are used.

|                   |                                          |
|-------------------|------------------------------------------|
| 192.168.1.1       | broadcast address                        |
| 192.168.1.20      | default gateway                          |
| 192.168.1.254     | host IP address                          |
| 192.168.1.255     | last assignable IP address in the subnet |
| B8-76-3F-7C-57-DF | MAC address                              |
| 1A-76-3F-7C-57-DF | network address                          |
| 192.168.1.0       |                                          |

**Answer:**

Explanation:

|                   |                   |
|-------------------|-------------------|
|                   | 192.168.1.255     |
|                   | 192.168.1.1       |
|                   | 192.168.1.20      |
|                   | 192.168.1.254     |
|                   | B8-76-3F-7C-57-DF |
| 1A-76-3F-7C-57-DF | 192.168.1.0       |

---

**Question: 308**

---

Which two actions influence the EIGRP route selection process? (Choose two)

- A. The router calculates the reported distance by multiplying the delay on the exiting Interface by 256.
- B. The router calculates the best backup path to the destination route and assigns it as the feasible successor.
- C. The router calculates the feasible distance of all paths to the destination route
- D. The advertised distance is calculated by a downstream neighbor to inform the local router of the bandwidth on the link
- E. The router must use the advertised distance as the metric for any given route

---

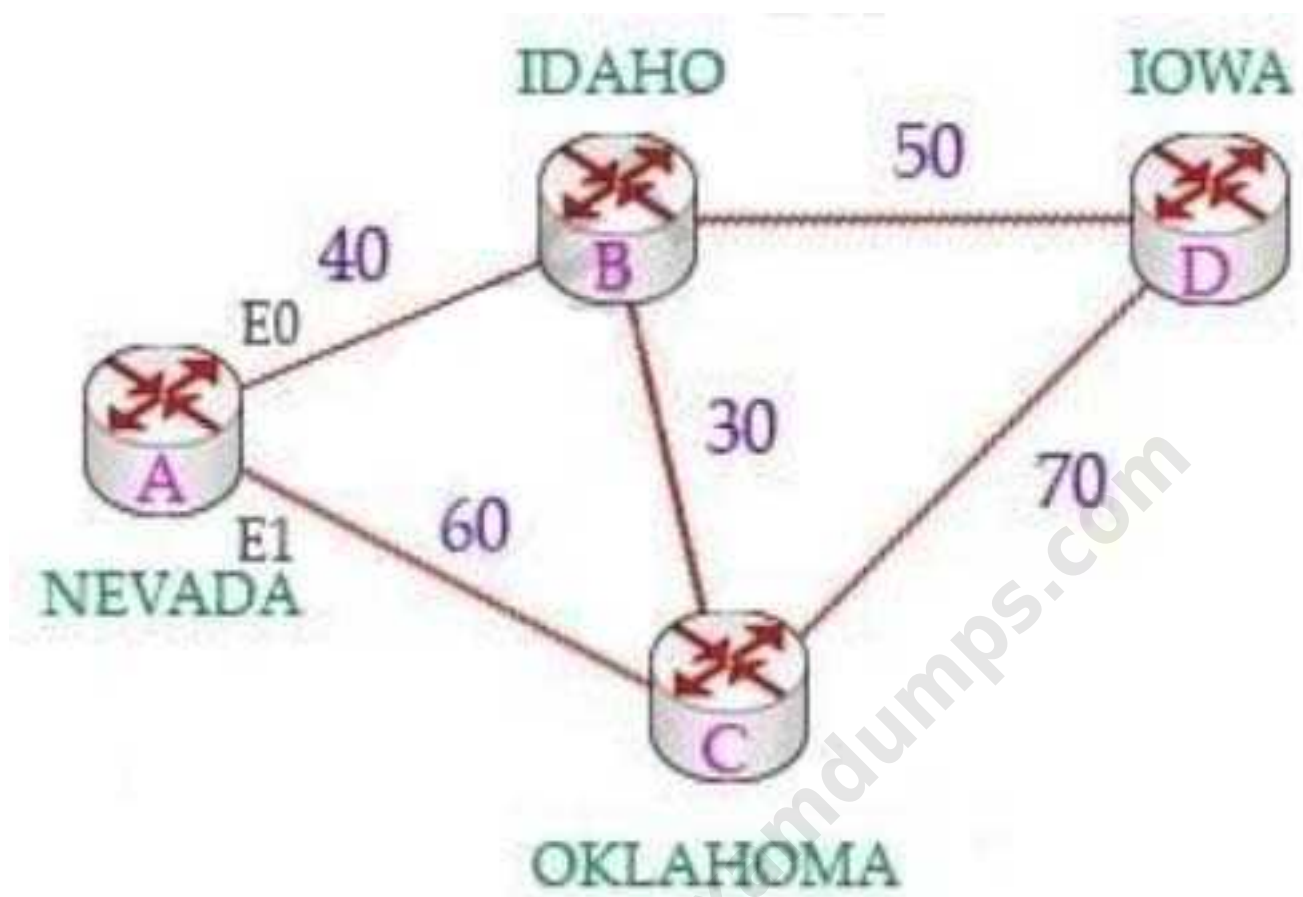
**Answer: B,C**

---

Explanation:

The reported distance (or advertised distance) is the cost from the neighbor to the destination. It is calculated from the router advertising the route to the network. For example in the topology below, suppose router A & B are exchanging their routing tables for the first time. Router B says "Hey, the best metric (cost) from me to IOWA is 50 and the metric from you to IOWA is 90" and advertises it to router A.

Router A considers the first metric (50) as the Advertised distance. The second metric (90), which is from NEVADA to IOWA (through IDAHO), is called the Feasible distance.



The reported distance is calculated in the same way of calculating the metric. By default ( $K1 = 1$ ,  $K2 = 0$ ,  $K3 = 1$ ,  $K4 = 0$ ,  $K5 = 0$ ), the metric is calculated as follows:

$$\text{metric} = \left[ \frac{10,000,000}{\text{slowest bandwidth[in kbps]}} + \frac{\text{sum of delay[in } \mu\text{sec}]}{10} \right] * 256$$

---

**Question: 309**

---

Which mode must be set for APs to communicate to a Wireless LAN Controller using the Control and Provisioning of Wireless Access Points (CAPWAP) protocol?

- A. bridge
- B. route
- C. autonomous
- D. lightweight

---

**Answer: D**

---

Explanation:

---

**Question: 310**

---

Which configuration is needed to generate an RSA key for SSH on a router?

- A. Configure the version of SSH
- B. Configure VTY access.
- C. Create a user with a password.
- D. Assign a DNS domain name

---

**Answer: D**

---

Explanation:

---

**Question: 311**

---

What makes Cisco DNA Center different from traditional network management applications and their management of networks?

- A. It omits supports auto-discovery of network elements in a greenfield deployment.
- B. It modular design allows someone to implement different versions to meet the specific needs of an organization
- C. It abstracts policy from the actual device configuration
- D. It does not support high availability of management functions when operating in cluster mode

---

**Answer: C**

---

Explanation:

---

**Question: 312**

---

Which two primary drivers support the need for network automation? (Choose two.)

- A. Eliminating training needs
- B. Increasing reliance on self-diagnostic and self-healing
- C. Policy-derived provisioning of resources
- D. Providing a ship entry point for resource provisioning
- E. Reducing hardware footprint

---

**Answer: C,D**

---

Explanation:

---

**Question: 313**

---

Using direct sequence spread spectrum, which three 2.4-GHz channels are used to limit collisions?

- A. 1,6,11
- B. 1,5,10
- C. 1,2,3
- D. 5,6,7

---

**Answer: A**

---

Explanation:

---

**Question: 314**

---

Which port type supports the spanning-tree portfast command without additional configuration?

- A. access ports
- B. Layer 3 main Interfaces
- C. Layer 3 subinterfaces
- D. trunk ports

---

**Answer: A**

---

Explanation:

---

**Question: 315**

---

Refer to the exhibit.



```
interface GigabitEthernet3/1/4
switchport voice vlan 50
!
```

An administrator is tasked with configuring a voice VLAN. What is the expected outcome when a Cisco phone is connected to the GigabitEthernet3/1/4 port on a switch?

- A. The phone and a workstation that is connected to the phone do not have VLAN connectivity
- B. The phone and a workstation that is connected to the phone send and receive data in VLAN 50.
- C. The phone sends and receives data in VLAN 50, but a workstation connected to the phone has no VLAN connectivity
- D. The phone sends and receives data in VLAN 50, but a workstation connected to the phone sends



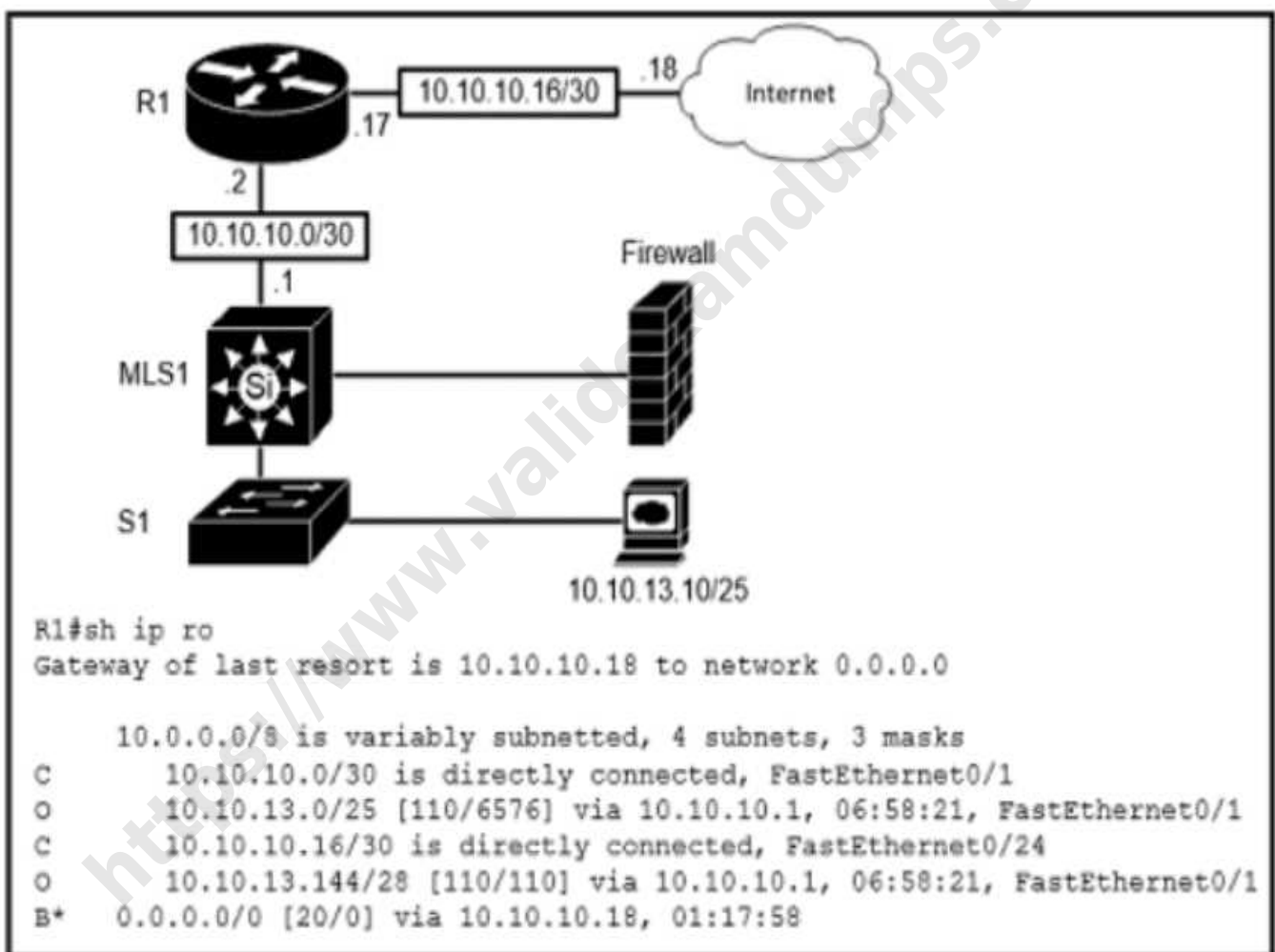
and receives data in VLAN 1

**Answer: D**

Explanation:

### Question: 316

Refer to the exhibit.



Which route type is configured to reach the internet?

- A. host route
- B. default route
- C. floating static route

D. network route

---

**Answer: B**

---

Explanation:

---

**Question: 317**

---

An engineer needs to configure LLDP to send the port description time length value (TLV). What command sequence must be implemented?

- A. switch(config-line)#lldp port-description
- B. switch(config)#lldp port-description
- C. switch(config-if)#lldp port-description
- D. switch#lldp port-description

---

**Answer: B**

---

Explanation:

---

**Question: 318**

---

A user configured OSPF and advertised the Gigabit Ethernet interface in OSPF. By default, which type of OSPF network does this interface belong to?

- A. point-to-multipoint
- B. point-to-point
- C. broadcast
- D. nonbroadcast

---

**Answer: C**

---

Explanation:

<https://www.oreilly.com/library/view/cisco-ios-cookbook/0596527225/ch08s15.html>

The Broadcast network type is the default for an OSPF enabled ethernet interface (while Point-toPoint is the default OSPF network type for Serial interface with HDLC and PPP encapsulation).

---

**Question: 319**

---

What benefit does controller-based networking provide versus traditional networking?

- A. moves from a two-tier to a three-tier network architecture to provide maximum redundancy
- B. provides an added layer of security to protect from DDoS attacks
- C. allows configuration and monitoring of the network from one centralized port
- D. combines control and data plane functionality on a single device to minimize latency

---

**Answer: C**

---

Explanation:

---

**Question: 320**

---

What are two benefits of FHRPs? (Choose two.)

- A. They prevent (oops in the Layer 2 network.
- B. They allow encrypted traffic.
- C. They are able to bundle multiple ports to increase bandwidth
- D. They enable automatic failover of the default gateway.

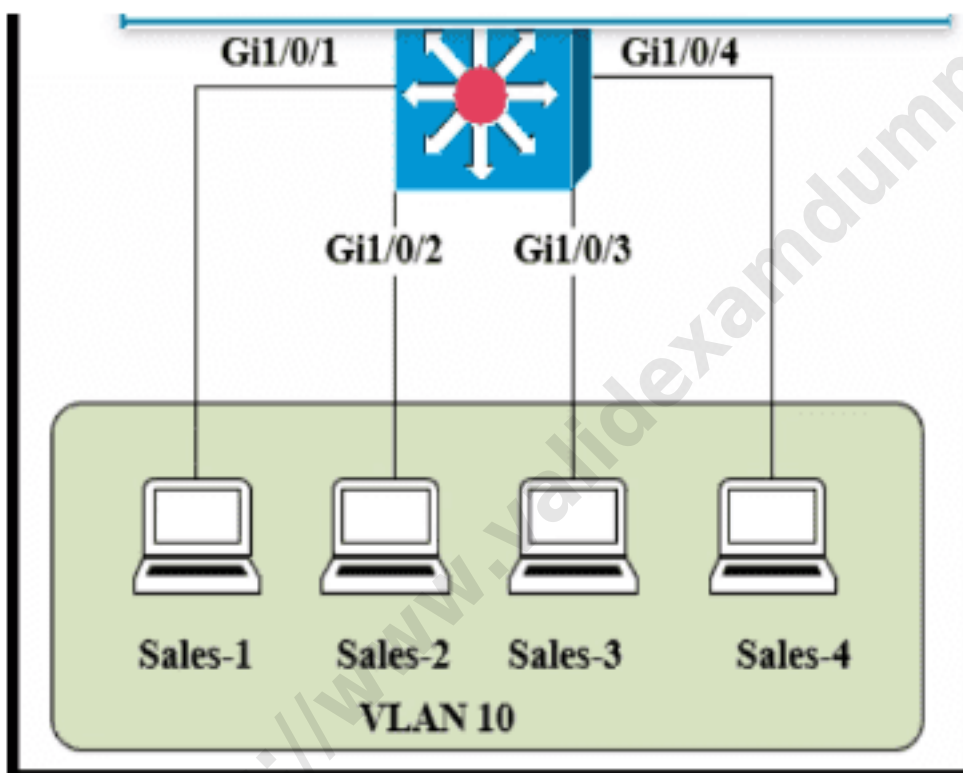
E. They allow multiple devices to serve as a single virtual gateway for clients in the network

**Answer: D,E**

Explanation:

### Question: 321

Refer to the exhibit.



The entire contents of the MAC address table are shown. Sales-4 sends a data frame to Sales-1.

```
Sales-SW#show mac-address-table
Mac Address Table
```

| VLAN | MAC Address    | Type    | Ports   |
|------|----------------|---------|---------|
| 10   | 000c.8590.bb7d | DYNAMIC | Gi1/0/1 |
| 10   | 3910.4161.9bb7 | DYNAMIC | Gi1/0/2 |
| 10   | 00d0.d3b6.957c | DYNAMIC | Gi1/0/3 |

```
Sales-SW#
```

What does the switch do as it receives the frame from Sales-4?

- A. Perform a lookup in the MAC address table and discard the frame due to a missing entry.
- B. Insert the source MAC address and port into the forwarding table and forward the frame to Sales-1.
- C. Map the Layer 2 MAC address to the Layer 3 IP address and forward the frame.
- D. Flood the frame out of all ports except on the port where Sales-1 is connected.

---

**Answer: B**

---

Explanation:

<https://www.ciscopress.com/articles/article.asp?p=3089352&seqNum=6>

---

**Question: 322**

---

How does CAPWAP communicate between an access point in local mode and a WLC?

- A. The access point must directly connect to the WLC using a copper cable
- B. The access point must not be connected to the wired network, as it would create a loop
- C. The access point must be connected to the same switch as the WLC
- D. The access point has the ability to link to any switch in the network, assuming connectivity to the WLC

---

**Answer: D**

---

Explanation:

---

**Question: 323**

---

Where does wireless authentication happen?

- A. SSID
- B. radio
- C. band
- D. Layer 2

---

**Answer: D**

---

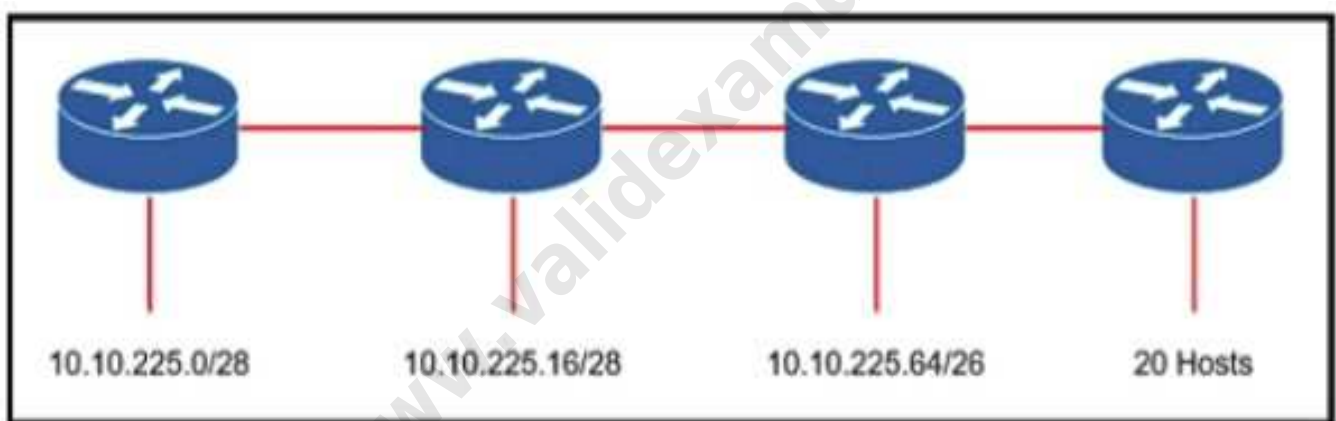
Explanation:

---

**Question: 324**

---

Refer to the exhibit.



Refer to the exhibit. An engineer must add a subnet for a new office that will add 20 users to the network. Which IPv4 network and subnet mask combination does the engineer assign to minimize wasting addresses?

- A. 10.10.225.48 255.255.255.240
- B. 10.10.225.32 255.255.255.240
- C. 10.10.225.48 255.255.255.224
- D. 10.10.225.32 255.255.255.224

---

**Answer: D**

---

Explanation:

---

**Question: 325**

---

Which WPA3 enhancement protects against hackers viewing traffic on the Wi-Fi network?

- A. TKiP encryption
- B. AES encryption
- C. scrambled encryption key
- D. SAE encryption

---

**Answer: D**

---

Explanation:

---

**Question: 326**

---

A Cisco IP phone receive untagged data traffic from an attached PC. Which action is taken by the phone?

- A. It allows the traffic to pass through unchanged
- B. It drops the traffic
- C. It tags the traffic with the default VLAN
- D. It tags the traffic with the native VLAN

---

**Answer: A**

---

Explanation:

[https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst2960x/software/15-0\\_2\\_EX/vlan/configuration\\_guide/b\\_vlan\\_152ex\\_2960-x\\_cg/b\\_vlan\\_152ex\\_2960-x\\_cg\\_chapter\\_0110.pdf](https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst2960x/software/15-0_2_EX/vlan/configuration_guide/b_vlan_152ex_2960-x_cg/b_vlan_152ex_2960-x_cg_chapter_0110.pdf)

Untagged traffic from the device attached to the Cisco IP Phone passes through the phone unchanged, regardless of the trust state of the access port on the phone.

---

**Question: 327**

---

Which statement about Link Aggregation when implemented on a Cisco Wireless LAN Controller is true?

- A. To pass client traffic two or more ports must be configured.
- B. The EtherChannel must be configured in "mode active"
- C. When enabled the WLC bandwidth drops to 500 Mbps
- D. One functional physical port is needed to pass client traffic

---

**Answer: D**

---

Explanation:

Reference: [https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-2/config-guide/b\\_cg82/b\\_cg82\\_chapter\\_010101011.html](https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-2/config-guide/b_cg82/b_cg82_chapter_010101011.html)

---

**Question: 328**

---

Which set of action satisfy the requirement for multifactor authentication?

- A. The user swipes a key fob, then clicks through an email link
- B. The user enters a user name and password, and then clicks a notification in an authentication app on a mobile device
- C. The user enters a PIN into an RSA token, and then enters the displayed RSA key on a login screen
- D. The user enters a user name and password and then re-enters the credentials on a second screen



---

**Answer: B**

---

Explanation:

This is an example of how two-factor authentication (2FA) works:

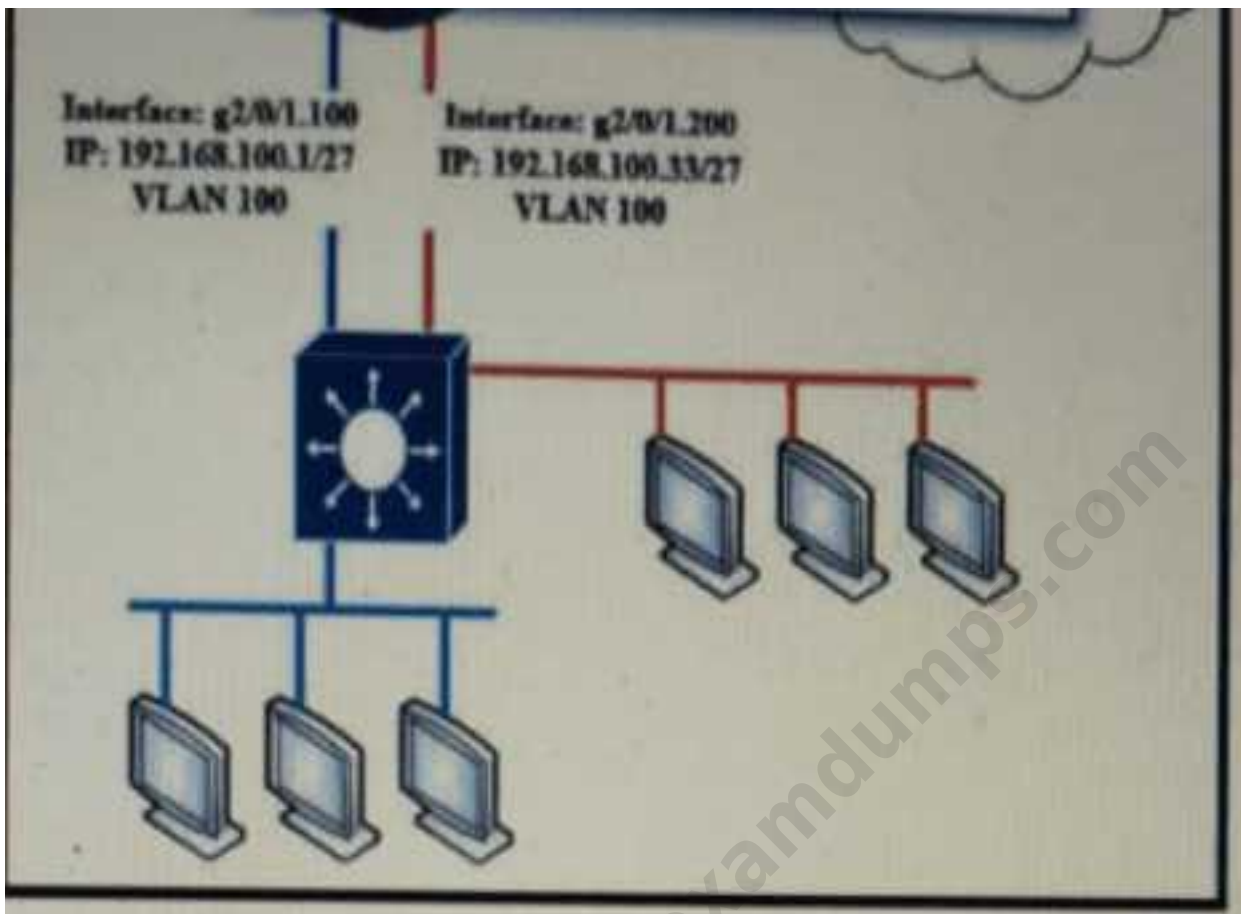
1. The user logs in to the website or service with their username and password.
2. The password is validated by an authentication server and, if correct, the user becomes eligible for the second factor.
3. The authentication server sends a unique code to the user's second-factor method (such as a smartphone app).
4. The user confirms their identity by providing the additional authentication for their second-factor method.

---

**Question: 329**

---

Refer to Exhibit.



Which configuration must be applied to the router that configures PAT to translate all addresses in VLAN 200 while allowing devices on VLAN 100 to use their own IP addresses?



A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: D**

---

Explanation:

---

**Question: 330**

---

Which two outcomes are predictable behaviors for HSRP? (Choose two)

- A. The two routers share a virtual IP address that is used as the default gateway for devices on the LAN.
- B. The two routers negotiate one router as the active router and the other as the standby router
- C. Each router has a different IP address both routers act as the default gateway on the LAN, and traffic is load balanced between them.
- D. The two routers synchronize configurations to provide consistent packet forwarding
- D. The two routers share the same IP address, and default gateway traffic is load-balanced between them

---

**Answer: A,B**

---

Explanation:

---

**Question: 331**

---

How do traditional campus device management and Cisco DNA Center device management differ in regards to deployment?

- A. Cisco DNA Center device management can deploy a network more quickly than traditional campus device management
- B. Traditional campus device management allows a network to scale more quickly than with Cisco DNA Center device management
- C. Cisco DNA Center device management can be implemented at a lower cost than most traditional campus device management options
- D. Traditional campus device management schemes can typically deploy patches and updates more quickly than Cisco DNA Center device management

---

**Answer: A**

---

Explanation:

---

**Question: 332**

---

A corporate office uses four floors in a building

- Floor 1 has 24 users
- Floor 2 has 29 users
- Floor 3 has 28 users
- Floor 4 has 22 users

Which subnet summarizes and gives the most efficient distribution of IP addresses for the router configuration?

- A. 192.168.0.0/26 as summary and 192.168.0.0/29 for each floor
- B. 192.168.0.0/24 as summary and 192.168.0.0/28 for each floor
- C. 192.168.0.0/23 as summary and 192.168.0.0/25 for each floor
- D. 192.168.0.0/25 as summary and 192.168.0.0/27 for each floor

---

**Answer: D**

---

Explanation:

---

**Question: 333**

---

An engineer configured an OSPF neighbor as a designated router. Which state verifies the designated router is in the proper mode?

- A. Exchange
- B. 2-way
- C. Full
- D. Init

---

**Answer: C**

---

Explanation:

---

**Question: 334**

---

What are two reasons for an engineer to configure a floating state route? (Choose two)

- A. to automatically route traffic on a secondary path when the primary path goes down
- B. to route traffic differently based on the source IP of the packet
- C. to enable fallback static routing when the dynamic routing protocol fails
- D. to support load balancing via static routing
- E. to control the return path of traffic that is sent from the router

---

**Answer: A,C**

---

Explanation:

---

**Question: 335**

---

The service password-encryption command is entered on a router. What is the effect of this configuration?

- A. restricts unauthorized users from viewing clear-text passwords in the running configuration
- B. encrypts the password exchange when a VPN tunnel is established
- C. prevents network administrators from configuring clear-text passwords
- D. protects the VLAN database from unauthorized PC connections on the switch

---

**Answer: A**

---

Explanation:

---

**Question: 336**

---

DRAG DROP

Drag and drop the application protocols from the left onto the transport protocols that it uses on the right

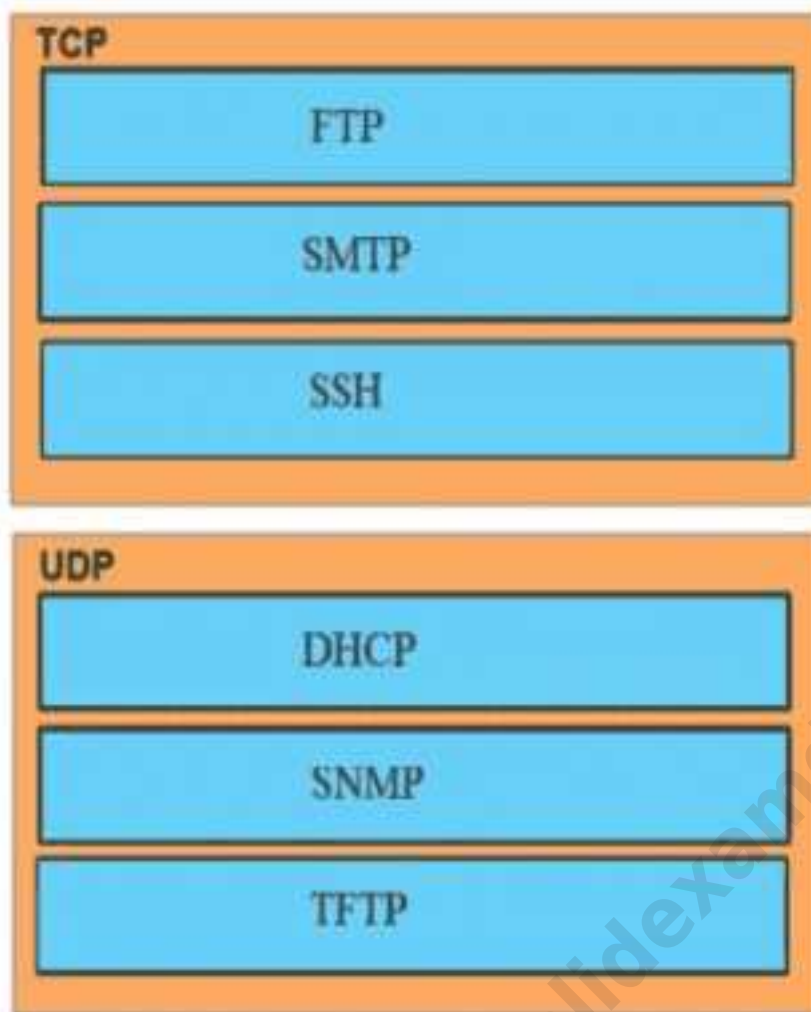
|      |     |
|------|-----|
| DHCP | TCP |
| FTP  |     |
| SMTP |     |
| SSH  |     |
| SNMP |     |
| TFTP | UDP |
|      |     |
|      |     |

---

**Answer:**

---

Explanation:



---

**Question: 337**

---

Refer to me exhibit.

```
Router1#show ip route
Gateway of last resort is not set
 209.165.200.0/27 is subnetted, 1 subnets
B    209.165.200.224 [20/0] via 10.10.12.2, 00:09:57
 10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C    10.10.10.0/28 is directly connected, GigabitEthernet0/0
C    10.10.11.0/30 is directly connected, FastEthernet2/0
O    10.10.13.0/24 [110/2] via 10.10.10.1, 00:08:34, GigabitEthernet0/0
C    10.10.12.0/30 is directly connected, GigabitEthernet0/1
```

Keystream Times  
<https://keystream.com>

Which action is taken by the router when a packet is sourced from 10.10.10.2 and destined for



10.10.10.16?

- A. It uses a route that is similar to the destination address
- B. It discards the packets.
- C. It floods packets to all learned next hops.
- D. It Queues the packets waiting for the route to be learned.

---

**Answer: A**

---

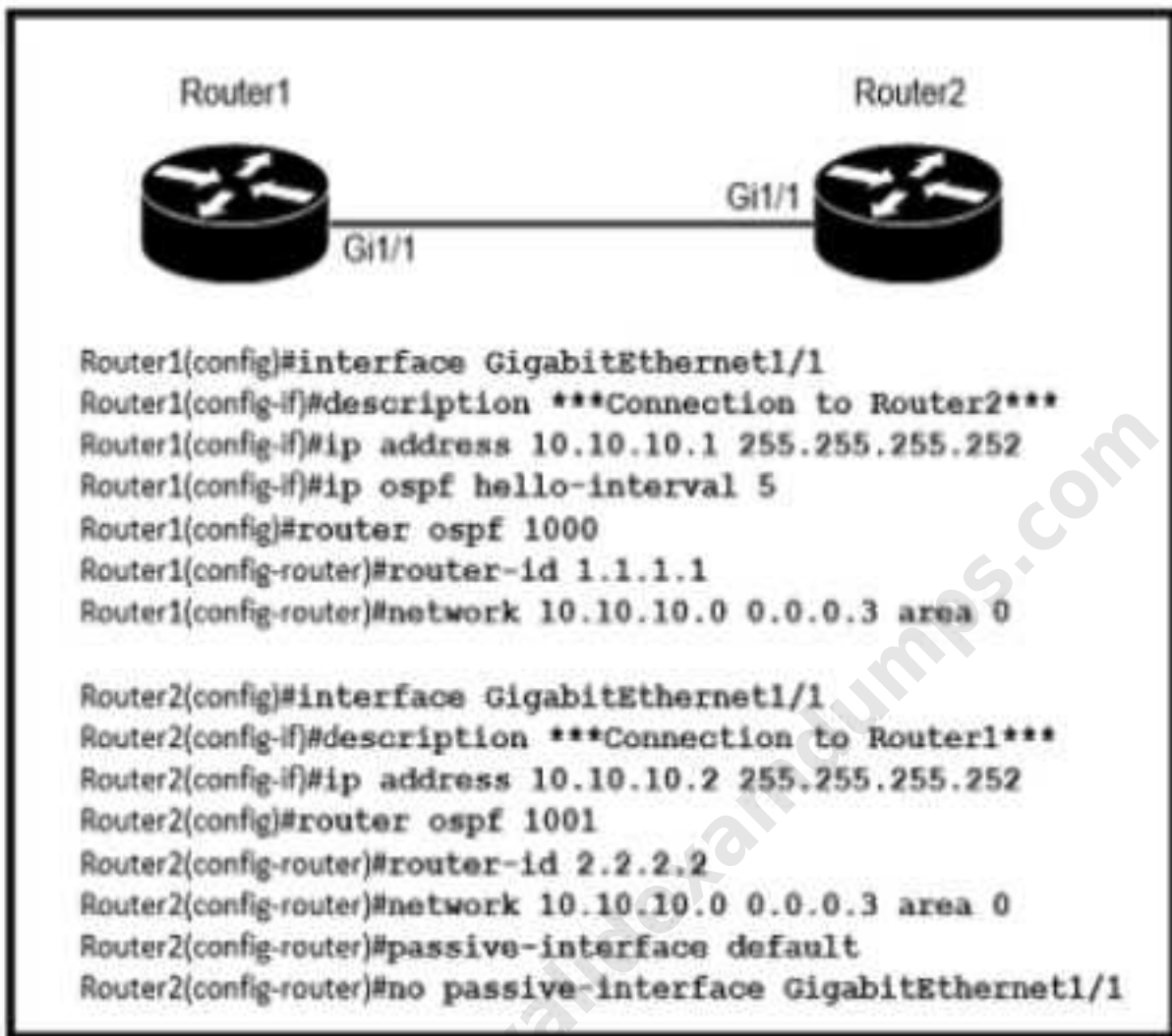
Explanation:

---

**Question: 338**

---

Refer to the exhibit.



Refer to the exhibit. After the configuration is applied, the two routers fail to establish an OSPF neighbor relationship. what is the reason for the problem?

- A. The OSPF router IDs are mismatched.
- B. Router2 is using the default hello timer.
- C. The network statement on Router1 is misconfigured.
- D. The OSPF process IDs are mismatched.

---

**Answer: B**

---

Explanation:

---

**Question: 339**

---

Which design element is a best practice when deploying an 802.11b wireless infrastructure?

- A. disabling TPC so that access points can negotiate signal levels with their attached wireless devices.
- B. setting the maximum data rate to 54 Mbps on the Cisco Wireless LAN Controller
- C. allocating nonoverlapping channels to access points that are in close physical proximity to one another
- D. configuring access points to provide clients with a maximum of 5 Mbps

---

**Answer: C**

---

Explanation:

---

**Question: 340**

---

Refer to the exhibit.

Gateway of last resort is 10.12.0.1 to network 0.0.0.0

```
O*E2  0.0.0.0/0 [110/1] via 10.12.0.1, 00:00:01, GigabitEthernet0/0
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      10.0.0.0/24 is directly connected, GigabitEthernet0/0
L      10.0.0.2/32 is directly connected, GigabitEthernet0/0
C      10.13.0.0/24 is directly connected, GigabitEthernet0/1
L      10.13.0.2/32 is directly connected, GigabitEthernet0/1
```

If configuring a static default route on the router with the ip route 0.0.0.0 0.0.0.0 10.13.0.1 120 command how does the router respond?

- A. It ignores the new static route until the existing OSPF default route is removed
- B. It immediately replaces the existing OSPF route in the routing table with the newly configured static route
- C. It starts load-balancing traffic between the two default routes
- D. It starts sending traffic without a specific matching entry in the routing table to GigabitEthernet0/1

---

**Answer: A**

---

Explanation:

Our new static default route has the Administrative Distance (AD) of 120, which is bigger than the AD of OSPF External route (O\*E2) so it will not be pushed into the routing table until the current OSPF External route is removed.

For your information, if you don't type the AD of 120 (using the command "ip route 0.0.0.0 0.0.0.0 10.13.0.1") then the new static default route would replace the OSPF default route as the default AD of static route is 1. You will see such line in the routing table:

S\* 0.0.0.0/0 [1/0] via 10.13.0.1

---

**Question: 341**

---

What does an SDN controller use as a communication protocol to relay forwarding changes to a southbound API?

- A. OpenFlow
- B. Java
- C. REST
- D. XML

---

**Answer: A**

---

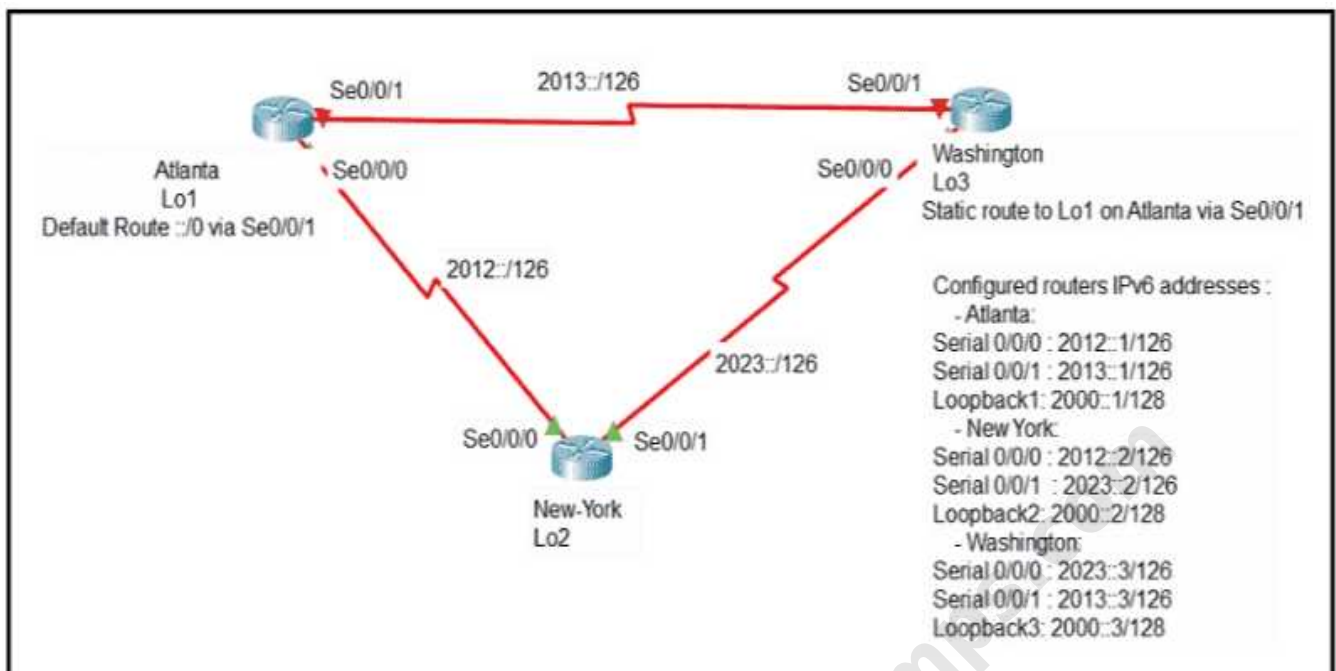
Explanation:

---

**Question: 342**

---

Refer to Exhibit.



An engineer is configuring the NEW York router to reach the Lo1 interface of the Atlanta router using interface Se0/0/0 as the primary path. Which two commands must be configured on the New York router so that it can reach the Lo1 interface of the Atlanta router via Washington when the link between New York and Atlanta goes down? (Choose two)

- A. ipv6 router 2000::1/128 2012::1
- B. ipv6 router 2000::1/128 2012::1 5
- C. ipv6 router 2000::1/128 2012::2
- D. ipv6 router 2000::1/128 2023::2 5
- E. ipv6 router 2000::1/128 2023::3 5

**Answer: A,E**

Explanation:

Floating static routes are static routes that have an administrative distance greater than the administrative distance (AD) of another static route or dynamic routes. By default a static route has an AD of 1 then floating static route must have the AD greater than 1. Floating static route has a manually configured administrative distance greater than that of the primary route and therefore would not be in the routing table until the primary route fails.

---

**Question: 343**

---

Which statement correctly compares traditional networks and controller-based networks?

- A. Only traditional networks offer a centralized control plane
- B. Only traditional networks natively support centralized management
- C. Traditional and controller-based networks abstract policies from device configurations
- D. Only controller-based networks decouple the control plane and the data plane

---

**Answer: D**

---

Explanation:

Most traditional devices use a distributed architecture, in which each control plane is resided in a networking device. Therefore they need to communicate with each other via messages to work correctly.

In contrast to distributed architecture, centralized (or controller-based) architectures centralizes the control of networking devices into one device, called SDN controller

---

**Question: 344**

---

How does the dynamically-learned MAC address feature function?

- A. The CAM table is empty until ingress traffic arrives at each port
- B. Switches dynamically learn MAC addresses of each connecting CAM table.
- C. The ports are restricted and learn up to a maximum of 10 dynamically-learned addresses
- D. It requires a minimum number of secure MAC addresses to be filled dynamically

---

**Answer: A**

---

Explanation:

---

**Question: 345**

---

Refer to the exhibit.

```
R1# show ip route

D    192.168.16.0/26 [90/2679326] via 192.168.1.1
R    192.168.16.0/24 [120/3] via 192.168.1.2
O    192.168.16.0/21 [110/2] via 192.168.1.3
i L1 192.168.16.0/27 [115/30] via 192.168.1.4
```

Which route does R1 select for traffic that is destined to 192.168.16.2?

- A. 192.168.16.0/21
- B. 192.168.16.0/24
- C. 192.168.26.0/26
- D. 192.168.16.0/27

---

**Answer: D**

---

Explanation:

The destination IP addresses match all four entries in the routing table but the 192.168.16.0/27 has the longest prefix so it will be chosen. This is called the “longest prefix match” rule.

---

**Question: 346**

---

Which technology can prevent client devices from arbitrarily connecting to the network without state remediation?

- A. 802.1x
- B. IP Source Guard

- C. MAC Authentication Bypass
- D. 802.11n

---

**Answer: A**

---

Explanation:

---

**Question: 347**

---

An engineer observes high usage on the 2.4GHz channels and lower usage on the 5GHz channels. What must be configured to allow clients to preferentially use 5GHz access points?

- A. Re- Anchor Roamed Clients
- B. 11ac MU-MIMO
- C. OEAP Split Tunnel
- D. Client Band Select

---

**Answer: D**

---

Explanation:

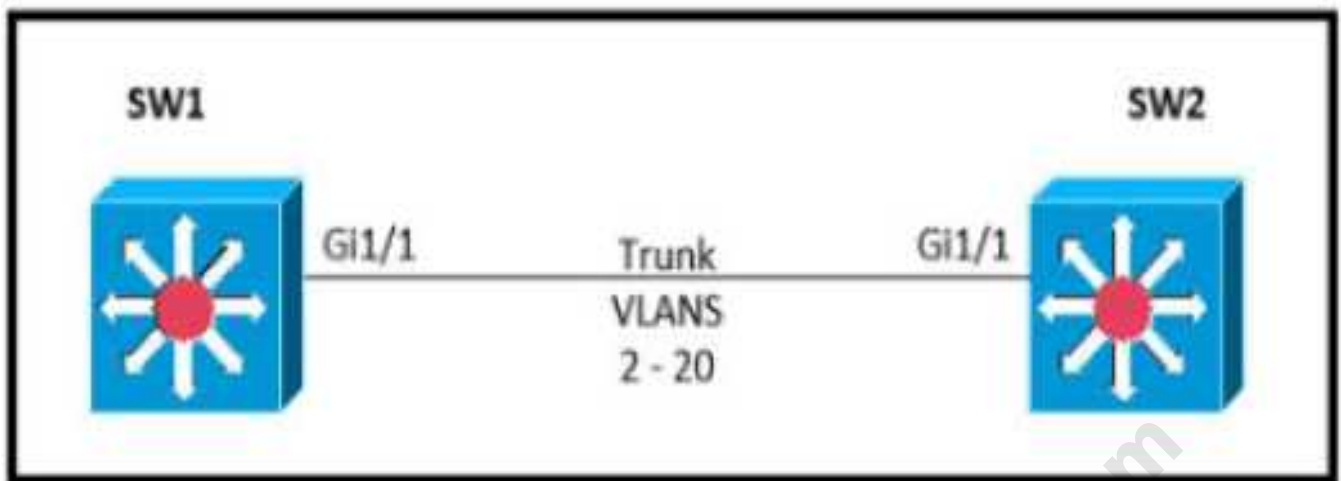
---

**Question: 348**

---

Refer to the exhibit.





Which command must be executed for Gi1.1 on SW1 to become a trunk port if Gi1/1 on SW2 is configured in desirable or trunk mode?

- A. switchport mode trunk
- B. switchport mode dot1-tunnel
- C. switchport mode dynamic auto
- D. switchport mode dynamic desirable

---

**Answer: C**

---

Explanation:

---

**Question: 349**

---

Which IPv6 address type provides communication between subnets and is unable to route on the Internet?

- A. global unicast
- B. unique local
- C. link-local
- D. multicast

---

**Answer: B**

---

Explanation:

---

**Question: 350**

---

What are two descriptions of three-tier network topologies? (Choose two)

- A. The core and distribution layers perform the same functions
- B. The access layer manages routing between devices in different domains
- C. The network core is designed to maintain continuous connectivity when devices fail.
- D. The core layer maintains wired connections for each host
- E. The distribution layer runs Layer 2 and Layer 3 technologies

---

**Answer: C,E**

---

Explanation:

---

**Question: 351**

---

Refer to the exhibit.

```
SW1(config-line)#line vty 0 15
SW1(config-line)#no login local
SW1(config-line)#password cisco

SW2(config)#username admin1 password abcd1234
SW2(config)#username admin2 password abcd1234
SW2(config-line)#line vty 0 15
SW2(config-line)#login local

SW3(config)#username admin1 secret abcd1234
SW3(config)#username admin2 secret abcd1234
SW3(config-line)#line vty 0 15
SW3(config-line)#login local

SW4(config)#username admin1 secret abcd1234
SW4(config)#username admin2 secret abcd1234
SW4(config-line)#line console 0
SW4(config-line)#login local
```

An administrator configures four switches for local authentication using passwords that are stored in a cryptographic hash. The four switches must also support SSH access for administrators to manage the network infrastructure. Which switch is configured correctly to meet these requirements?

- A. SW1
- B. SW2
- C. SW3
- D. SW4

---

**Answer: C**

---

Explanation:

---

**Question: 352**

---

What is a role of access points in an enterprise network?

- A. connect wireless devices to a wired network
- B. support secure user logins to devices or the network
- C. integrate with SNMP in preventing DDoS attacks
- D. serve as a first line of defense in an enterprise network

---

**Answer: A**

---

Explanation:

---

**Question: 353**

---

What is a function of TFTP in network operations?

- A. transfers a backup configuration file from a server to a switch using a username and password
- B. transfers files between file systems on a router
- C. transfers a configuration files from a server to a router on a congested link
- D. transfers IOS images from a server to a router for firmware upgrades

---

**Answer: D**

---

Explanation:

TFTP is mostly used (Firmware upgrade) whereby the admin have the IOS image on one device and

uses TFTP to load the image to all other devices quickly.

---

**Question: 354**

---

A network engineer must create a diagram of a multivendor network. Which command must be configured on the Cisco devices so that the topology of the network can be mapped?

- A. Device(Config)#lldp run
- B. Device(Config)#cdp run
- C. Device(Config-if)#cdp enable
- D. Device(Config)#flow-sampler-map topology

---

**Answer: A**

---

Explanation:

---

**Question: 355**

---

What is the same for both copper and fiber interfaces when using SFP modules?

- A. They support an inline optical attenuator to enhance signal strength
- B. They provide minimal interruption to services by being hot-swappable
- C. They offer reliable bandwidth up to 100 Mbps in half duplex mode
- D. They accommodate single-mode and multi-mode in a single module

---

**Answer: B**

---

Explanation:

---

**Question: 356**

---

When a WLAN with WPA2 PSK is configured in the Wireless LAN Controller GUI which format is

supported?

- A. Unicode
- B. base64
- C. decimal
- D. ASCII

---

**Answer: D**

---

Explanation:

---

**Question: 357**

---

DRAG DROP

Drag and drop the TCP/IP protocols from the left onto the transmission protocols on the right

|      |
|------|
| DNS  |
| SMTP |
| SNMP |

|        |
|--------|
| HTTP   |
| RTP    |
| Telnet |

|     |
|-----|
| TCP |
|     |

|     |
|-----|
| UDP |
|     |

---

**Answer:**

---

Explanation:



---

**Question: 358**

---

When deploying syslog, which severity level logs informational message?

- A. 0
- B. 2
- C. 4
- D. 6

---

**Answer: D**

---

Explanation:

<https://en.wikipedia.org/wiki/Syslog>

---

**Question: 359**

---

Refer to the exhibit.

```
R1#show ip interface brief
```

| Interface          | IP-Address   | OK? | Method | Status                | Protocol |
|--------------------|--------------|-----|--------|-----------------------|----------|
| FastEthernet0/0    | unassigned   | YES | NVRAM  | administratively down | down     |
| GigabitEthernet1/0 | 192.168.0.1  | YES | NVRAM  | up                    | up       |
| GigabitEthernet2/0 | 10.10.1.10   | YES | manual | up                    | up       |
| GigabitEthernet3/0 | 10.10.10.20  | YES | manual | up                    | up       |
| GigabitEthernet4/0 | unassigned   | YES | NVRAM  | administratively down | down     |
| Loopback0          | 172.16.15.10 | YES | manual |                       |          |

What does router R1 use as its OSPF router-ID?

- A. 10.10.1.10
- B. 10.10.10.20
- C. 172.16.15.10
- D. 192.168.0.1

---

**Answer: C**

---

Explanation:

OSPF uses the following criteria to select the router ID:

1. Manual configuration of the router ID (via the "router-id x.x.x.x" command under OSPF router configuration mode).
2. Highest IP address on a loopback interface.
3. Highest IP address on a non-loopback and active (no shutdown) interface.

---

**Question: 360**

---

Which protocol does an access point use to draw power from a connected switch?



- A. Internet Group Management Protocol
- B. Adaptive Wireless Path Protocol
- C. Cisco Discovery Protocol
- D. Neighbor Discovery Protocol

---

**Answer: C**

---

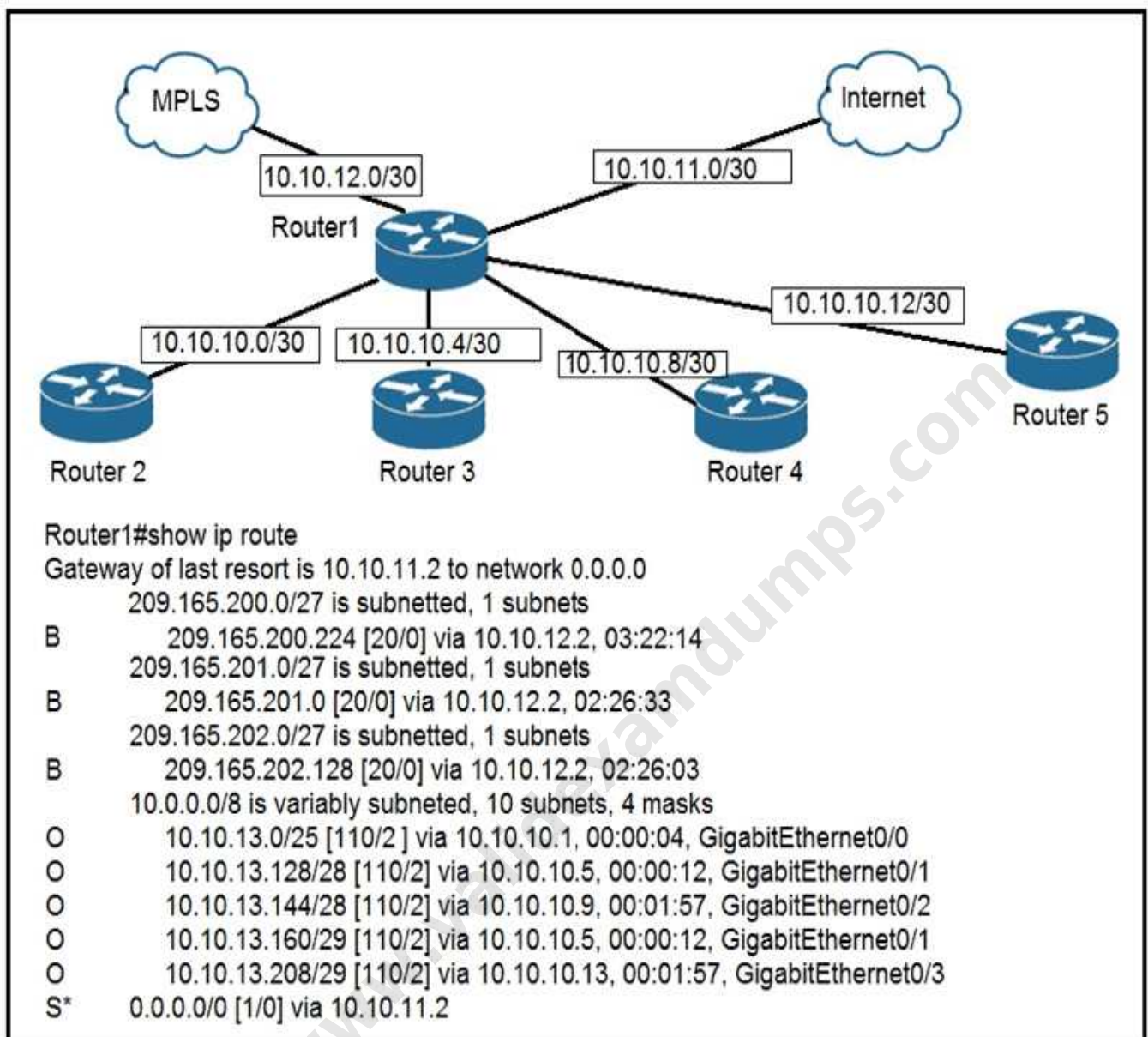
Explanation:

---

**Question: 361**

---

Refer to the exhibit.



To which device does Router1 send packets that are destined to host 10.10.13.165?

- A. Router2
- B. Router3
- C. Router4
- D. Router5

**Answer: B**

Explanation:

---

**Question: 362**

---

Which networking function occurs on the data plane?

- A. forwarding remote client/server traffic
- B. facilitates spanning-tree elections
- C. processing inbound SSH management traffic
- D. sending and receiving OSPF Hello packets

---

**Answer: A**

---

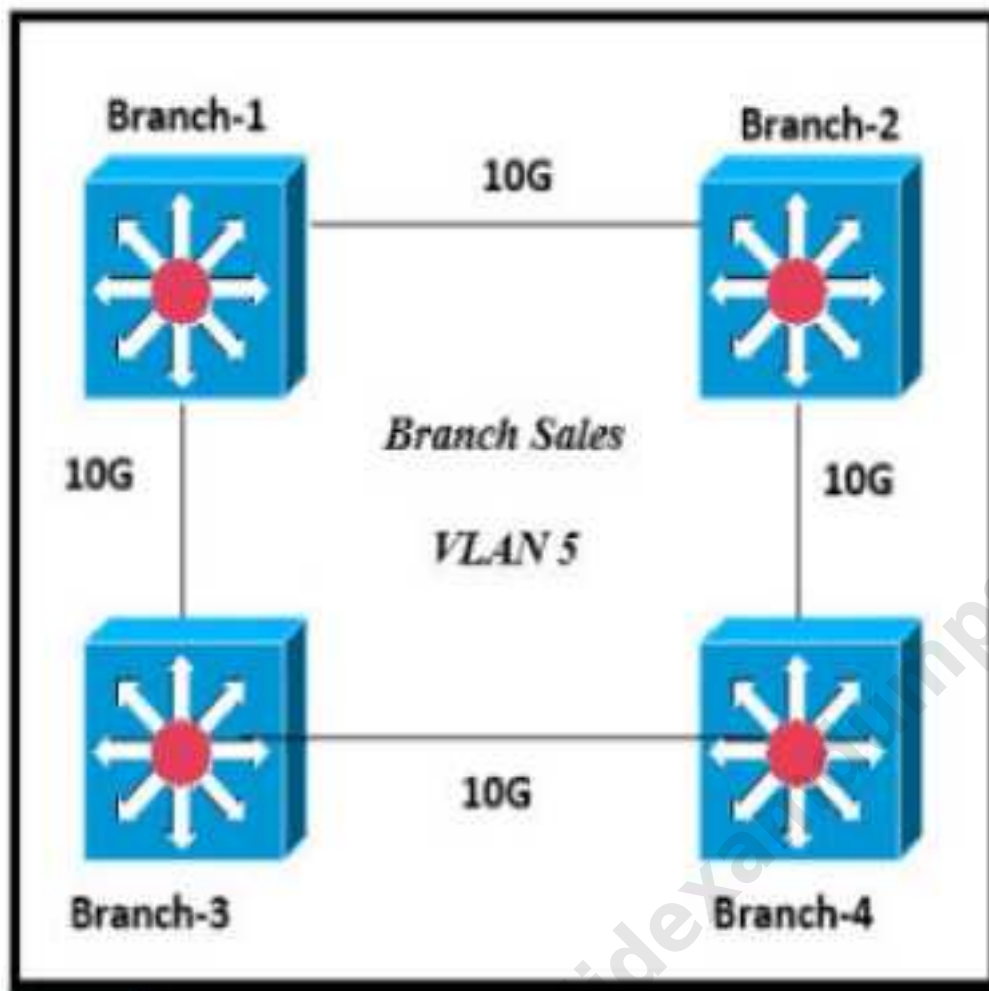
Explanation:

---

**Question: 363**

---

Refer to the exhibit.



Only four switches are participating in the VLAN spanning-tree process.

Branch-1 priority 614440

Branch-2: priority 39082416

Branch-3: priority 0

Branch-4: root primary

Which switch becomes the permanent root bridge for VLAN 5?

- A. Branch-1
- B. Branch-2
- C. Branch-3
- D. Branch-4

---

**Answer: C**

---

Explanation:

Dynamic ARP inspection is an ingress security feature; it does not perform any egress checking.

---

**Question: 364**

---

Which two tasks must be performed to configure NTP to a trusted server in client mode on a single network device? (Choose two)

- A. Enable NTP authentication.
- B. Verify the time zone.
- C. Disable NTP broadcasts
- D. Specify the IP address of the NTP server
- E. Set the NTP server private key

---

**Answer: A,D**

---

Explanation:

<https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst4000/8-2glx/configuration/guide/ntp.html>

To configure authentication, perform this task in privileged mode:

Step 1: Configure an authentication key pair for NTP and specify whether the key will be trusted or untrusted.

Step 2: Set the IP address of the NTP server and the public key.

Step 3: Enable NTP client mode.

Step 4: Enable NTP authentication.

Step 5: Verify the NTP configuration.

---

**Question: 365**

---

Refer to the exhibit.

```
SiteA#show interface TenGigabitEthernet0/1/0
TenGigabitEthernet0/1/0 is up, line protocol is up
  Hardware is BUILT-IN-EPA-8x10G, address is 780c.f02a.db91 (bia 780a.f02b.db91)
  Description: Connection to SiteB
  Internet address is 10.10.10.1/30
  MTU 8146 bytes, BW 10000000 Kbit/sec, DLY 10 usec,
    reliability 166/255, txload 1/255, rxload 1/255
  Full Duplex, 10000Mbps, link type is force-up, media type is SFP-LR
  5 minute input rate 264797000 bits/sec, 26672 packets/sec
  5 minute output rate 122464000 bits/sec, 15724 packets/sec

SiteB#show interface TenGigabitEthernet0/1/0
TenGigabitEthernet0/1/0 is up, line protocol is up
  Hardware is BUILT-IN-EPA-8x10G, address is 780c.f02c.db26 (bia 780c.f02c.db26)
  Description: Connection to SiteA
  Internet address is 10.10.10.2/30
  MTU 8146 bytes, BW 10000000 Kbit/sec, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Full Duplex, 10000Mbps, link type is force-up, media type is SFP-LR
  5 minute input rate 122464000 bits/sec, 15724 packets/sec
  5 minute output rate 264797000 bits/sec, 26672 packets/sec
```

Shortly after SiteA was connected to SiteB over a new single-mode fiber path users at SiteA report intermittent connectivity issues with applications hosted at SiteB. What is the cause of the intermittent connectivity issue?

- A. Interface errors are incrementing
- B. An incorrect SFP media type was used at SiteA
- C. High usage is causing high latency
- D. The sites were connected with the wrong cable type

---

**Answer: A**

---

Explanation:

---

**Question: 366**

---

Where does a switch maintain DHCP snooping information?

- A. in the MAC address table
- B. in the CAM table
- C. in the binding database
- D. in the frame forwarding database

---

**Answer: C**

---

Explanation:

---

**Question: 367**

---

Refer to the exhibit.

```
SW1#show run int gig 0/1
interface GigabitEthernet0/1
  switchport access vlan 11
  switchport trunk allowed vlan 1-10
  switchport trunk encapsulation dot1q
  switchport trunk native vlan 5
  switchport mode trunk
  speed 1000
  duplex full
```

Which action is expected from SW1 when the untagged frame is received on the GigabitEthernet0/1 interface?

- A. The frame is processed in VLAN 5.
- B. The frame is processed in VLAN 11
- C. The frame is processed in VLAN 1
- D. The frame is dropped

---

**Answer: A**

---

Explanation:

---

**Question: 368**

---

Refer to the exhibit.

```
SW1#sh lacp neighbor
Flags: S - Device is requesting Slow LACPDUs
       F - Device is requesting Fast LACPDUs
       A - Device is in Active mode           P - Device is in Passive mode

Channel group 35 neighbors

Partner's information:
```

| Port  | Flags | LACP port<br>Priority | Dev ID         | Age | Admin<br>key | Oper<br>Key | Port<br>Number | Port<br>State |
|-------|-------|-----------------------|----------------|-----|--------------|-------------|----------------|---------------|
| Et1/0 | SP    | 32768                 | aabb.cc80.7000 | 8s  | 0x0          | 0x23        | 0x101          | 0x3C          |
| Et1/1 | SP    | 32768                 | aabb.cc80.7000 | 8s  | 0x0          | 0x23        | 0x102          | 0x3C          |

Based on the LACP neighbor status, in which mode is the SW1 port channel configured?

- A. passive
- B. mode on
- C. auto



D. active

---

**Answer: D**

---

Explanation:

From the neighbor status, we notice the "Flags" are SP. "P" here means the neighbor is in Passive mode.

In order to create an Etherchannel interface, the (local) SW1 ports should be in Active mode.

Moreover, the "Port State" in the exhibit is "0x3c" (which equals to "00111100 in binary format).

Bit 3 is "1" which means the ports are synchronizing -> the ports are working so the local ports should be in Active mode.

---

**Question: 369**

---

DRAG DROP

Refer to the exhibit.

```
[root@HostTest ~]# ip route
default via 192.168.1.193 dev eth1 proto static
192.168.1.0/26 dev eth1 proto kernel scope link src 192.168.1.200 metric 1

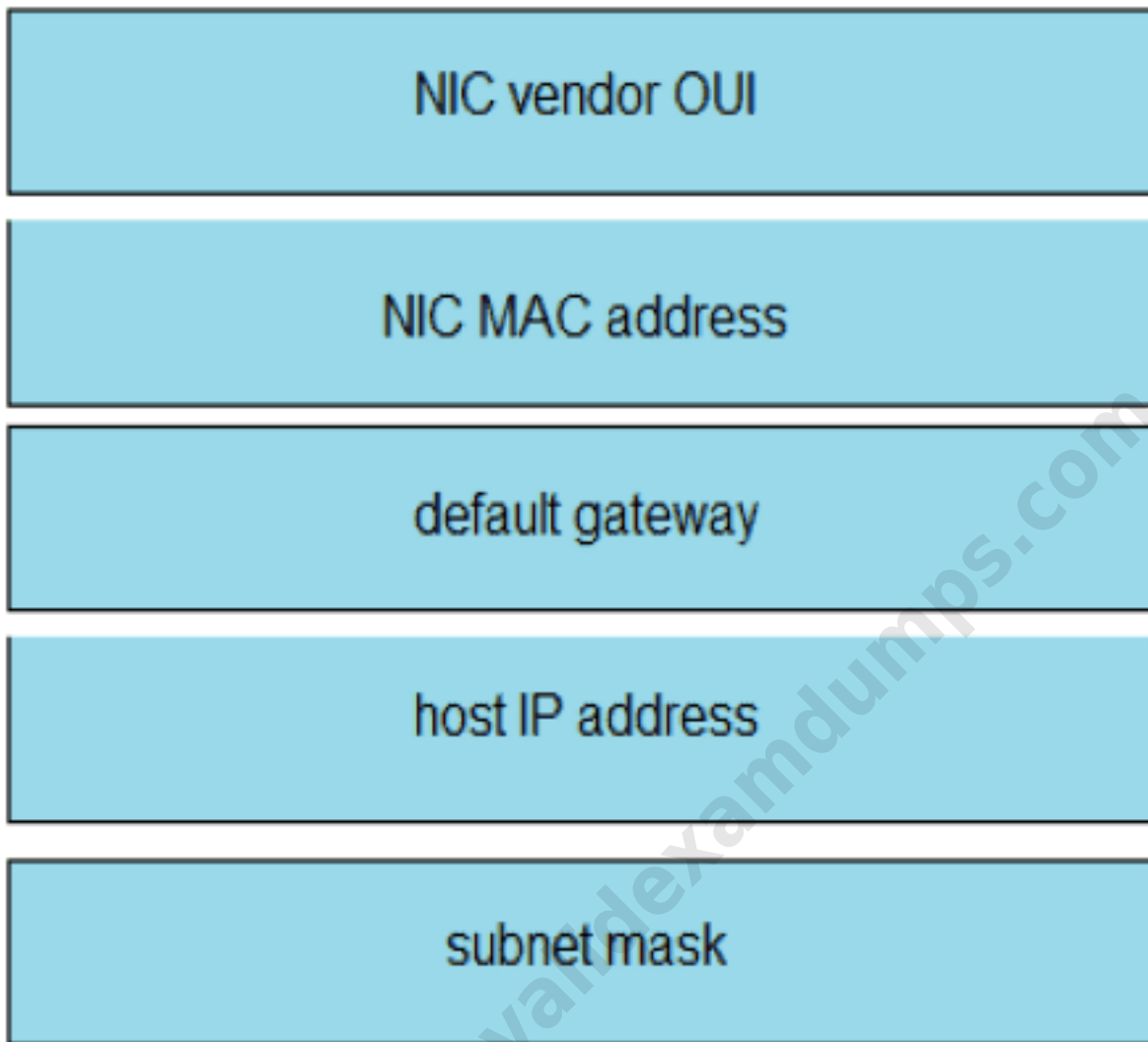
[root@HostTest ~]# ip addr show eth1
eth1: mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 00:0c:22:83:79:a3 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.200/26 brd 192.168.1.255 scope global eth1
    inet6 fe80::20c:29ff:fe89:79b3/64 scope link
    valid_lft forever preferred_lft forever
```

Drag and drop the networking parameters from the left onto the correct values on the right.

|                 |                   |
|-----------------|-------------------|
| default gateway | 00:0C:22          |
| host IP address | 00:0C:22:83:79:A3 |
| NIC MAC address | 192.168.1.193     |
| NIC vendor OUI  | 192.168.1.200     |
| subnet mask     | 255.255.255.192   |

**Answer:**

Explanation:



NIC vendor OUI → 00:0C:22

NIC MAC address → 00:0C:22:83:79:A3

default gateway → 192.168.1.193

host IP address → 192.168.1.200

subnet mask → 255.255.255.192

The “ip route” and “ip addr show eth1” are Linux commands.

+ “ip route”: display the routing table

+ “ip addr show eth1”: get depth information (only on eth1 interface) about your network interfaces like IP Address, MAC Address information

---

**Question: 370**

---

Which result occurs when PortFast is enabled on an interface that is connected to another switch?

- A. Spanning tree may fail to detect a switching loop in the network that causes broadcast storms
- B. VTP is allowed to propagate VLAN configuration information from switch to switch automatically.
- C. Root port choice and spanning tree recalculation are accelerated when a switch link goes down
- D. After spanning tree converges PortFast shuts down any port that receives BPDUs.

---

**Answer: A**

---

Explanation:

Enabling the PortFast feature causes a switch or a trunk port to enter the STP forwarding-state immediately or upon a linkup event, thus bypassing the listening and learning states.

Note: To enable portfast on a trunk port you need the trunk keyword "spanning-tree portfast trunk

---

**Question: 371**

---

What is the primary different between AAA authentication and authorization?

- A. Authentication verifies a username and password, and authorization handles the communication between the authentication agent and the user database.
- B. Authentication identifies a user who is attempting to access a system, and authorization validates the users password
- C. Authentication identifies and verifies a user who is attempting to access a system, and authorization controls the tasks the user can perform.
- D. Authentication controls the system processes a user can access and authorization logs the activities the user initiates

---

**Answer: C**

---

Explanation:

AAA stands for Authentication, Authorization and Accounting.

- + Authentication: Specify who you are (usually via login username & password)
- + Authorization: Specify what actions you can do, what resource you can access
- + Accounting: Monitor what you do, how long you do it (can be used for billing and auditing)

An example of AAA is shown below:

- + Authentication: "I am a normal user. My username/password is user\_tom/learnforever"
- + Authorization: "user\_tom can access LearnCCNA server via HTTP and FTP"
- + Accounting: "user\_tom accessed LearnCCNA server for 2 hours". This user only uses "show" commands.

---

**Question: 372**

---

A network administrator must to configure SSH for remote access to router R1 The requirement is to use a public and private key pair to encrypt management traffic to and from the connecting client.

Which configuration, when applied, meets the requirements?

```
R1#enable
R1#configure terminal
R1(config)#ip domain-name cisco.com
R1(config)#crypto key generate ec keysizes 2048
```

```
R1#enable
R1#configure terminal
R1(config)#ip domain-name cisco.com
R1(config)#crypto key generate rsa modulus 1024
```

```
R1#enable
R1#configure terminal
R1(config)#ip domain-name cisco.com
R1(config)#crypto key generate ec keysizes 1024
```

```
R1#enable
R1#configure terminal
R1(config)#ip domain-name cisco.com
R1(config)#crypto key encrypt rsa name myKey
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: C**

---

Explanation:

---

### Question: 373

---

A network engineer must configure the router R1 GigabitEthernet1/1 interface to connect to the router R2 GigabitEthernet1/1 interface. For the configuration to be applied the engineer must compress the address 2001:0db8:0000:0000:0500:000a:400F:583B. Which command must be issued on the interface?

- A. ipv6 address 2001:0db8::5: a: 4F 583B
- B. ipv6 address 2001:db8::500:a:400F:583B
- C. ipv6 address 2001 db8:0::500:a:4F:583B
- D. ipv6 address 2001::db8:0000::500:a:400F:583B

---

**Answer: B**

---

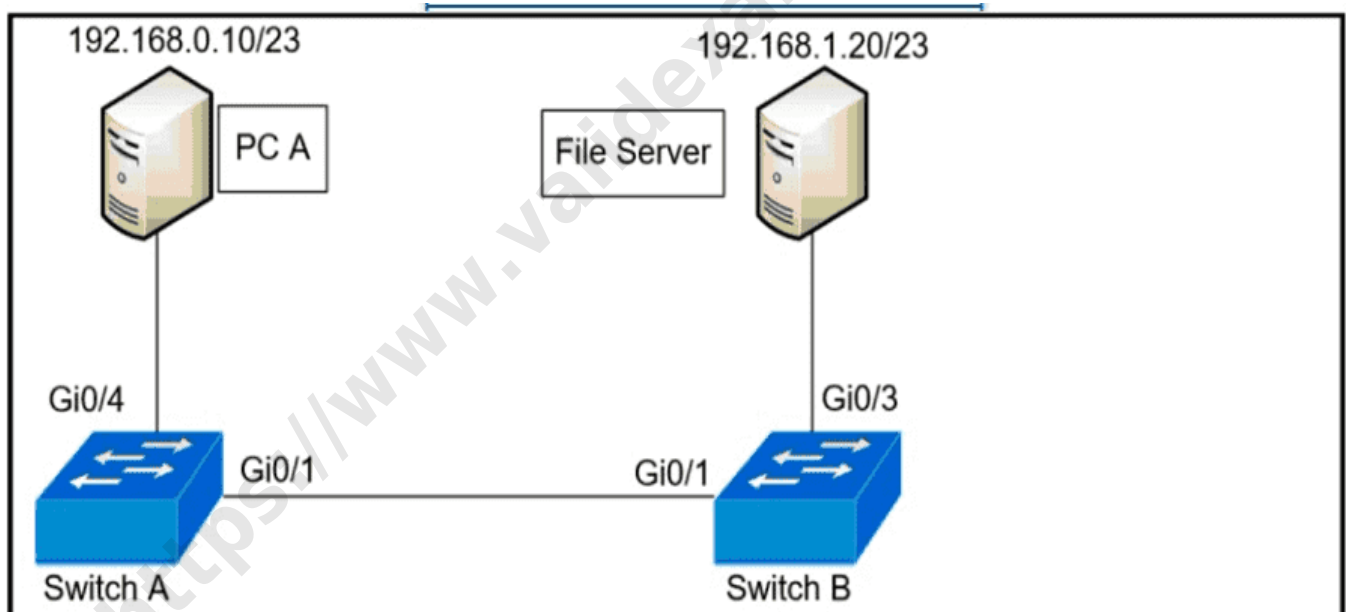
Explanation:

---

**Question: 374**

---

Refer to the exhibit.



|                                                                                                                                                                                                                                     |                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Switch A</b><br>Vlan 10,11,12,13<br><br>interface GigabitEthernet0/1<br>switchport mode trunk<br>switchport trunk allowed vlan 10-12<br>!<br>interface GigabitEthernet0/4<br>switchport access vlan 13<br>switchport mode access | <b>Switch B</b><br>Vlan 10,11,12,13<br><br>interface GigabitEthernet0/1<br>switchport mode trunk<br>!<br>interface GigabitEthernet0/3<br>switchport access vlan 13<br>switchport mode access |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

A network administrator assumes a task to complete the connectivity between PC A and the File Server. Switch A and Switch B have been partially configured with VLAN 10, 11, 12, and 13. What is the next step in the configuration?

- A. Add PC A to VLAN 10 and the File Server to VLAN 11 for VLAN segmentation
- B. Add VLAN 13 to the trunk links on Switch A and Switch B for VLAN propagation
- C. Add a router on a stick between Switch A and Switch B allowing for Inter-VLAN routing.
- D. Add PC A to the same subnet as the File Server allowing for intra-VLAN communication.

---

**Answer: B**

---

Explanation:

---

**Question: 375**

---

Which goal is achieved by the implementation of private IPv4 addressing on a network?

- A. provides an added level of protection against Internet exposure
- B. provides a reduction in size of the forwarding table on network routers
- C. allows communication across the Internet to other private networks
- D. allows servers and workstations to communicate across public network boundaries



---

**Answer: A**

---

Explanation:

---

**Question: 376**

---

What is a characteristic of spine-and-leaf architecture?

- A. Each device is separated by the same number of hops
- B. It provides variable latency
- C. It provides greater predictability on STP blocked ports.
- D. Each link between leaf switches allows for higher bandwidth.

---

**Answer: A**

---

Explanation:

---

**Question: 377**

---

A router running EIGRP has learned the same route from two different paths. Which parameter does the router use to select the best path?

- A. cost
- B. administrative distance
- C. metric
- D. as-path

---

**Answer: C**

---

Explanation:

If a router learns two different paths for the same network from the same routing protocol, it has to decide which route is better and will be placed in the routing table. Metric is the measure used to decide which route is better (lower number is better). Each routing protocol uses its own metric.

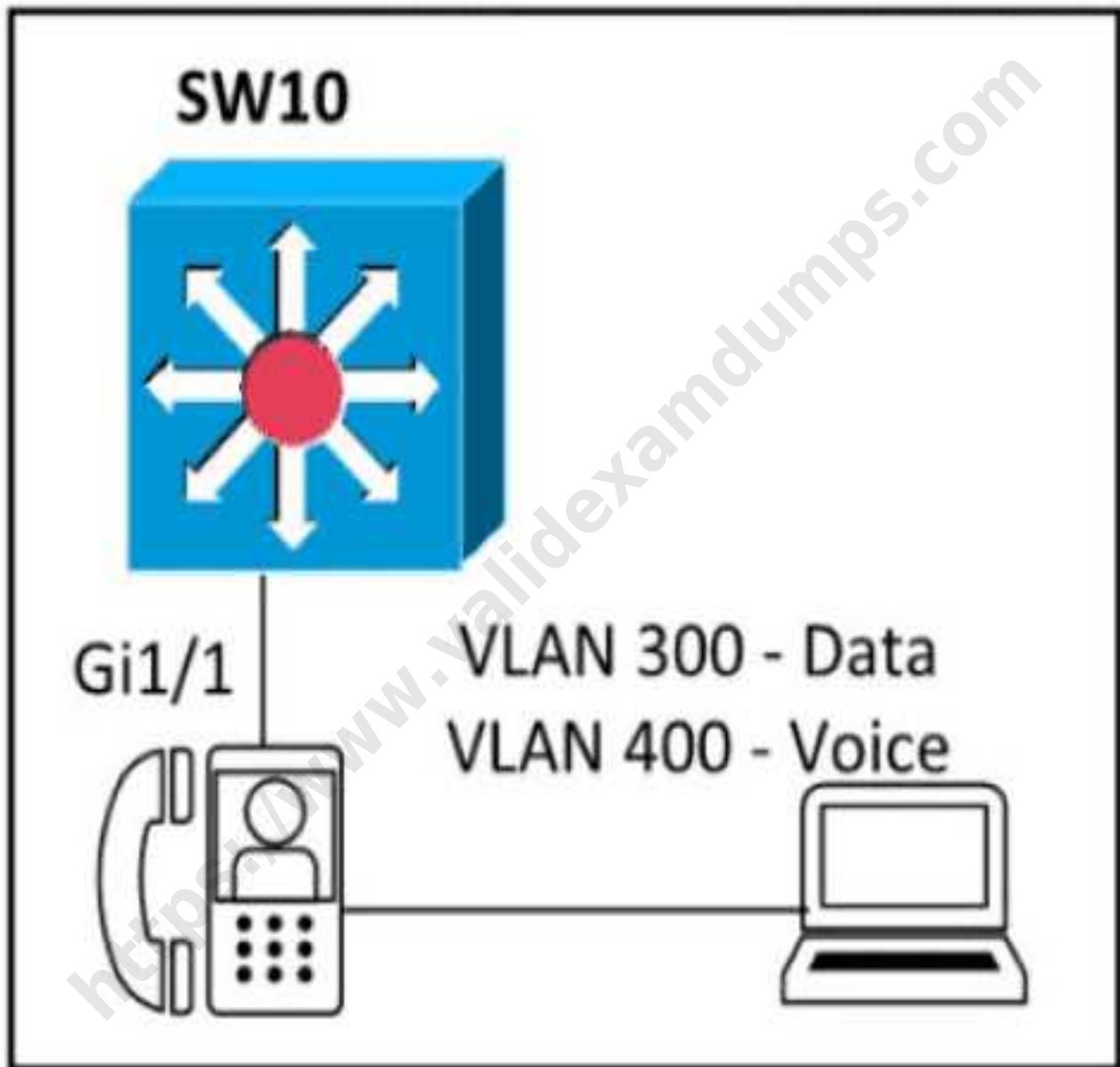
For example, RIP uses hop counts as a metric, while OSPF uses cost.

---

**Question: 378**

---

Refer to the exhibit.



An engineer must configure GigabitEthernet1/1 to accommodate voice and data traffic Which configuration accomplishes this task?

```
interface gigabitethernet1/1
switchport mode access
switchport access vlan 300
switchport voice vlan 400
```

```
interface gigabitethernet1/1
switchport mode trunk
switchport trunk vlan 300
switchport voice vlan 400
```

```
interface gigabitethernet1/1
switchport mode trunk
switchport trunk vlan 300
switchport trunk vlan 400
```

```
interface gigabitethernet1/1
switchport mode access
switchport voice vlan 300
switchport access vlan 400
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: A**

---

Explanation:

---

**Question: 379**

---

Refer to the exhibit.

```
Designated Router (ID) 10.11.11.11, Interface address 10.10.10.1
Backup Designated router (ID) 10.3.3.3, Interface address 10.10.10.3
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
oob-resync timeout 40
Hello due in 00:00:08
Supports Link-local Signaling (LLS)
Cisco NSF helper support enabled
IETF NSF helper support enabled
Index 1/1/1, flood queue length 0
Next 0x0(0)/0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 6
Last flood scan time is 0 msec, maximum is 1 msec
Neighbor Count is 3, Adjacent neighbor count is 3
Adjacent with neighbor 10.1.1.4
Adjacent with neighbor 10.2.2.2
Adjacent with neighbor 10.3.3.3 (Backup Designated Router)
Suppress hello for 0 neighbor(s)
```

The show ip ospf interface command has been executed on R1 How is OSPF configured?

- A. The interface is not participating in OSPF
- B. A point-to-point network type is configured
- C. The default Hello and Dead timers are in use
- D. There are six OSPF neighbors on this interface

---

**Answer: C**

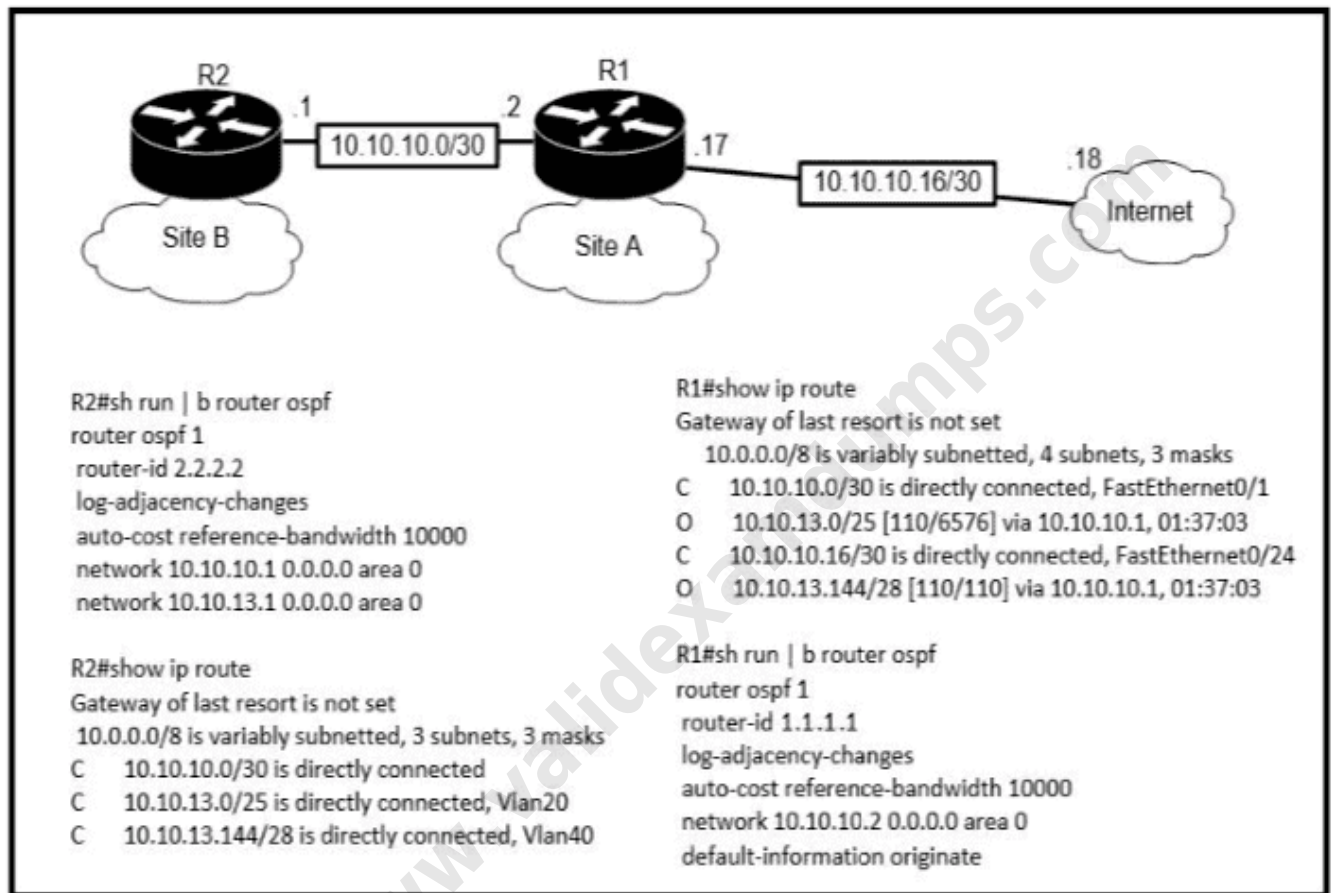
---

Explanation:

<https://www.cisco.com/c/en/us/support/docs/ip/open-shortest-path-first-ospf/13689-17.html>

**Question: 380**

Refer to the exhibit.



An engineer is bringing up a new circuit to the MPLS provider on the Gi0/1 interface of Router1. The new circuit uses eBGP and teams the route to VLAN25 from the BGP path. What is the expected behavior for the traffic flow for route 10.10.13.0/25?

- A. Traffic to 10.10.13.0.25 is load balanced out of multiple interfaces
- B. Route 10.10.13.0/25 is updated in the routing table as being learned from interface Gi0/1.
- C. Traffic to 10.10.13.0/25 is asymmetrical
- D. Route 10.10.13.0/25 learned via the Gi0/0 interface remains in the routing table

**Answer: D**

Explanation:

---

**Question: 381**

---

Refer to the exhibit.

```
R1# show ip route | begin gateway
Gateway of last resort is 209.165.200.246 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 209.165.200.246, Serial0/1/0
    is directly connected, Serial0/1/0
    172.16.0.0/16 is variably subnetted, 3 subnets, 3 masks
S    172.16.0.0/24 [1/0] via 207.165.200.250, Serial0/0/0
O    172.16.0.128/25 [110/38443] via 207.165.200.254, 00:00:23, Serial0/0/1
D    172.16.0.192/29 [90/3184439] via 207.165.200.254, 00:00:25, Serial0/0/1
    209.165.200.0/24 is variably subnetted, 4 subnets, 2 masks
C    209.165.200.248/30 is directly connected, Serial0/0/0
L    209.165.200.249/32 is directly connected, Serial0/0/0
C    209.165.200.252/30 is directly connected, Serial0/0/1
L    209.165.200.253/32 is directly connected, Serial0/0/1
```

With which metric was the route to host 172.16.0.202 learned?

- A. 0
- B. 110
- C. 38443
- D. 3184439

---

**Answer: C**

---

Explanation:

Both the line "O 172.16.0.128/25" and "S 172.16.0.0/24" cover the host 172.16.0.202 but with the "longest (prefix) match" rule the router will choose the first route.

---

**Question: 382**

---

Refer to the exhibit.

```
R1# show ip route
....
D      172.16.32.0/27 [90/2888597172] via 20.1.1.1
O      172.16.32.0/19 [110/292094] via 20.1.1.10
R      172.16.32.0/24 [120/2] via 20.1.1.3
```

Router R1 is running three different routing protocols. Which route characteristic is used by the router to forward the packet that it receives for destination IP 172.16.32.1?

- A. longest prefix
- B. metric
- C. cost
- D. administrative distance

---

**Answer: A**

---

Explanation:

<https://learningnetwork.cisco.com/s/question/0D53i00000KszSICAJ/administrative-distance-vs-longest-match-rule>

---

### Question: 383

---

What are two benefits of network automation? (Choose two)

- A. reduced operational costs
- B. reduced hardware footprint
- C. faster changes with more reliable results
- D. fewer network failures
- E. increased network security

---

**Answer: A,C**

---

Explanation:

---

**Question: 384**

---

An administrator must secure the WLC from receiving spoofed association requests. Which steps must be taken to configure the WLC to restrict the requests and force the user to wait 10 ms to retry an association request?

- A. Enable Security Association Teardown Protection and set the SA Query timeout to 10
- B. Enable MAC filtering and set the SA Query timeout to 10
- C. Enable 802.1x Layer 2 security and set the Comeback timer to 10
- D. Enable the Protected Management Frame service and set the Comeback timer to 10

---

**Answer: C**

---

Explanation:

---

**Question: 385**

---

Which function does an SNMP agent perform?

- A. it sends information about MIB variables in response to requests from the NMS
- B. it requests information from remote network nodes about catastrophic system events.
- C. it manages routing between Layer 3 devices in a network
- D. it coordinates user authentication between a network device and a TACACS+ or RADIUS server

---

**Answer: A**

---

Explanation:

---

**Question: 386**

---



What is the effect when loopback interfaces and the configured router ID are absent during the OSPF Process configuration?

- A. No router ID is set, and the OSPF protocol does not run.
- B. The highest up/up physical interface IP address is selected as the router ID.
- C. The lowest IP address is incremented by 1 and selected as the router ID.
- D. The router ID 0.0.0.0 is selected and placed in the OSPF process.

---

**Answer: B**

---

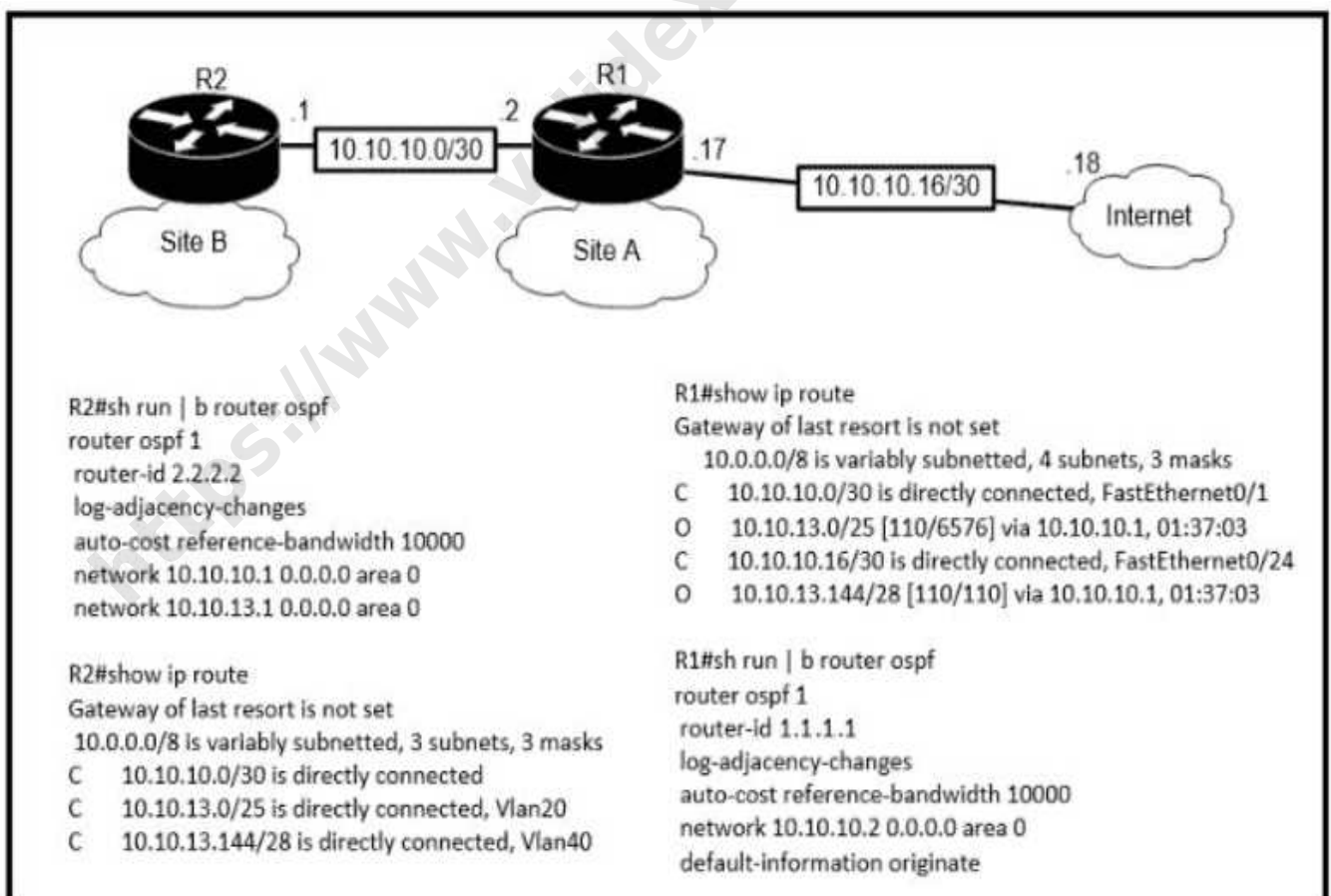
Explanation:

---

**Question: 387**

---

Refer to the exhibit.



The default-information originate command is configured under the R1 OSPF configuration. After testing, workstations on VLAN 20 at Site B cannot reach a DNS server on the Internet. Which action corrects the configuration issue?

- A. Add the default-information originate command on R2
- B. Configure the ip route 0.0.0.0 0.0.0.0 10.10.10.18 command on R1
- C. Configure the ip route 0.0.0.0 0.0.0.0 10.10.10.2 command on R2
- D. Add the always keyword to the default-information originate command on R1

---

**Answer: B**

---

Explanation:

---

**Question: 388**

---

Refer to Exhibit.

```
SW2
vtp domain cisco
vtp mode transparent
vtp password ciscotest
interface fastethernet0/1
  description connection to sw1
  switchport mode trunk
  switchport trunk encapsulation dot1q
```

How does SW2 interact with other switches in this VTP domain?

- A. It processes VTP updates from any VTP clients on the network on its access ports.
- B. It receives updates from all VTP servers and forwards all locally configured VLANs out all trunk ports
- C. It forwards only the VTP advertisements that it receives on its trunk ports.
- D. It transmits and processes VTP updates from any VTP Clients on the network on its trunk ports

---

**Answer: C**

---

Explanation:

Reference: <https://www.cisco.com/c/en/us/support/docs/lan-switching/vtp/10558-21.html>

The VTP mode of SW2 is transparent so it only forwards the VTP updates it receives to its trunk links without processing them.

---

**Question: 389**

---

Which condition must be met before an NMS handles an SNMP trap from an agent?

- A. The NMS software must be loaded with the MIB associated with the trap.
- B. The NMS must be configured on the same router as the SNMP agent
- C. The NMS must receive a trap and an inform message from the SNMP agent within a configured interval
- D. The NMS must receive the same trap from two different SNMP agents to verify that it is reliable.

---

**Answer: A**

---

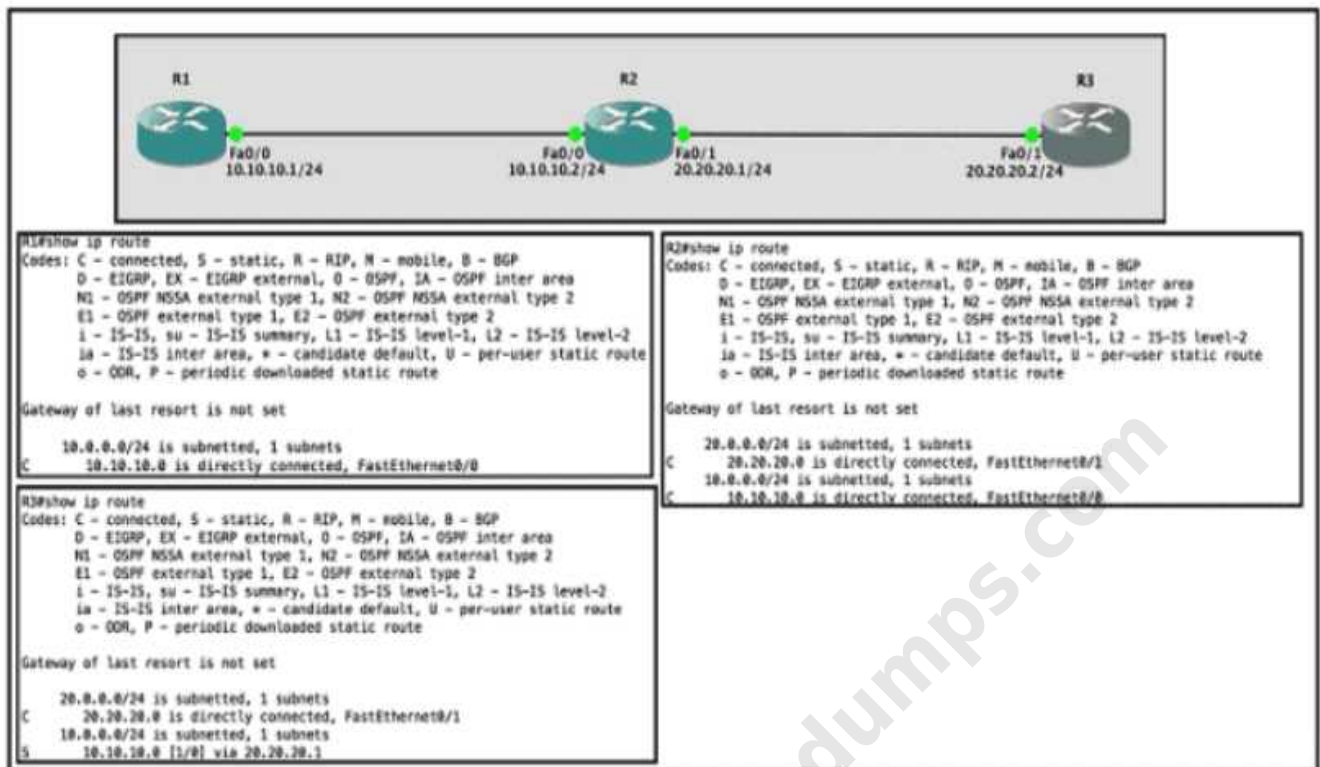
Explanation:

---

**Question: 390**

---

Refer to the exhibit.



Router R1 Fa0/0 is unable ping router R3 Fa0/1.

Which action must be taken in router R1 to help resolve the configuration issue?

- A. set the default network as 20.20.20.0/24
- B. set the default gateway as 20.20.20.2
- C. configure a static route with Fa0/1 as the egress interface to reach the 20.20.20.0/24 network
- D. configure a static route with 10.10.10.2 as the next hop to reach the 20.20.20.0/24 network

**Answer: D**

Explanation:

**Question: 391**

How does a Cisco Unified Wireless network respond to Wi-Fi channel overlap?

- A. It alternates automatically between 2.4 GHz and 5 GHz on adjacent access points
- B. It allows the administrator to assign channels on a per-device or per-interface basis.
- C. It segregates devices from different manufacturers onto different channels.
- D. It analyzes client load and background noise and dynamically assigns a channel.

---

**Answer: A**

---

Explanation:

---

**Question: 392**

---

When a site-to-site VPN is used, which protocol is responsible for the transport of user data?

- A. IKEv2
- B. IKEv1
- C. IPsec
- D. MD5

---

**Answer: C**

---

Explanation:

A site-to-site VPN allows offices in multiple fixed locations to establish secure connections with each other over a public network such as the Internet. A site-to-site VPN means that two sites create a VPN tunnel by encrypting and sending data between two devices. One set of rules for creating a site-to-site VPN is defined by IPsec.

---

**Question: 393**

---

An engineer is configuring NAT to translate the source subnet of 10.10.0.0/24 to any of three addresses 192.168.30.1, 192.168.3.2, 192.168.3.3 Which configuration should be used?

enable  
configure terminal  
ip nat pool mypool 192.168.3.1 192.168.3.3 prefix-length 30  
route-map permit 10.10.0.0 255.255.255.0  
ip nat outside destination list 1 pool mypool  
interface g1/1  
ip nat inside  
interface g1/2  
ip nat outside

enable  
configure terminal  
ip nat pool mypool 192.168.3.1 192.168.3.3 prefix-length 30  
access-list 1 permit 10.10.0.0 0.0.0.255  
ip nat inside source list 1 pool mypool  
interface g1/1  
ip nat inside  
interface g1/2  
ip nat outside

☐ enable  
configure terminal  
ip nat pool mypool 192.168.3.1 192.168.3.3 prefix-length 30  
access-list 1 permit 10.10.0.0 0.0.0.255  
ip nat outside destination list 1 pool mypool  
interface g1/1  
ip nat inside  
interface g1/2  
ip nat outside

☐ enable  
configure terminal  
ip nat pool mypool 192.168.3.1 192.168.3.3 prefix-length 30  
access-list 1 permit 10.10.0.0 0.0.0.254  
ip nat inside source list 1 pool mypool  
interface g1/1  
ip nat inside  
interface g1/2  
ip nat outside

- A. Option A  
B. Option B  
C. Option C  
D. Option D

---

**Answer: C**

---

Explanation:

---

**Question: 394**

---

What is the primary function of a Layer 3 device?

- A. to analyze traffic and drop unauthorized traffic from the Internet

- B. to transmit wireless traffic between hosts
- C. to pass traffic between different networks
- D. forward traffic within the same broadcast domain

---

**Answer: C**

---

Explanation:

---

**Question: 395**

---

Router A learns the same route from two different neighbors, one of the neighbor routers is an OSPF neighbor and the other is an EIGRP neighbor. What is the administrative distance of the route that will be installed in the routing table?

- A. 20
- B. 90
- C. 110
- D. 115

---

**Answer: B**

---

Explanation:

The Administrative distance (AD) of EIGRP is 90 while the AD of OSPF is 110 so EIGRP route will be chosen to install into the routing table.

---

**Question: 396**

---

When the active router in an HSRP group fails, what router assumes the role and forwards packets?



- A. backup
- B. standby
- C. listening
- D. forwarding

---

**Answer: B**

---

Explanation:

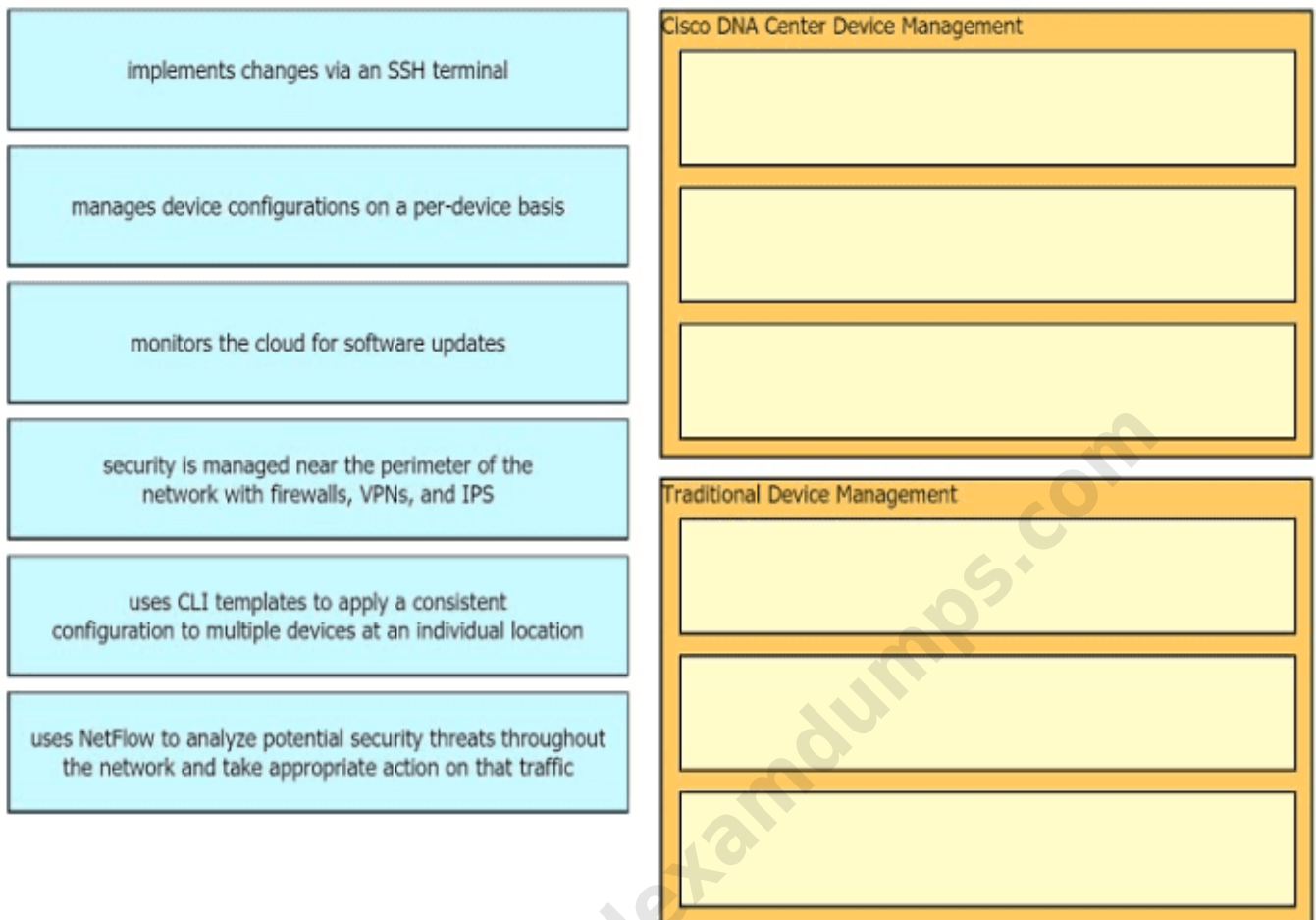
---

**Question: 397**

---

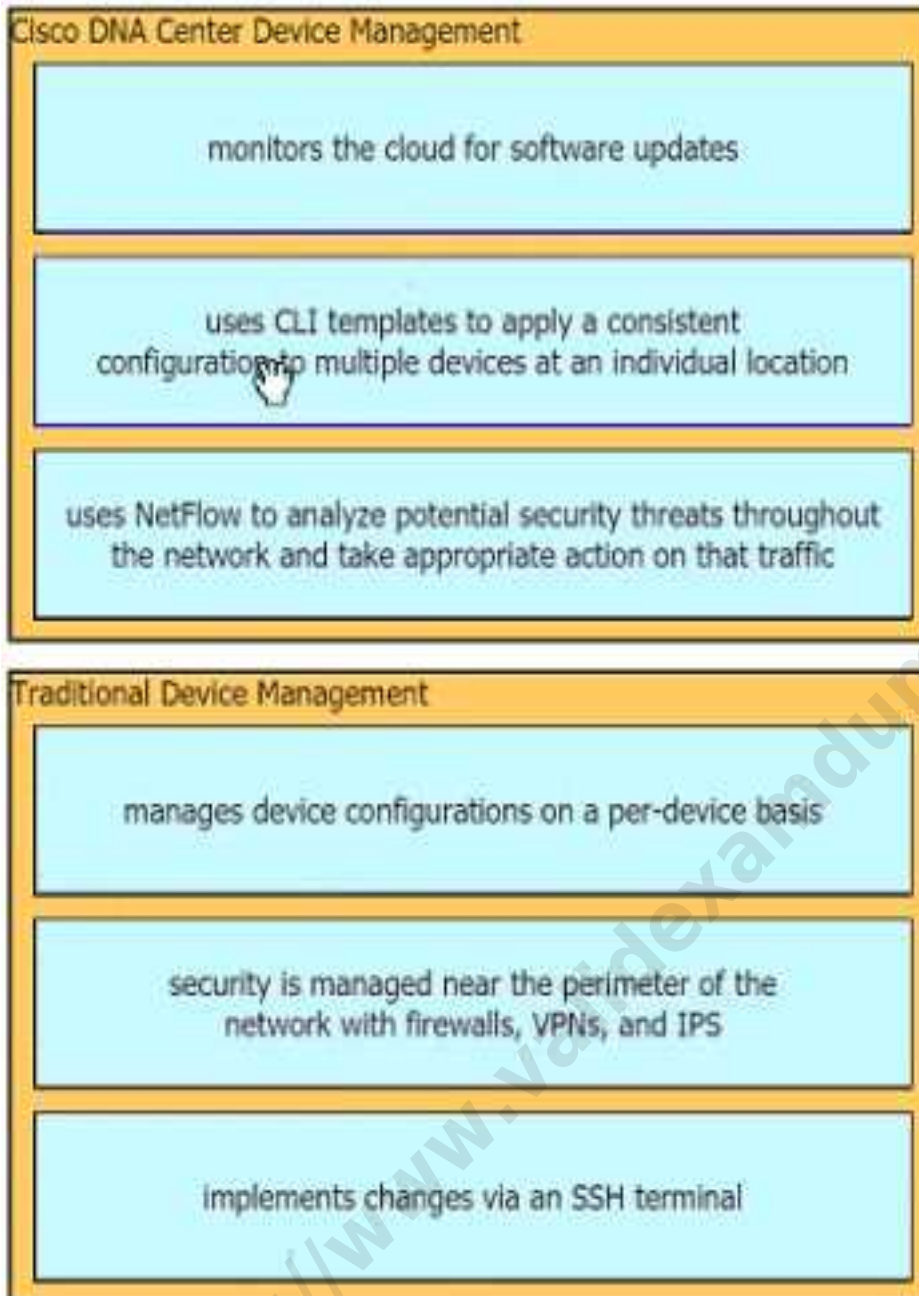
DRAG DROP

Drag the descriptions of device management from the left onto the types of device management on the right.



Explanation:

**Answer:**



---

**Question: 398**

---

Which two protocols must be disabled to increase security for management connections to a Wireless LAN Controller? (Choose two.)

- A. Telnet
- B. SSH
- C. HTTP
- D. HTTPS

E. TFTP

**Answer: A,C**

Explanation:

### Question: 399

DRAG DROP

Refer to the exhibit.

```
C:\>ipconfig/all

Windows IP Configuration

    Host Name . . . . . : Inspiron15
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Mixed
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Wireless LAN adapter Local Area Connection* 12:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
    Physical Address. . . . . : 1A-76-3F-7C-57-DF
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . :
    Description . . . . . : Dell Wireless 1703 802.11b/g/n (2.4GHz)
    Physical Address. . . . . : B8-76-3F-7C-57-DF
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::e09f:9839:6e86:f755%12<Preferred>
    . . . . . : 192.168.1.20<Preferred>
    . . . . . : 255.255.255.0
    . . . . . : 192.168.1.1
    DHCPv6 IAID . . . . . : 263747135
    DHCPv6 Client DUID. . . . . : 00-01-00-01-18-E6-32-43-B8-76-3F-7C-57-DF
    . . . . . : 192.168.1.15
    . . . . . : 192.168.1.16
    NetBIOS over Tcpip. . . . . : Enabled
```

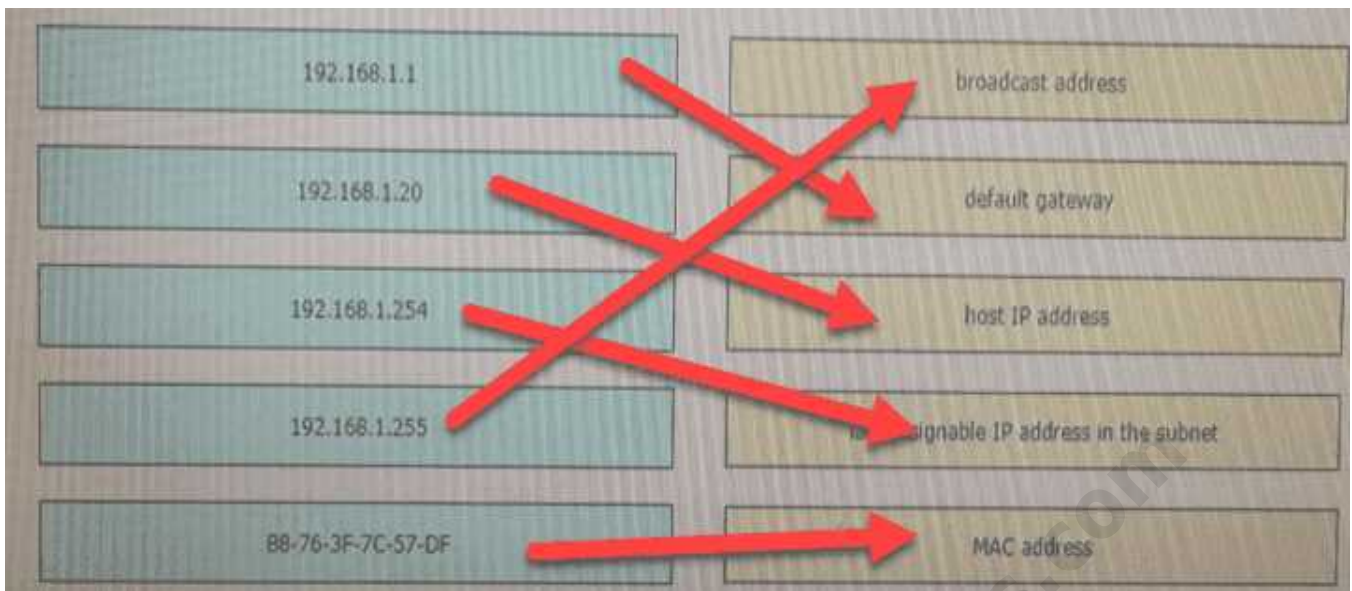
An engineer is tasked with verifying network configuration parameters on a client workstation to report back to the team lead. Drag and drop the node identifiers from the left onto the network parameters on the right.

|                   |                                          |
|-------------------|------------------------------------------|
| 192.168.1.1       | broadcast address                        |
| 192.168.1.20      | default gateway                          |
| 192.168.1.254     | host IP address                          |
| 192.168.1.255     | last assignable IP address in the subnet |
| B8-76-3F-7C-57-DF | MAC address                              |

**Answer:**

Explanation:

|                   |
|-------------------|
| 192.168.1.255     |
| 192.168.1.1       |
| 192.168.1.20      |
| 192.168.1.254     |
| B8-76-3F-7C-57-DF |



---

**Question: 400**

---

Which action does the router take as it forwards a packet through the network?

- A. The router replaces the source and destination labels with the sending router interface label as a source and the next hop router label as a destination
- B. The router encapsulates the source and destination IP addresses with the sending router IP address as the source and the neighbor IP address as the destination
- C. The router replaces the original source and destination MAC addresses with the sending router MAC address as the source and neighbor MAC address as the destination
- D. The router encapsulates the original packet and then includes a tag that identifies the source router MAC address and transmit transparently to the destination

---

**Answer: C**

---

Explanation:

---

**Question: 401**

---

Which function is performed by DHCP snooping?

- A. propagates VLAN information between switches
- B. listens to multicast traffic for packet forwarding
- C. provides DDoS mitigation
- D. rate-limits certain traffic

---

**Answer: A**

---

Explanation:

---

**Question: 402**

---

When a client and server are not on the same physical network, which device is used to forward requests and replies between client and server for DHCP?

- A. DHCP relay agent
- B. DHCP server
- C. DHCPDISCOVER
- D. DHCPOFFER

---

**Answer: A**

---

Explanation:

---

**Question: 403**

---

What is a similarity between 1000BASE-LX and 1000BASE-T standards?

- A. Both use the same data-link header and trailer formats
- B. Both cable types support LP connectors
- C. Both cable types support RJ-45 connectors

D. Both support up to 550 meters between nodes

---

**Answer: A**

---

Explanation:

“In computer networking, Gigabit Ethernet (GbE or 1 GigE) is the term applied to transmitting Ethernet frames at a rate of a gigabit per second.”

Both standards use Ethernet framing (same headers and trailers)

---

**Question: 404**

---

An organization secures its network with multi-factor authentication using an authenticator app on employee smartphone. How is the application secured in the case of a user's smartphone being lost or stolen?

- A. The application requires an administrator password to reactivate after a configured Interval.
- B. The application requires the user to enter a PIN before it provides the second factor.
- C. The application challenges a user by requiring an administrator password to reactivate when the smartphone is rebooted.
- D. The application verifies that the user is in a specific location before it provides the second factor.

---

**Answer: B**

---

Explanation:

---

**Question: 405**

---

Which protocol requires authentication to transfer a backup configuration file from a router to a remote server?

- A. DTP



- B. FTP
- C. SMTP
- D. TFTP

---

**Answer: B**

---

Explanation:

---

**Question: 406**

---

Refer to the exhibit.

```
Switch#show etherchannel summary
[output omitted]
```

| Group | Port-channel | Protocol | Ports               |
|-------|--------------|----------|---------------------|
| 10    | Po10 (SU)    | LACP     | Gi0/0 (P) Gi0/1 (P) |
| 20    | Po20 (SU)    | LACP     | Gi0/2 (P) Gi0/3 (P) |

Which two commands were used to create port channel 10? (Choose two )

- ☐ int range g0/0-1  
channel-group 10 mode active
- ☐ int range g0/0-1  
channel-group 10 mode desirable
- ☐ int range g0/0-1  
channel-group 10 mode passive
- ☐ int range g0/0-1  
channel-group 10 mode auto
- ☐ int range g0/0-1  
channel-group 10 mode on

A. Option A

B. Option B

C. Option C

D. Option D

E. Option E

---

**Answer: A,C**

---

Explanation:

---

**Question: 407**

---

Refer to the exhibit.

```
ip arp inspection vlan 2
interface fastethernet 0/1
  switchport mode access
  switchport access vlan 2
```

What is the effect of this configuration?

- A. The switch port interface trust state becomes untrusted
- B. The switch port remains administratively down until the interface is connected to another switch
- C. Dynamic ARP inspection is disabled because the ARP ACL is missing
- D. The switch port remains down until it is configured to trust or untrust incoming packets

---

**Answer: A**

---

Explanation:

Dynamic ARP inspection (DAI) is a security feature that validates ARP packets in a network. It intercepts, logs, and discards ARP packets with invalid IP-to-MAC address bindings. This capability protects the network from certain man-in-the-middle attacks. After enabling DAI, all ports become untrusted ports.

---

### Question: 408

---

An engineer must configure traffic for a VLAN that is untagged by the switch as it crosses a trunk link. Which command should be used?

- A. switchport trunk allowed vlan 10
- B. switchport trunk native vlan 10
- C. switchport mode trunk
- D. switchport trunk encapsulation dot1q

---

**Answer: B**

---

Explanation:

---

**Question: 409**

---

While examining excessive traffic on the network, it is noted that all incoming packets on an interface appear to be allowed even though an IPv4 ACL is applied to the interface.

Which two misconfigurations cause this behavior? (Choose two)

- A. The packets fail to match any permit statement
- B. A matching permit statement is too high in the access test
- C. A matching permit statement is too broadly defined
- D. The ACL is empty
- E. A matching deny statement is too high in the access list

---

**Answer: B,C**

---

Explanation:

---

**Question: 410**

---

Why does a switch flood a frame to all ports?

- A. The frame has zero destination MAC addresses.
- B. The source MAC address of the frame is unknown
- C. The source and destination MAC addresses of the frame are the same
- D. The destination MAC address of the frame is unknown.

---

**Answer: B**

---

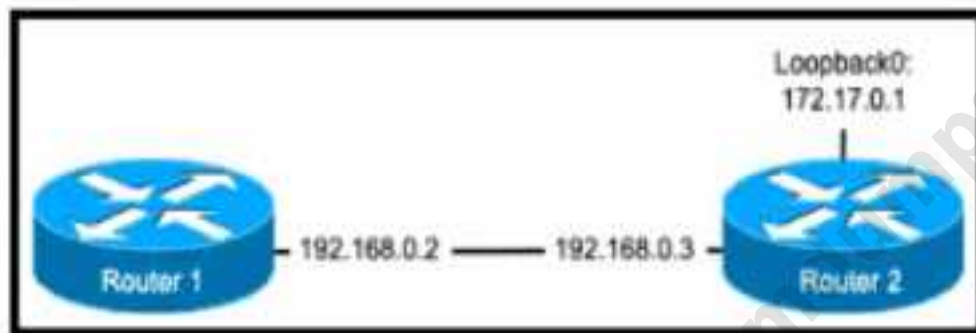
Explanation:

---

**Question: 411**

---

Refer to the exhibit.



The `ntp server 192.168.0.3` command has been configured on router 1 to make it an NTP client of router 2. Which command must be configured on router 2 so that it operates in server-only mode and relies only on its internal clock?

- A. Router2(config)#ntp passive
- B. Router2(config)#ntp server 172.17.0.1
- C. Router2(config)#ntp master 4
- D. Router2(config)#ntp server 192.168.0.2

---

**Answer: B**

---

Explanation:

- To use internal clock of this router, use any configured IP address in any interface of this router.

---

**Question: 412**

---

A network administrator enabled port security on a switch interface connected to a printer. What is the next configuration action in order to allow the port to learn the MAC address of the printer and insert it into the table automatically?

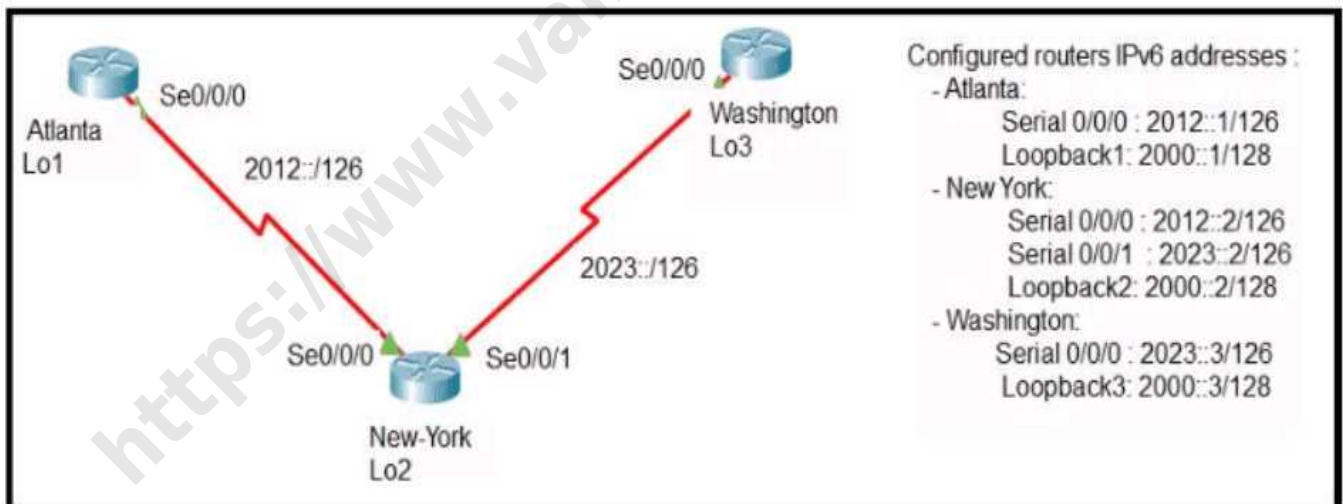
- A. enable dynamic MAC address learning
- B. implement static MAC addressing.
- C. enable sticky MAC addressing
- D. implement auto MAC address learning

**Answer: C**

Explanation:

### Question: 413

Refer to the exhibit.



The New York router is configured with static routes pointing to the Atlanta and Washington sites. Which two tasks must be performed so that the Serial0/0/0 interfaces on the Atlanta and Washington routers can reach one another?

(Choose two.)

- A. Configure the ipv6 route 2012::/126 2023::1 command on the Washington router.

- B. Configure the ipv6 route 2023::/126 2012::1 command on the Atlanta router.
- C. Configure the Ipv6 route 2012::/126 s0/0/0 command on the Atlanta router.
- D. Configure the ipv6 route 2023::/126 2012::2 command on the Atlanta router.
- E. Configure the ipv6 route 2012::/126 2023::2 command on the Washington router.

---

**Answer: D,E**

---

Explanation:

The short syntax of static IPv6 route is:

ipv6 route <destination-IPv6-address> {next-hop-IPv6-address | exit-interface}

---

**Question: 414**

---

DRAG DROP

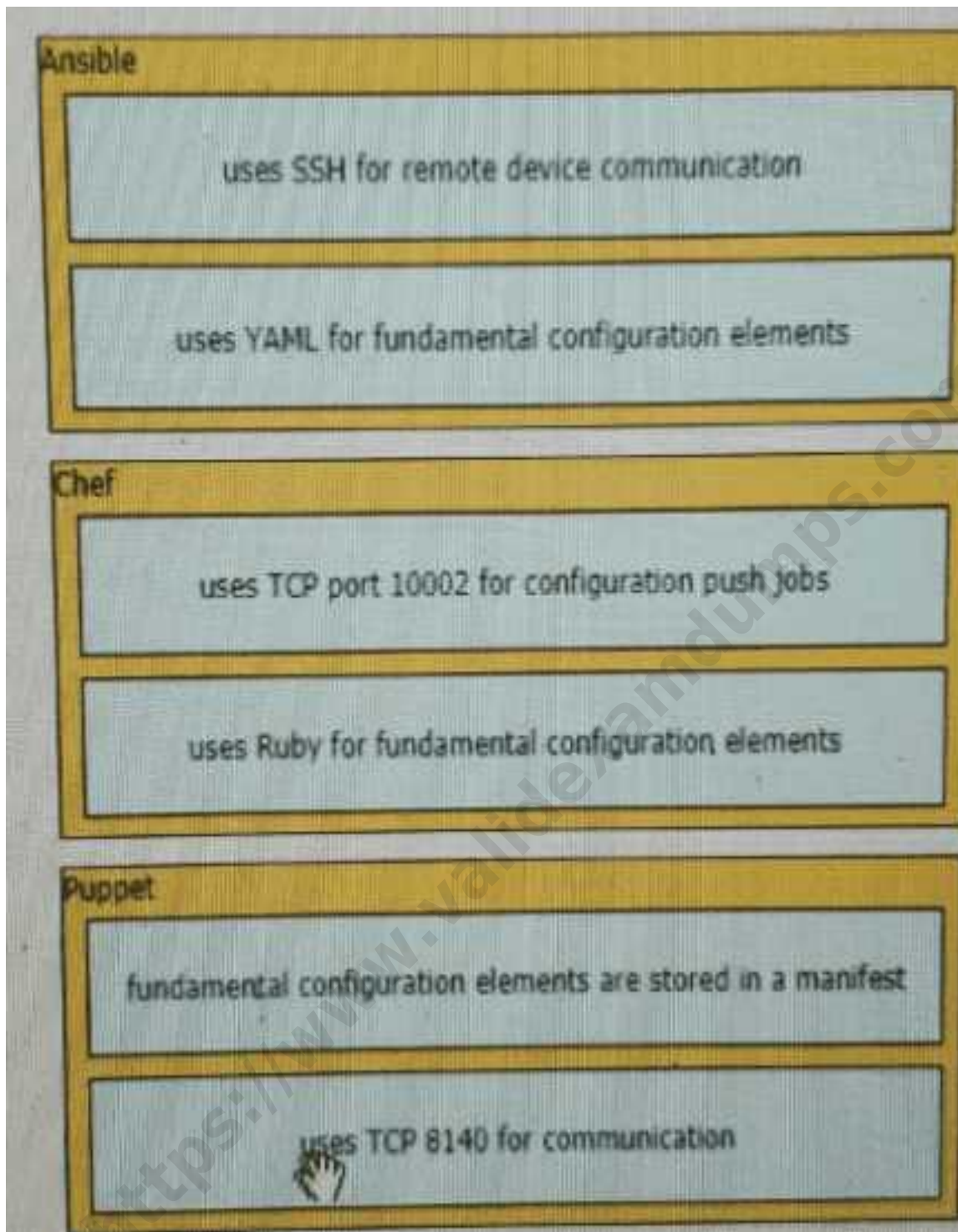
Drag and drop the descriptions from the left onto the configuration-management technologies on the right.

|                                                             |         |
|-------------------------------------------------------------|---------|
| fundamental configuration elements are stored in a manifest | Ansible |
| uses TCP port 10002 for configuration push jobs             |         |
| uses Ruby for fundamental configuration elements            | Chef    |
| uses SSH for remote device communication                    |         |
| uses TCP 8140 for communication                             | Puppet  |
| uses YAML for fundamental configuration elements            |         |

Explanation:

Answer:





Ansible:

- uses SSH for remote device communication
- uses YAML for fundamental configuration elements

Chef:

- uses TCP port 10002 for configuration push jobs

- uses Ruby for fundamental configuration elements

Puppet:

- fundamental configuration elements are stored in a manifest
- uses TCP 8140 for communication

The focus of Ansible is to be streamlined and fast, and to require no node agent installation.

Thus, Ansible performs all functions over SSH. Ansible is built on Python, in contrast to the Ruby foundation of Puppet and Chef.

TCP port 10002 is the command port. It may be configured in the Chef Push Jobs configuration file .

This port allows Chef Push Jobs clients to communicate with the Chef Push Jobs server.

Puppet is an open-source configuration management solution, which is built with Ruby and offers custom Domain Specific Language (DSL) and Embedded Ruby (ERB) templates to create custom Puppet language files, offering a declarative-paradigm programming approach.

A Puppet piece of code is called a manifest, and is a file with .pp extension.

---

**Question: 415**

---

What are two benefits of using the PortFast feature? (Choose two )

- A. Enabled interfaces are automatically placed in listening state
- B. Enabled interfaces come up and move to the forwarding state immediately
- C. Enabled interfaces never generate topology change notifications.
- D. Enabled interfaces that move to the learning state generate switch topology change notifications
- E. Enabled interfaces wait 50 seconds before they move to the forwarding state

---

**Answer: A,B**

---

Explanation:

---

**Question: 416**

---

What are two characteristics of an SSID? (Choose Two)

- A. It can be hidden or broadcast in a WLAN
- B. It uniquely identifies an access point in a WLAN
- C. It uniquely identifies a client in a WLAN
- D. It is at most 32 characters long.
- E. IT provides secured access to a WLAN

---

**Answer: B,E**

---

Explanation:

---

**Question: 417**

---

What is the purpose of an SSID?

- A. It provides network security
- B. It differentiates traffic entering access points
- C. It identifies an individual access point on a WLAN
- D. It identifies a WLAN

---

**Answer: D**

---

Explanation:

"In IEEE 802.11 wireless local area networking standards (including Wi-Fi), a service set is a group of wireless network devices which share a service set identifier (SSID)

...

A service set forms a logical network of nodes operating with shared link-layer networking parameters; they form one logical network segment.”

---

**Question: 418**

---

DRAG DROP

Drag and drop the AAA terms from the left onto the description on the right.

|                |                            |
|----------------|----------------------------|
| accounting     | tracks activity            |
| authentication | updates session attributes |
| authorization  | verifies access rights     |
| CoA            | verifies identity          |

---

**Answer:**

---

Explanation:

1-1, 2-4, 3-3, 4-2

---

**Question: 419**

---

Which plane is centralized by an SDN controller?

- A. management-plane
- B. control-plane
- C. data-plane
- D. services-plane

---

**Answer: B**

---

Explanation:

---

**Question: 420**

---

Refer to the exhibit.

```
access-list 101 permit ospf any any
access-list 101 permit tcp any any eq 179
access-list 101 permit tcp any eq 179 any
access-list 101 permit gre any any
access-list 101 permit esp any any

access-list 101 deny ospf any any
access-list 101 permit tcp 10.1.1.0 0.0.0.255 172.16.1.0 0.0.0.255 eq telnet
access-list 101 permit udp 10.1.1.0 0.0.0.255 172.16.1.0 0.0.0.255 eq 500
access-list 101 permit udp 10.1.1.0 0.0.0.255 172.16.1.0 0.0.0.255 eq 4500
access-list 101 deny ip any any log

interface Ethernet0/0
 ip address 10.1.1.25 255.255.255.0
 ip access-group 101 in
```

A network administrator has been tasked with securing VTY access to a router. Which access-list entry accomplishes this task?

- A. access-list 101 permit tcp 10.1.10 0.0.0.255 172.16.10 0.0.0.255 eq ssh
- B. access-list 101 permit tcp 10.11.0 0.0.0.255 172.16.10 0.0.0.255 eq scp
- C. access-list 101 permit tcp 10.11.0 0.0.0.255 172.16.10 0.0.0.255 eq telnet
- D. access-list 101 permit tcp 10.1.10 0.0.0.255 172.16.10 0.0.0.255 eq https

---

**Answer: A**

---

Explanation:

---

**Question: 421**

---

Which action is taken by a switch port enabled for PoE power classification override?

- A. When a powered device begins drawing power from a PoE switch port a syslog message is generated
- B. As power usage on a PoE switch port is checked data flow to the connected device is temporarily paused
- C. If a switch determines that a device is using less than the minimum configured power it assumes the device has failed and disconnects
- D. Should a monitored port exceeds the maximum administrative value for power, the port is shutdown and err-disabled

---

**Answer: D**

---

Explanation:

Reference: [https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst6500/ios/12-2SX/configuration/guide/book/power\\_over\\_ethernet.pdf](https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst6500/ios/12-2SX/configuration/guide/book/power_over_ethernet.pdf)

PoE monitoring and policing compares the power consumption on ports with the administrative maximum value (either a configured maximum value or the port's default value). If the power consumption on a monitored port exceeds the administrative maximum value, the following actions occur:– A syslog message is issued.– The monitored port is shut down and error-disabled.– The allocated power is freed.

---

**Question: 422**

---

R1 has learned route 192.168.12.0/24 via IS-IS, OSPF, RIP, and Internal EIGRP Under normal operating conditions, which routing protocol is installed in the routing table?

- A. IS-IS
- B. RIP
- C. Internal EIGRP

D. OSPF

---

**Answer: C**

---

Explanation:

With the same route (prefix), the router will choose the routing protocol with lowest Administrative Distance (AD) to install into the routing table. The AD of Internal EIGRP (90) is lowest so it would be chosen. The table below lists the ADs of popular routing protocols.

| Route Source        | Administrative Distance |
|---------------------|-------------------------|
| Directly Connected  | 0                       |
| Static              | 1                       |
| EIGRP               | 90                      |
| EIGRP Summary route | 5                       |
| OSPF                | 110                     |
| RIP                 | 120                     |

Note: The AD of IS-IS is 115. The "EIGRP" in the table above is "Internal EIGRP". The AD of "External EIGRP" is 170. An EIGRP external route is a route that was redistributed into EIGRP.

---

**Question: 423**

---

What are two reasons that cause late collisions to increment on an Ethernet interface? (Choose two)

- A. when the sending device waits 15 seconds before sending the frame again
- B. when the cable length limits are exceeded
- C. when one side of the connection is configured for half-duplex
- D. when Carrier Sense Multiple Access/Collision Detection is used

E. when a collision occurs after the 32nd byte of a frame has been transmitted

---

**Answer: B,C**

---

Explanation:

The usual possible causes are full-duplex/half-duplex mismatch, exceeded Ethernet cable length limits, or defective hardware such as incorrect cabling, non-compliant number of hubs in the network, or a bad NIC.

---

**Question: 424**

---

Which QoS tool is used to optimize voice traffic on a network that is primarily intended for data traffic?

- A. FIFO
- B. WFQ
- C. PQ
- D. WRED

---

**Answer: C**

---

Explanation:

---

**Question: 425**

---

What are two characteristics of a controller-based network? (Choose two)

- A. The administrator can make configuration updates from the CLI
- B. It uses northbound and southbound APIs to communicate between architectural layers
- C. It moves the control plane to a central point.



- D. It decentralizes the control plane, which allows each device to make its own forwarding decisions
- E. It uses Telnet to report system issues.

---

**Answer: B,C**

---

Explanation:

---

**Question: 426**

---

What is the benefit of configuring PortFast on an interface?

- A. After the cable is connected, the interface uses the fastest speed setting available for that cable type
- B. After the cable is connected, the interface is available faster to send and receive user data
- C. The frames entering the interface are marked with higher priority and then processed faster by a switch.
- D. Real-time voice and video frames entering the interface are processed faster

---

**Answer: B**

---

Explanation:

---

**Question: 427**

---

An engineer configures interface Gi1/0 on the company PE router to connect to an ISP Neighbor discovery is disabled

```
interface Gi1/0
description HQ_DC3978-87297
duplex full
speed 100
negotiation auto
lldp transmit
lldp receive
```

Which action is necessary to complete the configuration if the ISP uses third-party network devices?

- A. Enable LLDP globally
- B. Disable autonegotiation
- C. Disable Cisco Discovery Protocol on the interface
- D. Enable LLDP-MED on the ISP device

---

**Answer: D**

---

Explanation:

---

**Question: 428**

---

An implementer is preparing hardware for virtualization to create virtual machines on a host. What is needed to provide communication between hardware and virtual machines?

- A. hypervisor
- B. router
- C. straight cable
- D. switch

---

**Answer: A**

---

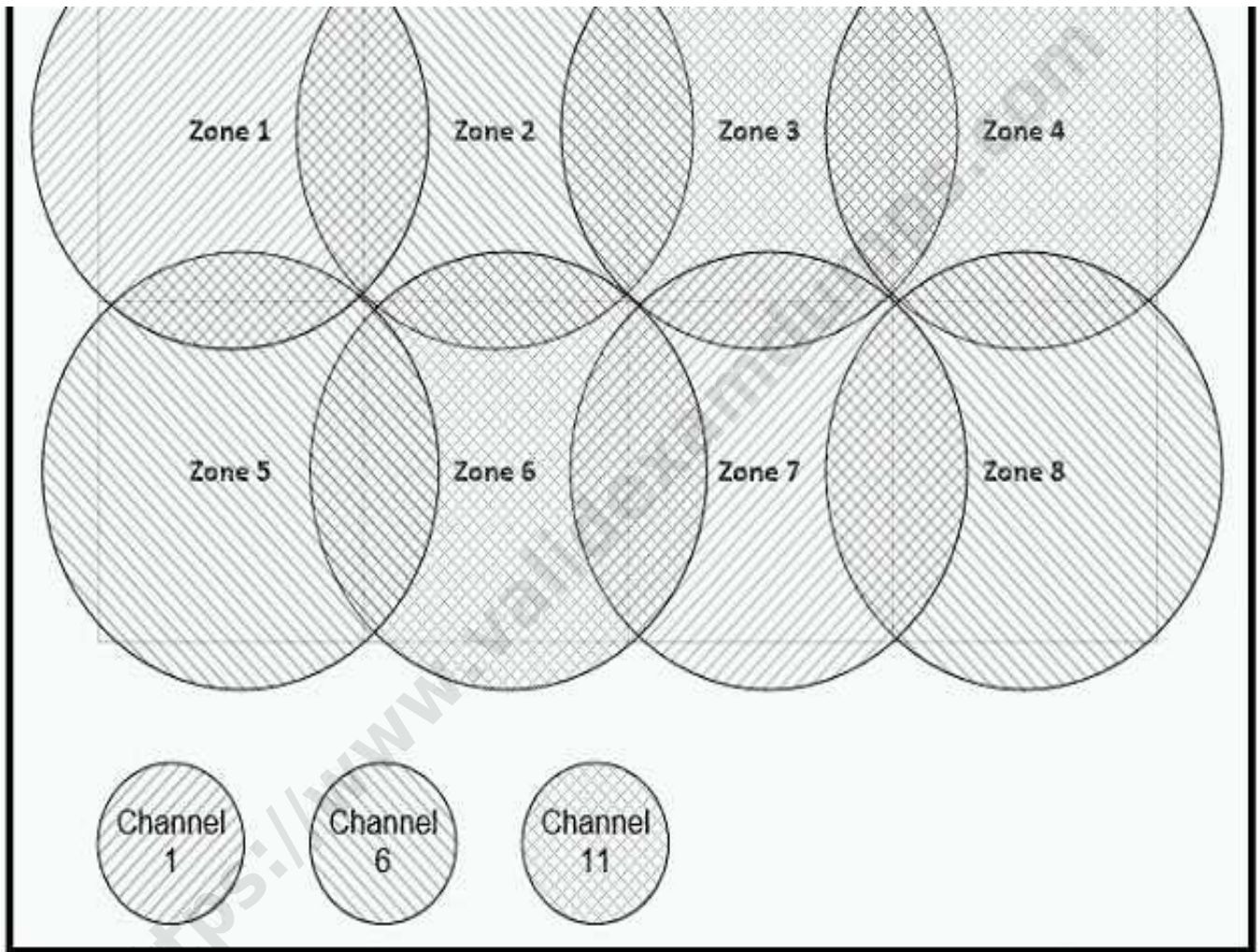
Explanation:

---

**Question: 429**

---

Refer to the exhibit.



Between which zones do wireless users expect to experience intermittent connectivity?

- A. between zones 1 and 2
- B. between zones 2 and 5
- C. between zones 3 and 4
- D. between zones 3 and 6

---

**Answer: D**

---

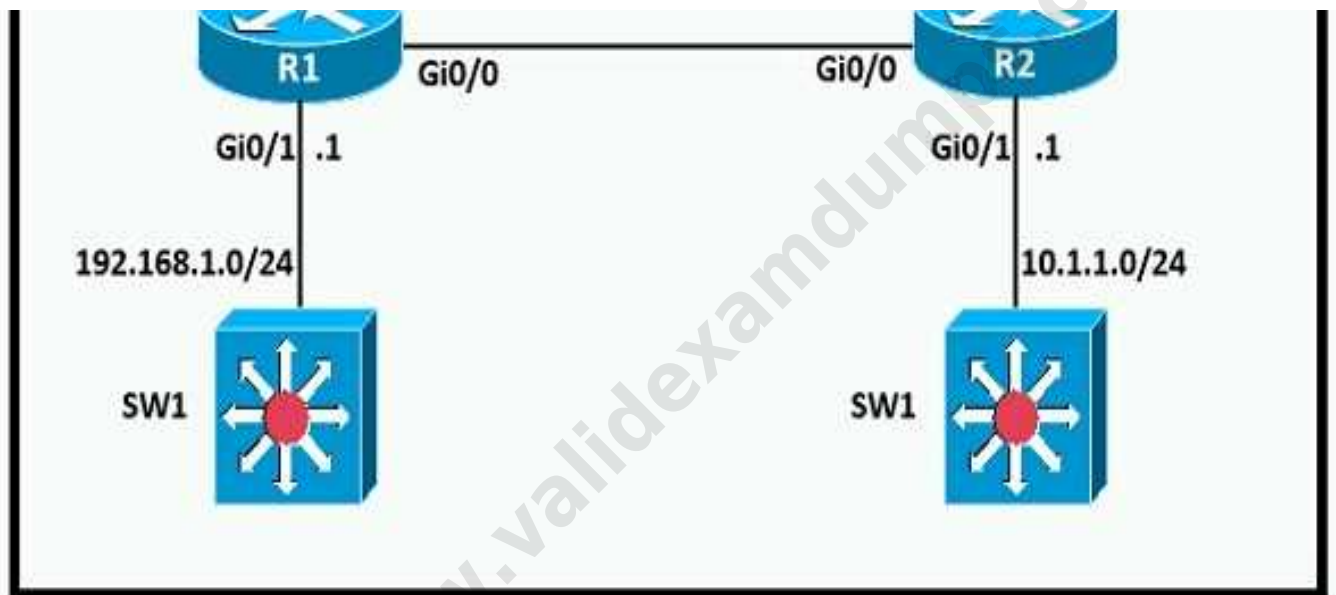
Explanation:

---

**Question: 430**

---

Refer to the exhibit.



A network engineer is in the process of establishing IP connectivity between two sites. Routers R1 and R2 are partially configured with IP addressing. Both routers have the ability to access devices on their respective LANs. Which command set configures the IP connectivity between devices located on both LANs in each site?

- ☐ R1  
ip route 192.168.1.0 255.255.255.0 GigabitEthernet0/0  
R2  
ip route 10.1.1.1 255.255.255.0 GigabitEthernet0/0
- ☐ R1  
ip route 0.0.0.0 0.0.0.0 209.165.200.225  
R2  
ip route 0.0.0.0 0.0.0.0 209.165.200.226
- ☐ R1  
ip route 192.168.1.1 255.255.255.0 GigabitEthernet0/1  
R2  
ip route 10.1.1.1 255.255.255.0 GigabitEthernet0/1
- ☐ R1  
ip route 0.0.0.0 0.0.0.0 209.165.200.226  
R2  
ip route 0.0.0.0 0.0.0.0 209.165.200.225

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: D**

---

Explanation:

---

**Question: 431**

---

DRAG DROP

Drag and drop the lightweight access point operation modes from the left onto the descriptions on the right

|                  |                                                                                          |
|------------------|------------------------------------------------------------------------------------------|
| bridge mode      | allows the access point to communicate with the WLC over a WAN link                      |
| local mode       | allows for packet captures of wireless traffic                                           |
| monitor mode     | rogue detector mode                                                                      |
| Flexconnect mode | preferred for connecting access points in a mesh environment                             |
|                  | receive only mode which acts as a dedicated sensor for RFID and IDS                      |
| sniffer mode     | transmits normally on one channel and monitors other channels for noise and interference |

---

**Answer:**

---

Explanation:

|                     |
|---------------------|
| local mode          |
| sniffer mode        |
| rogue detector mode |
| bridge mode         |
| Flexconnect mode    |
| monitor mode        |

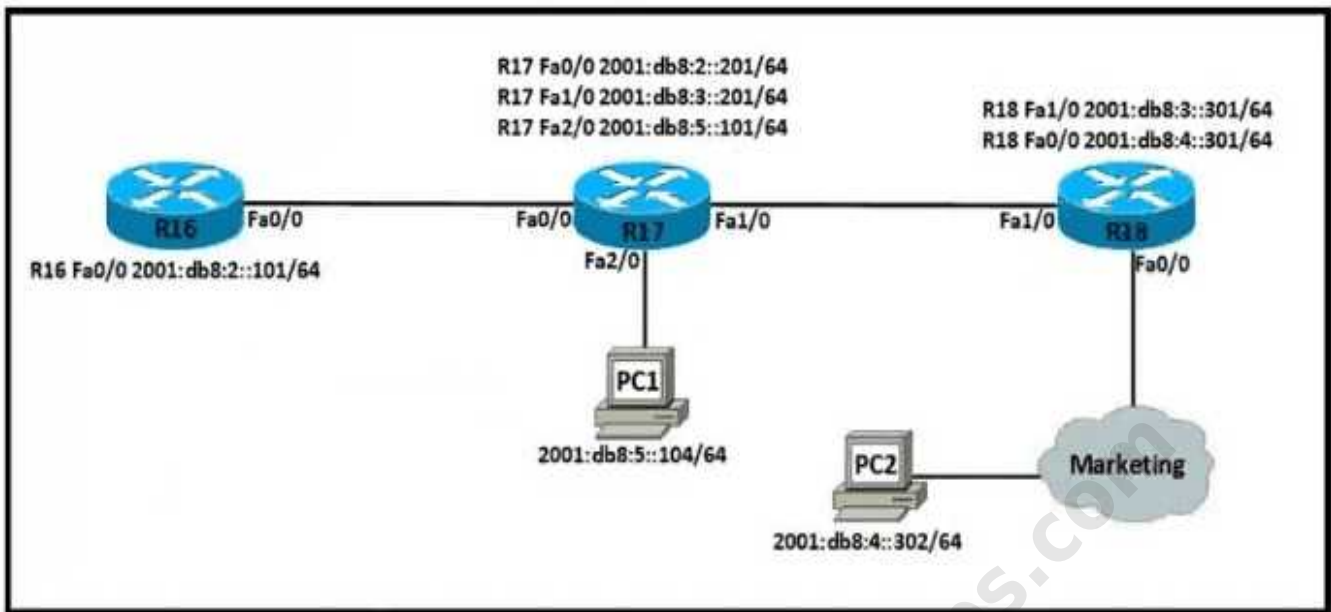
---

**Question: 432**

---

Refer to the exhibit.





Which IPv6 configuration is required for R17 to successfully ping the WAN interface on R18?

A)

```

R17#
!
no ip domain lookup
ip cef
!
interface FastEthernet0/0
no ip address
duplex auto
speed auto
ipv6 address 2001:DB8:3::201/64
!
interface FastEthernet1/0
no ip address
duplex auto
speed auto
ipv6 address 2001:DB8:2::201/64
!
no cdp log mismatch duplex
ipv6 route 2001:DB8:4::/64 2001:DB8:5::101

```

B)



```
R17#
!
no ip domain lookup
ip cef
ipv6 unicast-routing
!
interface FastEthernet0/0
no ip address
duplex auto
speed auto
ipv6 address 2001:DB8:2::201/64
!
interface FastEthernet1/0
no ip address
duplex auto
speed auto
ipv6 address 2001:DB8:3::201/64
!
no cdp log mismatch duplex
ipv6 route 2001:DB8:4::/64 2001:DB8:3::301
```

C)

```
R17#
!
no ip domain lookup
ip cef
ipv6 cef
!
interface FastEthernet0/0
no ip address
duplex auto
speed auto
ipv6 address 2001:DB8:2::201/64
!
interface FastEthernet1/0
no ip address
duplex auto
speed auto
ipv6 address 2001:DB8:3::201/64
!
no cdp log mismatch duplex
ipv6 route 2001:DB8:4::/64 2001:DB8:4::302
```

D)

```
R17#  
!  
no ip domain lookup  
ip cef  
ipv6 unicast-routing  
!  
interface FastEthernet0/0  
no ip address  
duplex auto  
speed auto  
ipv6 address 2001:DB8:2::201/64  
!  
interface FastEthernet1/0  
no ip address  
duplex auto  
speed auto  
ipv6 address 2001:DB8:3::201/64  
!  
no cdp log mismatch duplex  
ipv6 route 2001:DB8:4::/64 2001:DB8:2::201
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: B**

---

Explanation:

ipv6 unicast-routing statement included (IPv6 is enabled on the router).

Compared to the exhibit, Fa0/0 and Fa0/1 have correct configurations.

The route to subnet 2001:db8:4::/64 points to R18's Fa1/0 (correct next-hop).

---

**Question: 433**

---

Which type of organization should use a collapsed-core architecture?

- A. large and requires a flexible, scalable network design
- B. large and must minimize downtime when hardware fails
- C. small and needs to reduce networking costs currently
- D. small but is expected to grow dramatically in the near future

---

**Answer: C**

---

Explanation:

A collapsed-core architecture is a limited investment for a small company, and may be efficient and productive for a limited time.

---

**Question: 434**

---

What is a capability of FTP in network management operations?

- A. encrypts data before sending between data resources
- B. devices are directly connected and use UDP to pass file information
- C. uses separate control and data connections to move files between server and client
- D. offers proprietary support at the session layer when transferring data

---

**Answer: C**

---

Explanation:

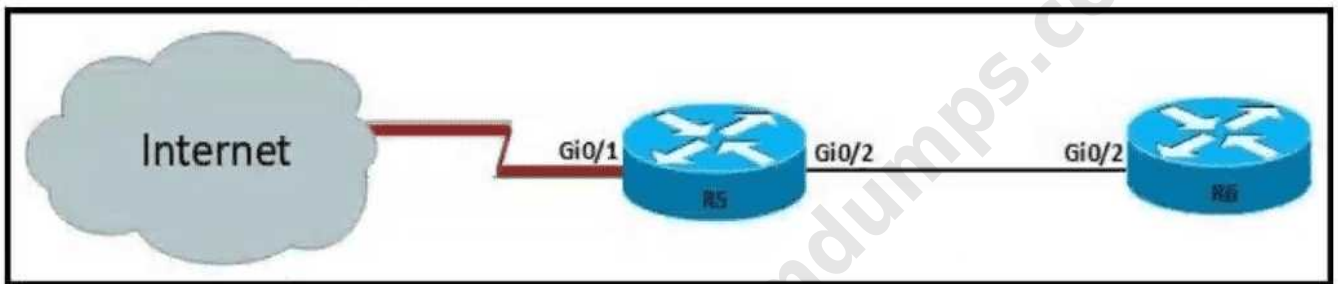
The File Transfer Protocol (FTP) is a standard communication protocol used for the transfer of computer files from a server to a client on a computer network. FTP is built on a client–server model architecture using separate control and data connections between the client and the server.

---

**Question: 435**

---

Refer to the exhibit.



For security reasons, automatic neighbor discovery must be disabled on the R5 Gi0/1 interface. These tasks must be completed:

- Disable all neighbor discovery methods on R5 interface Gi0/1.
- Permit neighbor discovery on R5 interface Gi0/2.
- Verify there are no dynamically learned neighbors on R5 interface Gi0/1.
- Display the IP address of R6's interface Gi0/2.

Which configuration must be used?

- ☒ R5(config)#int Gi0/1  
R5(config-if)#no cdp run  
R5(config-if)#exit  
R5(config)#lldp run  
R5(config)#cdp enable  
R5#sh cdp neighbor  
R5#sh lldp neighbor
- ☐ R5(config)#int Gi0/1  
R5(config-if)#no cdp enable  
R5(config-if)#exit  
R5(config)#no lldp run  
R5(config)#cdp run  
R5#sh cdp neighbor  
R5#sh lldp neighbor
- ☐ R5(config)#int Gi0/1  
R5(config-if)#no cdp enable  
R5(config-if)#exit  
R5(config)#no lldp run  
R5(config)#cdp run  
R5#sh cdp neighbor detail  
R5#sh lldp neighbor
- ☐ R5(config)#int Gi0/1  
R5(config-if)#no cdp enable  
R5(config-if)#exit  
R5(config)#lldp run  
R5(config)#no cdp run  
R5#sh cdp neighbor detail  
R5#sh lldp neighbor

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: C**

---

Explanation:

---

**Question: 436**

---

What is a function of a Layer 3 switch?

- A. move frames between endpoints limited to IP addresses
- B. transmit broadcast traffic when operating in Layer 3 mode exclusively
- C. forward Ethernet frames between VLANs using only MAC addresses
- D. flood broadcast traffic within a VLAN

---

**Answer: A**

---

Explanation:

---

**Question: 437**

---

Which type of API allows SDN controllers to dynamically make changes to the network?

- A. northbound API
- B. REST API
- C. SOAP API
- D. southbound API

---

**Answer: D**

---

Explanation:

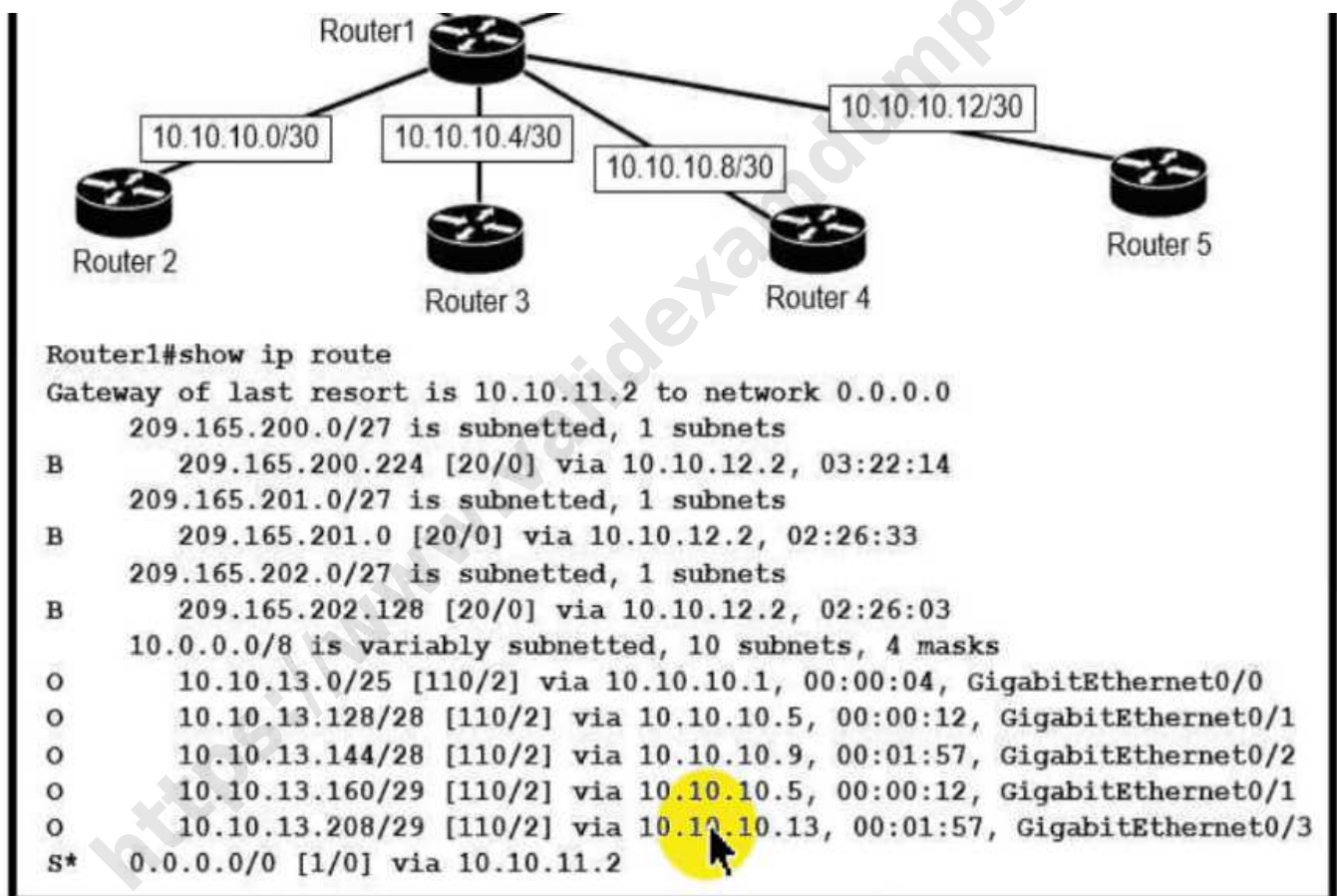
Cisco overview doc for SDN here:

[https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Data\\_Center/VMDC/SDN/SDN.html](https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Data_Center/VMDC/SDN/SDN.html)

Topic 3, Exam Pool C

### Question: 438

Refer to the exhibit.



Which next-hop IP address does Routed use for packets destined to host 10 10.13.158?

- A. 10.10.10.5
- B. 10.10.11.2
- C. 10.10.12.2
- D. 10.10.10.9

---

**Answer: A**

---

Explanation:

---

**Question: 439**

---

A Cisco engineer must configure a single switch interface to meet these requirements

- accept untagged frames and place them in VLAN 20
- accept tagged frames in VLAN 30 when CDP detects a Cisco IP phone

Which command set must the engineer apply?

A)

```
switchport mode dynamic desirable
switchport access vlan 20
switchport trunk allowed vlan 30
switchport voice vlan 30
```

B)

```
switchport mode dynamic auto
switchport trunk native vlan 20
switchport trunk allowed vlan 30
switchport voice vlan 30
```

C)

```
switchport mode access
switchport access vlan 20
switchport voice vlan 30
```

D)



```
switchport mode trunk  
switchport access vlan 20  
switchport voice vlan 30
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: C**

---

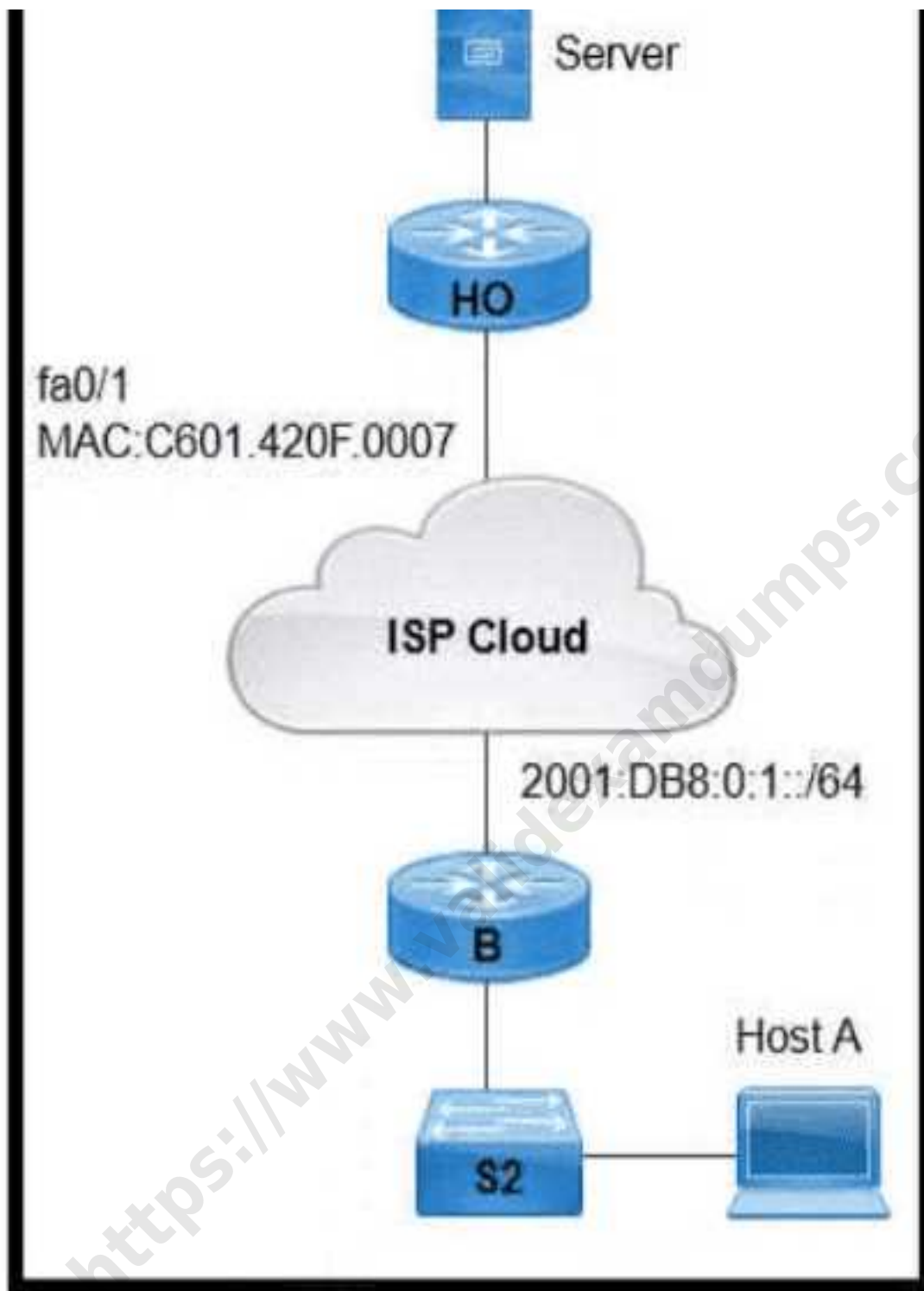
Explanation:

---

**Question: 440**

---

Refer to the exhibit.



An engineer is configuring the HO router. Which IPv6 address configuration must be applied to the router fa0/1 interface for the router to assign a unique 64-bit IPv6 address to itself?

- A. ipv6 address 2001:DB8:0:1:C601:42FF:FE0F:7/64
- B. ipv6 address 2001:DB8:0:1:C601:42FE:800F:7/64
- C. ipv6 address 2001 :DB8:0:1:FFFF:C601:420F:7/64
- D. iov6 address 2001 :DB8:0:1:FE80:C601:420F:7/64

---

**Answer: A**

---

Explanation:

---

**Question: 441**

---

Which WLC management connection type is vulnerable to man-in-the-middle attacks?

- A. SSH
- B. HTTPS
- C. Telnet
- D. console

---

**Answer: C**

---

Explanation:

---

**Question: 442**

---

Which action is taken by the data plane within a network device?

- A. forwards traffic to the next hop
- B. constructs a routing table based on a routing protocol
- C. provides CLI access to the network device
- D. looks up an egress interface in the forwarding information base

---

**Answer: A**

---

Explanation:

---

**Question: 443**

---

What is a function of a Next-Generation IPS?

- A. makes forwarding decisions based on learned MAC addresses
- B. serves as a controller within a controller-based network
- C. integrates with a RADIUS server to enforce Layer 2 device authentication rules
- D. correlates user activity with network events

---

**Answer: D**

---

Explanation:

---

**Question: 444**

---

Which characteristic differentiates the concept of authentication from authorization and accounting?

- A. user-activity logging
- B. service limitations
- C. consumption-based billing
- D. identity verification

---

**Answer: D**

---

Explanation:

---

**Question: 445**

---

Which value is the unique identifier that an access point uses to establish and maintain wireless connectivity to wireless network devices?

- A. VLANID
- B. SSID
- C. RFID

D. WLANID

---

**Answer: B**

---

Explanation:

---

**Question: 446**

---

An engineer is configuring remote access to a router from IP subnet 10.139.58.0/28. The domain name, crypto keys, and SSH have been configured. Which configuration enables the traffic on the destination router?

A)

```
interface FastEthernet0/0
  ip address 10.122.49.1 255.255.255.240
  access-group 120 in

ip access-list extended 120
  permit tcp 10.139.58.0 255.255.255.248 any eq 22
```

B)

```
interface FastEthernet0/0
  ip address 10.122.49.1 255.255.255.252
  ip access-group 110 in

ip access-list extended 110
  permit tcp 10.139.58.0 0.0.0.15 host 10.122.49.1 eq 22
```

C)

```
interface FastEthernet0/0
  ip address 10.122.49.1 255.255.255.248
  ip access-group 10 in

ip access-list standard 10
  permit udp 10.139.58.0 0.0.0.7 host 10.122.49.1 eq 22
```

D)

```
interface FastEthernet0/0
  ip address 10.122.49.1 255.255.255.252
  ip access-group 105 in

ip access-list standard 105
  permit tcp 10.139.58.0 0.0.0.7 eq 22 host 10.122.49.1
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: B**

---

Explanation:

---

**Question: 447**

---

Which QoS per-hop behavior changes the value of the ToS field in the IPv4 packet header?

- A. shaping
- B. classification
- C. policing
- D. marking

---

**Answer: D**

---

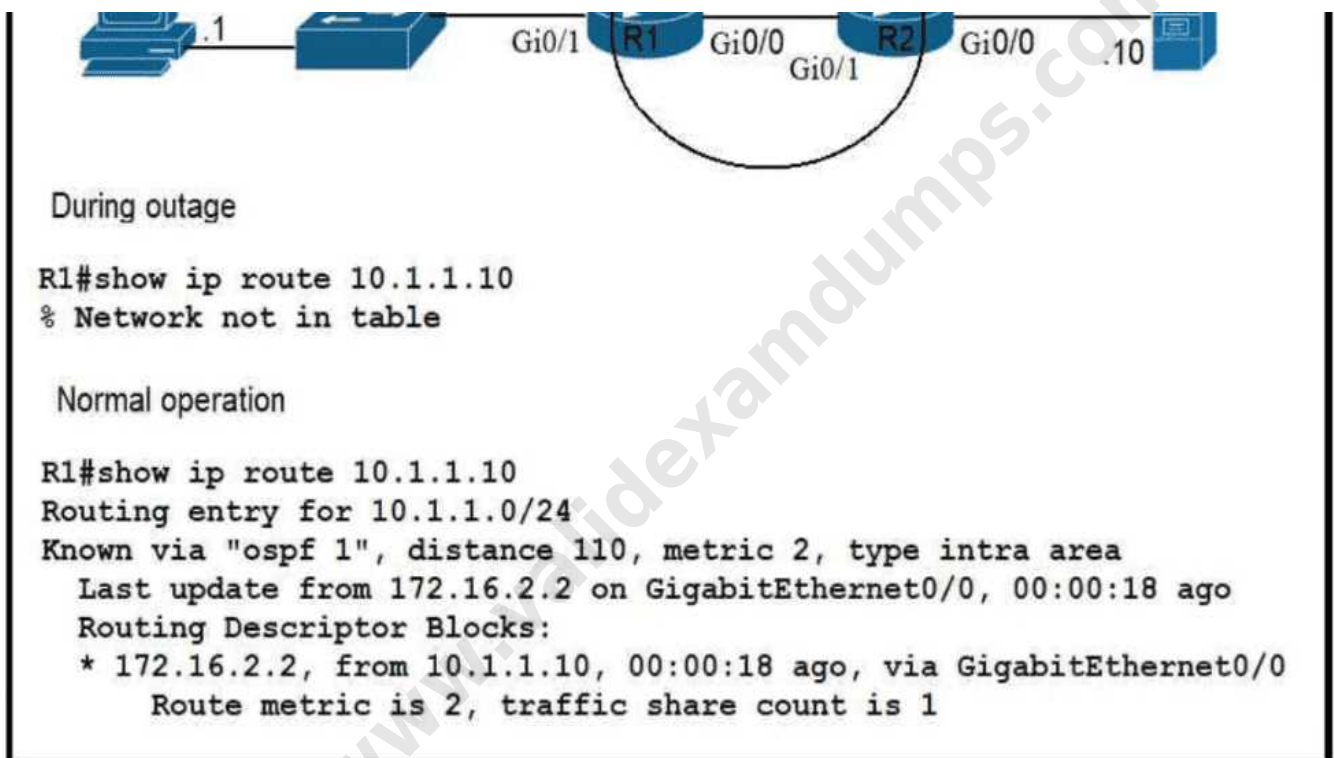
Explanation:

---

**Question: 448**

---

Refer to the exhibit.



Which route must be configured on R1 so that OSPF routing is used when OSPF is up, but the server is still reachable when OSPF goes down?

- A. ip route 10.1.1.10 255.255.255.255 172.16.2.2 100
- B. ip route 10.1.1.0 255.255.255.0 gi0/1 125
- C. ip route 10.1.1.0 255.255.255.0 172.16.2.2 100
- D. ip route 10.1.1.10 255.255.255.255 gi0/0 125

---

**Answer: D**

---

Explanation:

**Question: 449****DRAG DROP**

Drag and drop the statements about networking from the left onto the corresponding networking types on the right.

|                                                                                         |                                                                           |
|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| This type allows better control over how networks work and how networks are configured. | <b>Traditional Networking</b><br><div></div> <div></div>                  |
| This type enables networks to integrate with applications through APIs.                 |                                                                           |
| New devices are configured using the physical infrastructure.                           | <b>Controller-Based Networking</b><br><div></div> <div></div> <div></div> |
| This type provisions resources from a centralized location.                             |                                                                           |
| This type requires a distributed control plane.                                         |                                                                           |

Explanation:

**Answer:**



### Traditional Networking

New devices are configured using the physical infrastructure.

This type provisions resources from a centralized location.

### Controller-Based Networking

This type requires a distributed control plane.

This type enables networks to integrate with applications through APIs.

This type allows better control over how networks work and how networks are configured.

---

**Question: 450**

---

Refer to the exhibit.

The screenshot displays a configuration window for a wireless network. At the top, 'Fast Transition' is set to 'Disable'. Below it, the 'Protected Management Frame' section shows 'PMF' as 'Disabled'. The 'WPA+WPA2 Parameters' section includes 'WPA Policy' (unchecked), 'WPA2 Policy' (checked), 'WPA2 Encryption' (checked) with 'AES' selected over 'TKIP', 'CCMP256', 'GCMP128', and 'GCMP256', and 'OSN Policy' (unchecked). The 'Authentication Key Management' section, which is expanded, shows '802.1X', 'CCKM', and 'FT 802.1X' as 'Enable' (unchecked), while 'PSK' and 'FT PSK' are 'Enable' (checked).

Users need to connect to the wireless network with IEEE 802. 11r-compatible devices. The connection must be maintained as users travel between floors or to other areas in the building What must be the configuration of the connection?

- A. Select the WPA Policy option with the CCKM option.
- B. Disable AES encryption.
- C. Enable Fast Transition and select the FT 802.1x option.
- D. Enable Fast Transition and select the FT PSK option.

---

**Answer: D**

---

Explanation:

---

**Question: 451**

---

Refer to the exhibit.

```
Hardware is ISR4331-3xlGE, address is 5486.bc25.1f70 (bia 5486.bc25.1f70)
Description: << WAN Link >>
Internet address is 192.0.2.2/30
MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive not supported
Full Duplex, 1000Mbps, link type is auto, media type is RJ45
output flow-control is off, input flow-control is off
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:00, output 00:00:11, output hang never
Last clearing of "show interface" counters never
Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 7000 bits/sec, 4 packets/sec
5 minute output rate 4000 bits/sec, 4 packets/sec
  22579370 packets input, 8825545968 bytes, 0 no buffer
  Received 67 broadcasts (0 IP multicasts)
  0 runs, 0 giants, 0 throttles
  3612699 input errors, 3612699 CRC, 0 frame, 0 overrun, 0 ignored
  0 watchdog, 10747057 multicast, 0 pause input
  12072167 packets output, 1697953637 bytes, 0 underruns
  0 output errors, 0 collisions, 1 interface resets
  6 unknown protocol drops
  0 babbles, 0 late collision, 0 deferred
  5 lost carrier, 0 no carrier, 0 pause output
  0 output buffer failures, 0 output buffers swapped out
```

What is a reason for poor performance on the network interface?

- A. The interface is receiving excessive broadcast traffic.
- B. The cable connection between the two devices is faulty.
- C. The interface is operating at a different speed than the connected device.
- D. The bandwidth setting of the interface is misconfigured

---

**Answer: A**

---

Explanation:

---

**Question: 452**

---

Refer to the exhibit.

```
Switch2# show lldp
Global LLDP Information
  Status: ACTIVE
  LLDP advertisements are sent every 30 seconds
  LLDP hold time advertised is 120 seconds
  LLDP interface reinitialization delay is 2 seconds
```

A network engineer must update the configuration on Switch2 so that it sends LLDP packets every minute and the information sent via LLDP is refreshed every 3 minutes. Which configuration must the engineer apply?

A)

```
Switch2(config)#lldp timer 60
Switch2(config)#lldp holdtime 180
```

B)

```
Switch2(config)#lldp timer 60
Switch2(config)#lldp tlv-select 180
```

C)

```
Switch2(config)#lldp timer 1
Switch2(config)#lldp holdtime 3
```

D)

```
Switch2(config)#lldp timer 1
Switch2(config)#lldp tlv-select 3
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: A**

---

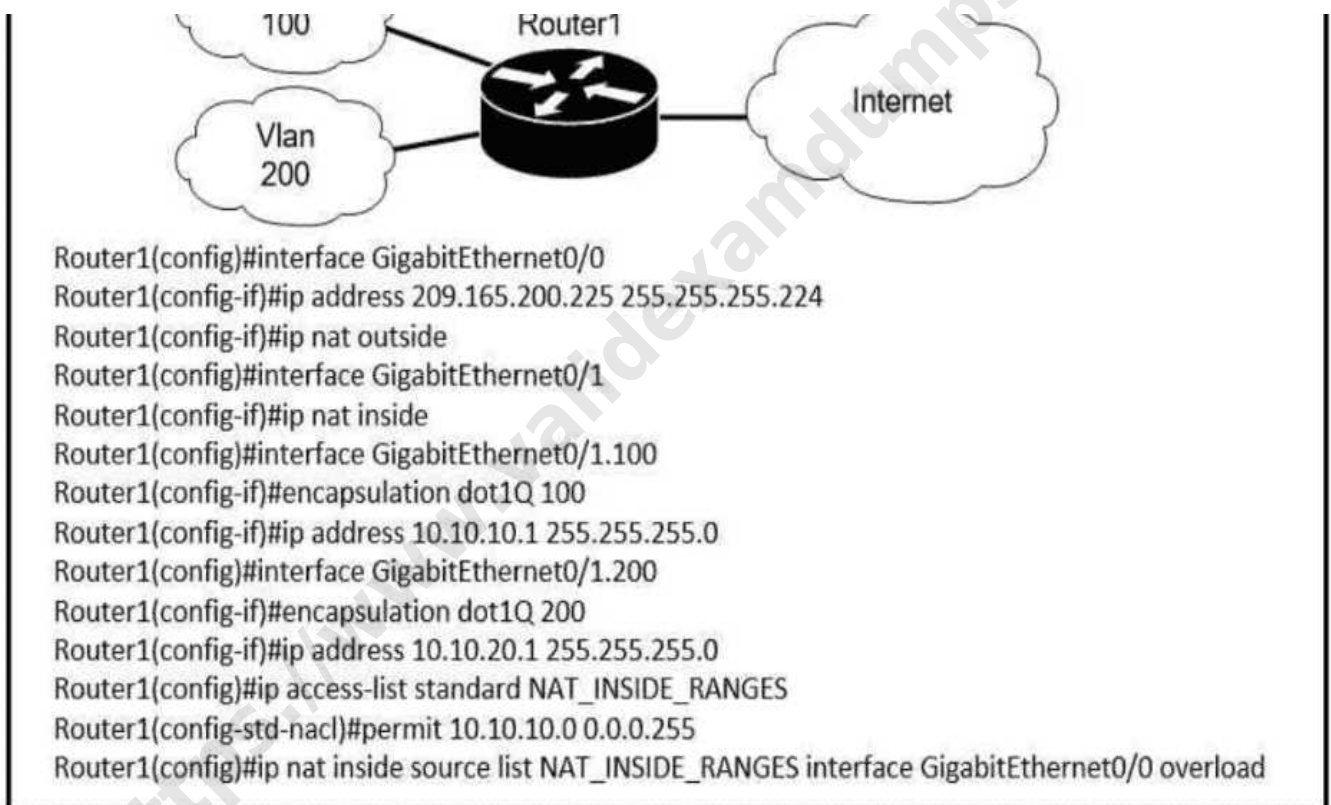
Explanation:

---

**Question: 453**

---

Refer to the exhibit.



Users on existing VLAN 100 can reach sites on the Internet. Which action must the administrator take to establish connectivity to the Internet for users in VLAN 200?

- A. Define a NAT pool on the router.
- B. Configure static NAT translations for VLAN 200.
- C. Configure the ip nat outside command on another interface for VLAN 200.
- D. Update the NAT INSIDE RANGES ACL

---

**Answer: B**

---

Explanation:

---

**Question: 454**

---

Refer to the exhibit.

|       |                     |     |      |
|-------|---------------------|-----|------|
| RIP   | 10.1.1.16/28[120/5] | via | F0/0 |
| OSPF  | 10.1.1.0/24[110/30] | via | F0/1 |
| OSPF  | 10.1.1.0/24[110/40] | via | F0/2 |
| EIGRP | 10.1.0.0/26[90/20]  | via | F0/3 |
| EIGRP | 10.0.0.0/8 [90/133] | via | F0/4 |

Packets received by the router from BGP enter via a serial interface at 209 165 201 1 Each route is present within the routing table Which interface is used to forward traffic with a destination IP of 10.1.1.19?

- A. F0/4
- B. F0/0
- C. F0/1
- D. F0/3

---

**Answer: B**

---

Explanation:

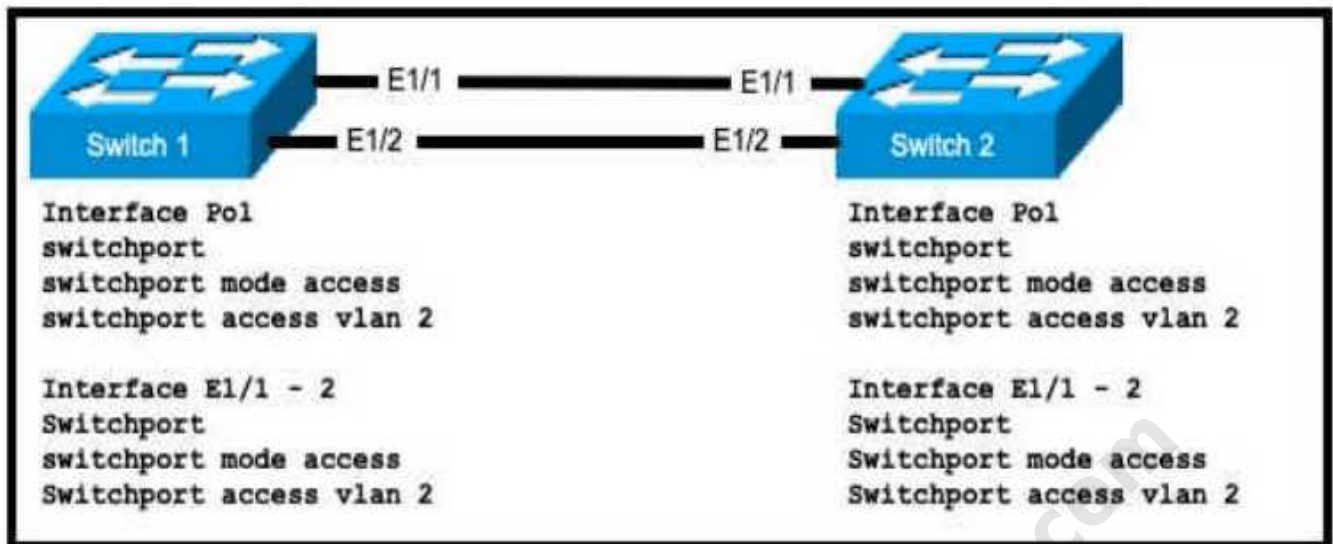
---

**Question: 455**

---

Refer to the exhibit.





An engineer is configuring an EtherChannel using LACP between Switches 1 and 2. Which configuration must be applied so that only Switch 1 sends LACP initiation packets?

- A. Switch 1 (config-if)#channel-group 1 mode onSwitch2(config-if)#channel-group 1 mode passive
- B. Switch1(config-if)#channel-group 1 mode passiveSwitch2(config-if)#channel-group 1 mode active
- C. Switch1(config-if)#channel-group 1 mode activeSwitch2(config-if)#channel-group 1 mode passive
- D. Switch1(config-if)#channel-group 1 mode onSwitch2(config-if)#channel-group 1 mode active

---

**Answer: C**

---

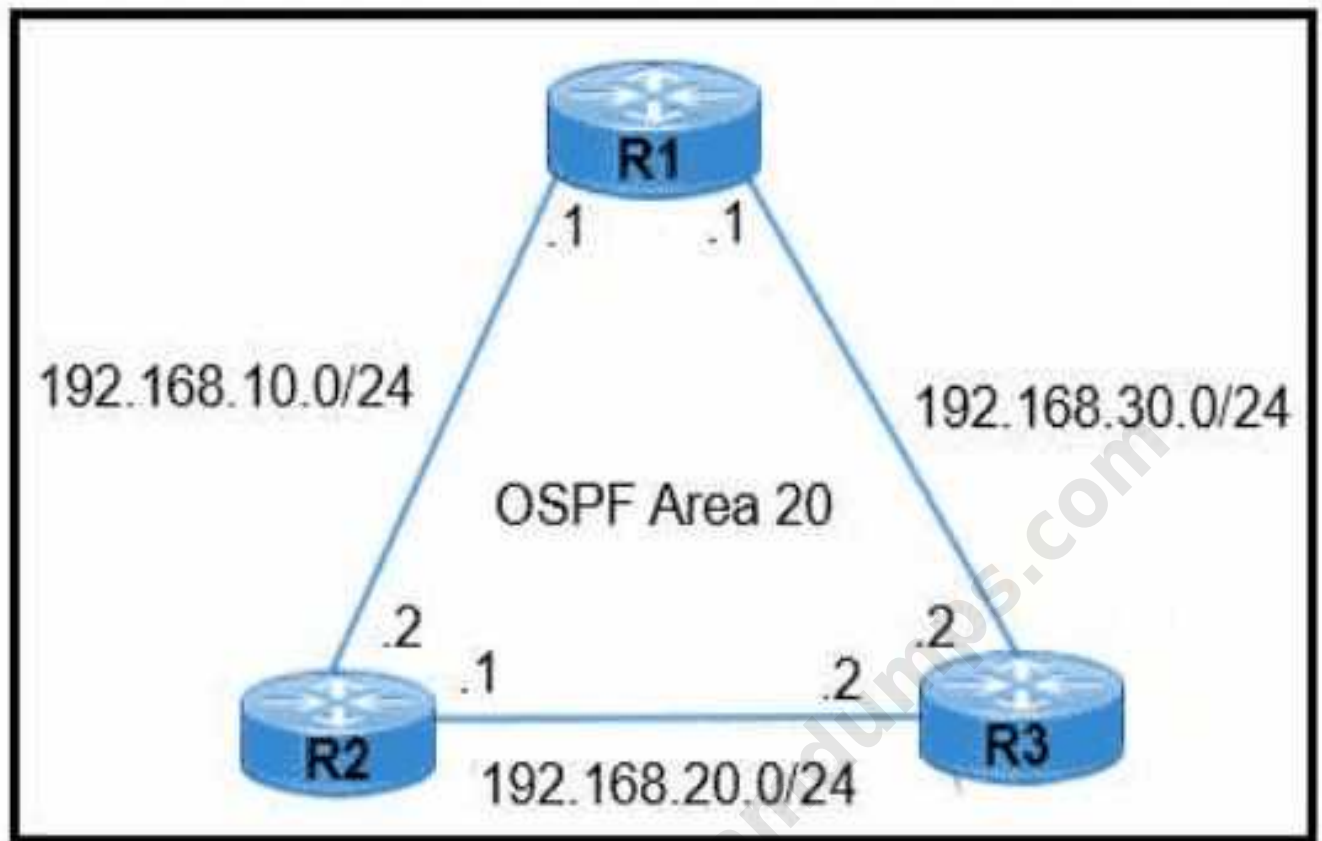
Explanation:

---

**Question: 456**

---

Refer to the exhibit.



R1 learns all routes via OSPF. Which command configures a backup static route on R1 to reach the 192.168.20.0/24 network via R3?

- A. R1(config)#ip route 192.168.20.0 255.255.0.0 192.168.30.2
- B. R1(config)#ip route 192.168.20.0 255.255.255.0 192.168.30.2 90
- C. R1(config)#ip route 192.168.20.0 255.255.255.0 192.168.30.2 111
- D. R1(config)#ip route 192.168.20.0 255.255.255.0 192.168.30.2

**Answer: C**

Explanation:

### Question: 457

What is the difference between IPv6 unicast and anycast addressing?

- A. IPv6 anycast nodes must be explicitly configured to recognize the anycast address, but IPv6 unicast nodes require no special configuration



- B. IPv6 unicast nodes must be explicitly configured to recognize the unicast address, but IPv6 anycast nodes require no special configuration
- C. An individual IPv6 unicast address is supported on a single interface on one node but an IPv6 anycast address is assigned to a group of interfaces on multiple nodes.
- D. Unlike an IPv6 anycast address, an IPv6 unicast address is assigned to a group of interfaces on multiple nodes

---

**Answer: C**

---

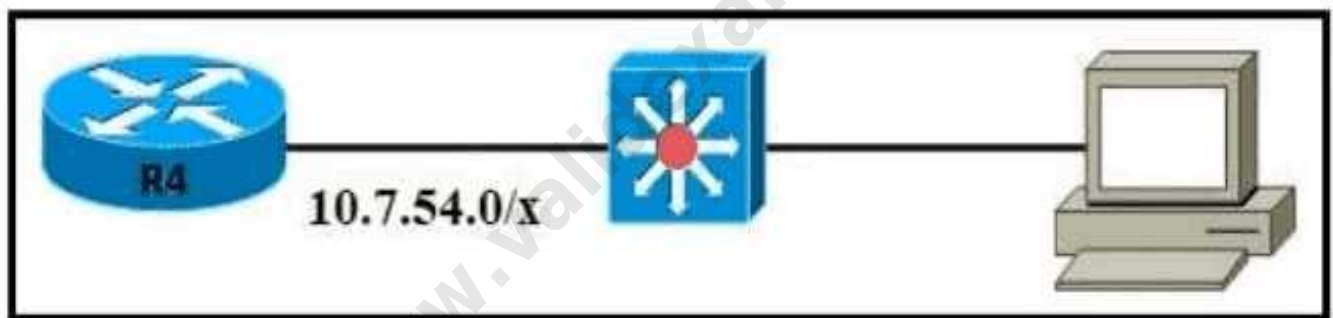
Explanation:

---

**Question: 458**

---

Refer to the exhibit.



The router has been configured with a supernet to accommodate the requirement for 380 users on a subnet. The requirement already considers 30% future growth. Which configuration verifies the IP subnet on router R4?

A)

Subnet: 10.7.54.0

Subnet mask: 255.255.254.0

Broadcast address: 10.7.54.255

Usable IP address range: 10.7.54.1 - 10.7.55.254

B)

Subnet: 10.7.54.0  
Subnet mask: 255.255.254.0  
Broadcast address: 10.7.55.255  
Usable IP address range: 10.7.54.1 - 10.7.55.254

C)

Subnet: 10.7.54.0  
Subnet mask: 255.255.128.0  
Broadcast address: 10.7.55.255  
Usable IP address range: 10.7.54.1 - 10.7.55.254

D)

Subnet: 10.7.54.0  
Subnet mask: 255.255.255.0  
Broadcast address: 10.7.54.255  
Usable IP address range: 10.7.54.1 - 10.7.55.254

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: B**

---

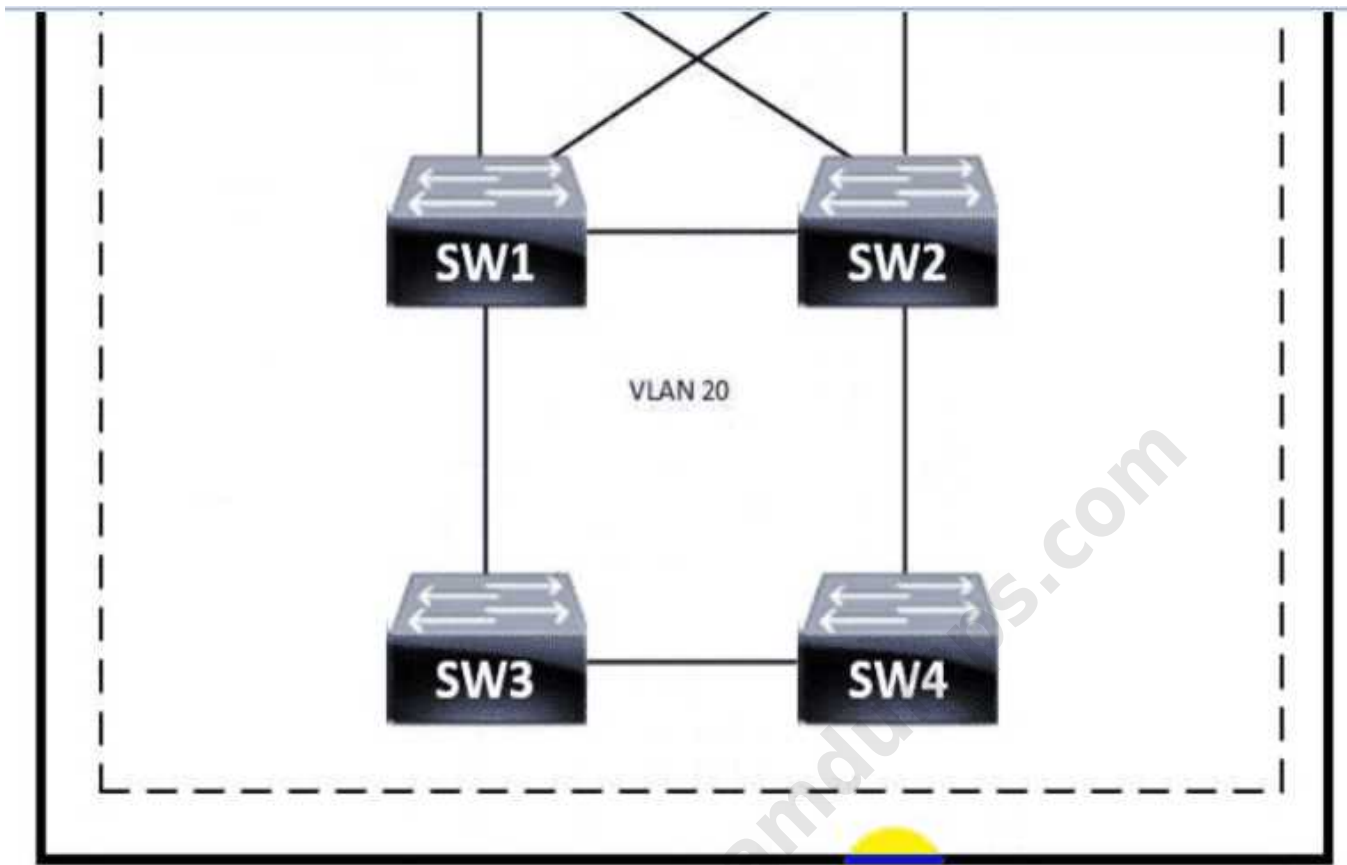
Explanation:

---

**Question: 459**

---

Refer to the exhibit.



Which switch becomes the root of a spanning tree for VLAN 20 if all links are of equal speed?

SW1 = 24596 0018.184e.3c00  
SW2 = 28692 004a.14e5.4077  
SW3 = 32788 0022.55cf.dd00  
SW4 = 64000 0041.454d.407f

- A. SW1
- B. SW2
- C. SW3
- D. SW4

**Answer: C**

Explanation:

**Question: 460**

Which protocol uses the SSL?

- A. HTTP
- B. SSH
- C. HTTPS
- D. Telnet

---

**Answer: C**

---

Explanation:

---

**Question: 461**

---

Which two spanning-tree states are bypassed on an interface running PortFast? (Choose two.)

- A. disabled
- B. listening
- C. forwarding
- D. learning
- E. blocking

---

**Answer: B,D**

---

Explanation:

---

**Question: 462**

---

A Cisco engineer is configuring a factory-default router with these three passwords:

- The user EXEC password for console access is p4ssw0rd1
- The user EXEC password for Telnet access is s3cr3t2
- The password for privileged EXEC mode is pnv4t3p4ss Which command sequence must the

engineer configured

A)

```
enable secret priv4t3p4ss
!  
line con 0  
password login p4ssw0rd1  
!  
line vty 0 15  
password login s3cr3t2  
login
```

B)

```
enable secret privilege 15 priv4t3p4ss
!  
line con 0  
password p4ssw0rd1  
login  
!  
line vty 0 15  
password s3cr3t2  
login
```

C)

```
enable secret priv4t3p4ss
!  
line con 0  
password p4ssw0rd1  
login  
!  
line vty 0 15  
password s3cr3t2  
login
```

D)

enable secret priv4t3p4ss  
!  
line con 0

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: B**

---

Explanation:

---

**Question: 463**

---

How does Rapid PVST+ create a fast loop-free network topology?

- A. It requires multiple links between core switches
- B. It generates one spanning-tree instance for each VLAN
- C. It maps multiple VLANs into the same spanning-tree instance
- D. It uses multiple active paths between end stations.

---

**Answer: A**

---

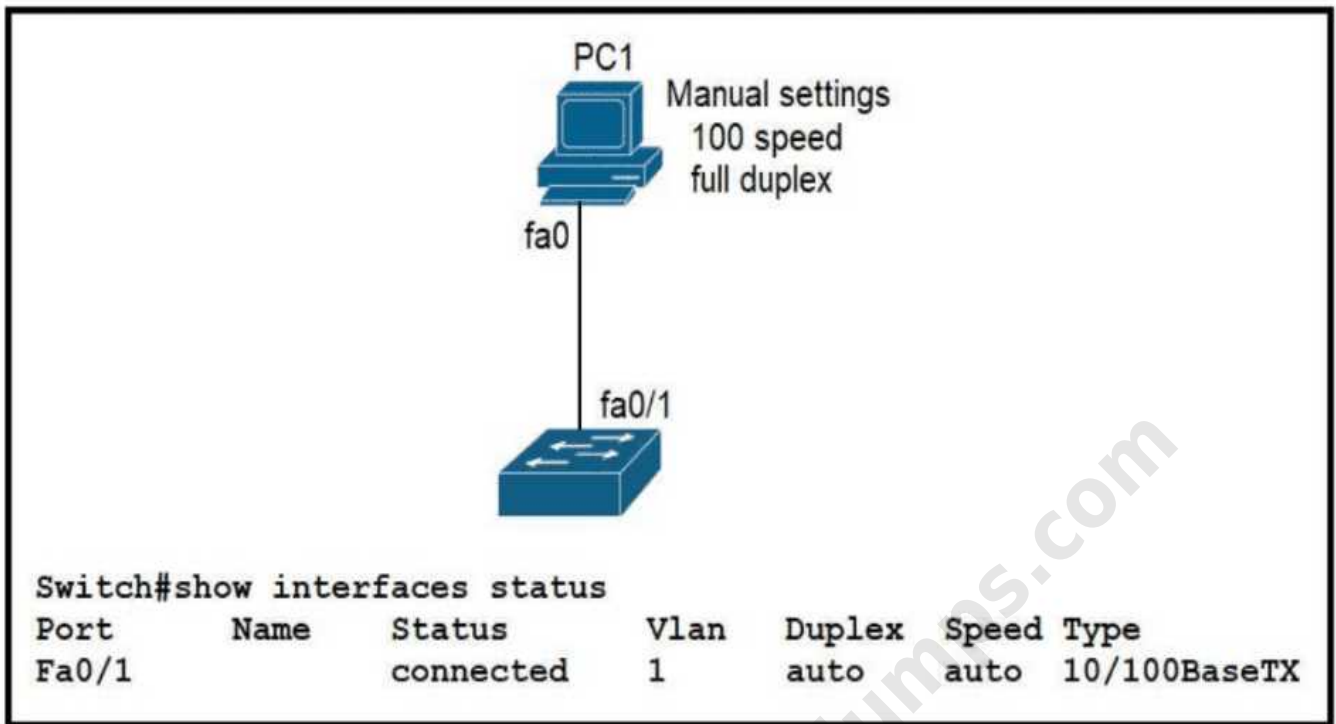
Explanation:

---

**Question: 464**

---

Refer to the exhibit.



The link between PC1 and the switch is up, but it is performing poorly. Which interface condition is causing the performance problem?

- A. There is a duplex mismatch on the interface
- B. There is an issue with the fiber on the switch interface.
- C. There is a speed mismatch on the interface.
- D. There is an interface type mismatch

**Answer: A**

Explanation:

### Question: 465

Which PoE mode enables powered-device detection and guarantees power when the device is detected?

- A. dynamic
- B. static
- C. active

D. auto

---

**Answer: B**

---

Explanation:

---

**Question: 466**

---

What is an expected outcome when network management automation is deployed?

- A. A distributed management plane must be used.
- B. Software upgrades are performed from a central controller
- C. Complexity increases when new device configurations are added
- D. Custom applications are needed to configure network devices

---

**Answer: B**

---

Explanation:

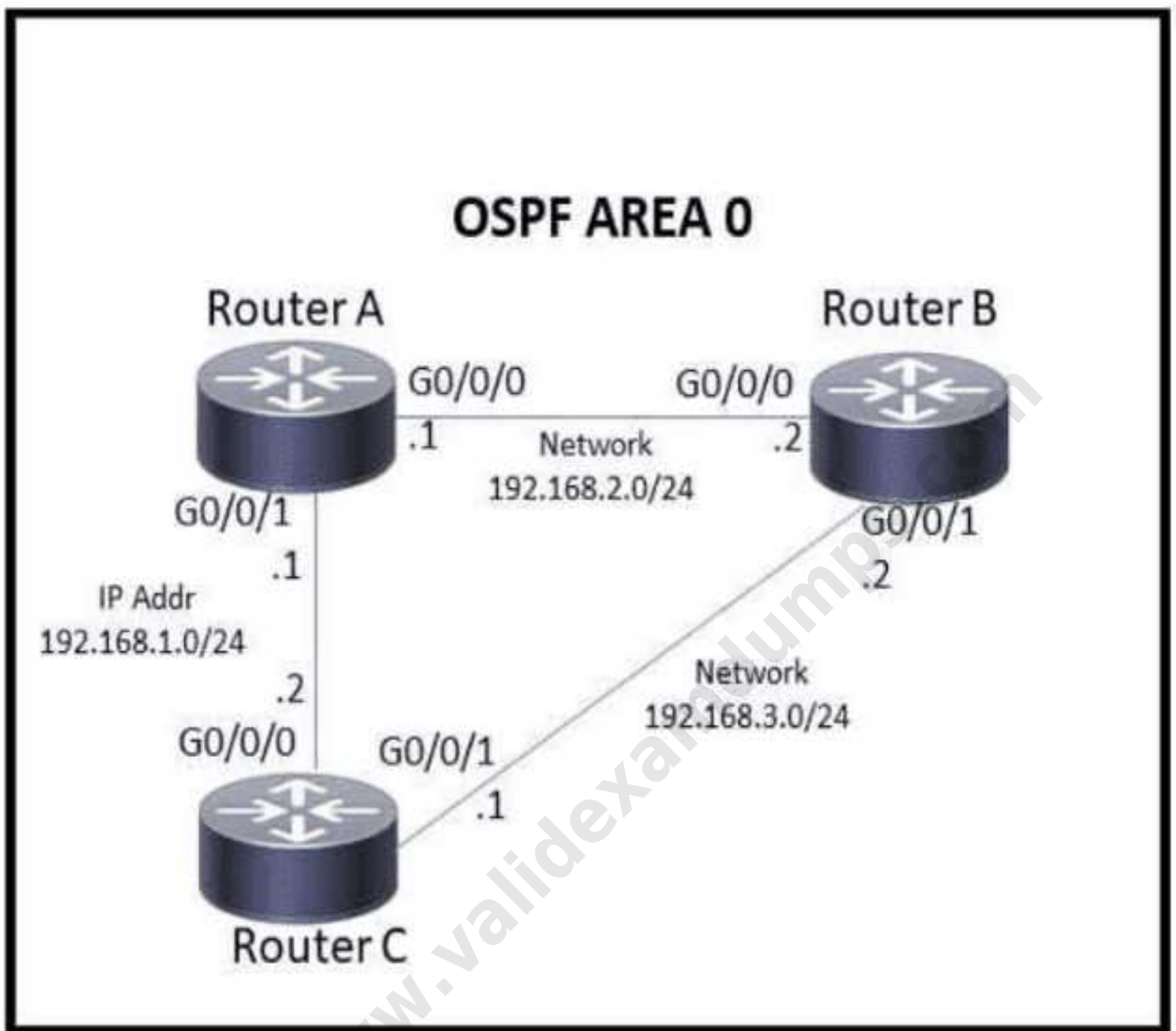
---

**Question: 467**

---

Refer to the exhibit.





Which action must be taken to ensure that router A is elected as the DR for OSPF area 0?

- A. Configure the OSPF priority on router A with the lowest value between the three routers.
- B. Configure router B and router C as OSPF neighbors of router A.
- C. Configure the router A interfaces with the highest OSPF priority value within the area.
- D. Configure router A with a fixed OSPF router ID

**Answer: C**

Explanation:

**Question: 468**

Refer to the exhibit.

```
R1# show ip route
Codes:
C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP, D -
EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA
external type 1, N2 - OSPF NSSA external type 2, E1 - OSPF external type
1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default,
U - per-user static route, o - ODR
Gateway of last resort is not set
C 10.0.0.0/8 is directly connected, Loopback0
  10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O 10.0.1.3/32 [110/100] via 10.0.1.100, 00:39:08, Serial0
C 10.0.1.0/24 is directly connected, Serial0
O 10.0.1.5/32 [110/5] via 10.0.1.50, 00:39:08, Serial0
O 10.0.10.0/24 [110/10] via 10.0.1.4, 00:39:08, Gigabit Ethernet 0/0
D 10.0.10.0/24 [90/10] via 10.0.1.5, 00:39:08, Gigabit Ethernet 0/1
```

Web traffic is coming in from the WAN interface. Which route takes precedence when the router is processing traffic destined for the LAN network at 10.0.10.0/24?

- A. via next-hop 10.0.1.5
- B. via next-hop 10.0.1.4
- C. via next-hop 10.0.1.50
- D. via next-hop 10.0.1.100

---

**Answer: A**

---

Explanation:

---

**Question: 469**

---

Which two components comprise part of a PKI? (Choose two.)

- A. preshared key that authenticates connections
- B. RSA token
- C. CA that grants certificates

- D. clear-text password that authenticates connections
- E. one or more CRLs

---

**Answer: B,C**

---

Explanation:

---

**Question: 470**

---

What are two benefits of FHRPs? (Choose two.)

- A. They enable automatic failover of the default gateway.
- B. They allow multiple devices to serve as a single virtual gateway for clients in the network.
- C. They are able to bundle multiple ports to increase bandwidth.
- D. They prevent loops in the Layer 2 network.
- E. They allow encrypted traffic.

---

**Answer: A,B**

---

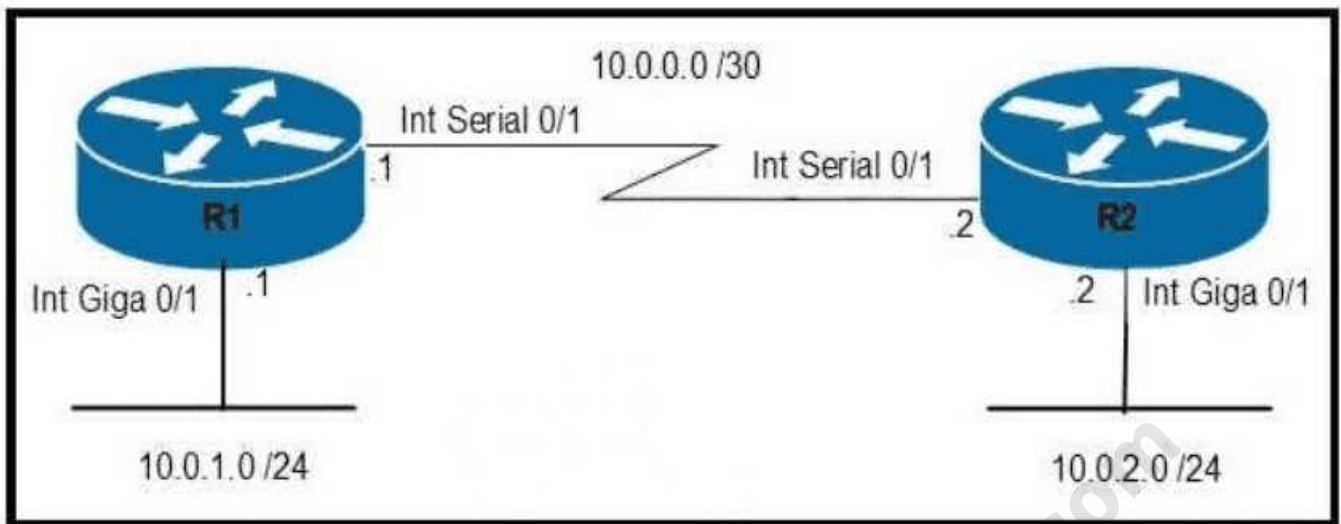
Explanation:

---

**Question: 471**

---

Refer to the exhibit.



Which command configures OSPF on the point-to-point link between routers R1 and R2?

- A. router-id 10.0.0.15
- B. neighbor 10.1.2.0 cost 180
- C. ipospf priority 100
- D. network 10.0.0.0 0.0.0.255 area 0

**Answer: D**

Explanation:

### Question: 472

What causes a port to be placed in the err-disabled state?

- A. nothing plugged into the port
- B. link flapping
- C. shutdown command issued on the port
- D. latency

**Answer: B**

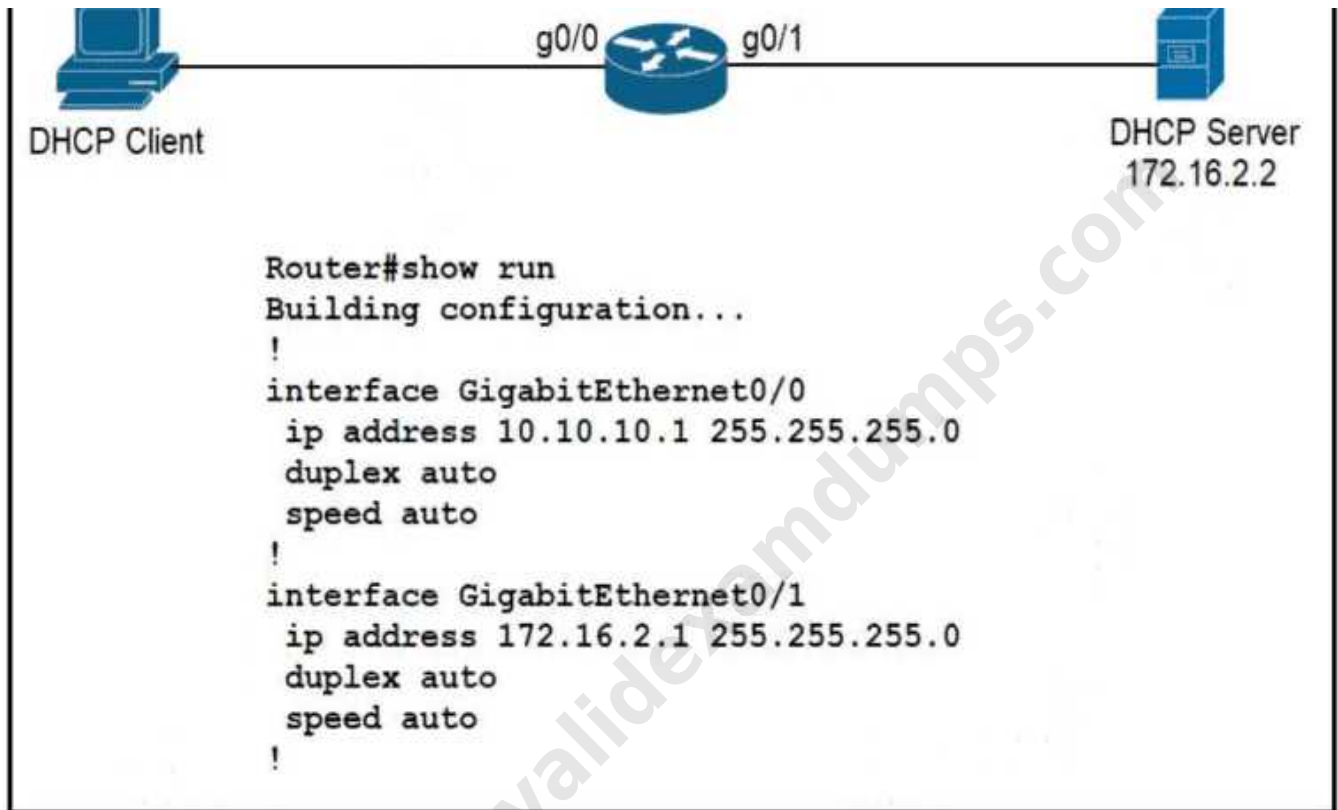
Explanation:

---

**Question: 473**

---

Refer to the exhibit.



An engineer is configuring a new router on the network and applied this configuration. Which additional configuration allows the PC to obtain its IP address from a DHCP server?

- A. Configure the ip dhcp relay information command under interface Gi0/1.
- B. Configure the ip dhcp smart-relay command globally on the router
- C. Configure the ip helper-address 172.16.2.2 command under interface Gi0/0
- D. Configure the ip address dhcp command under interface Gi0/0

---

**Answer: C**

---

Explanation:

---

**Question: 474**

---

Which Layer 2 switch function encapsulates packets for different VLANs so that the packets traverse the same port and maintain traffic separation between the VLANs?

- A. VLAN numbering
- B. VLAN DSCP
- C. VLAN tagging
- D. VLAN marking

---

**Answer: C**

---

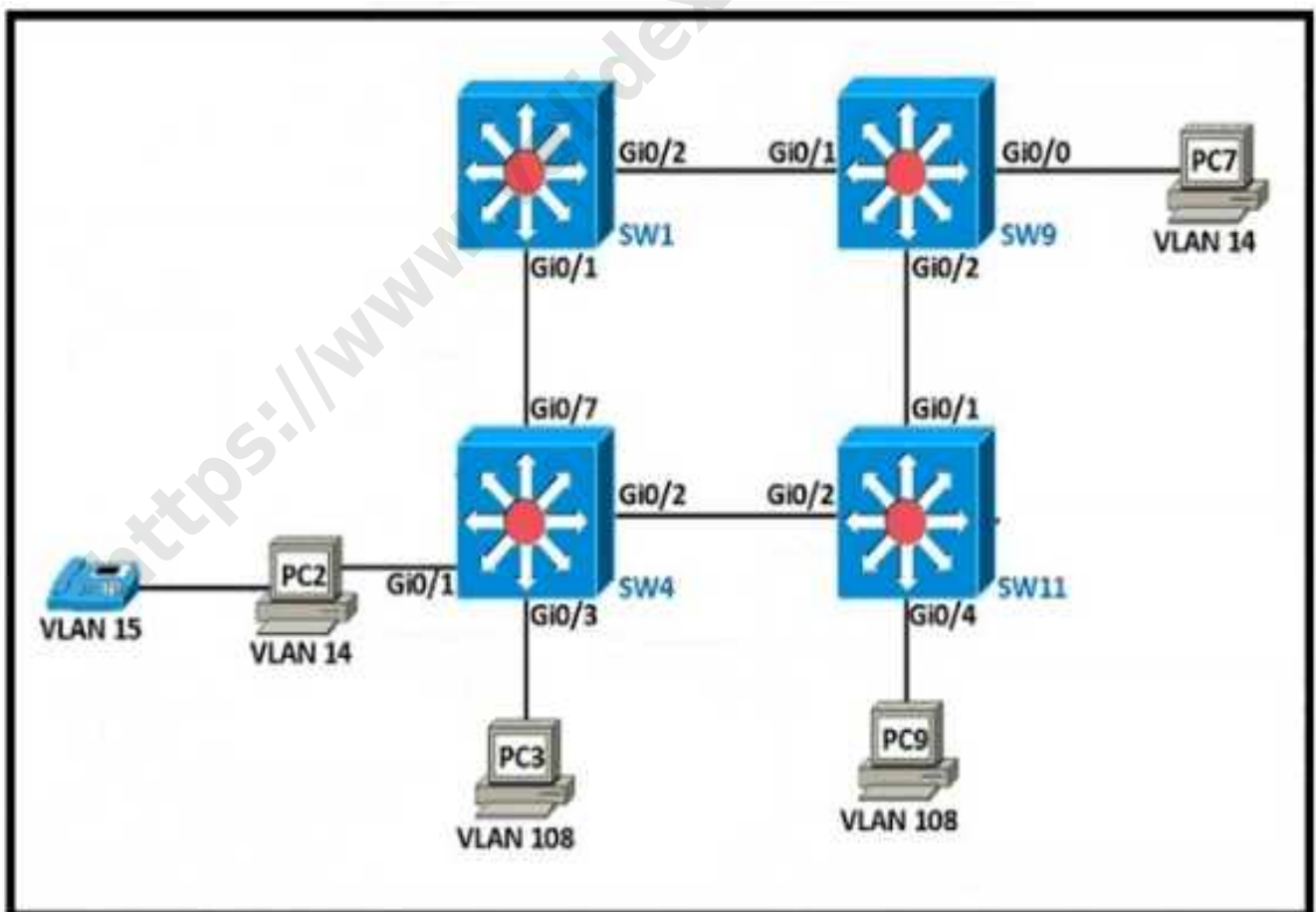
Explanation:

---

**Question: 475**

---

Refer to the exhibit.



The following must be considered:

- SW1 is fully configured for all traffic
- The SW4 and SW9 links to SW1 have been configured
- The SW4 interface Gi0/1 and Gi0/0 on SW9 have been configured
- The remaining switches have had all VLANs added to their VLAN database

Which configuration establishes a successful ping from PC2 to PC7 without interruption to traffic flow between other PCs?

A)

```
SW4#  
interface Gi0/2  
switchport mode trunk  
switchport trunk allowed vlan 14  
  
SW11#  
interface Gi0/1  
switchport mode trunk  
switchport trunk allowed vlan 14  
  
SW9#  
interface Gi0/2  
switchport mode trunk  
switchport trunk allowed vlan 108
```

B)

```
SW4#  
interface Gi0/2  
switchport mode trunk  
switchport trunk allowed vlan 14  
  
SW11#  
interface Gi0/1  
switchport mode trunk  
switchport trunk allowed vlan 14  
  
SW9#  
interface Gi0/2  
switchport mode trunk  
switchport trunk allowed vlan 108
```

C)

```
SW4#  
interface Gi0/2  
switchport mode trunk  
switchport trunk allowed vlan 14,108  
  
SW11#  
interface Gi0/2  
switchport mode trunk  
switchport trunk allowed vlan 14,108  
!  
interface Gi0/1  
switchport mode trunk  
switchport trunk allowed vlan 14,108  
  
SW9#  
interface Gi0/2  
switchport mode trunk  
switchport trunk allowed vlan 14
```

D)

```
SW4#  
interface Gi0/2  
switchport mode access  
switchport access vlan 14  
  
SW11#  
interface Gi0/2  
switchport mode access  
switchport access vlan 14  
!  
interface Gi0/0  
switchport mode access  
switchport access vlan 14  
!  
interface Gi0/1  
switchport mode trunk  
  
SW9#  
interface Gi0/2  
switchport mode access  
switchport access vlan 14
```

A. Option A



- B. Option B
- C. Option C
- D. Option D

---

**Answer: C**

---

Explanation:

---

**Question: 476**

---

R1 as an NTP server must have:

- NTP authentication enabled
- NTP packets sourced from Interface loopback 0
- NTP stratum 2
- NTP packets only permitted to client IP 209.165.200.225

How should R1 be configured?

A)

```
ntp authenticate
ntp authentication-key 2 md5 CISCO123
ntp source Loopback0
nntp access-group server-only 10
ntp master 2
!
access-list 10 permit 209.165.200.225
```

B)

```
ntp authenticate
ntp authentication-key 2 md5 CISCO123
ntp source Loopback0
ntp access-group server-only 10
ntp stratum 2
!
access-list 10 permit udp host 209.165.200.225 any eq 123
```

C)

```
ntp authenticate
ntp authentication-key 2 sha1 CISCO123
ntp source Loopback0
ntp access-group server-only 10
ntp master 2
!
access-list 10 permit udp host 209.165.200.225 any eq 123
```

D)

```
ntp authenticate
ntp authentication-key 2 md5 CISCO123
ntp interface Loopback0
ntp access-group server-only 10
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: B**

---

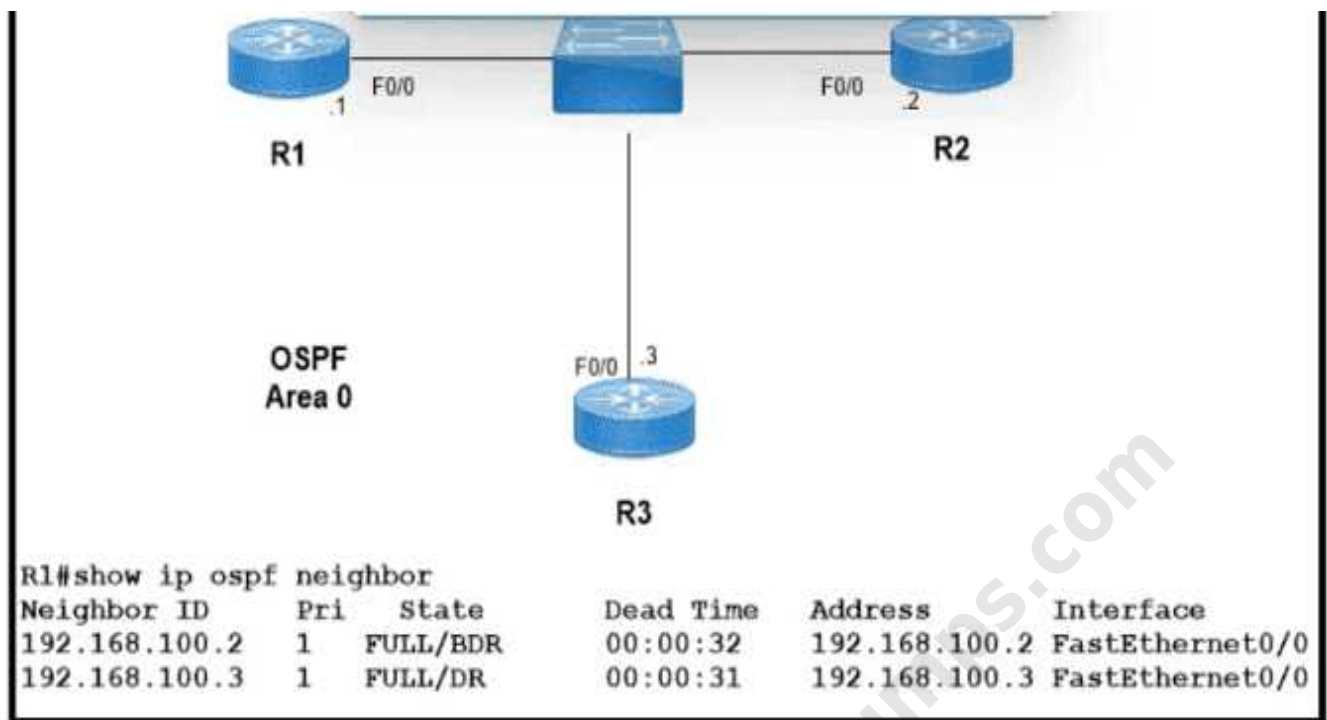
Explanation:

---

**Question: 477**

---

Refer to the exhibit.



Which two configurations must the engineer apply on this network so that R1 becomes the DR?  
(Choose two.)

A)

```
R1(config)#router ospf 1
R1(config-router)#router-id 192.168.100.1
```

B)

```
R1(config)#interface fastethernet 0/0
R1(config-if)#ip ospf priority 200
```

C)

```
R3(config)#interface fastethernet 0/0
R3(config-if)#ip ospf priority 0
```

D)

```
R1(config)#interface fastethernet 0/0
R1(config-if)#ip ospf priority 0
```

E)

```
R3(config)#interface fastethernet 0/0
R3(config-if)#ip ospf priority 200
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D
- E. Option E

---

**Answer: B,C**

---

Explanation:

---

**Question: 478**

---

Which type of IPv6 address is similar to a unicast address but is assigned to multiple devices on the same network at the same time?

- A. global unicast address
- B. anycast address
- C. multicast address
- D. link-local address

---

**Answer: B**

---

Explanation:

---

**Question: 479**

---

Which two network actions occur within the data plane? (Choose two.)

- A. Add or remove an 802.1Q trunking header.
- B. Make a configuration change from an incoming NETCONF RPC.
- C. Run routing protocols.

- D. Match the destination MAC address to the MAC address table.
- E. Reply to an incoming ICMP echo request.

---

**Answer: B,D**

---

Explanation:

---

**Question: 480**

---

Which QoS traffic handling technique retains excess packets in a queue and reschedules these packets for later transmission when the configured maximum bandwidth has been surpassed?

- A. weighted random early detection
- B. traffic policing
- C. traffic shaping
- D. traffic prioritization

---

**Answer: C**

---

Explanation:

---

**Question: 481**

---

Refer to the exhibit.

```
CPE# show ip route
192.168.1.0/24 is variably subnetted, 3 subnets, 3 masks
B   192.168.1.0/24 [20/1] via 192.168.12.2, 00:00:06
R   192.168.1.128/25 [120/5] via 192.168.13.3, 00:02:35, Ethernet0/1
O   192.168.1.192/26 [110/11] via 192.168.14.4, 00:02:23, Ethernet0/2
D   192.168.1.224/27 [90/1024640] via 192.168.15.5, 00:01:40, Ethernet0/3
```

All traffic enters the CPE router from interface Serial0/3 with an IP address of 192.168.50.1. Web traffic from the WAN is destined for a LAN network where servers are load-balanced. An IP packet

with a destination address of the HTTP virtual IP of 192.168.1.250 must be forwarded Which routing table entry does the router use?

- A. 192.168.1.0/24 via 192.168.12.2
- B. 192.168.1.128/25 via 192.168.13.3
- C. 192.168.1.192/26 via 192.168.14.4
- D. 192.168.1.224/27 via 192.168.15.5

---

**Answer: B**

---

Explanation:

---

**Question: 482**

---

Which interface mode must be configured to connect the lightweight APs in a centralized architecture?

- A. WLAN dynamic
- B. management
- C. trunk
- D. access

---

**Answer: D**

---

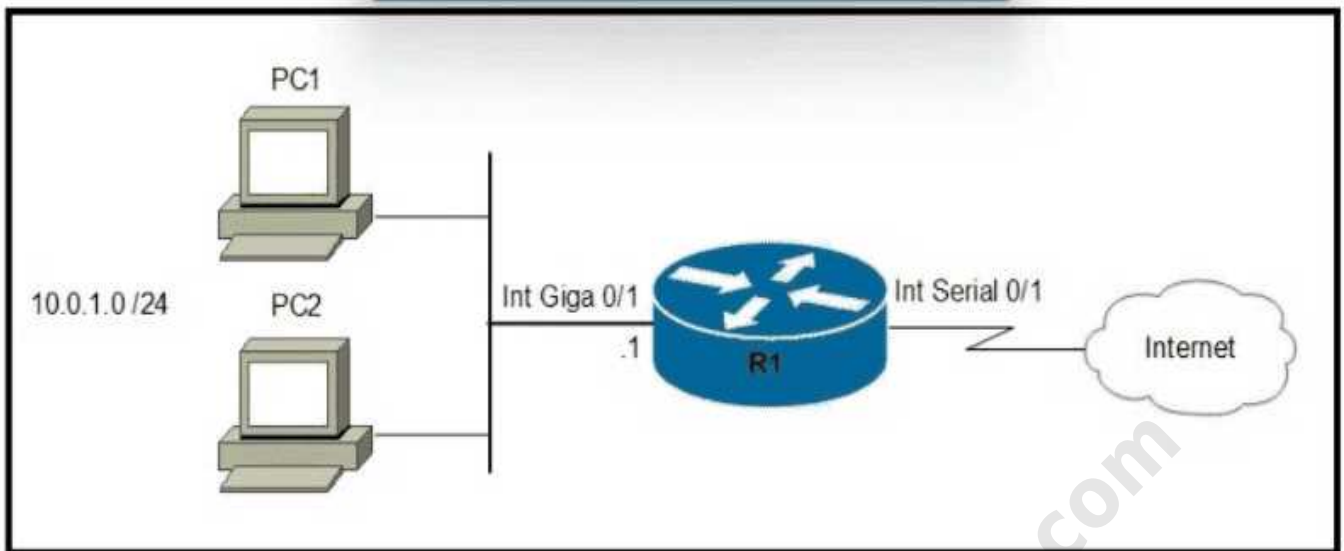
Explanation:

---

**Question: 483**

---

Refer to the exhibit.



Which two commands must be configured on router R1 to enable the router to accept secure remote-access connections? (Choose two)

- A. transport input telnet
- B. crypto key generate rsa
- C. ip ssh pubkey-chain
- D. login console
- E. username cisco password 0 Cisco

**Answer: B,E**

Explanation:

#### Question: 484

Which type of network attack overwhelms the target server by sending multiple packets to a port until the half-open TCP resources of the target are exhausted?

- A. SYIM flood
- B. reflection
- C. teardrop
- D. amplification

---

**Answer: A**

---

Explanation:

---

**Question: 485**

---

Refer to the exhibit.

```
A# show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
172.1.1.1 1 EXCHANGE/ - 00:00:36 172.16.32.1 Serial0.1
```

An engineer assumes a configuration task from a peer Router A must establish an OSPF neighbor relationship with neighbor 172.1.1.1. The output displays the status of the adjacency after 2 hours. What is the next step in the configuration process for the routers to establish an adjacency?

- A. Configure router A to use the same MTU size as router B.
- B. Set the router B OSPF ID to a nonhost address.
- C. Configure a point-to-point link between router A and router B.
- D. Set the router B OSPF ID to the same value as its IP address

---

**Answer: A**

---

Explanation:

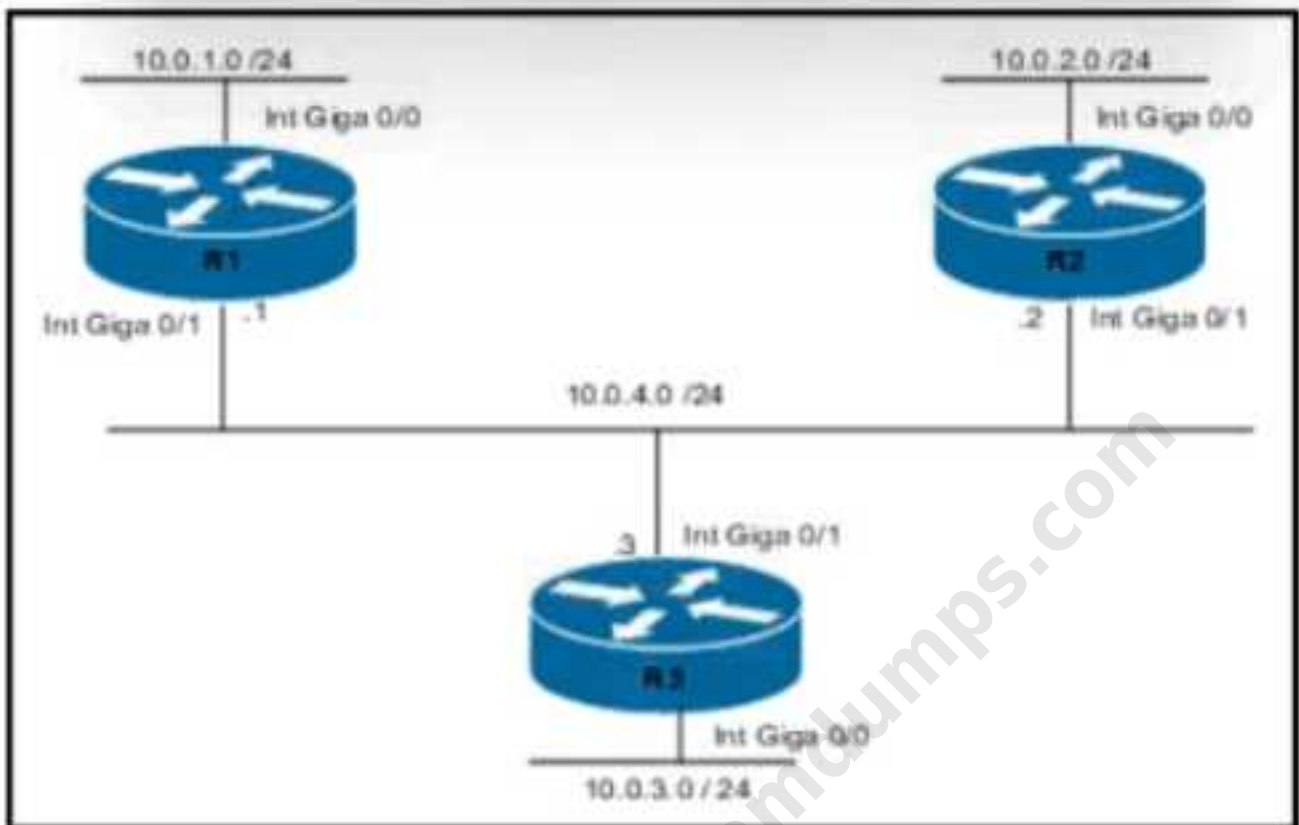
---

**Question: 486**

---

Refer to the exhibit.





Routers R1 and R3 have the default configuration. The router R2 priority is set to 99. Which commands on R3 configure it as the DR in the 10.0.4.0/24 network?

- A. R3(config)#interface Gig0/1 R3(config-if)#ip ospf priority 100
- B. R3(config)#interface Gig0/0 R3(config-if)#ip ospf priority 100
- C. R3(config)#interface Gig0/0 R3(config-if)#ip ospf priority 1
- D. R3(config)#interface Gig0/1 R3(config-if)#ip ospf priority 0

---

**Answer: B**

---

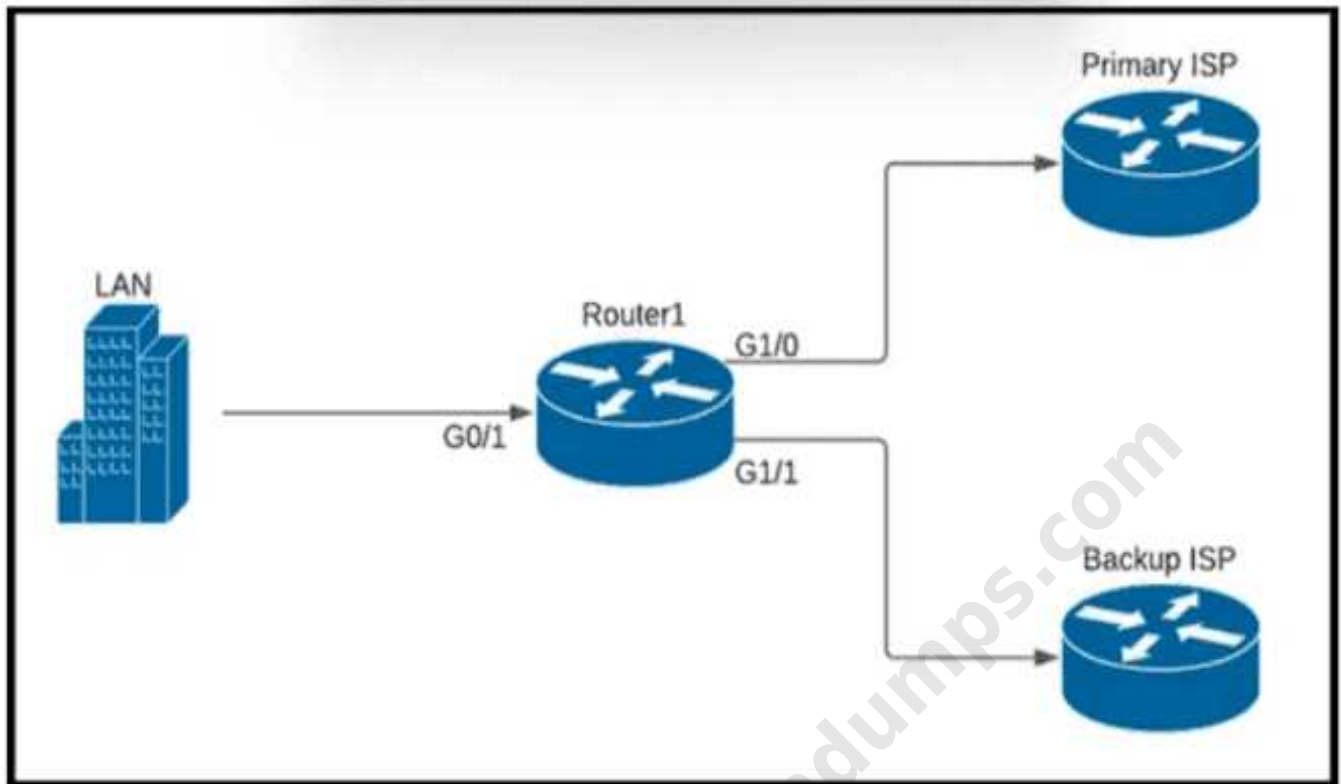
Explanation:

---

**Question: 487**

---

Refer to the exhibit.



A company is configuring a failover plan and must implement the default routes in such a way that a floating static route will assume traffic forwarding when the primary link goes down. Which primary route configuration must be used?

- A. `ip route 0.0.0.0 0.0.0.0 192.168.0.2 GigabitEthernet1/0`
- B. `ip route 0.0.0.0 0.0.0.0 192.168.0.2 tracked`
- C. `ip route 0.0.0.0 0.0.0.0 192.168.0.2 floating`
- D. `ip route 0.0.0.0 0.0.0.0 192.168.0.2`

---

**Answer: B**

---

Explanation:

---

**Question: 488**

---

What is one reason to implement LAG on a Cisco WLC?

- A. to increase security and encrypt management frames
- B. to provide link redundancy and load balancing

- C. to allow for stateful and link-state failover
- D. to enable connected switch ports to failover and use different VLANs

---

**Answer: B**

---

Explanation:

---

**Question: 489**

---

Which action implements physical access control as part of the security program of an organization?

- A. configuring a password for the console port
- B. backing up syslogs at a remote location
- C. configuring enable passwords on network devices
- D. setting up IP cameras to monitor key infrastructure

---

**Answer: A**

---

Explanation:

---

**Question: 490**

---

DRAG DROP

Drag and drop the facts about wireless architectures from the left onto the types of access point on the right. Not all options are used.

|                                                         |                          |
|---------------------------------------------------------|--------------------------|
| supports automatic deployment                           | Autonomous Access Point  |
| managed from a web-based dashboard                      |                          |
| accessible for management via Telnet, SSH, or a web GUI |                          |
| configured and managed by a WLC                         | Cloud-Based Access Point |
| requires a management IP address                        |                          |

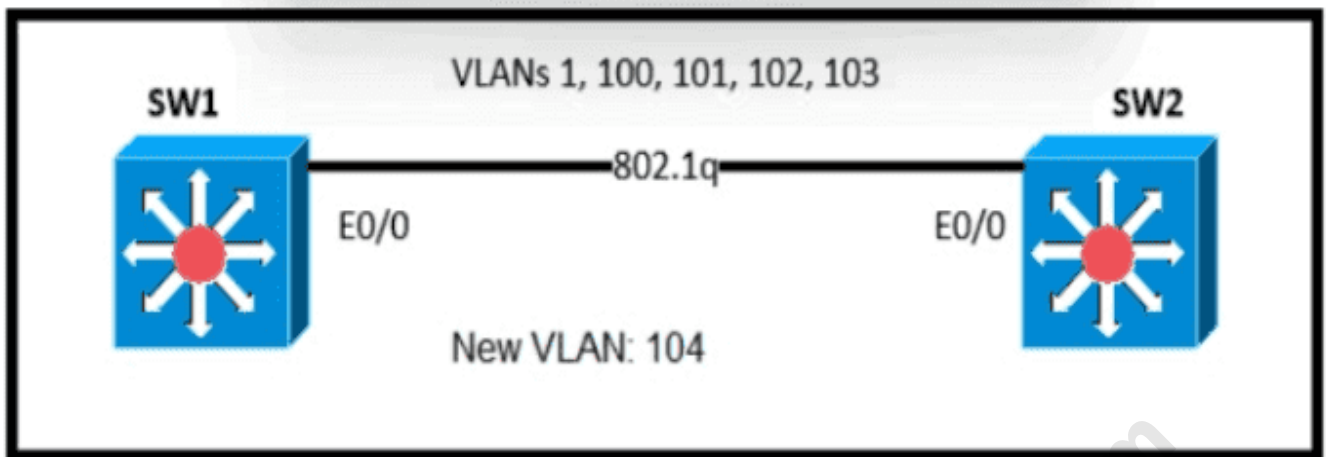
**Answer:**

Explanation:

|                                    |                                                         |
|------------------------------------|---------------------------------------------------------|
| managed from a web-based dashboard | Autonomous Access Point                                 |
|                                    | accessible for management via Telnet, SSH, or a web GUI |
|                                    | configured and managed by a WLC                         |
|                                    | Cloud-Based Access Point                                |
|                                    | requires a management IP address                        |
|                                    | supports automatic deployment                           |

**Question: 491**

Refer to the exhibit.



An engineer is asked to insert the new VLAN into the existing trunk without modifying anything previously configured. Which command accomplishes this task?

- A. switchport trunk allowed vlan 100-104
- B. switchport trunk allowed vlan add 104
- C. switchport trunk allowed vlan all
- D. switchport trunk allowed vlan 104

---

**Answer: D**

---

Explanation:

---

### Question: 492

---

DRAG DROP

An engineer is tasked to configure a switch with port security to ensure devices that forward unicasts, multicasts, and broadcasts are unable to flood the port. The port must be configured to permit only two random MAC addresses at a time. Drag and drop the required configuration commands from the left onto the sequence on the right. Not all commands are used.

|                                                     |
|-----------------------------------------------------|
| switchport mode access                              |
| switchport port-security                            |
| switchport port-security mac-address 0060.3EDD.77AB |
| switchport port-security mac-address 00D0.D3ED.622A |
| switchport port-security mac-address sticky         |
| switchport port-security maximum 2                  |
| switchport port-security violation shutdown         |

|   |
|---|
| 1 |
| 2 |
| 3 |
| 4 |

**Answer:**

Explanation:

|                                                     |
|-----------------------------------------------------|
| switchport mode access                              |
| switchport port-security mac-address 0060.3EDD.77AB |
| switchport port-security mac-address 00D0.D3ED.622A |

|                                             |
|---------------------------------------------|
| switchport port-security                    |
| switchport port-security mac-address sticky |
| switchport port-security maximum 2          |
| switchport port-security violation shutdown |

### Question: 493

Which field within the access-request packet is encrypted by RADIUS?

- A. authorized services
- B. authenticator
- C. username
- D. password

---

**Answer: D**

---

Explanation:

---

**Question: 494**

---

A network administrator is setting up a new IPv6 network using the 64-bit address 2001 0EB8 00C1 2200:0001 0000 0000 0331/64 To simplify the configuration the administrator has decided to compress the address Which IP address must the administrator configure?

- A. ipv6 address 21:EB8:C1:2200:1::331/64
- B. ipv6 address 2001:EB8:C1:22:1::331/64
- C. ipv6 address 2001 :EB8:C 1:2200.1 ::331-64
- D. ipv6 address 2001:EB8:C1:2200:1:0000:331/64

---

**Answer: C**

---

Explanation:

---

**Question: 495**

---

A network engineer is configuring a switch so that it is remotely reachable via SSH. The engineer has already configured the host name on the router. Which additional command must the engineer configure before entering the command to generate the RSA key?

- A. password password
- B. crypto key generate rsa modulus 1024
- C. ip domain-name domain
- D. ip ssh authentication-retries 2

---

**Answer: C**

---

Explanation:

<https://www.cisco.com/c/en/us/solutions/small-business/resource-center/networking/how-to-setup-network-switch.html>

---

**Question: 496**

---

Refer to the exhibit.

```
ip domain-name CNAC.com
!  
interface GigabitEthernet0/0/0  
  ip address 192.168.1.10 255.255.255.0  
  duplex auto  
  speed auto  
!  
line vty 0 15  
  login local  
  
R1#show crypto key mypubkey rsa  
  
R1#show ssh  
%No SSHv2 server connections running.  
%No SSHv1 server connections running.
```

Which two commands must be added to update the configuration of router R1 so that it accepts only encrypted connections? (Choose two )

- A. username CNAC secret R!41!4319115@
- B. ip ssh version 2
- C. line vty 0 4



- D. crypto key generate rsa 1024
- E. transport input ssh

---

**Answer: D,E**

---

Explanation:

---

**Question: 497**

---

A network engineer must configure two new subnets using the address block 10.70.128.0/19 to meet these requirements:

- The first subnet must support 24 hosts
- The second subnet must support 472 hosts
- Both subnets must use the longest subnet mask possible from the address block

Which two configurations must be used to configure the new subnets and meet a requirement to use the first available address in each subnet for the router interfaces? (Choose two )

- A. interface vlan 1234 ip address 10.70.159.1 255.255.254.0
- B. interface vlan 1148 ip address 10.70.148.1 255.255.254.0
- C. interface vlan 4722 ip address 10.70.133.17 255.255.255.192
- D. interface vlan 3002 ip address 10.70.147.17 255.255.255.224
- E. interface vlan 155 ip address 10.70.155.65 255.255.255.224

---

**Answer: B,D**

---

Explanation:

---

**Question: 498**

---

What is a function of Opportunistic Wireless Encryption in an environment?

- A. offer compression

- B. increase security by using a WEP connection
- C. provide authentication
- D. protect traffic on open networks

---

**Answer: D**

---

Explanation:

---

**Question: 499**

---

Refer to the exhibit.

```
Switch#show etherchannel summary
[output omitted]
```

| Group                   | Port-channel | Protocol | Ports     |           |
|-------------------------|--------------|----------|-----------|-----------|
| -----+-----+-----+----- |              |          |           |           |
| 10                      | Po10 (SU)    | LACP     | Gi0/0 (P) | Gi0/1 (P) |
| 20                      | Po20 (SU)    | LACP     | Gi0/2 (P) | Gi0/3 (P) |

Which two commands when used together create port channel 10? (Choose two.)

- A. int range g0/0-1channel-group 10 mode active
- B. int range g0/0-1 chanm.l-group 10 mode desirable
- C. int range g0/0-1channel-group 10 mode passive
- D. int range g0/0-1 channel-group 10 mode auto
- E. int range g0/0-1 channel-group 10 mode on

---

**Answer: A,C**

---

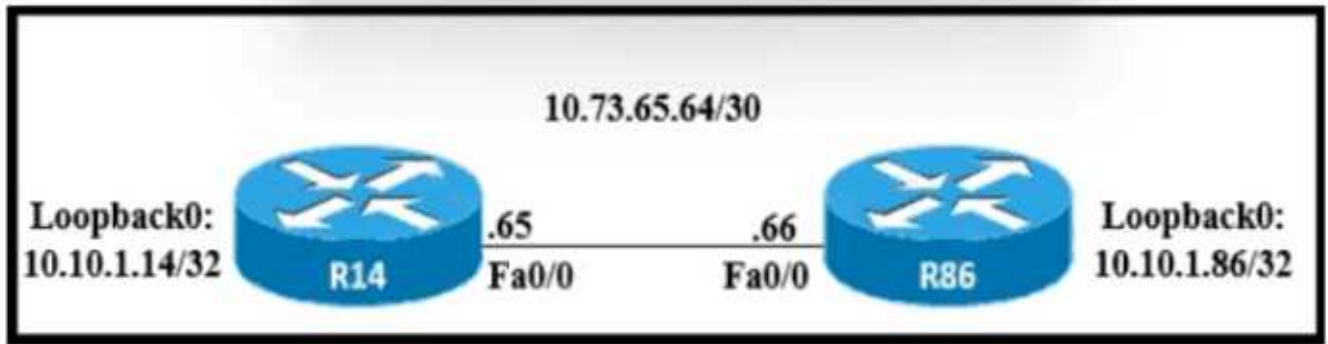
Explanation:

---

**Question: 500**

---

Refer to the exhibit.



A static route must be configured on R14 to forward traffic for the 172.21.34.0/25 network that resides on R86. Which command must be used to fulfill the request?

- A. ip route 172.21.34.0 255.255.255.192 10.73.65.65
- B. ip route 172.21.34.0 255.255.255.0 10.73.65.65
- C. ip route 172.21.34.0 255.255.128.0 10.73.65.64
- D. ip route 172.21.34.0 255.255.255.128 10.73.65.66

**Answer: D**

Explanation:

### Question: 501

A network engineer must implement an IPv6 configuration on the vlan 2000 interface to create a routable locally-unique unicast address that is blocked from being advertised to the internet. Which configuration must the engineer apply?

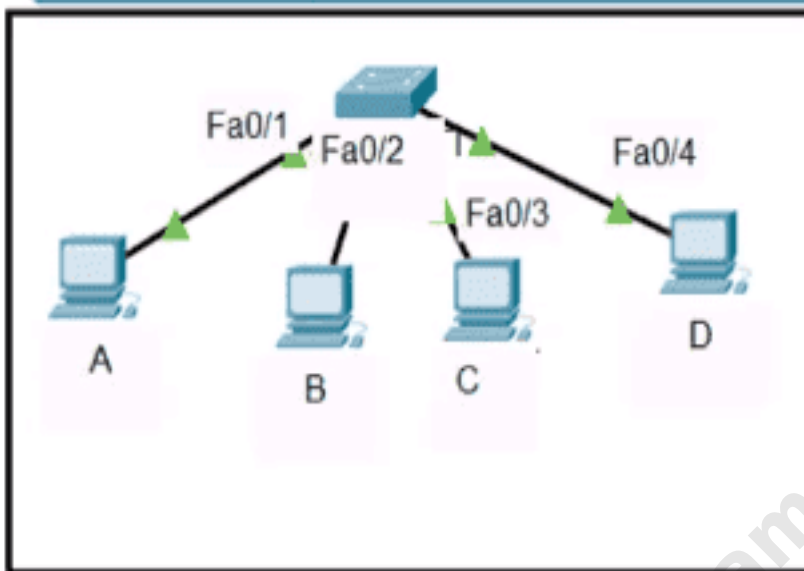
- A. interface vlan 2000 ipv6 address ffc0:0000:aaaa::1234:2343/64
- B. interface vlan 2000 ipv6 address fc00:0000:aaaa:a15d:1234:2343:8aca/64
- C. interface vlan 2000 ipv6 address fe80:0000:aaaa::1234:2343/64
- D. interface vlan 2000 ipv6 address fd00::1234:2343/64

**Answer: B**

Explanation:

**Question: 502**

Refer to the exhibit.



Host A sent a data frame destined for host D

```
SwitchA#show mac-address table  
Mac Address Table
```

| Vlan | Mac Address    | Type    | Ports |
|------|----------------|---------|-------|
| 2    | 000c.859c.bb7b | DYNAMIC | Fa0/1 |
| 2    | 0010.11dc.3e91 | DYNAMIC | Fa0/2 |
| 2    | 0041.45d7.c451 | DYNAMIC | Fa0/3 |

```
SwitchA#
```

What does the switch do when it receives the frame from host A?

- A. It drops the frame from the switch CAM table.
- B. It floods the frame out of all ports except port Fa0/1.
- C. It shuts down the port Fa0/1 and places it in err-disable mode.
- D. It experiences a broadcast storm.

**Answer: B**

Explanation:

---

**Question: 503**

---

What is the function of the controller in a software-defined network?

- A. multicast replication at the hardware level
- B. fragmenting and reassembling packets
- C. making routing decisions
- D. forwarding packets

---

**Answer: C**

---

Explanation:

---

**Question: 504**

---

Refer to the exhibit.



```
SW1#show run interface fastEthernet 0/1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 100,200,300
channel-group 1 mode on
```

```
SW1#show run interface fastEthernet 0/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 100,200,300
channel-group 1 mode on
```

```
SW2#show run interface fastEthernet 0/1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 100,200,300
channel-group 1 mode active
```

```
SW2#show run interface fastEthernet 0/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 100,200,300
channel-group 1 mode active
```

An engineer built a new L2 LACP EtherChannel between SW1 and SW2 and executed these show commands to verify the work. Which additional task allows the two switches to establish an LACP port channel?

- A. Change the channel-group mode on SW2 to auto
- B. Change the channel-group mode on SW1 to desirable.
- C. Configure the interface port-channel 1 command on both switches.
- D. Change the channel-group mode on SW1 to active or passive.

---

**Answer: D**

---

Explanation:

---

**Question: 505**

---

What is a requirement for nonoverlapping Wi-Fi channels?

- A. different security settings
- B. discontinuous frequency ranges
- C. different transmission speeds
- D. unique SSIDs

---

**Answer: B**

---

Explanation:

---

**Question: 506**

---

A network engineer is installing an IPv6-only capable device. The client has requested that the device IP address be reachable only from the internal network. Which type of IPv6 address must the engineer assign?

- A. unique local address
- B. link-local address
- C. aggregatable global address
- D. IPv4-compatible IPv6 address

---

**Answer: B**

---

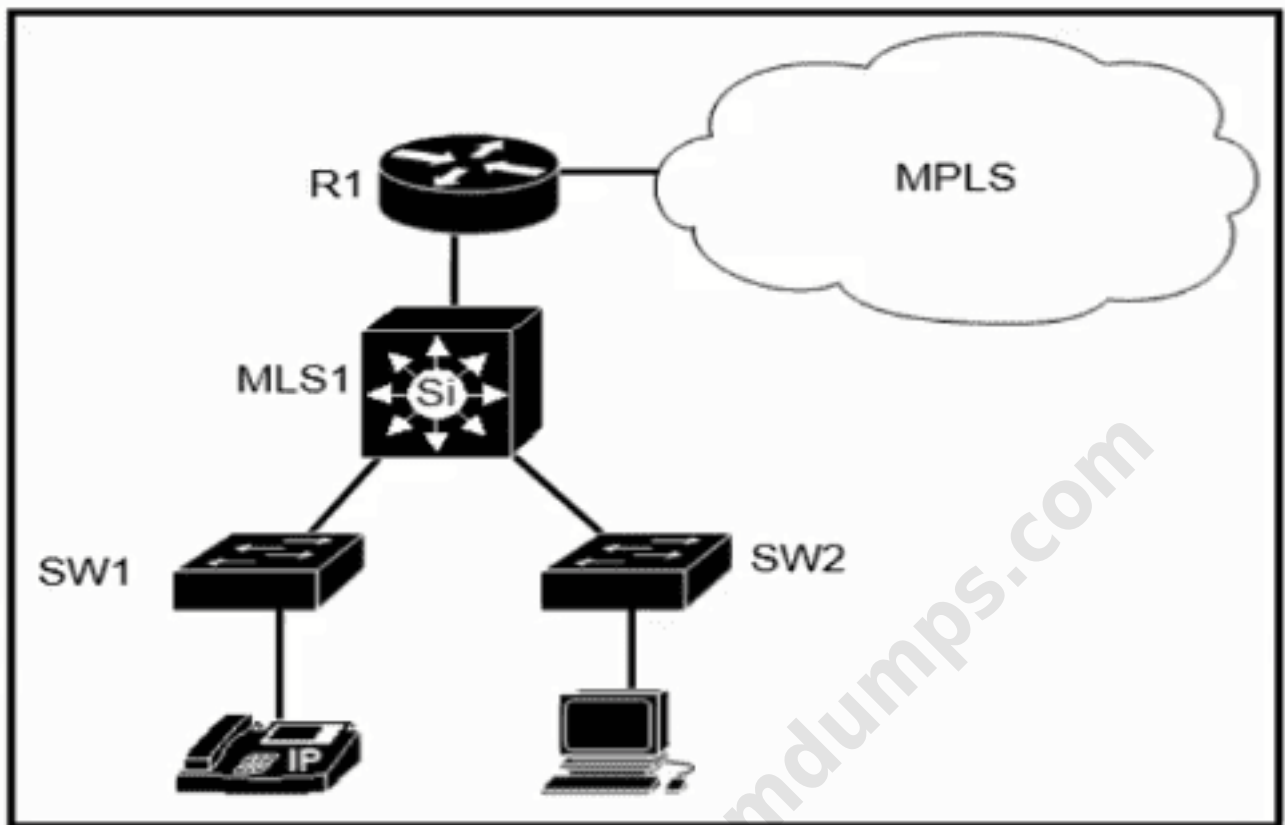
Explanation:

---

**Question: 507**

---

Refer to the exhibit.



Which plan must be Implemented to ensure optimal QoS marking practices on this network?

- A. As traffic traverses MLS1 remark the traffic, but trust all markings at the access layer.
- B. Trust the IP phone markings on SW1 and mark traffic entering SW2 at SW2.
- C. Remark traffic as it traverses R1 and trust all markings at the access layer.
- D. As traffic enters from the access layer on SW1 and SW2. trust all traffic markings.

**Answer: C**

Explanation:

**Question: 508**

Refer to the exhibit.



```
TenGigabitEthernet0/0/0 is up, line protocol is up
Hardware is BUILT-IN-2T+6X1GE, address is 74a0.2f7a.0123 (bia 74a0.2f7a.0123)
Description: Uplink
Internet address is 10.1.1.1/24
MTU 1500 bytes, BW 10000000 Kbit/sec, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive not supported
Full Duplex, 10000Mbps, link type is force-up, media type is unknown media type
output flow-control is on, input flow-control is on
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:00, output 00:05:40, output hang never
Last clearing of "show interface" counters never
Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 6160000 bits/sec, 1113 packets/sec
5 minute output rate 11213000 bits/sec, 1553 packets/sec
  12662416065 packets input, 12607032232894 bytes, 0 no buffer
    Received 14117163 broadcasts (0 IP multicasts)
      0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    0 watchdog, 26271385 multicast, 0 pause input
  7907779058 packets output, 5073750426832 bytes, 0 underruns
    0 output errors, 8662416065 collisions, 1 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier, 0 pause output
    0 output buffer failures, 0 output buffers swapped out
    1 carrier transitions
```

Traffic that is flowing over interface TenGigabitEthernet0/0 experiences slow transfer speeds. What is the reason for the issue?

- A. heavy traffic congestion
- B. a duplex incompatibility
- C. a speed conflict
- D. queuing drops

---

**Answer: C**

---

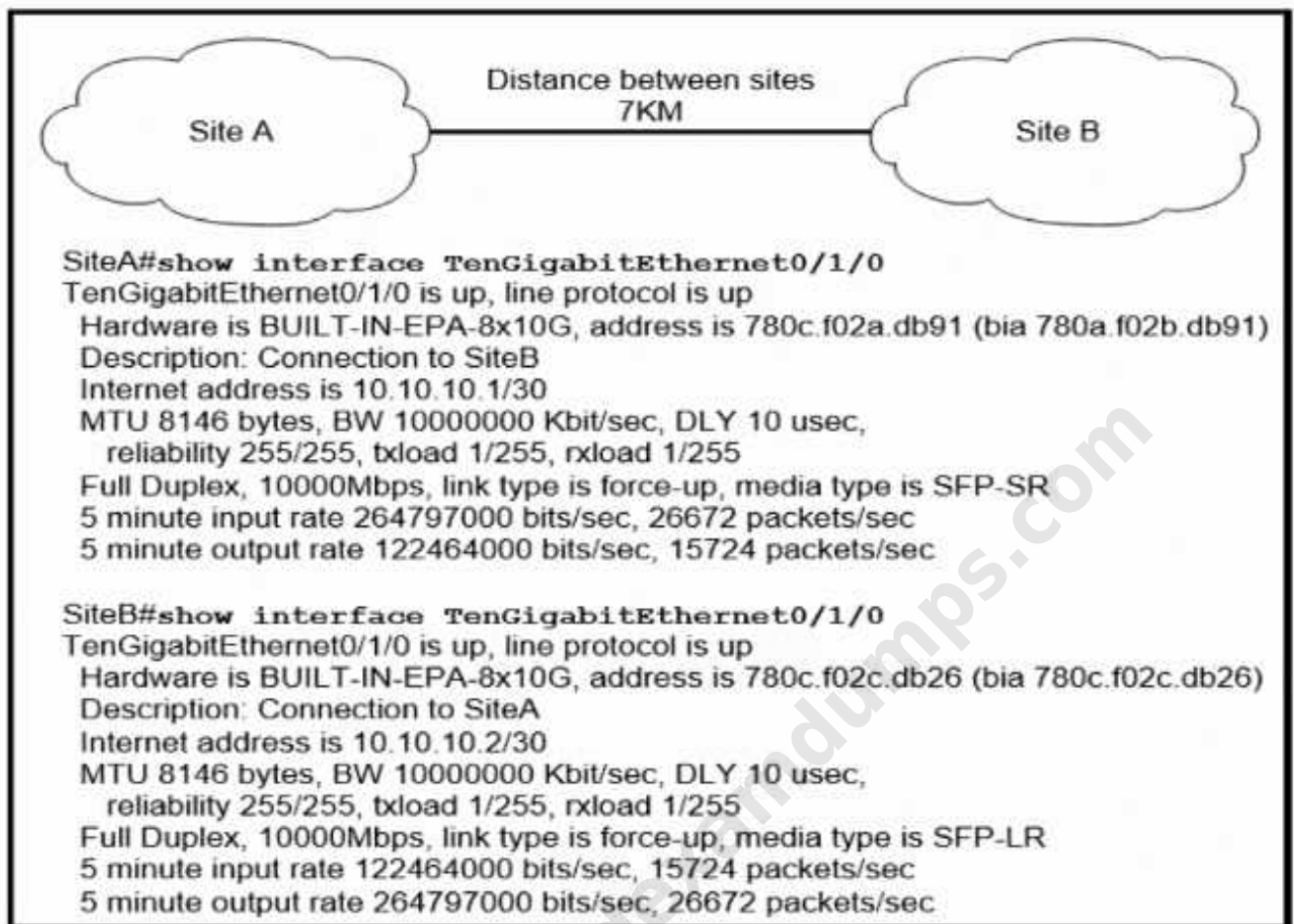
Explanation:

---

**Question: 509**

---

Refer to the exhibit.



Site A was recently connected to site B over a new single-mode fiber path. Users at site A report Intermittent connectivity Issues with applications hosted at site B. What is the reason for the problem?

- A. Heavy usage is causing high latency.
- B. An incorrect type of transceiver has been inserted into a device on the link.
- C. physical network errors are being transmitted between the two sites.
- D. The wrong cable type was used to make the connection.

---

**Answer: B**

---

Explanation:

---

**Question: 510**

---

Refer to the exhibit.



Switch A is newly configured. All VLANs are present in the VLAN database. The IP phone and PC A on Gi0/1 must be configured for the appropriate VLANs to establish connectivity between the PCs. Which command set fulfills the requirement?

A)

```
SwitchA(config-if)#switchport mode access
SwitchA(config-if)#switchport access vlan 50
SwitchA(config-if)#switchport voice vlan 51
```

B)

```
SwitchA(config-if)#switchport mode access
SwitchA(config-if)#switchport access vlan 50
SwitchA(config-if)#switchport voice vlan untagged
```

C)

```
SwitchA(config-if)#switchport mode trunk
SwitchA(config-if)#switchport trunk allowed vlan add 50, 51
SwitchA(config-if)#switchport voice vlan dot1p
```

D)

```
SwitchA(config-if)#switchport mode trunk
SwitchA(config-if)#switchport trunk allowed vlan 50, 51
SwitchA(config-if)#mls qos trust cos
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: A**

---

Explanation:

---

**Question: 511**

---

DRAG DROP

Drag and drop the characteristics of networking from the left onto the networking types on the right.

focused on network

focused on devices

user input is a configuration

user input is a policy

uses allow list security model

uses block list security model

Controller-Based Networking

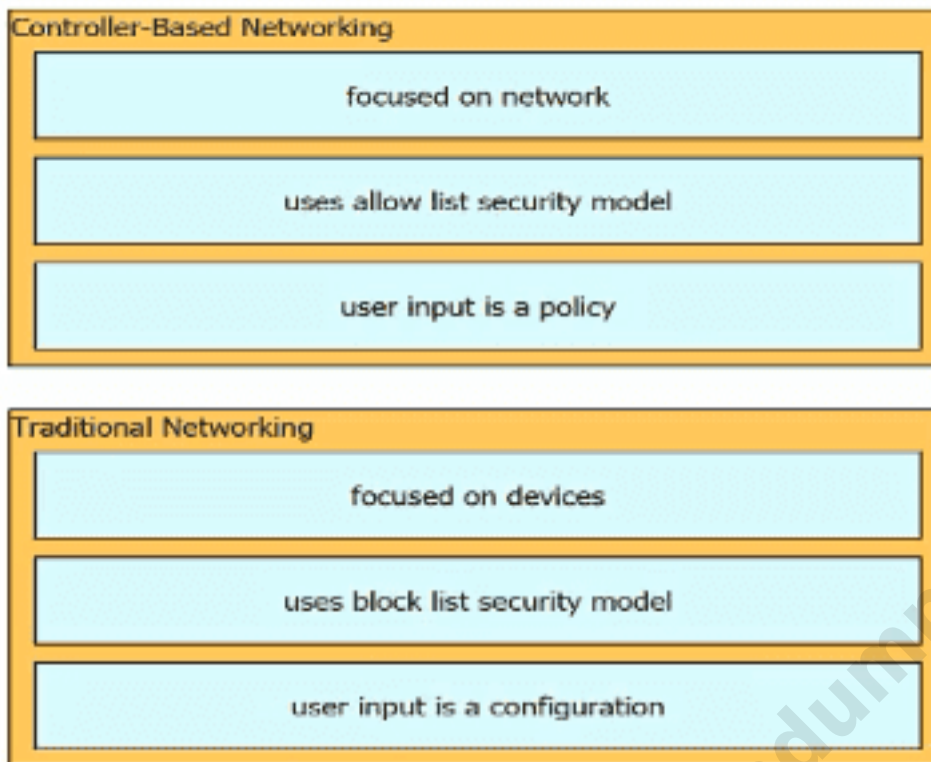
Traditional Networking

---

**Answer:**

---

Explanation:



---

**Question: 512**

---

An engineer must configure R1 for a new user account. The account must meet these requirements:

- \* It must be configured in the local database.
- \* The username is engineer.
- \* It must use the strongest password configurable. Which command must the engineer configure on the router?

- A. R1 (config)# username engineer2 algorithm-type scrypt secret test2021
- B. R1(config)# username engineer2 secret 5 password S1\$b1Ju\$kZbBS1Pyh4QzwXyZ
- C. R1(config)# username engineer2 privilege 1 password 7 test2021
- D. R1(config)# username engIneer2 secret 4 S1Sb1Ju\$kZbBS1Pyh4QzwXyZ

---

**Answer: B**

---

Explanation:

---

**Question: 513**

---

Refer to the exhibit.

```
R1#show run
!
router ospf 1
auto-cost reference-bandwidth 100000
!
interface GigabitEthernet0/0
bandwidth 10000000
!
interface GigabitEthernet0/1
bandwidth 100000000
!
interface GigabitEthernet0/2
ip ospf cost 100
!
interface GigabitEthernet0/3
ip ospf cost 1000
end
```

Router R1 resides in OSPF Area 0. After updating the R1 configuration to influence the paths that it will use to direct traffic, an engineer verified that each of the four Gigabit interfaces has the same route to 10.10.0.0/16. Which interface will R1 choose to send traffic to reach the route?

- A. GigabitEthernet0/0
- B. GigabitEthernet0/1
- C. GigabitEthernet0/2
- D. GigabitEthernet0/3

---

**Answer: B**

---

Explanation:

---

**Question: 514**

---



What is a requirement when configuring or removing LAG on a WLC?

- A. The Incoming and outgoing ports for traffic flow must be specified If LAG Is enabled.
- B. The controller must be rebooted after enabling or reconfiguring LAG.
- C. The management interface must be reassigned if LAG disabled.
- D. Multiple untagged interfaces on the same port must be supported.

---

**Answer: C**

---

Explanation:

---

**Question: 515**

---

Refer to the exhibit.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>Switch#show ip dhcp snooping Switch DHCP snooping is enabled Switch DHCP gleaning is disabled DHCP snooping is configured on following VLANs: 1 DHCP snooping is operational on following VLANs: 1 DHCP snooping is configured on the following L3 Interfaces: Insertion of option 82 is disabled circuit-id default format: vlan-mod-port remote-id: aabb.cc00.6500 (MAC) Option 82 on untrusted port is not allowed Verification of hwaddr field is enabled Verification of giaddr field is enabled DHCP snooping trust/rate is configured on the following Interfaces: Interface Trusted Allow option Rate limit (pps)</pre> | <pre>Switch#show ip dhcp snooping statistics detail Packets Processed by DHCP Snooping = 34 Packets Dropped Because IDB not known = 0 Queue full = 0 Interface is in errdisabled = 0 Rate limit exceeded = 0 Received on untrusted ports = 32 Nonzero giaddr = 0 Source mac not equal to chaddr = 0 No binding entry = 0 Insertion of opt82 fail = 0 Unknown packet = 0 Interface Down = 0 Unknown output interface = 0 Misdirected Packets = 0 Packets with Invalid Size = 0 Packets with Invalid Option = 0</pre> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The DHCP server and clients are connected to the same switch. What is the next step to complete the DHCP configuration to allow clients on VLAN 1 to receive addresses from the DHCP server?

- A. Configure the ip dhcp snooping trust command on the interlace that is connected to the DHCP client.
- B. Configure the ip dhcp relay information option command on the interface that is connected to the DHCP client.
- C. Configure the ip dhcp snooping trust command on the interface that is connected to the DHCP server.

D. Configure the Ip dhcp relay information option command on the interface that is connected to the DHCP server.

---

**Answer: C**

---

Explanation:

---

**Question: 516**

---

DRAG DROP

Drag and drop the Rapid PVST+ forwarding slate actions from the left to the right. Not all actions are used.

|                                                                           |        |
|---------------------------------------------------------------------------|--------|
| BPDUs received are forwarded to the system module.                        | action |
| BPDUs received from the system module are processed and transmitted.      | action |
| Frames received from the attached segment are discarded.                  | action |
| Frames received from the attached segment are processed.                  | action |
| Switched frames received from other ports are advanced.                   |        |
| The port in the forwarding state responds to network management messages. |        |

---

**Answer:**

---

Explanation:



BPDUs received are forwarded to the system module.

BPDUs received from the system module are processed and transmitted.

Frames received from the attached segment are discarded.

The port in the forwarding state responds to network management messages.

---

**Question: 517**

---

What provides centralized control of authentication and roaming In an enterprise network?

- A. a lightweight access point
- B. a firewall
- C. a wireless LAN controller
- D. a LAN switch

---

**Answer: C**

---

Explanation:

---

**Question: 518**

---

Refer to the exhibit.

```

R1# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate
       default
       U - per-user static route, o - ODR
Gateway of last resort is not set
C 192.168.3.5 is directly connected, Loopback0
  10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O   10.0.1.3/32 [110/100] via 192.168.0.40, 00:39:08, Serial0
C   10.0.1.0/24 is directly connected, Serial0
O   10.0.1.190/32 [110/5] via 192.168.0.35, 00:39:08, Serial0
O   10.0.1.0/24 [110/10] via 192.168.0.4, 00:39:08, Gigabit Ethernet 0/0
D   10.0.1.0/28 [90/10] via 192.168.0.7, 00:39:08, Gigabit Ethernet 0/0

```

Traffic sourced from the loopback0 Interface is trying to connect via ssh to the host at 10.0.1.15.  
What Is the next hop to the destination address?

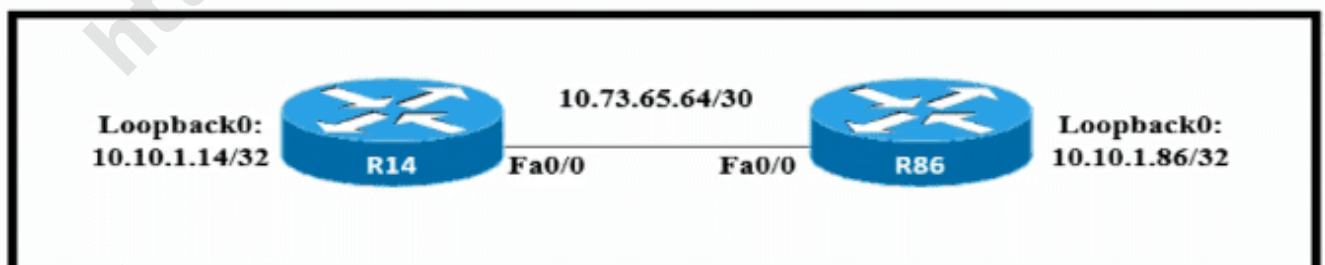
- A. 192.168.0.7
- B. 192.168.0.4
- C. 192.168.0.40
- D. 192.168.3.5

**Answer: B**

Explanation:

### Question: 519

Refer to the exhibit.



Which configuration allows routers R14 and R86 to form an OSPFv2 adjacency while acting as a central point for exchanging OSPF information between routers?

- A)



```
R14#
interface Loopback0
ip ospf 10 area 0

interface FastEthernet0/0
ip address 10.73.65.65 255.255.255.252
ip ospf network broadcast
ip ospf 10 area 0
ip mtu 1500

router ospf 10
ip ospf priority 255
router-id 10.10.1.14

R86#
interface Loopback0
ip ospf 10 area 0

interface FastEthernet0/0
ip address 10.73.65.66 255.255.255.252
ip ospf network broadcast
ip ospf 10 area 0
ip mtu 1500
```

B)

```
R14#
interface FastEthernet0/0
ip address 10.73.65.65 255.255.255.252
ip ospf network broadcast
ip ospf priority 255
ip mtu 1500

router ospf 10
router-id 10.10.1.14
network 10.10.1.14 0.0.0.0 area 0
network 10.73.65.64 0.0.0.3 area 0

R86#
interface FastEthernet0/0
ip address 10.73.65.66 255.255.255.252
ip ospf network broadcast
ip mtu 1500

router ospf 10
router-id 10.10.1.86
network 10.10.1.86 0.0.0.0 area 0
network 10.73.65.64 0.0.0.3 area 0
```

C)

```
R14#  
interface FastEthernet0/0  
ip address 10.73.65.65 255.255.255.252  
ip ospf network broadcast  
ip ospf priority 0  
ip mtu 1400  
  
router ospf 10  
router-id 10.10.1.14  
network 10.10.1.14 0.0.0.0 area 0  
network 10.73.65.64 0.0.0.3 area 0  
R86#  
interface Loopback0  
ip address 10.10.1.86 255.255.255.255
```

D)

```
R14#  
interface FastEthernet0/0  
ip address 10.73.65.65 255.255.255.252  
ip ospf network broadcast  
ip ospf priority 255  
ip mtu 1500  
  
router ospf 10  
router-id 10.10.1.14  
network 10.10.1.14 0.0.0.0 area 0  
network 10.73.65.64 0.0.0.3 area 0  
R86#  
interface FastEthernet0/0  
ip address 10.73.65.66 255.255.255.252  
ip ospf network broadcast  
ip mtu 1400  
  
router ospf 10  
router-id 10.10.1.86  
network 10.10.1.86 0.0.0.0 area 0  
network 10.73.65.64 0.0.0.3 area 0
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: D**

---

Explanation:

---

**Question: 520**

---

Refer to the exhibit.

```
R1# show ip route | begin gateway
Gateway of last resort is 209.165.200.254 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 209.165.200.254, Serial0/0/1
    is directly connected, Serial0/0/1
    172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
C    172.16.1.0/24 is directly connected, FastEthernet0/0
L    172.16.1.1/32 is directly connected, FastEthernet0/0
R    172.16.2.0/24 [120/2] via 207.165.200.250, 00:00:25, Serial0/0/0
O    192.168.1.0/24 [110/4437] via 207.165.200.254, 00:00:17, Serial0/0/1
D    192.168.2.0/24 [90/84437] via 207.165.200.254, 00:00:15, Serial0/0/1
    207.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
S    207.165.200.244/30 [1/1] via 207.165.200.254, Serial0/0/1
C    207.165.200.248/30 is directly connected, Serial0/0/0
L    207.165.200.249/32 is directly connected, Serial0/0/0
C    207.165.200.252/30 is directly connected, Serial0/0/1
L    207.165.200.253/32 is directly connected, Serial0/0/1
```

Which network prefix was learned via EIGRP?

- A. 172.16.0.0/16
- B. 192.168.2.0/24
- C. 207.165.200.0/24
- D. 192.168.1.0/24

---

**Answer: B**

---

Explanation:

---

**Question: 521**

---

DRAG DROP

Drag and drop the TCP or UDP details from the left onto their corresponding protocols on the right.

|                                                                                        |     |
|----------------------------------------------------------------------------------------|-----|
| transmitted based on data contained in the packet without the need for a data channel  | TCP |
| requires the client and the server to establish a connection before sending the packet |     |
| used to reliably share files between devices                                           | UDP |
| appropriate for streaming operations with minimal latency                              |     |

**Answer:**

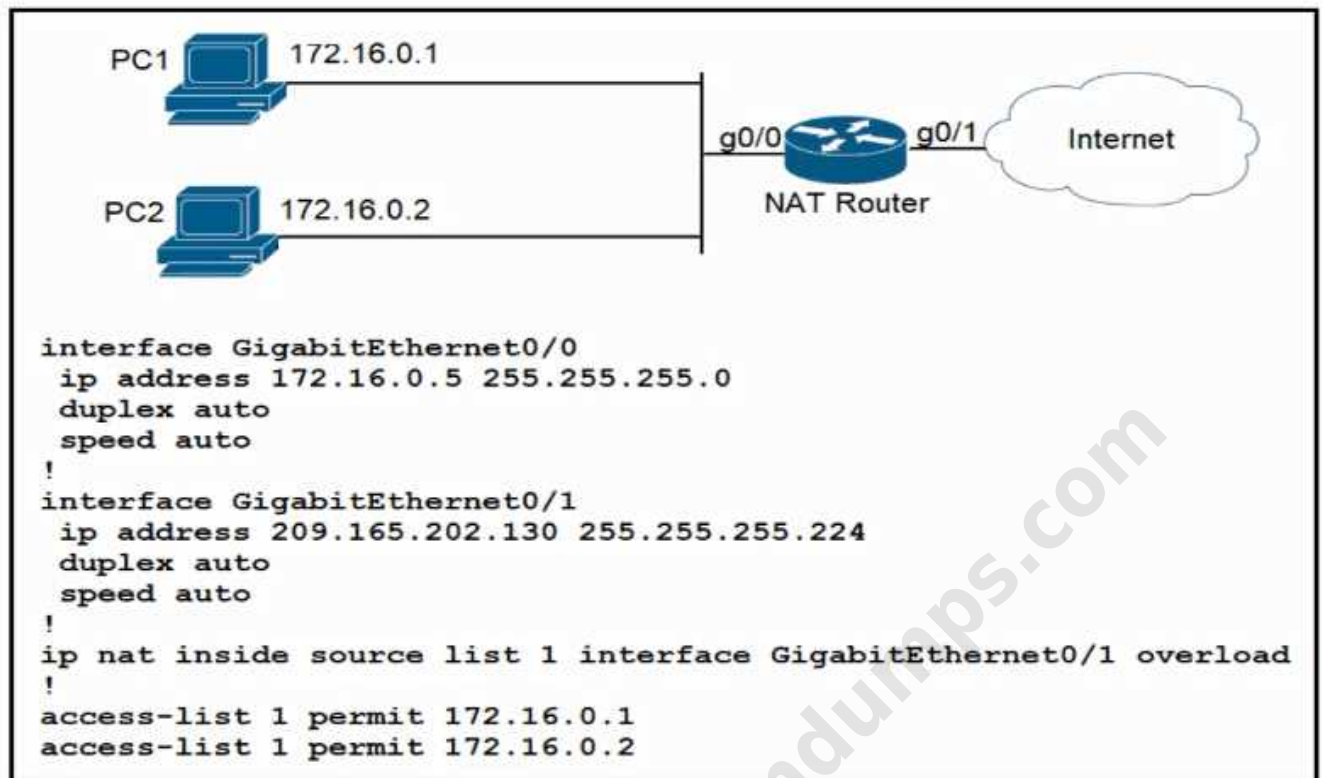
Explanation:

|                                                                                        |
|----------------------------------------------------------------------------------------|
| TCP                                                                                    |
| requires the client and the server to establish a connection before sending the packet |
| used to reliably share files between devices                                           |
| UDP                                                                                    |
| transmitted based on data contained in the packet without the need for a data channel  |
| appropriate for streaming operations with minimal latency                              |

**Question: 522**

Refer to the exhibit.





How should the configuration be updated to allow PC1 and PC2 access to the Internet?

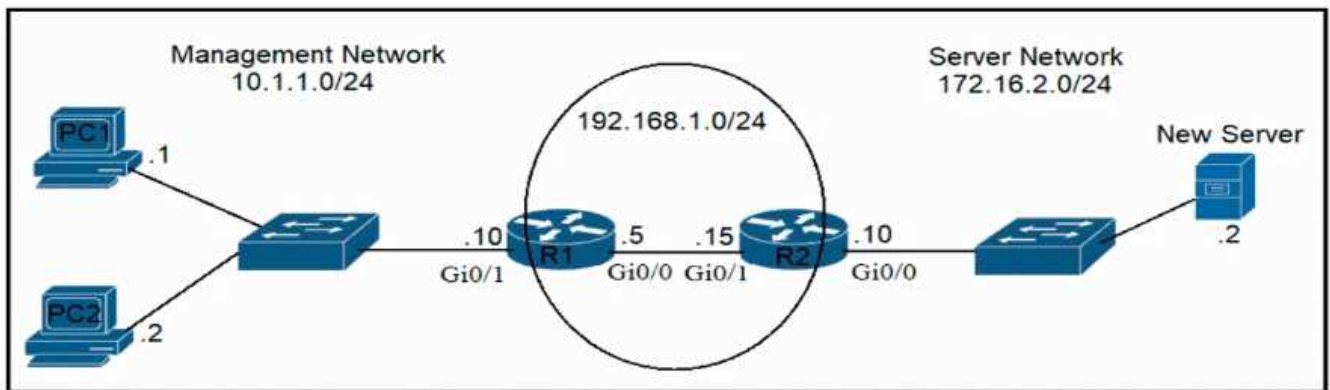
- A. Modify the configured number of the second access list.
- B. Add either the ip nat {inside|outside} command under both interfaces.
- C. Remove the overload keyword from the ip nat inside source command.
- D. Change the ip nat inside source command to use interface GigabitEthernet0/0.

**Answer: B**

Explanation:

**Question: 523**

Refer to the exhibit.



An engineer is updating the R1 configuration to connect a new server to the management network. The PCs on the management network must be blocked from pinging the default gateway of the new server. Which command must be configured on R1 to complete the task?

- A. R1(config)#ip route 172.16.2.2 255.255.255.248 gi0/1
- B. R1(config)#ip route 172.16.2.2 255.255.255.255 gi0/0
- C. R1(config)#ip route 172.16.2.0 255.255.255.0 192.168.1.15
- D. R1(config)#ip route 172.16.2.0 255.255.255.0 192.168.1.5

---

**Answer: C**

---

Explanation:

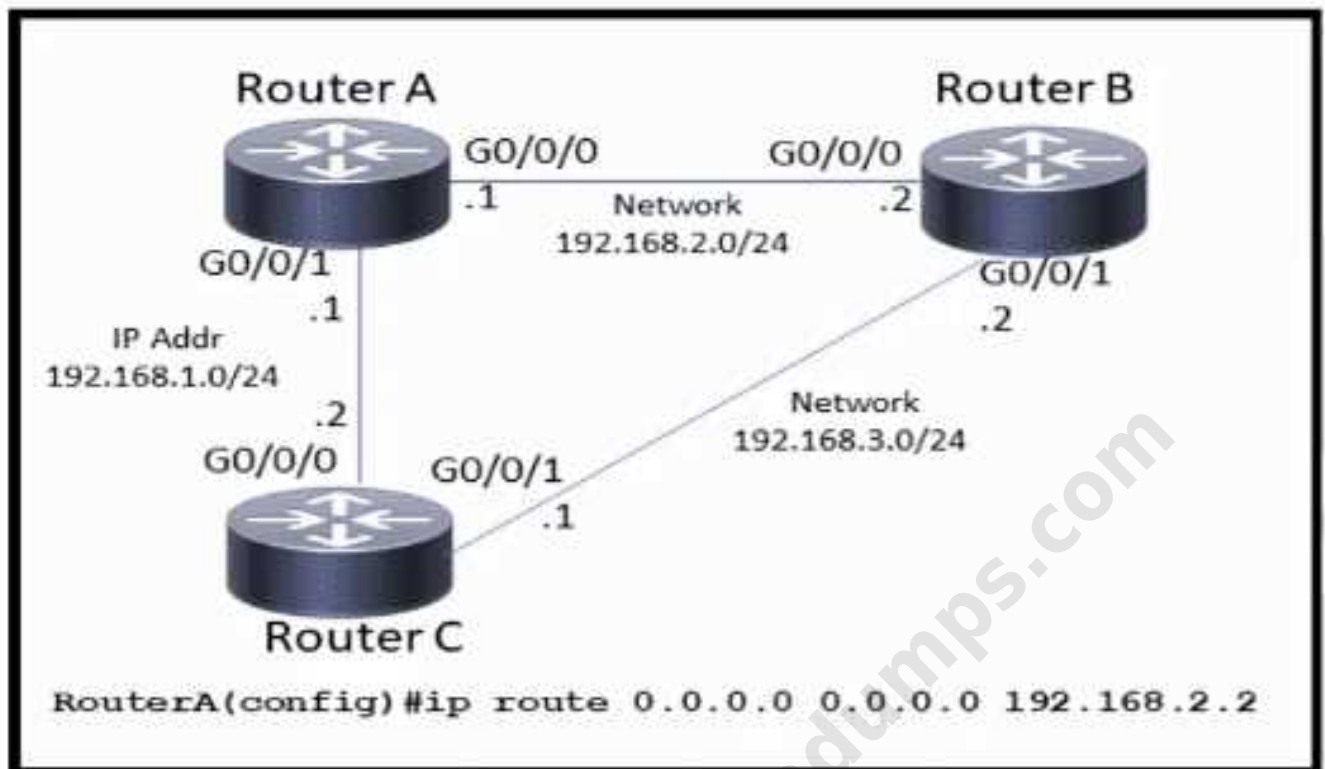
---

**Question: 524**

---

Refer to the exhibit.





Which command must be issued to enable a floating static default route on router A?

- A. `ip route 0.0.0.0 0.0.0.0 192.168.1.2`
- B. `ip default-gateway 192.168.2.1`
- C. `ip route 0.0.0.0 0.0.0.0 192.168.2.1 10`
- D. `ip route 0.0.0.0 0.0.0.0 192.168.1.2 10`

**Answer: D**

Explanation:

**Question: 525**

DRAG DROP

Drag and drop the descriptions of AAA services from the left onto the corresponding services on the right.

|                                          |                |
|------------------------------------------|----------------|
| allows the user to change to enable mode | Accounting     |
| limits the user's access permissions     |                |
| logs session statistics                  | Authentication |
| records user commands                    |                |
| secures access to routers                | Authorization  |
| validates user credentials               |                |

**Answer:**

Explanation:

|                                          |
|------------------------------------------|
| Accounting                               |
| records user commands                    |
| logs session statistics                  |
| Authentication                           |
| validates user credentials               |
| allows the user to change to enable mode |
| Authorization                            |
| limits the user's access permissions     |
| secures access to routers                |

**Question: 526**

Which protocol is used for secure remote CLI access?

- A. HTTPS
- B. HTTP
- C. Telnet
- D. SSH

---

**Answer: D**

---

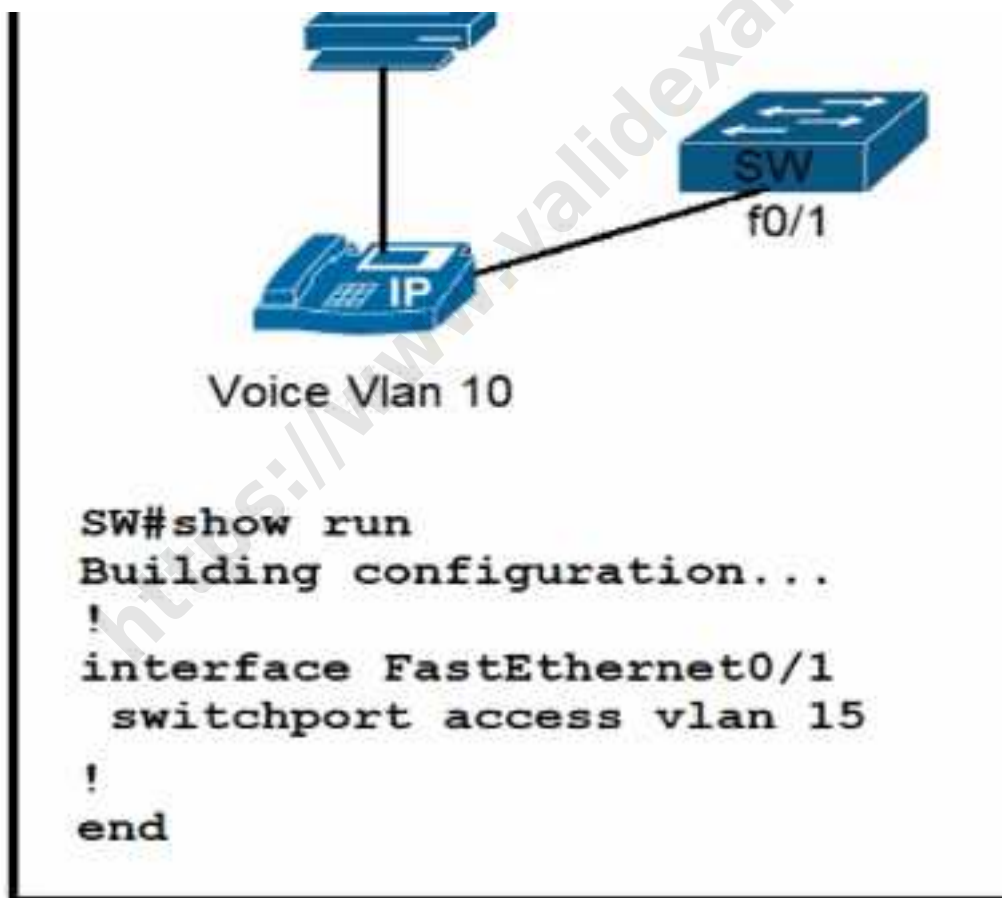
Explanation:

---

**Question: 527**

---

Refer to the exhibit.



All VLANs are present in the VLAN database. Which command sequence must be applied to complete the configuration?

- A. Interface FastEthernet0/1 switchport trunk native vlan 10 switchport trunk allowed vlan 10,15
- B. Interface FastEthernet0/1 switchport mode trunk switchport trunk allowed vlan 10,15
- C. interface FastEthernet0/1 switchport mode access switchport voice vlan 10
- D. Interface FastEthernet0/1 switchport trunk allowed vlan add 10 vlan 10 private-vlan isolated

---

**Answer: C**

---

Explanation:

---

**Question: 528**

---

Refer to the exhibit.

```
Router#show run
Building configuration...

Current configuration : 1530 bytes
!
! Last configuration change at 11:32:53 UTC Sat Oct 10 2020
upgrade fpd auto
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname Router
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
no ip icmp rate-limit unreachable
!
!
!
!
--More--
```

Which minimum configuration items are needed to enable Secure Shell version 2 access to R15?

A)

```
Router(config)#hostname R15
R15(config)#crypto key generate rsa general-keys modulus 1024
R15(config-line)#line vty 0 15
R15(config-line)# transport input ssh
R15(config)#ip ssh source-interface Fa0/0
R15(config)#ip ssh stricthostkeycheck
```

B)

```
Router(config)#crypto key generate rsa general-keys modulus 1024
Router(config)#ip ssh version 2
Router(config-line)#line vty 0 15
Router(config-line)# transport input ssh
Router(config)#ip ssh logging events
R15(config)#ip ssh stricthostkeycheck
```

C)

```
Router(config)#ip domain-name cisco.com
Router(config)#crypto key generate rsa general-keys modulus 1024
Router(config)#ip ssh version 2
Router(config-line)#line vty 0 15
Router(config-line)# transport input all
Router(config)#ip ssh logging events
```

D)

```
Router(config)#hostname R15
R15(config)#ip domain-name cisco.com
R15(config)#crypto key generate rsa general-keys modulus 1024
R15(config)#ip ssh version 2
R15(config-line)#line vty 0 15
R15(config-line)# transport input ssh
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: C**

---

Explanation:

---

### Question: 529

---

What is the purpose of the ip address dhcp command?

A. to configure an Interface as a DHCP server

B. to configure an interface as a DHCP helper

C. to configure an interface as a DHCP relay

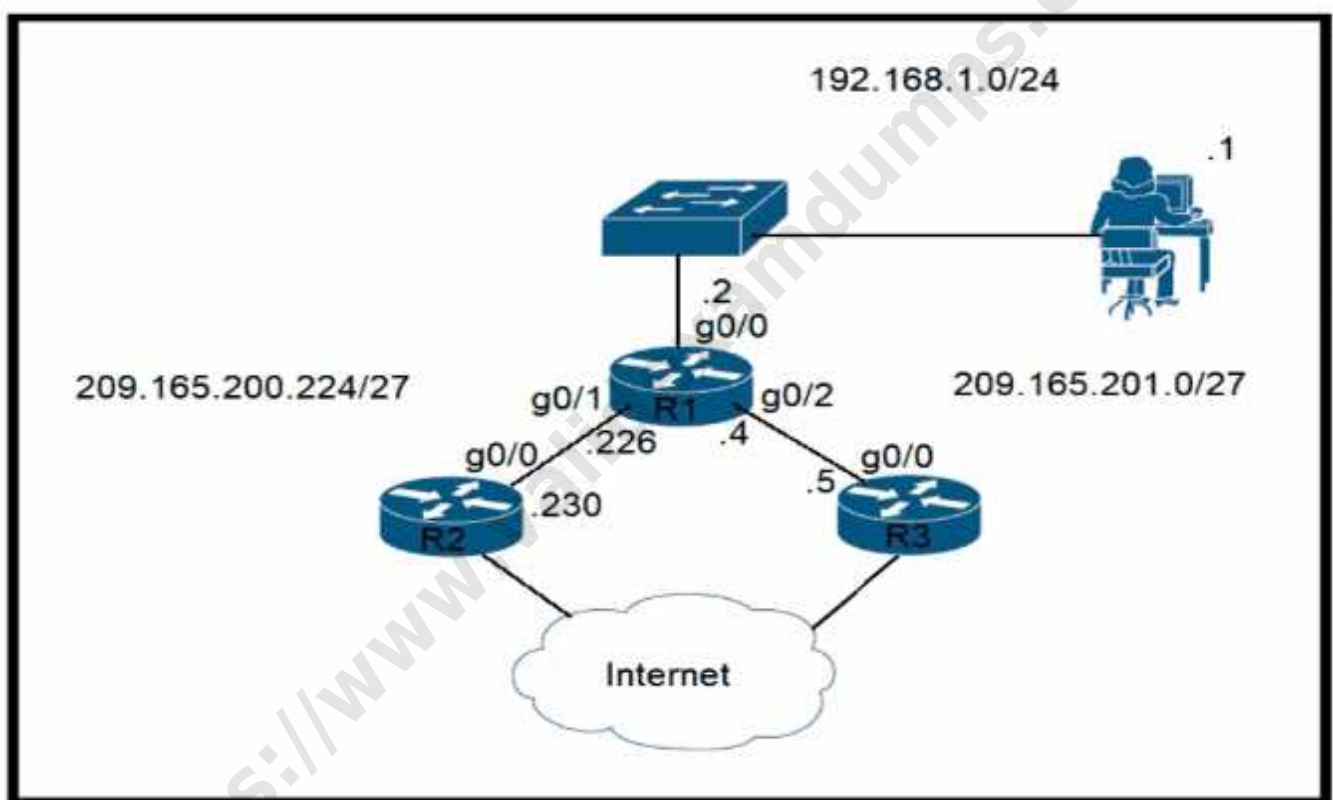
D. to configure an interface as a DHCP client

**Answer: D**

Explanation:

**Question: 530**

Refer to the exhibit.



Router R1 currently is configured to use R3 as the primary route to the Internet, and the route uses the default administrative distance settings. A network engineer must configure R1 so that it uses R2 as a backup, but only if R3 goes down. Which command must the engineer configure on R1 so that it correctly uses R2 as a backup route, without changing the administrative distance configuration on the link to R3?

- A. ip route 0.0.0.0 0.0.0.0 g0/1 1
- B. ip route 0.0.0.0 0.0.0.0 209.165.201.5 10
- C. ip route 0.0.0.0 0.0.0.0 209.165.200.226 1
- D. ip route 0,0.0.0 0.0.0.0 g0/1 6

---

**Answer: C**

---

Explanation:

---

**Question: 531**

---

Refer to the exhibit.

```
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname R4
!
boot-start-marker
boot-end-marker
!
ip cef
!
interface FastEthernet0/0
description WAN_INTERFACE
ip address 10.0.1.2 255.255.255.252
ip access-group 100 in
!
interface FastEthernet0/1
description LAN_INTERFACE
ip address 10.148.2.1 255.255.255.0
duplex auto
speed auto
!
ip forward-protocol nd
!
access-list 100 permit eigrp any any
access-list 100 permit icmp any any
access-list 100 permit tcp 10.149.3.0 0.0.0.255 host 10.0.1.2 eq 22
access-list 100 permit tcp any any eq 80
access-list 100 permit tcp any any eq 443
access-list 100 deny ip any any log
```

Which configuration enables DHCP addressing for hosts connected to interface FastEthernet0/1 on router R4?



- A. interface FastEthernet0/0ip helper-address 10.0.1.1access-list 100 permit udp host 10.0.1.1 eq bootps host 10.148.2.1
- B. interface FastEthernet0/1ip helper-address 10.0.1.1!access-list 100 permit tcp host 10.0.1.1 eq 67 host 10.148.2.1
- C. interface FastEthernet0/0ip helper-address 10.0.1.1access-list 100 permit host 10.0.1.1 host 10.148.2.1 eq bootps
- D. interface FastEthernet0/1ip helper-address 10.0.1.1!access-list 100 permit udp host 10.0.1.1 eq bootps host 10.148.2.1

---

**Answer: B**

---

Explanation:

---

**Question: 532**

---

OSPF must be configured between routers R1 and R2. Which OSPF configuration must be applied to router R1 to avoid a DR/BDR election?

- A. router ospf 1network 192.168.1.1 0.0.0.0 area 0interface e1/1ip address 192.168.1.1 255.255.255.252ip ospf network broadcast
- B. router ospf 1network 192.168.1.1 0.0.0.0 area 0interface e1/1ip address 192.168.1.1 255.255.255.252ip ospf network point-to-point
- C. router ospf 1network 192.168.1.1 0.0.0.0 area 0interface e1/1ip address 192.168.1.1 255.255.255.252ip ospf cost 0
- D. router ospf 1network 192.168.1.1 0.0.0.0 area 0hello interval 15interface e1/1ip address 192.168.1.1 255.255.255.252

---

**Answer: B**

---

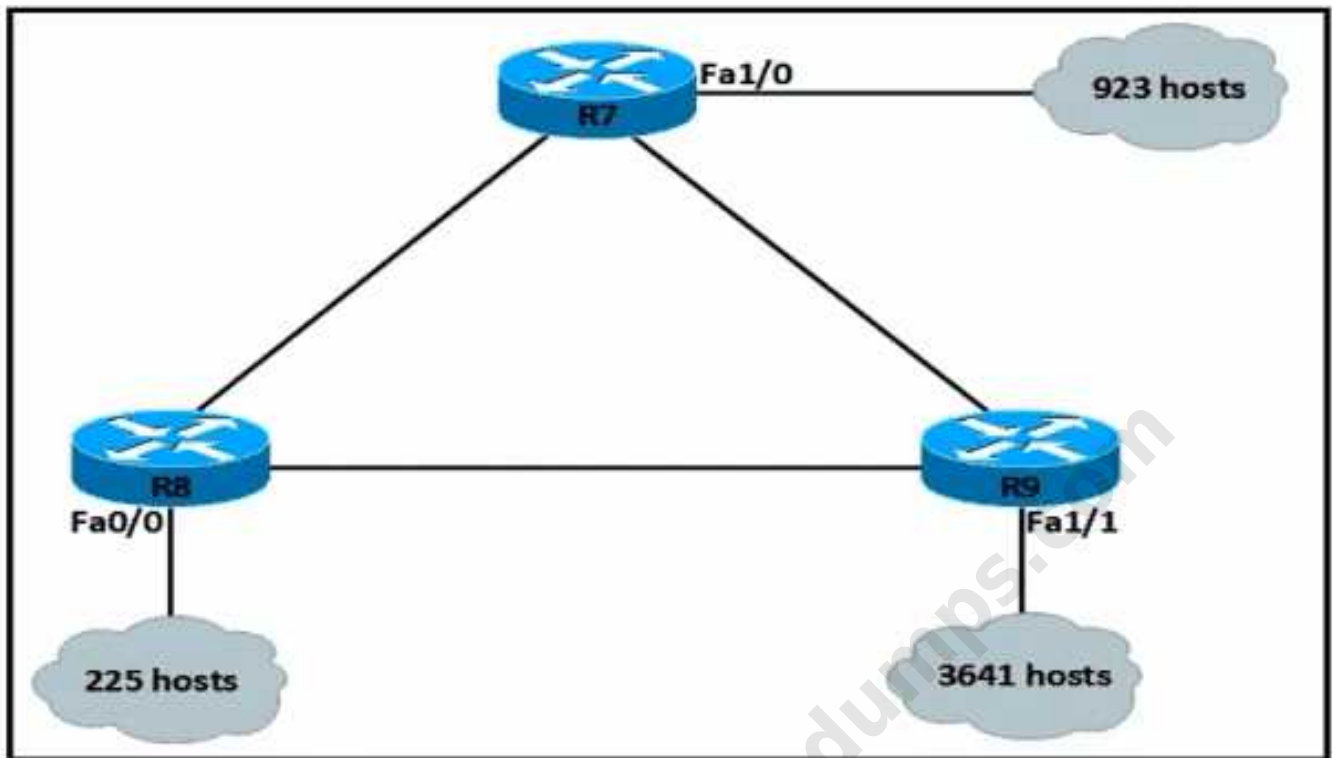
Explanation:

---

**Question: 533**

---

Refer to the exhibit.



An IP subnet must be configured on each router that provides enough addresses for the number of assigned hosts and anticipates no more than 10% growth for now hosts. Which configuration script must be used?

A)

```
R7#
configure terminal
interface Fa1/0
ip address 10.1.56.1 255.255.252.0
no shutdown

R8#
configure terminal
interface Fa0/0
ip address 10.9.32.1 255.255.255.0
no shutdown

R9#
configure terminal
interface Fa1/1
ip address 10.23.96.1 255.255.240.0
no shutdown
```

B)

```
R7#  
configure terminal  
interface Fa1/0  
ip address 10.1.56.1 255.255.248.0  
no shutdown
```

```
R8#  
configure terminal  
interface Fa0/0  
ip address 10.9.32.1 255.255.254.0  
no shutdown
```

```
R9#  
configure terminal  
interface Fa1/1  
ip address 10.23.96.1 255.255.248.0  
no shutdown
```

C)

```
R7#  
configure terminal  
interface Fa1/0  
ip address 10.1.56.1 255.255.240.0  
no shutdown
```

```
R8#  
configure terminal  
interface Fa0/0  
ip address 10.9.32.1 255.255.224.0  
no shutdown
```

```
R9#  
configure terminal  
interface Fa1/1  
ip address 10.23.96.1 255.255.192.0  
no shutdown
```

D)

```
R7#  
configure terminal  
interface Fa1/0  
ip address 10.1.56.1 255.255.192.0  
no shutdown  
  
R8#  
configure terminal  
interface Fa0/0  
ip address 10.9.32.1 255.255.224.0  
no shutdown  
  
R9#  
configure terminal  
interface Fa1/1  
ip address 10.23.96.1 255.255.128.0  
no shutdown
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: C**

---

Explanation:

---

**Question: 534**

---

Which wireless security protocol relies on Perfect Forward Secrecy?

- A. WPA3
- B. WPA
- C. WEP
- D. WPA2

---

**Answer: C**

---

Explanation:

---

**Question: 535**

---

What is a function of an endpoint on a network?

- A. forwards traffic between VLANs on a network
- B. connects server and client devices to a network
- C. allows users to record data and transmit to a file server
- D. provides wireless services to users in a building

---

**Answer: C**

---

Explanation:

An endpoint is a host that acts as the source or destination of data traffic flowing through a network.

When you are at your PC, editing your CV and uploading it to a file server, you are sitting at an endpoint.

---

**Question: 536**

---

Refer to the exhibit.

|              |                               |            |              |
|--------------|-------------------------------|------------|--------------|
| <b>EIGRP</b> | <b>10.10.10.0/24[90/1441]</b> | <b>via</b> | <b>F0/10</b> |
| <b>EIGRP</b> | <b>10.10.10.0/24[90/144]</b>  | <b>via</b> | <b>F0/11</b> |
| <b>EIGRP</b> | <b>10.10.10.0/24[90/1441]</b> | <b>via</b> | <b>F0/12</b> |
| <b>OSPF</b>  | <b>10.10.10.0/24[110/20]</b>  | <b>via</b> | <b>F0/13</b> |
| <b>OSPF</b>  | <b>10.10.10.0/24[110/30]</b>  | <b>via</b> | <b>F0/14</b> |

Packets received by the router from BGP enter via a serial interface at 209.165.201.10. Each route is present within the routing table. Which interface is used to forward traffic with a destination IP of 10.10.10.24?

- A. F0/10
- B. F0/11

C. F0/12

D. F0/13

---

**Answer: B**

---

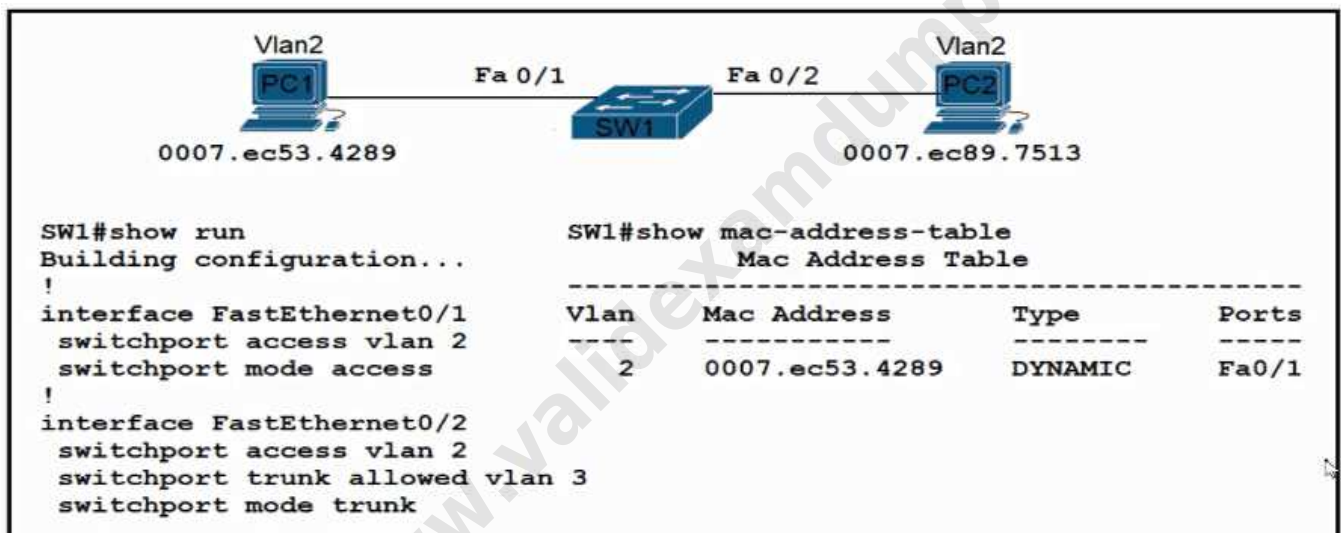
Explanation:

---

**Question: 537**

---

Refer to the exhibit.



An engineer has started to configure replacement switch SW1. To verify part of the configuration, the engineer issued the commands as shown and noticed that the entry for PC2 is missing. Which change must be applied to SW1 so that PC1 and PC2 communicate normally?

A)

```
SW1(config)#interface fa0/2
SW1(config-if)#no switchport mode trunk
SW1(config-if)#no switchport trunk allowed vlan 3
SW1(config-if)#switchport mode access
```

B)

```
SW1(config)#interface fa0/1
SW1(config-if)#no switchport access vlan 2
SW1(config-if)#switchport trunk native vlan 2
SW1(config-if)#switchport trunk allowed vlan 3
```

C)

```
SW1(config)#interface fa0/1
SW1(config-if)#no switchport access vlan 2
SW1(config-if)#switchport access vlan 3
SW1(config-if)#switchport trunk allowed vlan 2
```

D)

```
SW1(config)#interface fa0/2
SW1(config-if)#no switchport access vlan 2
SW1(config-if)#no switchport trunk allowed vlan 3
SW1(config-if)#switchport trunk allowed vlan 2
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: A**

---

Explanation:

---

**Question: 538**

---

DRAG DROP

Drag and drop the threat-mitigation techniques from the left onto the types of threat or attack they mitigate on the right.

|                                              |                                               |
|----------------------------------------------|-----------------------------------------------|
| configure the BPDU guard feature             | 802.1q double tagging                         |
| configure the dynamic ARP inspection feature | ARP spoofing                                  |
| configure the root guard feature             | unwanted superior BPDUs                       |
| configure a VLAN access control list         | unwanted BPDUs on PortFast-enabled interfaces |

---

**Answer:**

---

Explanation:

configure a VLAN access control list

configure the dynamic ARP inspection feature

configure the root guard feature

configure the BPDU guard feature

Topic 4, Exam Pool D (NEW)

---

**Question: 539**

---

What is a zero-day exploit?

- A. It is when a new network vulnerability is discovered before a fix is available
- B. It is when the perpetrator inserts itself in a conversation between two parties and captures or alters data.
- C. It is when the network is saturated with malicious traffic that overloads resources and bandwidth
- D. It is when an attacker inserts malicious code into a SQL server.

---

**Answer: A**

---

Explanation:

<https://www.kaspersky.com/resource-center/definitions/zero-day-exploit>



---

**Question: 540**

---

A network engineer is replacing the switches that belong to a managed-services client with new Cisco Catalyst switches. The new switches will be configured for updated security standards, including replacing Telnet services with encrypted connections and doubling the modulus size from 1024. Which two commands must the engineer configure on the new switches? (Choose two.)

- A. crypto key generate rsa general-keys modulus 1024
- B. transport input all
- C. crypto key generate rsa usage-keys
- D. crypto key generate rsa modulus 2048
- E. transport Input ssh

---

**Answer: A,E**

---

Explanation:

---

**Question: 541**

---

Which QoS queuing method discards or marks packets that exceed the desired bit rate of traffic flow?

- A. shaping
- B. policing
- C. CBWFQ
- D. LLQ

---

**Answer: B**

---

Explanation:

---

**Question: 542**

---

What is the role of disaggregation in controller-based networking?

- A. It divides the control-plane and data-plane functions.
- B. It summarizes the routes between the core and distribution layers of the network topology.
- C. It enables a network topology to quickly adjust from a ring network to a star network
- D. It streamlines traffic handling by assigning individual devices to perform either Layer 2 or Layer 3 functions.

---

**Answer: A**

---

Explanation:

---

**Question: 543**

---

DRAG DROP

Drag and drop the IPv6 address details from the left onto the corresponding types on the right.

identifies an interface on an IPv6 device

includes link-local and loopback addresses

provides one-to-many communications

used exclusively by a non-host device

assigned to more than one interface

derived from the FF00::/8 address range

Anycast

Multicast

Unicast

Explanation:

Answer:

**Anycast**

provides one-to-many communications

used exclusively by a non-host device

**Multicast**

assigned to more than one interface

derived from the FF00::/8 address range

**Unicast**

identifies an interface on an IPv6 device

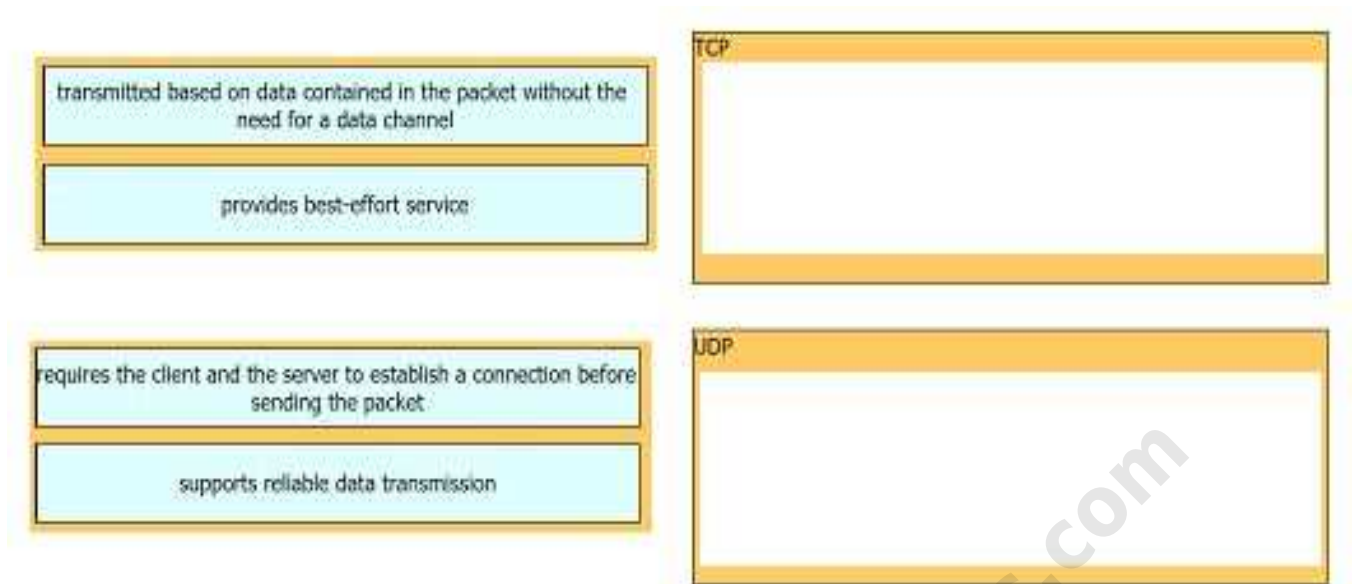
includes link-local and loopback addresses

---

**Question: 544**

DRAG DROP

Drag and drop the TCP or UDP details from the left onto their corresponding protocols on the right.

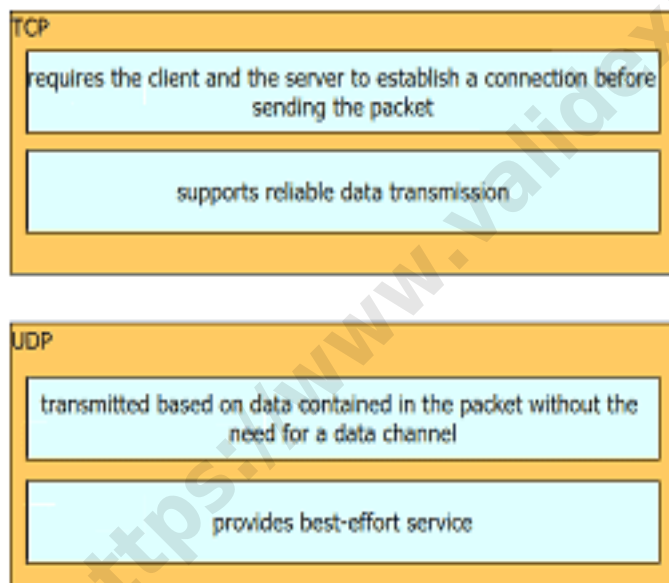


---

**Answer:**

---

Explanation:



---

**Question: 545**

---

Refer to the exhibit.

```
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set
 10.0.0.0/24 is subnetted, 5 subnets
D    10.1.2.0/24 [90/2170112] via 10.165.20.226, 00:01:30, Serial0/0
D    10.1.3.0/24 [90/2170112] via 10.165.20.226, 00:01:30, Serial0/0
D    10.1.2.0/25 [90/2170112] via 10.165.20.126, 00:01:30, Serial0/0
D    10.1.3.0/25 [90/2170112] via 10.165.20.146, 00:01:30, Serial0/0
D    10.1.4.0/25 [90/2170112] via 10.165.20.156, 00:01:30, Serial0/0
 192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.18.10.0/24 is directly connected, GigabitEthernet0/0
 192.168.21.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.11.0/24 is directly connected, GigabitEthernet0/1
 10.165.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    10.165.20.224/24 is directly connected, Serial0/0
S    10.1.2.112/28 [1/0] via 10.165.20.166
```

What is the next hop for traffic entering R1 with a destination of 10.1.2 126?

- A. 10.165 20.126
- B. 10.165.20.146
- C. 10.165.20.166
- D. 10.165 20.226

---

**Answer: D**

---

Explanation:

---

**Question: 546**

---

Refer to the exhibit.

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  : 
Description . . . . . : Realtek PCIe GBE Family
Controller
Physical Address. . . . . : 3C-52-82-33-F3-8F
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . : Yes

Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix  : arcep.se
Description . . . . . : Intel(R) Dual Band
Wireless-AC 7265
Physical Address. . . . . : C8-21-58-B4-F3-EF
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . : Yes
Link-local IPv6 Address . . . . : fe80::45a1:b3fa:2f37:bf37%2 (Preferred)
IPv4 Address. . . . . : 192.168.1.226 (Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : October 3, 2019 12:28:08 PM
Lease Expires . . . . . : October 3, 2019 7:18:37 PM
Default Gateway . . . . . : 192.168.1.100
DHCP Server . . . . . : 192.168.1.254
DHCPv6 IAID . . . . . : 46670168
DHCPv6 Client DUID. . . . . : 00-01-00-01-20-FF-05-55-3C-52-82-33-D3-84
DNS Servers . . . . . : 192.168.1.253
NetBIOS over Tcpip. . . . . : Enabled
Connection-specific DNS Suffix Search List :
arcep.se
```

The given Windows PC is requesting the IP address of the host at [www.cisco.com](http://www.cisco.com). To which IP address is the request sent?

- A. 192.168.1.226
- B. 192.168.1.100
- C. 192.168.1.254
- D. 192.168.1.253

---

**Answer: D**

---

Explanation:

---

**Question: 547**

---

Why would VRRP be implemented when configuring a new subnet in a multivendor environment?

- A. when a gateway protocol is required that support more than two Cisco devices for redundancy
- B. to enable normal operations to continue after a member failure without requiring a change in a host ARP cache
- C. to ensure that the spanning-tree forwarding path to the gateway is loop-free
- D. to interoperate normally with all vendors and provide additional security features for Cisco devices

---

**Answer: A**

---

Explanation:

---

**Question: 548**

---

An engineer has configured the domain name, user name, and password on the local router. What is the next step to complete the configuration for a Secure Shell access RSA key?

- A. crypto key Import rsa pem
- B. crypto key pubkey-chain rsa
- C. crypto key generate rsa
- D. crypto key zeroize rsa

---

**Answer: C**

---

Explanation:

---

**Question: 549**

---

An engineer is configuring SSH version 2 exclusively on the R1 router. What is the minimum configuration required to permit remote management using the cryptographic protocol?



<https://www.validexamdumps.com>

● hostname R1

ip domain name cisco

crypto key generate rsa general-keys modulus 1024

username cisco privilege 15 password 0 cisco123

ip ssh version 2

line vty 0 15

transport input ssh

login local

● hostname R1

crypto key generate rsa general-keys modulus 1024

username cisco privilege 15 password 0 cisco123

ip ssh version 2

line vty 0 15

transport input all

login local

● hostname R1

service password-encryption

crypto key generate rsa general-keys modulus 1024

username cisco privilege 15 password 0 cisco123

ip ssh version 2

line vty 0 15

transport input ssh

login local

● hostname R1

ip domain name cisco

crypto key generate rsa general-keys modulus 1024

username cisco privilege 15 password 0 cisco123

ip ssh version 2

line vty 0 15

transport input all

login local

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: C**

---

Explanation:

---

**Question: 550**

---

After a recent security breach and a RADIUS failure, an engineer must secure the console port of each enterprise router with a local username and password. Which configuration must the engineer apply to accomplish this task?

- ☐ `aaa new-model`  
`line con 0`  
`password plaintextpassword`  
`privilege level 15`
- ☐ `username localuser secret plaintextpassword`  
`line con 0`  
`login authentication default`  
`privilege level 15`
- ☐ `username localuser secret plaintextpassword`  
`line con 0`  
`no login local`  
`privilege level 15`
- ☐ `aaa new-model`  
`aaa authorization exec default local`  
`aaa authentication login default radius`  
`username localuser privilege 15 secret plaintextpassword`

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: B**

---

Explanation:

---

**Question: 551**

---

DRAG DROP

Drag and drop each characteristic of device-management technologies from the left onto the deployment type on the right.

The interface consists of two columns. The left column contains six light blue boxes with orange borders, each containing a characteristic of device-management technologies. The right column contains two larger yellow boxes with orange borders, representing deployment types. The characteristics are:

- relies on per-device management
- uses individual software management
- orchestrates background device configuration
- supports open APIs
- provides greater flexibility for custom and non-standard configurations
- supports centralized software management

The deployment types are:

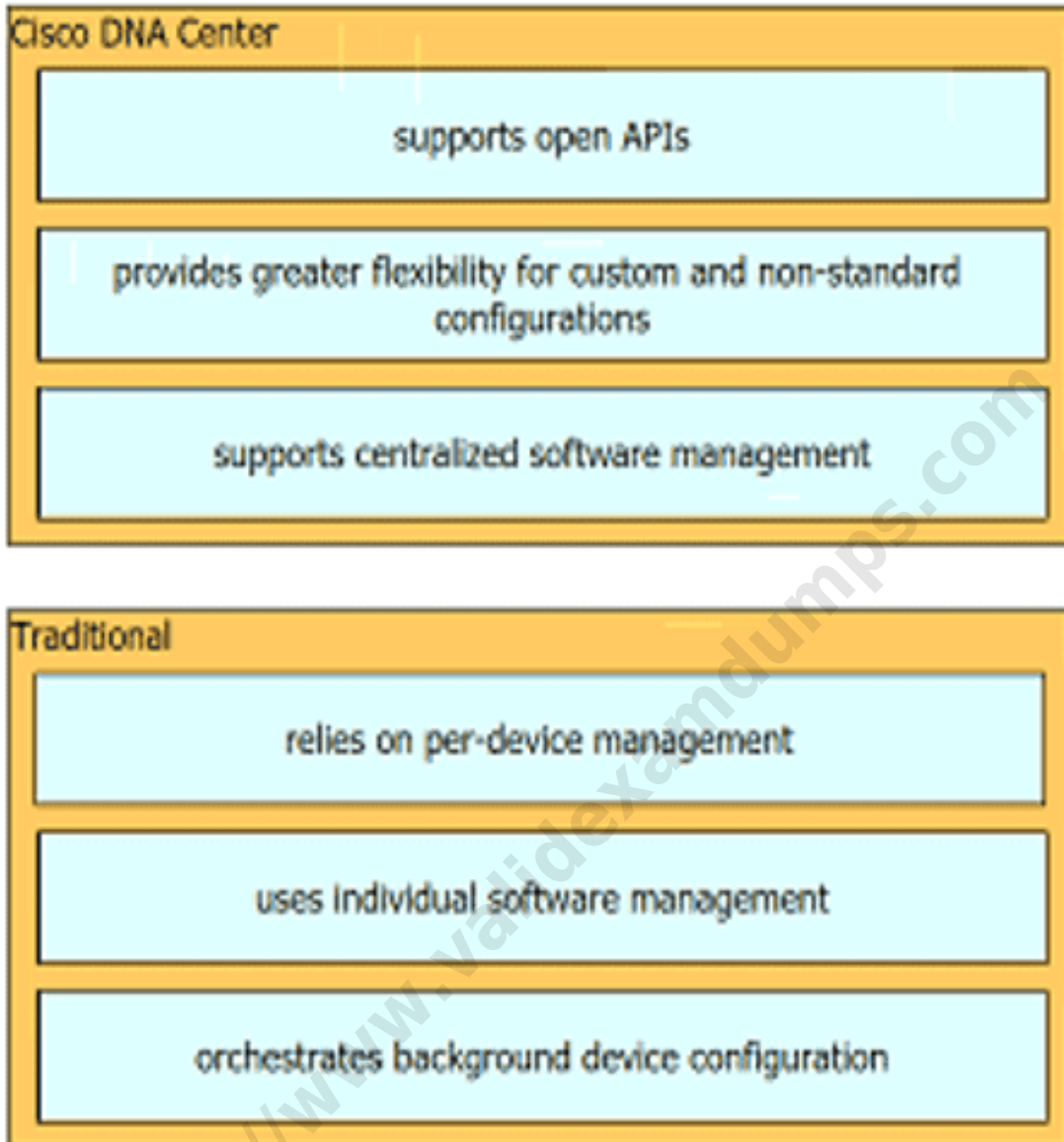
- Cisco DNA Center
- Traditional

---

**Answer:**

---

Explanation:



---

**Question: 552**

---

Which REST method updates an object in the Cisco DNA Center Intent API?

- A. CHANGE
- B. UPDATE
- C. POST
- D. PUT

---

**Answer: D**

---

Explanation:

PUT is most-often utilized for **\*\*update\*\*** capabilities, PUT-ing to a known resource URI with the request body containing the newly-updated representation of the original resource.

However, PUT can also be used to create a resource in the case where the resource ID is chosen by the client instead of by the server. In other words, if the PUT is to a URI that contains the value of a non-existent resource ID. Again, the request body contains a resource representation. Many feel this is convoluted and confusing. Consequently, this method of creation should be used sparingly, if at all.

Alternatively, use POST to create new resources and provide the client-defined ID in the body representation—presumably to a URI that doesn't include the ID of the resource (see POST below).

On successful update, return 200 (or 204 if not returning any content in the body) from a PUT. If using PUT for create, return HTTP status 201 on successful creation. A body in the response is optional—providing one consumes more bandwidth. It is not necessary to return a link via a Location header in the creation case since the client already set the resource ID.

PUT is not a safe operation, in that it modifies (or creates) state on the server, but it is idempotent. In other words, if you create or update a resource using PUT and then make that same call again, the resource is still there and still has the same state as it did with the first call.

If, for instance, calling PUT on a resource increments a counter within the resource, the call is no longer idempotent. Sometimes that happens and it may be enough to document that the call is not idempotent. However, it's recommended to keep PUT requests idempotent. It is strongly recommended to use POST for non-idempotent requests.

Examples:

<https://www.restapitutorial.com/lessons/httpmethods.html>

---

**Question: 553**

---

Which two practices are recommended for an acceptable security posture in a network? (Choose two)

- A. Backup device configurations to encrypted USB drives for secure retrieval
- B. maintain network equipment in a secure location

- C. Use a cryptographic keychain to authenticate to network devices
- D. Place internal email and file servers in a designated DMZ
- E. Disable unused or unnecessary ports, interfaces and services

---

**Answer: C,E**

---

Explanation:

---

**Question: 554**

---

An administrator must use the password complexity not manufacturer-name command to prevent users from adding "cisco" as a password. Which command must be issued before this command?

- A. Password complexity enable
- B. confreg 0x2142
- C. Login authentication my-auth-list
- D. service password-encryption

---

**Answer: A**

---

Explanation:

---

**Question: 555**

---

An engineer is configuring router R1 with an IPv6 static route for prefix 2019:C15C:0CAF:E001::/64. The next hop must be 2019:C15C:0CAF:E002::1 The route must be reachable via the R1 Gigabit 0/0 interface. Which command configures the designated route?

- A. R1(config)#ipv6 route 2019:C15C:0CAF:E001::/64 2019:C15C:0CAF:E002::1
- B. R1(config-if)#ipv6 route 2019:C15C:0CAF:E001::/64 2019:C15C:0CAF:E002::1

C. R1(config-if)#ip route 2019:C15C:0CAF:E001::/64 GigabitEthernet0/0

D. R1(config)#ip route 2019:C15C:0CAF:E001::/64 GigabitEthernet0/0

---

**Answer: C**

---

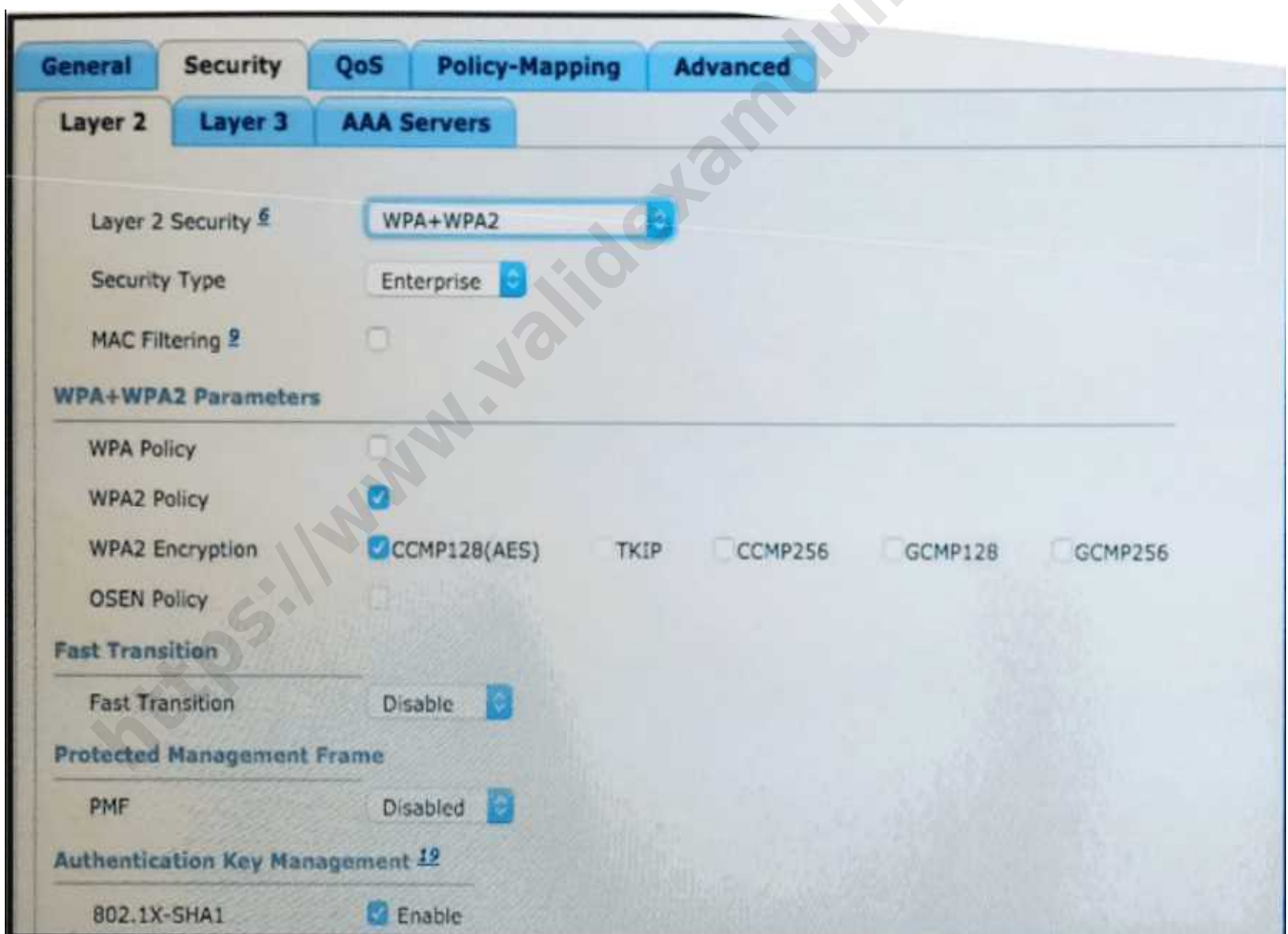
Explanation:

---

**Question: 556**

---

Refer to the exhibit.



What must be configured to enable 802.11w on the WLAN?

A. Set PMF to Required.



- B. Enable MAC Filtering.
- C. Enable WPA Policy.
- D. Set Fast Transition to Enabled

---

**Answer: A**

---

Explanation:

---

**Question: 557**

---

Refer to the exhibit.

Security

- AAA
  - General
  - RADIUS
    - Authentication
    - Accounting
    - Fallback
    - DNS
    - Downloaded AVP
  - TACACS+
    - LDAP
    - Local Net Users
    - MAC Filtering
  - Disabled Clients
    - User Login Policies
    - AP Policies
    - Password Policies
  - Local EAP
  - Advanced EAP
  - Priority Order
  - Certificate
  - Access Control Lists
  - Wireless Protection Policies

RADIUS Authentication Servers > New

Server Index (Priority): 1

Server IP Address (IPv4/IPv6): 192.168.25.2

Shared Secret Format: ASCII

Shared Secret: \*\*\*\*\*

Confirm Shared Secret: \*\*\*\*\*

Key Wrap: ☐ (Designed for FIPS customers and requires a key wrap compliant RADIUS server)

Port Number: 1812

Server Status: Enabled

Support for CoA: Disabled

Server Timeout: 2 seconds

Network User: ☐ Enable

Management: ☐ Enable

Management Retransmit Timeout: 2 seconds

Tunnel Proxy: ☐ Enable

IPSec: ☐ Enable

A network engineer configures the Cisco WLC to authenticate local wireless clients against a RADIUS server Which task must be performed to complete the process?

- A. Change the Server Status to Disabled
- B. Select Enable next to Management

- C. Select Enable next to Network User
- D. Change the Support for CoA to Enabled.

---

**Answer: C**

---

Explanation:

---

**Question: 558**

---

DRAG DROP

Drag and drop the elements of a security program from the left onto the corresponding descriptions on the right.

|                   |                                                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| awareness         | document that outlines an organization's security goals and practices and the roles and responsibilities of the organization's personnel |
| education         | tactical document that sets out specific tasks and methods to maintain security                                                          |
| security policy   | user-awareness learning level that focuses on learning about topics and practices beyond what is typically required by the user's job    |
| security standard | user-awareness learning level that focuses on security practices that all employees must understand and enforce                          |
| training          | user-awareness learning level that focuses on teaching employees how to perform tasks specifically required by their jobs                |

---

**Answer:**

---

Explanation:

|                   |                                                                                                                                          |                           |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| awareness         | document that outlines an organization's security goals and practices and the roles and responsibilities of the organization's personnel | <i>Security Standards</i> |
| education         | tactical document that sets out specific tasks and methods to maintain security                                                          | <i>Security Policy</i>    |
| security policy   | user-awareness learning level that focuses on learning about topics and practices beyond what is typically required by the user's job    | <i>Education</i>          |
| security standard | user-awareness learning level that focuses on security practices that all employees must understand and enforce                          | <i>Awareness</i>          |
| training          | user-awareness learning level that focuses on teaching employees how to perform tasks specifically required by their jobs                | <i>Training</i>           |

<https://www.ciscopress.com/articles/article.asp?p=1998559&seqNum=3>

### Question: 559

#### DRAG DROP

Drag and drop the Cisco IOS attack mitigation features from the left onto the types of network attack they mitigate on the right.

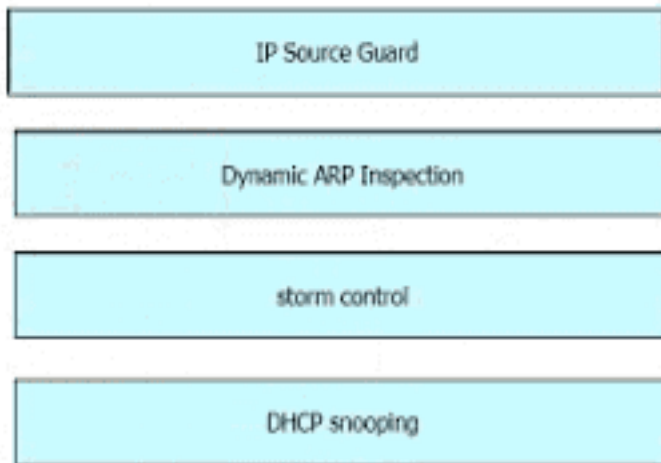
|                        |                                           |
|------------------------|-------------------------------------------|
| DHCP snooping          | rogue server that spoofs IP configuration |
| Dynamic ARP Inspection | cache poisoning                           |
| IP Source Guard        | flood attacks                             |
| storm control          | rogue clients on the network              |

---

**Answer:**

---

Explanation:



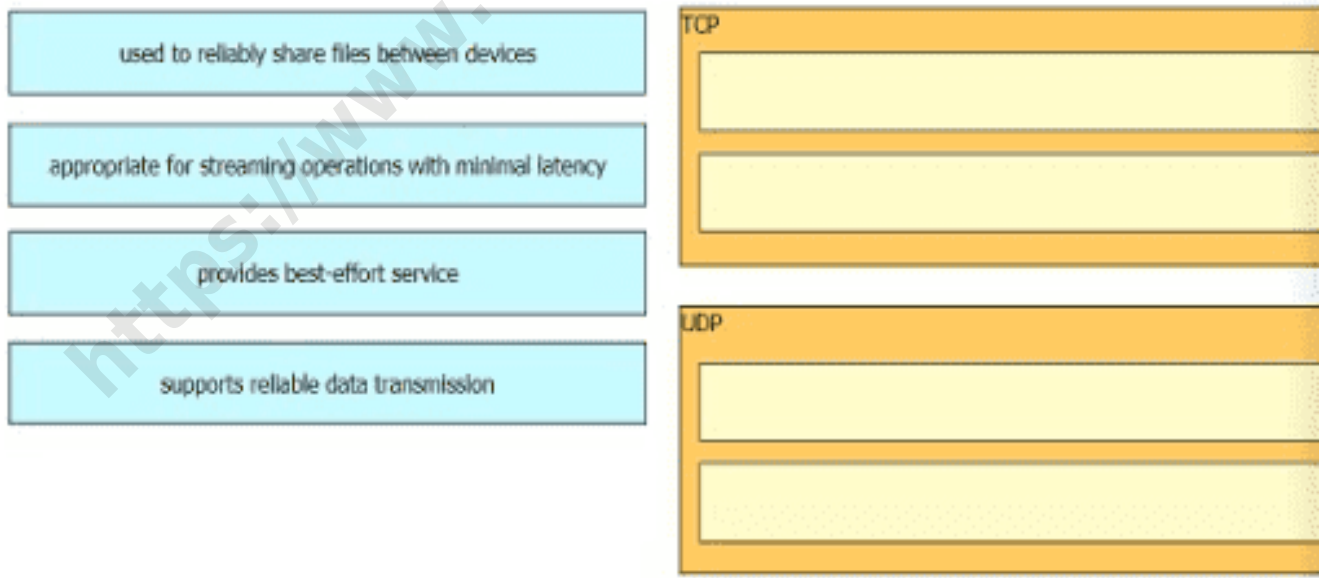
---

**Question: 560**

---

DRAG DROP

Drag and drop the TCP or UDP details from the left onto their corresponding protocols on the right.

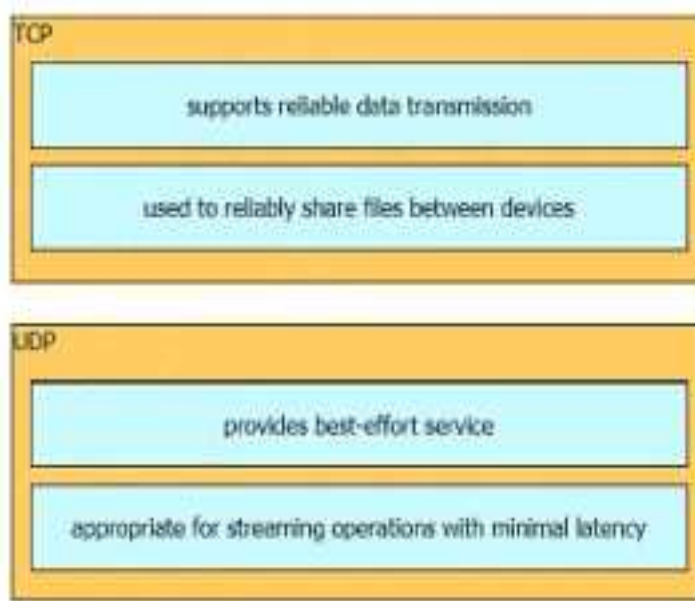


---

**Answer:**

---

Explanation:



---

**Question: 561**

---

What is a function of Cisco Advanced Malware Protection for a Next-Generation IPS?

- A. authorizing potentially compromised wireless traffic
- B. inspecting specific files and file types for malware
- C. authenticating end users
- D. URL filtering

---

**Answer: B**

---

Explanation:

AMP gives you real-time blocking of malware and advanced sandboxing, that is backed up by world class global threat intelligence, to provide rapid detection, containment and removal of advanced malware

<https://www.cisco.com/c/en/us/products/security/amp-appliances/index.html>

---

**Question: 562**

---

Refer to the exhibit.

```
{  
  "SW1" : ["Ten-GigabitEthernet0/0", "Ten-GigabitEthernet0/1"],  
  "SW2" : ["Ten-GigabitEthernet0/0", "Ten-GigabitEthernet0/1"],  
  "SW3" : ["Ten-GigabitEthernet0/0", "Ten-GigabitEthernet0/1"],  
  "SW4" : ["Ten-GigabitEthernet0/0", "Ten-GigabitEthernet0/1"]  
}
```

How many JSON objects are represented?

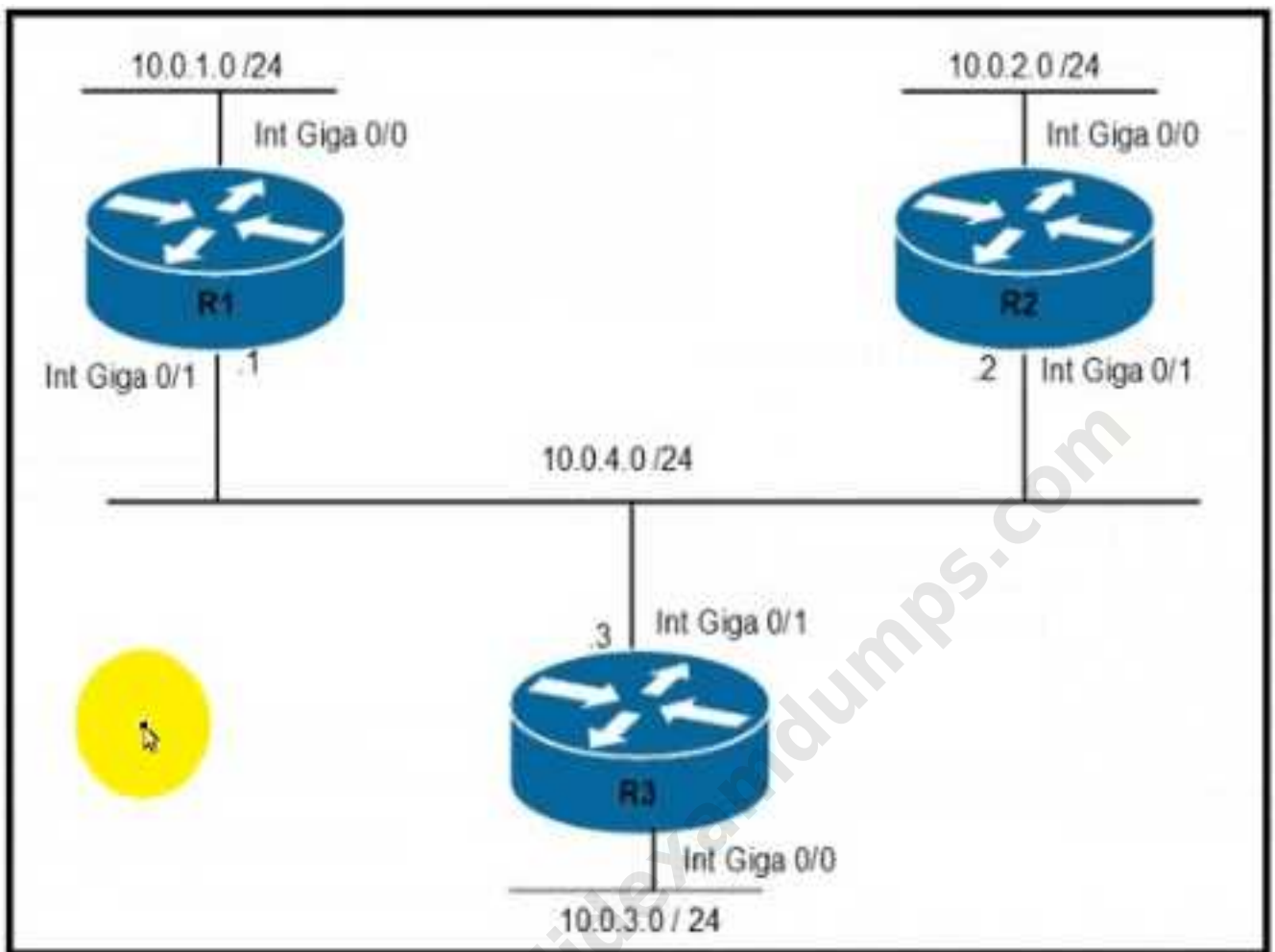
- A. 1
- B. 2
- C. 3
- D. 4

**Answer: D**

Explanation:

**Question: 563**

Refer to the exhibit.



Router R1 must be configured to reach the 10.0.3.0/24 network from the 10.0.1.0/24 segment.

Which command must be used to configure the route?

- A. ip route 10.0.3.0 0.255.255.255 10.0.4.2
- B. route add 10.0.3.0 mask 255.255.255.0 10.0.4.3
- C. ip route 10.0.3.0 255.255.255.0 10.0.4.3
- D. route add 10.0.3.0 0.255.255.255 10.0.4.2

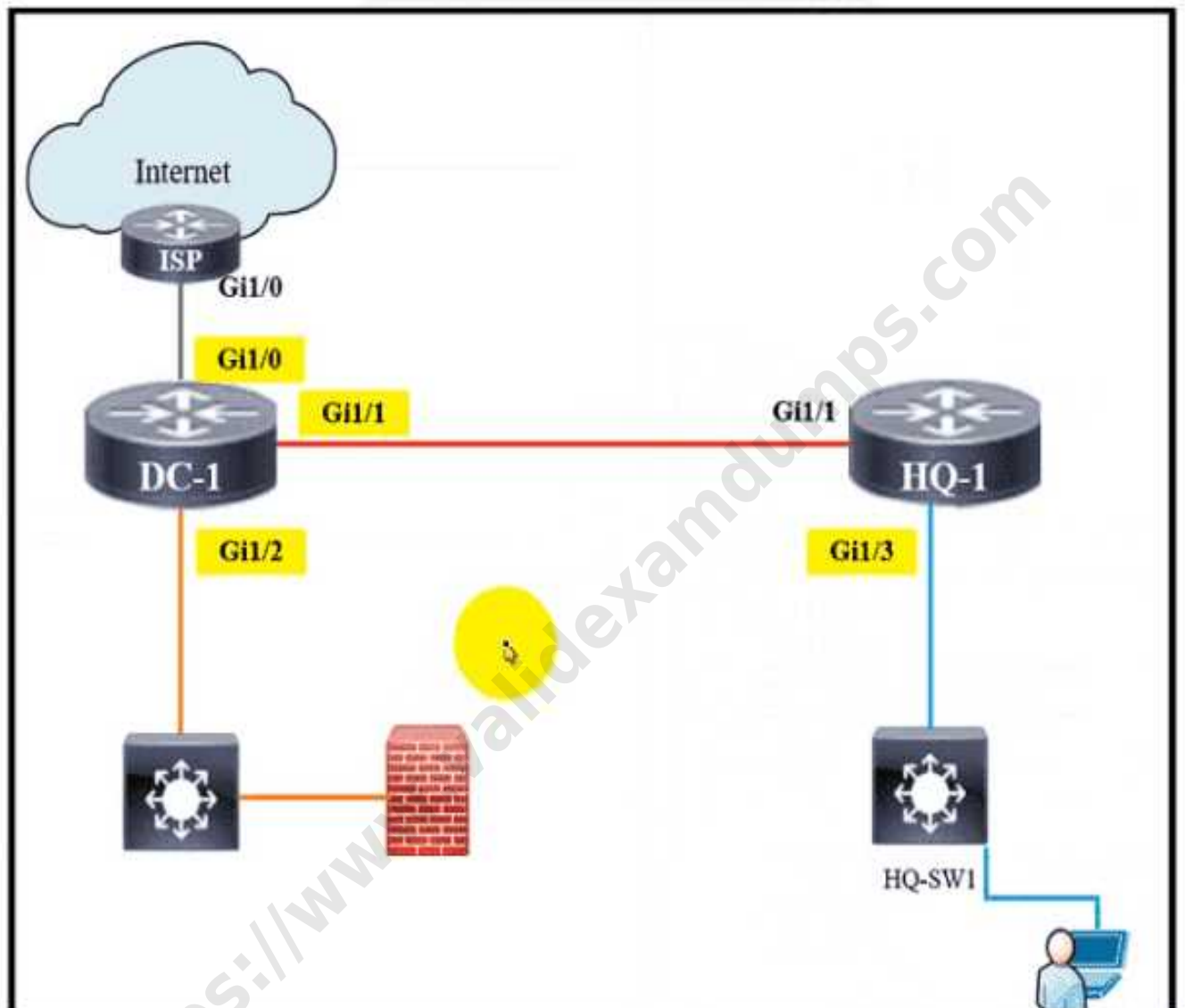
**Answer: C**

Explanation:

**Question: 564**

DRAG DROP

Refer to Exhibit.



Refer to the exhibit. The IP address configurations must be completed on the DC-1 and HQ-1 routers based on these requirements:

DC-1 Gi1/0 must be the last usable address on a /30

DC-1 Gi1/1 must be the first usable address on a /29

DC-1 Gi1/2 must be the last usable address on a /28

HQ-1 Gi1/3 must be the last usable address on a /29



Drag and drop the commands from the left onto the destination interfaces on the right. Not all commands are used

Left side (Commands):

- ip address 192.168.4.9 255.255.255.248
- ip address 192.168.3.14 255.255.255.240
- ip address 209.165.202.129 255.255.255.252
- ip address 192.168.4.13 255.255.255.240
- ip address 209.165.202.130 255.255.255.252
- ip address 209.165.202.131 255.255.255.252
- ip address 192.168.3.14 255.255.255.248

Right side (Interfaces):

DC-1

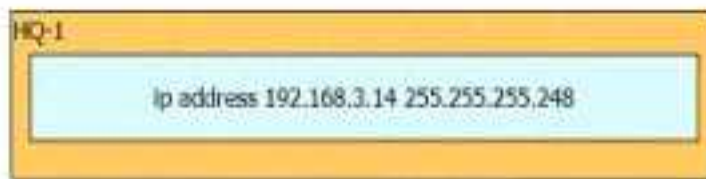
- Gi1/0
- Gi1/1
- Gi1/2

HQ-1

- Gi1/3

**Answer:**

Explanation:



---

**Question: 565**

---

DRAG DROP

Drag and drop the IPv6 address description from the left onto the IPv6 address types on the right.  
Not all options are used.

Diagram showing a drag-and-drop interface for IPv6 address types. On the left, there are six light blue boxes with descriptions. On the right, there are two orange boxes with empty slots for matching the descriptions.

**Left Side (Descriptions):**

- IPv6 addresses in the format FF02::5
- IPv6 addresses that begin with FD
- may be used by multiple organizations at the same time
- private IPv6 addresses
- serve as next-hop addresses
- unable to serve as destination addresses

**Right Side (Types):**

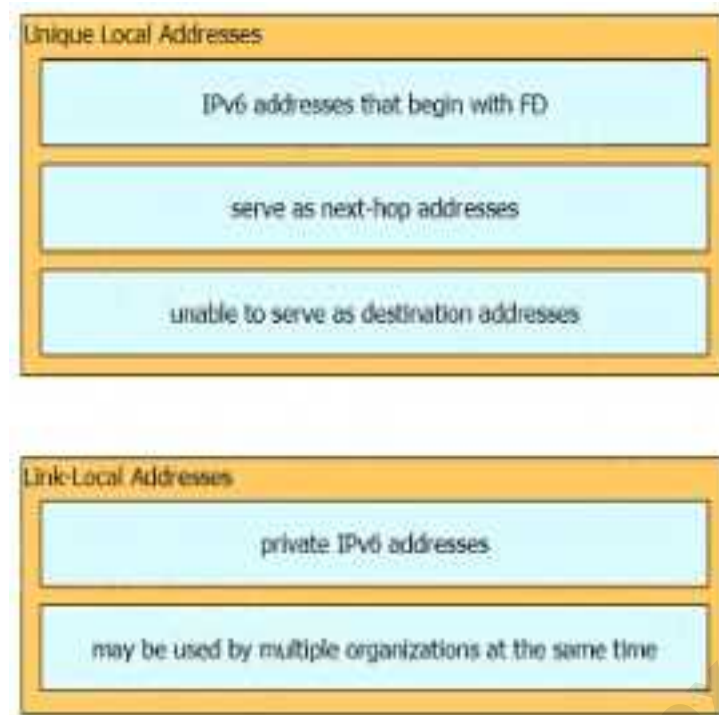
- Unique Local Addresses** (3 empty slots)
- Link-Local Addresses** (2 empty slots)

---

**Answer:**

---

Explanation:



---

**Question: 566**

---

DRAG DROP

|                                      |                    |
|--------------------------------------|--------------------|
| 2001:db8:600d:cafe::123              | Global Unicast     |
| fcba:926a:e8e:7a25:b1:c6d2:1a76:8fdc |                    |
| fe80::a00:27ff:feeb:89aa             | Link-Local Unicast |
| ff05::1:3                            | Multicast          |
|                                      | Unique Local       |

**Answer:**

Explanation:

|                                      |
|--------------------------------------|
| Global Unicast                       |
| 2001:db8:600d:cafe::123              |
| Link-Local Unicast                   |
| fe80::a00:27ff:feeb:89aa             |
| Multicast                            |
| ff05::1:3                            |
| Unique Local                         |
| fcba:926a:e8e:7a25:b1:c6d2:1a76:8fdc |

**Question: 567**

Refer to the exhibit.

```
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
       la - LISP away, le - LISP extranet-policy, lp - LISP publications

ND  ::/0 [2/0]
    via FE80::A8BB:CCFF:FE00:200, Ethernet0/0
NDp 2001:DB8:1234:1::/64 [2/0]
    via Ethernet0/0, directly connected
L   2001:DB8:1234:1:A8BB:CCFF:FE00:100/128 [0/0]
    via Ethernet0/0, receive
C   2001:DB8:1234:2::/64 [0/0]
    via Ethernet0/1, directly connected
L   2001:DB8:1234:2:A8BB:CCFF:FE00:110/128 [0/0]
    via Ethernet0/1, receive
L   FF00::/8 [0/0]
    via Null0, receive
```

The administrator must configure a floating static default route that points to 2001:db8:1234:2::1 and replaces the current default route only if it fails. Which command must the engineer configure on the CPE?

- A. `ipv6 route ::/0 2001:db8:1234:2::1 3`
- B. `ipv6 route ::/128 2001:db8:1234:2::1 3`
- C. `ipv6 route ::/0 2001:db8:1234:2::1 1`
- D. `ipv6 route ::/0 2001:db8:1234:2::1 2`

**Answer: B**

Explanation:

---

**Question: 568**

---

What is the function of "off-the-shell" switches in a controller-based network?

- A. providing a central view of the deployed network
- B. forwarding packets
- C. making routing decisions
- D. setting packet-handling policies

---

**Answer: D**

---

Explanation:

---

**Question: 569**

---

DRAG DROP

Drag and drop the QoS terms from the left onto the descriptions on the right.

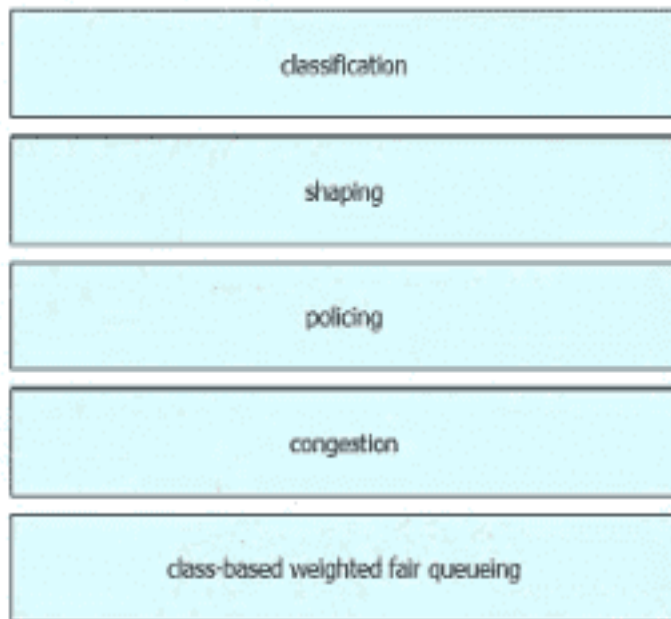
|                                   |                                                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------------|
| class-based weighted fair queuing | categorizes packets based on the value of a traffic descriptor                          |
| classification                    | guarantees minimum bandwidth to specific traffic classes when an interface is congested |
| congestion                        | prevents congestion by reducing the flow of outbound traffic                            |
| policing                          | outcome of overutilization                                                              |
| shaping                           | uses defined criteria to limit the transmission of one or more classes of traffic       |

---

**Answer:**

---

Explanation:



---

**Question: 570**

---

Which command do you enter so that a switch configured with Rapid PVST + listens and learns for a specific time period?

- A. switch(config)#spanning-tree vlan 1 max-age 6
- B. switch(config)#spanning-tree vlan 1 hello-time 10
- C. switch(config)#spanning-tree vlan 1 priority 4096
- D. switch(config)#spanning-tree vlan 1 forward-time 20

---

**Answer: D**

---

Explanation:

Forward time : Determines how long each of the listening and learning states last before the port begins forwarding.

Switch(config)# [ no ] spanning-tree vlan vlan\_ID forward-time forward\_time

Configures the forward time of a VLAN. The forward\_time value can be from 4 to 30 seconds.

<https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst4500/12-2/15->

02SG/configuration/guide/config/spantree.html#56177

---

**Question: 571**

---

Refer to the exhibit.

```
R1# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default
       U - per-user static route, o - ODR
Gateway of last resort is not set
C    172.16.0.0/16 is directly connected, Loopback0
     172.16.0/16 is variably subnetted, 4 subnets, 2 masks
O    172.16.1.3/24 [110/100] via 192.168.7.40, 00:39:08, Serial0
C    172.16.1.0/24 is directly connected, Serial0
O    172.16.1.184/29 [110/5] via 192.168.7.35, 00:39:08, Serial0
O    172.16.3.0/24 [110/10] via 192.168.7.4, 00:39:08, Gigabit Ethernet 0/0
D    172.16.1.0/28 [90/10] via 192.168.7.7, 00:39:08, Gigabit Ethernet 0/0
```

Load-balanced traffic is coming in from the WAN destined to a host at 172.16.1.190. Which next-hop is used by the router to forward the request?

- A. 192.168.7.4
- B. 192.168.7.7
- C. 192.168.7.35
- D. 192.168.7.40

---

**Answer: B**

---

Explanation:

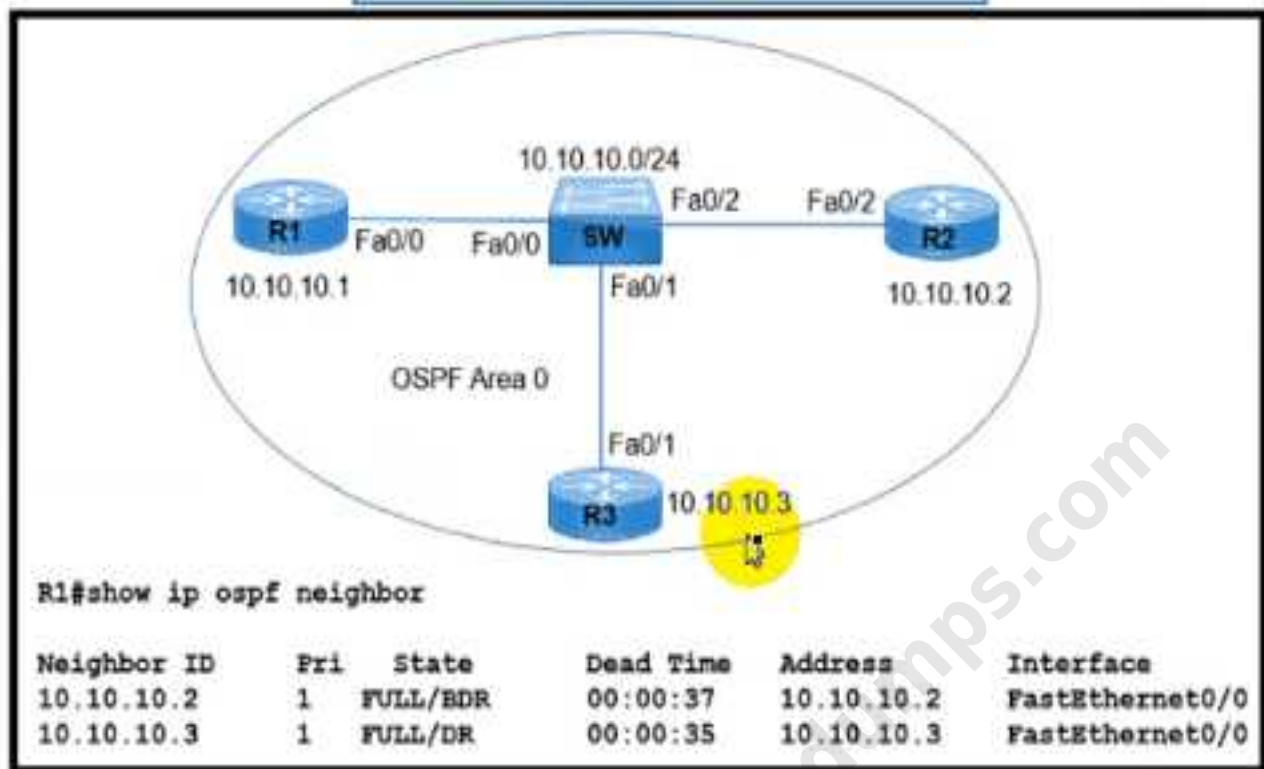
---

**Question: 572**

---

Refer to the exhibit.





R1 has taken the DROTHER role in the OSPF DR/BDR election process. Which configuration must an engineer implement so that R1 is elected as the DR?

- ❑ R1(config)#interface FastEthernet 0/0  
R1(config-if)#ip ospf priority 1  
R1#clear ip ospf process
- ❑ R1(config)#interface FastEthernet 0/0  
R1(config-if)#ip ospf priority 200  
R1#clear ip ospf process
- ❑ R3(config)#interface FastEthernet 0/1  
R3(config-if)#ip ospf priority 200  
R3#clear ip ospf process
- ❑ R2(config)#interface FastEthernet 0/2  
R2(config-if)#ip ospf priority 1  
R2#clear ip ospf process

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: B**

---

Explanation:

---

**Question: 573**

---

What is a feature of WPA?

- A. 802.1x authentication
- B. preshared key
- C. TKIP/MIC encryption
- D. small Wi-Fi application

---

**Answer: A**

---

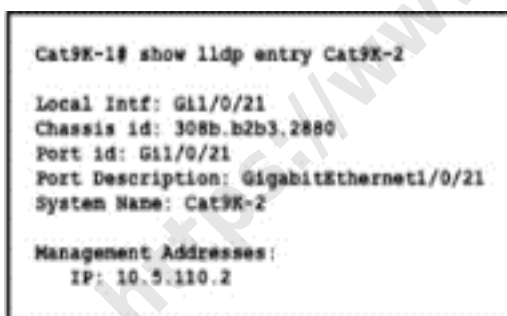
Explanation:

---

**Question: 574**

---

Refer to the exhibit.



```
Cat9K-1# show lldp entry Cat9K-2
Local Intf: Gi1/0/21
Chassis id: 308b.b2b3.2880
Port id: Gi1/0/21
Port Description: GigabitEthernet1/0/21
System Name: Cat9K-2

Management Addresses:
  IP: 10.5.110.2
```

The network administrator must prevent the switch Cat9K-2 IP address from being visible in LLDP without disabling the protocol. Which action must be taken must be taken to complete the task?

- A. Configure the no lldp tlv-select-management-address command globally on Cat9K-2
- B. Configure the no lldp transmit command on interface G1/0/21 in Cat9K-1
- C. Configure the no lldp receive command on interface G1/0/21 on Cat9K-1
- D. Configure the no lldp mac-phy-cfg command globally on Cat9K-2

---

**Answer: A**

---

Explanation:

---

**Question: 575**

---

Refer to the exhibit.



What are the two steps an engineer must take to provide the highest encryption and authentication using domain credentials from LDAP?

- A. Select PSK under Authentication Key Management
- B. Select WPA+WPA2 on Layer 2 Security
- C. Select Static-WEP + 802.1X on Layer 2 Security
- D. Select WPA Policy with TKIP Encryption
- E. Select 802.1X from under Authentication Key Management

---

**Answer: B,E**

---

Explanation:

---

**Question: 576**

---

Which WAN topology has the highest degree of reliability?

- A. full mesh
- B. Point-to-point
- C. hub-and-spoke
- D. router-on-a-stick

---

**Answer: A**

---

Explanation:

---

**Question: 577**

---

An engineer must configure neighbor discovery between the company router and an ISP

```
interface gigabitethernet0/0
description Circuit-ATT4203-21099
duplex full
speed 1000
media-type gbic
negotiation auto
lldp transmit
lldp receive
```

What is the next step to complete the configuration if the ISP uses a third-party router?

- A. Enable LLDP globally.
- B. Disable CDP on gi0/0.
- C. Enable LLDP TLVs on the ISP router.
- D. Disable auto-negotiation.

---

**Answer: A**

---

Explanation:

---

**Question: 578**

---

Refer to the exhibit.

```
SW1#show spanning-tree vlan 30

VLAN0030
Spanning tree enabled protocol rstp
Root ID    Priority          32798
           Address        0025.63e9.c800
           Cost           19
           Port           1 (FastEthernet 2/1)
           Hello Time     2 sec
           Max Age        30 sec
           Forward Delay  20 sec

[Output suppressed]
```

What are two conclusions about this configuration? {Choose two.}

- A. The spanning-tree mode is Rapid PVST+.
- B. This is a root bridge.
- C. The root port is FastEthernet 2/1.
- D. The designated port is FastEthernet 2/1.
- E. The spanning-tree mode is PVST+.

---

**Answer: A**

---

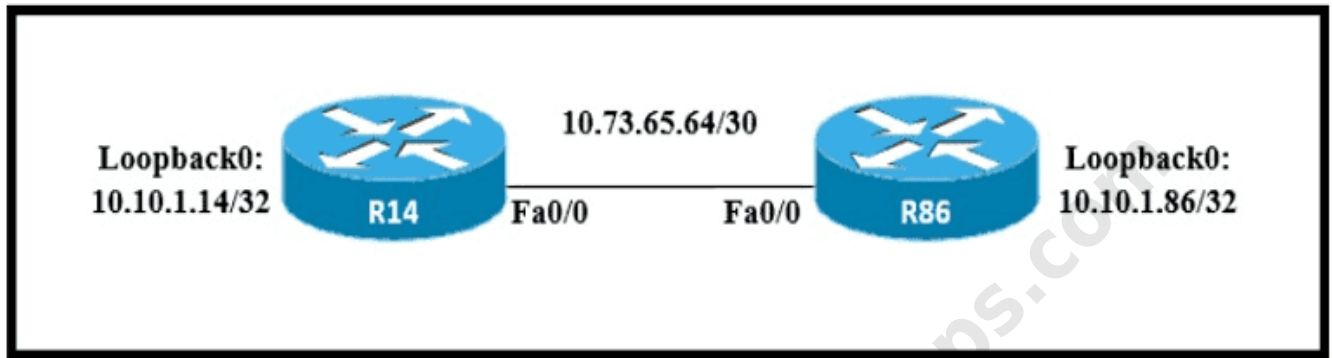
Explanation:

---

**Question: 579**

---

Refer to the exhibit.



All interfaces are configured with duplex auto and ip ospf network broadcast. Which configuration allows routers R14 and R86 to form an OSPFv2 adjacency and act as a central point for exchanging OSPF information between routers?

● R14#

```
interface FastEthernet0/0
ip address 10.73.65.65 255.255.255.252
ip ospf priority 0
ip mtu 1500
```

```
router ospf 10
router-id 10.10.1.14
network 10.10.1.14 0.0.0.0 area 0
network 10.73.65.64 0.0.0.3 area 0
```

R86#

```
interface FastEthernet0/0
ip address 10.73.65.66 255.255.255.252
ip mtu 1500
```

```
router ospf 10
router-id 10.10.1.86
network 10.10.1.86 0.0.0.0 area 0
network 10.73.65.64 0.0.0.3 area 0
```

● R14#

```
interface Loopback0  
ip ospf 10 area 0
```

```
interface FastEthernet0/0  
ip address 10.73.65.65 255.255.255.252  
ip ospf priority 255  
ip ospf 10 area 0  
ip mtu 1500
```

```
router ospf 10  
router-id 10.10.1.14
```

R86#

```
interface Loopback0  
ip ospf 10 area 0
```

```
interface FastEthernet0/0  
ip address 10.73.65.66 255.255.255.252  
ip ospf 10 area 0  
ip mtu 1500
```

```
router ospf 10  
router-id 10.10.1.86
```



● R14#  
interface FastEthernet0/0  
ip address 10.73.65.65 255.255.255.252  
ip ospf priority 255  
ip mtu 1500

router ospf 10  
router-id 10.10.1.14  
network 10.10.1.14 0.0.0.0 area 0  
network 10.73.65.64 0.0.0.3 area 0

R86#  
interface FastEthernet0/0  
ip address 10.73.65.66 255.255.255.252  
ip mtu 1400

router ospf 10  
router-id 10.10.1.86  
network 10.10.1.86 0.0.0.0 area 0  
network 10.73.65.64 0.0.0.3 area 0

● R14#

```
interface Loopback0  
ip ospf 10 area 0
```

```
interface FastEthernet0/0  
ip address 10.73.65.65 255.255.255.252  
ip ospf 10 area 0  
ip mtu 1500
```

```
router ospf 10  
ip ospf priority 255  
router-id 10.10.1.14
```

R86#

```
interface Loopback0  
ip ospf 10 area 0
```

```
interface FastEthernet0/0  
ip address 10.73.65.66 255.255.255.252  
ip ospf 10 area 0  
ip mtu 1500
```

```
router ospf 10  
router-id 10.10.1.86
```

---

A. Option A

B. Option B

C. Option C

D. option D

---

**Answer: B**

---

Explanation:

**Question: 580**

DRAG DROP

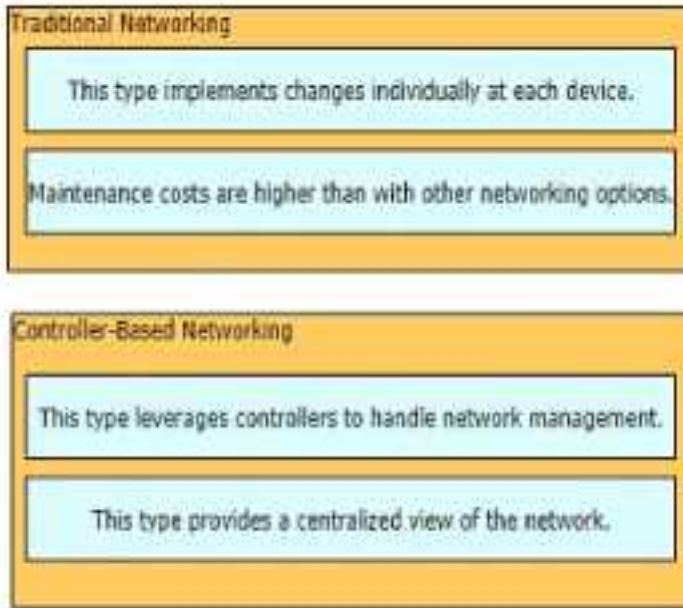
Drag and drop the statements about networking from the left onto the corresponding networking types on the right

|                                                                  |                                                               |
|------------------------------------------------------------------|---------------------------------------------------------------|
| This type implements changes individually at each device.        | <b>Traditional Networking</b><br><div></div> <div></div>      |
| This type leverages controllers to handle network management.    |                                                               |
| Maintenance costs are higher than with other networking options. | <b>Controller-Based Networking</b><br><div></div> <div></div> |
| This type provides a centralized view of the network.            |                                                               |

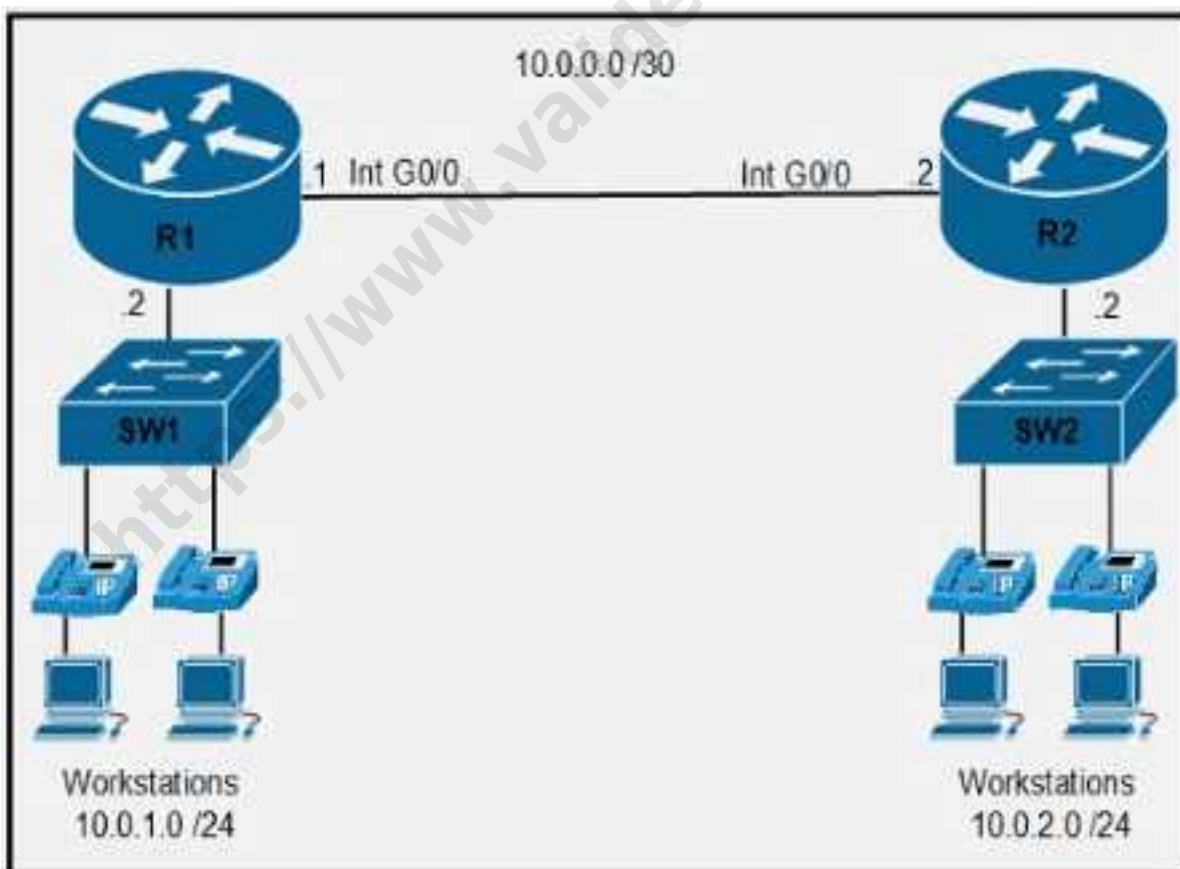
2/

Explanation:

**Answer:**

**Question: 581**

Refer to the exhibit.



Refer to the exhibit. An engineer is asked to configure router R1 so that it forms an OSPF single-area neighbor relationship with R2. Which command sequence must be implemented to configure the

router?

- router ospf 10  
network 10.0.0.0 0.0.0.3 area 0  
network 10.0.2.0 0.0.0.255 area 0
- router ospf 10  
network 10.0.0.0 0.0.0.3 area 0  
network 10.0.1.0 0.0.0.255 area 0
- router ospf 100  
network 10.0.0.0 0.0.0.3 area 0  
network 10.0.2.0 255.255.255.0 area 0
- router ospf 100  
network 10.0.0.0 0.0.0.252 area 0  
network 10.0.1.0 0.0.0.255 area 0

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: B**

---

Explanation:

---

**Question: 582**

---

What is a function performed by a web server?

- A. provide an application that is transmitted over HTTP
- B. send and retrieve email from client devices
- C. authenticate and authorize a user's identity
- D. securely store files for FTP access

---

**Answer: A**

---

Explanation:

---

**Question: 583**

---

What is the collapsed layer in collapsed core architectures?

- A. core and WAN
- B. access and WAN
- C. distribution and access
- D. core and distribution

---

**Answer: D**

---

Explanation:

---

**Question: 584**

---

Which set of 2.4 GHz nonoverlapping wireless channels is standard in the United States?

- A. channels 2, 7, 9, and 11
- B. channels 1, 6, 11, and 14
- C. channels 2, 7, and 11
- D. channels 1, 6, and 11

---

**Answer: D**

---

Explanation:

In the United States, while channels 1-13 can be used for 2.4 GHz WiFi, only three channels are considered non-overlapping (channels 12 and 13 are allowed under low powered conditions, but for most cases are not used). For best results, it is highly recommended to keep the 2.4 GHz channels to 1, 6, and 11, as these channel settings will allow for virtually no overlap in the WiFi signal. Shown below is a channel graph from WiFi Scanner showing three access points configured for channels 1, 6, and 11. As you can see, the center of each signal is at 1, 6, and 11 with the actual signal extending over several channels to the left and right.

---

**Question: 585**

---

An engineer is installing a new wireless printer with a static IP address on the Wi-Fi network. Which feature must be enabled and configured to prevent connection issues with the printer?

- A. client exclusion
- B. passive client
- C. DHCP address assignment
- D. static IP tunneling

---

**Answer: C**

---

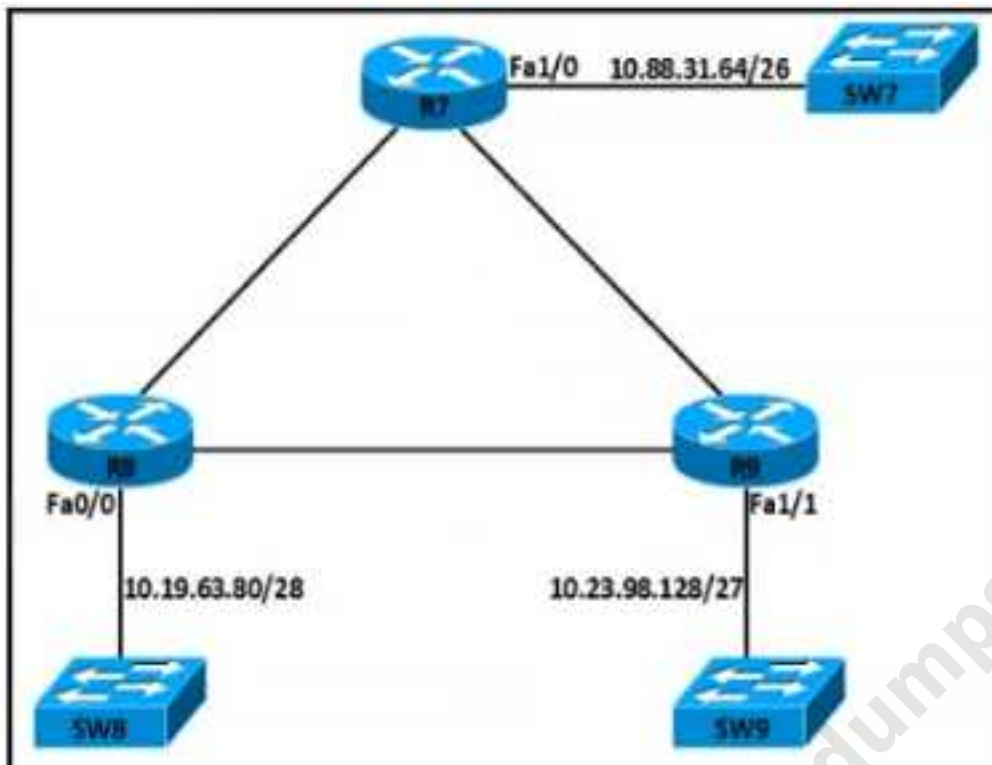
Explanation:

---

**Question: 586**

---

Refer to the exhibit.



Refer to the exhibit. Each router must be configured with the last usable IP address in the subnet.  
Which configuration fulfills this requirement?



☐ R7#  
interface FastEthernet1/0  
ip address 10.88.31.126 255.255.255.240

R8#  
interface FastEthernet0/0  
ip address 10.19.63.94 255.255.255.192

R9#  
interface FastEthernet1/1  
ip address 10.23.98.158 255.255.255.248

☐ R7#  
interface FastEthernet1/0  
ip address 10.88.31.127 255.255.255.240

R8#  
interface FastEthernet0/0  
ip address 10.19.63.95 255.255.255.192

R9#  
interface FastEthernet1/1  
ip address 10.23.98.159 255.255.255.248

☒ R7#  
interface FastEthernet1/0  
ip address 10.88.31.126 255.255.255.192

R8#  
interface FastEthernet0/0  
ip address 10.19.63.94 255.255.255.240

R9#  
interface FastEthernet1/1  
ip address 10.23.98.158 255.255.255.224

☐ R7#  
interface FastEthernet1/0  
ip address 10.88.31.127 255.255.255.192

R8#  
interface FastEthernet0/0  
ip address 10.19.63.95 255.255.255.240

R9#  
interface FastEthernet1/1  
ip address 10.23.98.159 255.255.255.224

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: C**

---

Explanation:

---

**Question: 587**

---

DRAG DROP

Drag and drop the functions of AAA supporting protocols from the left onto the protocols on the right.

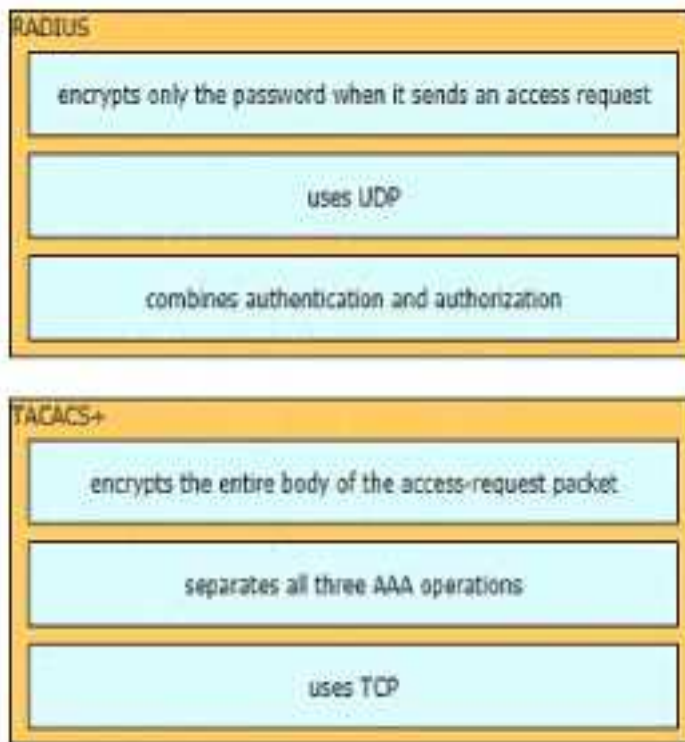
|                                                                                              |                                                |
|----------------------------------------------------------------------------------------------|------------------------------------------------|
| encrypts only the password when it sends an access request                                   | RADIUS<br><div></div> <div></div> <div></div>  |
| encrypts the entire body of the access-request packet                                        |                                                |
| separates all three AAA operations                                                           |                                                |
| combines authentication and authorization                                                    | TACACS+<br><div></div> <div></div> <div></div> |
| uses TCP  |                                                |
| uses UDP                                                                                     |                                                |

---

**Answer:**

---

Explanation:



---

**Question: 588**

---

DRAG DROP

Drag and drop the HTTP methods used with REST-Based APIs from the left onto the descriptions on the right.

|        |                                                                                          |
|--------|------------------------------------------------------------------------------------------|
| DELETE | creates a resource and returns its URI in the response header                            |
| GET    | creates or replaces a previously modified resource using information in the request body |
| POST   | removes a resource                                                                       |
| PATCH  | retrieves a list of a resource's URIs                                                    |
| PUT    | updates a resource using instructions included in the request body                       |

---

**Answer:**

---

Explanation:

|        |
|--------|
| POST   |
| DELETE |
| PATCH  |
| PUT    |
| GET    |

---

**Question: 589**

---

What is the MAC address used with VRRP as a virtual address?

A. 00-00-0C-07-AD-89

- B. 00-00-5E-00-01-0a
- C. 00-07-C0-70-AB-01
- D. 00-C6-41-93-90-91

---

**Answer: B**

---

Explanation:

---

**Question: 590**

---

DRAG DROP

Drag and drop the IPv6 addresses from the left onto the corresponding address types on the right.

|                                      |                                            |
|--------------------------------------|--------------------------------------------|
| 2001:db8:600d:cafe::123              | Global Unicast<br><input type="text"/>     |
| fcba:926a:e8e:7a25:b1:c6d2:1a76:8fdc | Link-Local Unicast<br><input type="text"/> |
| fe80::a00:27ff:feeb:89aa             | Multicast<br><input type="text"/>          |
| ff05::1:3                            | Unique Local<br><input type="text"/>       |

---

**Answer:**

---

Explanation:

|                    |                                      |
|--------------------|--------------------------------------|
| Global Unicast     | 2001:db8:600d:cafe::123              |
| Link-Local Unicast | fe80::a00:27ff:febc:89aa             |
| Multicast          | :::ff05::1:3                         |
| Unique Local       | fcba:926a:e8e:7a25:b1:c6d2:1a76:8f4c |

### Question: 591

Refer to the exhibit.

```

R1#sho ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 10.56.0.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.56.0.1
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      10.56.0.0/17 is directly connected, Vlan56
L      10.56.0.19/32 is directly connected, Vlan56
C      10.56.128.0/18 is directly connected, Vlan57
L      10.56.128.19/32 is directly connected, Vlan57

```

When router R1 is sending traffic to IP address 10.56.192.1, which interface or next hop address does it use to route the packet?

A. 0.0.0.0/0

- B. 10.56.0.1
- C. 10.56.128.19
- D. Vlan57

---

**Answer: B**

---

Explanation:

---

**Question: 592**

---

What is the difference between 1000BASE-LX/LH and 1000BASE-ZX interfaces?

- A. 1000BASE-ZX is supported on links up to 1000km, and 1000BASE-LX/LH operates over links up to 70 km.
- B. 1000BASE-LX/LH interoperates with multimode and single-mode fiber, and 1000BASE-ZX needs a conditioning patch cable with a multimode.
- C. 1000BASE-LX/LH is supported on links up to 10km, and 1000BASE-ZX operates over links up to 70 km
- D. 1000BASE-ZX interoperates with dual-rate 100M/1G 10Km SFP over multimode fiber, and 1000BASE-LX/LH supports only single-rate.

---

**Answer: C**

---

Explanation:

---

**Question: 593**

---

What is a benefit for external users who consume public cloud resources?

- A. implemented over a dedicated WAN
- B. located in the same data center as the users
- C. all hosted on physical servers
- D. accessed over the Internet

---

**Answer: D**

---

Explanation:

---

**Question: 594**

---

Refer to the exhibit.

```
R1# show ip route | begin gateway
Gateway of last resort is not set
    172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
C    172.16.1.0/24 is directly connected, FastEthernet0/0
L    172.16.1.1/32 is directly connected, FastEthernet0/0
EX   172.16.2.0/24 [170/2] via 207.165.200.250, 00:00:25, Serial0/0/0
O    192.168.1.0/24 [110/84437] via 207.165.200.254, 00:00:17, Serial0/0/1
D    192.168.2.0/24 [90/184437] via 207.165.200.254, 00:00:15, Serial0/0/1
E1   192.168.3.0/24 [110/1851437] via 207.165.200.254, 00:00:19, Serial0/0/1
    207.165.200.0/24 is variably subnetted, 4 subnets, 2 masks
C    207.165.200.248/30 is directly connected, Serial0/0/0
L    207.165.200.249/32 is directly connected, Serial0/0/0
C    207.165.200.252/30 is directly connected, Serial0/0/1
L    207.165.200.253/32 is directly connected, Serial0/0/1
```

Which prefix did router R1 learn from internal EIGRP?

- A. 192.168.10/24
- B. 192.168.3.0/24
- C. 192.168.2.0/24
- D. 172.16.1.0/24

---

**Answer: C**

---

Explanation:

---

**Question: 595**

---

An engineer is configuring switch SW1 to act an NTP server when all upstream NTP server



connectivity fails. Which configuration must be used?

A)

```
SW1# config t
SW1(config)#ntp peer 192.168.1.1
SW1(config)#ntp access-group peer accesslist1
```

B)

```
SW1# config t
SW1(config)#ntp master
SW1(config)#ntp server 192.168.1.1
```

C)

```
SW1# config t
SW1(config)#ntp server 192.168.1.1
SW1(config)#ntp access-group server accesslist1
```

D)

```
SW1# config t
SW1(config)#ntp backup
SW1(config)#ntp server 192.168.1.1
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: B**

---

Explanation:

---

### Question: 596

---

How does authentication differ from authorization?

A. Authentication verifies the identity of a person accessing a network, and authorization determines

what resource a user can access.

B. Authentication is used to record what resource a user accesses, and authorization is used to determine what resources a user can access

C. Authentication is used to determine what resources a user is allowed to access, and authorization is used to track what equipment is allowed access to the network

D. Authentication is used to verify a person's identity, and authorization is used to create syslog messages for logins.

---

**Answer: A**

---

Explanation:

---

**Question: 597**

---

When should an engineer implement a collapsed-core architecture?

A. for small networks with minimal need for growth

B. the access and distribution layers must be on the same device

C. for large networks that are connected to multiple remote sites

D. only when using VSS technology

---

**Answer: C**

---

Explanation:

---

**Question: 598**

---

Refer to the exhibit.



An OSPF neighbor relationship must be configured using these guidelines:

- R1 is only permitted to establish a neighbor with R2
- R1 will never participate in DR elections
- R1 will use a router-id of 101.1.1.1.

Which configuration must be used?

A)

```
interface Loopback0
  ip address 10.1.1.1 255.255.255.255

interface FastEthernet0/0
  ip address 10.100.1.1 255.255.255.252
  ip ospf priority 100
  ip access-group 102 in

router ospf 10
  log-adjacency-changes
  network 10.1.1.1 0.0.0.0 area 0
  network 10.100.1.0 0.0.0.3 area 0
  ospf router-id 10.1.1.1

access-list 102 permit 88 host 10.100.1.2 host 224.0.0.5
access-list 102 deny 88 any any
access-list 102 permit ip any any
```

B)

```
interface Loopback0
  ip address 10.1.1.1 255.255.255.255
```

```
interface FastEthernet0/0
  ip address 10.100.1.1 255.255.255.252
  ip ospf priority 0
  ip access-group 102 in
```

```
router ospf 10
  log-adjacency-changes
  network 10.1.1.1 0.0.0.0 area 0
  network 10.100.1.0 0.0.0.3 area 0
  router-id 10.1.1.1
```

```
access-list 102 permit 88 host 10.100.1.2 host 224.0.0.5
access-list 102 deny 88 any any
access-list 102 permit ip any any
```

C)

```
interface FastEthernet0/0
  ip address 10.100.1.1 255.255.255.252
  ip ospf priority 100
  ip access-group 102 in
```

```
router ospf 10
  log-adjacency-changes
  network 10.1.1.1 0.0.0.0 area 0
  network 10.100.1.0 0.0.0.3 area 0
  ospf router-id 10.1.1.1
```

```
access-list 102 permit 89 host 10.100.1.2 host 224.0.0.5
access-list 102 deny 89 any any
access-list 102 permit ip any any
```

D)

```
interface FastEthernet0/0
  ip address 10.100.1.1 255.255.255.252
  ip ospf priority 0
  ip access-group 102 in
```

```
router ospf 10
  log-adjacency-changes
  network 10.1.1.1 0.0.0.0 area 0
  network 10.100.1.0 0.0.0.3 area 0
  router-id 10.1.1.1
```

```
access-list 102 permit 89 host 10.100.1.2 host 224.0.0.5
access-list 102 deny 89 any any
access-list 102 permit ip any any
```

- A. Option
- B. Option
- C. Option
- D. Option

---

**Answer: C**

---

Explanation:

---

**Question: 599**

---

Which two wireless security stewards use Counter Mode Cipher Block Chaining Message Authentication Code Protocol for encryption and data integrity'? (Choose two.)

- A. WPA2
- B. WPA3
- C. Wi-Fi 6
- D. WEP
- E. WPA

---

**Answer: B,D**

---

Explanation:

---

**Question: 600**

---

What is the default port-security behavior on a trunk link?

- A. It causes a network loop when a violation occurs.
- B. It disables the native VLAN configuration as soon as port security is enabled.
- C. It places the port in the err-disabled state if it learns more than one MAC address.

D. It places the port in the err-disabled state after 10 MAC addresses are statically configured.

---

**Answer: A**

---

Explanation:

---

**Question: 601**

---

When a switch receives a frame for an unknown destination MAC address, how is the frame handled?

- A. broadcast to all ports on the switch
- B. flooded to all ports except the origination port
- C. forwarded to the first available port
- D. inspected and dropped by the switch

---

**Answer: B**

---

Explanation:

---

**Question: 602**

---

Refer to the exhibit.

```
interface g2/0/0
  channel-group 1 mode active
interface g4/0/0
  channel-group 1 mode active
interface Port-channel1
  ip address 203.0.113.65 255.255.255.252

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down
```

An engineer is configuring a Layer 3 port-channel interface with LACP. The configuration on the first device is complete, and it is verified that both interfaces have registered the neighbor device in the

CDP table. Which task on the neighbor device enables the new port channel to come up without negotiating the channel?

- A. Change the EtherChannel mode on the neighboring interfaces to auto.
- B. Configure the IP address of the neighboring device.
- C. Bring up the neighboring interfaces using the no shutdown command.
- D. Modify the static EtherChannel configuration of the device to passive mode.

---

**Answer: D**

---

Explanation:

---

**Question: 603**

---

Refer to the exhibit.

```
Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0

209.165.200.0/27 is subnetted, 1 subnets
B    209.165.200.224 [20/0] via 10.10.12.2, 03:22:14
209.165.201.0/27 is subnetted, 1 subnets
B    209.165.201.0 [20/0] via 10.10.12.2, 02:26:33
209.165.202.0/27 is subnetted, 1 subnets
B    209.165.202.128 [20/0] via 10.10.12.2, 02:26:03
10.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
C    10.10.10.0/28 is directly connected, GigabitEthernet0/0
C    10.10.11.0/30 is directly connected, FastEthernet2/0
C    10.10.12.0/30 is directly connected, GigabitEthernet0/1
O    10.10.13.0/25 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O    10.10.13.128/28 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O    10.10.13.144/28 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O    10.10.13.160/29 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O    10.10.13.208/29 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
S*  0.0.0.0/0 [1/0] via 10.10.11.2
```

What is the subnet mask of the route to the 10.10.13.160 prefix?

- A. 255.255.255.240
- B. 255.255.255.128
- C. 255.255.248.
- D. 255.255.255.248

---

**Answer: D**

---

Explanation:

---

**Question: 604**

---

What is the purpose of the Cisco DNA Center controller?

- A. to secure physical access to a data center
- B. to scan a network and generate a Layer 2 network diagram
- C. to securely manage and deploy network devices
- D. to provide Layer 3 services to autonomous access points

---

**Answer: C**

---

Explanation:

---

**Question: 605**

---

Which encryption method is used by WPA3?

- A. PSK
- B. TKIP
- C. SAE
- D. AES

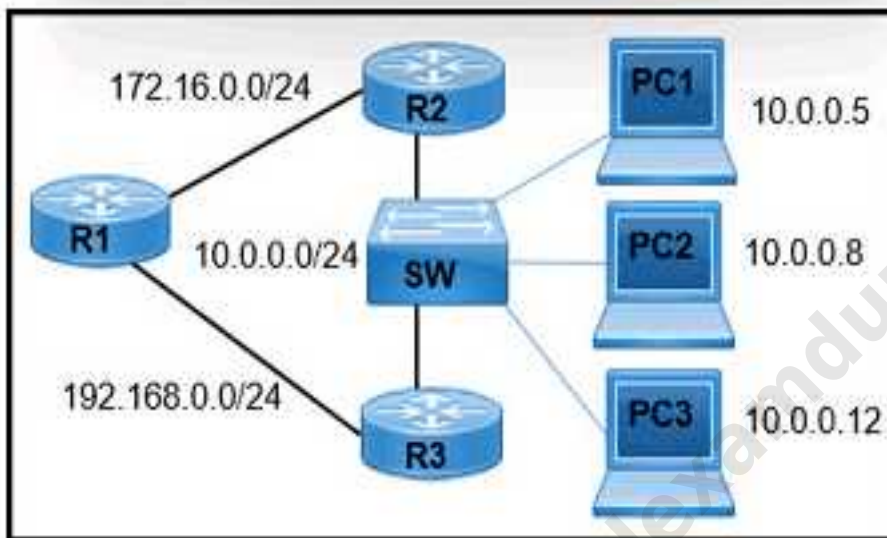


**Answer: D**

Explanation:

**Question: 606**

Refer to the exhibit.



A network engineer must configure R1 so that it sends all packets destined to the 10.0.0.0/24 network to R3, and all packets destined to PC1 to R2. Which configuration must the engineer implement?

A)

```
R1(config)#ip route 10.0.0.0 255.255.255.0 172.16.0.2  
R1(config)#ip route 10.0.0.5 255.255.255.255 192.168.0.2
```

B)

```
R1(config)#ip route 10.0.0.0 255.255.0.0 172.16.0.2  
R1(config)#ip route 10.0.0.5 255.255.255.255 192.168.0.2
```

C)

```
R1(config)#ip route 10.0.0.0 255.255.255.0 192.168.0.2  
R1(config)#ip route 10.0.0.5 255.255.255.255 172.16.0.2
```

D)

```
R1(config)#ip route 10.0.0.0 255.255.0.0 192.168.0.2  
R1(config)#ip route 10.0.0.0 255.255.255.0 172.16.0.2
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: C**

---

Explanation:

---

**Question: 607**

---

Which command implies the use of SNMPv3?

- A. snmp-server host
- B. snmp-server community
- C. snmp-server enable traps
- D. snmp-server user

---

**Answer: B**

---

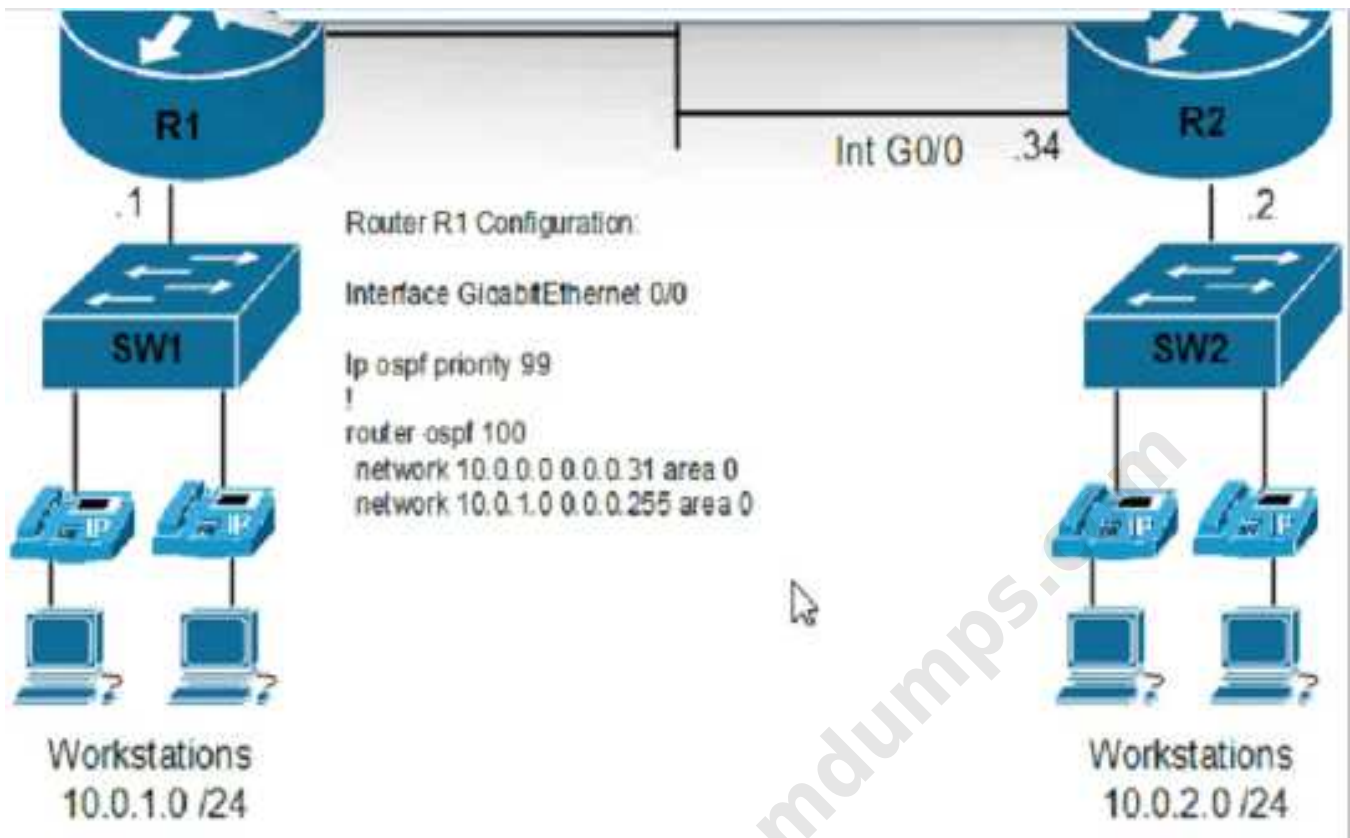
Explanation:

---

**Question: 608**

---

Refer to the exhibit.



An engineer must configure router R2 so it is elected as the DR on the WAN subnet. Which command sequence must be configured?

A)

```
interface gigabitethernet0/0
ip address 10.0.0.34 255.255.255.224
ip ospf priority 100
```

B)

```
interface gigabitethernet0/0
ip address 10.0.1.1 255.255.255.224
ip ospf priority 98
```

C)

```
interface gigabitethernet0/0
ip address 10.0.0.34 255.255.255.248
ip ospf priority 0
```

D)

```
interface gigabitethernet0/0
ip address 10.0.1.1 255.255.255.0
ip ospf priority 255
```

- A. Option
- B. Option
- C. Option
- D. Option

---

**Answer: A**

---

Explanation:

---

**Question: 609**

---

DRAG DROP

Drag and drop the Wi-Fi terms from the left onto the descriptions on the right.

|                               |                                                                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------|
| distribution system           | Wi-Fi option in which cells from different access points are linked together                         |
| extended service set          | Wi-Fi option that enables two or more clients to communicate directly without a central access point |
| independent basic service set | Wi-Fi option based around one or more access points                                                  |
| infrastructure mode           | alphanumeric text string that identifies a wireless network                                          |
| SSID                          | entire wireless cell of an access point and the linkage to the wired network                         |

---

**Answer:**

---

Explanation:

distribution system

independent basic service set

extended service set

SSID

infrastructure mode

---

**Question: 610**

---

Refer to the exhibit.

```
{
  "Routers": ["R1", "R2", "R3"],
  "Switches": ["SW1", "SW2", "SW3"]
}
```

What is represented by "R1" and "SW1" within the JSON output?

- A. key
- B. array

- C. value
- D. object

---

**Answer: C**

---

Explanation:

---

**Question: 611**

---

DRAG DROP

Drag and drop the statements about networking from the left onto the corresponding networking types on the right. Not all statements are used.

|                                                                       |                                                               |
|-----------------------------------------------------------------------|---------------------------------------------------------------|
| This type deploys a consistent configuration across multiple devices. | <b>Controller-based Networking</b><br><div></div> <div></div> |
| A distributed control plane is needed.                                |                                                               |
| This type requires a distributed management plane.                    | <b>Traditional Networking</b><br><div></div> <div></div>      |
| Southbound APIs are used to apply configurations.                     |                                                               |
| Northbound APIs interact with end devices                             |                                                               |

---

**Answer:**

---

Explanation:

Controller-based Networking :

- This type deploys a consistent configuration across multiple devices.
- Southbound APIs are used to apply configurations.

Traditional Networking :

- A distributed control plane is needed.
- This type requires a distributed management plane.

On a SDN network the control plane is centralized on the SDN controller not distributed on the networking devices.

Northbound APIs do not interact with end devices. They allow the SDN controller to interact with applications on the application plane

---

**Question: 612**

---

Refer to the exhibit.

```
"attributes": {
  "pwd": "password1",
  "firstName": "Abraham",
  "lastName": "Lincoln",
  "phone": "5555551212",
  "email": "test@cisco.com"
},
"children": [{
  "aaaUserDomain": {
    "attributes": {
      "name": "ExampleCisco"
    },
    "children": [{
      "aaaUserRole": {
        "attributes": {
          "name": "admin"
        }
      }
    ]
  }
}]
}
```

How many objects are present in the given JSON-encoded data?

- A. one
- B. four

C. seven

D. nine

---

**Answer: B**

---

Explanation:

---

**Question: 613**

---

What are two examples of multifactor authentication? (Choose two.)

A. single sign-on

B. unique user knowledge

C. passwords that expire

D. soft tokens

E. shared password responsibility

---

**Answer: B,D**

---

Explanation:

---

**Question: 614**

---

DRAG DROP

Drag and chop the TCP/IP protocols from the left onto their primary transmission protocols on the right.



|        | TCP |
|--------|-----|
| DNS    |     |
| HTTP   |     |
| RTP    |     |
| SMTP   |     |
| SNMP   |     |
| Telnet |     |

|  | UDP |
|--|-----|
|  |     |
|  |     |
|  |     |

**Answer:**

Explanation:

| TCP  |
|------|
| DNS  |
| SMTP |
| SNMP |

| UDP    |
|--------|
| HTTP   |
| RTP    |
| Telnet |

**Question: 615**

## DRAG DROP

Drag and drop the Rapid PVST+ forwarding state actions from the left to the right. Not all actions are used.

|                                                                           |        |
|---------------------------------------------------------------------------|--------|
| BPDUs received are forwarded to the system module.                        | action |
| BPDUs received from the system module are processed and transmitted.      | action |
| Frames received from the attached segment are discarded.                  | action |
| Frames received from the attached segment are processed.                  | action |
| Switched frames received from other ports are advanced.                   |        |
| The port in the forwarding state responds to network management messages. |        |

---

**Answer:**

---

Explanation:

1. BPDUs received are forwarded to the system module.
2. Frames received from the attached segment are processed.
3. Switched frames received from other ports are advanced.
4. The port in the forwarding state responds to network management messages.

[https://www.cisco.com/c/en/us/td/docs/switches/datacenter/nexus5000/sw/layer2/503\\_n1\\_1/Cisco\\_n5k\\_layer2\\_config\\_gd\\_rel\\_503\\_N1\\_1\\_chapter9.html](https://www.cisco.com/c/en/us/td/docs/switches/datacenter/nexus5000/sw/layer2/503_n1_1/Cisco_n5k_layer2_config_gd_rel_503_N1_1_chapter9.html)

---

**Question: 616**

---

Refer to the exhibit.

```
Known via "connected", distance 0, metric 0 (connected, via interface)
Routing Descriptor Blocks:
* directly connected, via Ethernet0/1
  Route metric is 0, traffic share count is 1

CPE# ping 203.0.113.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

CPE# show ip route
Gateway of last resort is 198.51.100.1 to network 0.0.0.0
B*    0.0.0.0/0 [20/0] via 198.51.100.1, 00:02:07
      198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C      198.51.100.0/30 is directly connected, Ethernet0/0
L      198.51.100.2/32 is directly connected, Ethernet0/0
      203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C      203.0.113.0/30 is directly connected, Ethernet0/1
L      203.0.113.2/32 is directly connected, Ethernet0/1
```

After configuring a new static route on the CPE, the engineer entered this series of commands to verify that the new configuration is operating normally. When is the static default route installed into the routing table?

- A. when 203.0.113.1 is no longer reachable as a next hop
- B. when the default route learned over external BGP becomes invalid
- C. when a route to 203.0.113.1 is learned via BGP
- D. when the default route over external BGP changes its next hop

---

**Answer: A**

---

Explanation:

---

### Question: 617

---

How does encryption protect the wireless network?

- A. via integrity checks to identify wireless forgery attacks in the frame

- B. via specific ciphers to detect and prevent zero-day network attacks
- C. via an algorithm to change wireless data so that only the access point and client understand it
- D. via a policy to prevent unauthorized users from communicating on the wireless network

---

**Answer: C**

---

Explanation:

---

**Question: 618**

---

Which action implements physical access control as part of the security program of an organization?

- A. backing up syslogs at a remote location
- B. configuring a password for the console port
- C. configuring enable passwords on network devices
- D. setting up IP cameras to monitor key infrastructure

---

**Answer: B**

---

Explanation:

---

**Question: 619**

---

Why implement VRRP?

- A. to provide end users with a virtual gateway in a multivendor network
- B. to leverage a weighting scheme to provide uninterrupted service
- C. to detect link failures without the overhead of Bidirectional Forwarding Detection
- D. to hand over to end users the autodiscovery of virtual gateways

---

**Answer: A**

---

Explanation:

---

**Question: 620**

---

A packet from a company's branch office is destined to host 172.31.0.1 at headquarters. The sending router has three possible matches in its routing table for the packet prefixes: 172.31.0.0/16, 172.31.0.0/24, and 172.31.0.0/25. How does the router handle the packet?

- A. It sends the traffic via prefix 172.31.0.0/16
- B. It sends the traffic via the default gateway 0.0.0.0.
- C. It sends the traffic via prefix 172.31.0.0/24
- D. It sends the traffic via prefix 172.31.0.0/25

---

**Answer: D**

---

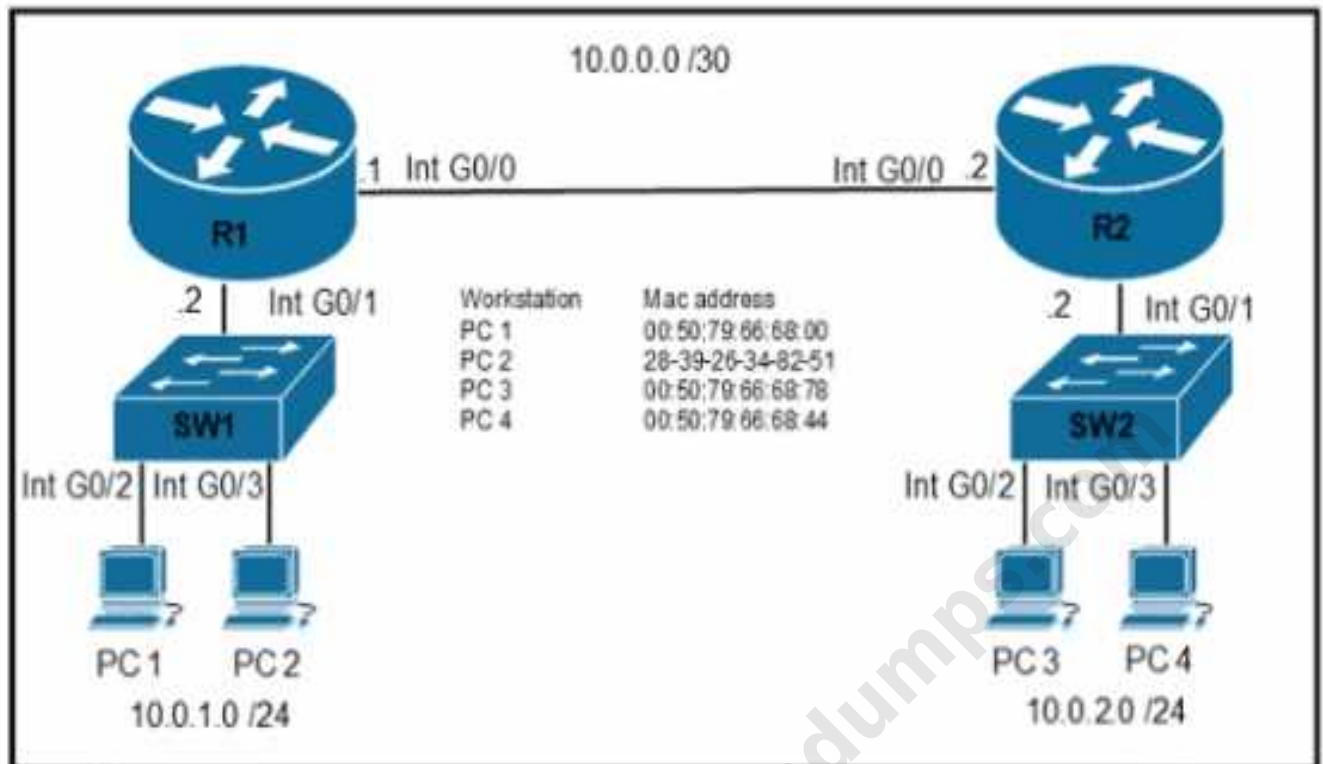
Explanation:

---

**Question: 621**

---

Refer to the exhibit.



An engineer must configure the interface that connects to PC 1 and secure it in a way that only PC1 is allowed to use the port No VLAN tagging can be used except for a voice VLAN. Which command sequence must be entered to configure the switch?

A)

```
SW1(config-if)#switchport mode nonegotiate
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security maximum 1
```

B)

```
SW1(config-if)#switchport mode access
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security mac-address 0050.7966.6800
```

C)

```
SW1(config-if)#switchport mode dynamic auto
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security violation restrict
```

D)

```
SW1(config-if)#switchport mode dynamic desirable
SW1(config-if)#switchport port-security mac-address 0050.7966.6800
SW1(config-if)#switchport port-security mac-address sticky
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: B**

---

Explanation:

---

**Question: 622**

---

DRAG DROP

Drag and drop the steps in a standard DNS lookup operation from the left into the order on the right.

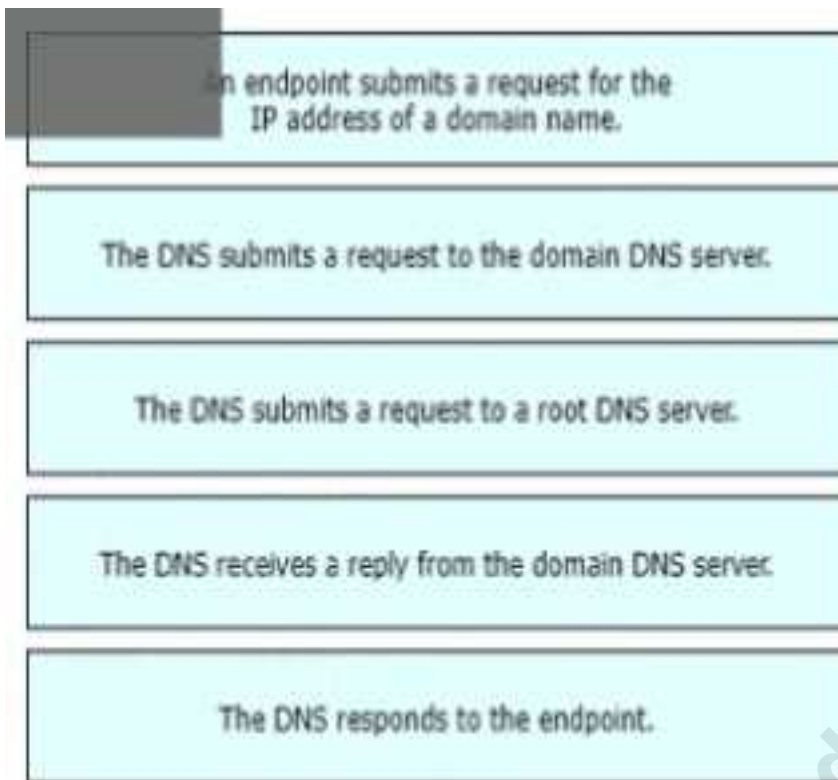
|                                                                    |   |
|--------------------------------------------------------------------|---|
| An endpoint submits a request for the IP address of a domain name. | 1 |
| The DNS receives a reply from the domain DNS server.               | 2 |
| The DNS responds to the endpoint.                                  | 3 |
| The DNS submits a request to a root DNS server.                    | 4 |
| The DNS submits a request to the domain DNS server.                | 5 |

---

**Answer:**

---

Explanation:



---

**Question: 623**

---

In a cloud-computing environment what is rapid elasticity?

- A. control and monitoring of resource consumption by the tenant
- B. automatic adjustment of capacity based on need
- C. pooling resources in a multitenant model based on need
- D. self-service of computing resources by the tenant

---

**Answer: B**

---

Explanation:

---

**Question: 624**

---

What is a reason to implement IPv4 private addressing?

- A. Reduce the risk of a network security breach



- B. Comply with PCI regulations
- C. Comply with local law
- D. Reduce the size of the forwarding table on network routers

---

**Answer: D**

---

Explanation:

---

**Question: 625**

---

DRAG DROP

Drag and drop the IPv6 address types from the left onto their description on the right.

|                                         |                                                                                      |
|-----------------------------------------|--------------------------------------------------------------------------------------|
| 2001:DB8::bced:1234:456d:aacc           | multicast address used only locally within the site                                  |
| FD00:0000:0000:1a2d:a153:3992:a19d:ccca | address that is automatically created on a link when IPv6 is enabled on an interface |
| FE80::abcf:ffff:12de:3992               | address that is prohibited from routing to the Internet                              |
| FF05::23:becf:22:1111                   | address that is unique and reserved for documentation purposes                       |

---

**Answer:**

---

Explanation:

FF05::23:bacf:22:1111

FE80::abcf:ffff:12de:3992

FD00:0000:0000:1a2d:a153:3992:a19d:ccca

2001:DB8::bced:1234:456d:aacc

---

**Question: 626**

---

What is the purpose of configuring different levels of syslog for different devices on the network?

- A. to rate-limit messages for different severity levels from each device
- B. to set the severity of syslog messages from each device
- C. to identify the source from which each syslog message originated
- D. to control the number of syslog messages from different devices that are stored locally

---

**Answer: B**

---

Explanation:

---

**Question: 627**

---

What is a function of MAC address learning?

- A. It is enabled by default on all VLANs and interfaces
- B. It increases the potential for MAC address flooding.
- C. It is disabled by default on all interfaces connected to trunks

D. It increases security on the management VLAN

---

**Answer: A**

---

Explanation:

---

**Question: 628**

---

Which two IPv6 addresses are used to provide connectivity between two routers on a shared link?  
(Choose two)

- A. ::ffif 1014 1011/96
- B. 2001 7011046:1111:1/64
- C. ;jff06bb43cd4dd111bbff02 4545234d
- D. 2002 5121204b 1111:1/64
- E. FF02::0WIFF00:0I)00/104

---

**Answer: B**

---

Explanation:

---

**Question: 629**

---

Refer to the exhibit.

When router R1 receives a packet with destination IP address 10.56.0 62. through which interface does it route the packet?

- A. Null0
- B. Vlan58
- C. Vlan60
- D. Vlan59

---

**Answer: B**

---

Explanation:

---

**Question: 630**

---

What are two features of the DHCP relay agent? (Choose two.)

- A. assigns DNS locally and then forwards request to DHCP server
- B. permits one IP helper command under an individual Layer 3 interface
- C. allows only MAC-to-IP reservations to determine the local subnet of a client
- D. minimizes the necessary number of DHCP servers
- E. configured under the Layer 3 interface of a router on the client subnet

---

**Answer: B,E**

---

Explanation:

---

**Question: 631**

---

Which is a fact related to FTP?

- A. It uses block numbers to identify and mitigate data-transfer errors
- B. It always operates without user authentication
- C. It relies on the well-known UDP port 69.
- D. It uses two separate connections for control and data traffic

---

**Answer: D**

---

Explanation:

---

**Question: 632**

---

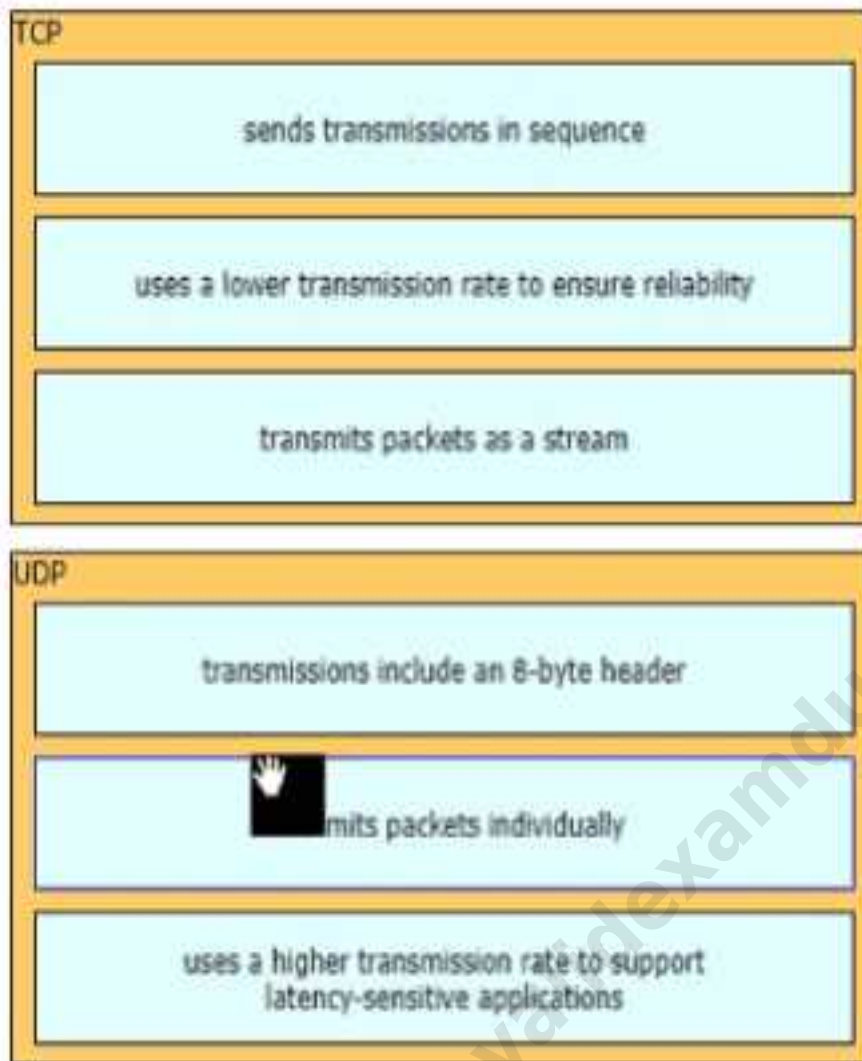
## DRAG DROP

Drag and drop the descriptions of IP protocol transmissions from the left onto the IP traffic types on the right.

|                                                                           |     |  |
|---------------------------------------------------------------------------|-----|--|
| sends transmissions in sequence                                           | TCP |  |
| transmissions include an 8-byte header                                    |     |  |
| transmits packets as a stream                                             |     |  |
| transmits packets individually                                            | UDP |  |
| uses a higher transmission rate to support latency-sensitive applications |     |  |
| uses a lower transmission rate to ensure reliability                      |     |  |

**Answer:**

Explanation:



---

**Question: 633**

---

Refer to the exhibit.

```
router# show ip route
```

```
....
```

```
D 172.18.32.0/26 [90/25789217] via 10.1.1.1
```

```
R 172.18.32.0/24 [120/4] via 10.1.1.2
```

```
O 172.18.32.0/19 [110/229840] via 10.1.1.3
```

```
C 172.18.32.32/32 is directly connected, Loopback0
```

```
C 172.18.32.36/30 is directly connected, GigabitEthernet0/0
```

```
L 172.18.32.37/32 is directly connected, GigabitEthernet0/0
```

A packet sourced from 172.18.33.2 is destined for 172.18.32.38. Where does the router forward the packet?

- A. GigabitEthernet0/0
- B. Loopback0
- C. 10.1.1.1
- D. 10.1.1.3

---

**Answer: A**

---

Explanation:

---

**Question: 634**

---

Which command configures the Cisco WLC to prevent a serial session with the WLC CLI from being automatical toggled out?

- A. config sessions maxsessions 0
- B. config sessions timeout 0
- C. config serial timeout 0
- D. config serial timeout 9600

---

**Answer: B**

---

Explanation:

---

**Question: 635**

---

Refer to the exhibit.

```
{  
  "SW1" : ["Ten-GigabitEthernet0/0", "Ten-GigabitEthernet0/1"],  
  "SW2" : ["Ten-GigabitEthernet0/0", "Ten-GigabitEthernet0/1"],  
  "SW3" : ["Ten-GigabitEthernet0/0", "Ten-GigabitEthernet0/1"],  
  "SW4" : ["Ten-GigabitEthernet0/0", "Ten-GigabitEthernet0/1"]  
}
```

How many JSON objects are presented?

- A. 1
- B. 2
- C. 3
- D. 4

---

**Answer: D**

---

Explanation:

---

**Question: 636**

---

Which enhancement is implemented in WPA3?

- A. applies 802.1x authentication



- B. uses TKIP
- C. employs PKI to identify access points
- D. protects against brute force attacks

---

**Answer: D**

---

Explanation:

---

**Question: 637**

---

What must be considered for a locally switched FlexConnect AP if the VLANs that are used by the AP and client access are different?

- A. The APs must be connected to the switch with multiple links in LAG mode
- B. The switch port mode must be set to trunk
- C. The native VLAN must match the management VLAN of the AP
- D. IEEE 802.1Q trunking must be disabled on the switch port.

---

**Answer: C**

---

Explanation:

---

**Question: 638**

---

Refer to the exhibit.

```
R1#show ip ospf interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Internet address is 192.168.1.2/24, Area 0
Process ID 1, Router ID 192.168.1.2, Network Type POINT-TO-POINT, Cost: 1
Transmit Delay is 1 sec, State POINT-TO-POINT,
Timer intervals configured, Hello 15, Dead 20, Wait 20, Retransmit 5
Hello due in 00:00:08
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Suppress hello for 0 neighbor(s)

R2#show ip ospf interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Internet address is 192.168.1.1/24, Area 0
Process ID 1, Router ID 10.1.1.1, Network Type POINT-TO-POINT, Cost: 1
Transmit Delay is 1 sec, State POINT-TO-POINT,
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:11
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Suppress hello for 0 neighbor(s)
```

The network engineer is configuring router R2 as a replacement router on the network. After the initial configuration is applied, it is determined that R2 failed to show R1 as a neighbor. Which configuration must be applied to R2 to complete the OSPF configuration and enable it to establish the neighbor relationship with R1?

A)

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.1.0 255.255.255.0 area 2
R2(config-router)#network 10.1.1.0 255.255.255.255 area 2
```

B)

```
R2(config)#router ospf 1
R2(config-router)#router-id 192.168.1.2
```

C)

```
R2(config)#interface g0/0/0
R2(config-if)#ip ospf dead-interval 20
```

D)

```
R2(config)#interface g0/0/0  
R2(config-if)#ip ospf hello-interval 15  
R2(config-if)#ip ospf dead-interval 20
```

- A. Option
- B. Option
- C. Option
- D. Option

---

**Answer: D**

---

Explanation:

---

**Question: 639**

---

What is an advantage of using auto mode versus static mode for power allocation when an access point is connected to a PoE switch port?

- A. All four pairs of the cable are used
- B. It detects the device is a powered device
- C. The default level is used for the access point
- D. Power policing is enabled at the same time

---

**Answer: D**

---

Explanation:

---

**Question: 640**

---

Which type of port is used to connect to the wired network when an autonomous AP maps two VLANs to its WLANs?

- A. LAG
- B. EtherChannel

- C. trunk
- D. access

---

**Answer: D**

---

Explanation:

---

**Question: 641**

---

What is a function of an endpoint?

- A. It is used directly by an individual user to access network services
- B. It passes unicast communication between hosts in a network
- C. It transmits broadcast traffic between devices in the same VLAN
- D. It provides security between trusted and untrusted sections of the network.

---

**Answer: A**

---

Explanation:

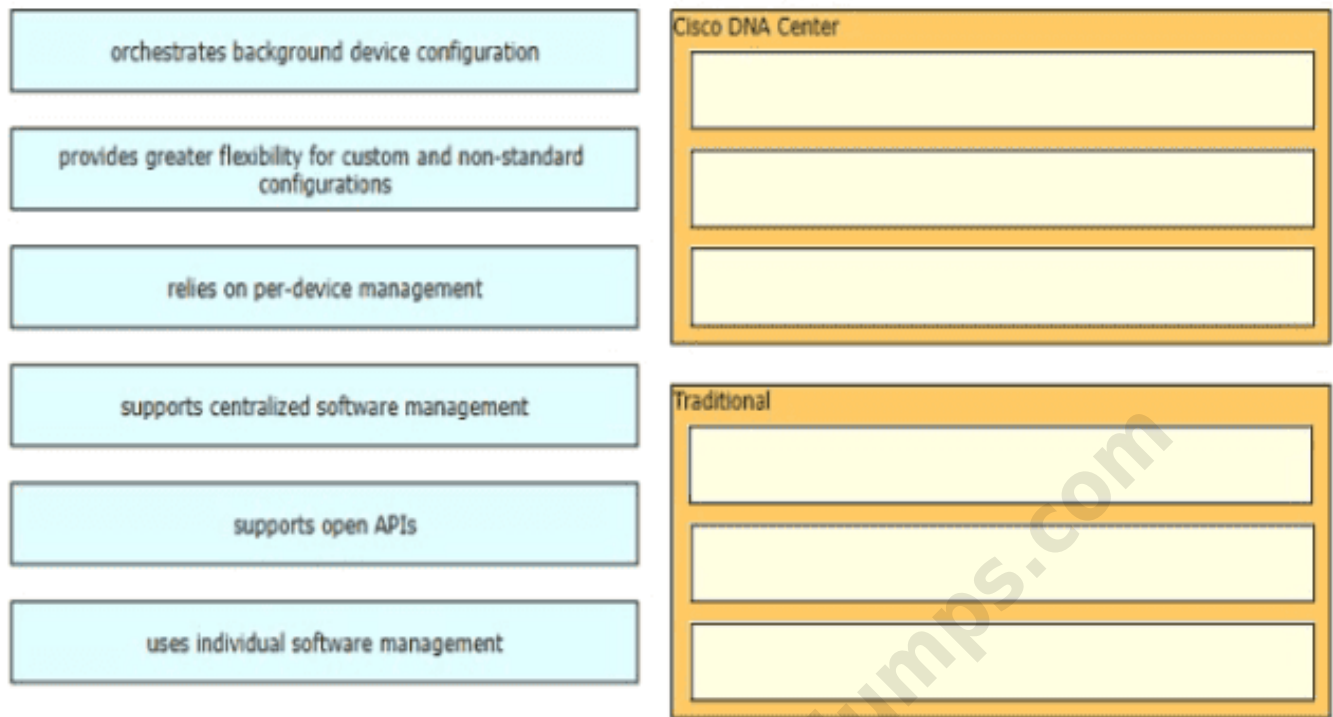
---

**Question: 642**

---

DRAG DROP

Drag and drop the characteristics of device-management technologies from the left onto the corresponding deployment types on the right.



**Answer:**

Explanation:



---

**Question: 643**

---

Refer to the exhibit.

The image shows the Cisco Wireless LAN Controller configuration interface. The top navigation bar includes tabs for MONITOR, WLANs, CONTROLLER, WIRELESS, SECURITY (highlighted), MANAGEMENT, and CONFIG. On the left, the 'Security' menu is expanded, showing options like AAA, RADIUS, TACACS+, LDAP, Local Net Users, MAC Filtering, Disabled Clients, User Login Policies, AP Policies, and Password Policies. The 'Local Net Users > New' configuration page is displayed. It contains the following fields: 'User Name' (NA-User), 'Password' (masked with dots), 'Confirm Password' (masked with dots), 'Guest User' (checked), 'Lifetime (seconds)' (86400), 'Guest User Role' (unchecked), 'WLAN Profile' (Any WLAN), and 'Description' (For NA WLAN Auth).

Wireless LAN access must be set up to force all clients from the NA WLAN to authenticate against the local database. The WLAN is configured for local EAP authentication. The time that users access the network must not be limited. Which action completes this configuration?

- A. Uncheck the Guest User check box
- B. Check the Guest User Role check box
- C. Set the Lifetime (seconds) value to 0
- D. Clear the Lifetime (seconds) value

---

**Answer: C**

---

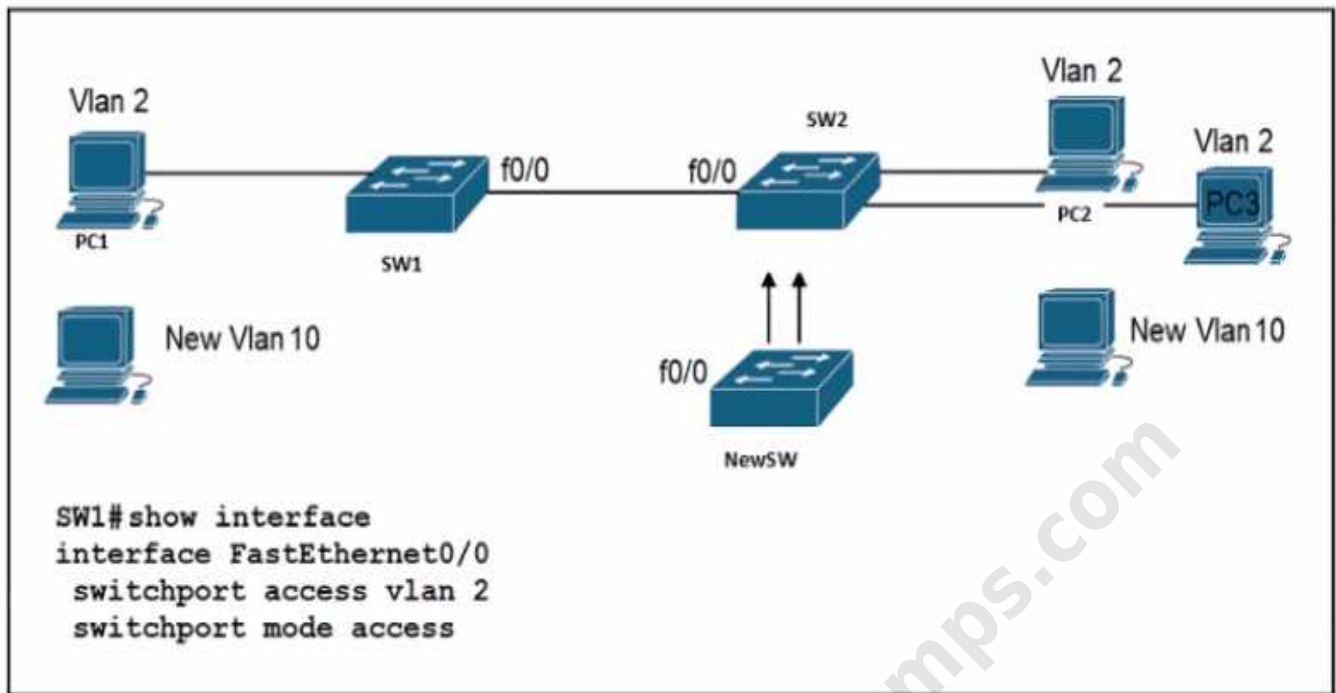
Explanation:

---

**Question: 644**

---

Refer to the exhibit.



An engineer is configuring a new Cisco switch NewSW, to replace SW2. The details have been provided

- Switches SW1 and SW2 are third-party devices without support for trunk ports
- The existing connections must be maintained between PC1, PC2, and PC3
- Allow the switch to pass traffic from future VLAN 10. Which configuration must be applied?

A)

```

NewSW(config)#interface f0/0
NewSW(config-if)#switchport mode access
NewSW(config-if)#switchport trunk allowed vlan 2,10
NewSW(config-if)#switchport trunk native vlan 10
  
```

B)

```

NewSW(config)#interface f0/0
NewSW(config-if)#switchport mode trunk
NewSW(config-if)#switchport trunk allowed vlan 2,10
NewSW(config-if)#switchport trunk native vlan 2
  
```

C)

```

NewSW(config)#interface f0/0
NewSW(config-if)#switchport mode trunk
NewSW(config-if)#switchport trunk allowed vlan 10
NewSW(config-if)#switchport trunk native vlan 10
  
```

D)

```
NewSW(config)#interface f0/0  
NewSW(config-if)#switchport mode access  
NewSW(config-if)#switchport trunk allowed vlan 2,10  
NewSW(config-if)#switchport trunk native vlan 2
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: B**

---

Explanation:

---

**Question: 645**

---

Refer to the exhibit.



General Security QoS Policy-Mapping Advanced

Layer 2 Layer 3 AAA Servers

Layer 2 Security <sup>6</sup> WPA+WPA2 ▼

MAC Filtering<sup>8</sup> ☐

Fast Transition

Fast Transition ☐

Protected Management Frame

PMF Required ▼

WPA+WPA2 Parameters

WPA Policy ☐

WPA2 Policy ☐

WPA2 Encryption ☐ AES ☐ TKIP

OSSEN Policy ☐

Authentication Key Management <sup>19</sup>

802.1X ☐ Enable

CCKM ☐ Enable

PSK ☐ Enable

FT 802.1X ☐ Enable

FT PSK ☐ Enable

PSK Format ASCII ▼

WPA gtk-randomize State <sup>14</sup> Disable ▼

A)

Select WPA Policy  
Select WPA2 Policy  
Enable FT PSK

B)

Select WPA2 Policy  
Disable PMF  
Enable PSK

C)

Select WPA Policy  
Enable CCKM  
Enable PSK

D)

Disable PMF  
Enable PSK  
Enable 802.1x

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: C**

---

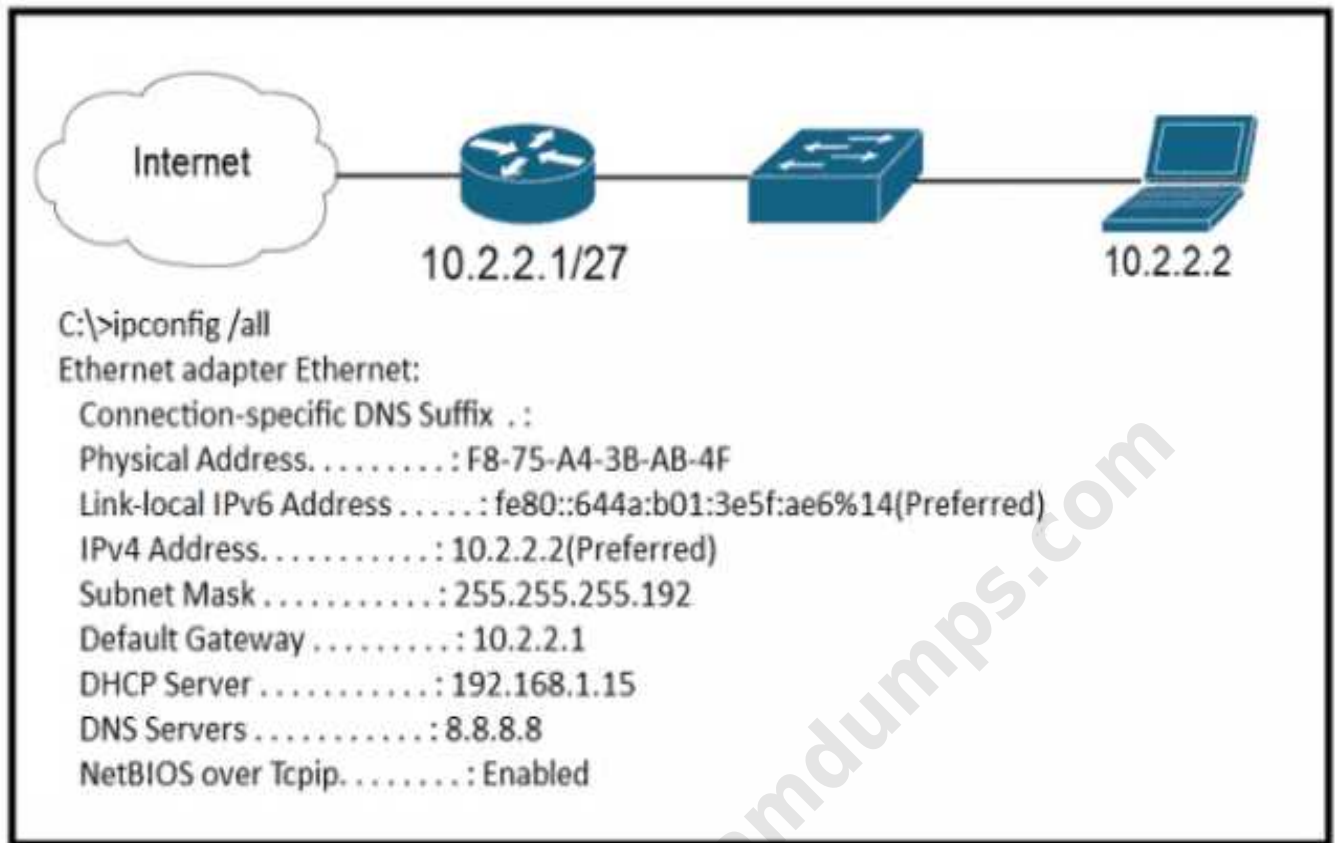
Explanation:

---

**Question: 646**

---

Refer to the exhibit.



A newly configured PC fails to connect to the internet using TCP port 80 to www.cisco.com. Which setting must be modified for the connection to work?

- A. Subnet Mask
- B. DNS Servers
- C. Default Gateway
- D. DHCP Server

**Answer: B**

Explanation:

**Question: 647**

Refer to the exhibit.

```
CPE1# show protocols e0/1
Ethernet0/1 is up, line protocol is up
Internet address is 10.0.12.2/24

CPE1# show ip access-list LAN
Standard IP access list LAN
 10 permit 10.0.12.0, wildcard bits 0.0.0.255

CPE1# show ip nat translations

CPE1# show ip nat statistics
Total active translations: 0 (0 static, 0 dynamic; 0 extended)
Peak translations: 0
Outside interfaces:
Inside interfaces:
  Ethernet0/1
Hits: 0 Misses: 0
CEF Translated packets: 0, CEF Punted packets: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list LAN pool NATPOOL refcount 0
  pool NATPOOL: netmask 255.255.255.0
    start 198.51.100.11 end 198.51.100.20
    type generic, total addresses 10, allocated 0 (0%), misses 0

Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
```

What is the next step to complete the implementation for the partial NAT configuration shown?

- A. Reconfigure the static NAT entries that overlap the NAT pool
- B. Configure the NAT outside interface
- C. Modify the access list for the internal network on e0/1
- D. Apply the ACL to the pool configuration

---

**Answer: B**

---

Explanation:

---

**Question: 648**

---

DRAG DROP

Drag and drop the Ansible features from the left to the right Not all features are used.

|                                                         |         |
|---------------------------------------------------------|---------|
| executes modules via SSH by default                     | feature |
| uses the YAML language                                  | feature |
| uses agents to manage hosts                             | feature |
| pushes configurations to the client                     | feature |
| requires clients to pull configurations from the server |         |
| operates without agents                                 |         |

---

**Answer:**

---

Explanation:

operates without agents

executes modules via SSH by default

uses agents to manage hosts

pushes configurations to the client

---

**Question: 649**

---

The address block 192.168.32.0/24 must be subnetted into smaller networks. The engineer must meet these requirements:

- Create 8 new subnets
- Each subnet must accommodate 30 hosts
- Interface VLAN 10 must use the last usable IP in the first new subnet
- A Layer 3 interface is used

Which configuration must be applied to the interface?

A)

```
no switchport mode access  
ip address 192.168.32.62 255.255.255.240
```

B)

```
switchport  
ip address 192.168.32.65 255.255.255.240
```

C)

```
no switchport mode trunk  
ip address 192.168.32.97 255.255.255.224
```

D)

```
no switchport
ip address 192.168.32.30 255.255.255.224
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: D**

---

Explanation:

---

### Question: 650

---

DRAG DROP

Refer to the exhibit.

```
R1# show ip route | begin gateway
Gateway of last resort is not set
  172.16.0.0/16 is variably subnetted, 5 subnets, 5 masks
O   172.16.2.128/25 [110/3184437] via 207.165.200.250, 00:00:25, Serial0/0/0
O   172.16.3.64/27 [110/3184437] via 207.165.200.250, 00:00:25, Serial0/0/0
O   172.16.3.128/28 [110/3184437] via 207.165.200.250, 00:00:25, Serial0/0/0
O   172.16.3.192/29 [110/3184437] via 207.165.200.250, 00:00:25, Serial0/0/0
O   172.16.4.0/23 [110/3184437] via 207.165.200.250, 00:00:25, Serial0/0/0
  207.165.200.0/24 is variably subnetted, 4 subnets, 2 masks
C   207.165.200.248/30 is directly connected, Serial0/0/0
L   207.165.200.249/32 is directly connected, Serial0/0/0
C   207.165.200.252/30 is directly connected, Serial0/0/1
L   207.165.200.253/32 is directly connected, Serial0/0/1
```

Drag and drop the learned prefixes from the left onto the subnet masks on the right

|              |                 |
|--------------|-----------------|
| 172.16.3.128 | 255.255.254.0   |
| 172.16.3.64  | 255.255.255.128 |
| 172.16.2.128 | 255.255.255.224 |
| 172.16.3.192 | 255.255.255.240 |
| 172.16.4.0   | 255.255.255.248 |

---

**Answer:**

---

Explanation:

|              |
|--------------|
| 172.16.4.0   |
| 172.16.2.128 |
| 172.16.3.64  |
| 172.16.3.128 |
| 172.16.3.192 |

---

**Question: 651**

---

What are two reasons a switch experiences frame flooding? (Choose two.)

- A. A defective patch cable is connected to the switch port
- B. Topology changes are occurring within spanning-tree



- C. An aged MAC (able entry is causing excessive updates
- D. Port-security is configured globally
- E. The forwarding table has overflowed

---

**Answer: A,B**

---

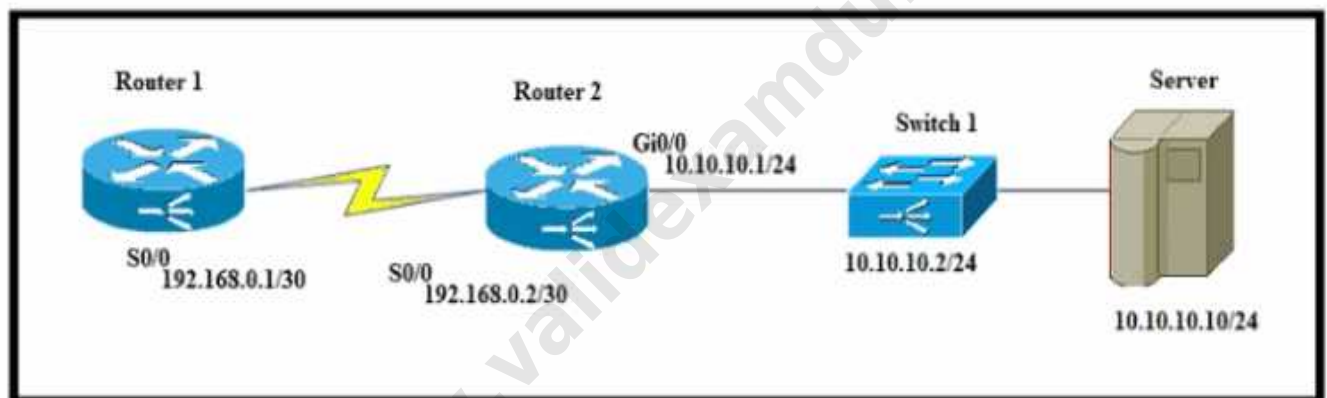
Explanation:

---

**Question: 652**

---

Refer to the exhibit.



A network engineer must configure router R1 with a host route to the server. Which command must the engineer configure?

- A. R1(config)#ip route 10.10.10.0 255.255.255.0 192.168.0.2
- B. R1(Config)#ip route 10.10.10.10 255.255.255.255 192 168.0.2
- C. R1(config)#ip route 192.168.0.2 255.255.255.255 10.10.10.10
- D. R1(config)#ip route 0.0.0.0 0.0.0.0 192 168.0.2

---

**Answer: B**

---

Explanation:

---

**Question: 653**

---

A Cisco engineer at a new branch office is configuring a wireless network with access points that connect to a controller that is based at corporate headquarters. Wireless client traffic must terminate at the branch office and access-point survivability is required in the event of a WAN outage. Which access point mode must be selected?

- A. Lightweight with local switching disabled
- B. Local with AP fallback enabled
- C. OfficeExtend with high availability disabled
- D. FlexConnect with local switching enabled

---

**Answer: D**

---

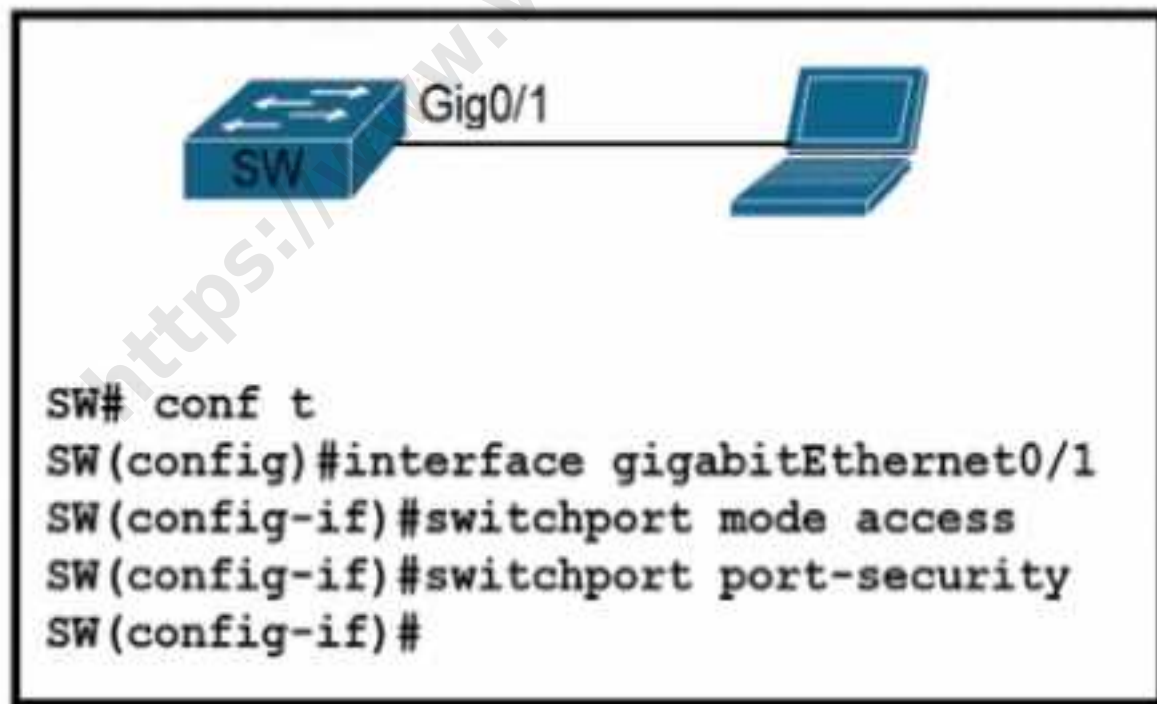
Explanation:

---

**Question: 654**

---

Refer to the exhibit



A network engineer started to configure port security on a new switch. These requirements must be met:

- \* MAC addresses must be learned dynamically
- \* Log messages must be generated without disabling the interface when unwanted traffic is seen

Which two commands must be configured to complete this task"? (Choose two)

- A. SW(ccnfig-if)=switchport port-security mac-address sticky
- B. SW(confKj-if)=switchport port-security violation restrict
- C. SW(config.if)sswitchport port-security mac-address 0010.7B84.45E6
- D. SW(config-if)aswitchport port-security maximum 2
- E. SW(ccnfig-if)=switchport port-security violation shutdown

---

**Answer: C**

---

Explanation:

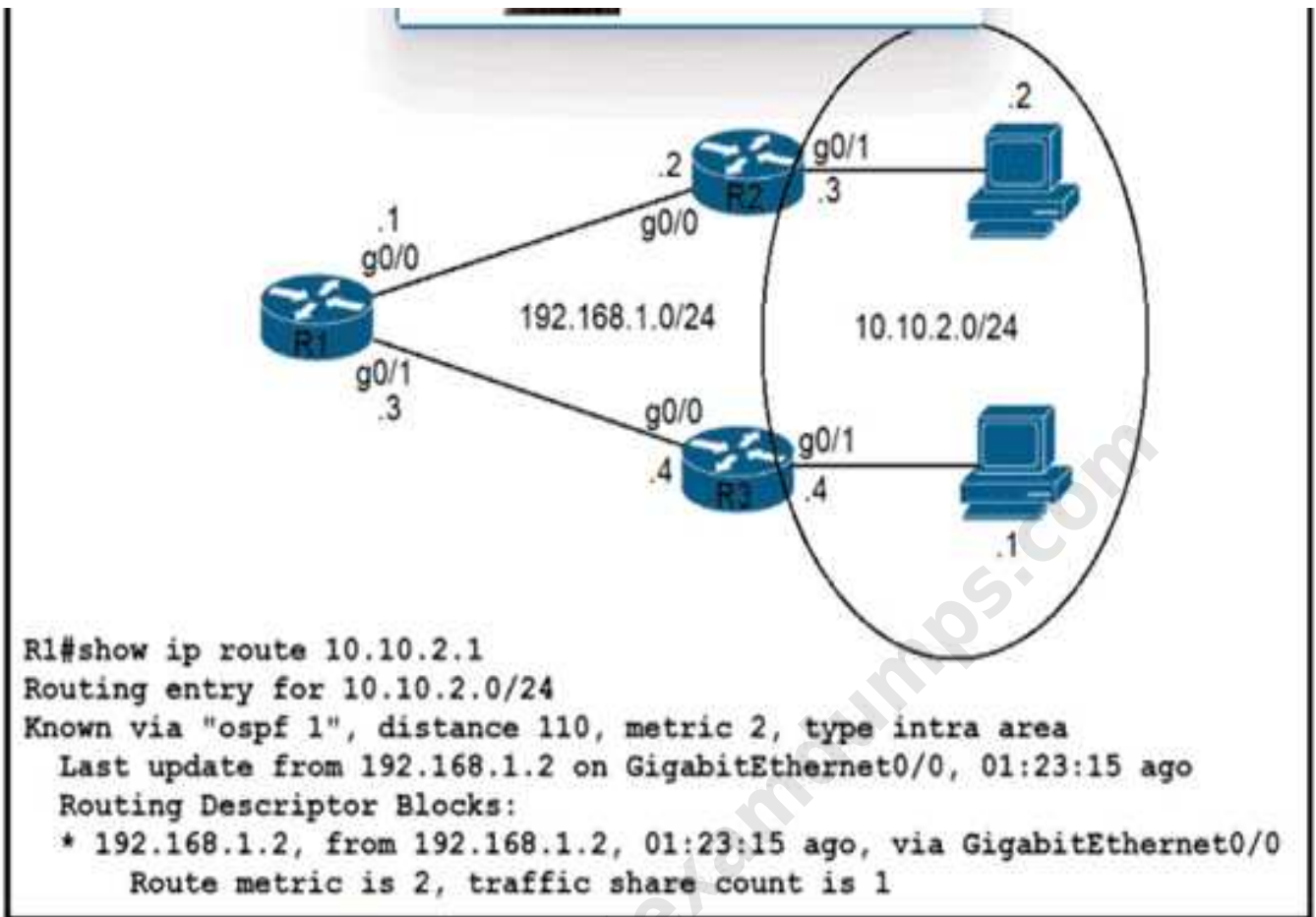
---

**Question: 655**

---

Refer to the exhibit.

Traffic from R1 to the 10.10.2.0/24 subnet uses 192.168.1.2 as its next hop. An network engineer wants to update the R1 configuration so that traffic with destination 10.10.2.1 passes through router R3, and all other traffic to the 10.10.20/24 subnet passes through r2. Which command must be used?



- A. Ip route 10.10.2.1 255.255.255.255 192.168.1.4 115
- B. Ip route 10.10.2.0 255.255.255.0 192.168.1.4 100
- C. Ip route 10.10.2.0 255.255.255.0 192.168.1.4 115
- D. Ip route 10.10.2.1 255.255.255.255 192.168.1.4 100

**Answer: D**

Explanation:

**Question: 656**

Refer the exhibit.

```
R19#sh int fa0/0
FastEthernet0/0 is up, line protocol is up
Hardware is DEC21140, address is ca02.7788.0000 (bia ca02.7788.0000)
Description: SALES_SUBNET
Internet address is 10.32.102.2/30
MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (60 sec)
Full-duplex, 100Mb/s, 100BaseTX/FX
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:01, output 00:00:00, output hang never
Last clearing of "show interface" counters never
Input queue: 0/300/0/0 (size/max/drops/flushes); Total output drops:
135298429
Queueing strategy: fifo
Output queue: 0/300 (size/max)
30 second input rate 0 bits/sec, 0 packets/sec
30 second output rate 0 bits/sec, 0 packets/sec
73310 packets input, 7101162 bytes
Received 73115 broadcasts (0 IP multicasts)
0 runs, 0 giants, 0 throttles
0 input errors, 4 CRC, 0 frame, 0 overrun, 0 ignored
0 watchdog
0 input packets with dribble condition detected
3927513096455 packets output, 14404034810952 bytes, 0 underruns
0 output errors. 11 collisions. 0 interface resets
```

What is the cause of poor performance on router R19?

- A. excessive collisions
- B. speed and duplex mismatch
- C. port oversubscription
- D. excessive CRC errors

---

**Answer: A**

---

Explanation:

---

**Question: 657**

---

Which two protocols are used by an administrator for authentication and configuration on access points?

- A. Kerberos
- B. 802.1Q
- C. 802.1x
- D. TACACS+
- E. RADIUS

---

**Answer: D,E**

---

Explanation:

---

**Question: 658**

---

What is a similarity OM3 and OM4 fiber optical cable?

- A. Both have a 62.5 micron core diameter.
- B. Both have a 50 micron core diameter.
- C. Both have a 100 micron core diameter.
- D. Both have a 9 micron core diameter.

---

**Answer: B**

---

Explanation:

**Question: 659**

Why is TCP desired over UDP for application that require extensive error checking, such as HTTPS?

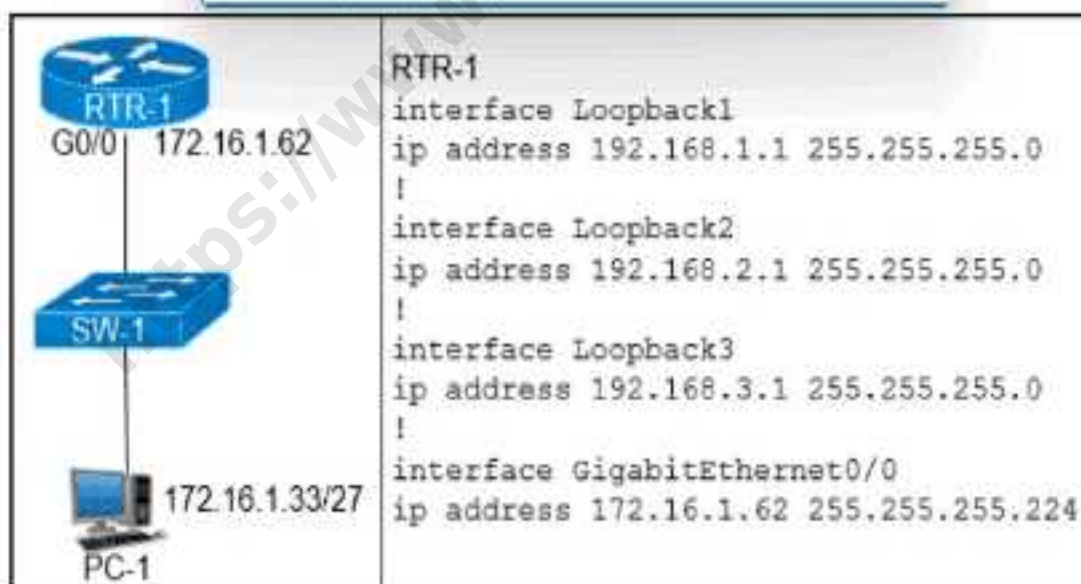
- A. UDP operates without acknowledgments, and TCP sends an acknowledgment for every packet received.
- B. UDP reliably guarantees delivery of all packets, and TCP drops packets under heavy load.
- C. UDP uses flow control mechanisms for the delivery of packets, and TCP uses congestion control for efficient packet delivery.
- D. UDP uses sequencing data for packets to arrive in order, and TCP offers the capability to receive packets in random order.

**Answer: A**

Explanation:

**Question: 660**

Refer to the exhibit.



Which configuration for RTR-1 denies SSH access from PC-1 to any RTR-1 interface and allows all other traffic?

A)



```
access-list 100 deny tcp host 172.16.1.33 any eq 22
access-list 100 permit ip any any
```

```
interface GigabitEthernet0/0
ip access-group 100 in
```

B)

```
access-list 100 deny tcp host 172.16.1.33 any eq 22
access-list 100 permit ip any any
```

```
line vty 0 15
access-class 100 in
```

C)

```
access-list 100 deny tcp host 172.16.1.33 any eq 23
access-list 100 permit ip any any
```

```
interface GigabitEthernet0/0
ip access-group 100 in
```

D)

```
access-list 100 deny tcp host 172.16.1.33 any eq 23
access-list 100 permit ip any any
```

```
line vty 0 15
access-class 100 in
```

A. Option A

B. Option B

C. Option C

D. Option D



---

**Answer: B**

---

Explanation:

---

**Question: 661**

---

An engineer is configuring a switch port that is connected to a VoIP handset. Which command must the engineer configure to enable port security with a manually assigned MAC address of abcd-bod on voice VLAN 4?

- A. switchport port-security mac-address abcd.abcd.abcd
- B. switchport port-security mac-address abed.abed.abed vlan 4
- C. switchport port-security mac-address sticky abcd.abcd.abcd vlan 4
- D. switchport port-security mac-address abcd.abcd.abcd vlan voice

---

**Answer: A**

---

Explanation:

---

**Question: 662**

---

DRAG DROP

Drag and drop the Ansible terms from the left onto the right.

|              |                                                                                            |
|--------------|--------------------------------------------------------------------------------------------|
| control node | collection of actions to perform on target devices, expressed in YAML format               |
| inventory    | device with Ansible installed that manages target devices                                  |
| managed node | network device, without Ansible installed, upon which commands can be executed             |
| module       | specific action to be performed on one or more target devices                              |
| playbook     | unit of Python code to be executed                                                         |
| task         | Ansible file that defines the target devices upon which commands and tasks can be executed |

**Answer:**

Explanation:



---

**Question: 663**

---

Refer to the exhibit.

| Entry # |                              |
|---------|------------------------------|
| 1       | 192.168.10.0 255.255.254.0   |
| 2       | 192.168.10.0 255.255.255.192 |
| 3       | 192.168.10.0 255.255.0.0     |
| 4       | 192.168.10.0 255.255.224.0   |

Which entry is the longest prefix match for host IP address 192.168.10.5?

- A. 1
- B. 2
- C. 3

D. 4

---

**Answer: B**

---

Explanation:

---

**Question: 664**

---

What is the function of northbound API?

- A. It upgrades software and restores files.
- B. It relies on global provisioning and configuration.
- C. It supports distributed processing for configuration.
- D. It provides a path between an SDN controller and network applications.

---

**Answer: D**

---

Explanation:

---

**Question: 665**

---

Which interface enables communication between a program on the controller and a program on the networking devices?

- A. northbound interface
- B. software virtual interface
- C. southbound interface
- D. tunnel Interface

---

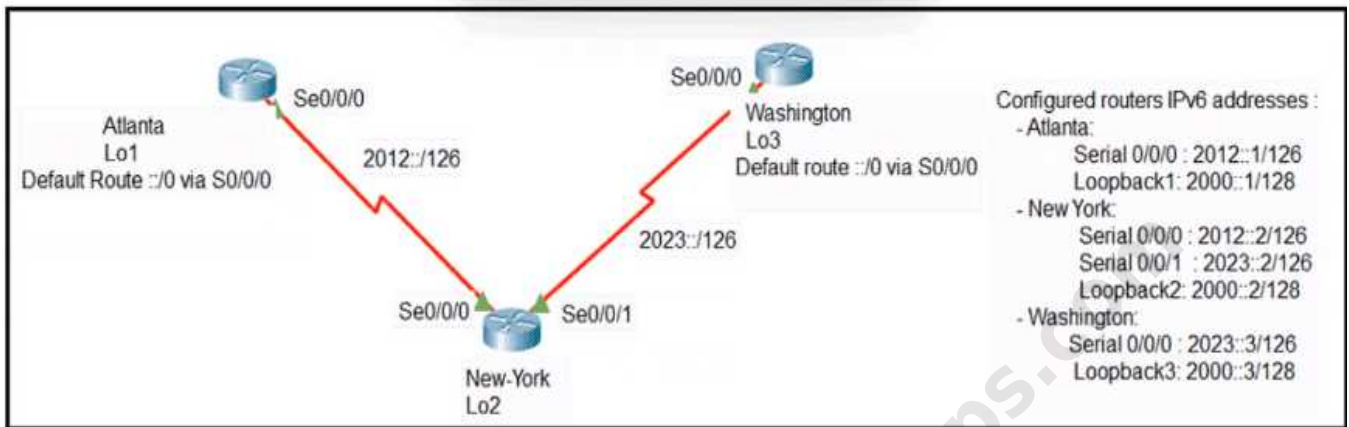
**Answer: B**

---

Explanation:

**Question: 666**

Refer to the exhibit.



The loopback1 interface of the Atlanta router must reach the lookback3 interface of the Washington router.

- A. ipv6 route 2000::1/128 2012::2
- B. ipv6 route 2000::1/128 2012::1
- C. ipv6 route 2000::3 123 s0/0/0
- D. ipv6 route 2000::3/128 2023::3
- E. ipv6 route 2000::1/128 s0/0/1

**Answer: B,D**

Explanation:

**Question: 667**

Refer to the exhibit.

```
Router1#show ip route
Gateway of last resort is not set
  209.165.200.0/27 is subnetted, 1 subnets
B       209.165.200.224 [20/0] via 10.10.12.2, 00:09:57
  10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C       10.10.10.0/28 is directly connected, GigabitEthernet0/0
C       10.10.11.0/30 is directly connected, FastEthernet2/0
O       10.10.13.0/24 [110/2] via 10.10.10.1, 00:08:34, GigabitEthernet0/0
C       10.10.12.0/30 is directly connected, GigabitEthernet0/1
```

Which action by the router when a packet is sourced from 10.10.10.2 and destined 10.10.10.16?

- A. It queues the packets waiting for the route to be learned.
- B. It floods packets to all learned next hops.
- C. It discards the packets.
- D. It uses a route that is similar to the destination address.

---

**Answer: D**

---

Explanation:

---

### Question: 668

---

What does a switch do when it receives a frame whose destination MAC address is missing from the MAC address table?

- A. It floods the frame unchanged across all remaining ports in the incoming VLAN.
- B. It appends the table with a static entry for the MAC and shuts down the port.
- C. It updates the CAM table with the destination MAC address of the frame.
- D. It changes the checksum of the frame to a value that indicates an invalid frame.

---

**Answer: A**

---

Explanation:

---

**Question: 669**

---

Which security method is used to prevent man-in-the-middle attack?

- A. authorization
- B. authentication
- C. anti-replay
- D. accounting

---

**Answer: B**

---

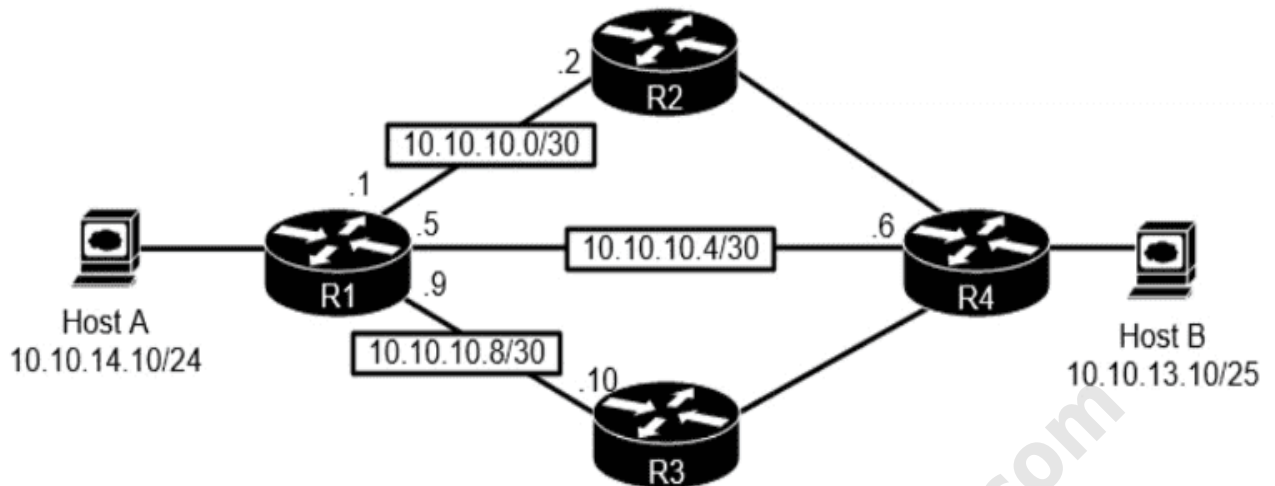
Explanation:

---

**Question: 670**

---

Refer to the exhibit.



```

R1(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.2
R1(config)#ip route 10.10.13.0 255.255.255.128 10.10.10.2 111
R1(config)#ip route 10.10.13.0 255.255.255.128 10.10.10.6 112
R1(config)#ip route 10.10.13.0 255.255.255.128 10.10.10.10 108
R1(config-router)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#network 10.10.10.5 0.0.0.0 area 0
R1(config-router)#network 10.10.10.1 0.0.0.0 area 0
R1(config-router)#network 10.10.14.1 0.0.0.0 area 0

```

R1 has just received a packet from host A that is destined to host B. Which route in the routing table is used by R1 to reach B?

- A. 10.10.13.0/25 [108/0] via 10.10.10.10
- B. 10.10.13.0/25 [110/2] via 10.10.10.2
- C. 10.10.13.0/25 [110/2] via 10.10.10.6
- D. 10.10.13.0/25 [1/0] via 10.10.10.2

**Answer: D**

Explanation:

### Question: 671

DRAG DROP

Drag and drop the WLAN components from the left onto the component details on the right.



|                         |                                                             |
|-------------------------|-------------------------------------------------------------|
| access point            | manages access points                                       |
| virtual interface       | provides Wi-Fi devices with a connection to a wired network |
| dynamic interface       | used for out-of-band management                             |
| service port            | used for guest authentication                               |
| wireless LAN controller | applied to the WLAN for wireless client communication       |

**Answer:**

Explanation:

wireless LAN controller

access point

service port

virtual interface

dynamic interface

---

**Question: 672**

---

Which IPsec transport mode encrypts the IP header and the payload?

- A. pipe
- B. control
- C. transport
- D. tunnel

---

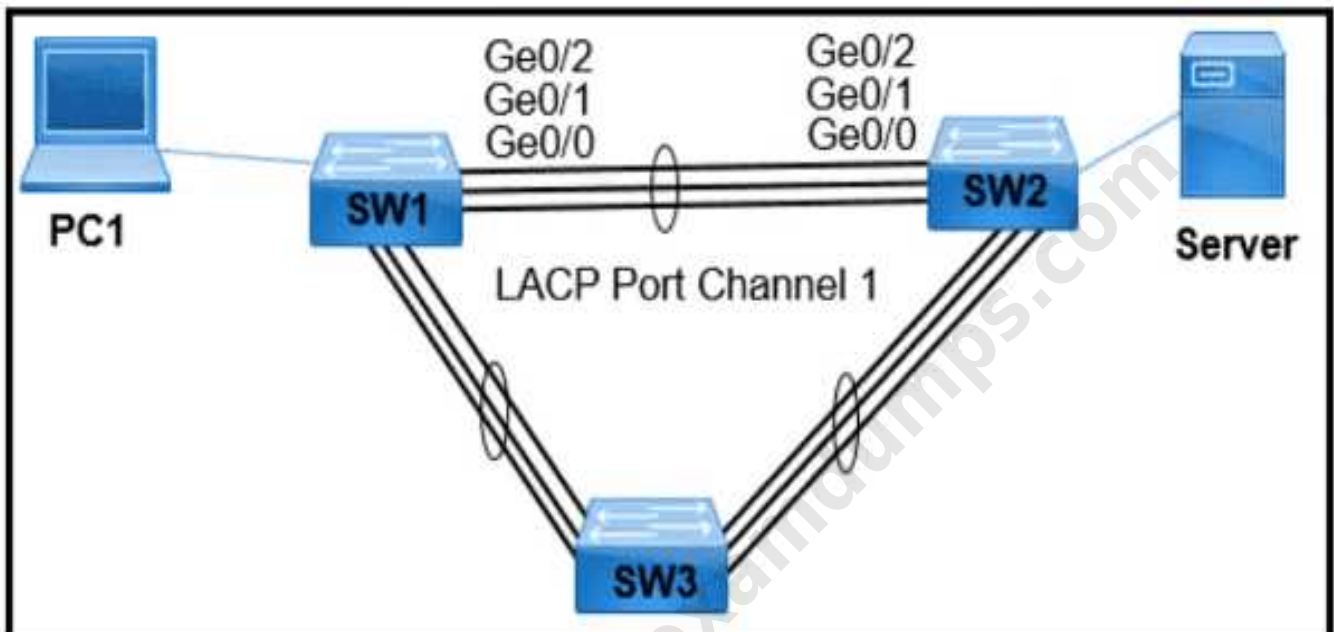
**Answer: D**

---

Explanation:

**Question: 673**

Refer to the exhibit.



PC1 regularly sends 1800 Mbps of traffic to the server. A network engineer needs to configure the EtherChannel to disable Port Channel 1 between SW1 and SW2 when the Ge0/0 and Ge0/1 ports on SW2 go down. Which configuration must the engineer apply to the switch?

A)

```
SW2# configure terminal
SW2(config)# interface port-channel 1
SW2(config-if)# lacp port-priority 32000
```

B)

```
SW2# configure terminal
SW2(config)# interface port-channel 1
SW2(config-if)# lacp max-bundle 2
```

C)

```
SW2# configure terminal
SW2(config)# lacp system-priority 32000
```

D)

```
SW2# configure terminal
SW2(config)# interface port-channel 1
SW2(config-if)# port-channel min-links 2
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: D**

---

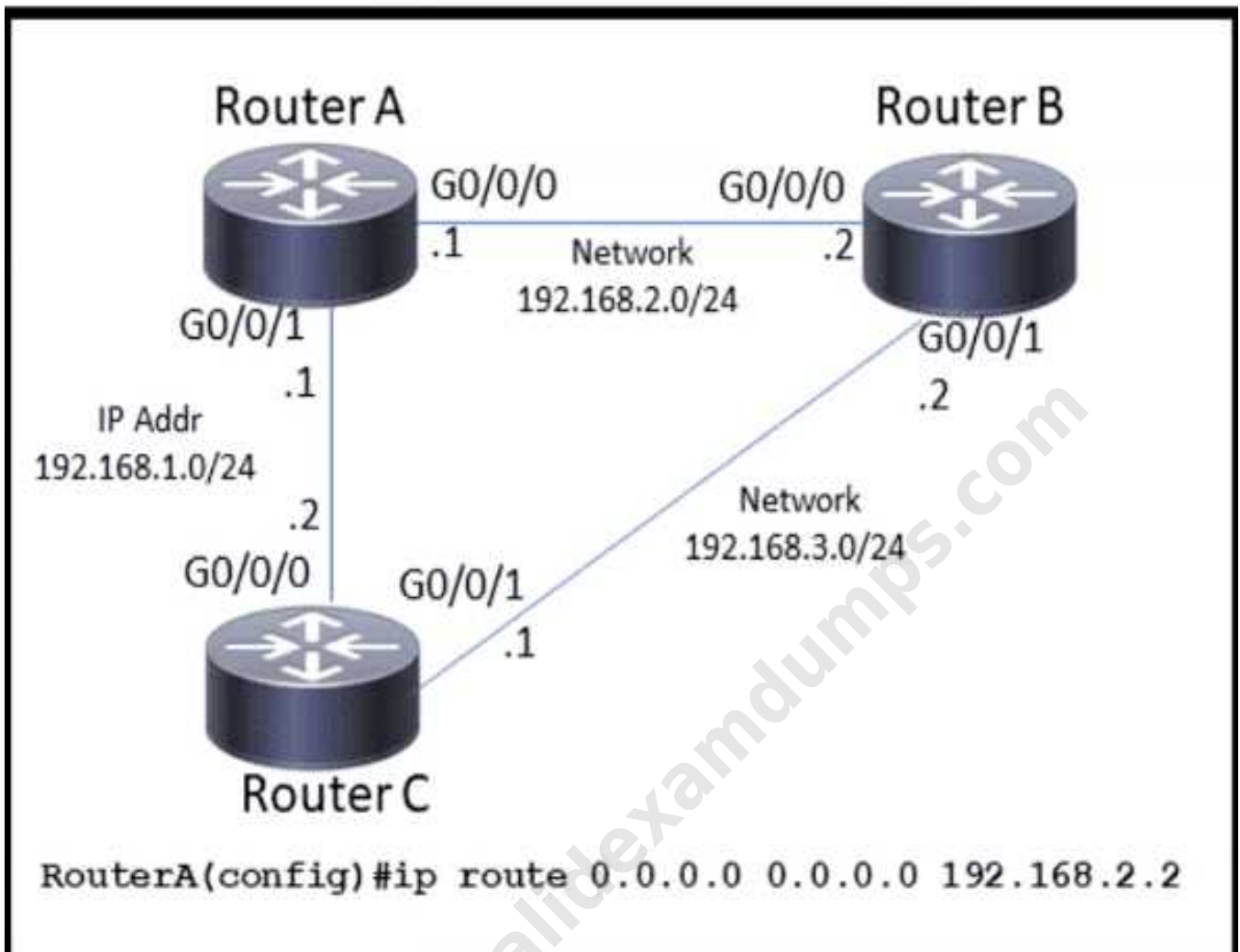
Explanation:

---

**Question: 674**

---

Refer to the exhibit.



Which command must be enable a floating default route on router A?

- A. ip route 0.0.0.0 0.0.0.0 192.168.1.2
- B. ip default-gateway 192.168.2.1
- C. ip route 0.0.0.0 0.0.0.0 192.168.1.2 10
- D. ip route 0.0.0.0 0.0.0.0 192.168.2.1 10

**Answer: C**

Explanation:

**Question: 675**

DRAG DROP

Drag and drop the DNS commands from the left onto their effects on the right.

Drag and drop the DNS commands from the left onto their effects on the right.

|                              |                                            |
|------------------------------|--------------------------------------------|
| ip domain-lookup             | adds an entry to the host table            |
| ip domain-name               | completes the FQDN of the DNS server       |
| ip host switch_1 192.168.0.1 | displays address-mapping information       |
| ip name-server               | enables host-to-IP-address translation     |
| show hosts                   | specifies the IP address of the DNS server |

**Answer:**

Explanation:

|                              |
|------------------------------|
| ip domain-name               |
| ip domain-lookup             |
| show hosts                   |
| ip host switch_1 192.168.0.1 |
| ip name-server               |

**Question: 676**

Refer to the exhibit.

Current Neighbor Relationship

| Neighbor ID | Pri | State   | Dead Time | Address     | Interface          |
|-------------|-----|---------|-----------|-------------|--------------------|
| 192.168.1.1 | 1   | FULL/DR | 00:00:33  | 192.168.1.1 | GigabitEthernet0/0 |

Desired Neighbor Relationship

| Neighbor ID | Pri | State   | Dead Time | Address     | Interface          |
|-------------|-----|---------|-----------|-------------|--------------------|
| 192.168.1.1 | 0   | FULL/ - | 00:00:31  | 192.168.1.1 | GigabitEthernet0/0 |

How must OSPF be configured on the GigabitEthernet0/0 interface of the neighbor device to achieve.

A)

```
Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip ospf priority 1
```

B)

```
Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip ospf 1 area 2
```

C)

```
Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip ospf cost 5
```

D)

```
Router(config)#interface GigabitEthernet 0/0  
Router(config-if)#ip ospf network point-to-point
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: A**

---

Explanation:

---

**Question: 677**

---

What is the purpose of using First Hop Redundancy Protocol on a specific subnet?

- A. ensures a loop-free physical topology
- B. filters traffic based on destination IP addressing
- C. sends the default route to the hosts on a network
- D. forwards multicast hello messages between routers

---

**Answer: D**

---

Explanation:

---

**Question: 678**

---

Which two features introduced in SNMPv2 provides the ability to retrieve large amounts of data in one request

- A. Get
- B. GetNext



- C. Set
- D. GetBulk
- E. Inform

---

**Answer: A,D**

---

Explanation:

---

**Question: 679**

---

Refer to the exhibit. A multivendor network exists and the company is implementing VoIP over the network for the first time.

A)

```
SW1(config)#no cdp enable
SW1(config)#interface gigabitethernet1/0/1
SW1(config-if)#cdp run
```

B)

```
SW1(config)#lldp enable
SW1(config)#interface gigabitethernet1/0/1
SW1(config-if)#lldp run
```

C)

```
SW1(config)#lldp run
SW1(config)#interface gigabitethernet1/0/1
SW1(config-if)#lldp enable
```

D)

```
SW1(config)#no cdp run
SW1(config)#interface gigabitethernet1/0/1
SW1(config-if)#lldp transmit
SW1(config-if)#lldp receive
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: B**

---

Explanation:

---

**Question: 680**

---

Refer to the exhibit.

```
Switch2# show lldp
Global LLDP Information
  Status: ACTIVE
  LLDP advertisements are sent every 30 seconds
  LLDP hold time advertised is 120 seconds
  LLDP interface reinitialization delay is 2 seconds
```

A network engineer must update the configuring on switch2 so that it sends LLDP packets.

A)

```
Switch2(config)#lldp timer 1
Switch2(config)#lldp tlv-select 3
```

B)

```
Switch2(config)#lldp timer 1
Switch2(config)#lldp holdtime 3
```

C)

```
Switch2(config)#lldp timer 60
Switch2(config)#lldp holdtime 180
```

D)

```
) Switch2(config)#lldp timer 60
Switch2(config)#lldp tlv-select 180
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: C**

---

Explanation:

---

**Question: 681**

---

Which IPv6 address range is suitable for anycast addresses for distributed services such DHCP or DNS?

A. FF00:1/12

B. 2001:db8:0234:ca3e::1/128

C. 2002:db84:3f37:ca98:be05:8/64

D. FE80::1/10

---

**Answer: A**

---

Explanation:

---

**Question: 682**

---

A WLC sends alarms about a rogue AP, and the network administrator verifies that the alarms are caused by a legitimate autonomous AP.

- A. Place the AP into manual containment.
- B. Remove the AP from WLC management.
- C. Manually remove the AP from Pending state.
- D. Set the AP Class Type to Friendly.

---

**Answer: B**

---

Explanation:

---

**Question: 683**

---

What is the primary purpose of private address space?

- A. conserve globally unique address space
- B. simplify the addressing in the network
- C. limit the number of nodes reachable via the Internet
- D. reduce network complexity

---

**Answer: D**

---

Explanation:

---

**Question: 684**

---

Refer to the exhibit.

The screenshot shows the 'Advanced' configuration tab for a Wireless LAN Controller (WLC). The 'P2P Blocking Action' is set to 'Disabled'. The 'DHCP Server' is set to 'Override' with a 'DHCP Server IP Addr' of '0.0.0.0'. The 'DHCP Addr. Assignment' is set to 'Required'. The 'Management Frame Protection (MFP)' is set to 'Optional'. The 'DTIM Period (in beacon intervals)' is set to '1' for both '802.11a/n (1 - 255)' and '802.11b/g/n (1 - 255)'. The 'NAC State' is set to 'None'. The 'Client Load Balancing' and 'Client Band Select' are both unchecked.

The P2P blocking action option is disabled on the WLC.

- A. Enable the Static IP Tunneling option.
- B. Disable the Coverage Hole Detection option.
- C. Check the DHCP Addr. Assignment check box.
- D. Set the P2P Blocking Action option to Forward-UpStream.

**Answer: A**

Explanation:

### Question: 685

What is a reason to configure a trunk port that connects to a WLC distribution port?

- A. Eliminate redundancy with a link failure in the data path.
- B. Allow multiple VLAN to be used in the data path.
- C. Provide redundancy if there is a link failure for out-of-band management.
- D. Permit multiple VLANs to provide out-of-band management.

---

**Answer: D**

---

Explanation:

---

**Question: 686**

---

What is a purpose of traffic shaping?

- A. It enables dynamic flow identification.
- B. It enables policy-based routing.
- C. It provides best-effort service.
- D. It limits bandwidth usage.

---

**Answer: A**

---

Explanation:

---

**Question: 687**

---

What is a characteristics of a collapsed-core network topology?

- A. It allows the core and distribution layers to run as a single combined layer.
- B. It enables the core and access layers to connect to one logical distribution device over an EtherChannel.
- C. It enables all workstations in a SOHO environment to connect on a single switch with internet access.
- D. It allows wireless devices to connect directly to the core layer, which enables faster data transmission.

---

**Answer: B**

---

Explanation:

---

**Question: 688**

---

To improve corporate security, an organization is planning to implement badge authentication to limit access to the data center. Which element of a security program is being deployed?

- A. user training
- B. user awareness
- C. vulnerability verification
- D. physical access control

---

**Answer: D**

---

Explanation:

---

**Question: 689**

---

DRAG DROP

Drag and drop the statement about AAA services from the left to the corresponding AAA services on the right.

|                                                               |               |
|---------------------------------------------------------------|---------------|
| It grants access to network assets, such as FTP servers.      | Accounting    |
| It restricts the CLI commands that a user is able to perform. |               |
| It performs user validation via TACACS+.                      | Authorization |
| It records the duration of each connection.                   |               |
| It supports User Access Reporting.                            |               |
| It verifies "who you are".                                    |               |

---

**Answer:**

---

Explanation:

**Accounting**

It supports User Access Reporting.

It restricts the CLI commands that a user is able to perform.

**Authorization**

It performs user validation via TACACS+.

It grants access to network assets, such as FTP servers.

---

**Question: 690**

---

When a WPA2-PSK WLAN is configured in the Wireless LAN Controller, what is the minimum number of characters that is required in ASCII format?

- A. 6
- B. 8
- C. 12
- D. 18

---

**Answer: B**

---

Explanation:



---

**Question: 691**

---

Refer to the exhibit.

```
R1# show ip route
....
D       172.16.32.0/27 [90/2888597172]  via 20.1.1.1
O       172.16.32.0/19  [110/292094]   via 20.1.1.10
R       172.16.32.0/24  [120/2]    via 20.1.1.3
```

An engineer executed the script and added commands that were not necessary for SSH and now must remove the commands.

- A. metric
- B. cost
- C. longest prefix
- D. administrative distance

---

**Answer: D**

---

Explanation:

---

**Question: 692**

---

Which cipher is supported for wireless encryption only with the WPA2 standard?

- A. AES256
- B. AES
- C. RC4
- D. SHA

---

**Answer: B**

---

Explanation:

---

**Question: 693**

---

What is a specification for SSIDS?

- A. They are a Cisco proprietary security feature.
- B. They must include one number and one letter.
- C. They define the VLAN on a switch.
- D. They are case sensitive.

---

**Answer: B**

---

Explanation:

---

**Question: 694**

---

DRAG DROP

An engineer must configure a core router with a floating static default route to the backup router at 10.200.0.2.

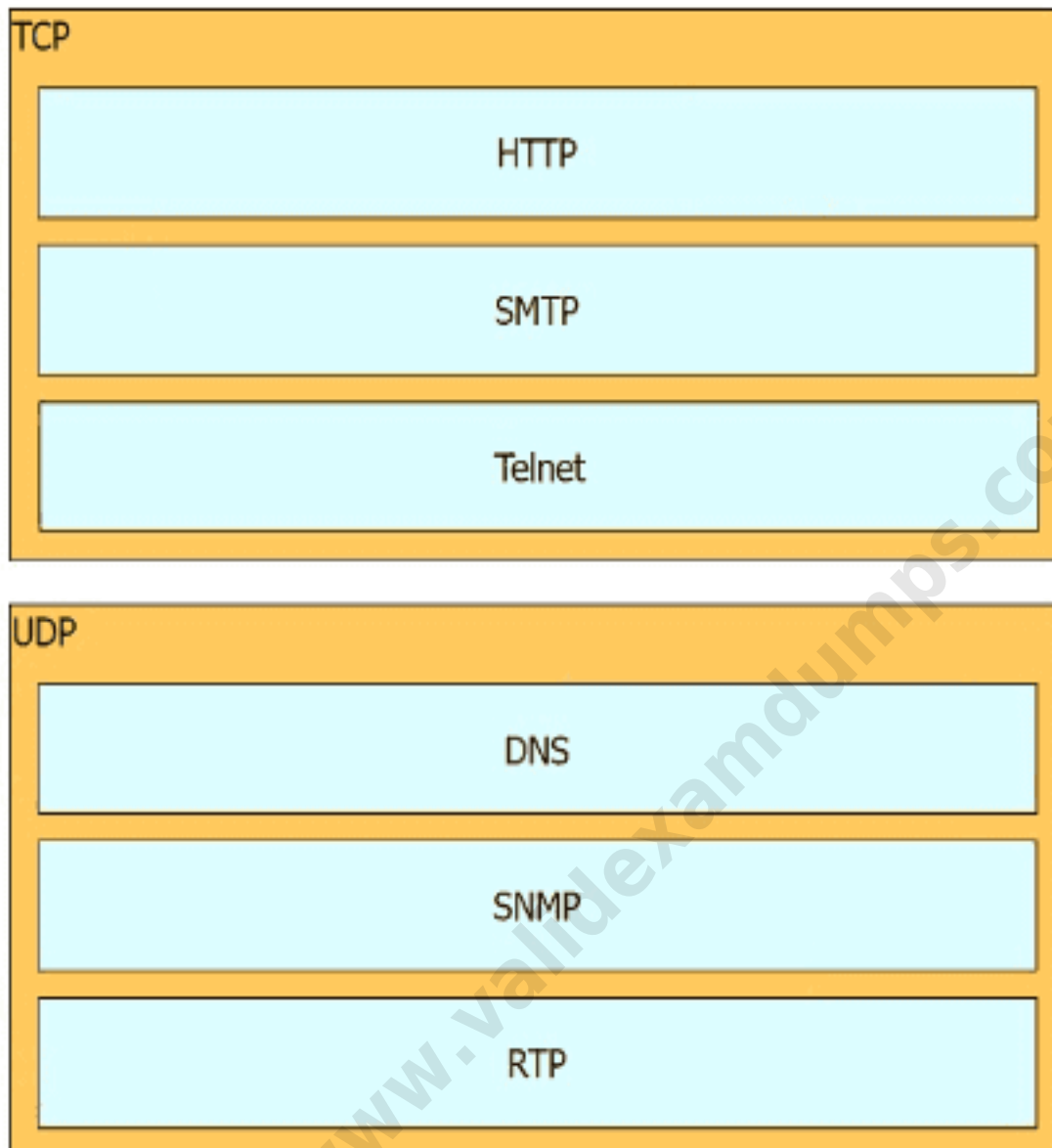
|        |
|--------|
| DNS    |
| HTTP   |
| RTP    |
| SMTP   |
| SNMP   |
| Telnet |

|     |
|-----|
| TCP |
|     |
|     |
|     |

|     |
|-----|
| UDP |
|     |
|     |
|     |

Answer:

Explanation:



---

**Question: 695**

---

Refer to the exhibit.

How does router R1 handle traffic to the 172.16.1.4/30 subnet?

- A. It sends all traffic over the path via 172.16.9.5 using 172.16.4.4 as a backup.
- B. It sends all traffic over the path via 10.0.1.100.
- C. It load-balances traffic over 172.16.9.5 and 172.16.4.4.

D. It sends all traffic over the path via 172.16.4.4.

---

**Answer: C**

---

Explanation:

---

**Question: 696**

---

Which type of address is shared by routers in a HSRP implementation and used by hosts on the subnet as their default gateway address?

- A. multicast address
- B. loopback IP address
- C. virtual IP address
- D. broadcast address

---

**Answer: C**

---

Explanation:

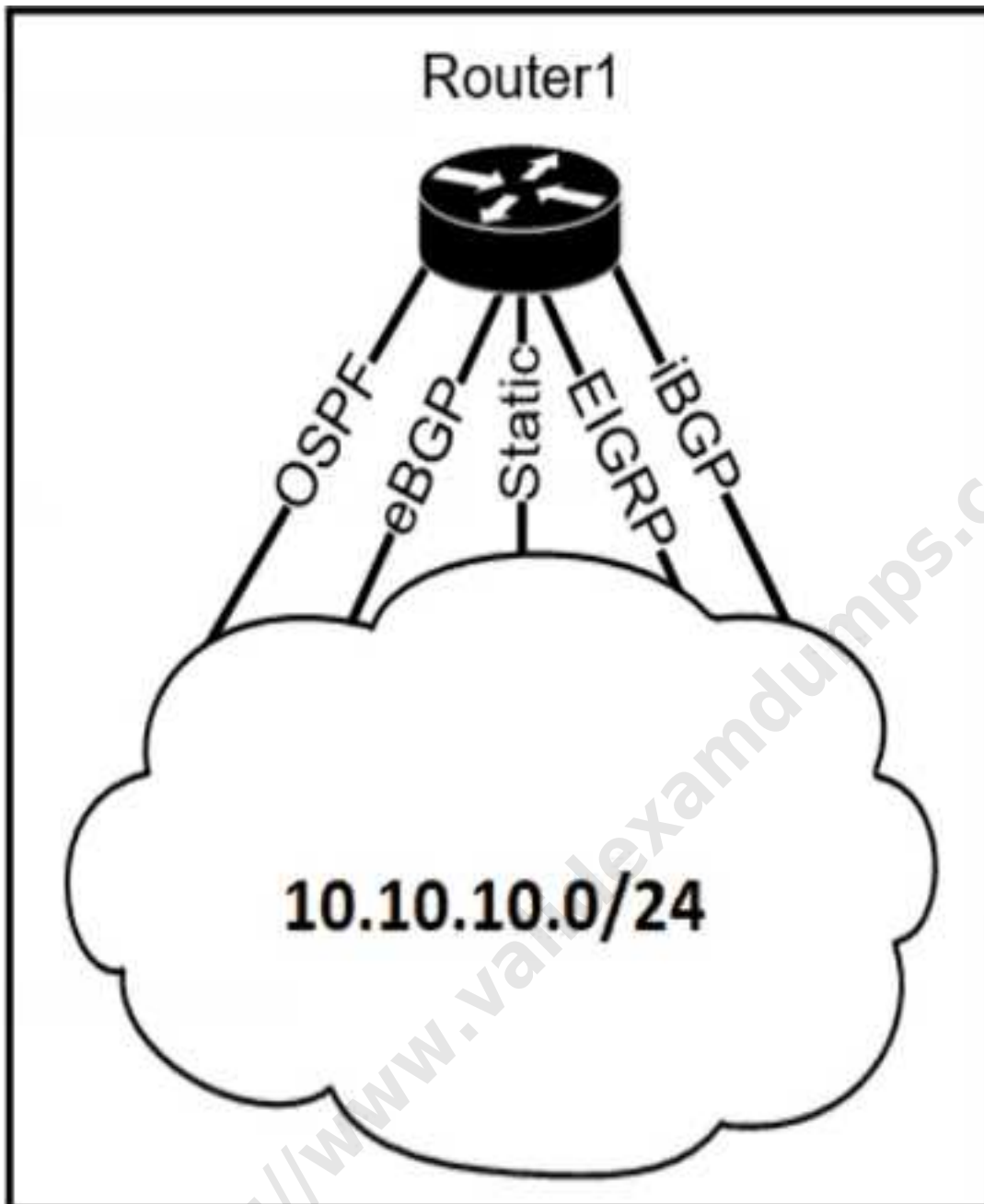
---

**Question: 697**

---

DRAG DROP

Refer to the exhibit.



The Router1 routing table has multiple methods to reach 10.10.10.0/24 as shown. The default Administrative Distance is used. Drag and drop the network conditions from the left onto the routing methods that Router1 uses on the right.

|                                      |                                                        |
|--------------------------------------|--------------------------------------------------------|
| All protocols are up.                | eBGP<br><input type="text"/><br><input type="text"/>   |
| OSPF and eBGP are down.              | <input type="text"/>                                   |
| The static route and eBGP are down.  | EIGRP<br><input type="text"/>                          |
| The static route and EIGRP are down. | <input type="text"/>                                   |
| The static route and OSPF are down.  | Static<br><input type="text"/><br><input type="text"/> |

**Answer:**

Explanation:

**eBGP**

All protocols are up.

OSPF and eBGP are down.

**EIGRP**

The static route and EIGRP are down.

**Static**

The static route and OSPF are down.

The static route and eBGP are down.

---

**Question: 698**

---

Which device segregates a network into separate zones that have their own security policies?

- A. IPS
- B. firewall
- C. access point
- D. switch

---

**Answer: C**

---



Explanation:

---

**Question: 699**

---

DRAG DROP

Drag and drop the wireless architecture benefits from the left onto the architecture types on the right.

Drag and drop the wireless architecture benefits from the left onto the architecture types on the right.

Appropriate for a small-business environment.

Work is divided between the access point and the controller.

The access points transmit beacon frames.

Supports per device configuration and management.

Uses the CAPWAP tunneling protocol.

Split-MAC

Autonomous

---

**Answer:**

---

Explanation:

**Split-MAC**

Work is divided between the access point and the controller.

Supports per device configuration and management.

Uses the CAPWAP tunneling protocol.

**Autonomous**

The access points transmit beacon frames.

Appropriate for a small-business environment.

---

**Question: 700**

---

A network analyst is tasked with configuring the date and time on a router using EXEC mode. The date must be set to January 1, 2020 and the time must be set to 12:00 am. Which command should be used?

- A. clock summer-time recurring
- B. clock timezone
- C. clock summer-time date
- D. clock set

---

**Answer: D**

---

Explanation:

---

**Question: 701**

---

DRAG DROP

Drag and drop the REST API call method for HTTP from the left onto the action they perform on the right.

|        |                                    |
|--------|------------------------------------|
| DELETE | creates a resource on the server   |
| GET    | reads data from the server         |
| POST   | removes a resource from the server |
| PUT    | updates an entry in the database   |
| PATCH  |                                    |

---

**Answer:**

---

Explanation:

POST

GET

DELETE

PUT

---

**Question: 702**

---

Which type of IPv4 address type helps to conserve the globally unique address classes?

- A. multicast
- B. private
- C. loopback
- D. public

---

**Answer: B**

---

Explanation:

---

**Question: 703**

---

Which functionality is provided by the console connection on a Cisco WLC?

- A. out-of-band management

- B. secure in-band connectivity for device administration
- C. unencrypted in-band connectivity for file transfers
- D. HTTP-based GUI connectivity

---

**Answer: A**

---

Explanation:

---

**Question: 704**

---

What determines the sequence in which materials are planned during the material requirements planning (MRP) run?

- A. The control parameters of the MRP run
- B. The creation date of the materials
- C. The low-level code of the materials
- D. The replenishment lead time of the materials

---

**Answer: C**

---

Explanation:

---

**Question: 705**

---

Which advantage does the network assurance capability of Cisco DNA Center provide over traditional campus management?

- A. Cisco DNA Center correlates information from different management protocols to obtain insights, and traditional campus management requires manual analysis.
- B. Cisco DNA Center handles management tasks at the controller to reduce the load on infrastructure devices, and traditional campus management uses the data backbone.
- C. Cisco DNA Center leverages YANG and NETCONF to assess the status of fabric and nonfabric devices, and traditional campus management uses CLI exclusively.

D. Cisco DNA Center automatically compares security postures among network devices, and traditional campus management needs manual comparisons.

---

**Answer: C**

---

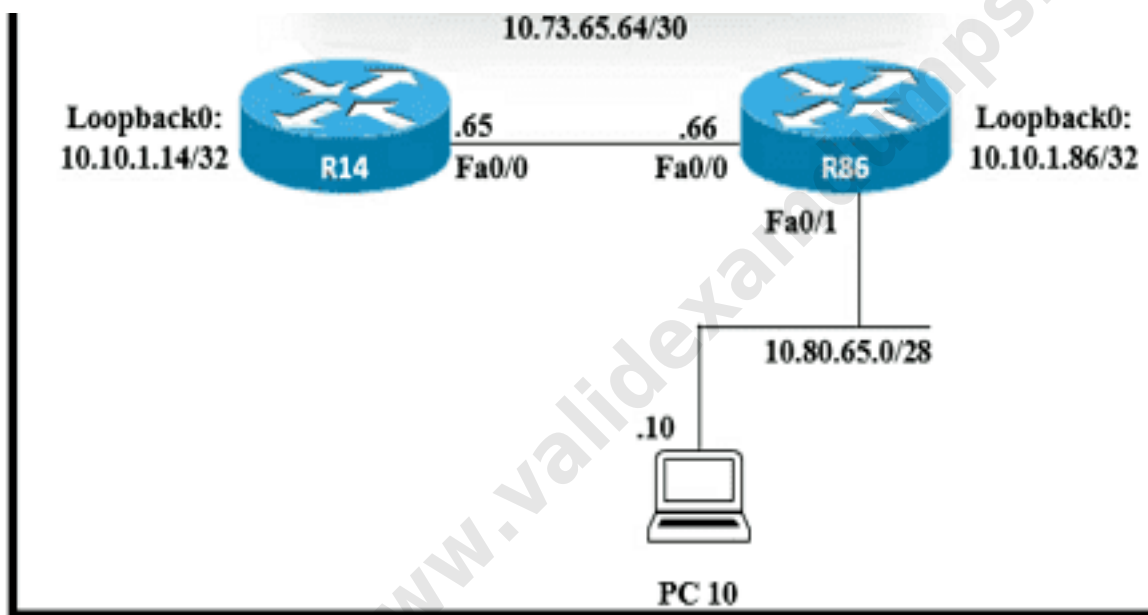
Explanation:

---

**Question: 706**

---

Refer to the exhibit.



Router R14 is in the process of being configured. Which configuration must be used to establish a host route to PC 10?

- A. ip route 10.80.65.10 255.255.255.254 10.80.65.1
- B. ip route 10.8065.10 255.255.255.255 10.73.65.66
- C. ip route 1073.65.65 255.0.0.0 10.80.65.10
- D. ip route 10.73.65.66 0.0.0.255 10.80.65.10

---

**Answer: B**

---

Explanation:

---

**Question: 707**

---

What is the put method within HTTP?

- A. It is a read-only operation.
- B. It is a nonidempotent operation.
- C. It replaces data at the destination.
- D. It displays a web site.

---

**Answer: D**

---

Explanation:

---

**Question: 708**

---

How do TCP and UDP fit into a query-response model?

- A. TCP establishes a connection prior to sending data, and UDP sends immediately.
- B. TCP uses error detection for packets, and UDP uses error recovery.
- C. TCP avoids using sequencing, and UDP avoids using acknowledgments.
- D. TCP encourages out-of-order packet delivery, and UDP prevents re-ordering.

---

**Answer: A**

---

Explanation:

---

**Question: 709**

---

What does the implementation of a first-hop redundancy protocol protect against on a network?

- A. root-bridge loss
- B. spanning-tree loops

- C. default gateway failure
- D. BGP neighbor flapping

---

**Answer: C**

---

Explanation:

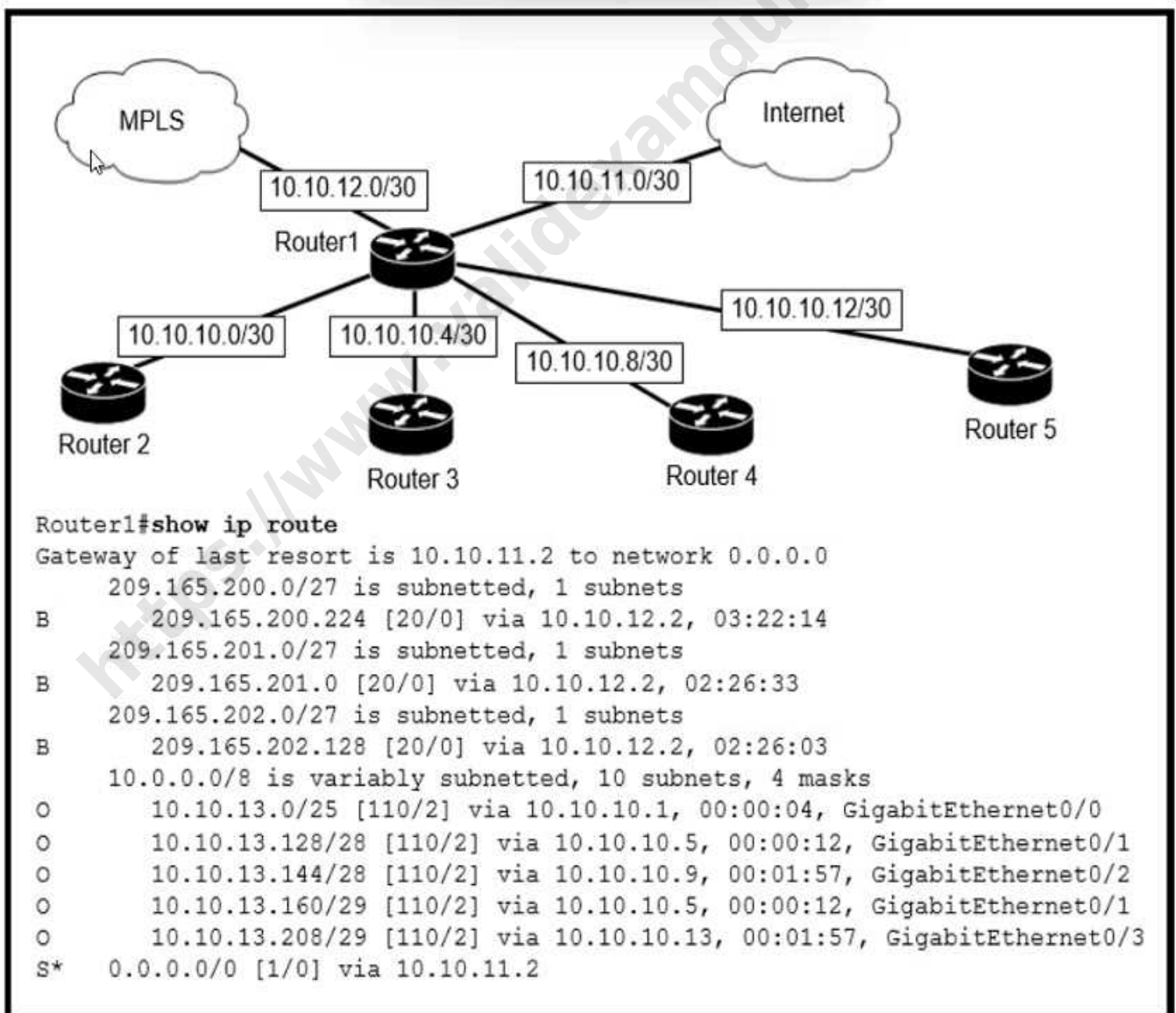
---

**Question: 710**

---

DRAG DROP

Refer to the exhibit.





Drag and drop the destination IPs from the left onto the paths to reach those destinations on the right.

|                |                |
|----------------|----------------|
| 1.1.1.1        | Router2        |
| 10.10.13.126   | Router3        |
| 10.10.13.129   | Router4        |
| 10.10.13.150   | Router5        |
| 10.10.13.209   | Internet cloud |
| 209.165.200.30 | MPLS cloud     |

**Answer:**

Explanation:

10.10.13.126

10.10.13.129

10.10.13.150

10.10.13.209

209.165.200.30

1.1.1.1

---

**Question: 711**

Which benefit does Cisco DNA Center provide over traditional campus management?

- A. Cisco DNA Center leverages SNMPv3 for encrypted management, and traditional campus management uses SNMPv2.
- B. Cisco DNA Center automates HTTPS for secure web access, and traditional campus management uses HTTP.
- C. Cisco DNA Center leverages APIs, and traditional campus management requires manual data gathering.
- D. Cisco DNA Center automates SSH access for encrypted entry, and SSH is absent from traditional campus management.

---

**Answer: B**

---

Explanation:

---

**Question: 712**

---

Refer to the exhibit.

```
Last clearing of "show interface" counters never
Input queue: 1/75/1/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: random early detection (RED)
Output queue : 0/40 (size/max)
5 minute input rate 1000 bits/sec, 2 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
 7558065 packets input, 783768942 bytes, 1 no buffer
Received 8280963 broadcasts, 0 runts, 0 giants, 1 throttles
15 input errors, 14278 CRC, 0 frame, 0 overrun, 3 ignored
0 input packets with dribble condition detected
798092 packets output, 50280266 bytes, 0 underruns
0 output errors, 15000 collisions, 0 interface resets
0 babbles, 0 late collision, 179 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
```

An administrator received a call from a branch office regarding poor application performance hosted at the headquarters. Ethernet 1 is connected between Router1 and the LAN switch. What identifies the issue?

- A. The QoS policy is dropping traffic.
- B. There is a duplex mismatch.
- C. The link is over utilized.
- D. The MTU is not set to the default value.

---

**Answer: B**

---

Explanation:

---

**Question: 713**

---

Refer to the exhibit.

```
1 [
2   { "switch": "3750", "port": e2 },
3   { "router": "2951", "port": e20 }.
4   { "switch": "3750", "port": e23 }
5 ]
```

What is represented by the word "switch" in line 2 of the JSON schema?

- A. array
- B. key
- C. value
- D. object

---

**Answer: C**

---

Explanation:

---

### Question: 714

---

The clients and OHCP server reside on different subnets. Which command must be used to forward requests and replies between clients on the 10.10.0.1/24 subnet and the DHCP server at 192.168.10.1?

- A. ip route 192.168.10.1
- B. ip default-gateway 192.168.10.1
- C. ip helper-address 192.168.10.1
- D. ip dhcp address 192.168.10.1

---

**Answer: C**

---

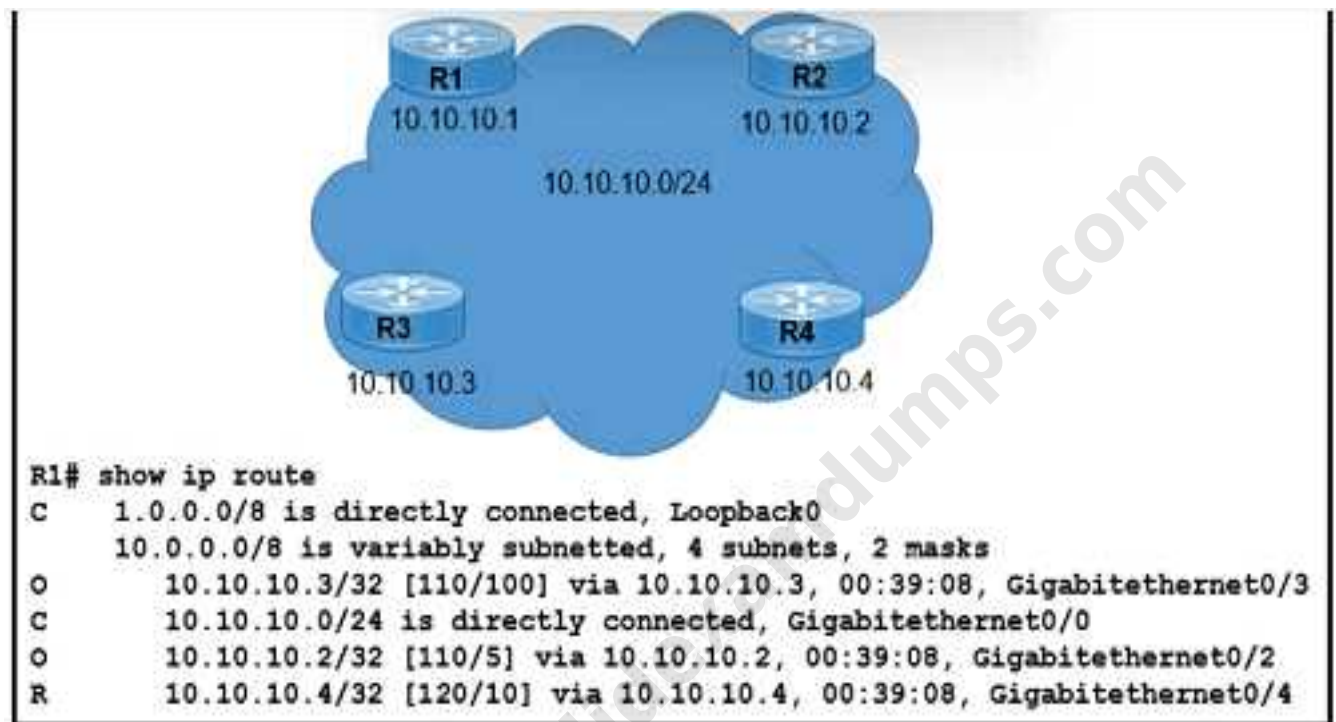
Explanation:

---

**Question: 715**

---

Refer to the exhibit.



Which next-hop IP address has the least desirable metric when sourced from R1?

- A. 10.10.10.5
- B. 10.10.10.3
- C. 10.10.10.4
- D. 10.10.10.2

---

**Answer: C**

---

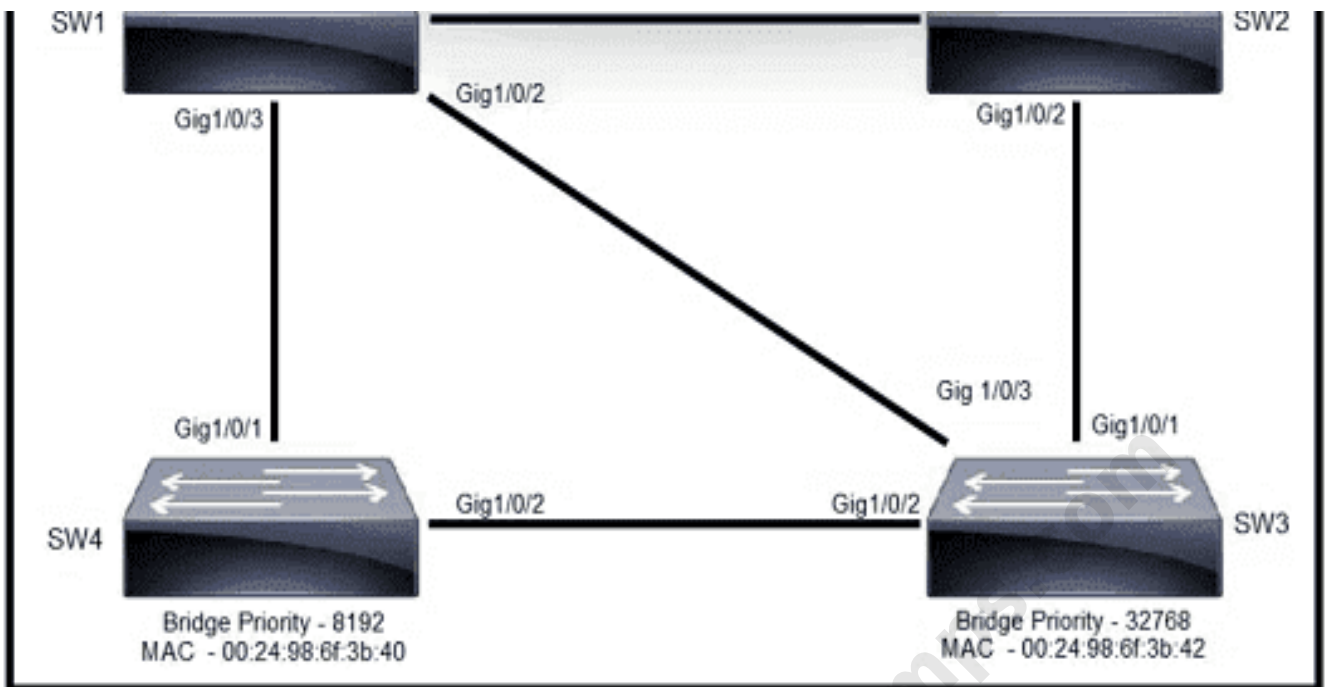
Explanation:

---

**Question: 716**

---

Refer to the exhibit.



Rapid PVST+ mode is on the same VLAN on each switch. Which switch becomes the root bridge and why?

- A. SW2, because its MAC address is the highest
- B. SW3, because its priority is the highest
- C. SW4, because its priority is highest and its MAC address is lower
- D. SW1, because its priority is the lowest and its MAC address is higher

---

**Answer: B**

---

Explanation:

---

**Question: 717**

---

DRAG DROP

Drag and drop the statements about access-point modes from the left onto the corresponding modes on the right.

Monitor

Sensor

Sniffer

It provides air-quality data and interference detection across all enabled channels.

It enables enhanced RFID-tag location tracking.

It supports analytics for wireless performance testing.

It supports real-time Wi-Fi client troubleshooting when network engineers are offsite.

It supports software that analyzes wireless frames on a remote device.

It captures and forwards packets on a specific wireless channel.

Explanation:

Answer:

**Monitor**

It supports analytics for wireless performance testing.

It supports real-time Wi-Fi client troubleshooting when network engineers are offsite.

**Sensor**

It provides air-quality data and interference detection across all enabled channels.

It enables enhanced RFID-tag location tracking.

**Sniffer**

It supports software that analyzes wireless frames on a remote device.

It captures and forwards packets on a specific wireless channel.

---

**Question: 718**

---

PC1 tries to send traffic to newly installed PC2. The PC2 MAC address is not listed in the MAC address table of the switch, so the switch sends the packet to all ports in the same VLAN Which switching concept does this describe?

- A. MAC address aging
- B. MAC address table
- C. frame flooding



D. spanning-tree protocol

---

**Answer: A**

---

Explanation:

---

**Question: 719**

---

A network engineer is upgrading a small data center to host several new applications, including server backups that are expected to account for up to 90% of the bandwidth during peak times. The data center connects to the MPLS network provider via a primary circuit and a secondary circuit. How does the engineer inexpensively update the data center to avoid saturation of the primary circuit by traffic associated with the backups?

- A. Assign traffic from the backup servers to a dedicated switch.
- B. Configure a dedicated circuit for the backup traffic.
- C. Place the backup servers in a dedicated VLAN.
- D. Advertise a more specific route for the backup traffic via the secondary circuit.

---

**Answer: A**

---

Explanation:

---

**Question: 720**

---

DRAG DROP

Drag and drop the characteristics of northbound APIs from the left onto any position on the right. Not all characteristics are used.

- supports automation
- communicates between the SDN controller and the application plane
- supports data sharing between systems
- communicates between the SDN controller and the data plane
- supports network virtualization protocols
- supports REST-based requirements
- uses OpenFlow to interface between the data and control planes

---

**Answer:**

---

Explanation:

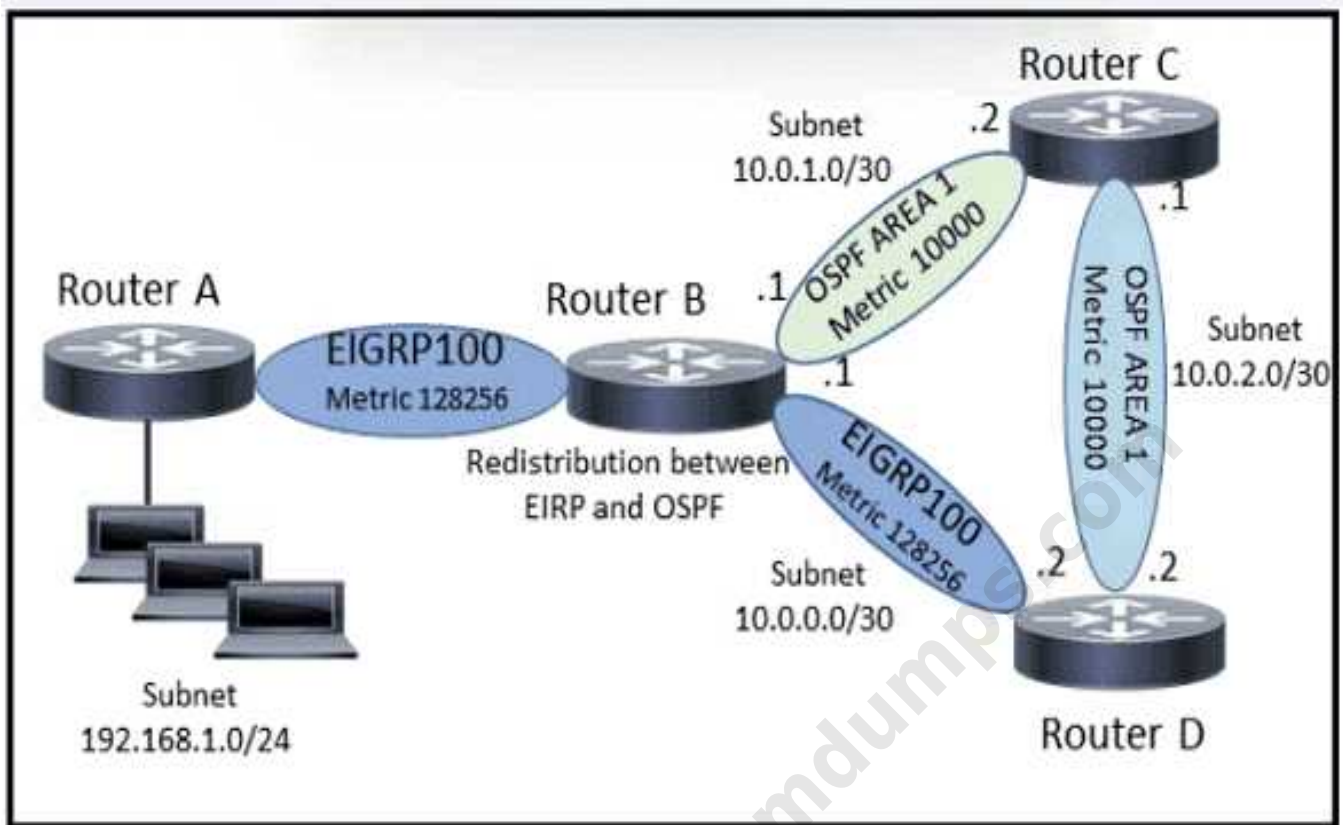
- supports data sharing between systems
- communicates between the SDN controller and the data plane
- supports network virtualization protocols
- supports REST-based requirements

---

**Question: 721**

---

Refer to the exhibit.



A network engineer executes the show ip route command on router D. What is the next hop to network 192.168.1.0/24 and why?

- A. The next hop is 10.0.2.1 because it uses distance vector routing
- B. The next hop is 10.0.2.1 because it is a link-state routing protocol
- C. The next hop is 10.0.0.1 because it has a better administrative distance
- D. The next hop is 10.0.0.1 because it has a higher metric.

**Answer: B**

Explanation:

**Question: 722**

Refer to the exhibit.

The screenshot shows the configuration page for a new WLAN named 'lantest'. The configuration is as follows:

| Configuration Item           | Value                                                                                                   |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| SSID                         | lantest                                                                                                 |
| Status                       | <input type="checkbox"/> Enabled                                                                        |
| Security Policies            | [WPA2][Auth(802.1X)]<br>(Modifications done under security tab will appear after applying the changes.) |
| Radio Policy                 | All                                                                                                     |
| Interface/Interface Group(G) | guest                                                                                                   |
| Multicast Vlan Feature       | <input type="checkbox"/> Enabled                                                                        |
| Broadcast SSID               | <input type="checkbox"/> Enabled                                                                        |
| NAS-ID                       | none                                                                                                    |

A Cisco engineer creates a new WLAN called lantest. Which two actions must be performed so that only high-speed 2.4-Ghz clients connect? (Choose two.)

- A. Enable the Broadcast SSID option
- B. Enable the Status option.
- C. Set the Radio Policy option to 802 11g Only.
- D. Set the Radio Policy option to 802.11a Only.
- E. Set the Interface/Interface Group(G) to an interface other than guest

---

**Answer: A,B**

---

Explanation:

---

### Question: 723

---

What is the role of nonoverlapping channels in a wireless environment?

- A. to reduce interference
- B. to allow for channel bonding
- C. to stabilize the RF environment
- D. to increase bandwidth

---

**Answer: A**

---

Explanation:

---

**Question: 724**

---

A router has two static routes to the same destination network under the same OSPF process. How does the router forward packets to the destination if the next-hop devices are different?

- A. The router chooses the route with the oldest age.
- B. The router load-balances traffic over all routes to the destination.
- C. The router chooses the next hop with the lowest MAC address.
- D. The router chooses the next hop with the lowest IP address.

---

**Answer: B**

---

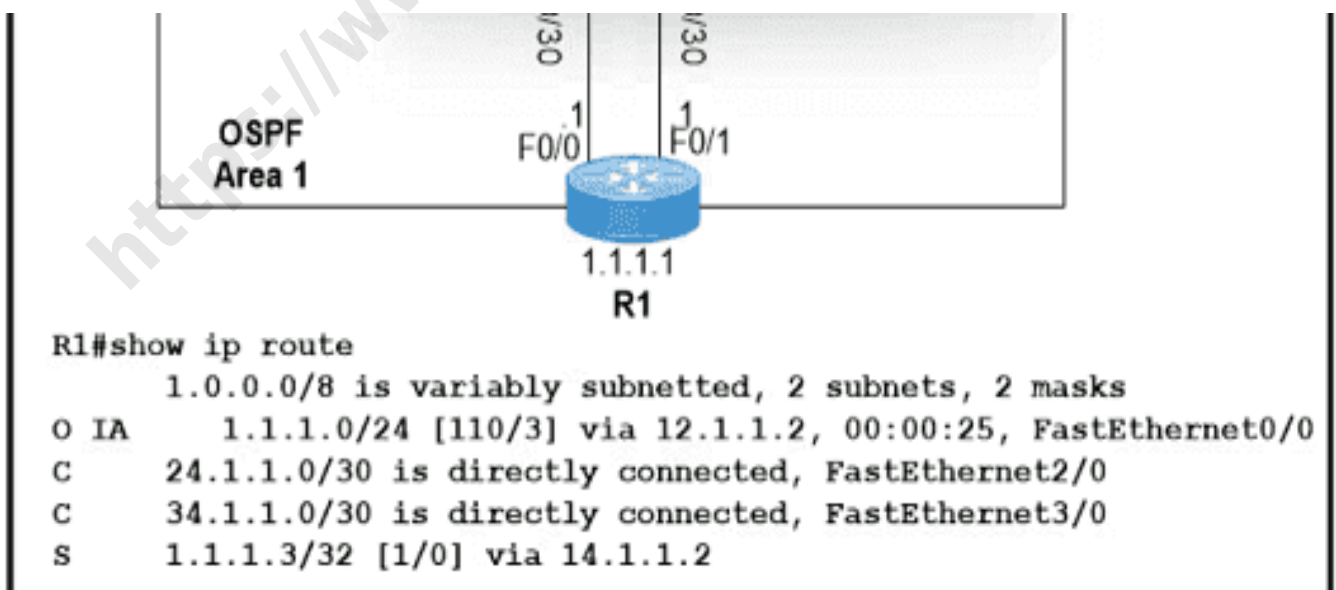
Explanation:

---

**Question: 725**

---

Refer to the exhibit.



Which two values does router R1 use to determine the best path to reach destinations in network

1,0.0.0/8? (Choose two.)

- A. longest prefix match
- B. highest administrative distance
- C. highest metric
- D. lowest metric
- E. lowest cost to reach the next hop

---

**Answer: A,D**

---

Explanation:

---

**Question: 726**

---

Refer to the exhibit.



Host A switch interface is configured in VLAN 2. Host D sends a unicast packet destined for the IP address of host A.

```
Sw1#show mac-address table
Mac Address Table
```

| Vlan | Mac Address    | Type    | Ports |
|------|----------------|---------|-------|
| 2    | 000c.859c.bb7b | DYNAMIC | e0/1  |
| 3    | 000c.859c.bb7b | DYNAMIC | e0/1  |
| 2    | 0010.11dc.3e91 | DYNAMIC | e0/2  |
| 3    | 0010.11dc.3e91 | DYNAMIC | e0/2  |
| 2    | 0043.49d4.c383 | DYNAMIC | e0/3  |

Sw1#

What does the switch do when it receives the frame from host D?

- A. It creates a broadcast storm.

- B. It drops the frame from the MAC table of the switch.
- C. It shuts down the source port and places It In err-disable mode.
- D. It floods the frame out of every port except the source port.

---

**Answer: C**

---

Explanation:

---

**Question: 727**

---

Which properly is shared by 10GBase-SR and 10GBase-LR interfaces?

- A. Both require fiber cable media for transmission.
- B. Both require UTP cable media for transmission.
- C. Both use the single-mode fiber type.
- D. Both use the multimode fiber type.

---

**Answer: A**

---

Explanation:

---

**Question: 728**

---

Refer to the exhibit.

```
R1
interface GigabitEthernet0/1
 ip address 192.168.12.1 255.255.255.128
 no shutdown
router ospf 1
 network 192.168.12.1 0.0.0.0 area 1

R2
interface GigabitEthernet0/1
 ip address 192.168.12.2 255.255.255.128
 no shutdown
```

A network engineer started to configure two directly-connected routers as shown. Which command sequence must the engineer configure on R2 so that the two routers become OSPF neighbors?

A)

```
router ospf 1
 network 192.168.12.1 0.0.0.0 area 1
```

B)

```
interface GigabitEthernet0/1
 ip ospf 1 area 1
```

C)

```
interface GigabitEthernet0/1
 ip ospf 1 area 0
```

D)

```
router ospf 1
 network 192.168.12.0 0.0.0.127 area 0
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: D**

---



Explanation:

---

**Question: 729**

---

Refer to the exhibit.

```
MacOs$ ifconfig

en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether f0:18:98:64:60:32
    inet6 fe80::492:c09f:57cf:8c36%en0 prefixlen 64 secured scopeid 0x6
    inet 10.8.138.14 netmask 0xffffe000 broadcast 10.8.159.255
    nd6 options=201<PERFORMNUD,DAD>
    media: autoselect
    status: active
```

A network engineer must provide configured IP addressing details to investigate a firewall rule Issue. Which subnet and mask Identify what is configured on the en0 interface?

- A. 10.8.0.0/16
- B. 10.8.64.0/18
- C. 10.8.128.0/19
- D. 10.8.138.0/24

---

**Answer: D**

---

Explanation:

---

**Question: 730**

---

Refer to the exhibit.

```
R1# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate
       default
       U - per-user static route, o - ODR
Gateway of last resort is not set
C    10.0.0.0/8 is directly connected, Loopback0
     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O    10.0.1.3/32 [110/100] via 10.0.1.100, 00:39:08, Serial0
C    10.0.1.0/24 is directly connected, Serial0
O    10.0.1.5/32 [110/5] via 10.0.1.50, 00:39:08, Gigabit Ethernet 0/0
D    10.0.1.4/32 [110/10] via 10.0.1.4, 00:39:08, Gigabit Ethernet 0/0
```

What does route 10.0.1.3/32 represent in the routing table?

- A. the 10.0.0.0 network
- B. a single destination address
- C. the source 10.0.1.100
- D. all hosts in the 10.0.1.0 subnet

---

**Answer: A**

---

Explanation:

---

**Question: 731**

---

Refer to the exhibit.

```
access-list 10 permit 10.0.0.0 0.0.0.255  
  
interface Serial0  
  
ip access-list 10 in
```

A network administrator must permit traffic from the 10.10.0.0/24 subnet to the WAN on interface Serial0. What is the effect of the configuration as the administrator applies the command?

- A. The permit command fails and returns an error code.
- B. The router accepts all incoming traffic to Serial0 with the last octet of the source IP set to 0.
- C. The sourced traffic from IP range 10.0.0.0 -10.0.0.255 is allowed on Serial0.
- D. The router fails to apply the access list to the interface.

---

**Answer: C**

---

Explanation:

---

**Question: 732**

---

What is the role of community strings in SNMP operations?

- A. It serves as a sequence tag on SNMP traffic messages.
- B. It serves as a password to protect access to MIB objects.
- C. It passes the Active Directory username and password that are required for device access
- D. It translates alphanumeric MIB output values to numeric values.

---

**Answer: B**

---

Explanation:

---

**Question: 733**

---

Which access point mode relies on a centralized controller for management, roaming, and SSID configuration?

- A. repeater mode
- B. autonomous mode
- C. bridge mode
- D. lightweight mode

---

**Answer: D**

---

Explanation:

---

**Question: 734**

---

What are two characteristics of a small office / home office connection environment? (Choose two.)

- A. It requires 10Gb ports on all uplinks.
- B. It supports between 50 and 100 users.
- C. It supports between 1 and 50 users.
- D. It requires a core, distribution, and access layer architecture.
- E. A router port connects to a broadband connection.

---

**Answer: C,E**

---

Explanation:

---

**Question: 735**

---

Which syslog severity level is considered the most severe and results in the system being considered unusable?

- A. Alert
- B. Error
- C. Emergency
- D. Critical

---

**Answer: C**

---

Explanation:

---

**Question: 736**

---

What is the definition of backdoor malware?

- A. malicious code that is installed onto a computer to allow access by an unauthorized user
- B. malicious code with the main purpose of downloading other malicious code
- C. malicious program that is used to launch other malicious programs
- D. malicious code that infects a user machine and then uses that machine to send spam

---

**Answer: A**

---

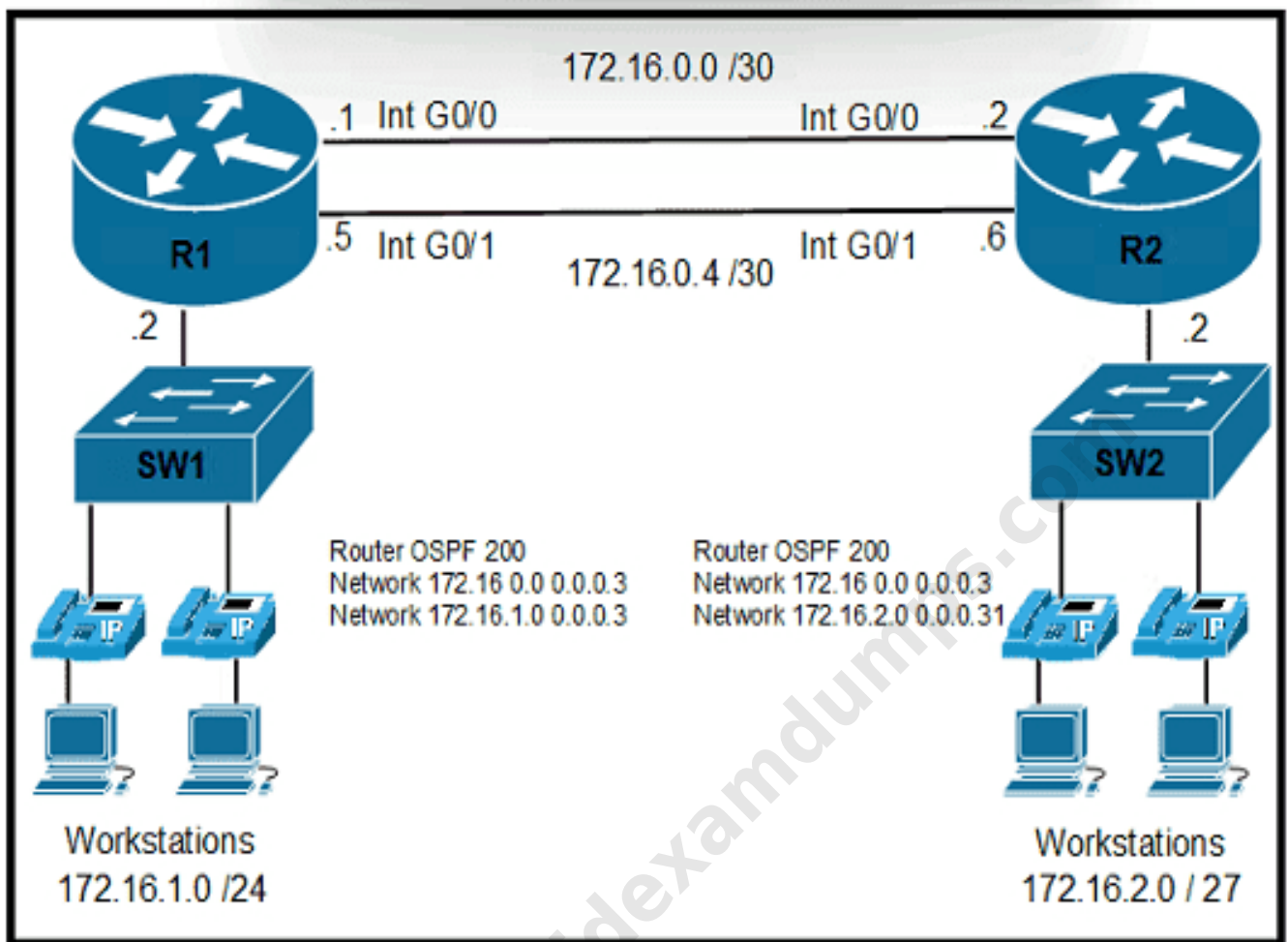
Explanation:

---

**Question: 737**

---

Refer to the exhibit.



The primary route across Gi0/0 is configured on both routers. A secondary route must be configured to establish connectivity between the workstation networks. Which command set must be configured to complete this task?

A)

```

R1
ip route 172.16.2.0 255.255.255.240 172.16.0.2 113

```

```

R2
ip route 172.16.1.0 255.255.255.0 172.16.0.1 114

```

B)

```

R1
ip route 172.16.2.0 255.255.255.240 172.16.0.5 89

```

```

R2
ip route 172.16.1.0 255.255.255.0 172.16.0.6 89

```

C)

>R1

**ip route 172.16.2.0 255.255.255.248 172.16.0.5 110**

R2

**ip route 172.16.1.0 255.255.255.0 172.16.0.6 110**

D)

R1

**ip route 172.16.2.0 255.255.255.224 172.16.0.6 111**

R2

**ip route 172.16.1.0 255.255.255.0 172.16.0.5 112**

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: D**

---

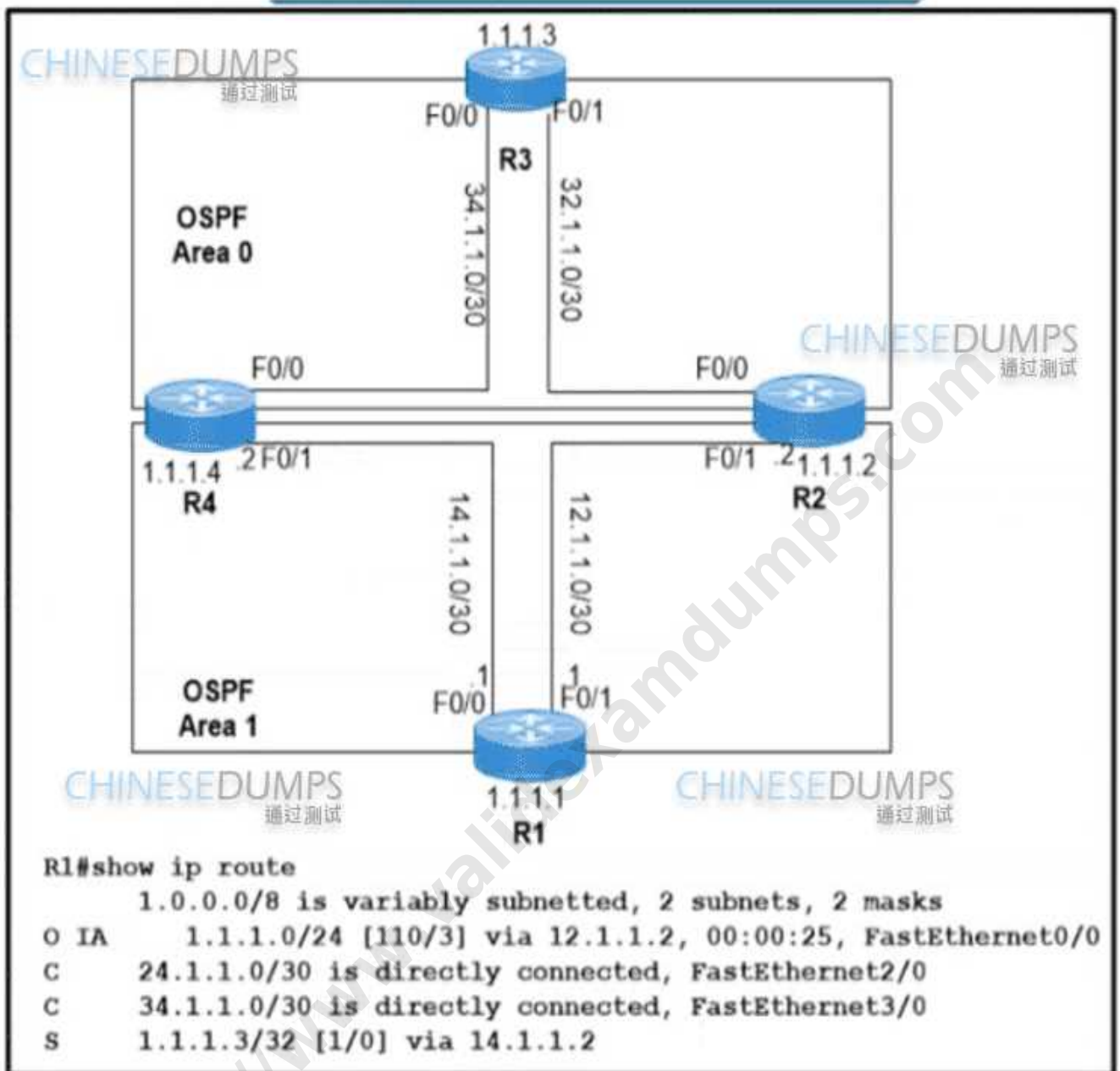
Explanation:

---

**Question: 738**

---

Refer to the exhibit.



Which two values does router R1 use to identify valid routes for the R3 loopback address 1.1.1.3/32? (Choose two.)

- A. lowest cost to reach the next hop
- B. highest metric
- C. highest administrative distance
- D. lowest metric
- E. lowest administrative distance



---

**Answer: D,E**

---

Explanation:

---

**Question: 739**

---

Refer to the exhibit.

```
SW1#show run
Building configuration...
!
hostname SW1
!
ip domain-name CCNA-test
!
username CCNA privilege 1 password 0 cisco123
!
interface FastEthernet0/1
  switchport access vlan 10
!
interface Vlan10
  ip address 192.168.1.2 255.255.255.0
!
line vty 0 4
  login local
  transport input telnet
line vty 5 15
  login local
  transport input telnet

SW1#show crypto key mypubkey rsa
% Key pair was generated at: 0:1:23 UTC Mar 1 2020
Key name: SW1.CCNA-test
```

An engineer is updating the management access configuration of switch SW1 to allow secured, encrypted remote configuration. Which two commands or command sequences must the engineer

apply to the switch? (Choose two.)

- A. SW1(config)#enable secret ccnaTest123
- B. SW1(config)#username NEW secret R3mote123
- C. SW1(config)#line vty 0 15 SW1(config-line)#transport input ssh
- D. SW1(config)# crypto key generate rsa
- E. SW1(config)# interface f0/1 SW1(config-if)# switchport mode trunk

---

**Answer: A,C**

---

Explanation:

---

**Question: 740**

---

SIP-based Call Admission Control must be configured in the Cisco WLC GUI. SIP call-snooping ports are configured. Which two actions must be completed next? (Choose two.)

- A. Set the QoS level to silver or greater for voice traffic.
- B. Set the QoS level to platinum for voice traffic.
- C. Enable Media Session Snooping on re WLAN.
- D. Enable traffic shaping for the LAN interlace of the WLC.
- E. Configure two different QoS rotes tor data and voice traffic.

---

**Answer: D,E**

---

Explanation:

---

**Question: 741**

---

What is a benefit of using private IPv4 addressing?

- A. Multiple companies can use the same addresses without conflicts.

- B. Direct connectivity is provided to internal hosts from outside an enterprise network.
- C. Communication to the internet is reachable without the use of NAT.
- D. All external hosts are provided with secure communication to the Internet.

---

**Answer: A**

---

Explanation:

---

**Question: 742**

---

Which cable type must be used to interconnect one switch using 1000 BASE-SX GBiC modules and another switch using 1000 BASE-SX SFP modules?

- A. LC to SC
- B. SC to ST
- C. SC to SC
- D. LC to LC

---

**Answer: D**

---

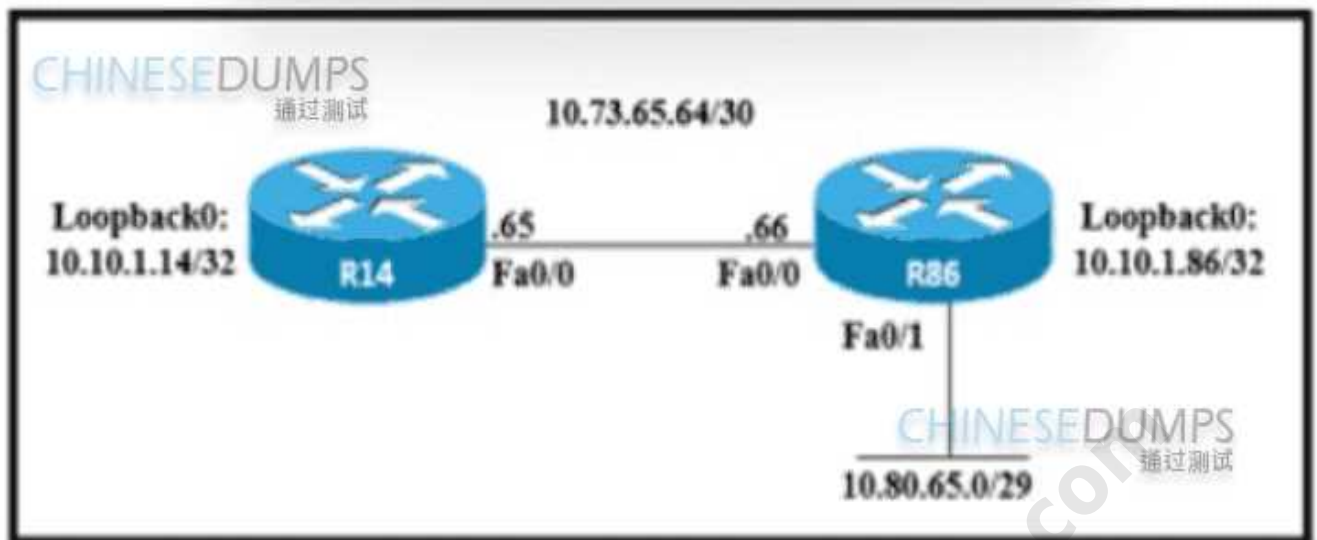
Explanation:

---

**Question: 743**

---

Refer to the exhibit.



An engineer must configure a floating static route on an external EIGRP network. The destination subnet is the /29 on the LAN Interface of R86. Which command must be executed on R14?

- A. ip route 10.80.65.0.255.255.248.0.10.73.65.66.1
- B. ip route 10.80.65.0.255.255.255..240 fa0/1 89
- C. ip route 10.80.65.0.255.255.248.0.10.73.65.66.171
- D. ip route 10.80.65.0.0.0.224.10.80.65.0. 255

---

**Answer: C**

---

Explanation:

---

**Question: 744**

---

Which channel-group mode must be configured when multiple distribution interfaces connected to a WLC are bundled?

- A. Channel-group mode passive.
- B. Channel-group mode on.
- C. Channel-group mode desirable.
- D. Channel-group mode active.

---

**Answer: B**

---

Explanation:

---

**Question: 745**

---

Which two server types support domains name to IP address resolution? (Choose two &gt;

- A. ESX host
- B. resolver
- C. web
- D. file transfer
- E. authentication

---

**Answer: A,C**

---

Explanation:

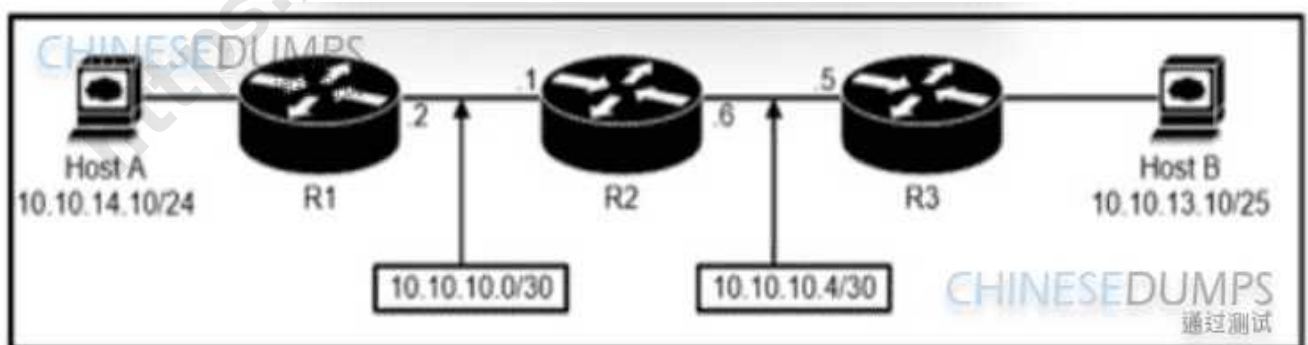
---

**Question: 746**

---

DRAG DROP

Refer to the exhibit.



Refer to the exhibit. An engineer must configure a static network route between two networks so that host A communicates with host B. Drag and drop the commands from the left onto the routers where they must be configured on the right. Not all commands are used.

ip route 10.10.13.0 255.255.255.128 10.10.10.1

ip route 10.10.13.0 255.255.255.128 10.10.10.5

ip route 10.10.13.10 255.255.255.255 10.10.10.1

ip route 10.10.14.0 255.255.255.0 10.10.10.2

ip route 10.10.14.0 255.255.255.0 10.10.10.6

ip route 10.10.14.10 255.255.255.255 10.10.10.6

**Answer:**

Explanation:

3 - R1

2 & 4 - R2

5 - R3

### Question: 747

What is the functionality of the Cisco DNA Center?

- A. data center network pokey con
- B. console server that permits secure access to all network devices
- C. IP address cool distribution scheduler
- D. software-defined controller for automaton of devices and services

**Answer: D**

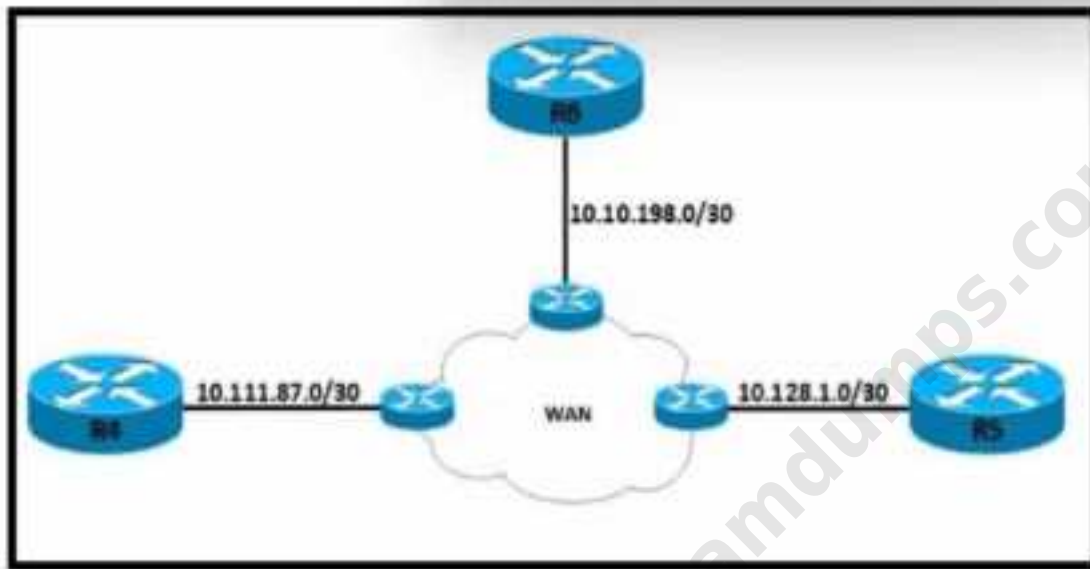
Explanation:

---

**Question: 748**

---

Refer to the exhibit.



Refer to the exhibit. Local access for R4 must be established and these requirements must be met:

- Only Telnet access is allowed.
- The enable password must be stored securely.
- The enable password must be applied in plain text.
- Full access to R4 must be permitted upon successful login.

Which configuration script meets the requirements?

A)

! CHINESEDUMPS  
通过测试  
conf t  
!  
username test1 password testpass1  
enable secret level 15 0 Test123  
!  
line vty 0 15  
login local  
transport input telnet

B)

! CHINESEDUMPS  
通过测试  
config t  
!  
username test1 password testpass1  
enable password level 15 0 Test123  
!  
line vty 0 15  
password Test123  
transport input all

C)

! CHINESEDUMPS  
通过测试  
config t  
!  
username test1 password testpass1  
enable password level 1 7 Test123  
!  
line vty 0 15  
accounting exec default  
transport input all



D)

```
!
config t
!
username test1 password testpass1
enable password level 1 7 Test123
!
line vty 0 15
  accounting exec default
  transport input all
```

A. Option

B. Option

C. Option

D. Option

---

**Answer: D**

---

Explanation:

---

**Question: 749**

---

Refer to the exhibit.

```
{
  "Routers": ["R1", "R2", "R3"],
  "Switches": ["SW1", "SW2", "SW3"]
}
```

Refer to the exhibit. What is represented by "R1" and "SW1" within the JSON output?

object

value

A. key

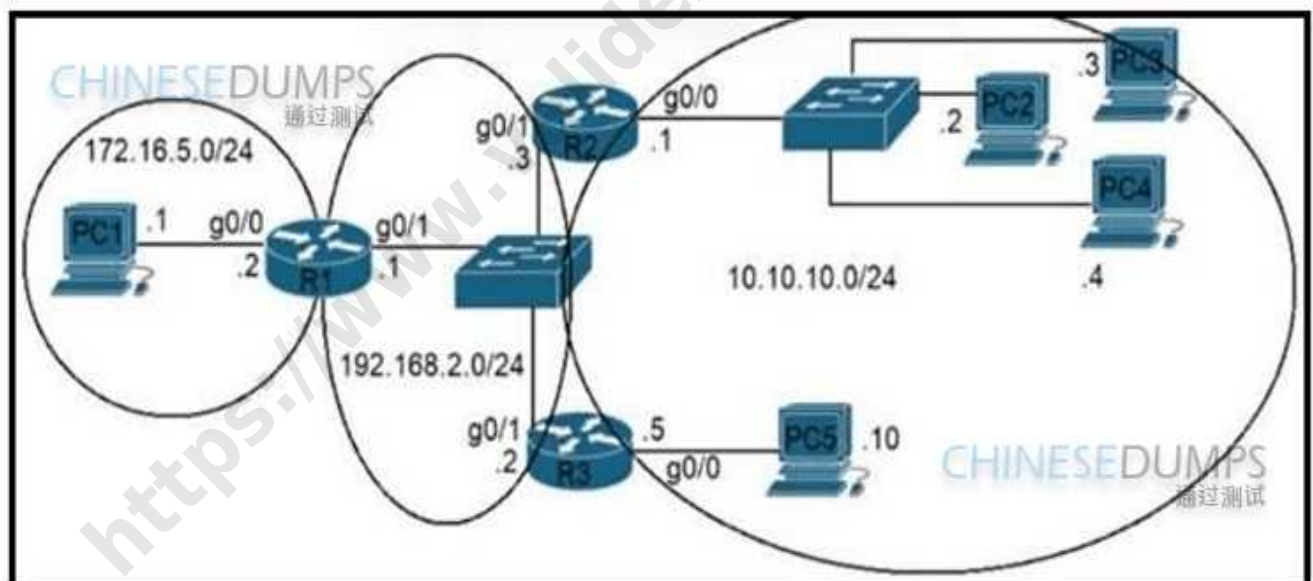
B. array

**Answer: B**

Explanation:

**Question: 750**

Refer to the exhibit.



Refer to the exhibit. The router R1 is in the process of being configured. Routers R2 and R3 are configured correctly for the new environment. Which two commands must be configured on R1 for PC1 to communicate to all PCs on the 10.10.10.0/24 network? (Choose two.)

A. ip route 10.10.10.0 255.255.255.0 192.168.2.3

- B. ip route 10.10.10.10 255.255.255.255 192.168.2.2
- C. ip route 10.10.10.10 255.255.255.255 g0/1
- D. ip route 10.10.10.8 255.255.255.248 g0/1
- E. ip route 10.10.10.0 255.255.255.248 192.168.2.2

---

**Answer: A,E**

---

Explanation:

---

**Question: 751**

---

What are two protocols within the IPsec suite? (Choose two)

- A. AH
- B. 3DES
- C. ESP
- D. TLS
- E. AES

---

**Answer: D,E**

---

Explanation:

---

**Question: 752**

---

When an access point is seeking to join wireless LAN controller, which message is sent to the AP-Manager interface?

- A. Discovery response
- B. DHCP request
- C. DHCP discover

D. Discovery request

**Answer: D**

Explanation:

### Question: 753

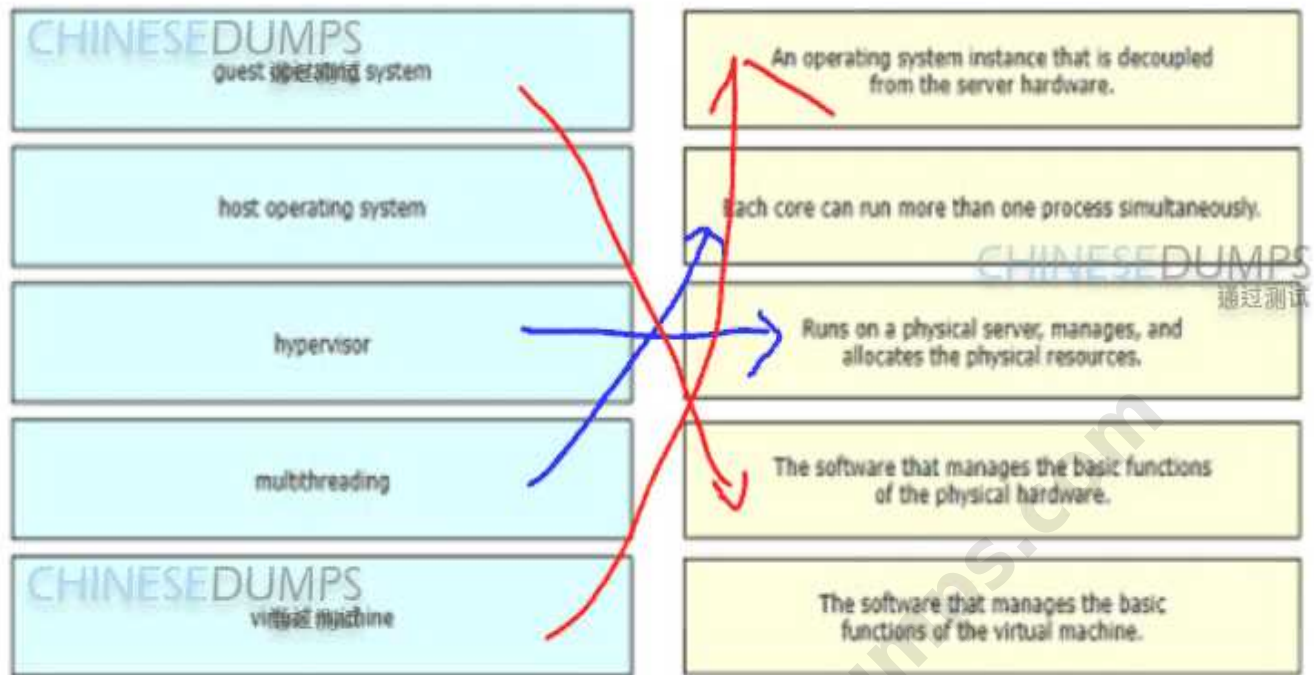
DRAG DROP

Drag and drop the virtualization concepts from the left onto the matching statements on the right.

|                                                |                                                                                               |
|------------------------------------------------|-----------------------------------------------------------------------------------------------|
| CHINESEDUMPS<br>通过测试<br>guest operating system | An operating system instance that is decoupled from the server hardware.                      |
| host operating system                          | Each core can run more than one process simultaneously.                                       |
| hypervisor                                     | Runs on a physical server, manages, and allocates the physical resources.                     |
| CHINESEDUMPS<br>通过测试<br>multithreading         | The software that manages the basic functions of the physical hardware.                       |
| virtual machine                                | CHINESEDUMPS<br>通过测试<br>The software that manages the basic functions of the virtual machine. |

**Answer:**

Explanation:



---

**Question: 754**

---

Which protocol is used in Software Defined Access (SDA) to provide a tunnel between two edge nodes in different fabrics?

- A. Generic Router Encapsulation (GRE)
- B. Virtual Local Area Network (VLAN)
- C. Virtual Extensible LAN (VXLAN)
- D. Point-to-Point Protocol

---

**Answer: C**

---

Explanation:

---

**Question: 755**

---

Which IP header field is changed by a Cisco device when QoS marking is enabled?

- A. Header Checksum

- B. Type of service
- C. DSCP
- D. ECN

---

**Answer: B**

---

Explanation:

---

**Question: 756**

---

DRAG DROP

Drag and drop the statements about networking from me left onto the corresponding networking types on the right

Statements to be dragged:

- This type implements changes individually at each device.
- This type leverages controllers to handle network management.
- Maintenance costs are higher than with other networking options.
- This type provides a centralized view of the network.

Target categories:

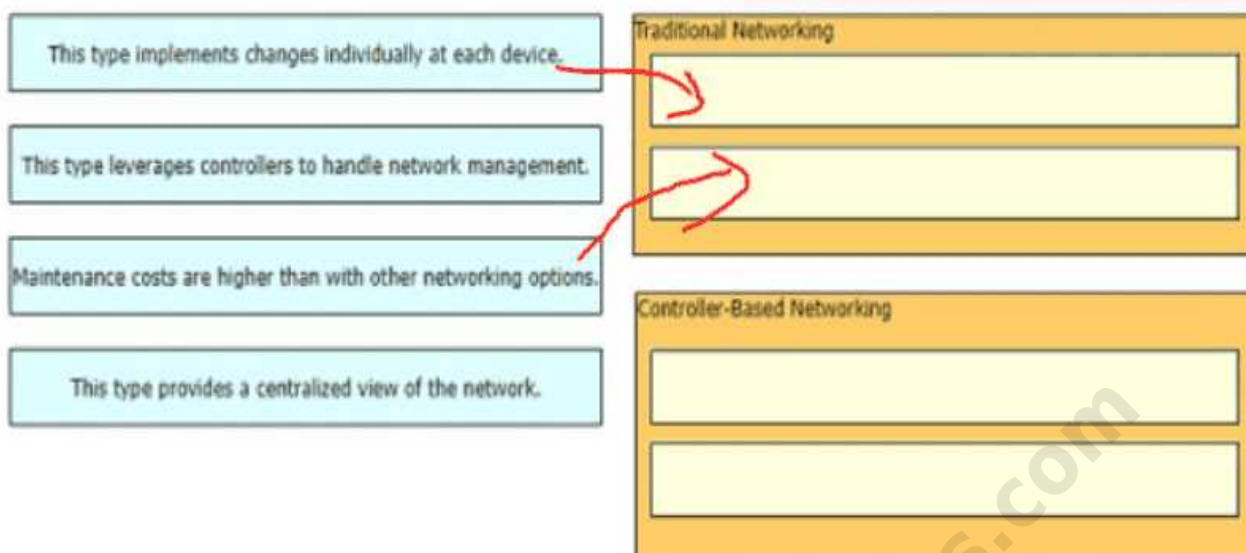
- Traditional Networking**
  - 
  -
- Controller-Based Networking**
  - 
  -

---

**Answer:**

---

Explanation:



---

**Question: 757**

---

Refer to the exhibit.

**General** **Security** **QoS** **Policy-Mapping** **Advanced**

**Layer 2** **Layer 3** **AAA Servers**

Layer 2 Security **WPA+WPA2**

MAC Filtering ☐

**Fast Transition**

Fast Transition ☐

**Protected Management Frame**

PMF **Required**

**WPA+WPA2 Parameters**

WPA Policy ☐

WPA2 Policy ☐

WPA2 Encryption ☐ AES ☐ TKIP

OSEN Policy ☐

**Authentication Key Management**

802.1X ☐ Enable

CCKM ☐ Enable

PSK ☐ Enable

FT 802.1X ☐ Enable

FT PSK ☐ Enable

PSK Format **ASCII**

WPA gtk-randomize State **Disable**

The network engineer is configuring a new WLAN and is told to use a setup password for authentication instead of the RADIUS servers. Which additional set of tasks must the engineer perform to complete the configuration?



- A. Disable PMF Enable PSK Enable 802.1x
- B. Select WPA Policy Enable CCKM Enable PSK
- C. Select WPA Policy Select WPA2 Policy Enable FT PSK
- D. Select WPA2 Policy Disable PMF Enable PSK

---

**Answer: D**

---

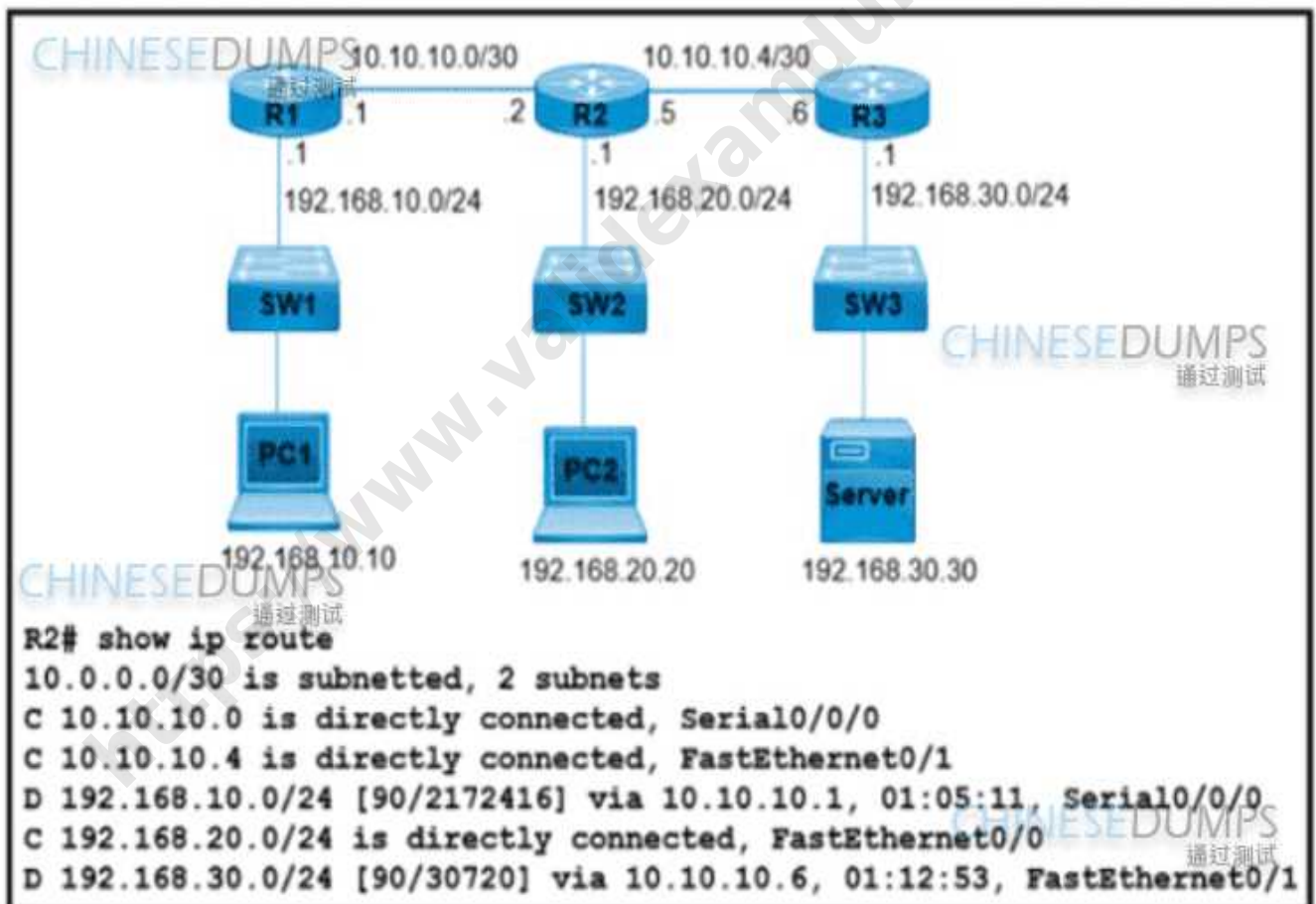
Explanation:

---

**Question: 758**

---

Refer to the exhibit.



Refer to the exhibit. What is the next-hop IP address for R2 so that PC2 reaches the application server via EIGRP?

- A. 192.168.30.1

- B. 10.10.105
- C. 10.10.10.6
- D. 192.168.201

---

**Answer: D**

---

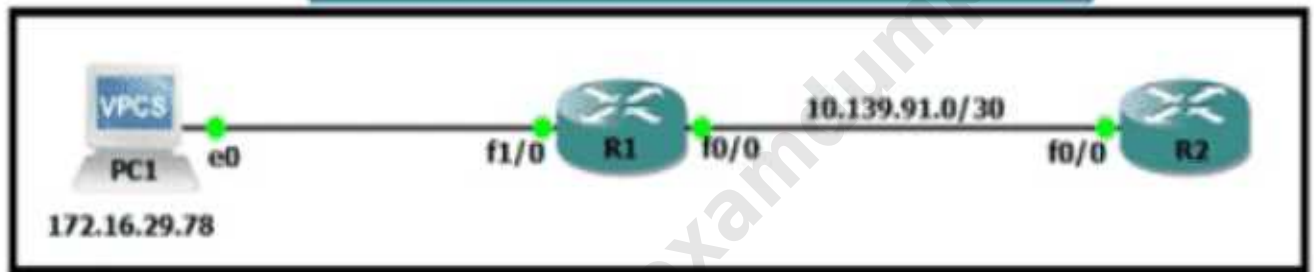
Explanation:

---

**Question: 759**

---

Refer to the exhibit.



Refer to the exhibit. An engineer must translate the PC1 IP address to 10.199.77.100 and permit PC1 to ping the loopback 0 on router R2. What command set must be used?

A)

**R1#**

```
!  
interface Loopback0  
ip address 10.1.1.1 255.255.255.255  
!  
interface FastEthernet0/0  
ip address 10.139.91.1 255.255.255.252  
ip nat inside  
ip virtual-reassembly in  
!  
interface FastEthernet1/0  
ip address 172.16.29.1 255.255.255.0  
ip nat outside  
ip virtual-reassembly in  
!  
router eigrp 100  
network 10.1.1.1 0.0.0.0  
network 10.139.91.0 0.0.0.3  
!  
ip nat inside source static 10.199.77.100 172.16.29.78
```

**R2#**

```
ip route 10.199.77.100 255.255.255.255 10.139.91.1
```

B)

```
○ R1#  
!  
interface Loopback0  
ip address 10.1.1.1 255.255.255.255  
!  
interface FastEthernet0/0  
ip address 10.139.91.1 255.255.255.252  
ip nat outside  
ip virtual-reassembly in  
!  
interface FastEthernet1/0  
ip address 172.16.29.1 255.255.255.0  
ip nat inside  
ip virtual-reassembly in  
!  
router eigrp 100  
network 10.1.1.1 0.0.0.0  
network 10.139.91.0 0.0.0.3  
!  
ip nat inside source static 172.16.29.78 10.199.77.100  
R2#  
ip route 10.199.77.100 255.255.255.255 10.139.91.1
```

c)

```
○ R1#  
!  
interface Loopback0  
ip address 10.1.1.1 255.255.255.255  
!  
interface FastEthernet0/0  
ip address 10.139.91.1 255.255.255.252  
ip nat outside  
ip virtual-reassembly in  
!  
interface FastEthernet1/0  
ip address 172.16.29.1 255.255.255.0  
ip nat inside  
ip virtual-reassembly in  
!  
router eigrp 100  
network 10.1.1.1 0.0.0.0  
network 10.139.91.0 0.0.0.3  
!  
ip nat inside source static 172.16.29.78 10.199.77.100  
  
R2#  
ip route 172.16.29.78 255.255.255.255 10.139.91.1
```

D)

```
○ R1#  
!  
interface Loopback0  
ip address 10.1.1.1 255.255.255.255  
!  
interface FastEthernet0/0  
ip address 10.139.91.1 255.255.255.252  
ip nat outside  
ip virtual-reassembly in  
!  
interface FastEthernet1/0  
ip address 172.16.29.1 255.255.255.0  
ip nat inside  
ip virtual-reassembly in  
!  
router eigrp 100  
network 10.1.1.1 0.0.0.0  
network 10.139.91.0 0.0.0.3  
!  
ip nat inside source static 172.16.29.78 10.199.77.100  
R2#  
ip route 172.16.29.78 255.255.255.255 10.139.91.1
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: B**

---

Explanation:

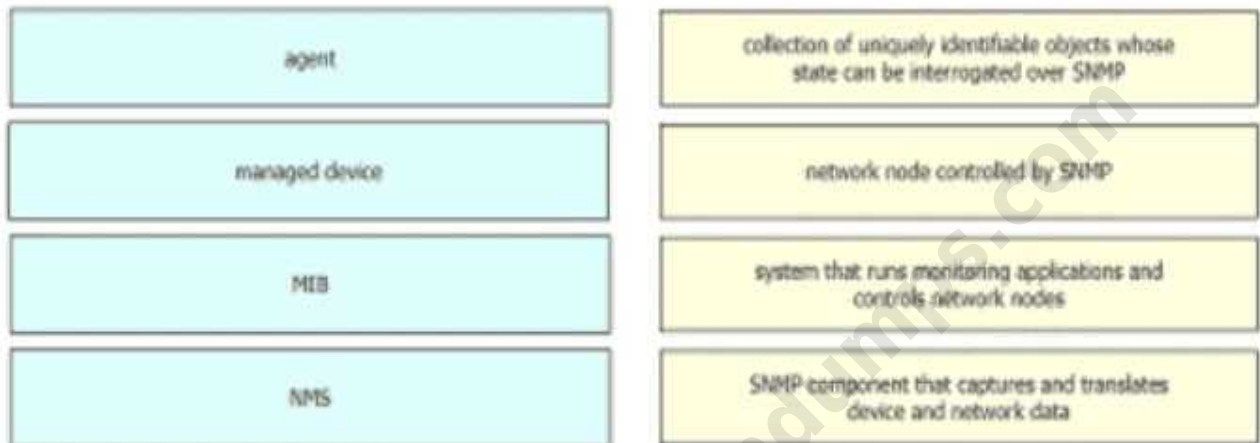
---

**Question: 760**

---

## DRAG DROP

Drag and drop the SNMP components from the left onto the description on the right.



**Answer:**

Explanation:

1-2

2-3

3-1

**Question: 761**

Refer to the exhibit. User traffic originating within site 0 is failing to reach an application hosted on IP address 192.168.0.10. Which is located within site A. What is determined by the routing table?

- A. The default gateway for site B is configured incorrectly
- B. The lack of a default route prevents delivery of the traffic
- C. The traffic is blocked by an implicit deny in an ACL on router2
- D. The traffic to 192.168.0.10 requires a static route to be configured in router 1.

---

**Answer: B**

---

Explanation:

---

**Question: 762**

---

A network engineer must configure an interface with IP address 10.10.10.145 and a subnet mask equivalent to 11111111.11111111.11111111.1111000. Which subnet mask must the engineer use?

- A. /29
- B. /30
- C. /27
- D. /28

---

**Answer: A**

---

Explanation:

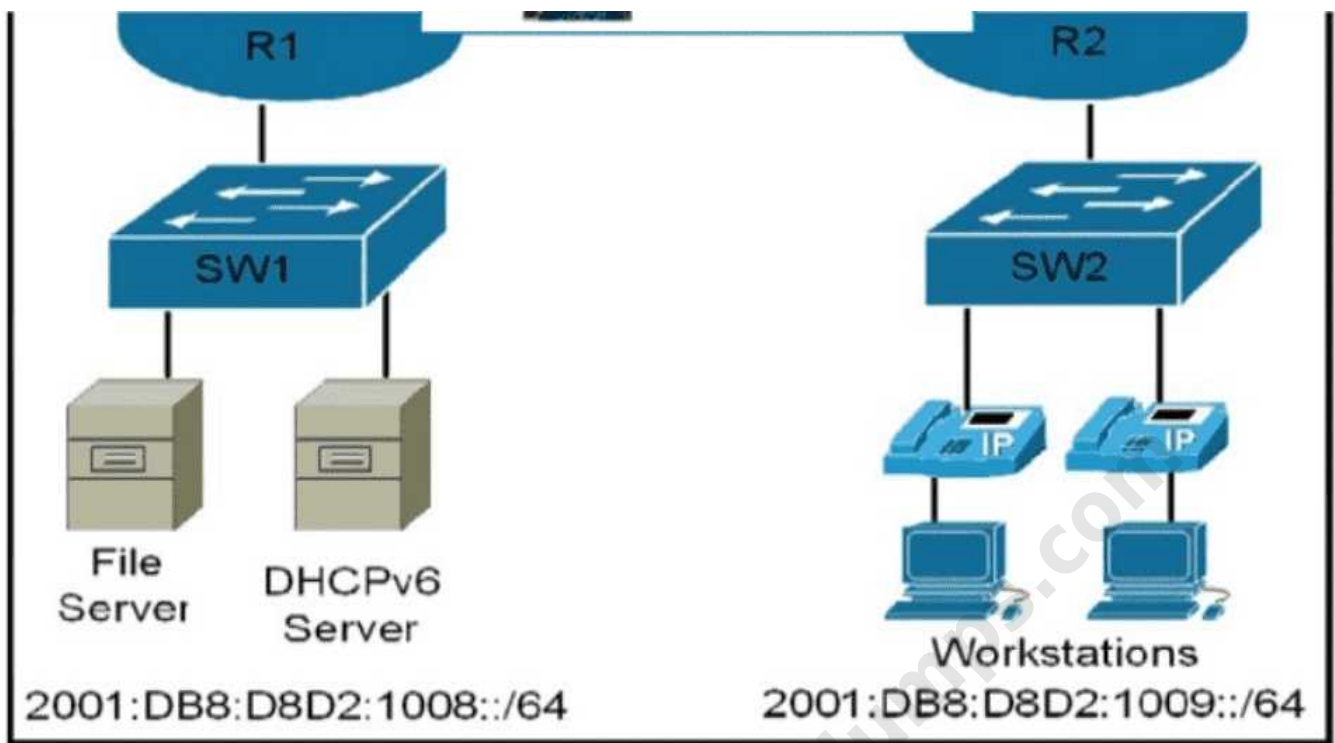
---

**Question: 763**

---

Refer to the exhibit An IPv6 address must be obtained automatically on the LAN interface on R1  
Which command must be implemented to accomplish the task?





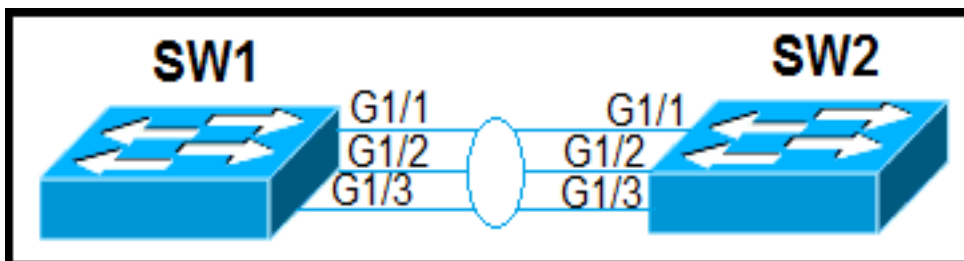
- A. Ipv6 address 2001:db8:d8d2:1008:4343:61:0010::/64
- B. Ipv6 address autoconfig
- C. Ipv6 address fe80::/10
- D. Ipv6 address dhcp

**Answer: B**

Explanation:

**Question: 764**

Refer to the exhibit.



Which configuration establishes a Layer 2 LACP EtherChannel when applied to both switches?

- A. Interface range G1/1 – 1/3 switchport mode trunk channel-group 1 mode active no shutdown
- B. Interface range G1/1 – 1/3 switchport mode access channel-group 1 mode passive no shutdown
- C. Interface range G1/1 – 1/3 switchport mode trunk channel-group 1 mode desirable no shutdown
- D. Interface range G1/1 – 1/3 switchport mode access channel-group 1 mode on no shutdown

---

**Answer: A**

---

Explanation:

---

**Question: 765**

---

What must a network administrator consider when deciding whether to configure a new wireless network with APs in autonomous mode or APs running in cloud-based mode?

Autonomous mode APs are less dependent on an underlay but more complex to maintain than APs in cloud-based mode.

Cloud-based mode APs rely on underlays and are more complex to maintain than APs in autonomous mode.

- A. Cloud-based mode APs are easy to deploy but harder to automate than APs in autonomous mode.
- B. Autonomous mode APs are easy to deploy and automate than APs in cloud-based mode.

---

**Answer: A**

---

Explanation:

---

**Question: 766**

---

Under which condition is TCP preferred over UDP?

- A. UDP is used when low latency is optimal, and TCP is used when latency is tolerable.

- B. TCP is used when dropped data is more acceptable, and UDP is used when data is accepted out-of-order.
- C. TCP is used when data reliability is critical, and UDP is used when missing packets are acceptable.
- D. UDP is used when data is highly interactive, and TCP is used when data is time-sensitive.

---

**Answer: C**

---

Explanation:

---

**Question: 767**

---

Which protocol must be implemented to support separate authorization and authentication solutions for wireless APs?

- A. RADIUS
- B. TACACS+
- C. 802.1X
- D. Kerberos

---

**Answer: A**

---

Explanation:

---

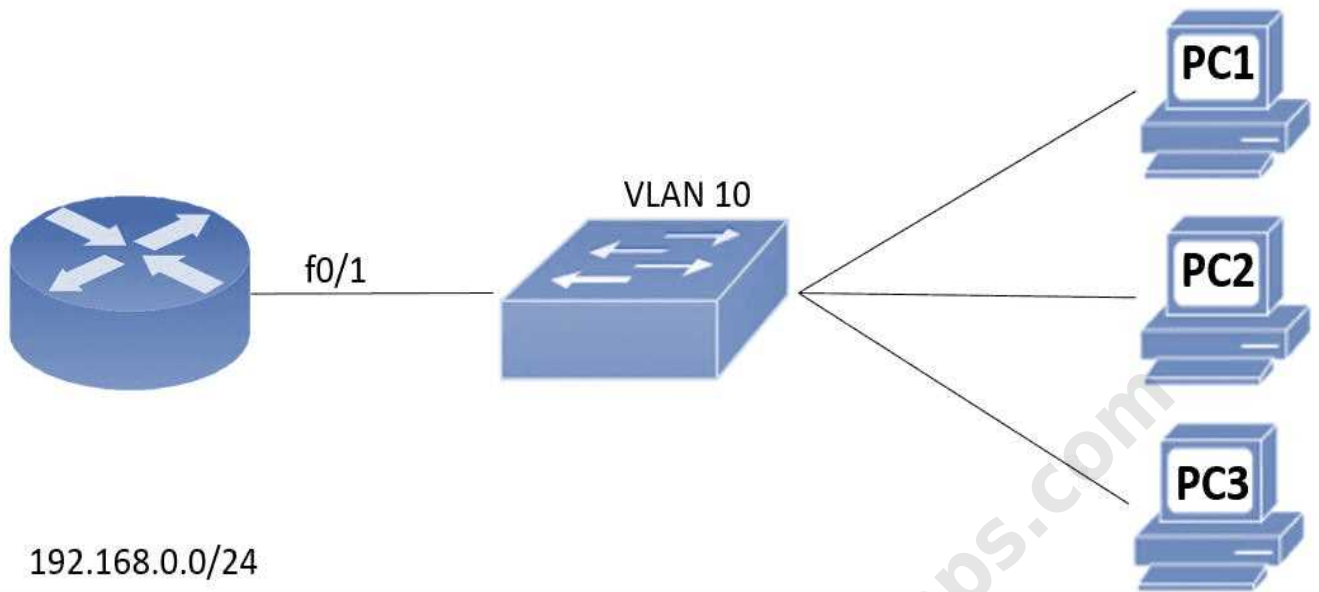
**Question: 768**

---

Refer to the exhibit.

An engineer assigns IP addressing to the current VLAN with three PCs. The configuration must also account for the expansion of 30 additional VLANS using the same Class C subnet for subnetting and host count. Which command set fulfills the request while reserving address space for the expected

growth?



192.168.0.0/24

- A. Switch(config)#interface vlan 10Switch(config-if)#ip address 192.168.0.1 255 255.255.252
- B. Switch(config)#interface vlan 10Switch(config-if)#ip address 192.168.0.1 255 255.255.248
- C. Switch(config)#interface vlan 10Switch(config-if)#ip address 192.168.0.1 255 255.255.0
- D. Switch(config)#interface vlan 10Switch(config-if)#ip address 192.168.0.1 255.255.255.128

**Answer: B**

Explanation:

### Question: 769

DRAG DROP

Drag and drop the AAA features from the left onto the corresponding AAA security services on the right. Not all options are used.

**Answer Area**

It enables the device to allow user- or group-based access.

It leverages a RADIUS server to grant user access to a reverse Telnet session.

It records the amount of time for which a user accesses the network on a remote server.

It restricts the CLI commands that a user can perform.

It uses TACACS+ to log the configuration commands entered by a network administrator.

It verifies the user and password before granting access to the device.

**Accounting****Authorization**

Explanation:

**Answer:**

### Accounting

It records the amount of time for which a user accesses the network on a remote server.

It uses TACACS+ to log the configuration commands entered by a network administrator.

### Authorization

It leverages a RADIUS server to grant user access to a reverse Telnet session.

It restricts the CLI commands that a user can perform.

---

#### Question: 770

---

Which WPA mode uses PSK authentication?

- A. Local
- B. Client
- C. Enterprise
- D. Personal

---

**Answer: D**

---

Explanation:

---

**Question: 771**

---

When the LAG configuration is updated on a Cisco WLC which additional task must be performed when changes are complete?

- A. Flush all MAC addresses from the WLC
- B. Re-associate the WLC with the access point.
- C. Re-enable the WLC interfaces
- D. Reboot the WLC

---

**Answer: C**

---

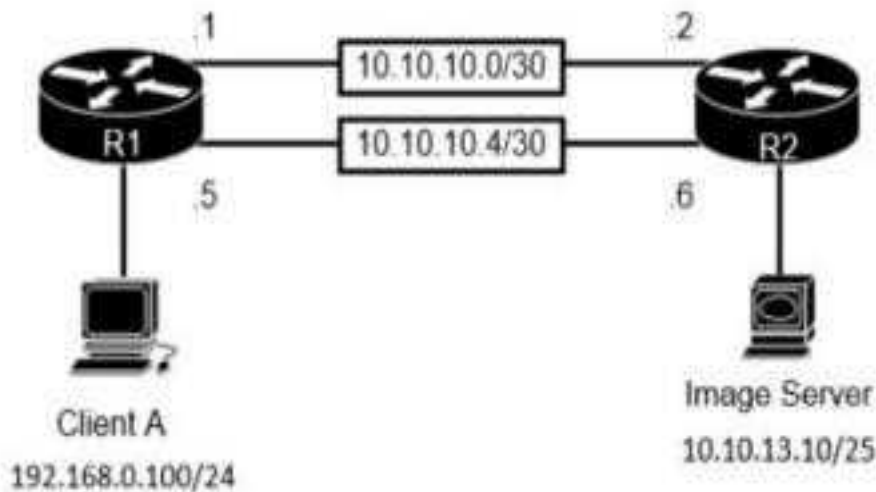
Explanation:

---

**Question: 772**

---

Refer to the exhibit.



```
R1#show ip route
Gateway of last resort is 10.10.10.2 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 10.10.10.2
```

```
R2#show ip route
Gateway of last resort is 10.10.10.1 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 10.10.10.1
```

The image server and client A are running an application that transfers an extremely high volume of data between the two. An engineer is configuring a dedicated circuit between R1 and R2. Which set of commands must the engineer apply to the routers so that only traffic between the image server and client A is forced to use the new circuit?

- A. R1(config)#ip route 10.10.13.10 255.255.255.255 10.10.10.6 R2(config)#ip route 192.168.0.100 255.255.255.255 10.10.10.5
- B. R1(config)#ip route 10.10.13.10 255.255.255.128 10.10.10.6 R2(config)#ip route 192.168.0.100 255.255.255.0 10.10.10.5
- C. R1(config)#ip route 10.10.13.10 255.255.255.252 10.10.10.6 R2(config)#ip route 192.168.0.100 255.255.255.252 10.10.10.5
- D. R1(config)#ip route 10.10.13.10 255.255.255.255 10.10.10.2 R2(config)#ip route 192.168.0.100 255.255.255.255 10.10.10.1

---

**Answer: D**

---

Explanation:



---

**Question: 773**

---

What provides connection redundancy increased bandwidth and load sharing between a wireless LAN controller and a Layer 2 switch?

- A. VLAN trunking
- B. tunneling
- C. first hop redundancy
- D. link aggregation

---

**Answer: D**

---

Explanation:

---

**Question: 774**

---

By default, how long will the switch continue to know a workstation MAC address after the workstation stops sending traffic?

- A. 200 seconds
- B. 300 seconds
- C. 600 seconds
- D. 900 seconds

---

**Answer: B**

---

Explanation:

---

**Question: 775**

---

Refer to the exhibit.

The screenshot shows a configuration page for Layer 2 security. At the top, there are tabs for 'Layer 2', 'Layer 3', and 'AAA Servers'. The 'Layer 2' tab is selected. Below the tabs, there is a 'Layer 2 Security' section with a dropdown menu set to 'WPA+WPA2'. Below this is a 'MAC Filtering' checkbox which is unchecked. The 'Fast Transition' section has a dropdown set to 'Adaptive', a checked 'Over the DS' checkbox, and a 'Reassociation Timeout' of '20' seconds. The 'Protected Management Frame' section has a 'PMF' dropdown set to 'Disabled'. The 'WPA+WPA2 Parameters' section has checkboxes for 'WPA Policy' (unchecked), 'WPA2 Policy' (checked), 'WPA2 Encryption' (checked), and 'OSN Policy' (unchecked). Under 'WPA2 Encryption', there are radio buttons for 'AES' (selected), 'TKIP', 'CCMP256', 'GCMP128', and 'GCMP256'. The 'Authentication Key Management' section has checkboxes for '802.1X' (checked), 'CCKM' (unchecked), 'PSK' (unchecked), 'FT 802.1X' (unchecked), 'FT PSK' (unchecked), 'SUITEB-1X' (unchecked), and 'SUITEB192-1X' (unchecked). At the bottom, there is a 'WPA gtk-randomize State' dropdown set to 'Disable'.

**Layer 2** **Layer 3** **AAA Servers**

Layer 2 Security **WPA+WPA2**

MAC Filtering ☐

**Fast Transition**

Fast Transition **Adaptive**

Over the DS ☒

Reassociation Timeout **20** Seconds

**Protected Management Frame**

PMF **Disabled**

**WPA+WPA2 Parameters**

WPA Policy ☐

WPA2 Policy ☒

WPA2 Encryption ☒ **AES** ☐ TKIP ☐ CCMP256 ☐ GCMP128 ☐ GCMP256

OSN Policy ☐

**Authentication Key Management**

802.1X ☒ **Enable**

CCKM ☐ Enable

PSK ☐ Enable

FT 802.1X ☐ Enable

FT PSK ☐ Enable

SUITEB-1X ☐ Enable

SUITEB192-1X ☐ Enable

WPA gtk-randomize State **Disable**

Clients on the WLAN are required to use 802.11r. What action must be taken to meet the requirement?

- A. Under Protected Management Frames, set the PMF option to Required.
- B. Enable CCKM under Authentication Key Management.
- C. Set the Fast Transition option and the WPA gtk-randomize State to disable.
- D. Set the Fast Transition option to Enable and enable FT 802.1X under Authentication Key Management.

---

**Answer: D**

---

Explanation:

---

**Question: 776**

---

Which Cisco proprietary protocol ensures traffic recovers immediately, transparently, and automatically when edge devices or access circuits fail?

- A. SLB
- B. FHRP
- C. VRRP
- D. HSRP

---

**Answer: D**

---

Explanation:

---

**Question: 777**

---

What is a link-local all-nodes IPv6 multicast address?

- A. ff02:0:0:0:0:0:0:1
- B. 2004:31c:73d9:683e:255::
- C. fffe:034:0dd:45d6:789e::
- D. fe80:4433:034:0dd::2

---

**Answer: D**

---

Explanation:

---

**Question: 778**

---

What is a characteristic of RSA?

- A. It uses preshared keys for encryption
- B. It requires both sides to have identical keys
- C. It is a private-key encryption algorithm
- D. It is a public-key cryptosystem

---

**Answer: D**

---

Explanation:

---

**Question: 779**

---

What is used as a solution for protecting an individual network endpoint from attack?

- A. Router
- B. Wireless controller
- C. Anti software
- D. Cisco DNA Center

---

**Answer: C**

---

Explanation:

---

**Question: 780**

---

What is the advantage of separating the control plane from the data plane within an SDN network?

- A. decreases overall network complexity
- B. limits data queries to the control plane
- C. reduces cost

D. offloads the creation of virtual machines to the data plane

---

**Answer: A**

---

Explanation:

---

**Question: 781**

---

DRAG DROP

Drag and drop the use cases for device-management technologies from the left onto the corresponding.

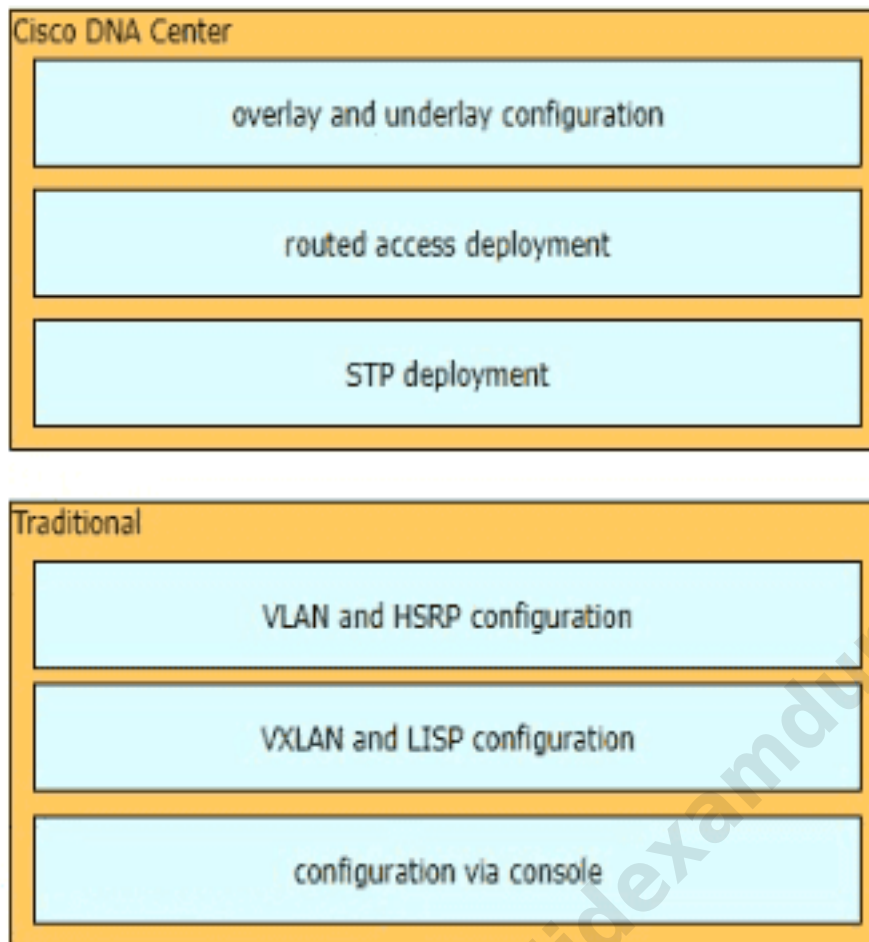
|                                    |                  |
|------------------------------------|------------------|
| overlay and underlay configuration | Cisco DNA Center |
| routed access deployment           |                  |
| STP deployment                     |                  |
| VLAN and HSRP configuration        | Traditional      |
| VXLAN and LISP configuration       |                  |
| configuration via console          |                  |

---

**Answer:**

---

Explanation:

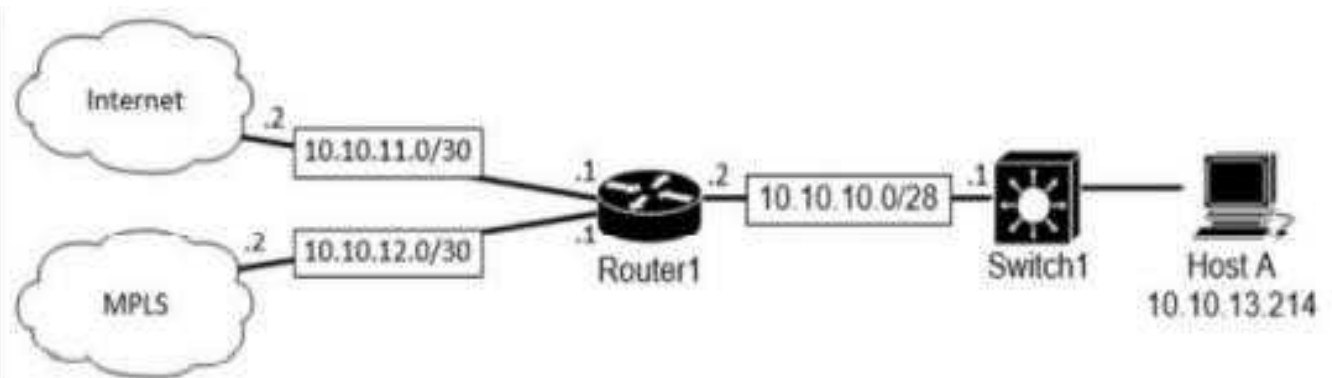


---

**Question: 782**

---

Refer to the exhibit.



```
Router1#show ip route
```

```
Gateway of last resort is 10.10.11.2 to network 0.0.0.0
```

```

    209.165.200.0/27 is subnetted, 1 subnets
B       209.165.200.224 [20/0] via 10.10.12.2, 03:22:14
    209.165.201.0/27 is subnetted, 1 subnets
B       209.165.201.0 [20/0] via 10.10.12.2, 02:26:33
    209.165.202.0/27 is subnetted, 1 subnets
B       209.165.202.128 [20/0] via 10.10.12.2, 02:26:03
    10.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
C       10.10.10.0/28 is directly connected, GigabitEthernet0/0
C       10.10.11.0/30 is directly connected, FastEthernet2/0
C       10.10.12.0/30 is directly connected, GigabitEthernet0/1
O       10.10.13.0/25 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O       10.10.13.128/28 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O       10.10.13.144/28 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O       10.10.13.160/29 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O       10.10.13.208/29 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
S*     0.0.0.0/0 [1/0] via 10.10.11.2

```

What is the prefix length for the route that router1 will use to reach host A?

- A. /25
- B. /27
- C. /28
- D. /29

---

**Answer: D**

---

Explanation:

---

**Question: 783**

---

Refer to the exhibit.

```
Device# configure terminal
Device(config)# netconf ssh acl 1
Device(config)# netconf lock-time 100
Device(config)# netconf max-sessions 1
Device(config)# netconf ma-message 10
```

A network engineer must configure NETCONF. After creating the configuration, the engineer gets output from the command show line but not from show running- config. Which command completes the configuration?

- A. Device(config)# netconf lock-time 500
- B. Device(config)# netconf max-message 1000
- C. Device(config)# no netconf ssh acl 1
- D. Device(config)# netconf max-sessions 100

---

**Answer: B**

---

Explanation:

---

**Question: 784**

---

Refer to the exhibit.



```
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 100,200,300
channel-group 1 mode active

SW1#show run interface fastEthernet 0/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 100,200
channel-group 1 mode active

SW2#show run interface fastEthernet 0/1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 100,200,300
channel-group 1 mode active

SW2#show run interface fastEthernet 0/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 100,200,300
channel-group 1 mode active
```

Refer to the exhibit. An engineer is building a new Layer 2 LACP EtherChannel between SW1 and SW2. and they executed the given show commands to verify the work Which additional task must be performed so that the switches successfully bundle the second member in the LACP port-channel?

- A. Configure the switchport trunk allowed vlan 300 command on SW1 port-channel 1
- B. Configure the switchport trunk allowed vlan 300 command on interface Fa0/2 on SW1.
- C. Configure the switchport trunk allowtd vlan add 300 command on interface FaO 2 on SW2.
- D. Configure the switchport trunk allowtd vlan add 300 command on SW1 port-channel 1

---

**Answer: B**

---

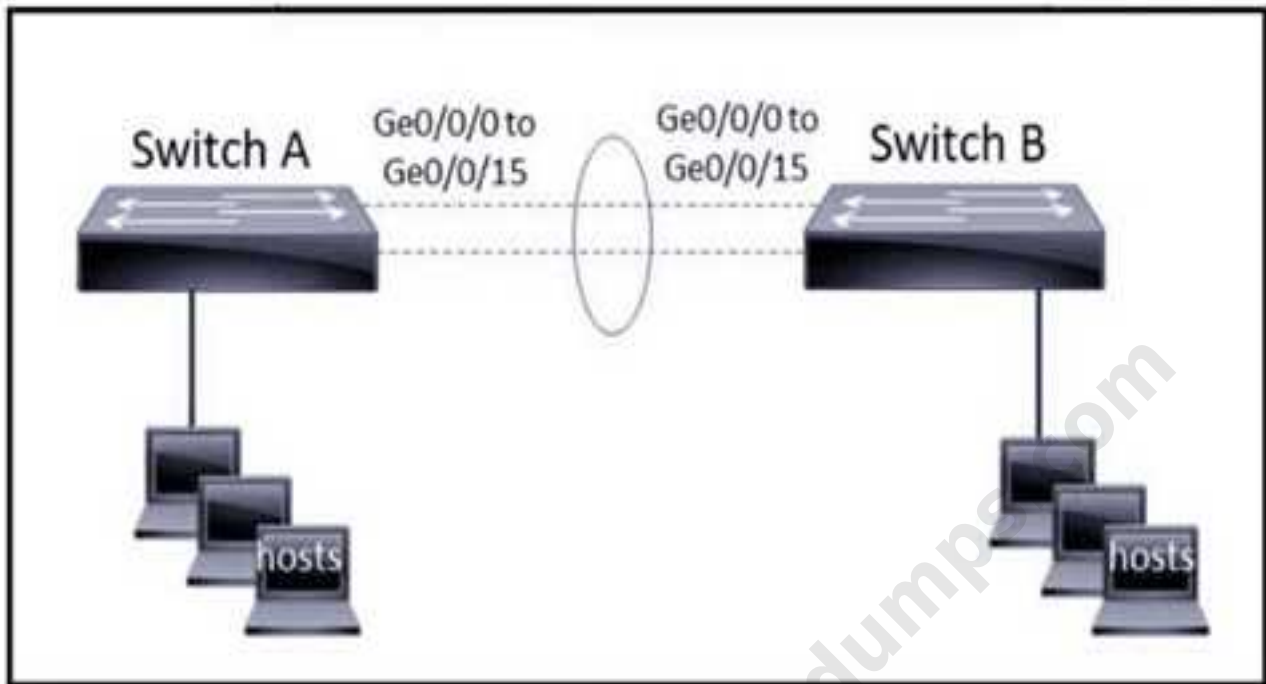
Explanation:

---

**Question: 785**

---

Refer to the exhibit.



The EtherChannel is configured with a speed of 1000 and duplex as full on both ends of channel group 1. What is the next step to configure the channel on switch A to respond to but not initiate LACP communication?

- A. interface range gigabitethernet0/0/0-15 channel-group 1 mode on
- B. interface range gigabitethernet0/0/0-15 channel-group 1 mode desirable
- C. interface port-channel 1 channel-group 1 mode auto
- D. interface port-channel 1 channel-group 1 mode passive

**Answer: D**

Explanation:

### Question: 786

Refer to the exhibit.

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.1.1.0/30 is directly connected, GigabitEthernet0/0
L    10.1.1.2/32 is directly connected, GigabitEthernet0/0
S    192.168.0.0/20 [1/0] via 10.1.1.1
     192.168.1.0/30 is subnetted, 1 subnets
S    192.168.1.0/30 [1/0] via 10.1.1.1
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
S    192.168.2.0/28 [1/0] via 10.1.1.1
S    192.168.2.0/29 [1/0] via 10.1.1.1
```

An engineer is checking the routing table in the main router to identify the path to a server on the network. Which route does the router use to reach the server at 192.168.2.2?

- A. S 192.168.0.0/20 [1/0] via 10.1.1.1
- B. S 192.168.2.0/29 [1/0] via 10.1.1.1
- C. S 192.168.2.0/28 [1/0] via 10.1.1.1
- D. S 192.168.1.0/30 [1/0] via 10.1.1.1

---

**Answer: B**

---

Explanation:

---

### Question: 787

---

Why would a network administrator choose to implement automation in a network environment?

- A. To simplify the process of maintaining a consistent configuration state across all devices
- B. To centralize device information storage
- C. To implement centralized user account management
- D. To deploy the management plane separately from the rest of the network

---

**Answer: A**

---

Explanation:

Answer: A

Explanation:

---

**Question: 788**

---

DRAG DROP

Drag and drop the statements about AAA services from the left onto the corresponding AAA services on the right. Not all options are used.

|                                                          |                |
|----------------------------------------------------------|----------------|
| It grants access to network assets, such as FTP servers. | Accounting     |
| It limits the services available to a user.              |                |
| It performs user validation via TACACS+.                 |                |
| It records the duration of each connection.              | Authentication |
| It supports User Access Reporting.                       |                |
| It verifies "who you are".                               |                |

---

**Answer:**

---

Explanation:

1&6 authentication

2&4 authorization

---

**Question: 789**

---

Refer to the exhibit.

```
Gateway of last resort is 172.16.2.2 to network 0.0.0.0
```

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.10.8.0/28 is directly connected, GigabitEthernet0/0/2
C 10.10.10.0/24 is directly connected, GigabitEthernet0/0/0
L 10.10.10.3.32 is directly connected, GigabitEthernet0/0/0

172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
S 172.16.1.33/32 is directly connected, GigabitEthernet0/0/1
C 172.16.2.0/23 is directly connected, GigabitEthernet0/0/1
L 172.16.2.1/32 is directly connected, GigabitEthernet0/0/1
S* 0.0.0.0/0 [1/0] via 172.16.2.2
```

A packet sourced from 10.10.10.1 is destined for 10.10.8.14. What is the subnet mask of the destination route?

- A. 255.255.254.0
- B. 255.255.255.240
- C. 255.255.255.248
- D. 255.255.255.252

---

**Answer: B**

---

Explanation:

---

**Question: 790**

---

Which two VPN technologies are recommended by Cisco for multiple branch offices and large-scale deployments? (Choose two.)

- A. site-to-site VPN
- B. DMVPN
- C. GETVPN
- D. IPsec remote access
- E. clientless VPN

---

**Answer: B,C**

---

Explanation:

---

**Question: 791**

---

Two switches have been implemented and all interfaces are at the default configuration level. A trunk link must be implemented between two switches with these requirements:

- using an industry-standard trunking protocol
- permitting VLANs 1 -10 and denying other VLANs

How must the interconnecting ports be configured?

A)

```
switchport mode dynamic
channel-protocol lacp
switchport trunk allowed vlags 1-10
```

B)

```
switchport mode trunk
switchport trunk encapsulation dot1q
switchport trunk allowed vlags 1-10
```

C)

```
switchport mode trunk
switchport trunk allowed vlags 1-10
switchport trunk native vlag 11
```

D)

```
switchport mode dynamic desirable
channel-group 1 mode desirable
switchport trunk encapsulation isl
switchport trunk allowed vlan except 11-4094
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: D**

---

Explanation:

---

**Question: 792**

---

Why is a first-hop redundancy protocol implemented?

- A. to protect against default gateway failures
- B. to prevent loops in a network
- C. to enable multiple switches to operate as a single unit
- D. to provide load-sharing for a multilink segment

---

**Answer: A**

---

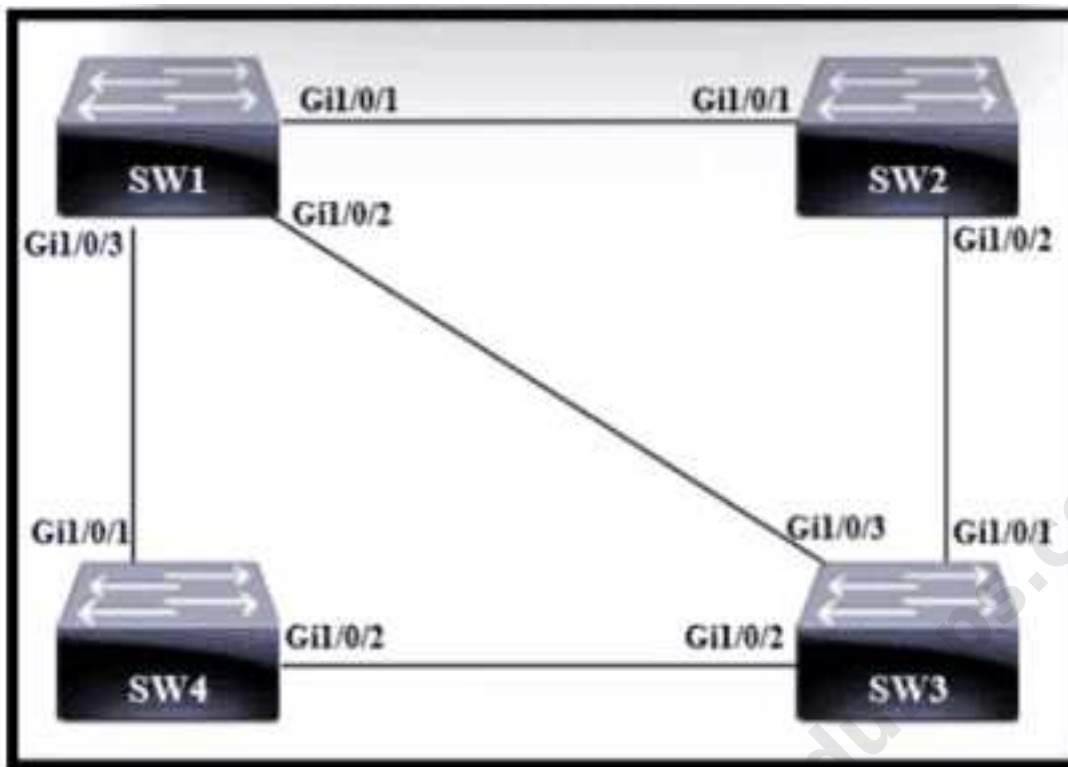
Explanation:

---

**Question: 793**

---

Refer to the exhibit.



A)

SW 4  
Bridge Priority - 40960  
mac-address 07:24:86:84:82:18

B)

SW 3  
Bridge Priority - 40960  
mac-address 08:71:50:67:61:38

C)

SW 2  
Bridge Priority - 32768  
mac-address 08:fd:b1:d7:78:39

D)

SW 1  
Bridge Priority - 32768  
mac-address 05:48:19:51:3e:49

A. Option A



- B. Option B
- C. Option C
- D. Option D

---

**Answer: D**

---

Explanation:

---

**Question: 794**

---

DRAG DROP

Drag and drop the characteristics of transport layer protocols from the left onto the corresponding protocols on the right.

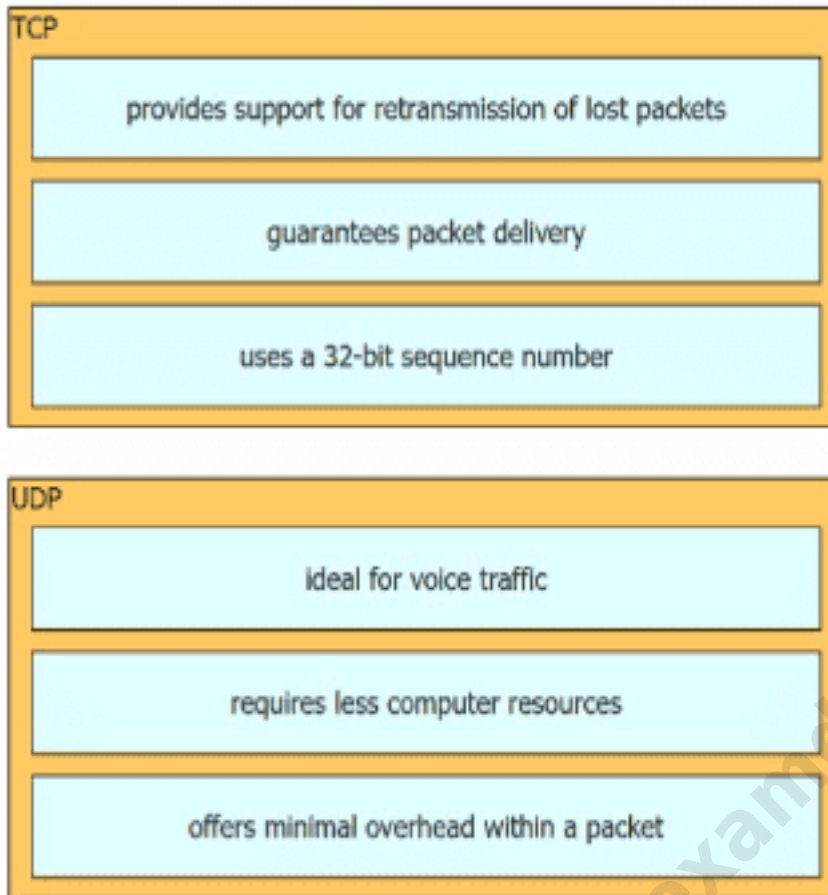
|                                                     |     |  |
|-----------------------------------------------------|-----|--|
| guarantees packet delivery                          | TCP |  |
| uses a 32-bit sequence number                       |     |  |
| ideal for voice traffic                             |     |  |
| provides support for retransmission of lost packets | UDP |  |
| offers minimal overhead within a packet             |     |  |
| requires less computer resources                    |     |  |

---

**Answer:**

---

Explanation:



---

**Question: 795**

---

What are two disadvantages of a full-mesh topology? (Choose two.)

- A. It needs a high MTU between sites.
- B. It has a high implementation cost.
- C. It must have point-to-point communication.
- D. It requires complex configuration.
- E. It works only with BGP between sites.

---

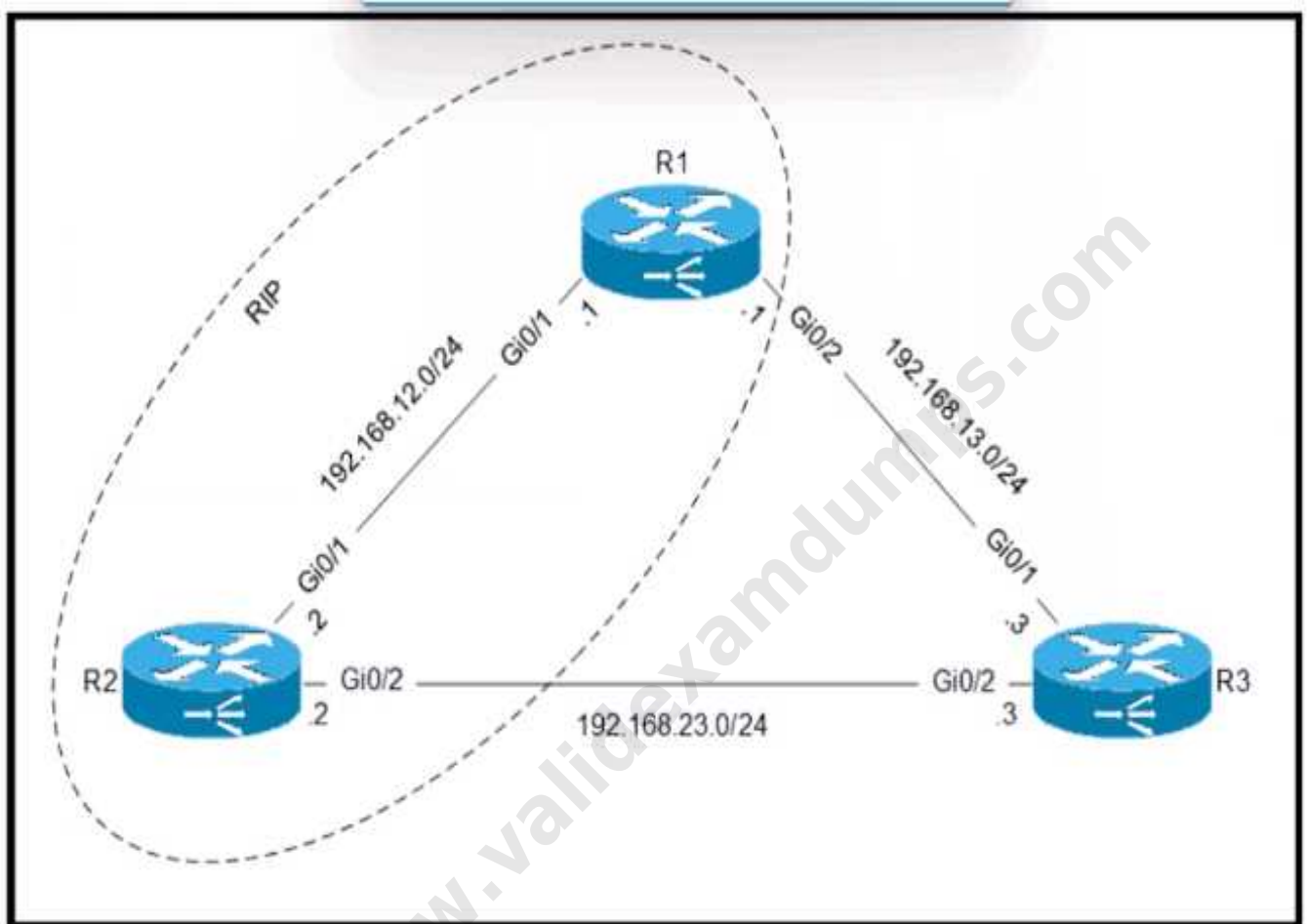
**Answer: B,D**

---

Explanation:

**Question: 796**

Refer to the exhibit.



Routers R1 and R2 are configured with RIP as the dynamic routing protocol. A network engineer must configure R1 with a floating static route to serve as a backup route to network 192.168.23. Which command must the engineer configure on R1?

- A. ip route 192.168.23.0 255.255.255.0 192.168.13.3 100
- B. ip route 192.168.23.0 255.255.255.0 192.168.13.3 121
- C. ip route 192.168.23.0 255.255.255.255 192.168.13.3 121
- D. ip route 192.168.23.0 255.255.255.0 192.168.13.3

**Answer: B**

Explanation:

---

**Question: 797**

---

Which remote access protocol provides unsecured remote CLI access?

- A. console
- B. Telnet
- C. Bash
- D. SSH

---

**Answer: B**

---

Explanation:

---

**Question: 798**

---

A Cisco engineer notices that two OSPF neighbors are connected using a crossover Ethernet cable. The neighbors are taking too long to become fully adjacent. Which command must be issued under the interface configuration on each router to reduce the time required for the adjacency to reach the FULL state?

- A. ip ospf network broadcast
- B. ip ospf dead-interval 40
- C. ip ospf network point-to-point
- D. ip ospf priority 0

---

**Answer: C**

---

Explanation:

---

**Question: 799**

---

DRAG DROP

Drag and drop the configuration management terms from the left onto the descriptions on the right.

Not all terms are used.

|           |                                                                                |
|-----------|--------------------------------------------------------------------------------|
| agent     | daemon that determines when the central authority has updates available        |
| agentless | model in which the central server sends updates to nodes on an as-needed basis |
| provision | easy-to-manage deployment option that may lack scalability                     |
| pull      | device hardware that runs without embedded management features                 |
| push      | to automatically install or deploy a configuration or update                   |
| post      |                                                                                |

---

**Answer:**

---

Explanation:

pull

push

agent

agentless

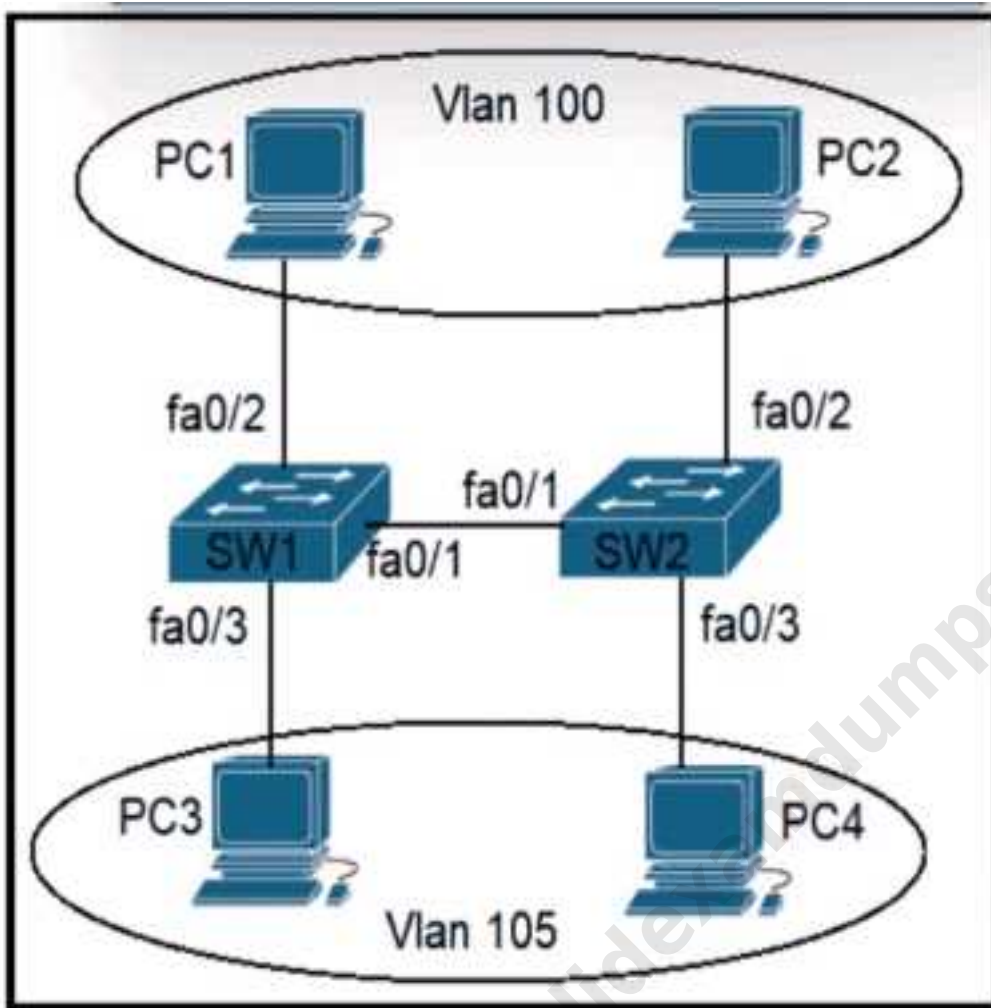
provision

---

**Question: 800**

---

Refer to the exhibit.



A)

```
Switch(config-if)#switchport mode dynamic
Switch(config-if)#switchport access vlan 100,105
Switch(config-if)#switchport trunk native vlan 1
```

B)

```
Switch(config-if)#switchport mode access
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport access vlan 100,105
Switch(config-if)#switchport trunk native vlan 3
```

C)

```
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk encapsulation isl
Switch(config-if)#switchport trunk allowed vlan 100,105
Switch(config-if)#switchport trunk native vlan 1
```

D)

```
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport trunk allowed vlan 100,105
Switch(config-if)#switchport trunk native vlan 3
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: B**

---

Explanation:

---

**Question: 801**

---

How does frame switching function on a switch?

- A. forwards frames to a neighbor port using CDP
- B. modifies frames that contain a known source VLAN
- C. inspects and drops frames from unknown destinations
- D. forwards known destinations to the destination port

---

**Answer: D**

---

Explanation:

---

**Question: 802**

---

DRAG DROP

Drag and drop the AAA features from the left onto the corresponding AAA security services on the right. Not all options are used.



| Authentication                                                                          |
|-----------------------------------------------------------------------------------------|
| It enables the device to allow user- or group-based access.                             |
| It leverages a RADIUS server to grant user access to a reverse Telnet session.          |
| It records the amount of time for which a user accesses the network on a remote server. |
| It restricts the CLI commands that a user is able to perform.                           |
| It uses TACACS+ to log the configuration commands entered by a network administrator.   |
| It verifies the user before granting access to the device.                              |

| Authorization |
|---------------|
|               |
|               |

**Answer:**

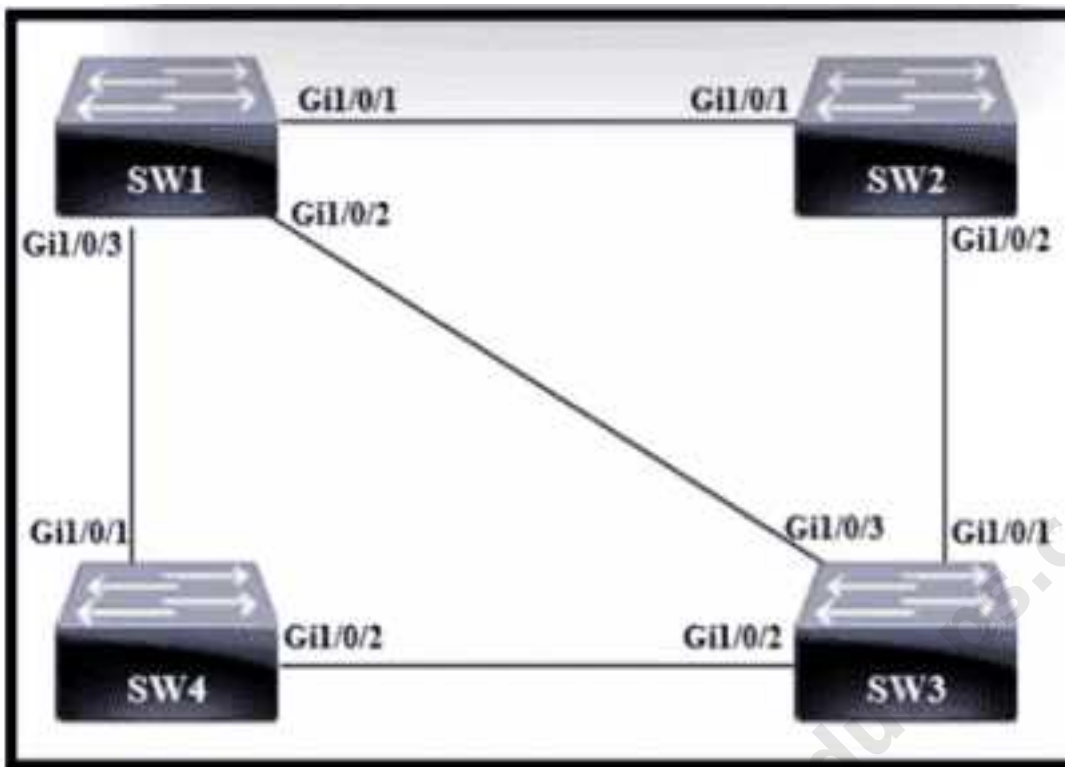
Explanation:

| Authentication                                                                          |
|-----------------------------------------------------------------------------------------|
| It records the amount of time for which a user accesses the network on a remote server. |
| It uses TACACS+ to log the configuration commands entered by a network administrator.   |

| Authorization                                                                  |
|--------------------------------------------------------------------------------|
| It leverages a RADIUS server to grant user access to a reverse Telnet session. |
| It restricts the CLI commands that a user is able to perform.                  |

**Question: 803**

Refer to the exhibit.



A)

SW 1  
Bridge Priority - 32768  
mac-address 0d:ca:8e:7f:a0:24

B)

SW 2  
Bridge Priority - 53248  
mac-address 02:3e:ee:61:5b:21

C)

SW 4  
Bridge Priority - 32768  
mac-address 07:c1:b7:27:dd:73

D)

SW 3  
Bridge Priority - 53248  
mac-address 02:aa:03:d3:05:87

A. Option A

- B. Option B
- C. Option C
- D. Option D

---

**Answer: D**

---

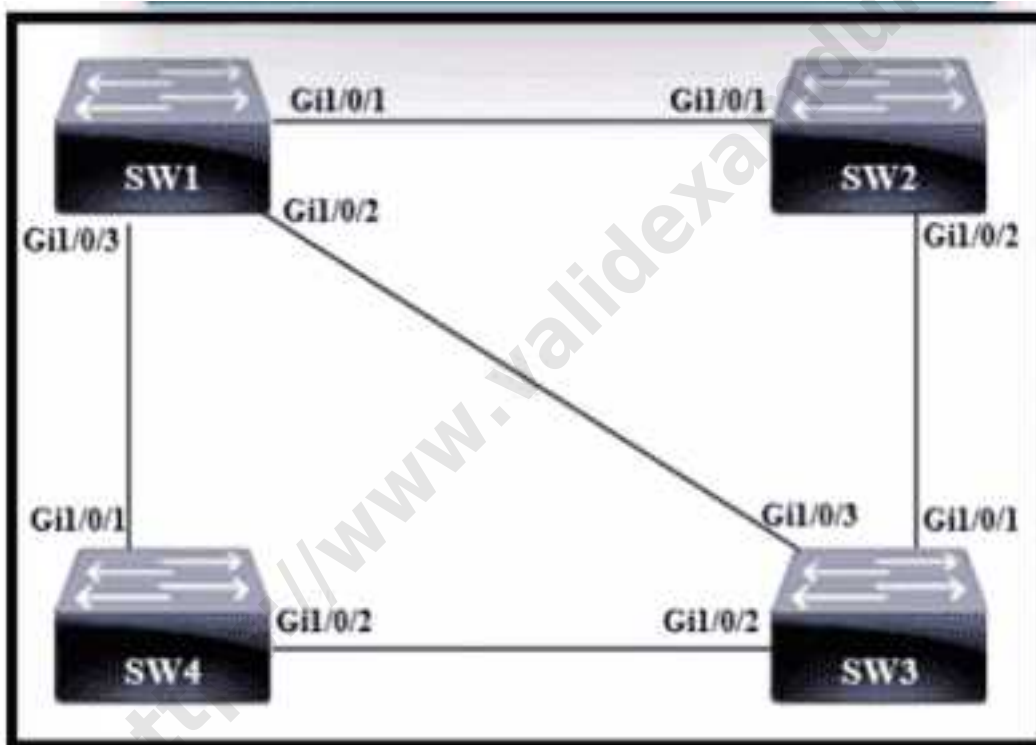
Explanation:

---

**Question: 804**

---

Refer to the exhibit.



Which switch becomes the root bridge?

A)

SW 1  
Bridge Priority - 32768  
mac-address 0d:ca:8e:7f:a0:24

B)

SW 2  
Bridge Priority - 53248  
mac-address 02:3e:ee:61:5b:21

C)

SW 4  
Bridge Priority - 32768  
mac-address 07:c1:b7:27:dd:73

D)

SW 3  
Bridge Priority - 53248  
mac-address 02:aa:03:d3:05:87

A. Option A

B. Option B

C. Option C

D. Option D

**Answer: B**

Explanation:

**Question: 805**

Refer to the exhibit.

```
GigabitEthernet1 is up, line protocol is up
  Hardware is CSR vNIC, address is 5000.0004.0000 (bia 5000.0004.0000)
  Internet address is 192.168.1.1/24
  MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full Duplex, 1000Mbps, link type is auto, media type is RJ45
```

Which format matches the Modified EUI-64 IPv6 interface address for the network 2001:db8::/64?

- A. 2001 :db8::5000:0004:5678:0090/64
- B. 2001 :db8:4425:5400:77ft:fe07:/64
- C. 2001 :db8::5000:00ff:fe04 0000/64
- D. 2001 :db8::5200:00ff:fe04:0000/64

---

**Answer: C**

---

Explanation:

---

**Question: 806**

---

Which WLC interface provides out-of-band management in the Cisco Unified Wireless Network Architecture?

- A. service port
- B. virtual
- C. AP-Manager
- D. dynamic

---

**Answer: A**

---

Explanation:

---

**Question: 807**

---

How do UTP and STP cables compare?

- A. STP cables are cheaper to procure and easier to install and UTP cables are more expensive and harder to install.
- B. UTP cables are less prone to crosstalk and interference and STP cables are more prone to crosstalk and interference.

C. UTP cables provide faster and more reliable data transfer rates and STP cables are slower and less reliable.

D. STP cables are shielded and protect against electromagnetic interference and UTP lacks the same protection against electromagnetic interference.

---

**Answer: D**

---

Explanation:

---

**Question: 808**

---

How does MAC learning function on a switch?

- A. broadcasts frames to all ports without queueing
- B. adds unknown source MAC addresses to the address table
- C. sends a retransmission request when a new frame is received
- D. sends frames with unknown destinations to a multicast group

---

**Answer: B**

---

Explanation:

---

**Question: 809**

---

What are two advantages of implementing a controller-based architecture instead of a traditional network architecture? (Choose two.)

- A. It allows for seamless connectivity to virtual machines.
- B. It supports complex and high-scale IP addressing schemes.
- C. It enables configuration task automation.
- D. It provides increased scalability and management options.
- E. It increases security against denial-of-service attacks.

---

**Answer: C,D**

---

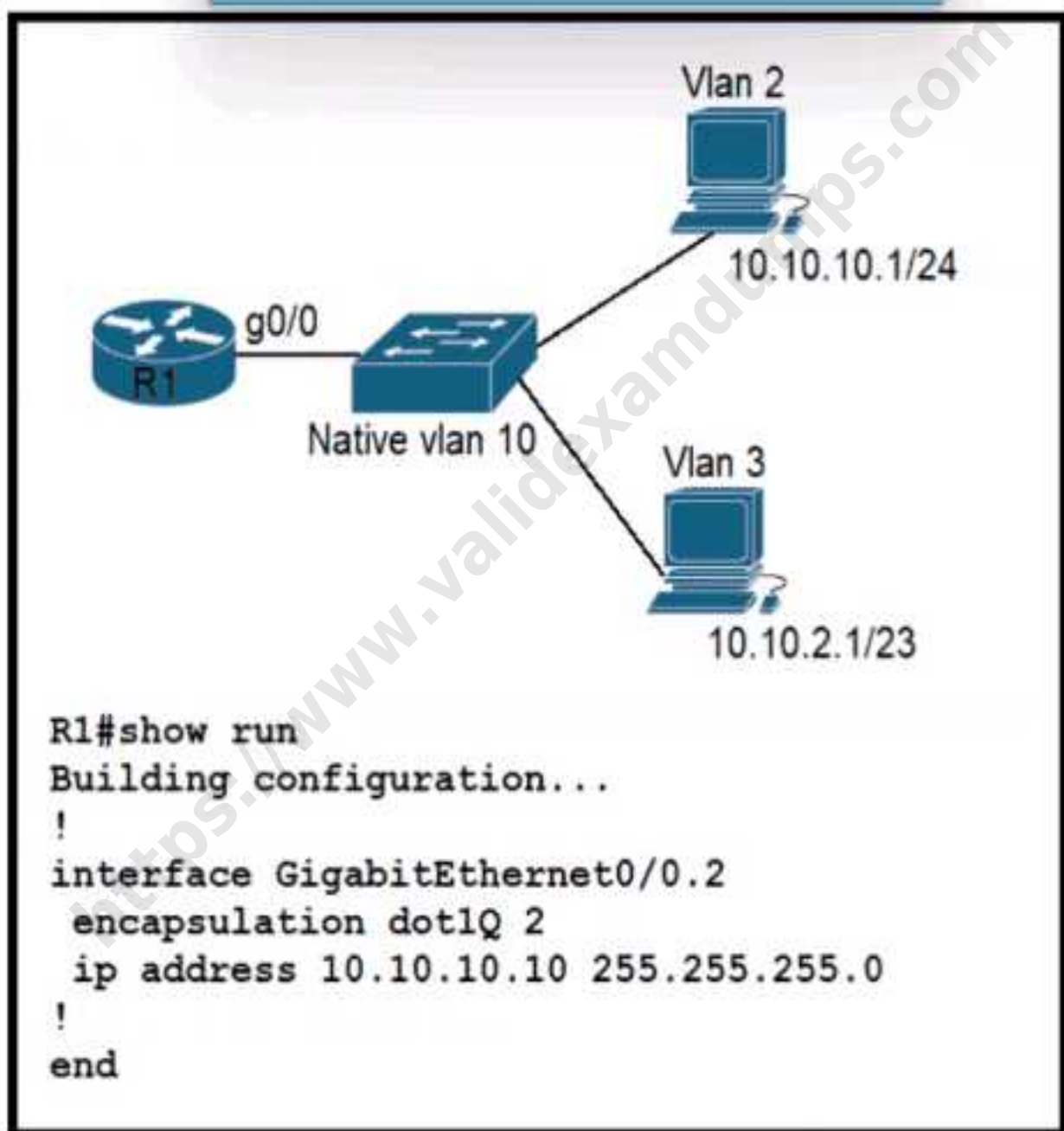
Explanation:

---

**Question: 810**

---

Refer to the exhibit.



A)

```
interface GigabitEthernet0/0  
ip address 10.10.2.10 255.255.252.0
```

B)

```
interface GigabitEthernet0/0.3  
encapsulation dot1Q 10  
ip address 10.10.2.10 255.255.255.252
```

C)

```
interface GigabitEthernet0/0.10  
encapsulation dot1Q 3  
ip address 10.10.2.10 255.255.254.0
```

D)

```
interface GigabitEthernet0/0.3  
encapsulation dot1Q 3 native  
ip address 10.10.2.10 255.255.252.0
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: C**

---

Explanation:

---

**Question: 811**

---

What is an enhancement implemented in WPA3?

- A. employs PKI and RADIUS to identify access points
- B. applies 802.1x authentication and AES-128 encryption
- C. uses TKIP and per-packet keying
- D. defends against deauthentication and disassociation attacks



---

**Answer: D**

---

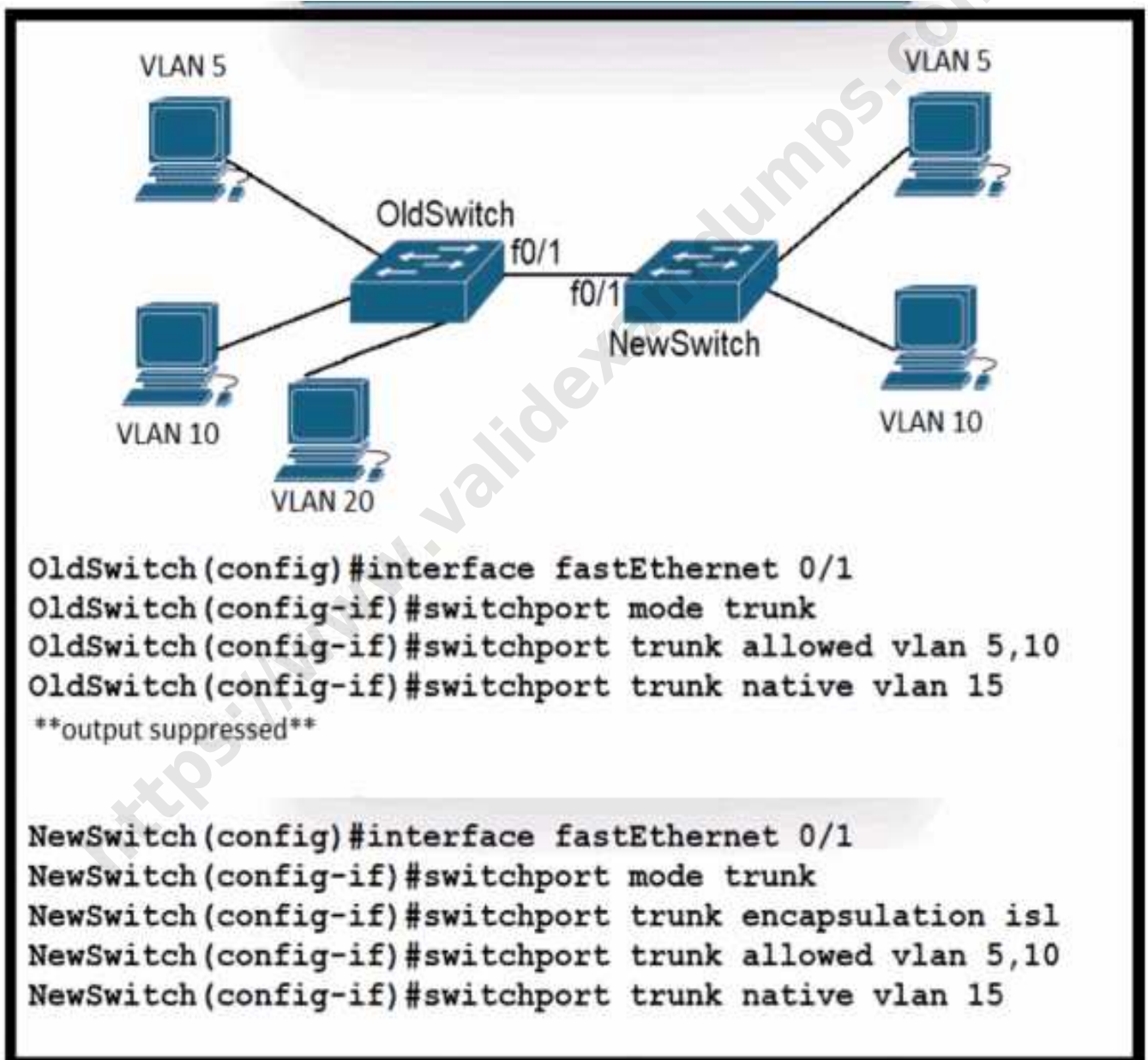
Explanation:

---

**Question: 812**

---

Refer to the exhibit.



A new VLAN and switch are added to the network. A remote engineer configures OldSwitch and must ensure that the configuration meets these requirements:

- accommodates current configured VLANs

- expands the range to include VLAN 20
- allows for IEEE standard support for virtual LANs

Which configuration on the NewSwitch side of the link meets these requirements?

A)

```
no switchport trunk encapsulation isl  
switchport trunk encapsulation dot1q  
switchport trunk allowed vlan add 20
```

B)

```
switchport nonegotiate  
no switchport trunk allowed vlan 5,10  
switchport trunk allowed vlan 5,10,15,20
```

C)

```
no switchport mode trunk  
switchport trunk encapsulation isl  
switchport mode access vlan 20
```

D)

```
switchport mode dynamic  
channel-group 1 mode active  
switchport trunk allowed vlan 5,10,15, 20
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: C**

---

Explanation:

---

**Question: 813**

---

A client experiences slow throughput from a server that is directly connected to the core switch in a data center. A network engineer finds minimal latency on connections to the server, but data transfers are unreliable, and the output of the show Interfaces counters errors command shows a high FCS-Err count on the interface that is connected to the server. What is the cause of the throughput issue?

- A. high bandwidth usage
- B. a physical cable fault
- C. a speed mismatch
- D. a cable that is too long

---

**Answer: B**

---

Explanation:

---

**Question: 814**

---

What is the purpose of classifying network traffic in QoS?

- A. services traffic according to its class
- B. identifies the type of traffic that will receive a particular treatment
- C. writes the class identifier of a packet to a dedicated field in the packet header
- D. configures traffic-matching rules on network devices

---

**Answer: B**

---

Explanation:

---

**Question: 815**

---

Which two HTTP methods are suitable for actions performed by REST-based APIs? (Choose two.)

- A. REMOVE
- B. REDIRECT

- C. OPOST
- D. GET
- E. UPOP

---

**Answer: C,D**

---

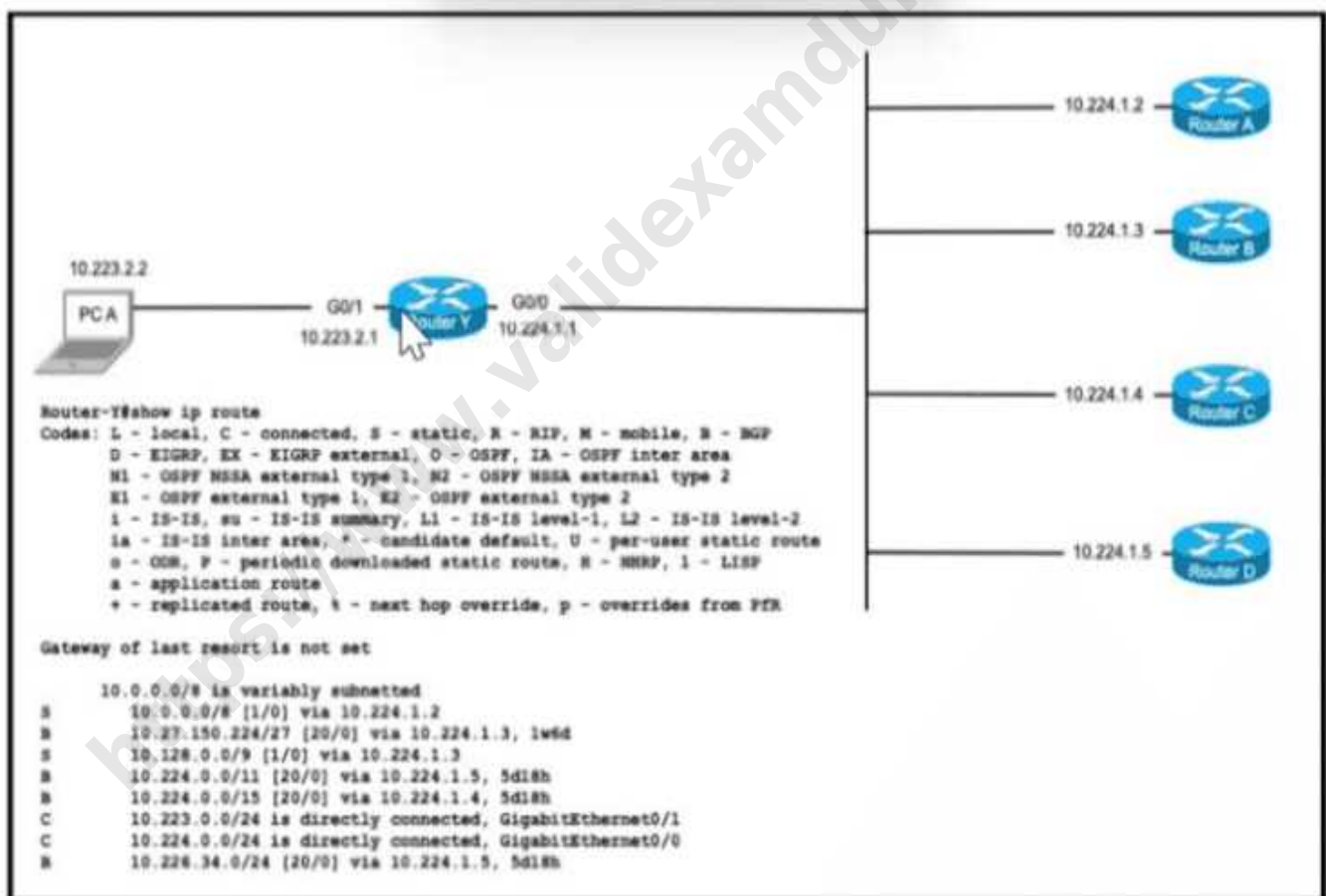
Explanation:

---

**Question: 816**

---

Refer to the exhibit.



PC A is communicating with another device at IP address 10.227.225.255. Through which router does router Y route the traffic?

- A. router A
- B. router B

C. router C

D. router D

---

**Answer: C**

---

Explanation:

---

**Question: 817**

---

Which component controls and distributes physical resources for each virtual machine?

A. OS

B. hypervisor

C. CPU

D. physical enclosure

---

**Answer: B**

---

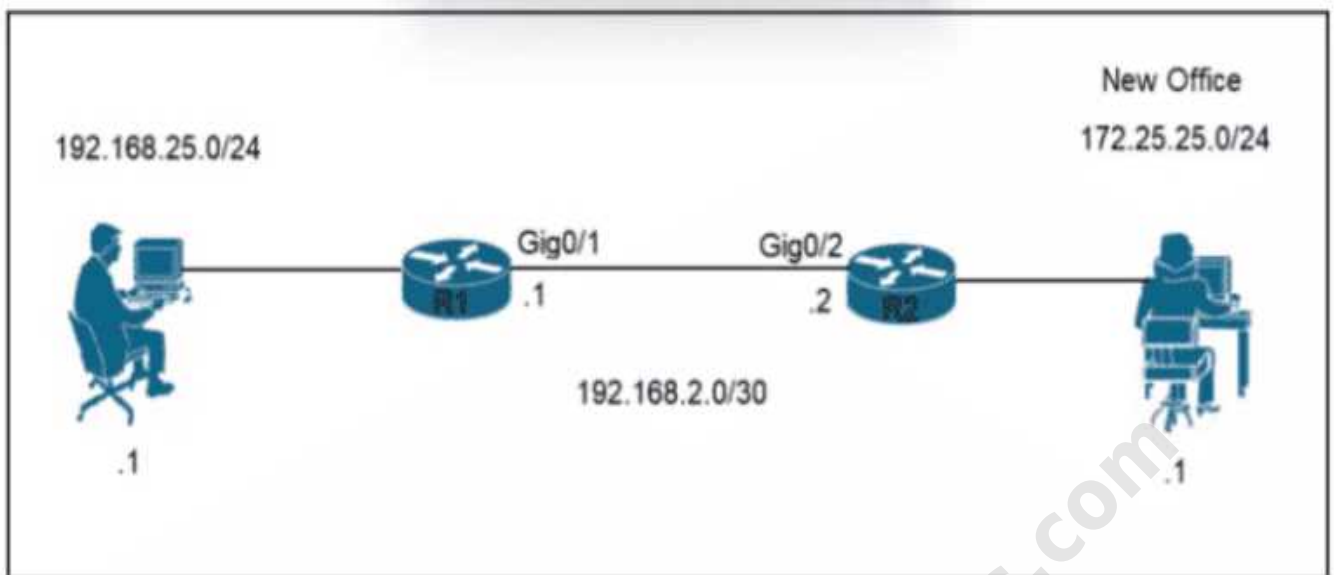
Explanation:

---

**Question: 818**

---

Refer to the exhibit.



A network engineer is updating the configuration on router R1 to connect a new branch office to the company network R2 has been configured correctly. Which command must the engineer configure so that devices at the new site communicate with the main office?

- A. ip route 172.25.25 0 255 255 255.0 192.168.2.1
- B. ip route 172.25.25 1 255 255 255 255 g0/1
- C. ip route 172.25.25.0.255.255.255.0.192.168.2.2

---

**Answer: C**

---

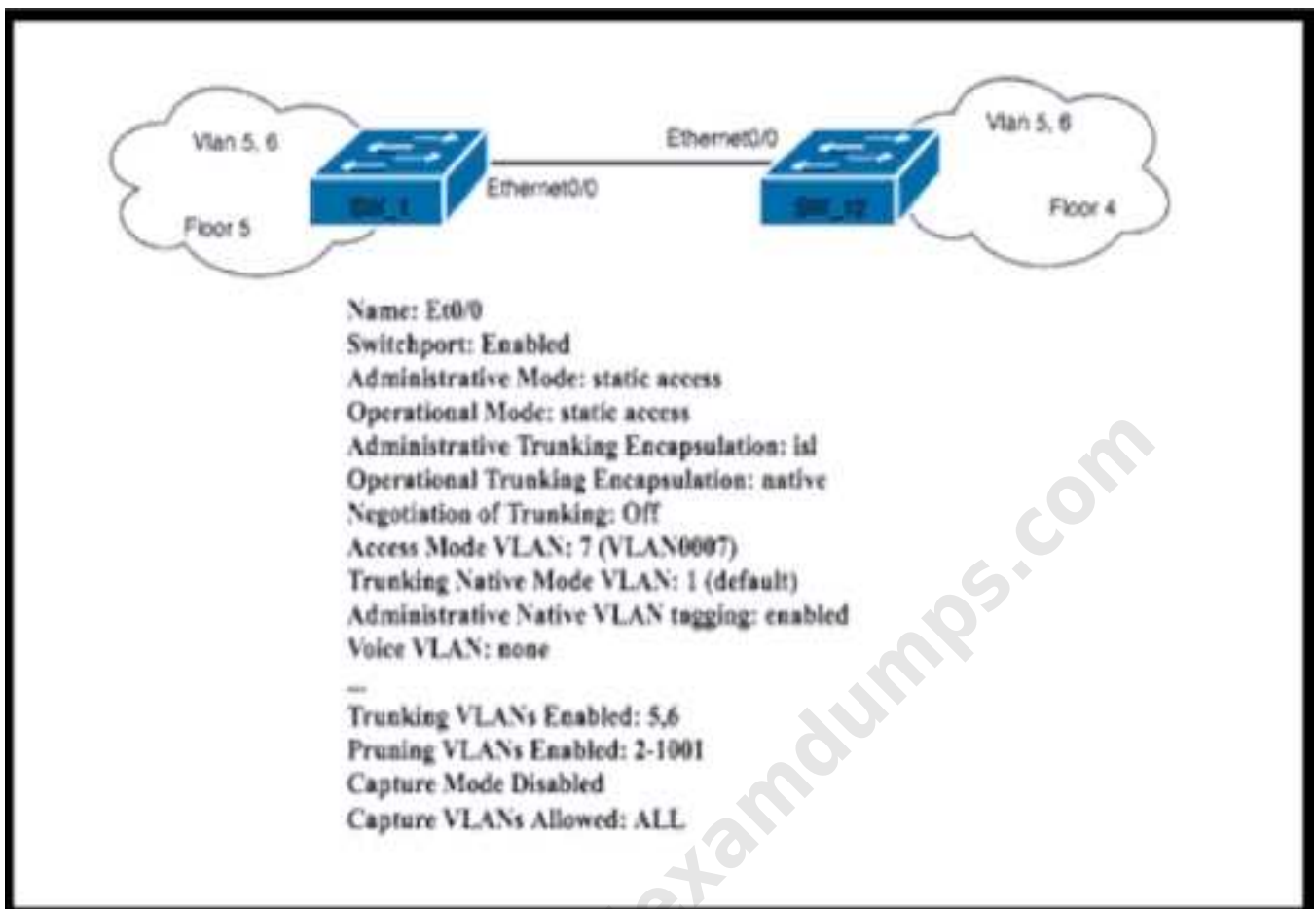
Explanation:

---

**Question: 819**

---

Refer to the exhibit.



SW\_1 and SW\_12 represent two companies that are merging. They use separate network vendors. The VLANs on both Sides have been migrated to share IP subnets. Which command sequence must be issued on both sides to join the two companies and pass all VLANs between the companies?

- A. switchport mode trunkswitchport trunk encapsulation dot1q
- B. switchport mode trunkswitchport trunk allowed vlan allswitchport dot1q ethertype 0800
- C. switchport mode dynamic desirableswitchport trunk allowed vlan allswitchport trunk native vlan 7
- D. switchport dynamic autoswitchport nonegotiate

---

**Answer: A**

---

Explanation:

---

### Question: 820

---

Refer to the exhibit.

```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - OGM, F - periodic downloaded static route, H - NHRP, I - IGRP
       a - application route
       * - replicated route, ! - next hop override, p - overrides from PFM
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
S*   0.0.0.0/0 is directly connected, Null0
      10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C     10.0.12.0/24 is directly connected, GigabitEthernet0/1
L     10.0.12.1/32 is directly connected, GigabitEthernet0/1
C     10.0.13.0/24 is directly connected, GigabitEthernet0/2
L     10.0.13.1/32 is directly connected, GigabitEthernet0/2
C     10.0.14.0/24 is directly connected, GigabitEthernet0/3
L     10.0.14.1/32 is directly connected, GigabitEthernet0/3
D     192.168.0.0/16 [90/130816] via 10.0.13.3, 00:10:09, GigabitEthernet0/2
O     192.168.0.0/23 [110/2] via 10.0.14.4, 00:00:46, GigabitEthernet0/3
S     192.168.0.0/24 [100/0] via 10.0.12.2

```

Which interface is chosen to forward traffic to the host at 192.168.0.55?

- A. GigabitEthernet0
- B. GigabitEthernet0/1
- C. Null0
- D. GigabitEthernet0/3

---

**Answer: B**

---

Explanation:

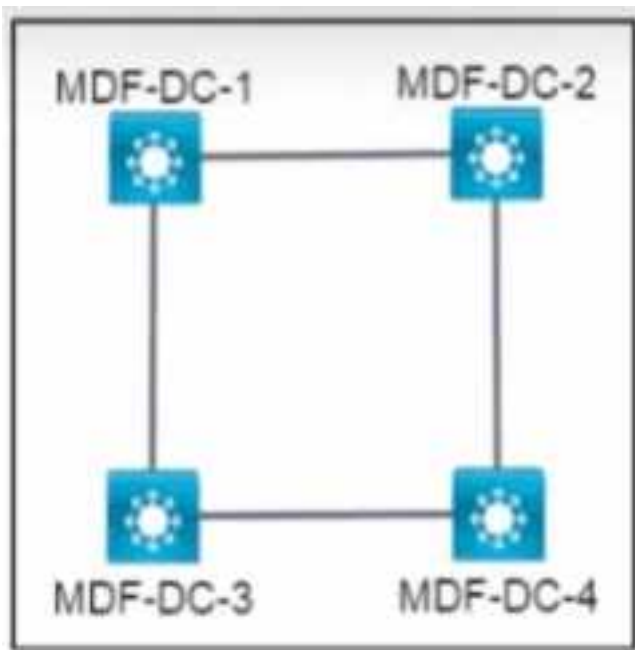
---

**Question: 821**

---

Refer to the exhibit.





All interfaces are in the same VLAN. All switches are configured with the default STP priorities. During the STP elections, which switch becomes the root bridge?

- A. MDF-DC-4:08:E0:19: 08:B3:19
- B. MDF-DC-3:08:0E:18::1A:3C:9D
- C. MDF-DC-08:0E:18:22:05:97
- D. MDF-DC-1:DB:E:44:02:54:79

---

**Answer: C**

---

Explanation:

---

**Question: 822**

---

Which command enables HTTP access to the Cisco WLC?

- A. config network secureweb enable
- B. config certificate generate web admin
- C. config network webmode enable
- D. config network telnet enable

---

**Answer: C**

---

Explanation:

---

**Question: 823**

---

What is the function of a controller in a software-defined network?

- A. multicast replication at the hardware level
- B. forwarding packets
- C. fragmenting and reassembling packets
- D. setting packet-handling policies

---

**Answer: B**

---

Explanation:

---

**Question: 824**

---

A network engineer is implementing a corporate SSID for WPA3-Personal security with a PSK. Which encryption cipher must be configured?

- A. GCMP256
- B. GCMP128
- C. CCMP256
- D. CCMP128

---

**Answer: D**

---

Explanation:

---

**Question: 825**

---

When is the PUT method used within HTTP?

- A. when a nonidempotent operation is needed
- B. to update a DNS server
- C. to display a web site
- D. when a read-only operation it required

---

**Answer: B**

---

Explanation:

---

**Question: 826**

---

Refer to the exhibit.

```
{  
  "Test_Questions" : [  
    "Automation",  
    "Configuration",  
  ],  
  "Test_Exam_Level" : [  
    "CCNA",  
    "CCNP",  
  ],  
  "Test_Response" : [  
    "Correct",  
    "Incorrect",  
  ],  
}
```

How many objects, Keys and JSON list values are present?

- A. three objects, two Keys, and three JSON list values
- B. three objects, three keys and two JSON MI values
- C. one object, three keys, and three JSON list values

D. one object, three keys and two JSON list values

---

**Answer: C**

---

Explanation:

---

**Question: 827**

---

A network engineer must migrate a router loopback interface to the IPv6 address space. If the current IPv4 address of the interface is 10.54.73.1/32, and the engineer configures IPv6 address 0.0.0.0:ffff:a36:4901, which prefix length must be used?

- A. /64
- B. /96
- C. /124
- D. /128

---

**Answer: D**

---

Explanation:

---

**Question: 828**

---

Which port type does a lightweight AP use to connect to the wired network when it is configured in local mode?

- A. EtherChannel
- B. LAG
- C. trunk
- D. access

---

**Answer: C**

---

Explanation:

---

**Question: 829**

---

What is the primary purpose of a console port on a Cisco WLC?

- A. In-band management via an asynchronous transport
- B. out-of-band management via an IP transport
- C. in-band management via an IP transport
- D. out-of-band management via an asynchronous transport

---

**Answer: D**

---

Explanation:

---

**Question: 830**

---

What describes a northbound REST API for SON?

- A. application-facing interface for SNMP GET requests
- B. network-element-facing interface for GET POST PUT and DELETE methods
- C. network-element-facing interface for the control and data planes
- D. application-facing interface far GET, POST, PUT, and DELETE methods

---

**Answer: D**

---

Explanation:

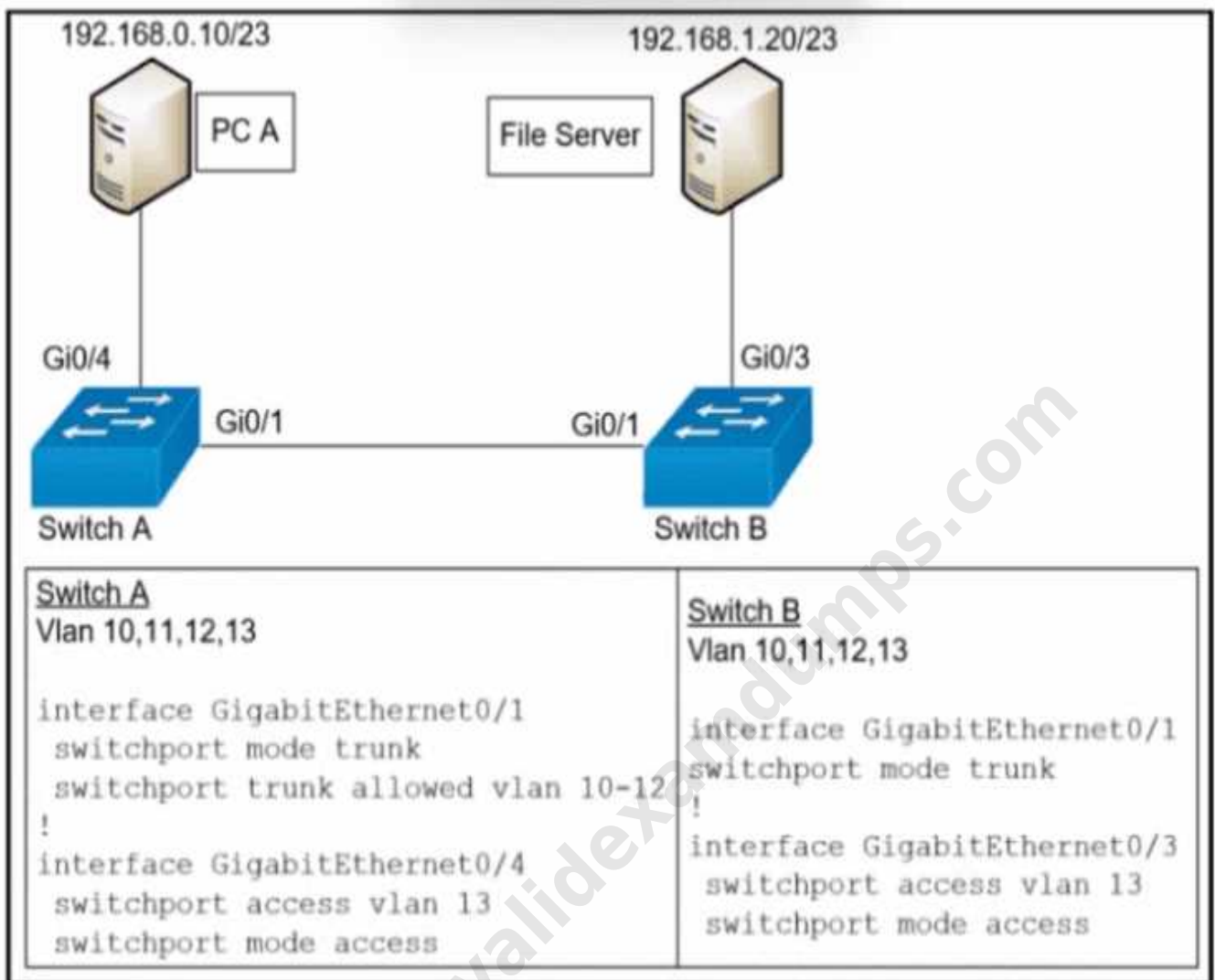
---

**Question: 831**

---

Refer to the exhibit.

PC A and the file server. Which command must be configured on switch A to prevent interruption of other communications?



- A. switch port trunk allowed vlan 12
- B. switchport trunk allowed vlan none
- C. switchport trunk allowed vlan add 13
- D. switch port trunk allowed vlan remove 10-11

**Answer: C**

Explanation:

### Question: 832

A switch is forwarding a frame out of an interface except the interface that received the frame. What is the technical term for this process?

- A. ARP
- B. CDP
- C. flooding
- D. multicast

---

**Answer: C**

---

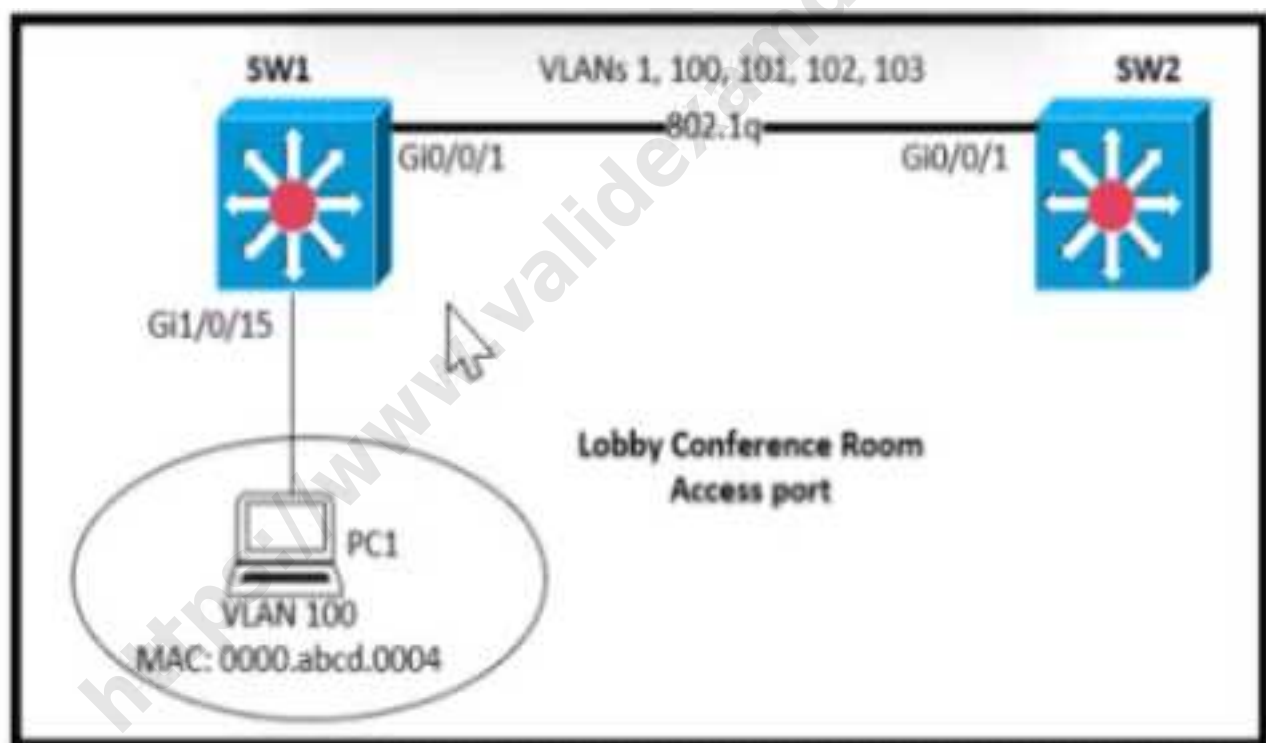
Explanation:

---

**Question: 833**

---

Refer to the exhibit.



SW1 supports connectivity for a lobby conference room and must be secured. The engineer must limit the connectivity from PC1 to the SW1 and SW2 network. The MAC addresses allowed must be limited to two. Which configuration secures the conference room connectivity?

A)

```
> interface gi1/0/15
  switchport port-security mac-address 0000.abcd.0004 vlan 100
```

B)

```
interface gi1/0/15
switchport port-security
switchport port-security maximum 2
```

C)

```
interface gi1/0/15
switchport port-security mac-address 0000.abcd.0004 vlan 100
interface switchport secure-mac limit 2
```

D)

```
interface gi1/0/15
switchport port-security
switchport port-security mac-address 0000.abcd.0004 vlan 100
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: B**

---

Explanation:

---

**Question: 834**

---

Refer to the exhibit.



```
Cat9300-1# show interface gi1/0/1 switchport
Name: Gi1/0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 321 (VLAN0321)
Administrative Native VLAN tagging: enabled
Trunking VLANs Enabled: 100,200,300
Pruning VLANs Enabled: 2-1001
```

Refer to the exhibit.

A network administrator configures an interface control re switch so that it connects to interface Gi1/0/1 on switch Cat9300-1. Which configuration must be applied to the new interface?

A)

```
switchport mode trunk
switchport trunk native vlan 321
switchport trunk allowed vlan 100,200,300
```

B)

```
switchport trunk encapsulation dot1q
switchport trunk native vlan 321
switchport trunk allowed vlan 100-300
```

C)

```
switchport mode dynamic desirable
switchport trunk native vlan 321
switchport trunk allowed vlan 100,200,300
```

D)

```
switchport nonegotiate
switchport access vlan 321
switchport trunk allowed vlan except 2-1001
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: A**

---

Explanation:

---

**Question: 835**

---

What are two differences between WPA2 and WPA3 wireless security? (Choose two.)

- A. WPA3 um AES for stronger protection than WPA2 which uses SAE
- B. WPA2 uses 1 M-bit key encryption and WPA3 requires 256-brt key encryption
- C. WPA3 uses AES for stronger protection than WPA2 which uses TKIP WPA3 uses
- D. SAE tor stronger protection than WPA2 which uses AES
- E. WPA2 uses 12B-M key encryption and WPA3 supports 128 bit and 192 bit key encryption

---

**Answer: C,E**

---

Explanation:

---

**Question: 836**

---

DRAG DROP

Drag and drop the device behaviors from the left onto the matching HSRP slate on the right.

|                                                                                      |         |
|--------------------------------------------------------------------------------------|---------|
| has heard from the neighbor device and is receiving hello packets                    | Active  |
| is forwarding packets                                                                | Learn   |
| is ready to forward packets if the device that is currently forwarding packets fails | Listen  |
| is transmitting and receiving hello packets                                          | Speak   |
| is waiting to hear from the neighbor device                                          | Standby |

---

**Answer:**

---

Explanation:

|                                                                                      |
|--------------------------------------------------------------------------------------|
| is forwarding packets                                                                |
| has heard from the neighbor device and is receiving hello packets                    |
| is waiting to hear from the neighbor device                                          |
| is transmitting and receiving hello packets                                          |
| is ready to forward packets if the device that is currently forwarding packets fails |

---

**Question: 837**

---

Which signal frequency appears 60 times per minute?

- A. 1 Hz signal
- B. 1 GHz signal
- C. 60 Hz signal
- D. 60 GHz signal

---

**Answer: B**

---

Explanation:

---

**Question: 838**

---

What is a benefit of a point-to-point leased line?

- A. flexibility of design
- B. simplicity of configurator
- C. low cost
- D. full-mesh capability

---

**Answer: C**

---

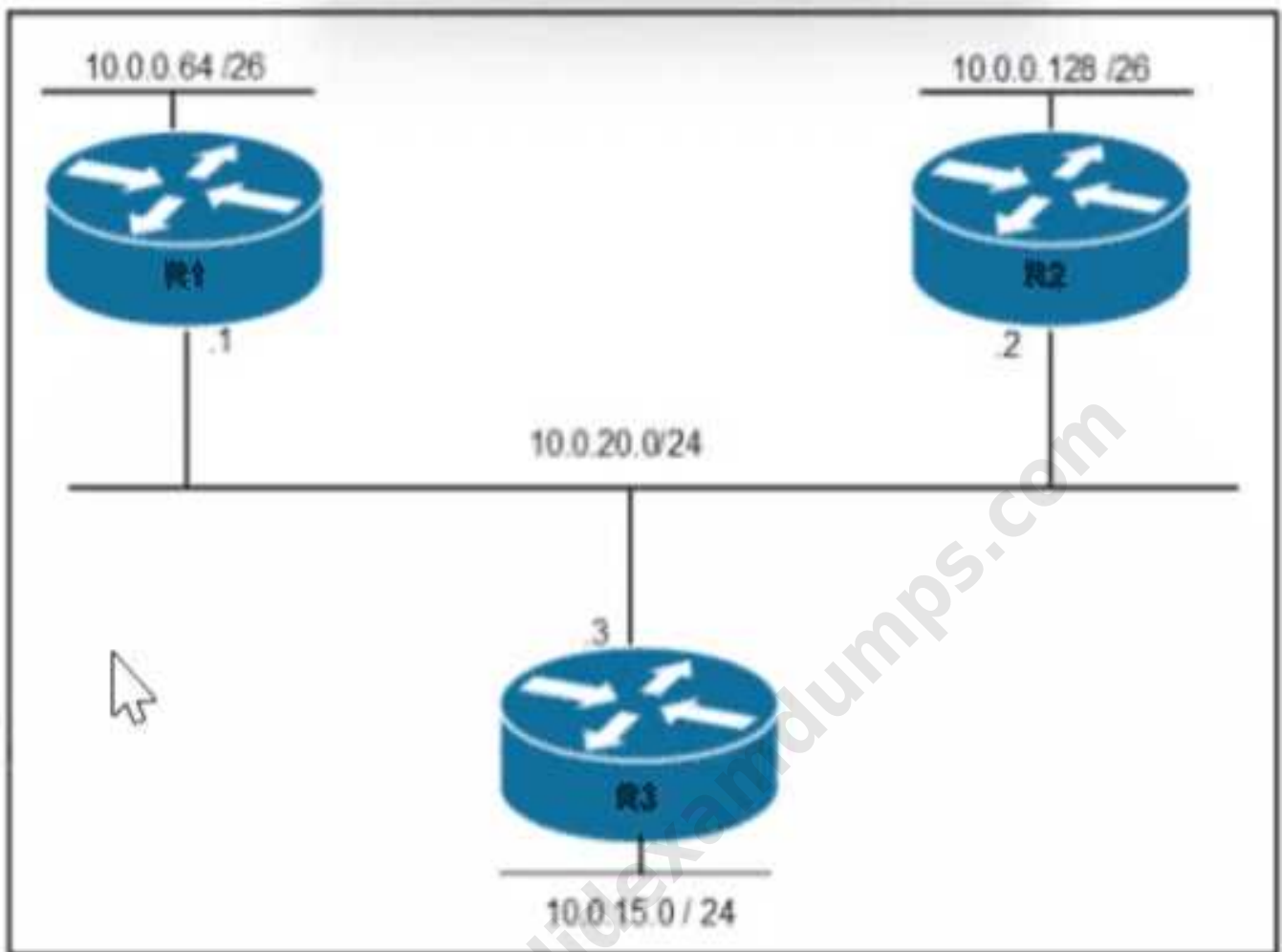
Explanation:

---

**Question: 839**

---

Refer to the exhibit.



Router R1 is added to the network and configured with the 10.0.0.64/26 and 10.0.20.0/24 subnets. However, traffic destined for the LAN on R3 is not accessible. Which command, when executed on R1, defines a tunnel route to reach the R3 LAN?

A)

`ip route 10.0.15.0 255.255.255.192 10.0.20.1`

B)

`ip route 10.0.15.0 255.255.255.0 10.0.20.1`

C)

`ip route 10.0.0.64 255.255.255.192 10.0.20.3`

A. Option

B. Option

C. Option

---

**Answer: C**

---

Explanation:

---

**Question: 840**

---

What is a function of the core and distribution layers in a collapsed-core architecture?

- A. The router must use IPv4 and IPv6 addresses at Layer 3.
- B. The core and distribution layers are deployed on two different devices to enable failover.
- C. The router can support HSRP for Layer 2 redundancy in an IPv6 network.
- D. The router operates on a single device or a redundant pair.

---

**Answer: D**

---

Explanation:

The core and distribution layers are collapsed into one layer in a collapsed-core architecture, and this layer operates on a single device or a redundant pair. This layer is responsible for the routing between the access layer and the WAN, as well as providing redundancy.

---

**Question: 841**

---

DRAG DROP

Drag and drop the characteristic from the left onto the IPv6 address type on the right.

|                                     |
|-------------------------------------|
| confined to a single link           |
| provides one-to-many communications |
| serves as the next-hop addresses    |
| cannot be used as a source address  |

|                    |
|--------------------|
| Multicast          |
|                    |
|                    |
| Link-Local Address |
|                    |
|                    |

**Answer:**

Explanation:

|                                     |
|-------------------------------------|
| Multicast                           |
| confined to a single link           |
| provides one-to-many communications |
| Link-Local Address                  |
| serves as the next-hop addresses    |
| cannot be used as a source address  |

**Question: 842**

## DRAG DROP

Drag and drop the characteristic from the left onto the IPv6 address type on the right.

|                                      |
|--------------------------------------|
| provides one-to-many communications  |
| has a unicast source sent to a group |

|                                            |
|--------------------------------------------|
| enables aggregation of routing prefixes    |
| is routable and reachable via the Internet |

|                        |
|------------------------|
| Global Unicast Address |
|                        |

|           |
|-----------|
| Multicast |
|           |

**Answer:**

Explanation:



### Global Unicast Address

enables aggregation of routing prefixes

is routable and reachable via the Internet

### Multicast

provides one-to-many communications

has a unicast source sent to a group

---

**Question: 843**

---

DRAG DROP

Drag and drop the characteristic from the left onto the IPv6 address type on the right.

|                                                        |                        |
|--------------------------------------------------------|------------------------|
| provides for one-to-one communication                  | Global Unicast Address |
| is a counterpart of private IPv4 addresses             |                        |
| is publicly routable in the same way as IPv4 addresses | Unique Local           |
| allows sites to be combined without address conflicts  |                        |

---

**Answer:**

---

Explanation:

**Global Unicast Address**

is publicly routable in the same way as IPv4 addresses

allows sites to be combined without address conflicts

**Unique Local**

provides for one-to-one communication

is a counterpart of private IPv4 addresses

**Question: 844**

DRAG DROP

Drag and drop the characteristic from the left onto the IPv6 address type on the right.

|                                            |                        |
|--------------------------------------------|------------------------|
| provides for one-to-one communication      | Global Unicast Address |
| confined to a single link                  |                        |
| serves as the next-hop addresses           | Link-Local Address     |
| is routable and reachable via the Internet |                        |

**Answer:**

Explanation:

**Global Unicast Address**

serves as the next-hop addresses

is routable and reachable via the Internet

**Link-Local Address**

provides for one-to-one communication

confined to a single link

---

**Question: 845**

DRAG DROP

Drag and drop the characteristic from the left onto the IPv6 address type on the right.

|                                                        |
|--------------------------------------------------------|
| is publicly routable in the same way as IPv4 addresses |
| serves as the next-hop addresses                       |
| required on all IPv6 devices                           |
| provides for one-to-one communication                  |

Global Unicast Address

Link-Local Address

---

**Answer:**

---

Explanation:

|                                                        |
|--------------------------------------------------------|
| Global Unicast Address                                 |
| is publicly routable in the same way as IPv4 addresses |
| serves as the next-hop addresses                       |
| Link-Local Address                                     |
| required on all IPv6 devices                           |
| provides for one-to-one communication                  |

---

**Question: 846**

---

## DRAG DROP

Drag and drop the characteristic from the left onto the IPv6 address type on the right.

|                                                               |
|---------------------------------------------------------------|
| is used exclusively by a non-host device                      |
| sends packets to a group address rather than a single address |

|                                                         |
|---------------------------------------------------------|
| has a unicast source sent to a group                    |
| is routed to the nearest interface that has the address |

|           |
|-----------|
| Multicast |
|           |

|         |
|---------|
| Anycast |
|         |

**Answer:**

Explanation:

**Multicast**

has a unicast source sent to a group

is routed to the nearest interface that has the address

**Anycast**

is used exclusively by a non-host device

sends packets to a group address rather than a single address

---

**Question: 847**

---

DRAG DROP

Drag and drop the characteristic from the left onto the IPv6 address type on the right.

|                                            |                        |
|--------------------------------------------|------------------------|
| is unable to route on the internet         | Global Unicast Address |
| is a counterpart of private IPv4 addresses |                        |
| enables aggregation of routing prefixes    | Unique Local           |
| is routable and reachable via the Internet |                        |

Answer:

Explanation:

|                                            |
|--------------------------------------------|
| Global Unicast Address                     |
| enables aggregation of routing prefixes    |
| is routable and reachable via the Internet |

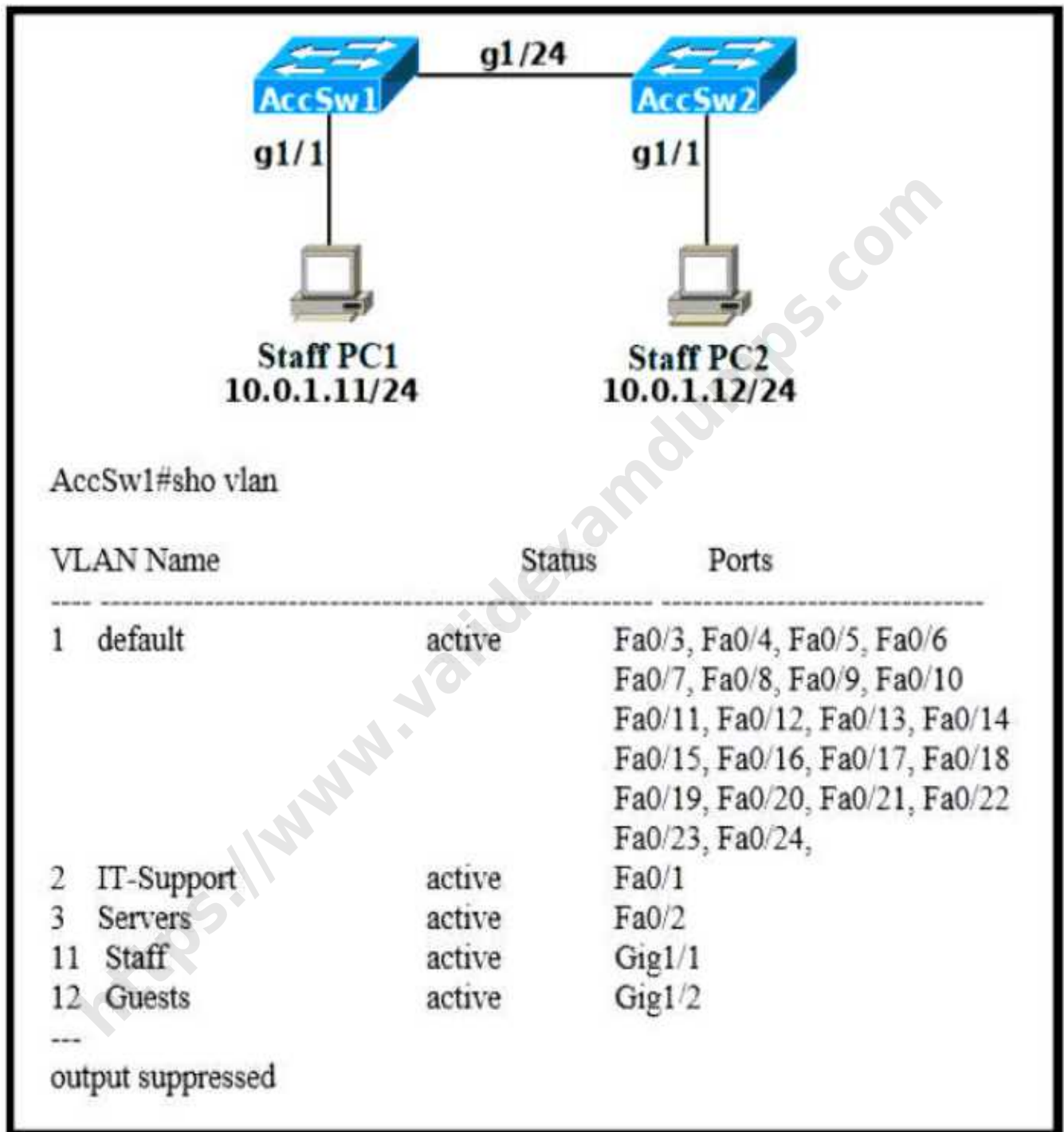
  

|                                            |
|--------------------------------------------|
| Unique Local                               |
| is unable to route on the internet         |
| is a counterpart of private IPv4 addresses |



**Question: 848**

Refer to the exhibit.



Switch AccSw2 has just been added to the network along with PC2. All VLANs have been implemented on AccSw2. How must the ports on AccSw2 be configured to establish Layer 2 connectivity between PC1 and PC2?

A.

☐ interface GigabitEthernet1/1  
switchport mode access  
switchport access vlan 11  
!  
interface GigabitEthernet1/24  
switchport mode trunk

B.

☐ interface GigabitEthernet1/2  
switchport mode access  
switchport access vlan 12  
!  
interface GigabitEthernet1/24  
switchport mode trunk  
switchport trunk allowed vlan 11,12

C.

☐ interface GigabitEthernet1/24  
switchport mode trunk  
switchport trunk allowed vlan 11,12  
!  
interface GigabitEthernet1/1  
switchport access vlan 11

D.

☐ interface GigabitEthernet1/2  
switchport mode access  
switchport access vlan 2  
!  
interface GigabitEthernet1/24  
switchport mode trunk

A. Option A

- B. Option B
- C. Option C
- D. Option D

---

**Answer: A**

---

Explanation:

---

**Question: 849**

---

Which device separates networks by security domains?

- A. firewall
- B. access point
- C. intrusion protection system
- D. wireless controller

---

**Answer: A**

---

Explanation:

Firewalls are devices that are used to separate networks into different security domains. They act as a barrier between two networks and control the flow of traffic between them. Firewalls use a set of rules to determine what types of traffic are allowed to pass through and what is blocked. This helps protect a network from malicious traffic and unauthorized access. Additionally, firewalls can be configured to log traffic and provide additional security measures such as packet filtering and stateful inspection.

---

**Question: 850**

---

DRAG DROP

Drag and drop the characteristic from the left onto the IPv6 address type on the right.

|                                                               |                        |
|---------------------------------------------------------------|------------------------|
| enables aggregation of routing prefixes                       | Global Unicast Address |
| provides for one-to-one communication                         |                        |
| provides one-to-many communications                           | Multicast              |
| sends packets to a group address rather than a single address |                        |

**Answer:**

Explanation:

|                                         |
|-----------------------------------------|
| Global Unicast Address                  |
| enables aggregation of routing prefixes |
| provides for one-to-one communication   |

|                                                               |
|---------------------------------------------------------------|
| Multicast                                                     |
| provides one-to-many communications                           |
| sends packets to a group address rather than a single address |

---

**Question: 851**

---

DRAG DROP

Drag and drop the characteristic from the left onto the IPv6 address type on the right.

|                                                               |
|---------------------------------------------------------------|
| cannot be used as a source address                            |
| sends packets to a group address rather than a single address |

|                              |
|------------------------------|
| confined to a single link    |
| required on all IPv6 devices |

|           |
|-----------|
| Multicast |
|           |

|                    |
|--------------------|
| Link-Local Address |
|                    |

---

**Answer:**

---

Explanation:

**Multicast**

cannot be used as a source address

sends packets to a group address rather than a single address

**Link-Local Address**

confined to a single link

required on all IPv6 devices

---

**Question: 852**

---

DRAG DROP

Drag and drop the characteristic from the left onto the IPv6 address type on the right.

|                              |
|------------------------------|
| confined to a single link    |
| required on all IPv6 devices |

|                        |
|------------------------|
| Global Unicast Address |
|                        |

|                                                        |
|--------------------------------------------------------|
| is publicly routable in the same way as IPv4 addresses |
| provides for one-to-one communication                  |

|                    |
|--------------------|
| Link-Local Address |
|                    |

Answer:

Explanation:

|                                                        |
|--------------------------------------------------------|
| Global Unicast Address                                 |
| is publicly routable in the same way as IPv4 addresses |
| provides for one-to-one communication                  |

|                              |
|------------------------------|
| Link-Local Address           |
| confined to a single link    |
| required on all IPv6 devices |



**Question: 853**

Refer to the exhibit.

```
R2#show ip ospf interface
GigabitEthernet0/0/0 is up, line protocol is up
Internet address is 192.168.1.1/24, Area 0
Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DROTHER, Priority 1
Designated Router (ID) 192.168.1.1, Interface address 192.168.1.2
Backup Designated Router (ID) 192.168.1.1, Interface address 192.168.1.2
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:02
Index 2/2, flood queue length 0
Neighbor Count is 1, Adjacent neighbor count is 2
```

Router OldR is replacing another router on the network with the intention of having OldR and R2 exchange routes\_ After the engineer applied the initial OSPF

configuration: the routes were still missing on both devices. Which command sequence must be issued before the clear IP ospf process command is entered to enable the neighbor relationship?

- ☐ OldR(config)#interface g0/0/0  
OldR(config-if)#ip ospf dead-interval 15
- ☐ OldR(config)#router ospf 1  
OldR(config-router)#no router-id 192.168.1.1
- ☒ OldR(config)#router ospf 1  
OldR(config-router)#network 192.168.1.0 255.255.255.0 area 2
- ☐ OldR(config)#interface g0/0/0  
OldR(config-if)#ip ospf hello-interval 15

- A. Option A
- B. Option B
- C. Option C



D. Option D

---

**Answer: C**

---

Explanation:

---

**Question: 854**

---

What is a reason to implement LAG on a Cisco WLC?

- A. Increase the available throughput on the link.
- B. Increase security by encrypting management frames
- C. Allow for stateful failover between WLCs
- D. Enable the connected switch ports to use different Layer 2 configurations

---

**Answer: A**

---

Explanation:

Link Aggregation Group (LAG) is a feature that allows you to bundle multiple physical Ethernet links into a single logical link, and is used to increase the available throughput on the link. LAG is supported on the Cisco Wireless LAN Controller (WLC) and the connected switch ports [1], and can be used to provide greater bandwidth and increased redundancy. It also enables the connected switch ports to use different Layer 2 configurations, such as Spanning Tree Protocol (STP) and Hot Standby Router Protocol (HSRP).

---

**Question: 855**

---

DRAG DROP

Drag and drop the characteristic from the left onto the IPv6 address type on the right.

|                                                        |
|--------------------------------------------------------|
| provides for one-to-one communication                  |
| allows sites to be combined without address conflicts  |
| is a counterpart of private IPv4 addresses             |
| is publicly routable in the same way as IPv4 addresses |

|                        |
|------------------------|
| Global Unicast Address |
|                        |
| Unique Local           |
|                        |

Answer:

Explanation:

|                                                        |
|--------------------------------------------------------|
| Global Unicast Address                                 |
| provides for one-to-one communication                  |
| allows sites to be combined without address conflicts  |
| Unique Local                                           |
| is a counterpart of private IPv4 addresses             |
| is publicly routable in the same way as IPv4 addresses |

---

**Question: 856**

---

What differentiates the Cisco OfficeExtend AP mode from FlexConnect AP mode?

- A. FlexConnect allows a personal SSID to be configured on the AP, and personal SSIDs are not supported with OfficeExtend.
- B. OfficeExtend does not support DTLS tunneling of traffic to the WLC, and FlexConnect tunnels traffic to the WLC with DTLS.
- C. OfficeExtend tunnels all traffic through the WLC, and FlexConnect terminates client traffic at the AP switch port.
- D. FlexConnect must be deployed behind a router that NATs the client traffic, and OfficeExtend uses public IP sources.

---

**Answer: C**

---

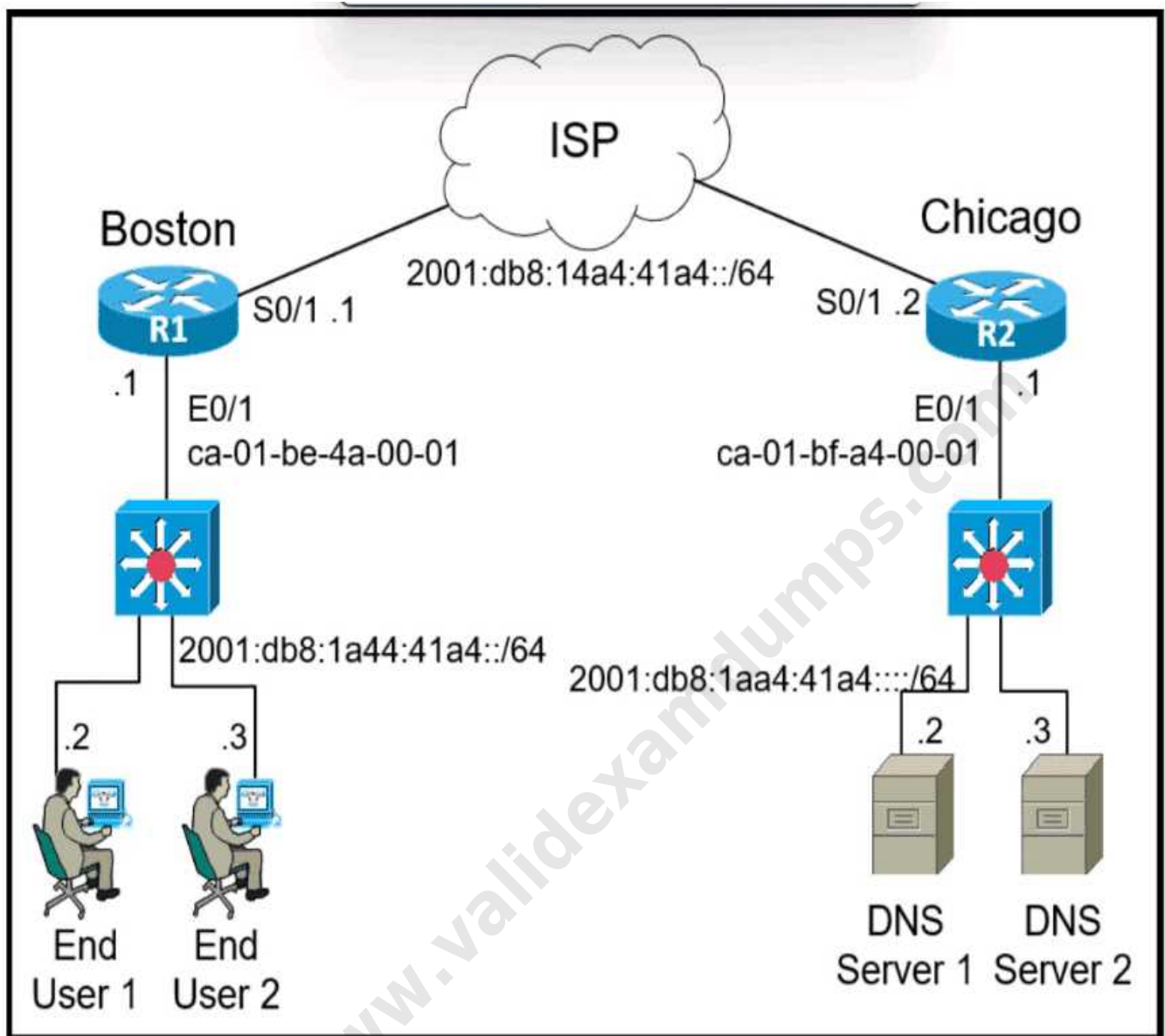
Explanation:

---

**Question: 857**

---

Refer to the exhibit.



Refer to the exhibit. The IPv6 address for the LAN segment on router R1 must be configured using the EUI-64 format. When configured which ipv6 address is produced by the router?

- A. 2001:db8:1a44:41a4:C801:BEFF:FE4A:1
- B. 2001:db8:1a44:41a4:C081:BFFF:FE4A:1
- C. 2001:db8:1a44:41a4:4562:098F:FE36:1
- D. 2001:db8:1a44:41a4:C800:BAFE:FF00:1

**Answer: B**

Explanation:

---

**Question: 858**

---

Refer to the exhibit.

The screenshot shows the Cisco WLC configuration page, Advanced tab. The 'FlexConnect' section is expanded, showing 'FlexConnect Local Switching' enabled. The 'Local Client Profiling' section is also expanded, showing 'HTTP Profiling' enabled. The 'BSS Transition' section is expanded, showing 'BSS Transition' enabled.

An architect is managing a wireless network with APs from several branch offices connecting to the WLC in the data center. There is a new requirement for a single WLAN to process the client data traffic without sending it to the WLC. Which action must be taken to complete the request?

- A. Enable local HTTP profiling.
- B. Enable Disassociation Imminent.
- C. Enable FlexConnect Local Switching.
- D. Enable local DHCP Profiling.

---

**Answer: C**

---

Explanation:

---

**Question: 859**

---

What is the role of SNMP in the network?

- A. to monitor network devices and functions using a TCP underlay that operates on the presentation layer

- B. to collect data directly from network devices using an SSL underlay that operates on the transport layer
- C. to monitor and manage network devices using a UDP underlay that operates on the application layer
- D. to collect telemetry and critical information from network devices using an SSH underlay that operates on the network layer

---

**Answer: C**

---

Explanation:

---

**Question: 860**

---

What are two purposes of HSRP? (Choose two.)

- A. It groups two or more routers to operate as one virtual router.
- B. It improves network availability by providing redundant gateways.
- C. It passes configuration information to hosts in a TCP/IP network.
- D. It helps hosts on the network to reach remote subnets without a default gateway.
- E. It provides a mechanism for diskless clients to autoconfigure their IP parameters during boot.

---

**Answer: A,B**

---

Explanation:

---

**Question: 861**

---

DRAG DROP

```
R1# show ip route | begin gateway
Gateway of last resort is not set
172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
172.16.1.0/24 is directly connected, FastEthernet0/0
172.16.1.1/32 is directly connected, FastEthernet0/0
172.16.2.0/24 [120/2] via 207.165.200.250, 00:00:25, Serial0/0/0
192.168.1.0/24 [110/84437] via 207.165.200.254, 00:00:17, Serial0/0/1
192.168.2.0/24 [90/3184437] via 207.165.200.254, 00:00:15, Serial0/0/1
207.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
207.165.200.244/30 [1/1] via 207.165.200.254, Serial0/0/1
207.165.200.248/30 is directly connected, Serial0/0/0
207.165.200.249/32 is directly connected, Serial0/0/0
207.165.200.252/30 is directly connected, Serial0/0/1
207.165.200.253/32 is directly connected, Serial0/0/1
```

Refer to the exhibit. Drag and drop the learned prefixes from the left onto the preferred route methods from which they were learned on the right.

|                    |        |
|--------------------|--------|
| 172.16.2.0/24      | static |
| 192.168.1.0/24     | EIGRP  |
| 192.168.2.0/24     | OSPF   |
| 207.165.200.244/30 | RIP    |
| 207.165.200.248/30 |        |

**Answer:**

Explanation:

192.168.1.0/24

172.16.2.0/24

192.168.2.0/24

207.165.200.244/30

---

**Question: 862**

---

Refer to the exhibit.

```
SW1#show ip interface brief
Interface                IP-Address      OK? Method Status  Protocol
FastEthernet0/1          unassigned      YES manual  down    down

SW1#show interface fa0/1 status
Port      Name      Status      Vlan    Duplex  Speed  Type
Fa0/1     Fa0/1     notconnect  1       a-full  a-100  10/100BaseTX
```

Refer to the exhibit. What is the cause of the issue?

- A. STP
- B. port security
- C. wrong cable type
- D. shutdown command

---

**Answer: C**

---

Explanation:



---

**Question: 863**

---

DRAG DROP

Drag and drop the AAA terms from the left onto the descriptions on the right.

|                |                            |
|----------------|----------------------------|
| accounting     | tracks activity            |
| authentication | updates session attributes |
| authorization  | verifies access rights     |
| CoA            | verifies identity          |

---

**Answer:**

---

Explanation:

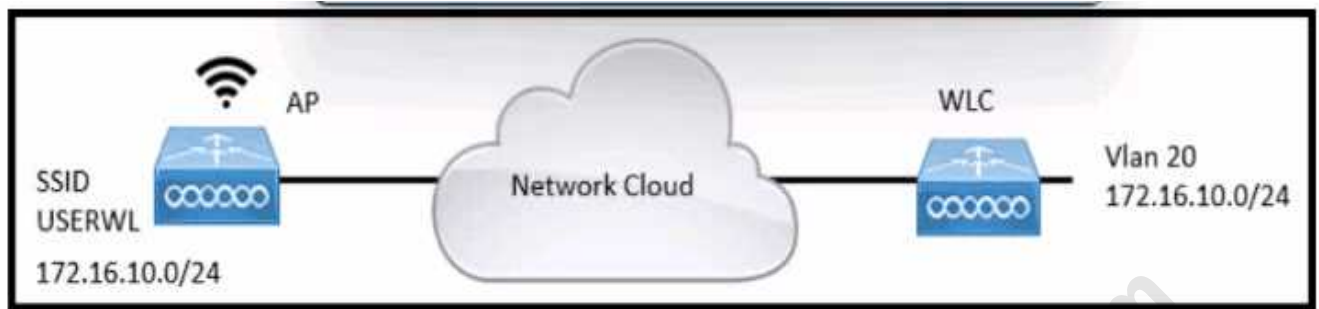
|                |
|----------------|
| accounting     |
| CoA            |
| authorization  |
| authentication |

---

**Question: 864**

---

Refer to the exhibit.



A network engineer is configuring a WLAN to connect with the 172.16.10.0/24 network on VLAN 20. The engineer wants to limit the number of devices that connect to the WLAN on the USERWL SSID to 125. Which configuration must the engineer perform on the WLC?

- A. In the Management Software activation configuration, set the Clients value to 125.
- B. In the Controller IPv6 configuration, set the Throttle value to 125.
- C. In the WLAN configuration, set the Maximum Allowed Clients value to 125.
- D. In the Advanced configuration, set the DTIM value to 125.

---

**Answer: C**

---

Explanation:

---

### Question: 865

---

Which interface or port on the WLC is the default for in-band device administration and communications between the controller and access points?

- A. virtual interface
- B. management interface
- C. console port
- D. service port

---

**Answer: B**

---

Explanation:

---

**Question: 866**

---

Which function generally performed by a traditional network device is replaced by a software-defined controller?

- A. encryption and decryption for VPN link processing
- B. building route tables and updating the forwarding table
- C. changing the source or destination address during NAT operations
- D. encapsulation and decapsulation of packets in a data-link frame

---

**Answer: B**

---

Explanation:

---

**Question: 867**

---

Refer to the exhibit.

```
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
```

```
10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
```

```
S    10.0.0.0/8 is directly connected, GigabitEthernet0/0
```

```
C    10.1.1.0/24 is directly connected, GigabitEthernet0/0
```

```
L    10.1.1.1/32 is directly connected, GigabitEthernet0/0
```

```
S    10.10.0.0/22 is directly connected, GigabitEthernet0/0
```

```
S    10.10.10.0/28 is directly connected, GigabitEthernet0/0
```

```
S    10.10.10.1/32 is directly connected, GigabitEthernet0/0
```

```
S*   0.0.0.0/0 is directly connected, GigabitEthernet0/0
```

Which IP route command created the best path for a packet destined for 10.10.10.3?

- A. ip route 10.10.0.0 255.255.252.0 g0/0
- B. ip route 10.0.0.0 255.0.0.0 g0/0
- C. ip route 10.10.10.1 255.255.255.255 g0/0

D. ip route 10.10.10.0 255.255.255.240 g0/0

---

**Answer: D**

---

Explanation:

---

**Question: 868**

---

Refer to the exhibit.

```
{  
  "Test_Questions" : [  
    "Automation",  
    "Configuration",  
  ],  
  "Test_Exam_Level" : [  
    "CCNA",  
    "CCNP",  
  ],  
  "Test_Response" : [  
    "Correct",  
    "Incorrect",  
  ],  
}
```

How many arrays are present in the JSON data?

- A. one
- B. three
- C. six
- D. nine

---

**Answer: C**

---

Explanation:

---

**Question: 869**

---

A network engineer must configure an access list on a new Cisco IOS router. The access list must deny HTTP traffic to network 10.125.128.32/27 from the 192.168.240.0/20 network, but it must allow the 192.168.240.0/20 network to reach the rest of the 10.0.0.0/8 network. Which configuration must the engineer apply?

A)

```
ip access-list extended deny_outbound
 10 deny tcp 10.125.128.32 255.255.255.224 192.168.240.0 255.255.240.0 eq 443
 20 deny tcp 192.168.240.0 255.255.240.0 10.125.128.32 255.255.255.224 eq 443
 30 permit ip 192.168.240.0 255.255.240.0 10.0.0.0 255.0.0.0
```

B)

```
ip access-list extended deny_outbound
 10 permit ip 192.168.240.0 255.255.240.0 10.0.0.0 255.0.0.0
 20 deny tcp 192.168.240.0 255.255.240.0 10.125.128.32 255.255.255.224 eq 443
 30 permit ip any any
```

C)

```
ip access-list extended deny_outbound
 10 deny tcp 192.168.240.0 0.0.15.255 10.125.128.32 0.0.0.31 eq 80
 20 permit ip 192.168.240.0 0.0.15.255 10.0.0.0 0.255.255.255
 30 deny ip any any log
```

D)

```
ip access-list extended deny_outbound
10 deny tcp 192.168.240.0 0.0.15.255 any eq 80
20 deny tcp 192.168.240.0 0.0.15.255 10.125.128.32 0.0.0.31 eq 80
30 permit ip 192.168.240.0 0.0.15.255 10.0.0.0 0.255.255.255
```

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: C**

---

Explanation:

---

**Question: 870**

---

What describes the functionality of southbound APIs?

- A. They use HTTP messages to communicate.
- B. They enable communication between the controller and the network device.
- C. They convey information from the controller to the SDN applications.
- D. They communicate with the management plane.

---

**Answer: B**

---

Explanation:

---

**Question: 871**

---

Which type of IPv4 address must be assigned to a server to protect it from external access and allow only internal users access while restricting internet access?

- A. global unicast
- B. public
- C. private
- D. multicast

---

**Answer: C**

---

Explanation:

---

**Question: 872**

---

A DHCP pool has been created with the name NOCC. The pool is using 192.168.20.0/24 and must use the next to last usable IP address as the default gateway for the DHCP clients. What is the next step in the process?

- A. default-router 192.168.20.253
- B. network 192.168.20.254 255.255.255.0 secondary
- C. ip default-gateway 0.0.0.0 0.0.0.0 192.168.20.253
- D. next-server 192.168.20.254

---

**Answer: A**

---

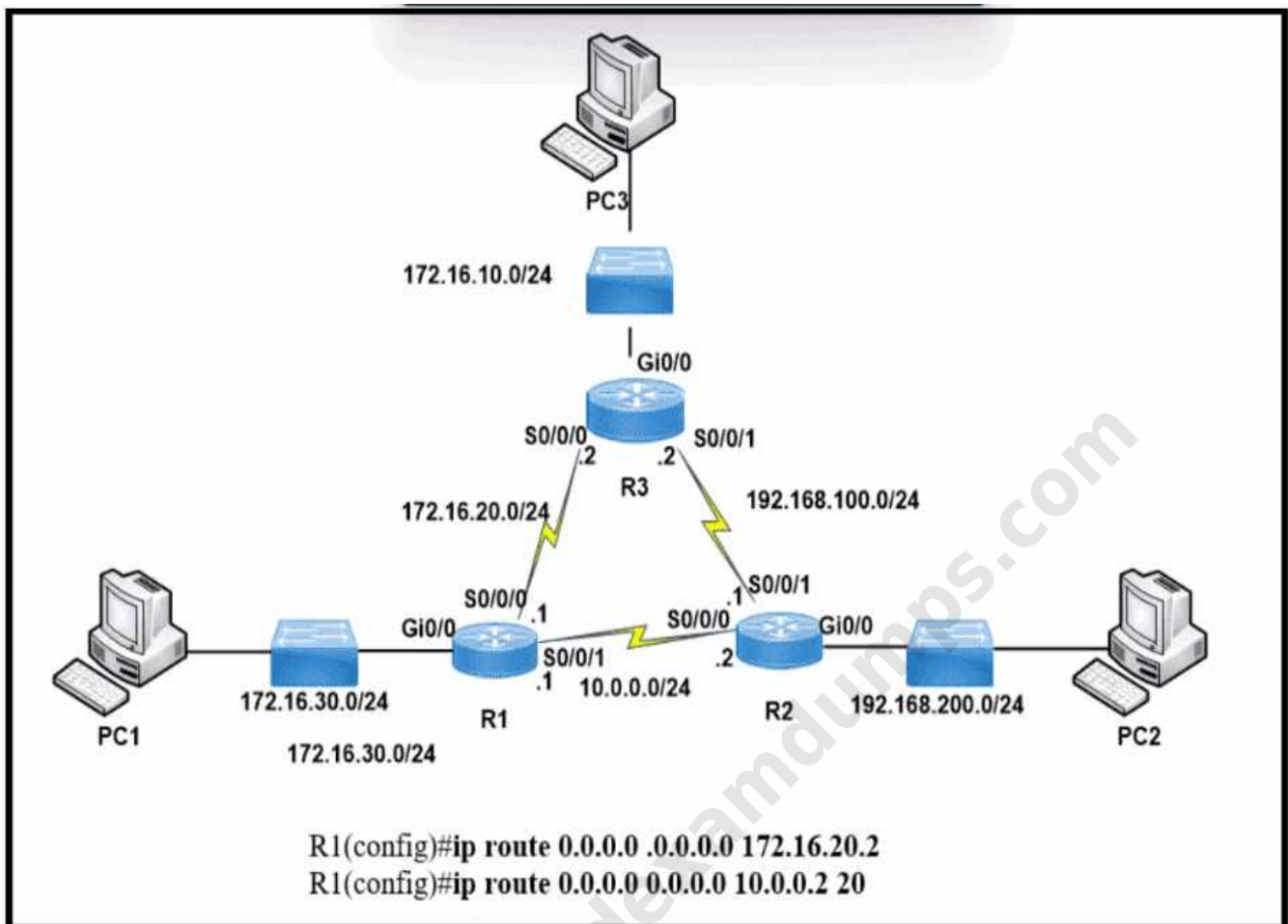
Explanation:

---

**Question: 873**

---

Refer to the exhibit.



Refer to the exhibit. After applying this configuration to router R1, a network engineer is verifying the implementation. If all links are operating normally, and the engineer sends a series of packets from PC1 to PC3, how are the packets routed?

- A. They are routed to 172.16.20.2.
- B. They are routed to 192.168.100.2.
- C. They are distributed sent round robin to interfaces S0/0/0 and S0/0/1.
- D. They are routed to 10.0.0.2.

**Answer: A**

Explanation:

#### Question: 874

Which Windows command is used instead of the route print command to display the contents of the



IP routing table?

- A. netstat-n
- B. ipconfig
- C. ifconfig
- D. netstat-r

---

**Answer: D**

---

Explanation:

---

**Question: 875**

---

Refer to the exhibit.

```
R1# show ip route | begin Gateway
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
S* 0.0.0.0/0 is directly connected, Serial0/0/1
    172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C      172.16.2.0/24 is directly connected, GigabitEthernet0/0
L      172.16.2.2/32 is directly connected, GigabitEthernet0/0
C      172.16.4.0/21 is directly connected, Serial0/0/1
L      172.16.8.2/26 is directly connected, Serial0/0/1
```

What is the subnet mask for route 172.16.4.0?

- A. 255.255.248.0
- B. 255.255.254.0
- C. 255.255.255.192
- D. 255.255.240.0

---

**Answer: A**

---

Explanation:

---

**Question: 876**

---

Refer to the exhibit.

```
{
  "myCar": {
    "name": "thunder",
    "wheels": ["good", "good", "pressureLow", "warning"],
    "gasLight": false
  },
  "oldCar": {
    "name": "sleepy",
    "wheels": ["pressureLow", "pressureLow", "pressureLow", "pressureLow"],
    "color": "rust"
    "gasLight": true
  },
  "newCar": {
    "name": "lightning",
    "wheels": ["pressureLow", "good", "pressureLow", "good"],
    "color": "blue"
    "gasLight": true
  }
}
```

In which structure does the word "warning" directly reside?

- A. array
- B. object
- C. Boolean
- D. string

---

**Answer: A**

---

Explanation:

---

**Question: 877**

---

Which components are contained within a virtual machine?

- A. physical resources, including the NIC, RAM, disk, and CPU

- B. configuration files backed by physical resources from the Hypervisor
- C. applications running on the Hypervisor
- D. processes running on the Hypervisor and a guest OS

---

**Answer: B**

---

Explanation:

A virtual machine is a software-based computer that runs on a physical host computer or a remote server. It has its own operating system, applications, and virtual hardware devices that are configured by the user. The virtual hardware devices are backed by the physical resources of the Hypervisor, which is a software layer that manages the virtual machines and allocates the CPU, memory, network, and storage resources to them. The configuration files of a virtual machine contain information such as the name, UUID, BIOS settings, hardware settings, and resource settings of the virtual machine. These files are stored on a datastore, which is a logical container for files and virtual disks<sup>12</sup>.

Reference:

- 1: VMware vSphere 7.0 Documentation - Virtual Machine Configuration Files
- 2: Cisco CCNA Certification Guide - Chapter 10: Virtualization Fundamentals

---

**Question: 878**

---

Refer to the exhibit.

```
R_1# show ip route
....
D   192.168.20.0/26 [90/24513456] via 10.10.10.1
R   192.168.20.0/24 [120/5] via 10.10.10.2
O   192.168.0.0/19 [110/219414] via 10.10.10.13
B   192.168.0.0/16 is variably subnetted, 4 subnets, 4 masks
D   192.168.20.0/27 [90/4123710] via 10.10.10.12
D   192.168.20.0/25 [90/14464211] via 10.10.10.11
S.   0.0.0.0/0 [1/0] via 10.10.10.14
```

Packets are flowing from 192.168.10.1 to the destination at IP address 192.168.20.75. Which next hop will the router select for the packet?

- A. 10.10101
- B. 10.10.10.11
- C. 10.10.10.12
- D. 10.101014

---

**Answer: B**

---

Explanation:

The router will select the next hop based on the longest prefix match in the routing table. The destination IP address 192.168.20.75 belongs to the network 192.168.0.0/19, which is a classless network created by subnetting the classful network 192.168.0.0/16. The routing table has two entries for the network 192.168.0.0/19, one with a metric of 219414 and another with a metric of 5. The router will choose the entry with the lower metric, which is 5, and forward the packet to the next hop 10.10.10.11.

---

**Question: 879**

---

Which interface IP address serves as the tunnel source for CAPWAP packets from the WLC to an AP?

- A. service
- B. trunk
- C. AP-manager
- D. virtual AP connection

---

**Answer: C**

---

Explanation:

The AP-manager interface is used by the WLC to communicate with access points using CAPWAP packets. The AP-manager interface has an IP address that serves as the tunnel source for CAPWAP

packets from the WLC to an AP3. The service interface is used for out-of-band management of the WLC, such as Telnet and SSH4. The trunk interface is used to connect the WLC to a switch and carry multiple VLANs5. The virtual AP connection is not an interface, but a logical connection between an AP and a WLC that allows multiple SSIDs to be supported by a single AP6.

---

**Question: 880**

---

Which IPsec encryption mode is appropriate when the destination of a packet differs from the security termination point?

- A. tunnel
- B. transport
- C. aggressive
- D. main

---

**Answer: B**

---

Explanation:

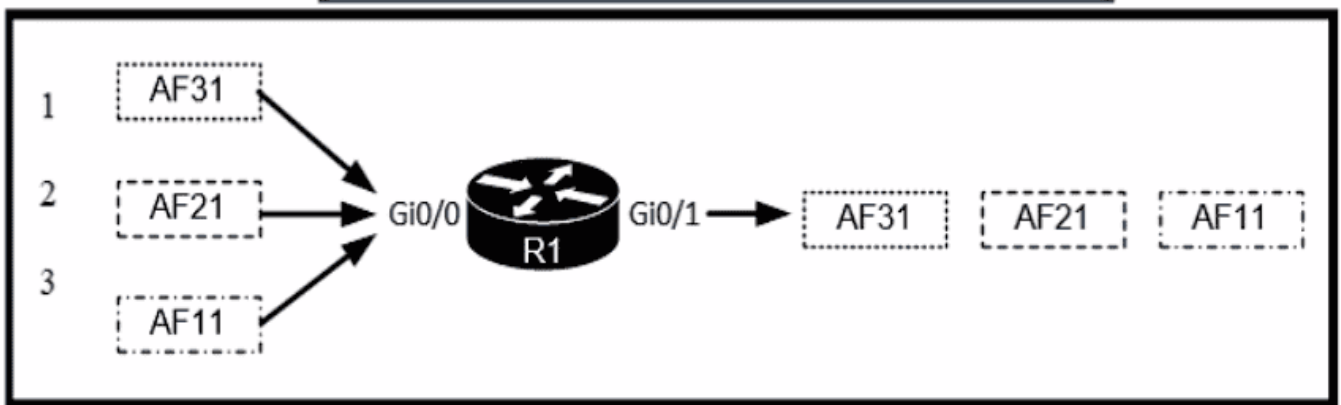
IPsec encryption mode is the way IPsec secures the data packets that are sent over an IP network. There are two main modes of IPsec encryption: tunnel mode and transport mode1. Tunnel mode encrypts the entire IP packet, including the original header, and adds a new IP header with the source and destination addresses of the security gateways (routers, firewalls, or VPN servers) that perform the encryption and decryption2. Transport mode encrypts only the payload (data) of the IP packet, leaving the original header intact, and uses the original source and destination addresses of the endpoints that generate and consume the data3. Therefore, transport mode is appropriate when the destination of a packet differs from the security termination point, as it does not change the original IP header information. Tunnel mode is more suitable when the security termination point is also the destination of the packet, as it provides more protection for the original IP header information.

---

**Question: 881**

---

Refer to the exhibit.



Which per-hop QoS behavior is R1 applying to incoming packets?

- A. queuing
- B. marking
- C. shaping
- D. policing

---

**Answer: D**

---

Explanation:

R1 is applying policing to incoming packets. Policing is a QoS mechanism that limits the rate of traffic flow by dropping or remarking packets that exceed the configured rate limit. In this case, R1 is applying policing to incoming packets on interface G0/0 and G0/1. The exhibit shows that R1 is configured to police traffic at a rate of AF31, AF21, and AF11. This means that R1 is limiting the rate of traffic flow for these three traffic classes .

Cisco CCNA Certification Guide - Chapter 16: Quality of Service (QoS)

Cisco IOS Quality of Service Solutions Configuration Guide - Configuring Class-Based Policing

---

### Question: 882

---

What is the role of the root port in a switched network?

- A. It replaces the designated port when the designated port fails

- B. It is the best path to the root from a nonroot switch.
- C. It replaces the designated port when the root port fails.
- D. It is administratively disabled until a failover occurs.

---

**Answer: B**

---

Explanation:

The root port is the port on a nonroot switch that has the lowest cost to reach the root bridge in a switched network. The root bridge is the switch that is elected as the central point of the network by the spanning tree protocol (STP). The root port is responsible for forwarding traffic to the root bridge, and it is always in a forwarding state. The cost of a port is determined by the speed and bandwidth of the link, and the lower the cost, the better the path. If there are multiple paths to the root bridge with the same cost, the switch will choose the one with the lowest bridge ID as the tiebreaker. The root port does not replace or replace any designated port, which is the port that forwards traffic on a network segment. The root port is also not administratively disabled or enabled, but it is dynamically selected by STP1234.

Reference:

- 1: Solved: STP: Root port vs. designated port - Cisco Community
- 2: Difference between Root Port and Designated Port - GeeksforGeeks
- 3: What is a Root Port - Snabay Networking
- 4: What is a Root Port - omniseu.com

---

**Question: 883**

---

What is the temporary state that switch ports always enter immediately after the boot process when Rapid PVST+ is used?

- A. discarding
- B. listening
- C. forwarding
- D. learning

---

**Answer: A**

---

Explanation:

When Rapid PVST+ is used, switch ports always enter the blocking state immediately after the boot process 1. The blocking state is the first of the five possible port states in the Rapid PVST+ protocol. In this state, the port does not forward frames but listens to BPDUs to determine the location of the root bridge and the best path to reach it 12. After a port enters the blocking state, it transitions to the listening state and then to the learning state before finally entering the forwarding state. When Rapid PVST+ (Per-VLAN Spanning Tree Plus) is used, switch ports go through a specific state immediately after the boot process. This state is known as the "discarding" state.

---

**Question: 884**

---

What does a switch search for in the CAM table when forwarding a frame?

- A. source MAC address and aging time
- B. destination MAC address and flush time
- C. source MAC address and source port
- D. destination MAC address and destination port

---

**Answer: D**

---

Explanation:

A switch searches for the destination MAC address and the destination port in the CAM table when forwarding a frame. The CAM table, or content addressable memory table, is a data structure that stores the MAC addresses of the devices connected to the switch ports and their associated VLANs. The switch uses the CAM table to make layer 2 forwarding decisions based on the destination MAC address of a frame. When a frame arrives at a switch port, the switch first learns the source MAC address and the source port of the frame and updates the CAM table accordingly. Then, the switch looks up the destination MAC address of the frame in the CAM table and finds the corresponding destination port. If there is a match, the switch forwards the frame out of that port only. If there is no match, the switch floods the frame out of all ports except the source port 123.



Reference:

1: Why is the CAM table in a switch called CAM table and not MAC table even though it holds MAC addresses?

2: ARP and CAM Table

3: The CAM Table or MAC address Table

---

**Question: 885**

---

What are two capabilities provided by VRRP within a LAN network? (Choose two.)

- A. dynamic routing updates
- B. bandwidth optimization
- C. granular QoS
- D. load sharing
- E. redundancy

---

**Answer: A,C**

---

Explanation:

Redundancy— VRRP enables you to configure multiple routers as the default gateway router, which reduces the possibility of a single point of failure in a network.

Load Sharing—You can configure VRRP in such a way that traffic to and from LAN clients can be shared by multiple routers, thereby sharing the traffic load more equitably among available router

---

**Question: 886**

---

A wireless access point is needed and must meet these requirements:

- "zero-touch" deployed and managed by a WLC

- process only real-time MAC functionality
- used in a split-MAC architecture. Which access point type must be used?

- A. autonomous
- B. lightweight
- C. mesh
- D. cloud-based

---

**Answer: B**

---

Explanation:

<https://www.cisco.com/c/en/us/support/docs/wireless/aironet-1200-series/70278-lap-faq.html>

---

**Question: 887**

---

What is used to identify spurious DHCP servers?

- A. DHCPREQUEST
- B. DHCPDISCOVER
- C. DHCPACK
- D. DHCPOFFER

---

**Answer: D**

---

Explanation:

DHCPOFFER is used to identify spurious DHCP servers. A spurious DHCP server is any device that is configured to act as a DHCP server without the network administrator's knowledge or permission. A spurious DHCP server can cause network problems by assigning incorrect or duplicate IP addresses to clients, or by redirecting traffic to malicious gateways. To prevent such attacks, the DHCP snooping feature can be enabled on switches to filter out invalid or unauthorized DHCP messages from untrusted sources<sup>1</sup>.

DHCP snooping works by intercepting and validating DHCP messages on a per-VLAN basis. The switch maintains a DHCP snooping binding database that contains information about the trusted hosts with leased IP addresses, such as MAC address, IP address, lease time, binding type, VLAN number, and interface information<sup>2</sup>. The switch also classifies its ports as trusted or untrusted. Trusted ports are those that connect to authorized DHCP servers or other trusted switches. Untrusted ports are those that connect to untrusted hosts or devices. The switch only allows DHCP messages from trusted ports, and drops any DHCP messages from untrusted ports that do not match the information in the binding database<sup>3</sup>.

The switch uses DHCP OFFER messages to identify spurious DHCP servers. A DHCP OFFER message is a response from a DHCP server to a client's request for an IP address. The message contains the offered IP address, subnet mask, default gateway, and other configuration parameters for the client<sup>4</sup>. When the switch receives a DHCP OFFER message from an untrusted port, it compares the source MAC address and the offered IP address with the binding database. If there is no match, the switch considers the message as coming from a spurious DHCP server and drops it. The switch also logs an error message and increments a counter for the number of dropped messages<sup>5</sup>.

Reference:

1: Configuring DHCP Snooping - Cisco

2: Catalyst 6500 Release 12.2SX Software Configuration Guide - DHCP Snooping Binding Database

3: What is DHCP Snooping? - IONOS

4: Dynamic Host Configuration Protocol (DHCP) and Bootstrap Protocol (BOOTP) Parameters

5: Configuring DHCP Snooping - Cisco

---

**Question: 888**

---

What happens when a switch receives a frame with a destination MAC address that recently aged out?

- A. The switch references the MAC address aging table for historical addresses on the port that received the frame.
- B. The switch floods the frame to all ports in all VLANs except the port that received the frame
- C. The switch drops the frame and learns the destination MAC address again from the port that received the frame
- D. The switch floods the frame to all ports in the VLAN except the port that received the frame.

---

**Answer: D**

---

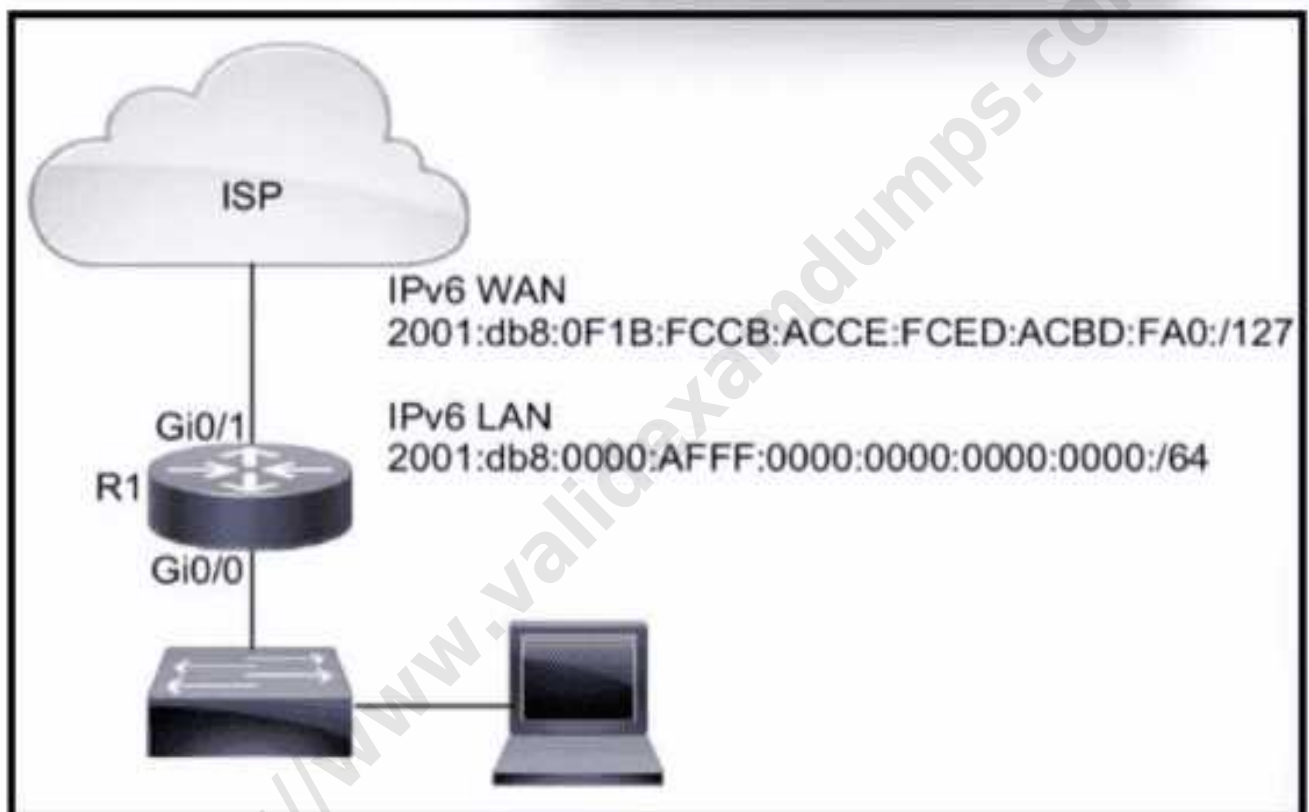
Explanation:

---

**Question: 889**

---

Refer to the exhibit.



Refer to the exhibit. IPv6 must be implemented on R1 to the ISP. The uplink between R1 and the ISP must be configured with a manual assignment, and the LAN interface must be self-provisioned. Both connections must use the applicable IPv6 networks. Which two configurations must be applied to R1? (Choose two.)

- A. interface Gi0/1 ipv6 address 2001:db8:0F1B:FCCB:ACCE:FCED:ABCD:FA02:/127
- B. interface Gi0/0 ipv6 address 2001:db8:1:AFFF::/64 eui-64
- C. interface Gi0/1 ipv6 address 2001:db8:0F1B:FCCB:ACCE:FCED:ABCD:FA00:/127
- D. interface Gi0/0 ipv6 address 2001:db8:0:AFFF::/64 eui-64
- E. interface Gi0/0 ipv6 address 2001:db8:0F1B:FCCB:ACCE:FCED:ABCD:FA03:/127

---

**Answer: C,D**

---

Explanation:

---

**Question: 890**

---

In which circumstance would a network architect decide to implement a global unicast subnet instead of a unique local unicast subnet?

- A. when the subnet must be available only within an organization
- B. when the subnet does not need to be routable
- C. when the addresses on the subnet must be equivalent to private IPv4 addresses
- D. when the subnet must be routable over the internet

---

**Answer: D**

---

Explanation:

---

**Question: 891**

---

Which type of hypervisor operates without an underlying OS to host virtual machines?

- A. Type 1
- B. Type 2
- C. Type 3
- D. Type 12

---

**Answer: A**

---

Explanation:

---

**Question: 892**

---

Refer to the exhibit.

```
Connection-specific DNS Suffix . . :  
Description . . . . . : Intel(R) Ethernet Connection (2) I218-V  
Physical Address. . . . . : D0-50-99-47-A9-7F  
DHCP Enabled. . . . . : Yes  
Autoconfiguration Enabled . . . . : Yes  
Link-local IPv6 Address . . . . . : fe80::8809:9772:c583:6b18%15 (Preferred)  
IPv4 Address. . . . . : 192.168.69.132 (Preferred)  
Subnet Mask . . . . . : 255.255.255.0  
Lease Obtained. . . . . : Thursday, January 21, 2021 11:10:46 PM  
Lease Expires . . . . . : Wednesday, February 3, 2021 11:27:29 AM  
Default Gateway . . . . . : 192.168.69.1  
DHCP Server . . . . . : 192.168.69.1  
DHCPv6 IAID . . . . . : 231755929  
DHCPv6 Client DUID. . . . . : 00-01-00-01-26-D7-BB-3F-D0-50-99-47-A9-7F  
DNS Servers . . . . . : 192.168.69.1  
NetBIOS over Tcpip. . . . . : Enabled
```

What does the host do when using the IPv4 Preferred function?

- A. It continues to use a statically assigned IPv4 address
- B. It forces the DNS server to provide the same IPv4 address at each renewal.
- C. It requests the same IPv4 address when it renews its lease with the DHCP server.
- D. It prefers a pool of addresses when renewing the IPv4 host IP address

**Answer: C**

Explanation:

**Question: 893**

Which type of encryption does WPA1 use for data protection?

- A. AES
- B. TKIP
- C. PEAP

D. EAP

---

**Answer: B**

---

Explanation:

---

**Question: 894**

---

Which two capabilities of Cisco DNA Center make it more extensible as compared to traditional campus device management? (Choose two.)

- A. REST APIs that allow for external applications to interact natively
- B. adapters that support all families of Cisco IOS software
- C. SDKs that support interaction with third-party network equipment
- D. customized versions for small, medium, and large enterprises
- E. modular design that is upgradable as needed

---

**Answer: A,C**

---

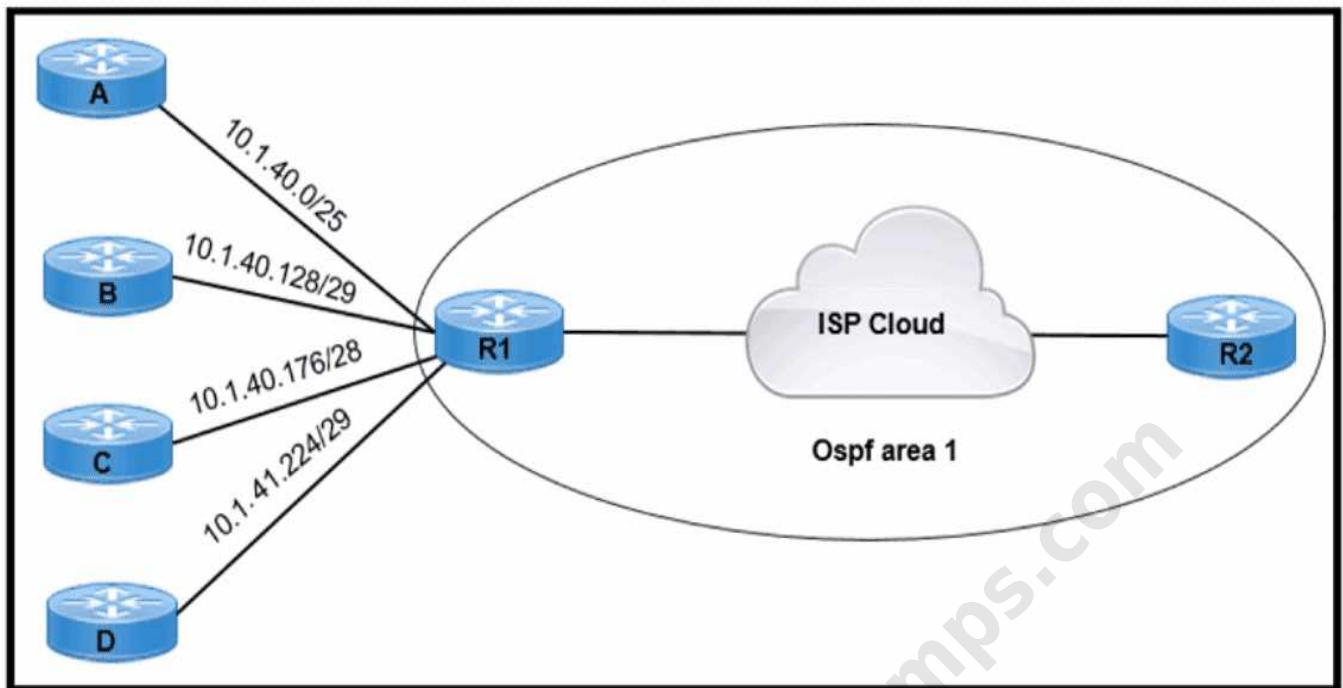
Explanation:

---

**Question: 895**

---

Refer to the exhibit.



Router R1 receives static routing updates from routers A, B, C, and D. The network engineer wants R1 to advertise static routes in OSPF area 1. Which nary address must be advertised in OSPF?

- A. 10.1.40.0/25
- B. 10.1.40.0/24
- C. 10.1.40.0/23
- D. 10.1.41.0/25

**Answer: C**

Explanation:

### Question: 896

A network architect is deciding whether to implement Cisco autonomous access points or lightweight access points. Which fact about firmware updates must the architect consider? Unlike lightweight access points, which require

- A. Unlike lightweight access points, which require redundant WLCs to support firmware upgrades, autonomous access points require only one WLC.



- B. Unlike autonomous access points, lightweight access points store a complete copy of the current firmware for backup.
- C. Unlike lightweight access points, autonomous access points can recover automatically from a corrupt firmware update.
- D. Unlike autonomous access points, lightweight access points require a WLC to implement remote firmware updates.

---

**Answer: D**

---

Explanation:

---

**Question: 897**

---

Refer to the exhibit.

Output from R1

```
GigabitEthernet0/0/1 is up, line protocol is down
Hardware is SPA-10X1GE-V2, address is 0023.33ee.7c00 (bia 0023.33ee.7c00)
MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive not supported
Half Duplex, 1000Mbps, link type is auto, media type is LX
output flow-control is off, input flow-control is off
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:01, output 00:02:31, output hang never

10 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 watchdog, 314 multicast, 0 pause input
1 packets output, 77 bytes, 0 underruns
0 output errors, 50 collisions, 6 interface resets
17 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
```

What is the issue with the interface GigabitEthernet0/0/1?

- A. Port security
- B. High throughput
- C. Cable disconnect

D. duplex mismatch

---

**Answer: C**

---

Explanation:

---

**Question: 898**

---

What must be considered before deploying virtual machines?

- A. location of the virtual machines within the data center environment
- B. whether to leverage VSM to map multiple virtual processors to two or more virtual machines
- C. resource limitations, such as the number of CPU cores and the amount of memory
- D. support for physical peripherals, such as monitors, keyboards, and mice

---

**Answer: C**

---

Explanation:

---

**Question: 899**

---

What is a reason why an administrator would choose to implement an automated network management approach?

- A. Reduce inconsistencies in the network configuration.
- B. Enable "box by box" configuration and deployment.
- C. Decipher simple password policies.
- D. Increase recurrent management costs.

---

**Answer: A**

---

Explanation:

---

**Question: 900**

---

Which capability does TFTP provide?

- A. loads configuration files on systems without data storage devices
- B. provides authentication for data communications over a private data network
- C. provides encryption mechanisms for file transfer across a WAN
- D. provides secure file access within the LAN

---

**Answer: A**

---

Explanation:

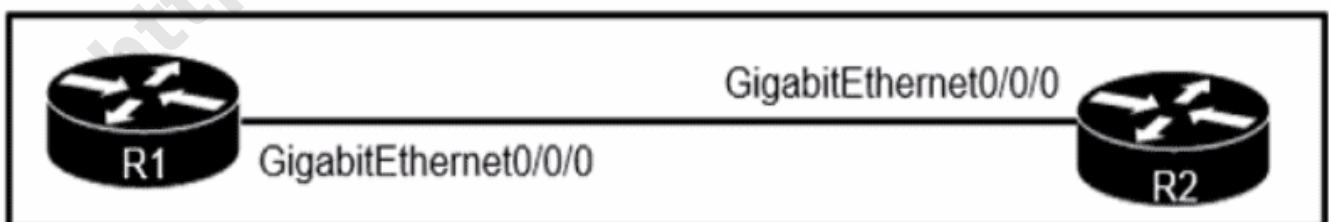
TFTP (Trivial File Transfer Protocol) is a simple file transfer protocol that is often used to transfer configuration files or firmware to network devices, such as routers or switches, during the boot process. TFTP is lightweight and lacks advanced features like authentication or encryption, making it suitable for basic file transfer operations in situations where security is not a primary concern. Therefore, option A accurately describes one of the primary capabilities of TFTP. It is commonly used to load configuration files onto systems, particularly those without extensive data storage devices, during the initialization or configuration process.

---

**Question: 901**

---

Refer to the exhibit.



A network engineer must configure the link with these requirements:

- Consume as few IP addresses as possible.
- Leave at least two additional useable IP addresses for future growth.

Which set of configurations must be applied?

A)

```
R1(config-if)#ip address 10.10.10.1 255.255.255.252  
R2(config-if)#ip address 10.10.10.2 255.255.255.252
```

B)

```
R1(config-if)#ip address 10.10.10.1 255.255.255.248  
R2(config-if)#ip address 10.10.10.4 255.255.255.248
```

C)

```
R1(config-if)#ip address 10.10.10.1 255.255.255.0  
R2(config-if)#ip address 10.10.10.5 255.255.255.0
```

D)

```
R1(config-if)#ip address 10.10.10.1 255.255.255.240  
R2(config-if)#ip address 10.10.10.12 255.255.255.240
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: B**

---

Explanation:

We have to configure the link which will need 2 IP addresses, 1 for each port on each Router. We also

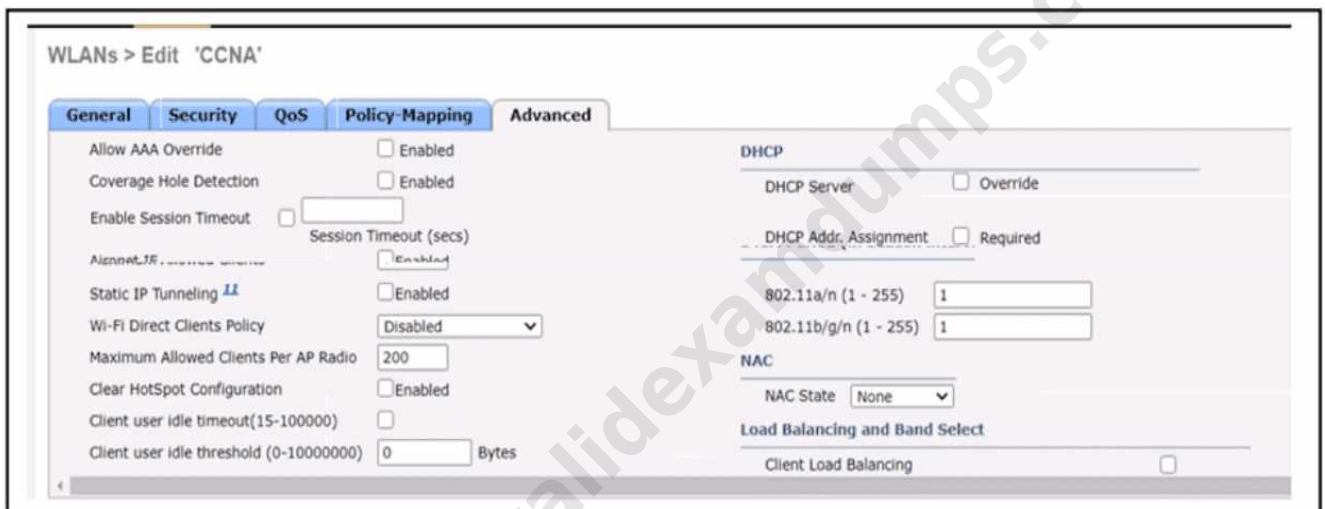
need 2 spare IPs for future growth, so overall we need 4 usable IP addresses. If we consider using the /30 (255.255.255.252) mask, it will give us  $2^2 (=4)$  i.e., total 4 IPs and 2 usable IPs, which doesn't fulfil the given requirements. So, we can consider using the next /29 (255.255.255.248) mask, which gives us  $2^3 (=8)$  i.e., total 8 IP address and 6 usable IP addresses, which perfectly fulfil the given requirements.

---

**Question: 902**

---

Refer to the exhibit.



A network engineer configures the CCNA WLAN so that clients must authenticate hourly and to limit the number of simultaneous connections to the WLAN to Which two actions complete this configuration? (Choose two.)

- A. Enable the Enable Session Timeout option and set the value to 3600.
- B. Set the Maximum Allowed Clients value to 10.
- C. Enable the Client Exclusion option and set the value to 3600.
- D. Enable the Wi-Fi Direct Clients Policy option.
- E. Set the Maximum Allowed Clients Per AP Radio value to 10.

---

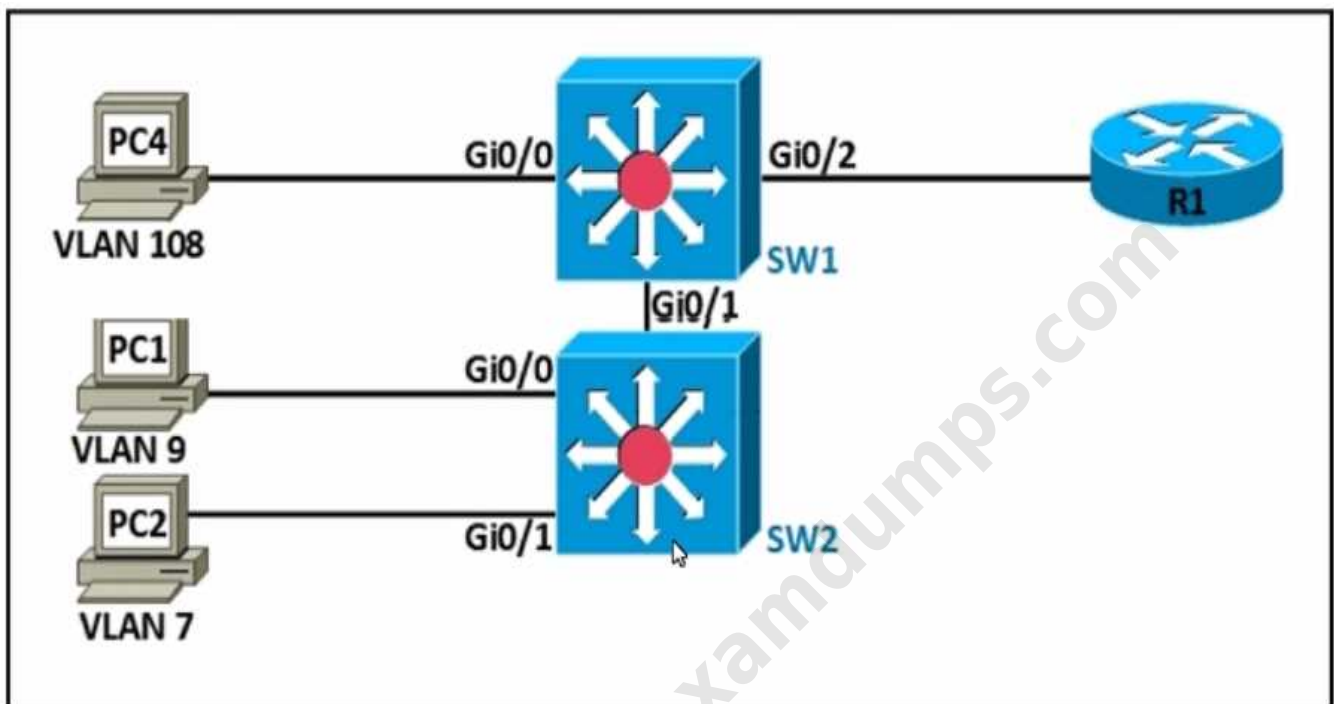
**Answer: A,B**

---

Explanation:

**Question: 903**

Refer to the exhibit.



The SW1 and SW2 Gi0/0 ports have been preconfigured. An engineer is given these requirements:

- Allow all PCs to communicate with each other at Layer 3.
- Configure untagged traffic to use VLAN 5.
- Disable VLAN 1 from being used.

Which configuration set meets these requirements?

A. SW1#interface Gi0/1switchport mode trunkswitchport trunk allowed vlan 5,7,9,108switchport trunk native vlan 5interface Gi0/2switchport mode trunkswitchport trunk allowed vlan 5,7,9,108SW2#interface Gi0/1switchport mode accessswitchport access vlan 7interface Gi0/7switchport mode trunkswitchport trunk allowed vlan 7,9,108

B. SW1#interface Gi0/1switchport mode trunkswitchport trunk allowed vlan 5,7,9,108switchport trunk native vlan 5interface Gi0/2switchport mode accessswitchport trunk allowed vlan 7,9,108SW2#interface Gi0/1switchport mode accessno switchport access vlan 1switchport access vlan 7interface Gi0/7switchport mode trunkswitchport trunk allowed vlan 7,9,108switchport trunk native vlan 5

C. SW#1 -interface Gi0/1switchport mode trunkswitchport trunk allowed vlan 5,7,9,108switchport trunk native vlan 5interface Gi0/2switchport mode trunkswitchport trunk allowed vlan 5,7,9,108SW2#interface Gi0/1switchport mode accessswitchport access vlan 7interface

Gi0/7switchport mode trunkswitchport trunk allowed vlan 5,7,9,108switchport trunk native vlan 5

D. SW1#interface Gi0/1switchport mode trunkswitchport trunk allowed vian 5,7,9,108interface Gi0/2switchport mode trunkswitchport trunk allowed vlan 7,9,108SW2#interface Gi0/1switchport mode trunkswitchport trunk allowed vlan 7interface Gi0/7switchport mode trunkswitchport trunk allowed vlan 5,7,9,108

---

**Answer: C**

---

Explanation:

---

**Question: 904**

---

What should a network administrator consider when deciding to implement automation?

- A. Automated systems may have difficulty expanding network changes at scale.
- B. Network automation typically is limited to the configuration and management of virtual devices within a network.
- C. Network automation typically increases enterprise management operating costs.
- D. Manual changes frequently lead to configuration errors and inconsistencies.

---

**Answer: D**

---

Explanation:

When deciding to implement automation, a network administrator should consider the benefits and challenges associated with automation. Option D highlights one of the key reasons for implementing automation—manual changes often result in configuration errors and inconsistencies. Automating repetitive and error-prone tasks can help improve the accuracy and reliability of network configurations.

---

**Question: 905**

---

Company has decided to require multifactor authentication for all systems. Which set of parameters meets the requirement?

- A. personal 10-digit PIN and RSA certificate
- B. complex password and personal 10-digit PIN
- C. password of 8 to 15 characters and personal 12-digit PIN
- D. fingerprint scanning and facial recognition

---

**Answer: A**

---

Explanation:

the set of parameters that meets the requirement for MFA is option A, personal 10-digit PIN and RSA certificate. A personal 10-digit PIN is something the user knows, and an RSA certificate is something the user has. An RSA certificate is a digital certificate that uses the RSA algorithm for encryption and digital signatures<sup>3</sup>.

---

**Question: 906**

---

DRAG DROP

Drag and drop the management connection types from the left onto the definitions on the right.

|         |                                                                         |
|---------|-------------------------------------------------------------------------|
| console | supports clear-text connections to the controller CLI                   |
| HTTPS   | supports encrypted access to CLI and a secure channel for data transfer |
| SSH     | supports physical connections over a serial cable                       |
| Telnet  | supports secure web access for management of the device                 |

---

**Answer:**

---

Explanation:



Telnet

SSH

console

HTTPS

---

**Question: 907**

---

How does IPsec provide secure networking for applications within an organization?

- A. It takes advantage of FTP to secure file transfers between nodes on the network.
- B. It provides GRE tunnels to transmit traffic securely between network nodes.
- C. It enables sets of security associations between peers.
- D. It leverages TFTP providing secure file transfers among peers on the network.

---

**Answer: C**

---

Explanation:

IPsec (Internet Protocol Security) is a protocol suite that provides secure communication over Internet Protocol (IP) networks. It achieves this by authenticating and encrypting each IP packet within a communication session. One of the key concepts in IPsec is the establishment of security associations (SAs) between peers. Security associations are the combination of algorithms and keys used to secure communication between two devices. They define the security parameters for the communication, including the encryption algorithm, integrity algorithm, and keying information. By establishing these security associations, IPsec ensures confidentiality, integrity, and authenticity of the data being transmitted between network nodes.

---

**Question: 908**

---

What are two functions of DHCP servers? (Choose two.)

- A. prevent users from assigning their own IP addresses to hosts
- B. assign dynamic IP configurations to hosts in a network
- C. support centralized IP management
- D. issue DHCPDISCOVER messages when added to the network
- E. respond to client DHCPOFFER requests by issuing an IP address

---

**Answer: B,C**

---

Explanation:

---

**Question: 909**

---

It work security team noticed that an increasing number of employees are becoming victims of phishing attacks. Which security program should be implemented to mitigate the problem?

- A. email system patches
- B. physical access control
- C. software firewall enabled on all PCs
- D. user awareness training

---

**Answer: D**

---

Explanation:

---

**Question: 910**

---

What does WPA3 provide in wireless networking?

- A. safeguards against brute force attacks with SAE
- B. optional Protected Management Frame negotiation
- C. backward compatibility with WPA and WPA2
- D. increased security and requirement of a complex configuration

---

**Answer: A**

---

Explanation:

<https://www.swascan.com/wi-fi-security/>

---

**Question: 911**

---

What is the operating mode and role of a backup port on a shared LAN segment in Rapid PVST+?

- A. forwarding mode and provides the lowest-cost path to the root bridge for each VLAN
- B. learning mode and provides the shortest path toward the root bridge handling traffic away from the LAN
- C. blocking mode and provides an alternate path toward the designated bridge
- D. listening mode and provides an alternate path toward the root bridge

---

**Answer: C**

---

Explanation:

---

**Question: 912**

---

An on-site service desk technician must verify the IP address and DNS server information on a users Windows computer. Which command must the technician enter at the command prompt on the user's computer?

- A. ipconfig /all
- B. ifconfig -a
- C. show interface
- D. netstat -r

---

**Answer: A**

---

Explanation:

The ipconfig /all command displays the configuration information of all the network adapters on a Windows computer, including the IP address, subnet mask, default gateway, and DNS server information<sup>12</sup>. This command can help troubleshoot network connectivity and DNS resolution issues.

---

**Question: 913**

---

Which enhancements were implemented as part of WPA3?

- A. 802.1x authentication and AES-128 encryption
- B. TKIP encryption improving WEP and per-packet keying
- C. AES-64 m personal mode and AES-128 in enterprise mode
- D. forward secrecy and SAE in personal mode for secure initial key exchange

---

**Answer: D**

---

Explanation:

---

**Question: 914**

---

How is noise defined in Wi-Fi?

- A. ratio of signal-to-noise rating supplied by the wireless device
- B. signals from other Wi-Fi networks that interfere with the local signal
- C. measured difference between the desired Wi-Fi signal and an interfering Wi-Fi signal
- D. any interference that is not Wi-Fi traffic that degrades the desired signal

---

**Answer: A**

---

Explanation:

---

**Question: 915**

---

What is the difference between an IPv6 link-local address and a unique local address?

- A. The scope of an IPv6 link-local address is limited to a loopback address, and an IPv6 unique local address is limited to a directly attached interface.
- B. The scope of an IPv6 link-local address can be used throughout a company site or network, but an IPv6 unique local address is limited to a loopback address.
- C. The scope of an IPv6 link-local address is global, but the scope of an IPv6 unique local address is limited to a loopback address.
- D. The scope of an IPv6 link-local address is limited to a directly attached interface, but an IPv6 unique local address is used throughout a company site or network.

---

**Answer: B**

---

Explanation:

---

**Question: 916**

---

What is a difference between an IPv6 multicast address and an IPv6 anycast address?

- A. A packet sent to an IPv6 multicast address is delivered to one or more destinations at once, but a packet sent to an IPv6 anycast address is routed to the closest interface with that address.
- B. An IPv6 multicast address uses the prefix 2002::/15 and forwards to one destination, and an IPv6 anycast address uses the prefix ff00::/8 and forwards to any destination in a group.
- C. IPv6 multicast addresses are used to transition from IPv4 to IPv6, and IPv6 anycast addresses are used for address aggregation in an IPv6-only environment.
- D. An IPv6 multicast address is assigned to numerous interfaces within a subnet, but an IPv6 anycast address is used for a predefined group of nodes in an all-IPv6 routers group.

---

**Answer: A**

---

Explanation:

---

**Question: 917**

---

Refer to the exhibit.

```
1  [  
2    { "switch": "3750", "port": e2 },  
3    { "router": "2951", "port": e20 },  
4    { "switch": "3750", "port": e23 }  
5  ]
```

What is represented beginning with line 1 and ending with line 5?

- A. value
- B. object
- C. key
- D. array

---

**Answer: D**

---

Explanation:

---

**Question: 918**

---

DRAG DROP

Drag and drop the statements about AAA from the left onto the corresponding AAA services on the right. Not all options are used.

It supports local, PPP, RADIUS, and TACACS+ options

It tracks the services that a user is using.

It records the amount of network resources consumed by the user.

It assigns per-user attributes.

It permits and denies login attempts.

**Accounting****Authentication**

---

**Answer:**

---

Explanation:

It assigns per-user attributes.

#### Accounting

It records the amount of network resources consumed by the user.

It tracks the services that a user is using.

#### Authentication

It permits and denies login attempts.

It supports local, PPP, RADIUS, and TACACS+ options.

---

**Question: 919**

---

What does traffic shaping do?

- A. It modifies the QoS attributes of a packet
- B. It queues excess traffic
- C. It organizes traffic into classes.
- D. It sets QoS attributes within a packet.

---

**Answer: A**

---

Explanation:

---

**Question: 920**

---

Which Rapid PVST+ feature should be configured on a switch port to immediately send traffic to a connected server as soon as it is active?

- A. BPDU guard
- B. loop guard
- C. portfast



D. uplinkfast

---

**Answer: C**

---

Explanation:

---

**Question: 921**

---

Which interface is used for out-of-band management on a WLC?

- A. dynamic
- B. service port
- C. virtual
- D. management

---

**Answer: B**

---

Explanation:

---

**Question: 922**

---

How is a configuration change made to a wireless AP in lightweight mode?

- A. SSH connection to the management IP of the AP
- B. EoIP connection via the parent WLC
- C. CAPWAP/LWAPP connection via the parent WLC
- D. HTTPS connection directly to the out-of-band address of the AP

---

**Answer: C**

---

Explanation:

**Question: 923**

How do traditional campus device management and Cisco DNA Center device management differ in regards to deployment?

- A. Traditional campus device management schemes typically deploy patches and updates quicker than Cisco DNA Center device management.
- B. Cisco DNA Center device management is implemented at a lower cost than most traditional campus device management options.
- C. Traditional campus device management allows a network to scale quicker than using Cisco DNA Center device management.
- D. Cisco DNA Center device management deploys a network faster than traditional campus device management.

**Answer: D**

Explanation:

**Question: 924**

Refer to the exhibit.



Which set of commands must be applied to the two switches to configure an LACP Layer 2 EtherChannel?

A)

```
SW1(config)#interface range f0/13 -14
SW1(config-if-range)#channel-group 1 mode active
```

```
SW2(config)#interface range f0/13 -14
SW2(config-if-range)#channel-group 1 mode passive
```

B)

```
SW1(config)#interface range f0/13 -14
SW1(config-if-range)#channel-group 1 mode desirable
```

```
SW2(config)#interface range f0/13 -14
SW2(config-if-range)#channel-group 1 mode passive
```

C)

```
SW1(config)#interface range f0/13 -14
SW1(config-if-range)#channel-group 1 mode auto
```

```
SW2(config)#interface range f0/13 -14
SW2(config-if-range)#channel-group 1 mode passive
```

D)

```
SW1(config)#interface range f0/13 -14
SW1(config-if-range)#channel-group 1 mode on
```

```
SW2(config)#interface range f0/13 -14
SW2(config-if-range)#channel-group 1 mode passive
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: A**

---

Explanation:

---

**Question: 925**

---

What are two reasons to implement DHCP in a network? (Choose two.)

- A. reduce administration time in managing IP address ranges for clients
- B. control the length of time an IP address is used by a network device
- C. manually control and configure IP addresses on network devices
- D. dynamic control over the best path to reach an IP address
- E. access a website by name instead of by IP address

---

**Answer: D,E**

---

Explanation:

---

**Question: 926**

---

What are two reasons to configure PortFast on a switch port attached to an end host? (Choose two.)

- A. to enable the number of MAC addresses learned on the port to 1
- B. to protect the operation of the port from topology change processes
- C. to enable the port to enter the forwarding state immediately when the host boots up
- D. to prevent the port from participating in Spanning Tree Protocol operations
- E. to block another switch or host from communicating through the port

---

**Answer: B,C**

---

Explanation:

---

**Question: 927**

---

Which action must be taken when password protection is Implemented?

- A. Use less than eight characters in length when passwords are complex.
- B. Store passwords as contacts on a mobile device with single-factor authentication.
- C. Include special characters and make passwords as long as allowed.
- D. Share passwords with senior IT management to ensure proper oversight.

---

**Answer: D**

---

Explanation:

---

**Question: 928**

---

Which group of channels in the 802.11b/g/n/ac/ax 2.4 GHz frequency Bands are nonoverlapping channels?

- A. channels 1, 5, and 10
- B. channels 1,6, and 11
- C. channels 1,5, and 11
- D. channels 1,6, and 10

---

**Answer: B**

---

Explanation:

---

**Question: 929**

---

Refer to the exhibit.

```
{
  "Cisco Devices": [
    {
      "name": "ASA - Security Device",
      "name": "Cisco 1100 ASR Router",
      "name": "Cisco 6800 Switch"
    }
  ]
}
```

What is missing from this output for it to be executed?

- A. double quotes C") around the "Cisco Devices" string
- B. curly brace (}) at the end
- C. exclamation point (!) at the beginning of each line
- D. square bracket ([) at the beginning

---

**Answer: B**

---

Explanation:

---

### Question: 930

---

A network engineer starts to implement a new wireless LAN by configuring the authentication server and creating the dynamic Interface. What must be performed next to complete the basic configuration?

- A. Install the management interface and add the management IP.
- B. Configure high availability and redundancy for the access points.
- C. Enable Telnet and RADIUS access on the management interface.
- D. Create the new WLAN and bind the dynamic interface to it.

---

**Answer: D**

---

Explanation:

---

**Question: 931**

---

Which EtherChannel mode must be configured when using LAG on a WLC?

- A. on
- B. active
- C. auto
- D. passive

---

**Answer: A**

---

Explanation:

---

**Question: 932**

---

Refer to the exhibit.

```
Cat9300# show cdp
Global CDP information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled
```

Which action must be taken so that neighboring devices rapidly discover switch Cat9300?

- A. Configure the cdp timer 10 command on switch Cat9300.
- B. Enable portfast on the ports that connect to neighboring devices.
- C. Configure the cdp holdtime 10 command on switch Cat9300.
- D. Configure the cdp timer 10 command on the neighbors of switch Cat9300.

---

**Answer: A**

---

Explanation:

---

**Question: 933**

---

Which physical component is distributed among multiple virtual machines running on the same hypervisor?

- A. external storage
- B. hardware resources
- C. network interfaces
- D. backplane network

---

**Answer: B**

---

Explanation:

---

**Question: 934**

---

Why choose Cisco DNA Center for automated lifecycle management?

- A. To perform upgrades without service interruption
- B. To provide fast and accurate deployment of patches and updates
- C. To allow SSH access to all nodes in the network.
- D. To provide software redundancy in the network.

---

**Answer: B**

---

Explanation:

---

**Question: 935**

---

DRAG DROP



```

Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0

 209.165.200.0/27 is subnetted, 1 subnets
B    209.165.200.224 [20/0] via 10.10.12.2, 06:08:59
 209.165.201.0/27 is subnetted, 1 subnets
B    209.165.201.0 [20/0] via 10.10.12.2, 05:13:18
 209.165.202.0/27 is subnetted, 1 subnets
B    209.165.202.128 [20/0] via 10.10.12.2, 05:12:48
10.0.0.0/8 is variably subnetted, 9 subnets, 4 masks
C    10.10.10.0/28 is directly connected, GigabitEthernet0/0
C    10.10.11.0/30 is directly connected, FastEthernet2/0
C    10.10.12.0/30 is directly connected, GigabitEthernet0/1
O    10.10.13.0/25 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O    10.10.13.128/28 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O    10.10.13.144/28 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O    10.10.13.160/29 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O    10.10.13.208/29 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O    10.10.13.252/30 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
S*   0.0.0.0/0 [1/0] via 10.10.11.2

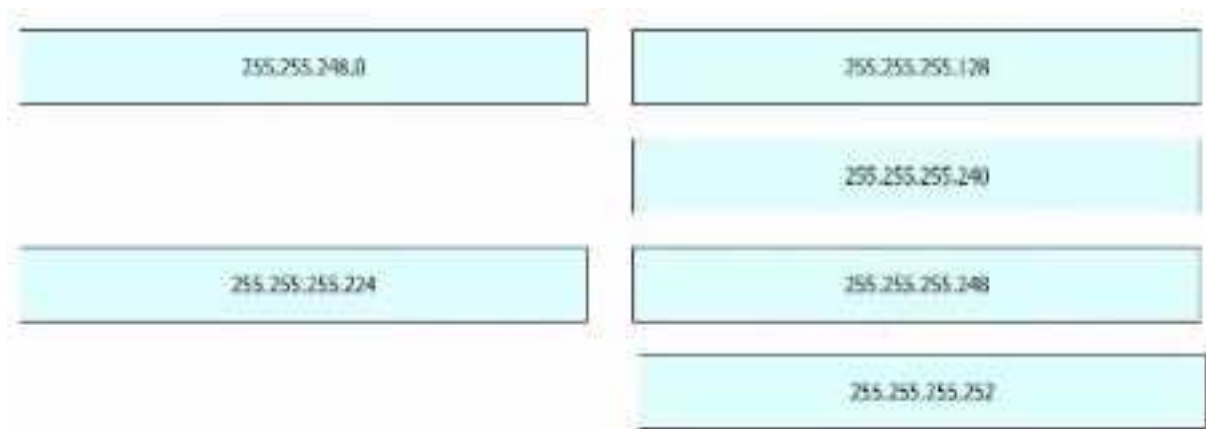
```

Refer to the exhibit. Drag and drop the subnet masks from the left onto the corresponding subnets on the right. Not all subnet masks used.

|                 |              |
|-----------------|--------------|
| 255.255.255.0   | 10.10.13.0   |
| 255.255.255.128 | 10.10.13.128 |
| 255.255.255.224 | 10.10.13.160 |
| 255.255.255.240 | 10.10.13.252 |
| 255.255.255.248 |              |
| 255.255.255.252 |              |

**Answer:**

Explanation:



---

**Question: 936**

---

Which protocol does Ansible use to push modules to nodes in a network?

- A. SSH
- B. SNMP
- C. Kerberos
- D. Telnet

---

**Answer: A**

---

Explanation:

---

**Question: 937**

---

Refer to the exhibit.

```
{  
  "interfaces": [ "ethernet0/3", "ethernet0/4", "ethernet0/5" ]  
}
```

Which types of JSON data is shown

- A. Object
- B. Sequence

- C. String
- D. boolean

---

**Answer: C**

---

Explanation:

---

**Question: 938**

---

A router received three destination prefixes: 10.0.0/18, and 10.0.0/24. When the show ip route command is executed, which output does it return?

A)

Gateway of last resort is 172.16.1.1 to network 0.0.0.0  
o E2 10.0.0.0/8 [110/5] via 192.168.1.1, 0:01:00, Ethernet0

B)

Gateway of last resort is 172.16.1.1 to network 0.0.0.0  
o E2 10.0.0.0/8 [110/5] via 192.168.1.1, 0:01:00, Ethernet0  
o E2 10.0.0.0/16 [110/5] via 192.168.2.1, 0:01:00, Ethernet1  
o E2 10.0.0.0/24 [110/5] via 192.168.3.1, 0:01:00, Ethernet2

C)

Gateway of last resort is 172.16.1.1 to network 0.0.0.0  
o E2 10.0.0.0/24 [110/5] via 192.168.3.1, 0:01:00, Ethernet2

D)

Gateway of last resort is 172.16.1.1 to network 0.0.0.0  
o E2 10.0.0.0/16 [110/5] via 192.168.2.1, 0:01:00, Ethernet1  
o E2 10.0.0.0/24 [110/5] via 192.168.3.1, 0:01:00, Ethernet2

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: A**

---

Explanation:

---

**Question: 939**

---

Which mechanism allows WPA3 to provide a higher degree of security than its predecessors?

- A. special-character support in preshared Keys
- B. SAE password-based key exchange
- C. automatic device pairing
- D. certificate-based authentication

---

**Answer: B**

---

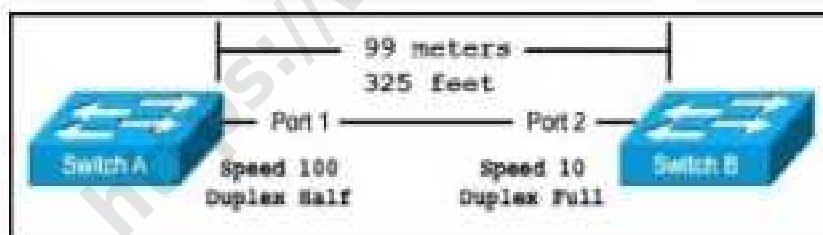
Explanation:

---

**Question: 940**

---

Exhibit.



The switches are connected via a Cat5 Ethernet cable that was successfully tested. The Interfaces are configured as access ports and are both in a 'down' status. What is the cause of this issue?

- A. The switches are configured with incompatible duplex settings.
- B. The distance between the two switches is not supported by Cat5.
- C. The speed settings on the switches are mismatched.
- D. The portfast command is missing from the configuration.

---

**Answer: C**

---

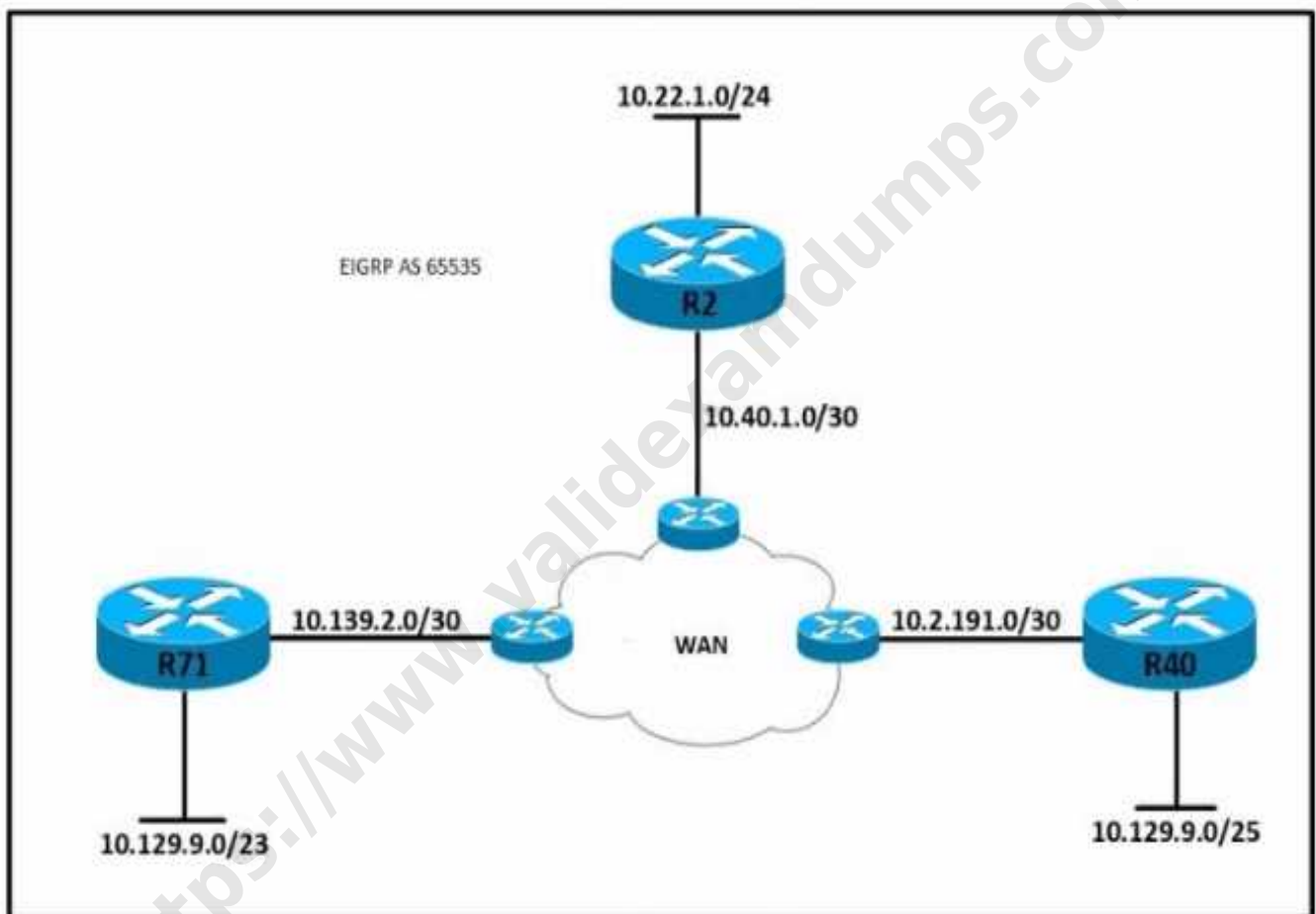
Explanation:

---

**Question: 941**

---

Refer to the exhibit.



All routers in the network are configured correctly, and the expected routes are being exchanged among the routers. Which set of routes are learned from neighbors and installed on router 2?

A)

10.40.1.0/30  
10.139.2.0/30  
10.2.191.0/30  
10.129.9.0/25

B)

10.129.9.0/23  
10.139.2.0/30  
10.2.191.0/30  
10.129.9.0/25

C)

10.129.9.0/23  
10.139.2.0/30  
10.129.9.0/25  
10.22.1.0/24

D)

10.129.9.0/23  
10.40.1.0/30  
10.2.191.0/30  
10.129.9.0/25

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: A**

---

Explanation:

---

**Question: 942**

---

Refer to the exhibit.

```
router# show ip route
```

```
****
```

```
D 172.16.32.0/26 [90/25789217] via 10.0.0.1
```

```
R 172.16.32.0/24 [120/4] via 10.0.0.2
```

```
O 172.16.32.0/19 [110/229840] via 10.0.0.3
```

```
C 172.16.32.32/32 is directly connected, Loopback0
```

```
C 172.16.32.4/30 is directly connected, GigabitEthernet0/0
```

A packet sourced from 172.16.32.254 is destined for 172.16.32.8. What is the subnet mask of the preferred destination route?

- A. 255.255.224.0
- B. 255.255.255.0
- C. 255.255.255.192
- D. 255.255.255.252

---

**Answer: C**

---

Explanation:

---

**Question: 943**

---

Which cable type must be used when connecting a router and switch together using these criteria?

- Pins 1 and 2 are receivers and pins 3 and 6 are transmitters
- Auto detection MDi-X is unavailable

- A. straight-through
- B. rollover
- C. crossover
- D. console

---

**Answer: C**

---

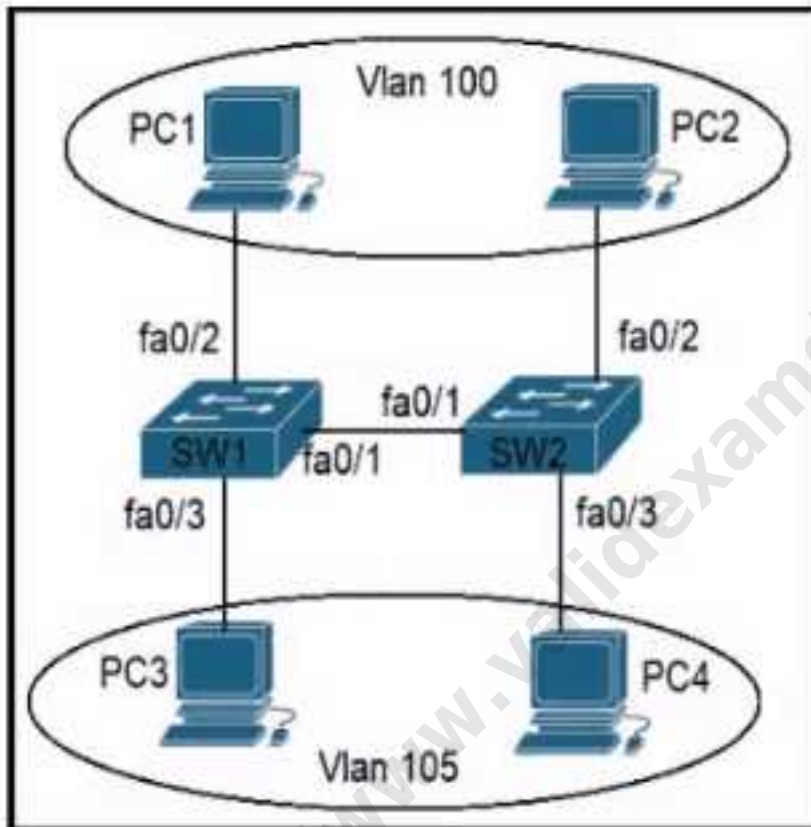
Explanation:

---

**Question: 944**

---

Refer to the exhibit.



An engineer configures interface fa0/1 on SW1 and SW2 to pass traffic from two different VLANs. For security reasons, company policy requires the native VLAN to be set to a nondefault value. Which configuration meets this requirement?

A)

```
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk encapsulation isl
Switch(config-if)#switchport trunk allowed vlan 100,105
Switch(config-if)#switchport trunk native vlan 1
```

B)



```
Switch(config-if)#switchport mode access
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport access vlan 100,105
Switch(config-if)#switchport trunk native vlan 3
```

C)

```
Switch(config-if)#switchport mode dynamic
Switch(config-if)#switchport access vlan 100,105
Switch(config-if)#switchport trunk native vlan 1
```

D)

```
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport trunk allowed vlan 100,105
Switch(config-if)#switchport trunk native vlan 3
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: D**

---

Explanation:

---

**Question: 945**

---

A HCP pool has been created with the name CONTROL. The pool uses the next to last usable IP address as the default gateway for the DHCP clients. The server is located at 172.16.32.15. What is the step in the process for clients on the 192.168.52.0/24 subnet to reach the DHCP server?

A. ip forward-protocol udp 137

B. ip default-network 192.168.52.253

C. ip helper-address 172.16.32.15

D. ip default-gateway 192.168.52.253

---

**Answer: C**

---

Explanation:

DHCP and broadcasts: DHCP clients use broadcasts to find a DHCP server, but broadcasts are typically limited to the local subnet. Since the DHCP server (172.16.32.15) is on a different subnet than the clients (192.168.52.0/24), the clients' broadcast requests won't reach it.

ip helper-address: This command is used on a router (or layer 3 switch) to forward DHCP broadcasts to a specific IP address (the DHCP server) on a different subnet. This allows clients to discover and obtain IP address information from a DHCP server that is not on their local network.

---

**Question: 946**

---

DRAG DROP

Drag and drop the TCP and UDP characteristics from the left onto the supporting protocols on the right. Not all options are used.

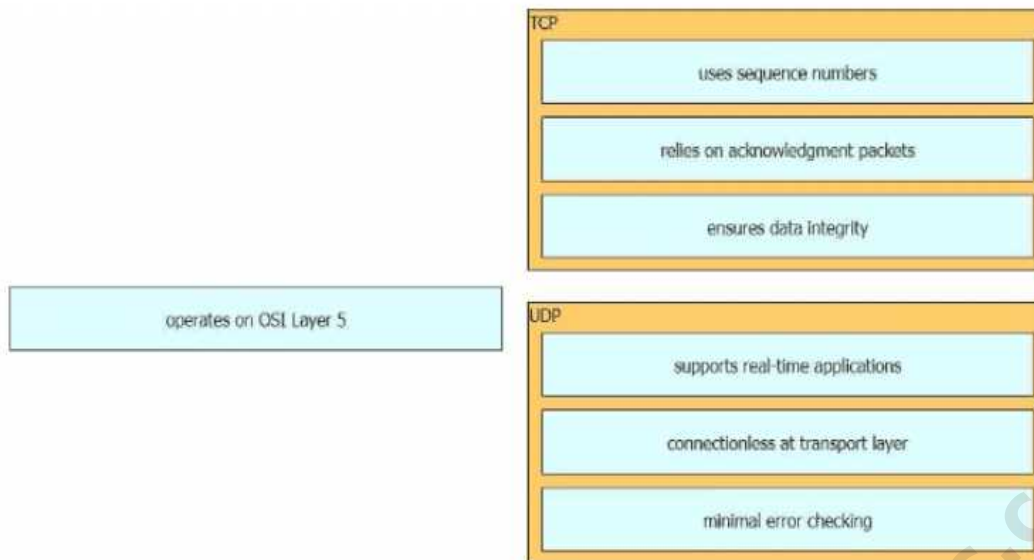
|                                   |     |
|-----------------------------------|-----|
| uses sequence numbers             | TCP |
| supports real-time applications   |     |
| relies on acknowledgment packets  |     |
| operates on OSI Layer 5           | UDP |
| connectionless at transport layer |     |
| minimal error checking            |     |
| ensures data integrity            |     |

---

**Answer:**

---

Explanation:



---

**Question: 947**

---

Refer to the exhibit. The user has connectivity to devices on network 192.168.3 0/24 but cannot reach users on the network 10.10.1.0/24.

What is the first step to verify connectivity?

```
C:\Users\VBOUL>ipconfig
```

# Windows IP Configuration

Ethernet adapter Ethernet:

```
Physical Address. . . . . : 04-42-1A-EE-AA-5E
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::8a79:b8fc:5a93:c119%25(Preferred)
IPv4 Address. . . . . : 192.168.3.20(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, 16 March 2023
6:25:01 AM
Lease Expires . . . . . : Sunday, 26 March 2023
4:17:26 PM
Default Gateway . . . . . : 192.168.3.1
DHCP Server . . . . . : 192.168.3.1
DHCPv6 IAID . . . . . : 201605658
DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-4F-56-37-04-
42-1A-EE-AA-5E
DNS Servers . . . . . : 10.10.1.254
NetBIOS over Tcpip. . . . . : Enabled
```

- A. Is the internet reachable?
- B. Is the default gateway reachable?
- C. Is the DNS server reachable?

---

**Answer: B**

---

Explanation:

---

**Question: 948**

---

Which CRUD operation corresponds to the HTTP GET method?

- A. delete
- B. create
- C. update
- D. read

---

**Answer: D**

---

Explanation:

---

**Question: 949**

---

What is the main difference between traditional networks and controller-based networking?

- A. Controller-based networks increase TCO for the company, and traditional networks require less investment.
- B. Controller-based networks provide a framework for Innovation, and traditional networks create efficiency.
- C. Controller-based networks are open for application requests, and traditional networks operate manually.
- D. Controller-based networks are a closed ecosystem, and traditional networks take advantage of programmability.

---

**Answer: C**

---

Explanation:

---

**Question: 950**

---

Which WLC management connection type is vulnerable to man-in-the-middle attacks?

- A. Telnet
- B. console
- C. HTTPS
- D. SSH

---

**Answer: A**

---

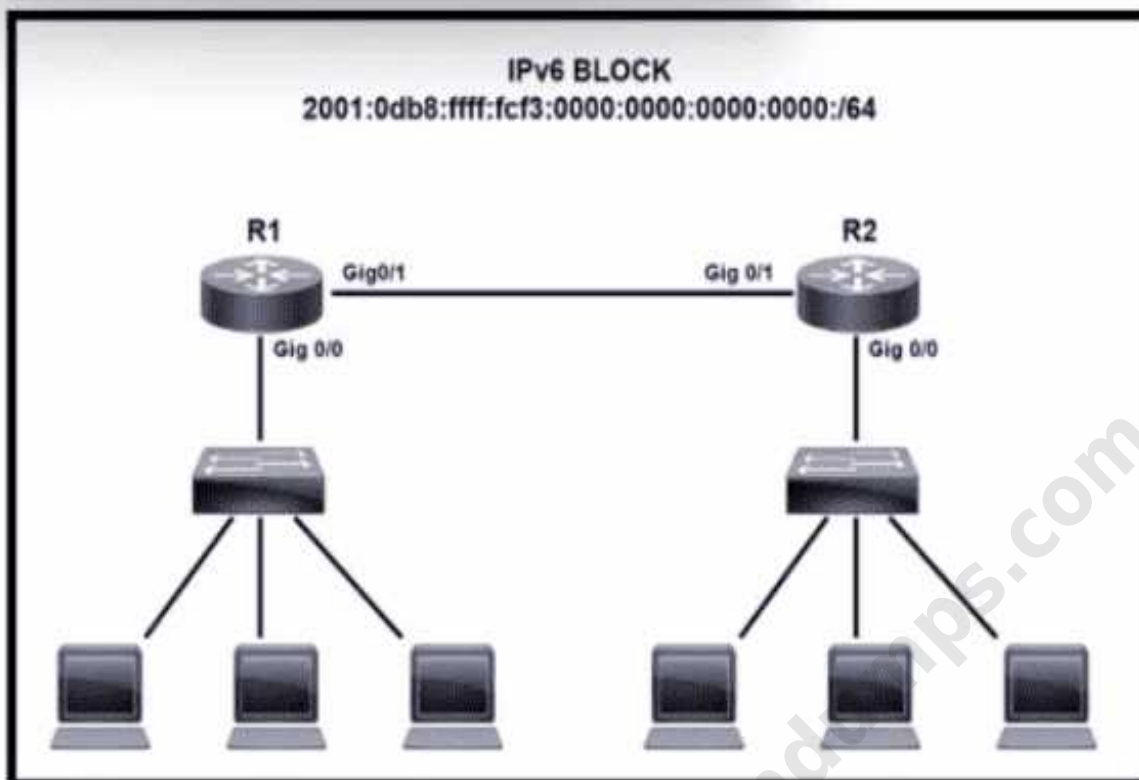
Explanation:

---

**Question: 951**

---

Refer to the exhibit.



Refer to the exhibit. IPv6 is being Implemented within the enterprise. The command `Ipv6 unicast-routing` is configure. Interlace `GlgO/0` on R1 must be configured to provide a dynamic assignment using the assigned IPv6 blk Which command accomplishes this task?

- A. `ipv6 address 2001:DB8:FFFF:FCF3::1/64`
- B. `ipv6 address autoconfig 2001:DB8:FFFF:FCF2::/64`
- C. `ipv6 address 2001:DB8:FFFF:FCF3::/64 eui-64`
- D. `ipv6 address 2001:DB8:FFFF:FCF3::/64 link-local`

---

**Answer: C**

---

Explanation:

---

### Question: 952

---

Which alternative to password authentication Is Implemented to allow enterprise devices to log in to the corporate network?

- A. magic links
- B. one-time passwords
- C. digital certificates
- D. 90-day renewal policies

---

**Answer: C**

---

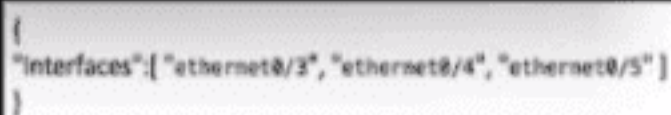
Explanation:

---

**Question: 953**

---

Refer to the exhibit.



```
{  
  "interfaces": [ "ethernet0/3", "ethernet0/4", "ethernet0/5" ]  
}
```

Refer to the exhibit. Which type of JSON data is shown?

- A. sequence
- B. string
- C. object
- D. Boolean

---

**Answer: C**

---

Explanation:

---

**Question: 954**

---

Refer to the exhibit.



```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#ip domain-name CC-Net.com
R1(config)#enable secret Passfornewuser
R1(config)#line vty 0 15
R1(config-line)#transport input ssh
R1(config-line)#login local
```

Refer to the exhibit. A network administrator is configuring a router for user access via SSH. The service-password encryption command has been issued. The configuration must meet these requirements:

- Create the username as CCUser.
- Create the password as NA!2ScC.
- Encrypt the user password.

What must be configured to meet the requirements?

- A. username CCUser privilege 10 password NA!2ScC
- B. username CCUser password NA!2ScC enable password level 5 NA!2\$cc
- C. username CCUser secret NA!2ScC
- D. username CCUser privilege 15 password NA!2ScC enable secret 0 NA!2\$cc

---

**Answer: B**

---

Explanation:

---

### Question: 955

---

What is the RFC 4627 default encoding for JSON text?

- A. UCS-2
- B. UTF-8
- C. Hex
- D. GB18030

---

**Answer: B**

---

Explanation:

---

**Question: 956**

---

A network engineer is configuring a new router at a branch office. The router is connected to an upstream WAN network that allows the branch to communicate with the head office. The central time server with IP address 172.24.54.8 is located behind a firewall at the head office. Which command must the engineer configure so that the software clock of the new router synchronizes with the time server?

- A. ntp master 172.24.54.8
- B. ntp client 172.24.54.8
- C. ntp peer 172.24.54.8
- D. ntp server 172.24.54.8

---

**Answer: D**

---

Explanation:

---

**Question: 957**

---

A network administrator wants the syslog server to filter incoming messages into different files based on their Importance. Which filtering criteria must be used?

- A. level
- B. message body
- C. process ID
- D. facility

---

**Answer: A**

---

Explanation:

---

**Question: 958**

---

Refer to the exhibit.

|                                                       |                                                                     |
|-------------------------------------------------------|---------------------------------------------------------------------|
| 100.0.0.0/8 is variably subnetted, 4 subnets, 4 masks |                                                                     |
| R                                                     | 100.0.0.0/8 [120/2] via 192.168.3.1, 00:00:13, Ethernet0/3          |
| S                                                     | 100.100.0.0/16 [1/0] via 192.168.4.1                                |
| D                                                     | 100.100.100.0/24 [90/435200] via 192.168.2.1, 00:00:13, Ethernet0/2 |
| O                                                     | 100.100.100.100/32 [110/21] via 192.168.1.1, 00:05:57, Ethernet0/1  |

Refer to the exhibit. How will the device handle a packet destined to IP address 100.100.100.100?

- A. It will choose the route with the longest match. O 100.100.100.100/32 (110/21) via 192.168.1.1. 00:05:57. Ethernet0/1.
- B. It will always prefer the static route over dynamic routes and choose the route S 100.100.0.0/16 (1/0) via 192.168.4.1.
- C. It will choose the route with the highest metric. D 100.100.100.0/24 (90/435200) via 192.168.2.1. 00:00:13. Ethernet0/2.
- D. It will choose the route with the lowest metric. R 100.0.0.0/8 [120/2] via 192.168.3.1. 00:00:13. Ethernet0/3.

---

**Answer: A**

---

Explanation:

---

**Question: 959**

---

An engineer must configure a core router with a floating static default route to the backup router at 10.200.0.2. Which command meets the requirements?

- A. ip route 0.0.0.0 0.0.0.0 10.200.0.2 1
- B. ip route 0.0.0.0 0.0.0.0 10.200.0.2 floating
- C. ip route 0.0.0.0 0.0.0.0 10.200.0.2
- D. ip route 0.0.0.0 0.0.0.0 10.200.0.2 10

---

**Answer: D**

---

Explanation:

---

**Question: 960**

---

Refer to the exhibit.

|                      |                                     |                          |
|----------------------|-------------------------------------|--------------------------|
| WPA2 Policy          | <input checked="" type="checkbox"/> |                          |
| WPA3 Policy          | <input type="checkbox"/>            |                          |
| WPA2/WPA3 Encryption | AES(CCMP128)                        | <input type="checkbox"/> |
|                      | CCMP256                             | <input type="checkbox"/> |
|                      | GCMP128                             | <input type="checkbox"/> |
|                      | GCMP256                             | <input type="checkbox"/> |
| MPSK                 | <input type="checkbox"/>            |                          |
| Auth Key Mgmt        | 802.1x                              | <input type="checkbox"/> |
|                      | PSK                                 | <input type="checkbox"/> |
|                      | CCKM                                | <input type="checkbox"/> |
|                      | SAE                                 | <input type="checkbox"/> |
|                      | OWE                                 | <input type="checkbox"/> |
|                      | FT + 802.1x                         | <input type="checkbox"/> |
|                      | FT + PSK                            | <input type="checkbox"/> |

Refer to the exhibit. An engineer is creating a secure preshared key based SSID using WPA2 for a wireless network running on 2.4 GHz and 5 GHz. Which two tasks must the engineer perform to complete the process? (Choose two.)

A. Select the 802.1 x option for Auth Key Management.

- B. Select the WPA Policy option.
- C. Select the PSK option for Auth Key Management.
- D. Select the AES option for Auth Key Management.
- E. Select the AES (CCMP128) option for WPA2/WPA3 Encryption.

---

**Answer: D,E**

---

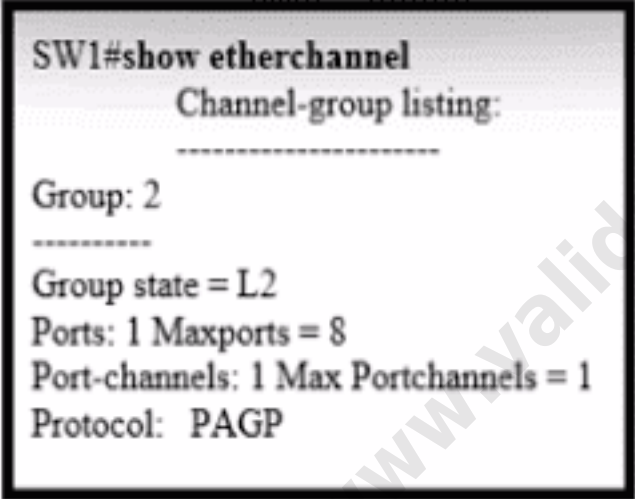
Explanation:

---

**Question: 961**

---

Refer to the exhibit.



```
SW1#show etherchannel
      Channel-group listing:
      -----
Group: 2
-----
Group state = L2
Ports: 1 Maxports = 8
Port-channels: 1 Max Portchannels = 1
Protocol: PAGP
```

Refer to the exhibit. A network engineer updates the existing configuration on interface fastethernet1/1 switch SW1. It must establish an EtherChannel by using the same group designation with another vendor switch. Which configuration must be performed to complete the process?

- A. interface port-channel 2channel-group 2 mode desirable
- B. interface fastethernet 1/1channel-group 2 mode active
- C. interface fastethernet 1/1channel-group 2 mode on
- D. interface port-channel 2channel-group 2 mode auto

---

**Answer: B**

---

Explanation:

---

**Question: 962**

---

Refer to the exhibit.

```
Gateway of last resort is 172.16.2.2 to network 0.0.0.0
```

```
10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
```

```
10.10.100.0/26 is directly connected, GigabitEthernet0/0/6
```

```
C 10.10.10.0/24 is directly connected, GigabitEthernet0/0/0
```

```
L 10.10.10.3/32 is directly connected, GigabitEthernet0/0/0
```

```
172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
```

```
S 172.16.1.33/32 is directly connected, GigabitEthernet0/0/1
```

```
C 172.16.2.0/23 is directly connected, GigabitEthernet0/0/1
```

```
L 172.16.2.1/32 is directly connected, GigabitEthernet0/0/1
```

```
S* 0.0.0.0/0 [1/0] via 172.16.2.2
```

Refer to the exhibit. A packet sourced from 10.10.10.32 is destined for the Internet. What is the administrative distance for the destination route?

- A. 0
- B. 1
- C. 2
- D. 32

---

**Answer: B**

---

Explanation:

---

**Question: 963**

---

Which solution is appropriate when mitigating password attacks where the attacker was able to sniff the clear-text password of the system administrator?

- A. next-generation firewall to keep stateful packet inspection
- B. multifactor authentication using two separate authentication sources

- C. ACL to restrict incoming Telnet sessions "admin" accounts
- D. IPS with a block list of known attack vectors

---

**Answer: A**

---

Explanation:

---

**Question: 964**

---

What are two facts that differentiate optical-fiber cabling from copper cabling? (Choose two.)

- A. It is less expensive when purchasing patch cables.
- B. It has a greater sensitivity to changes in temperature and moisture.
- C. It provides greater throughput options.
- D. It carries signals for longer distances.
- E. It carries electrical current further distances for PoE devices.

---

**Answer: C,D**

---

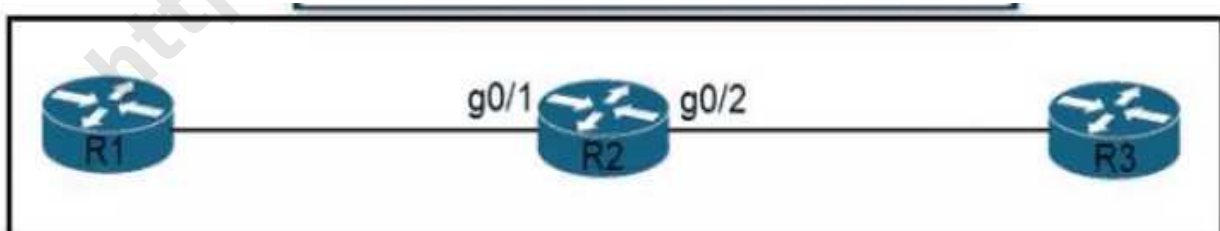
Explanation:

---

**Question: 965**

---

Refer to the exhibit.



Refer to the exhibit Routers R1 R2 and R3 use a protocol to identify their neighbors' IP addresses hardware platforms, and software versions. A network engineer must configure R2 to avoid sharing any neighbor information with R3, and maintain its relationship with R1. What action meets this requirement?

- A. Configure the no cdp enable command on g0/2.
- B. Configure the no cdp run command globally.
- C. Configure the no lldp run command globally.
- D. Configure the no lldp receive command on gQV1.

---

**Answer: A**

---

Explanation:

---

**Question: 966**

---

What is the difference between controller-based networks and traditional networks as they relate to control-plane and/or data-plane functions?

- A. Controller-based networks centralize all important data-plane functions, and traditional networks distribute data-plane functions.
- B. Controller-based networks centralize all important control-plane functions, and traditional networks distribute control-plane functions.
- C. Traditional networks centralize all important control-plane functions, and controller-based networks distribute control-plane functions.
- D. Traditional networks centralize all important data-plane functions, and controller-based networks distribute data-plane functions.

---

**Answer: A**

---

Explanation:

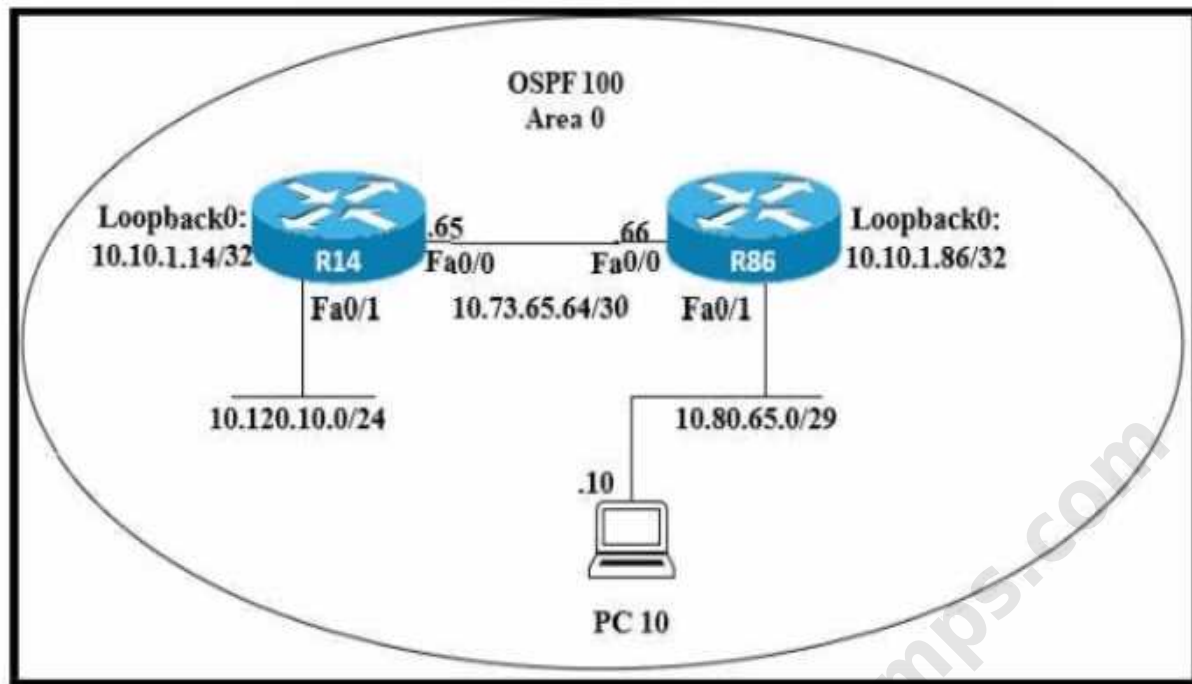
---

**Question: 967**

---

Refer to the exhibit.





Refer to the exhibit. An engineer just installed network 10.120.10.0/24. Which configuration must be applied to the R14 router to add the new network to its OSPF routing table?

- A. router ospf 100network 10.120.10.0 255.255.255.0 area 0
- B. router ospf 120network 10.120.10.0 255.255.255.0 area 0ip route 10.120.10.0 255.255.255.0 fa0/1
- C. router ospf 100 area 0network 10.120.10.0 0.0.0.255

---

**Answer: B**

---

Explanation:

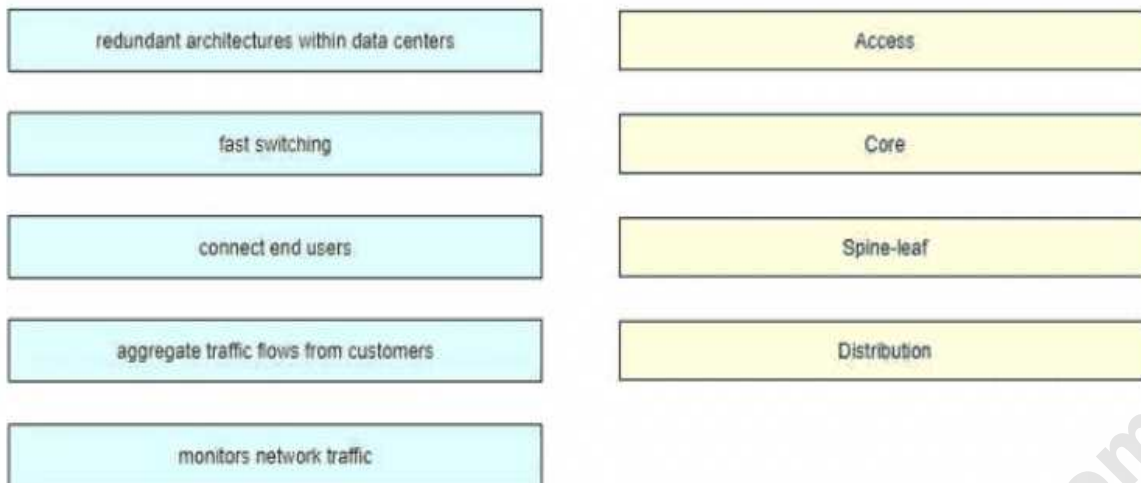
---

**Question: 968**

---

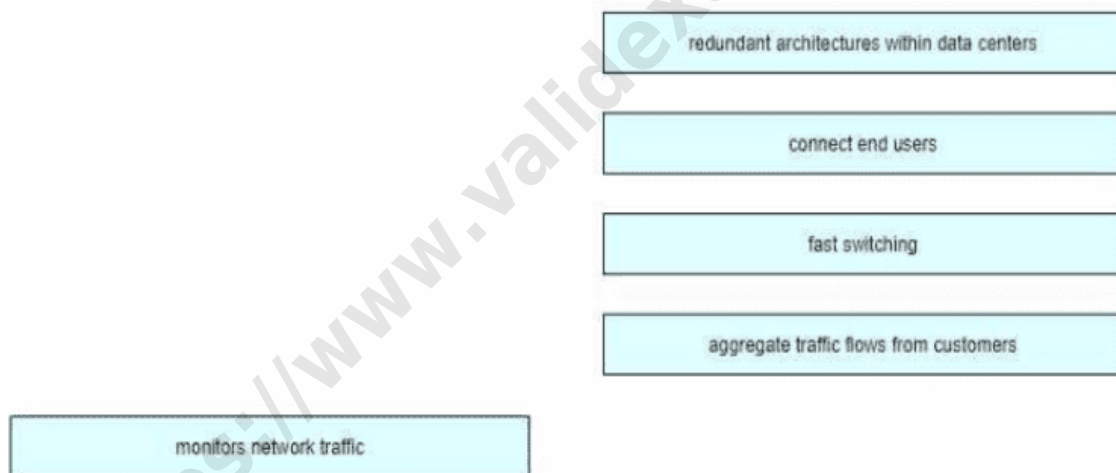
DRAG DROP

Drag and drop the common functions from the left onto the corresponding network topology architecture layer on the right. Not all common functions are used.



**Answer:**

Explanation:



### Question: 969

How does machine learning improve the detection of unauthorized network access?

- A. It monitors for outdated software.
- B. It dictates security policy updates.

- C. It identifies patterns indicating intrusions.
- D. It assigns security clearance levels.

---

**Answer: A**

---

Explanation:

---

**Question: 970**

---

Which two statements distinguish authentication from accounting? (Choose two.)

- A. Only authentication records the duration of a user's connection.
- B. Only authentication supports user-activity audits.
- C. Only authentication provides supporting information for billing users.
- D. Only authentication challenges users for their credentials and returns a response.
- E. Only authentication validates "who you are."

---

**Answer: D,E**

---

Explanation:

---

**Question: 971**

---

What is an Ansible inventory?

- A. file that defines the target devices upon which commands and tasks are executed
- B. unit of Python code to be executed within Ansible
- C. collection of actions to perform on target devices, expressed in YAML format
- D. device with Ansible installed that manages target devices

---

**Answer: C**

---

Explanation:

---

**Question: 972**

---

Which plane is centralized in software-defined networking?

- A. application
- B. services
- C. control
- D. data

---

**Answer: C**

---

Explanation:

---

**Question: 973**

---

Which authentication method requires the user to provide a physical attribute to authenticate successfully?

- A. password
- B. multifactor
- C. biometric
- D. certificate

---

**Answer: A**

---

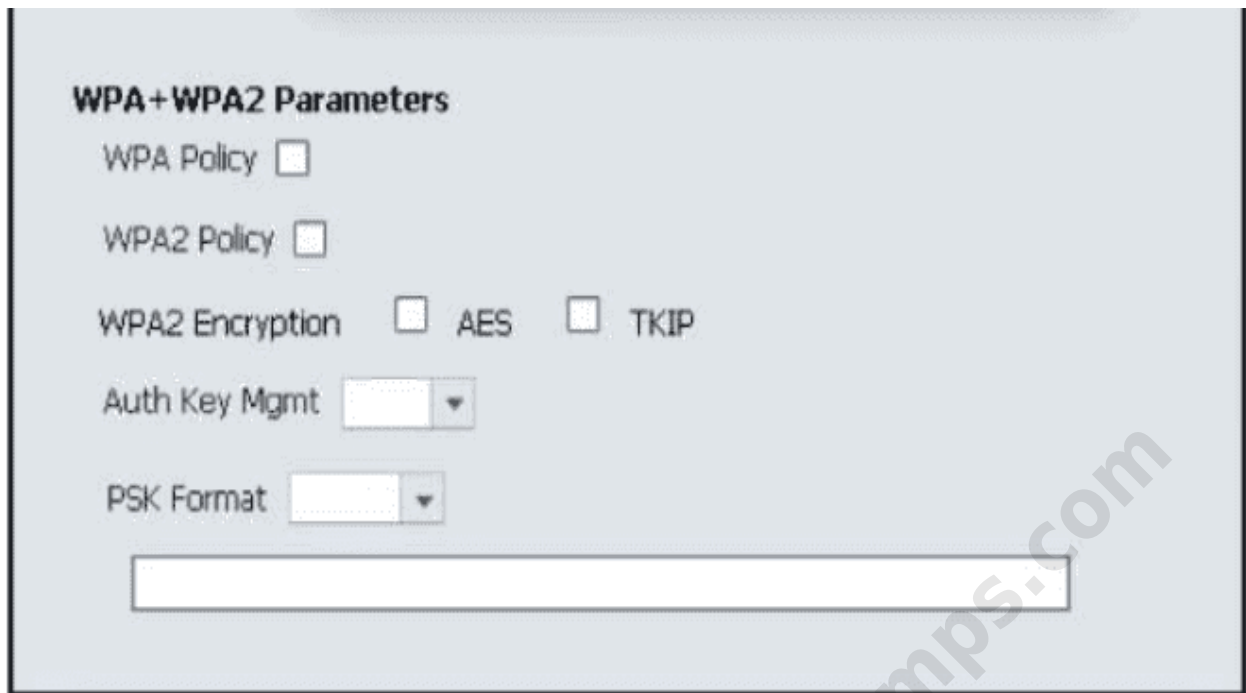
Explanation:

---

**Question: 974**

---

Refer to the exhibit.



The image shows a screenshot of the Cisco Wireless LAN Controller (WLC) GUI, specifically the 'WPA+WPA2 Parameters' configuration page. The page has a light blue background. At the top, the title 'WPA+WPA2 Parameters' is displayed. Below the title, there are several configuration options: 'WPA Policy' with an unchecked checkbox, 'WPA2 Policy' with an unchecked checkbox, 'WPA2 Encryption' with two unchecked checkboxes for 'AES' and 'TKIP', 'Auth Key Mgmt' with a dropdown menu showing a blank field, and 'PSK Format' with a dropdown menu showing a blank field. At the bottom of the configuration area, there is a long, empty text input field. A large, diagonal watermark reading 'https://www.validexamdumps.com' is visible across the entire screenshot.

Refer to the exhibit. An engineer is using the Cisco WLC GUI to configure a WLAN for WPA2 encryption with AES and preshared key Cisc0123456. After the engineer selects the WPA + WPA2 option from the Layer 2 Security drop-down list, which two tasks must they perform to complete the process? (Choose two.)

- A. Select the WPA2 Policy, AES, and TKIP check boxes.
- B. Select ASCII from the PSK Format drop-down list, enter the key, and leave the Auth Key Mgmt setting blank.
- C. Select PSK from the Auth Key Mgmt drop-down list, set the PSK Format to ASCII, and enter the key.
- D. Select the WPA2 Policy and AES check boxes.

---

**Answer: C,D**

---

Explanation:

---

**Question: 975**

---

Refer to the exhibit.

```
CPE# show ip access-list Services
Extended IP access list Services
 10 permit tcp 10.0.0.0 0.255.255.255 any eq www
 20 permit tcp 10.0.0.0 0.255.255.255 any eq 443
 30 permit udp 10.0.0.0 0.255.255.255 host 198.51.100.11 eq domain
 40 deny ip any any log
```

Refer to the exhibit. This ACL is configured to allow client access only to HTTP, HTTPS, and DNS services via UDP. The new administrator wants to add TCP access to the DNS service. Which configuration updates the ACL efficiently?

- A. ip access-list extended Services35 permit tcp 10.0.0.0 0.255.255.255 host 198.51.100.11 eq domain
- B. no ip access-list extended Servicesip access-list extended Services30 permit tcp 10.0.0.0 0.255.255.255 host 198.51.100.11 eq domain
- C. ip access-list extended Servicespermit tcp 10.0.0.0 0.255.255.255 host 198.51.100.11 eq domain
- D. no ip access-list extended Servicesip access-list extended Servicespermit udp 10.0.0.0 0.255.255.255 any eq 53permit tcp 10.0.0.0 0.255.255.255 host 198.51.100.11 eq domaindeny ip any any log

---

**Answer: A**

---

Explanation:

---

**Question: 976**

---

Why is UDP more suitable than TCP for applications that require low latency, such as VoIP?

- A. UDP reliably guarantees delivery of all packets and TCP drops packets under heavy load.
- B. TCP sends an acknowledgment for every packet that is received and UDP operates without acknowledgments.
- C. UDP uses sequencing data for packets to arrive in order, and TCP offers the capability to receive packets in random order.
- D. TCP uses congestion control for efficient packet delivery and UDP uses flow control mechanisms for the delivery of packets.

---

**Answer: B**

---

Explanation:

---

**Question: 977**

---

Refer to the exhibit.

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile B -
BGP, D - EIGRP EX - EIGRP external, O - OSPF, IA - OSPF inter area,
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2,
E1 - OSPF external type 1, E2 - OSPF external type 2, I - IS-IS,
su - IS-IS summary, L1 - IS-IS level-1, L2- IS-IS level-2, ia - IS-IS
inter area, * - candidate default, I - per-user static route, o - ODR,
P - periodic download static route, H - NHRP, I - LISP,
a - application route, + - replicated route, % - next hop override,
p - overrides from Pfr

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
C 10.10.10.16/28 is directly connected, FastEthernet 0/0
L 10.10.10.17/32 is directly connected, FastEthernet 0/0
C 10.10.10.8/29 is directly connected, FastEthernet 0/1
L 10.10.10.9/32 is directly connected, FastEthernet 0/1
C 10.10.10.4/30 is directly connected, FastEthernet 0/2
C 10.10.10.5/32 is directly connected, FastEthernet 0/2
C 10.10.10.0/30 is directly connected, Serial 0/0
C 10.10.10.1/32 is directly connected, Serial 0/0
```

Refer to the exhibit. Which interface does a packet take to reach the destination address of 10.10.10.147?

- A. FastEthernet 0/0
- B. Serial0/0
- C. FastEthernet 0/1

---

**Answer: B**

---

Explanation:

---

**Question: 978**

---

What does the term "split MAC" refer to in a wireless architecture?

- A. divides data link layer functions between the AP and WLC
- B. combines the management and control functions from the data-forwarding functions
- C. uses different MAC addresses for 2.4 GHz and 5 GHz bands on the same AP
- D. leverages two APs to handle control and data traffic

---

**Answer: A**

---

Explanation:

---

**Question: 979**

---

Refer to the exhibit.

```
hostname CPE
service password-encryption

ip domain name ccna.cisco.com
ip name-server 198.51.100.210

crypto key generate rsa modulus 1024

username admin privilege 15 secret S0m3s3cr3t

line vty 0 4
  transport input ssh
  login local
```

Refer to the exhibit. An engineer executed the script and added commands that were not necessary for SSH and must now remove the commands. Which two commands must be executed to correct the configuration? (Choose two.)

- A. no ip domain name ccna.cisco.com
- B. no login local



- C. no service password-encryption
- D. no ip name-server 198.51.100.210
- E. no hostname CPE

---

**Answer: C,D**

---

Explanation:

---

**Question: 980**

---

Refer to the exhibit.

The screenshot shows the Cisco IOS configuration interface for Layer 2 Security. The 'Security' tab is selected, and the 'Layer 2' sub-tab is active. The configuration shows WPA+WPA2 security type, Personal security type, and WPA+WPA2 parameters. The PSK Format is set to ASCII, and the WPA GTK-randomize State is set to Disable.

Refer to the exhibit. Company A wants to use a RADIUS server to service all user and device authentication attempts with a more secure and granular authentication approach. Not all client devices support dot1x authentication. Which two configuration changes must be made to accomplish the task? (Choose two.)

- A. Enable AutoConfig IPSK under the Layer 2 tab.
- B. Select Authentication server under the AAA servers tab.
- C. Configure Enterprise Security type under the Layer 2 tab.
- D. Set Authentication under the Layer 3 tab.
- E. Enable WPA2 Policy under the Layer 2 tab.

---

**Answer: B,C**

---

Explanation:

---

**Question: 981**

---

Refer to the exhibit.

```
S* 0.0.0.0/0 [1/0] via 10.12.12.2, GigabitEthernet0/0
    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C    10.12.12.0/24 is directly connected, GigabitEthernet0/0
C    10.14.14.0/24 is directly connected, GigabitEthernet0/2
C    10.255.1.1/32 is directly connected, Loopback0
    172.16.0.0/32 is subnetted, 1 subnets
C    172.16.1.1 is directly connected, Loopback1
    172.31.0.0/32 is subnetted, 2 subnets
D EX 172.31.3.3 [170/2586112] via 10.12.12.2, 01:35:59, GigabitEthernet0/0
D EX 172.31.5.5 [170/2586112] via 10.12.12.2, 00:25:32, GigabitEthernet0/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, Loopback10
L    192.168.1.1/32 is directly connected, Loopback10
    192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
D EX 192.168.5.0/24
    [170/2586112] via 10.12.12.2, 00:25:32, GigabitEthernet0/0
O    192.168.5.5/32 [110/3] via 10.14.14.4, 01:34:27, GigabitEthernet0/2
    209.165.200.0/27 is subnetted, 1 subnets
D    209.165.200.224
    [90/3072] via 10.12.12.2, 01:36:18, GigabitEthernet0/0
    209.165.201.0/24 is variably subnetted, 3 subnets, 2 masks
D    209.165.201.3/32
    [90/131072] via 10.12.12.2, 01:35:59, GigabitEthernet0/0
D    209.165.201.16/28
    [90/131072] via 10.12.12.2, 01:35:59, GigabitEthernet0/0
O    209.165.201.30/32
    [110/3] via 10.14.14.4, 00:25:26, GigabitEthernet0/2
    [110/3] via 10.12.12.2, 01:35:06, GigabitEthernet0/0
    209.165.202.0/27 is subnetted, 1 subnets
O    209.165.202.128 [110/2] via 10.14.14.4, 00:54:45, GigabitEthernet0/2
```

Refer to the exhibit. Four load-balancing servers are reachable through this router; however, the company is removing all static and default routes on the router.

Server 1 - 10.12.14.14

Server 2 - 192.168.4.4

Server 3 - 209.165.200.5

Server 4 - 209.165.201.26

Which server will handle all traffic after the policy changes take effect?

A. Server 1 - 10.12.14.14

B. Server 2 - 192.168.4.4

C. Server 3 - 209.165.200.5

D. Server 4 - 209.165.201.26

---

**Answer: B**

---

Explanation:

---

**Question: 982**

---

What is the difference between SNMP traps and SNMP polling?

A. SNMP traps are used for proactive monitoring, and SNMP polling is used for reactive monitoring.

B. SNMP traps send periodic updates via the MIB, and SNMP polling sends data on demand.

C. SNMP traps are initiated by the network management system, and network devices initiate SNMP polling.

D. SNMP traps are initiated using a push model at the network device, and SNMP polling is initiated at the server.

---

**Answer: B**

---

Explanation:

---

**Question: 983**

---

Refer to the exhibit.

```
router# show ip route
```

```
....
```

```
D 172.16.32.0/26 [90/25789217] via 10.0.0.1
```

```
R 172.16.32.0/24 [120/4] via 10.0.0.2
```

```
O 172.16.32.0/19 [110/229840] via 10.0.0.3
```

```
C 172.16.32.32/32 is directly connected, Loopback0
```

```
C 172.16.32.4/30 is directly connected, GigabitEthernet0/0
```

Refer to the exhibit. A packet sourced from 172.16.32.254 is destined for 172.16.32.8. What is the subnet mask of the preferred destination route?

- A. 255.255.224.0
- B. 255.255.255.0
- C. 255.255.255.192
- D. 255.255.255.252

---

**Answer: C**

---

Explanation:

---

### Question: 984

---

Which protocol should be used to transfer large files on a company intranet that allows TCP ports 20 and 21 through the firewall?

- A. REST API
- B. FTP
- C. SMTP
- D. TFTP

---

**Answer: B**

---

Explanation:

---

**Question: 985**

---

DRAG DROP

Drag and drop the network topology architecture types from the left onto the corresponding function on the right. Not all architecture types are used.

|                |                                            |
|----------------|--------------------------------------------|
| spine-leaf     | end user connectivity                      |
| access         | top-of-rack                                |
| distribution   | provide routing, filtering, and WAN access |
| core           | switch packets as fast as possible         |
| perimeter edge |                                            |

---

**Answer:**

---

Explanation:

|            |                |
|------------|----------------|
| spine-leaf | access         |
|            | distribution   |
|            | core           |
|            | perimeter edge |

---

**Question: 986**

---

A network architect planning a new Wi-Fi network must decide between autonomous, cloud-based, and split MAC architectures. Which two facts should the architect consider? (Choose two.)

- A. Lightweight access points are solely used by split MAC architectures.
- B. Cloud-based architectures uniquely use the CAPWAP protocol to communicate between access points and clients.
- C. Each of the three architectures must use WLCs to manage their access points.
- D. All three architectures use access points to manage the wireless devices connected to the wired infrastructure.
- E. Autonomous architectures exclusively use tunneling protocols to manage access points remotely.

---

**Answer: A,E**

---

Explanation:

---

**Question: 987**

---

Which protocol is implemented when an organization must verify network performance, troubleshoot issues, and use an agent to communicate between monitoring tools and end devices?

- A. SNMP
- B. FTP
- C. NTP
- D. NFS

---

**Answer: D**

---

Explanation:

---

**Question: 988**

---

Which command creates a static NAT binding for a PC address of 10.1.1.1 to the public routable address 209.165.200.225 assigned to the PC?

- A. R1(config)#ip nat inside source static 10.1.1.1 209.165.200.225
- B. R1(config)#ip nat inside source static 209.165.200.225 10.1.1.1
- C. R1(config)#ip nat outside source static 10.1.1.1 209.165.200.225
- D. R1(config)#ip nat outside source static 209.165.200.225 10.1.1.1

---

**Answer: A**

---

Explanation:

---

**Question: 989**

---

Refer to the exhibit.

The screenshot shows the 'Add WLAN' configuration window with the 'Security' tab selected. Under the 'Layer2' tab, the 'WPA2 + WPA3' radio button is selected. The 'WPA Parameters' section shows 'WPA2 Policy' and 'WPA3 Policy' both set to 'Required'. The 'WPA2/WPA3 Encryption' section shows 'AES/CCMP(128)' and 'CCMP(256)' both set to 'Required'. The 'Protected Management Frame' section shows 'PMF' set to 'Required'. The 'Fast Transition' section shows 'Status' set to 'Enabled', 'Over the DS' set to 'Disabled', and 'Reassociation Timeout' set to '30'. The 'Auth Key Mgmt' section shows 'SAE' set to 'Required', 'FT + SAE' set to 'Required', 'OWE' set to 'Required', 'FT + OWE' set to 'Required', '802.11- SHA256' set to 'Required', 'Anti Clogging Threshold' set to '1500', and 'Max Retries' set to '5'. The 'Cancel' button is at the bottom left and the 'Apply to Device' button is at the bottom right.

Refer to the exhibit. Which functionalities will this SSID have while being used by wireless clients?

- A. decreases network security against offline dictionary attacks and encourages easy access to the network
- B. increases network security against offline dictionary attacks and discourages time-consuming brute force attacks
- C. increases network security against man in the middle attacks and discourages denial of service attacks
- D. decreases network security against air sniffing attacks and discourages the use of complex passwords

---

**Answer: D**

---

Explanation:

---

**Question: 990**

---

Refer to the exhibit.



```
Router# show interface FastEthernet0/0
FastEthernet0/0 is up, line protocol is up
Hardware is Gt96k FE, address is 0017.59b2.7fb2 (bia 0017.59b2.7fb2)
Internet address is 10.0.0.2/30
MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
  reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Half-duplex, 100Mb/s, 100baseTX/FX
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:04, output 00:00:04, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 516000 bits/sec, 45 packets/sec
5 minute output rate 516000 bits/sec, 46 packets/sec
  13282 packets input, 20075670 bytes
    Received 25 broadcasts, 0 runs, 0 giants, 0 throttles
  383 input errors, 383 CRC, 0 frame, 0 overrun, 0 ignored
  0 watchdog
  0 input packets with dribble condition detected
  13438 packets output, 20084258 bytes, 0 underruns
  0 output errors, 831 collisions, 5 interface resets
  11 unknown protocol drops
  0 babbles, 0 late collision, 0 deferred
  0 lost carrier, 0 no carrier
  0 output buffer failures, 0 output buffers swapped out
```

Refer to the exhibit. Users at a branch office are experiencing application performance issues, poor VoIP audio quality, and slow downloads. What is the cause of the issues?

- A. broadcast storm
- B. overutilization
- C. interface configuration
- D. QoS queuing

---

**Answer: D**

---

Explanation:

---

### Question: 991

---

Which selections must be used on the WLC when implementing a RADIUS server for wireless authentication?

- A. Client Exclusion and SSH
- B. 802.1x and the MAC address of the server
- C. Network Access Control State and SSH
- D. AAA Override and the IP address of the server

---

**Answer: B**

---

Explanation:

---

**Question: 992**

---

Which IPsec mode encapsulates the entire IP packet?

- A. tunnel
- B. Q-in-Q
- C. SSL VPN
- D. transport

---

**Answer: A**

---

Explanation:

---

**Question: 993**

---

Refer to the exhibit. A network administrator configures the CPE to provide internet access to the company headquarters. Traffic must be load-balanced via ISP1 and ISP2 to ensure redundancy. Which two command sets must be configured on the CPE router? (Choose two.)

A.

```
ip route 0.0.0.0 0.0.0.0 198.51.100.1  
ip route 0.0.0.0 0.0.0.0 203.0.113.1 2
```

B.

```
ip route 0.0.0.0 128.0.0.0 198.51.100.1  
ip route 128.0.0.0 128.0.0.0 203.0.113.1
```

C.

```
ip route 0.0.0.0 0.0.0.0 198.51.100.1  
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

D.

```
ip route 0.0.0.0 0.0.0.0 198.51.100.1 255  
ip route 0.0.0.0 0.0.0.0 203.0.113.1 255  
ip route 128.0.0.0 128.0.0.0 203.0.113.1
```

E.

```
ip route 0.0.0.0 128.0.0.0 198.51.100.1  
ip route 128.0.0.0 128.0.0.0 203.0.113.1  
ip route 0.0.0.0 0.0.0.0 198.51.100.1  
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

A. Option A

B. Option B

C. Option C

D. Option D

E. Option E

---

**Answer: A,C**

---

Explanation:

---

**Question: 994**

---

Refer to the exhibit.

MONITOR WLANs CONTROLLER WIRELESS SECURITY MANAGEMENT COMMANDS HELP FEEDBACK

WLANs > Edit 'my\_WLAN'

General Security QoS Policy-Mapping Advanced

Profile Name: my\_WLAN

Type: WLAN

SSID: my\_WLAN

Status: ☒ Enabled

Security Policies: [WPA2][Auth(PSK)]  
(Modifications done under security tab will appear after applying the changes.)

Radio Policy: All

Interface/Interface Group(G): guest

Multicast Vlan Feature: ☒ Enabled

Broadcast SSID: ☒ Enabled

NAS-ID: none

Refer to the exhibit. The My\_WLAN wireless LAN was configured with WPA2 Layer 2 PSK security. Which additional configuration must the administrator perform to allow users to connect to this WLAN on a different subnet called Data?

- A. Enable Broadcast SSID and select data from the Interface/Interface Group drop-down list.
- B. Enable Status and select data from the Interface/Interface Group drop-down list.
- C. Enable Status and set the NAS-ID to data.
- D. Enable Status and enable Broadcast SSID.

---

**Answer: B**

---

Explanation:

---

**Question: 995**

---

What is the role of syslog level 7 in network device health monitoring?

- A. It provides information about error conditions visible on the network device.
- B. It shares normal operational messages from the network equipment.
- C. It sends outputs from various debug commands on the device.
- D. It warns about emergency conditions on the network appliance.

---

**Answer: C**

---

Explanation:

---

**Question: 996**

---

Refer to the exhibit.

```
SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#enable password test!2E
SW1#(config)# line con 0
SW1#(config-line)# password Labtest32!
SW1#(config-line)# exit
SW1#(config)#
```

Refer to the exhibit. A network engineer started to change default settings on SW1 to allow remote access and has entered the following in the configuration mode:

SW1(config)#line vty 0 15

SW1(config-line)#password Labtest32!

Which set of commands are needed to allow only SSH access and hide passwords in the running configuration?

- A. SW1(config-line)#exitSW1(config)#aaa new-model
- B. SW1(config-line)#login localSW1(config-line)#exitSW1(config)#enable secret test!2E
- C. SW1(config-line)#transport input sshSW1(config-line)#exitSW1(config)#service password-encryption
- D. SW1(config-line)#login localSW1(config-line)#exitSW1(config)#crypto key generate rsa

---

**Answer: C**

---

Explanation:

---

**Question: 997**

---

Which protocol does a REST API use to communicate?

- A. SSH
- B. SNMP
- C. HTTP
- D. STP

---

**Answer: C**

---

Explanation:

---

**Question: 998**

---

How does network automation help reduce network downtime?

- A. Changes can be implemented in parallel across multiple devices at once, which increases the speed of the change rate.
- B. By using automation platforms with intent-based configuration, all changes are checked for possible outages before being implemented.
- C. Emails can be generated based on when a network admin performs a network change, which increases visibility.
- D. Configuration templates and testing can be built into implementation, which increases the success rate of a network change.

---

**Answer: D**

---

Explanation:

---

**Question: 999**

---

What are two benefits for using private IPv4 addressing? (Choose two.)

- A. They alleviate the shortage of public IPv4 addresses.
- B. They supply redundancy in the case of failure.
- C. They offer Internet connectivity to endpoints on private networks.
- D. They allow for Internet access from IoT devices.
- E. They provide a layer of security from Internet threats.

---

**Answer: A,E**

---

Explanation:

---

**Question: 1000**

---

Which set of methods is supported with the REST API?

- A. GET, POST, MOD, ERASE
- B. GET, POST, ERASE, CHANGE
- C. GET, PUT, POST, DELETE
- D. GET, PUT, ERASE, CHANGE

---

**Answer: C**

---

Explanation:

---

**Question: 1001**

---

Which AP mode serves as the primary hub in a point-to-multipoint network topology.

- A. FlexConnect
- B. SE-Connect
- C. bridge
- D. local

---

**Answer: C**

---

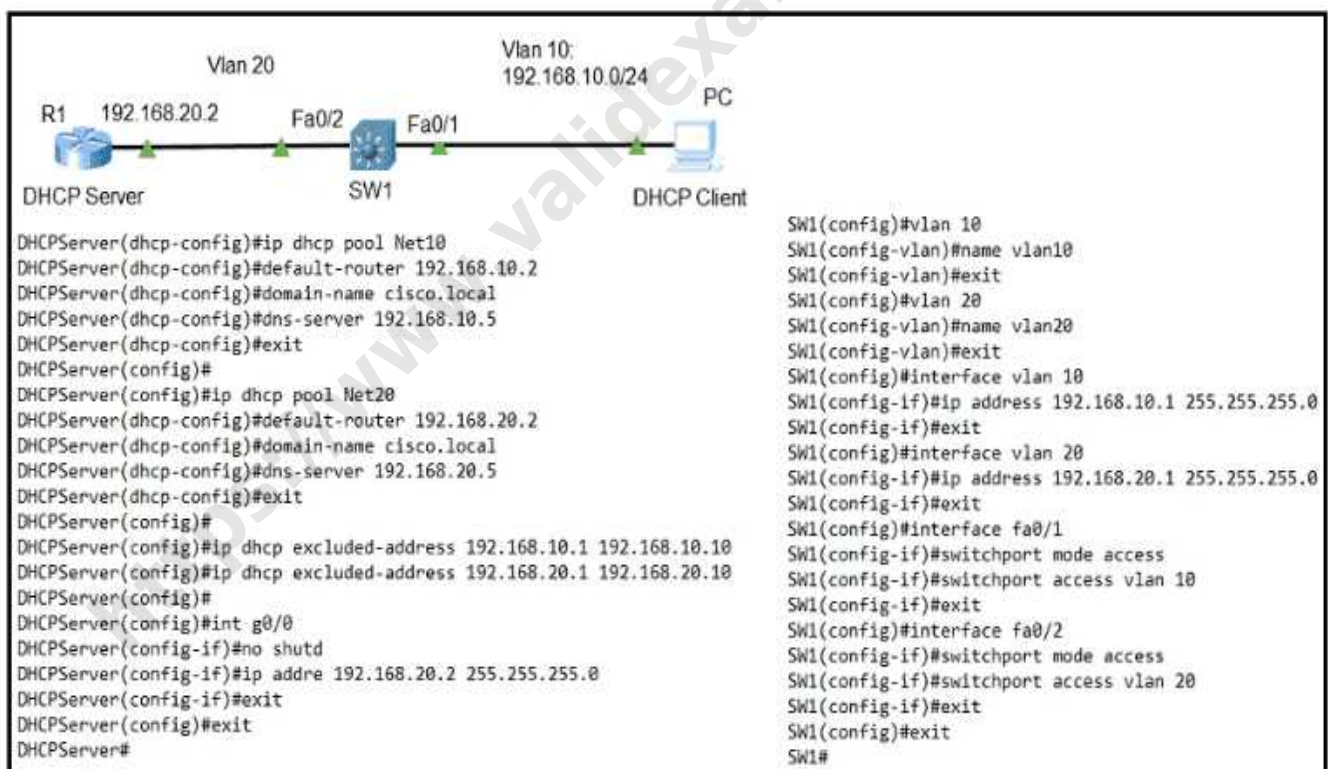
Explanation:

---

**Question: 1002**

---

Refer to the exhibit.



The DHCP server is configured with a DHCP pool for each of the subnets represented. Which command must be configured on switch SW1 to allow DHCP clients on VLAN 10 to receive dynamic IP addresses from the DHCP server?



- A. SW1(config-if)#ip helper-address 192.168.20.2
- B. SW1(config-if)#ip helper-address 192.168.20.1
- C. SW1(config-if)#ip helper-address 192.168.10.2
- D. SW1(config-if)#ip helper-address 192.168.10.1

---

**Answer: C**

---

Explanation:

---

**Question: 1003**

---

Which type of DNS record is used to specify the mail server responsible for accepting email messages on behalf of a recipient's domain?

- A. MX record
- B. TXT record
- C. A record
- D. SRV record

---

**Answer: C**

---

Explanation:

---

**Question: 1004**

---

What is a function of a southbound API?

- A. Automate configuration changes between a server and a switching fabric.
- B. Use orchestration to provision a virtual server configuration from a web server.
- C. Manage flow control between an SDN controller and a switching fabric.
- D. Facilitate the information exchange between an SDN controller and application.

---

**Answer: C**

---

Explanation:

---

**Question: 1005**

---

What is the purpose of the URI string in a REST request?

- A. To identify a resource on a target server.
- B. To transport data or payload to a remote resource.
- C. To specify the way in which a remote resource is modified.
- D. To respond with the data content encoding for a request.

---

**Answer: C**

---

Explanation:

---

**Question: 1006**

---

What is a functionality of the control plane in the network?

- A. It exchanges topology information with other routers.
- B. It forwards traffic to the next hop.
- C. It looks up an egress interface in the forwarding information base.
- D. It provides CLI access to the network device.

---

**Answer: A**

---

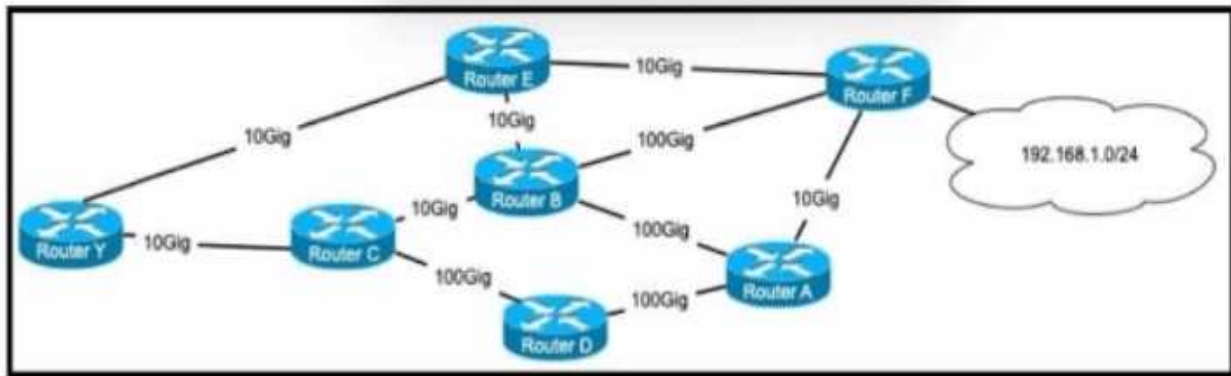
Explanation:

---

**Question: 1007**

---

Refer to the exhibit.



Refer to the exhibit. With a reference bandwidth of 100 Gb on all routers, which path does router Y use to get to network 192.168.1.0/24?

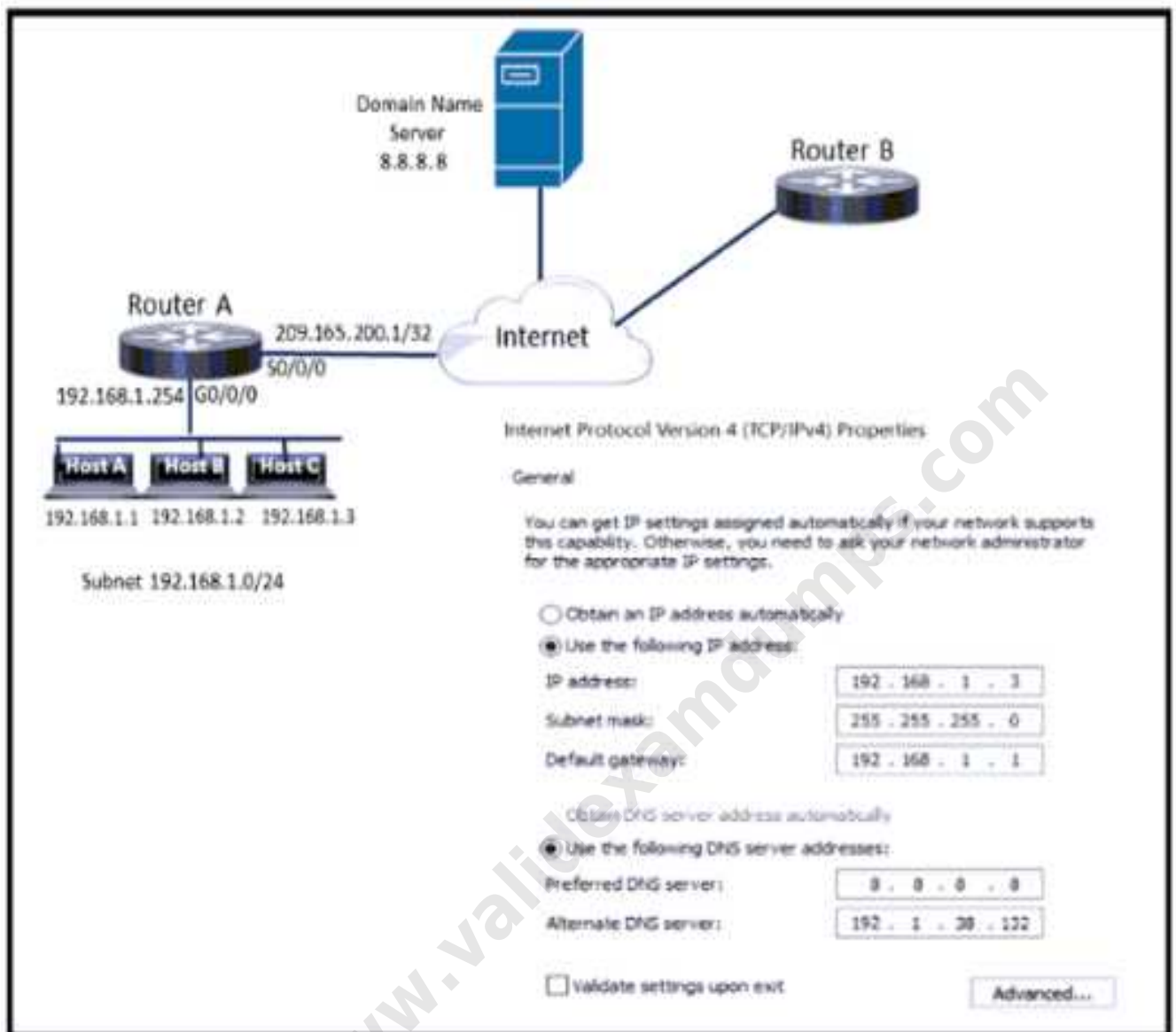
- A. router C > D > A > B > F
- B. router C > D > A > F
- C. router E > F
- D. router E > B > F

**Answer: A**

Explanation:

**Question: 1008**

Refer to the exhibit.



Which configuration parameter is preventing host C from reaching the internet?

- A. automatic DNS
- B. default gateway
- C. IP network mask
- D. IP address assignment

**Answer: B**

Explanation:

---

**Question: 1009**

---

Which part of the HTTP request is used in REST API basic authentication to carry credentials?

- A. cookie
- B. content-type
- C. authorization
- D. accept

---

**Answer: D**

---

Explanation:

---

**Question: 1010**

---

A user configured OSPF and advertised the Gigabit Ethernet interface in OSPF. By default, to which type of OSPF network does this interface belong?

- A. point-to-point
- B. non-broadcast
- C. point-to-multipoint
- D. broadcast

---

**Answer: D**

---

Explanation:

---

**Question: 1011**

---

A manager asks a network engineer to recommend a cloud service model so that employees do not spend time installing, managing, and updating software that is only used occasionally. Which cloud service model does the engineer recommend?

- A. Infrastructure-as-a-service

- B. software-as-a-service
- C. business process as a service
- D. platform-as-a-service

---

**Answer: B**

---

Explanation:

---

**Question: 1012**

---

Refer to the exhibit.

```
Designated Router (ID) 10.11.11.11, Interface address 10.10.10.1
Backup Designated router (ID) 10.3.3.3, Interface address 10.10.10.3
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
oob-resync timeout 40
Hello due in 00:00:08
Supports Link-local Signaling (LLS)
Cisco NSF helper support enabled
IETF NSF helper support enabled
Index 1/1/1, flood queue length 0
Next 0x0(0)/0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 6
Last flood scan time is 0 msec, maximum is 1 msec
Neighbor Count is 3, Adjacent neighbor count is 3
Adjacent with neighbor 10.1.1.4
Adjacent with neighbor 10.2.2.2
Adjacent with neighbor 10.3.3.3 (Backup Designated Router)
Suppress hello for 0 neighbor(s)
```

The show ip ospf interface command has been executed on R1. How is OSPF configured?

- A. A point-to-point network type is configured.
- B. The default Hello and Dead timers are in use.

- C. There are six OSPF neighbors on this interface.
- D. The interface is not participating in OSPF.

---

**Answer: B**

---

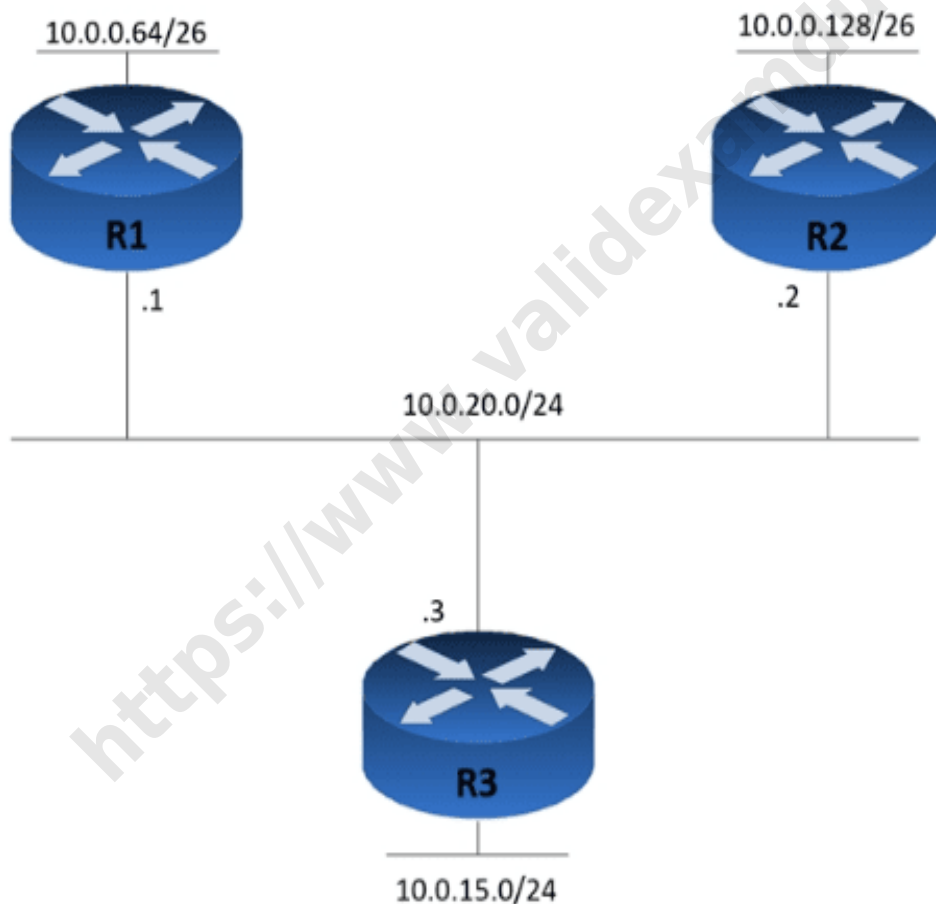
Explanation:

---

**Question: 1013**

---

Refer to the exhibit.



Router R1 is added to the network and configured with the 10.0.0.64/26 and 10.0.20.0/24 subnets. However, traffic destined for the LAN on R3 is not accessible. Which command when executed on R1 defines a static route to reach the R3 LAN?

- A. ip route 10.0.15.0 255.255.255.0 10.0.20.3
- B. ip route 10.0.15.0 255.255.255.0 10.0.20.1
- C. ip route 10.0.0.64 255.255.255.192 10.0.20.3
- D. ip route 10.0.15.0 255.255.255.192 10.0.20.1

---

**Answer: A**

---

Explanation:

---

**Question: 1014**

---

What is the benefit of using FHRP (First Hop Redundancy Protocol)?

- A. Reduced ARP traffic on the network
- B. Balancing traffic across multiple gateways in proportion to their loads
- C. Reduced management overhead on network routers
- D. Higher degree of availability

---

**Answer: D**

---

Explanation:

---

**Question: 1015**

---

Which group of channels in the 802.11b/g/n/ax 2.4 GHz frequency bands are non-overlapping channels?

- A. channels 1, 5, and 10
- B. channels 1, 5, and 11
- C. channels 1, 6, and 10
- D. channels 1, 6, and 11



---

**Answer: D**

---

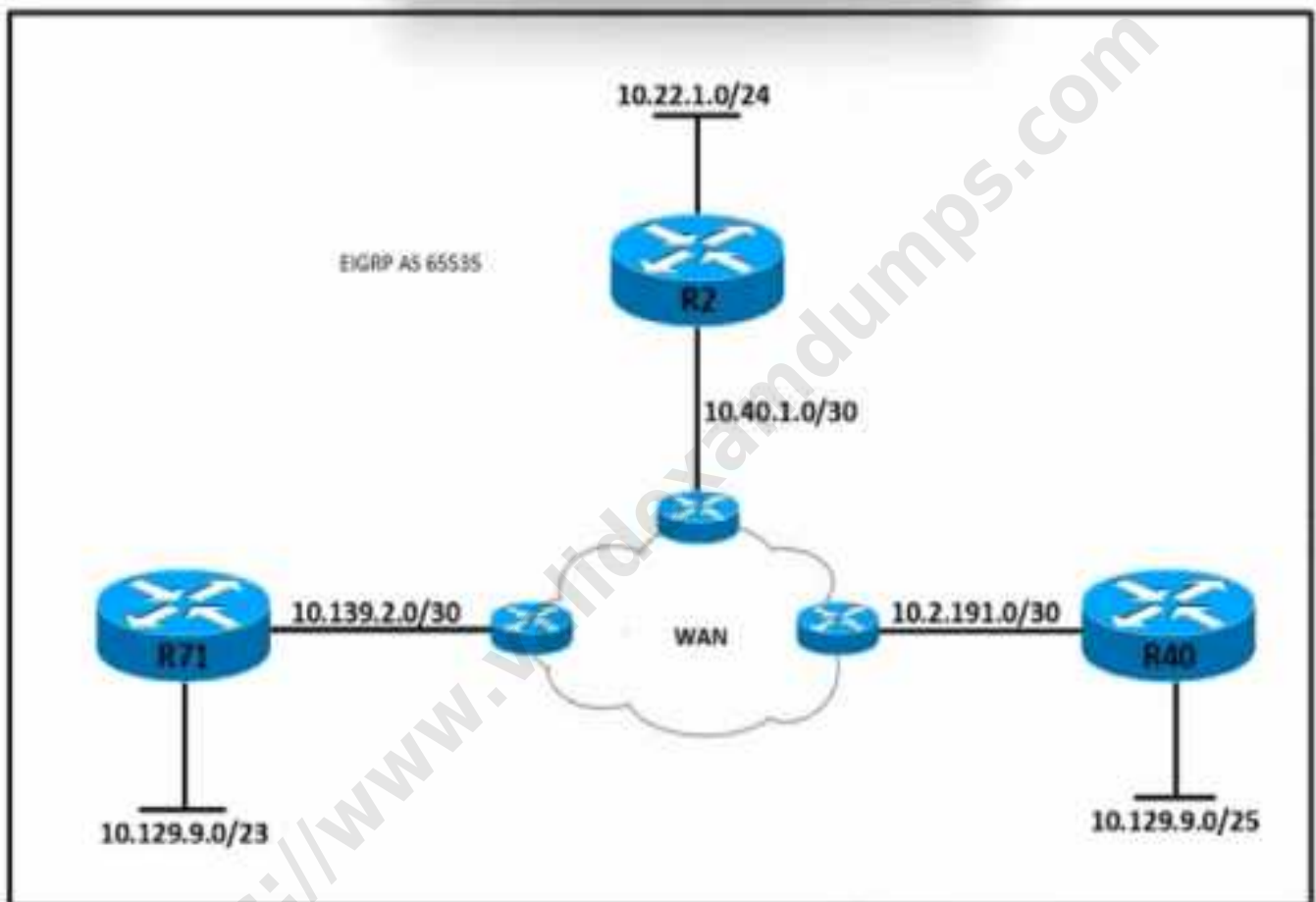
Explanation:

---

**Question: 1016**

---

Refer to the exhibit.



Refer to the exhibit. All routers in the network are configured correctly, and the expected routes are being exchanged among the routers. Which set of routes are learned from neighbors and installed on router 2?

- A. 10.129.9.0/23 10.139.2.0/30 10.2.191.0/30 10.129.9.0/25
- B. 10.129.9.0/23 10.40.1.0/30 10.2.191.0/30 10.129.9.0/25
- C. 10.40.1.0/30 10.139.2.0/30 10.2.191.0/30 10.129.9.0/25
- D. 10.129.9.0/23 10.139.2.0/30 10.129.9.0/25 10.22.1.0/24

---

**Answer: A**

---

Explanation:

---

**Question: 1017**

---

Refer to the exhibit.

```
R1# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate
       default
       U - per-user static route, o - ODR
Gateway of last resort is not set
C    10.0.0.0/8 is directly connected, Loopback0
     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O    10.0.1.3/32 [110/100] via 10.0.1.100, 00:39:08, Serial0
C    10.0.1.0/24 is directly connected, Serial0
O    10.0.1.5/32 [110/5] via 10.0.1.50, 00:39:08, Gigabit Ethernet 0/0
D    10.0.1.4/32 [110/10] via 10.0.1.4, 00:39:08, Gigabit Ethernet 0/0
```

What does route 10.0.1.3/32 represent in the routing table?

- A. a single destination address
- B. the source 10.0.1.100
- C. all hosts in the 10.0.1.0 subnet
- D. the 10.0.0.0 network

---

**Answer: A**

---

Explanation:

---

**Question: 1018**

---

Refer to the exhibit.

```
R1# show ip route

D    192.168.16.0/26 [90/2679326] via 192.168.1.1
R    192.168.16.0/24 [120/3] via 192.168.1.2
O    192.168.16.0/21 [110/2] via 192.168.1.3
i L1 192.168.16.0/27 [115/30] via 192.168.1.4
```

Which route does R1 select for traffic that is destined to 192.168.16.2?

- A. 192.168.16.0/21
- B. 192.168.16.0/24
- C. 192.168.16.0/26
- D. 192.168.16.0/27

---

**Answer: D**

---

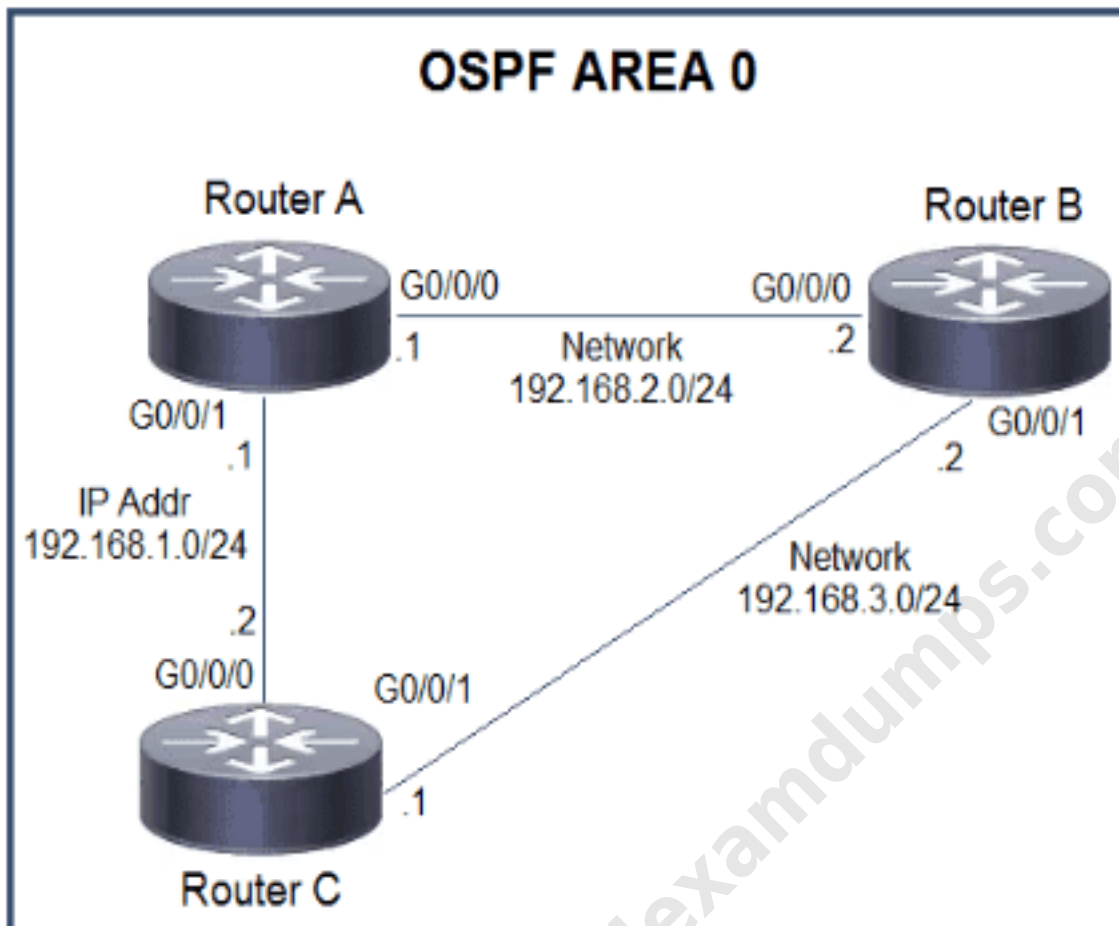
Explanation:

---

**Question: 1019**

---

Refer to the exhibit.



Which action must be taken to ensure that router A is elected as the DR for OSPF area 0?

- A. Configure the router A interfaces with the highest OSPF priority value within the area.
- B. Configure router B and router C as OSPF neighbors of router A.
- C. Configure the OSPF priority on router A with the lowest value between the three routers.
- D. Configure router A with a fixed OSPF router ID.

**Answer: A**

Explanation:

### Question: 1020

A network engineer is configuring a switch so that it is remotely reachable via SSH. The engineer has already configured the host name. Which additional command must the engineer configure before

entering the command to generate the RSA key?

- A. ip domain-name domain
- B. password password
- C. crypto key generate rsa modulus 1024
- D. ip ssh authentication-retries 2

---

**Answer: A**

---

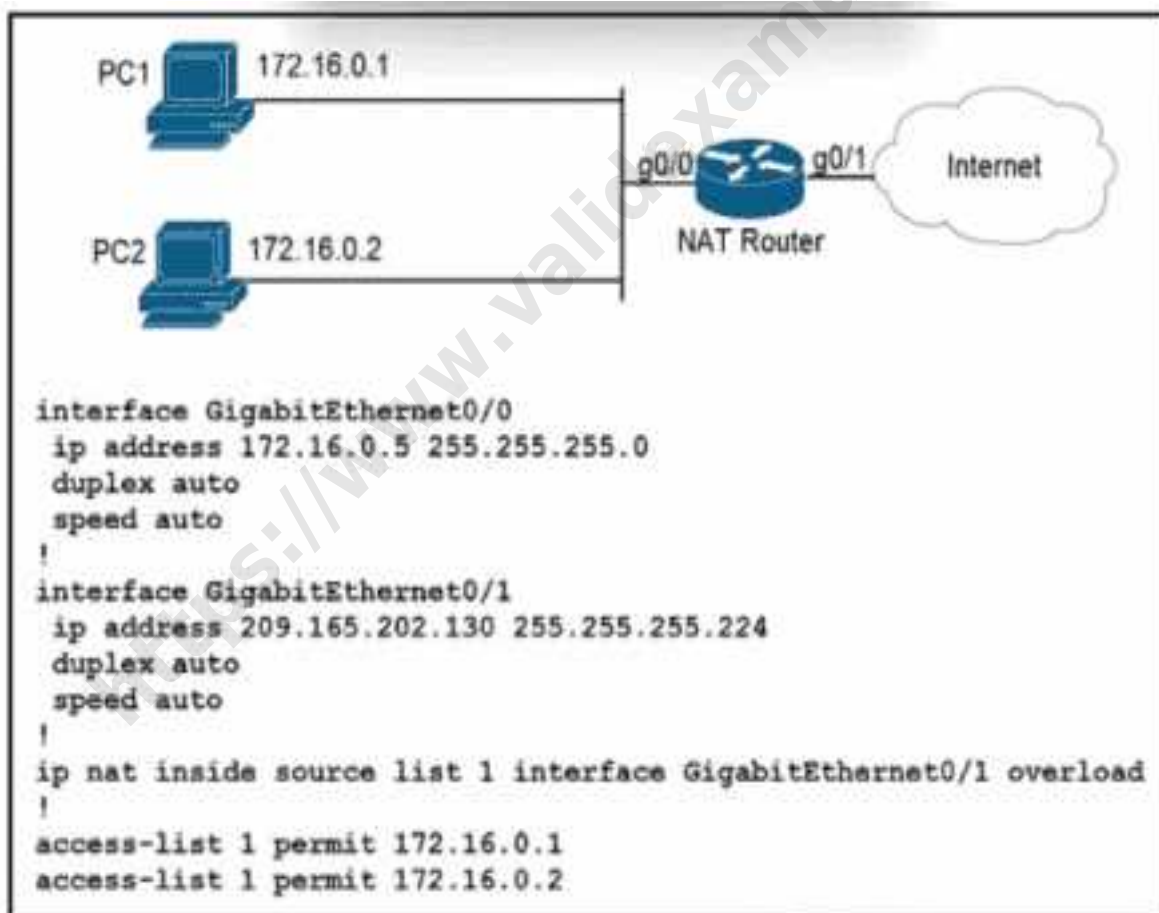
Explanation:

---

**Question: 1021**

---

Refer to the exhibit.



Refer to the exhibit. How should the configuration be updated to allow PC1 and PC2 access to the Internet?

- A. Change the ip nat inside source command to use interface GigabitEthernet0/0.

- B. Remove the overload keyword from the ip nat inside source command.
- C. Modify the configured number of the second access list.
- D. Add the ip nat inside and ip nat outside commands under both interfaces.

---

**Answer: A**

---

Explanation:

---

**Question: 1022**

---

Refer to the exhibit.

```
Router-WAN1#show interface g0/0
GigabitEthernet0/0 is up, line protocol is up
  Hardware is CSM NIC, address is 3000.0001.0000 (bia 3000.0001.0000)
  Internet address is 192.168.0.0/31
  MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full Duplex, 1000Mbps, link type is auto, media type is NIC
  output flow-control is unsupported, input flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input never, output 00:00:03, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate: 1000 bits/sec, 0 packets/sec
  5 minute output rate: 2000 bits/sec, 1 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 110 broadcasts (0 IP multicasts)
    0 runts, 0 giants, 0 throttles
    100 input errors, 100 CRC, 100 frame, 0 overrun, 0 ignored
    0 watchdog, 0 multicast, 0 pause input
    260 packets output, 49070 bytes, 0 underruns
    Output 0 broadcasts (0 IP multicasts)
    0 output errors, 100 collisions, 0 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    1 lost carrier, 0 no carrier, 0 pause output
```

Refer to the exhibit. Router-WAN1 has a new connection via Gi0/0 to the ISP. Users running the web applications indicate that connectivity is unstable to the internet. What is causing the interface issue?

- A. Small frames less than 64 bytes are rejected due to size.
- B. The receive buffer is full due to a broadcast storm.
- C. Frames are discarded due to a half-duplex negotiation.
- D. Broadcast packets are rejected because ARP timeout is enabled.

---

**Answer: C**

---

Explanation:

---

**Question: 1023**

---

Which architecture is best for small offices with minimal wireless needs and no central management?

- A. cloud-based AP
- B. split MAC
- C. autonomous AP
- D. mesh network

---

**Answer: C**

---

Explanation:

---

**Question: 1024**

---

Refer to the exhibit.

**["red", "one"]**

Which type of JSON data is represented?

- A. string
- B. array
- C. number

D. object

---

**Answer: B**

---

Explanation:

---

**Question: 1025**

---

Refer to the exhibit.

```
R_1# show ip route
.....
D   192.168.20.0/26 [90/24513456] via 10.10.10.1
R   192.168.20.0/24 [120/5] via 10.10.10.2
O   192.168.0.0/19 [110/219414] via 10.10.10.13
B   192.168.0.0/16 is variably subnetted, 4 subnets, 4 masks
D   192.168.20.0/27 [90/4123710] via 10.10.10.12
D   192.168.20.0/25 [90/14464211] via 10.10.10.11
S.  0.0.0.0/0 [1/0] via 10.10.10.14
```

Packets are flowing from 192.168.10.1 to the destination at IP address 192.168.20.75. Which next hop will the router select for the packet?

- A. 10.10.10.1
- B. 10.10.10.11
- C. 10.10.10.12
- D. 10.10.10.14

---

**Answer: B**

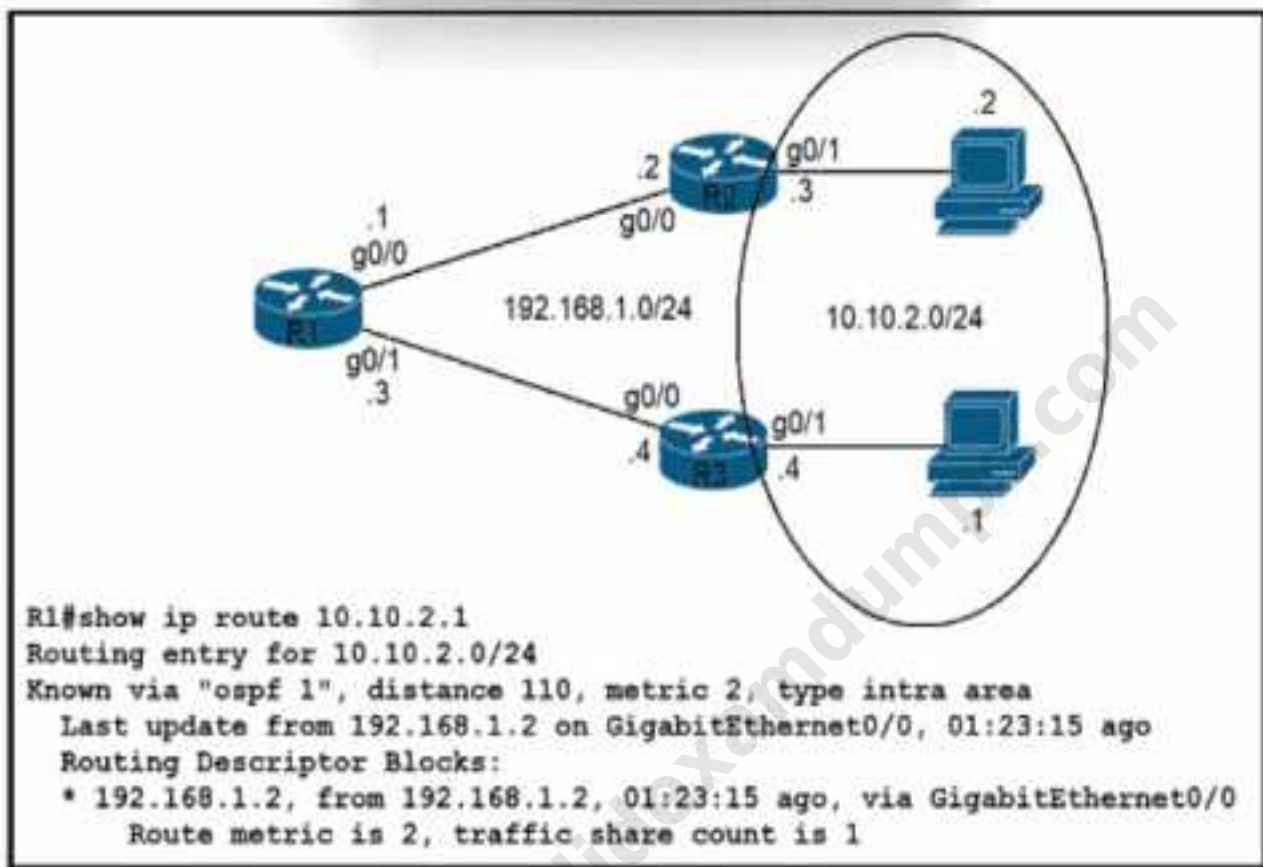
---

Explanation:



**Question: 1026**

Refer to the exhibit.



Refer to the exhibit. Traffic from R1 to the 10.10.2.0/24 subnet uses 192.168.1.2 as its next hop. A network engineer wants to update the R1 configuration so that traffic with destination 10.10.2.1 passes through router R3, and all other traffic to the 10.10.2.0/24 subnet passes through R2.

Which command must be used?

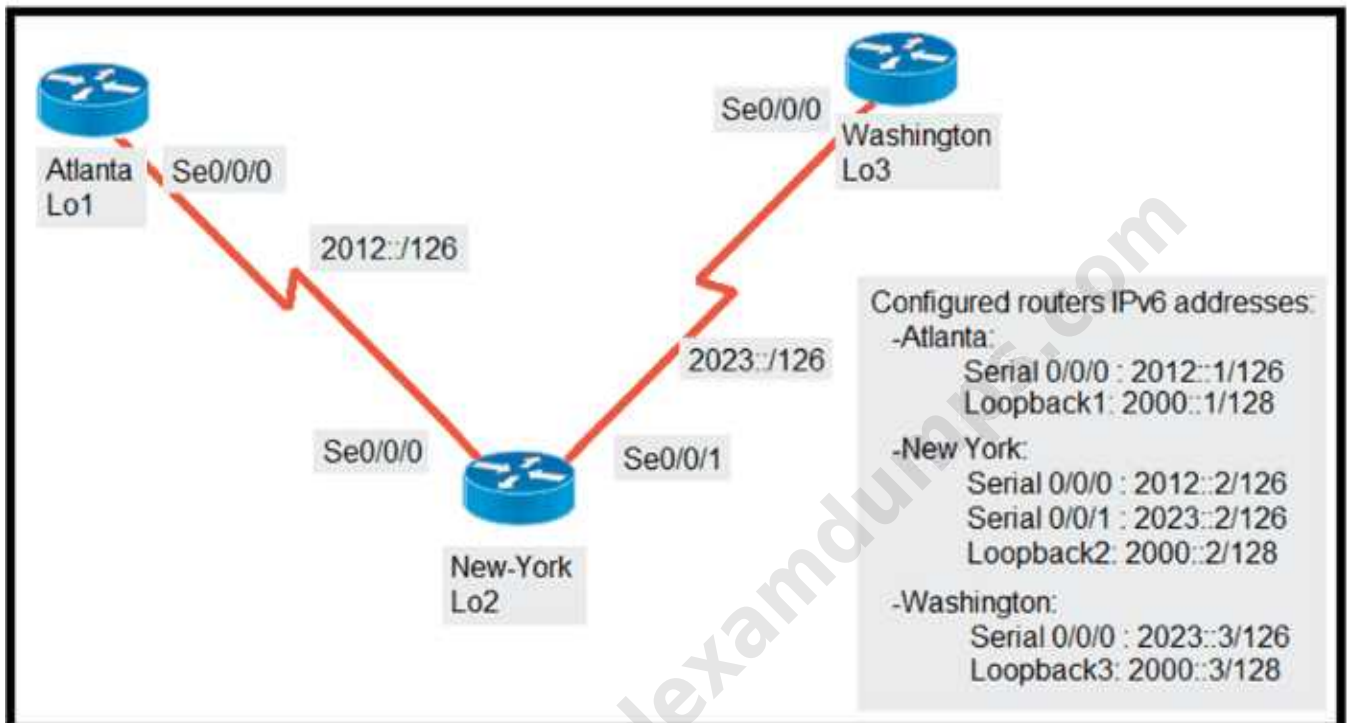
- A. ip route 10.10.2.0 255.255.255.0 192.168.1.4 100
- B. ip route 10.10.2.0 255.255.255.0 192.168.1.4 115
- C. ip route 10.10.2.1 255.255.255.255 192.168.1.4 100
- D. ip route 10.10.2.1 255.255.255.255 192.168.1.4 115

**Answer: D**

Explanation:

**Question: 1027**

Refer to the exhibit.



The New York router is configured with static routes pointing to the Atlanta and Washington sites. Which two tasks must be performed so that the Se0/0/0 interfaces on the Atlanta and Washington routers reach one another? (Choose two.)

- A. Configure the ipv6 route 2023::/126 2012::2 command on the Atlanta router.
- B. Configure the ipv6 route 2012::/126 2023::1 command on the Washington router.
- C. Configure the ipv6 route 2023::/126 2012::1 command on the Atlanta router.
- D. Configure the ipv6 route 2012::/126 s0/0/0 command on the Atlanta router.
- E. Configure the ipv6 route 2012::/126 2023::2 command on the Washington router.

**Answer: A,D**

Explanation:

**Question: 1028**

Why is UDP more suitable than TCP for applications that require low latency, such as VoIP?

- A. TCP uses congestion control for efficient packet delivery, and UDP uses flow control mechanisms for the delivery of packets.
- B. TCP sends an acknowledgment for every packet that is received, and UDP operates without acknowledgments.
- C. UDP reliably guarantees delivery of all packets, and TCP drops packets under heavy load.
- D. UDP uses sequencing data for packets to arrive in order, and TCP offers the capability to receive packets in random order.

---

**Answer: B**

---

Explanation:

---

**Question: 1029**

---

Which mechanism carries multicast traffic between remote sites and supports encryption?

- A. GRE over IPsec
- B. IPsec over ISATAP
- C. GRE
- D. ISATAP

---

**Answer: A**

---

Explanation:

---

**Question: 1030**

---

The address block 192.168.32.0/24 must be subnetted into smaller networks. The engineer must meet these requirements:

Create 8 new subnets

Each subnet must accommodate 30 hosts

Interface VLAN 10 must use the last usable IP in the first new subnet.

A Layer 3 interface is used.

Which configuration must be applied to the interface?

- A. no switchport ip address 192.168.32.30 255.255.255.224
- B. no switchport mode trunk ip address 192.168.32.97 255.255.255.224
- C. no switchport mode access ip address 192.168.32.62 255.255.255.240
- D. switchport ip address 192.168.32.65 255.255.255.240

---

**Answer: A**

---

Explanation:

---

**Question: 1031**

---

What are two facts that differentiate optical fiber cabling from copper cabling? (Choose two.)

- A. It has a greater sensitivity to changes in temperature and moisture.
- B. It carries signals for longer distances.
- C. It carries electrical current further distances for PoE devices.
- D. It is less expensive when purchasing patch cables.
- E. It provides greater throughput options.

---

**Answer: A,B**

---

Explanation:

---

**Question: 1032**

---

DRAG DROP

Drag and drop the TCP and UDP characteristics from the left onto the corresponding protocols on the right.

## Answer Area

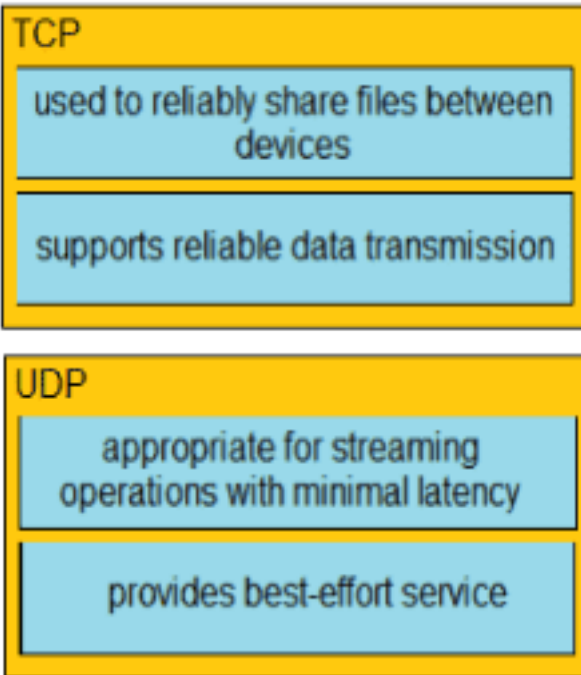
|                                                           | TCP |
|-----------------------------------------------------------|-----|
| used to reliably share files between devices              |     |
| appropriate for streaming operations with minimal latency |     |

|                                     | UDP |
|-------------------------------------|-----|
| provides best-effort service        |     |
| supports reliable data transmission |     |

**Answer:**

Explanation:



TCP: Sends data in a specific order, Requires an established connection, Supports web browsing

UDP: Suited for live streaming, Retransmission is unsupported, Tolerates packet loss

---

**Question: 1033**

---

Refer to the exhibit.



The router has been configured with a subnet to accommodate the requirement for 380 users on a subnet. The requirement already considers 30% future growth. Which configuration verifies the IP subnet on router R4?

A. Subnet: 10.7.54.0 Subnet mask: 255.255.254.0 Broadcast address: 10.7.54.255 Usable IP address range: 10.7.54.1 - 10.7.55.254

B. Subnet: 10.7.54.0 Subnet mask: 255.255.128.0 Broadcast address: 10.7.55.255 Usable IP address range: 10.7.54.1 - 10.7.55.254

C. Subnet: 10.7.54.0 Subnet mask: 255.255.255.0 Broadcast address: 10.7.54.255 Usable IP address range: 10.7.54.1 - 10.7.55.254

D. Subnet: 10.7.54.0 Subnet mask: 255.255.254.0 Broadcast address: 10.7.55.255 Usable IP address range: 10.7.54.1 - 10.7.55.254

---

**Answer: D**

---

Explanation:

---

**Question: 1034**

---

Refer to the exhibit.

```
R1# show ip route | begin Gateway
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
S* 0.0.0.0/0 is directly connected, Serial0/0/1
    172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C    172.16.2.0/24 is directly connected, GigabitEthernet0/0
L    172.16.2.2/32 is directly connected, GigabitEthernet0/0
C    172.16.4.0/21 is directly connected, Serial0/0/1
L    172.16.8.2/26 is directly connected, Serial0/0/1
```

What is the subnet mask for route 172.16.4.0?

A. 255.255.255.192

B. 255.255.254.0

C. 255.255.240.0

D. 255.255.248.0

---

**Answer: D**

---

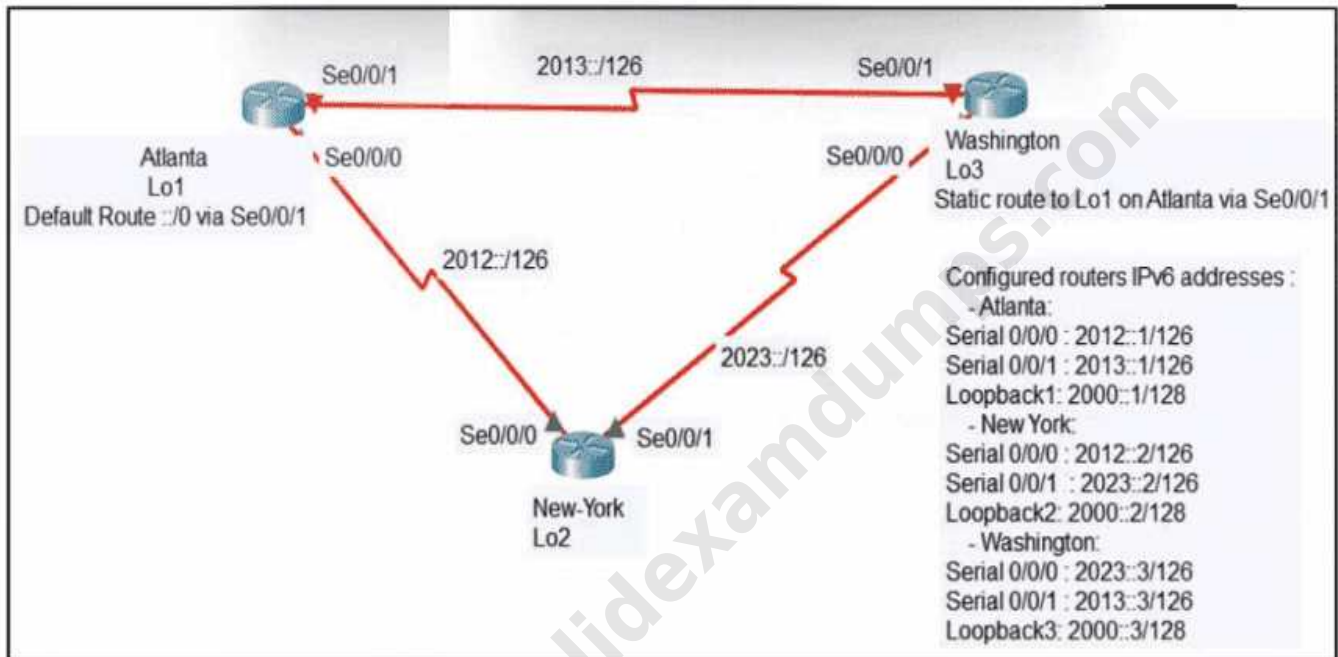
Explanation:

---

**Question: 1035**

---

Refer to the exhibit.



The New York router must be configured so that traffic to 2000::1 is sent primarily via the Atlanta site, with a secondary path via Washington that has an administrative distance of 2. Which two commands must be configured on the New York router? (Choose two.)

- A. ipv6 route 2000::1/128 2012::1
- B. ipv6 route 2000::1/128 2023::2 5
- C. ipv6 route 2000::1/128 2012::1 5
- D. ipv6 route 2000::1/128 2023::3 2
- E. ipv6 route 2000::1/128 2012::2

---

**Answer: D,E**

---

Explanation:



---

**Question: 1036**

---

What is a valid IPv6 address record in DNS?

- A. A
- B. MX
- C. AAAA
- D. CNAME

---

**Answer: C**

---

Explanation:

---

**Question: 1037**

---

A network engineer must configure the router R1 GigabitEthernet1/1 interface to connect to the router R2 GigabitEthernet1/1 interface. For the configuration to be applied, the engineer must compress the address 2001:0db8:0000:0000:0500:000a:400F:583B. Which command must be issued on the interface?

- A. ipv6 address 2001:db8 :: 500:a:400F:583B
- B. ipv6 address 2001:0db8 :: 5:a:4F:583B
- C. ipv6 address 2001 :: db8:0000 :: 500:a:400F:583B
- D. ipv6 address 2001:db8:0 :: 500:a:4F:583B

---

**Answer: A**

---

Explanation:

---

**Question: 1038**

---

Which event has occurred if a router sends a notice level message to a syslog server?

- A. An ICMP connection has been built

- B. A certificate has expired
- C. An interface line has changed status
- D. A TCP connection has been torn down

---

**Answer: C**

---

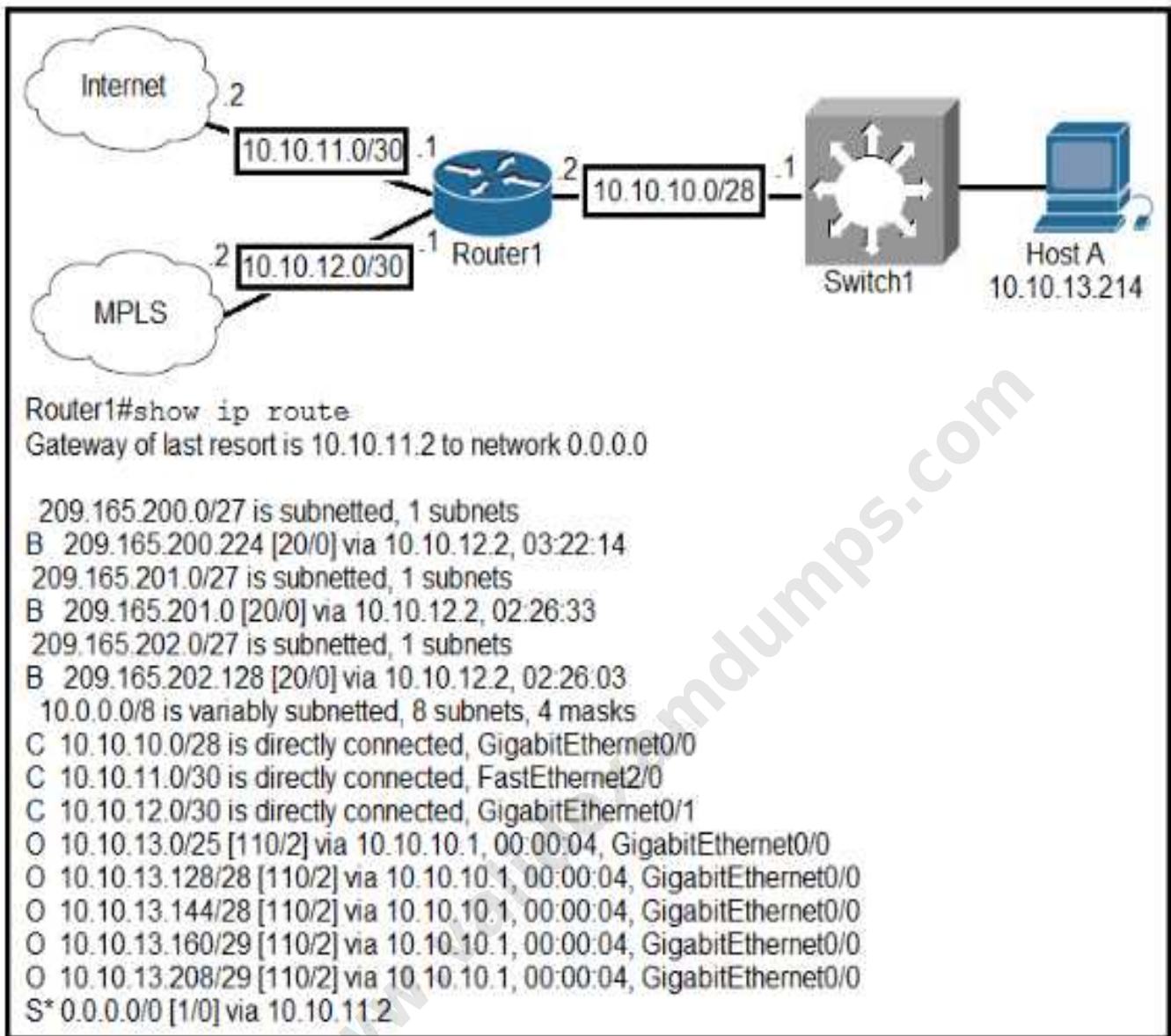
Explanation:

---

**Question: 1039**

---

Refer to the exhibit.



What is the subnet mask of the route to the 10.10.13.160 prefix?

- A. 255.255.255.240
- B. 255.255.255.128
- C. 255.255.255.248
- D. 255.255.248.0

**Answer: C**

Explanation:

---

**Question: 1040**

---

What is a DHCP client?

- A. a server that dynamically assigns IP addresses to hosts
- B. a router that statically assigns IP addresses to hosts
- C. a host that is configured to request an IP address automatically
- D. a workstation that requests a domain name associated with its IP address

---

**Answer: C**

---

Explanation:

---

**Question: 1041**

---

Refer to the exhibit.

The screenshot shows a network configuration interface with the following sections:

- General** | **Security** | **QoS** | **Policy-Mapping** | **Advanced**
- Layer 2** | **Layer 3** | **AAA Servers**
- Layer 2 Security** 6: WPA+WPA2 (dropdown)
- MAC Filtering 9: ☐
- Fast Transition**
- Fast Transition: ☐
- Protected Management Frame**
- PMF: Required (dropdown)
- WPA+WPA2 Parameters**
- WPA Policy: ☐
- WPA2 Policy: ☐
- WPA2 Encryption: ☐ AES ☐ TKIP
- OSN Policy: ☐
- Authentication Key Management** 19
- 802.1X: ☐ Enable
- CCKM: ☐ Enable
- PSK: ☐ Enable
- FT 802.1X: ☐ Enable
- FT PSK: ☐ Enable
- PSK Format: ASCII (dropdown)
- WPA gtk-randomize State 14: Disable (dropdown)

The network engineer is configuring a new WLAN and is told to use a static password for authentication instead of the RADIUS servers. Which additional set of tasks must the engineer perform to complete the configuration?

- A. Crable PSKEnable iod 1s
- B. Select WPA2 PolicyDisable PMFEnable PSK
- C. Select WPA PolicyEnable PSK
- D. Select WPA PolicySelect WRAS Policy

---

**Answer: B**

---

Explanation:

---

**Question: 1042**

---

What are two benefits of network automation? (Choose two.)

- A. Faster changes with more reliable results
- B. Reduced operational costs
- C. Fewer network failures
- D. Reduced hardware footprint
- E. Increased network security

---

**Answer: A,B**

---

Explanation:

---

**Question: 1043**

---

An engineer observes high usage on the 2.4GHz channels and lower usage on the 5GHz channels. What must be configured to allow clients to preferentially use 5GHz access points?

- A. Re-Anchor Roamed Clients
- B. OEAP Split Tunnel
- C. 11ac MU-MIMO
- D. Client Band Select

---

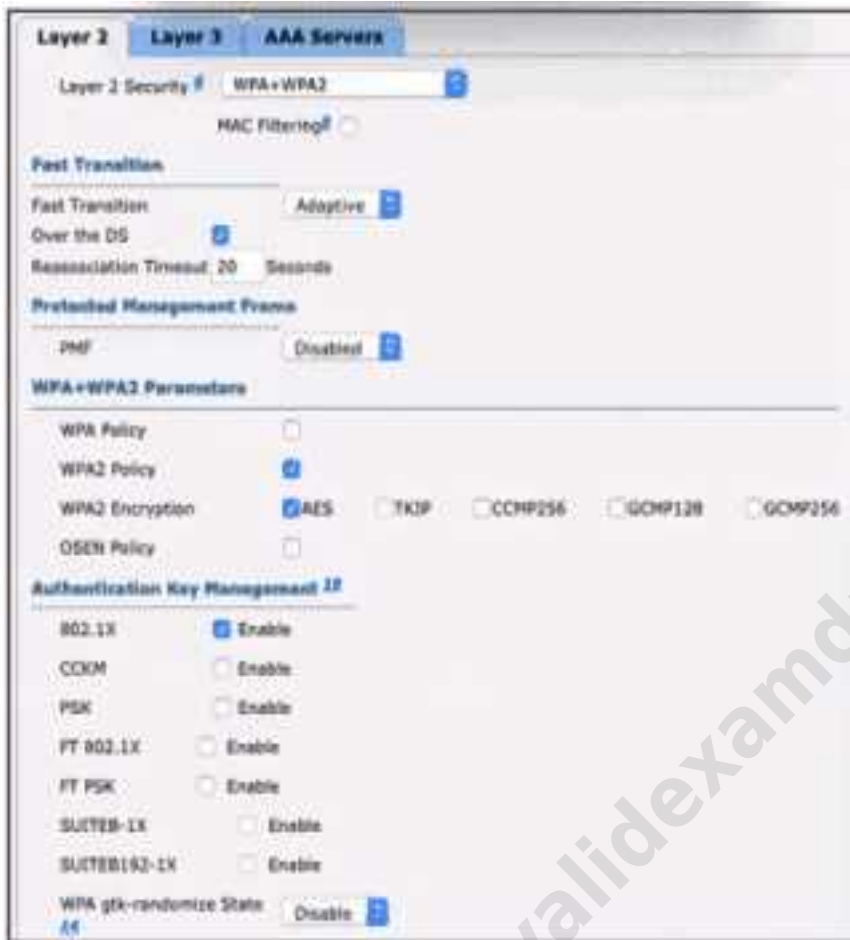
**Answer: D**

---

Explanation:

**Question: 1044**

Refer to the exhibit.



Refer to the exhibit. Clients on the WLAN are required to use 802.11r. What action must be taken to meet the requirement?

- A. Set the Fast Transition option to Enable and enable FT 802.1X under Authentication Key Management
- B. Set the Fast Transition option and the WPA gtk-randomize State to disable
- C. Under Protected Management Frames, set the PMF option to Required
- D. Enable CCKM under Authentication Key Management

**Answer: A**

Explanation:

**Question: 1045**

What are two characteristics of a controller-based network? (Choose two.)

- A. It uses Telnet to report system issues
- B. It uses northbound and southbound APIs to communicate between architectural layers
- C. It decentralizes the control plane, which allows each device to make its own forwarding decisions
- D. It moves the control plane to a central point
- E. The administrator can make configuration updates from the CLI

---

**Answer: B,D**

---

Explanation:

---

**Question: 1046**

---

Refer to the exhibit.

```
R1# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default
       U - per-user static route, o - ODR
Gateway of last resort is not set
C    172.16.0.0/16 is directly connected, Loopback0
     172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
O    172.16.1.3/32 [110/100] via 10.0.1.100, 00:39:08, Serial0
O    172.16.1.9/32 [110/5] via 172.16.1.50, 00:43:01, Gigabit Ethernet 0/0
D    172.16.1.4/30 [90/7445] via 172.16.9.5, 00:39:08, Gigabit Ethernet 0/0
     [90/7445] via 172.16.4.4, 00:39:08, Gigabit Ethernet 0/4
```

Refer to the exhibit. How does router R1 handle traffic to the 172.16.1.4/30 subnet?

- A. It sends all traffic over the path via 172.16.9.5 using 172.16.4.4 as a backup.
- B. It sends all traffic over the path via 172.16.4.4.
- C. It load-balances traffic over 172.16.9.5 and 172.16.4.4.
- D. It sends all traffic over the path via 10.0.1.100

---

**Answer: C**

---



Explanation:

---

**Question: 1047**

---

A switch is forwarding a frame out of all interfaces except the interface that received the frame.  
What is the technical term for this process?

- A. ARP
- B. Multicast
- C. CDP
- D. Flooding

---

**Answer: D**

---

Explanation:

---

**Question: 1048**

---

Refer to the exhibit.

```
cat9k-acc-1# show interfaces gigabitethernet 1/0/1
gigabitethernet 1/0/1 is up, line protocol is up
Hardware is gigabitethernet, address is aa00.0400.0134 (via 0000.0c00.4369)
MTU 1500 bytes, BW 1000 Kbit, DLY 1000 usec, rely 255/255, load 1/255
Encapsulation ARPA, loopback not set, keepalive set (10 sec)
ARP type: ARPA, PROBE, ARP Timeout 4:00:00
Last input 0:00:00, output 0:00:00, output hang never
Output queue 1/1, 1 drops; input queue 0/0, 0 drops
Five minute input rate 61000 bits/sec, 200 packets/sec
Five minute output rate 1000 bits/sec, 200 packets/sec
2295197 packets input, 305539992 bytes, 0 no buffer
Received 1925500 broadcasts, 0 runts, 0 giants
0 input errors, 1790 CRC, 1790 frame, 0 overrun, 0 ignored, 0 abort
0 input packets with dribble condition detected
3594664 packets output, 436549843 bytes, 1 underruns
0 output errors, 1 collisions, 1 interface resets, 0 restarts
```

The switch cat9k acc.1 connects users to the campus LAN. Printing services are inaccessible through the network. Which interface issue is causing the connectivity problems?

- A. A large number of broadcast packets are resulting in a port reset
- B. Excessive collisions are causing dropped frames.
- C. The interface output queue cannot process the Ethernet frames.
- D. A bad checksum is causing Ethernet frames to drop.

---

**Answer: D**

---

Explanation:

---

**Question: 1049**

---

Which plane is centralized in software-defined networking (SDN)?

- A. Data
- B. Control
- C. Application
- D. Services

---

**Answer: B**

---

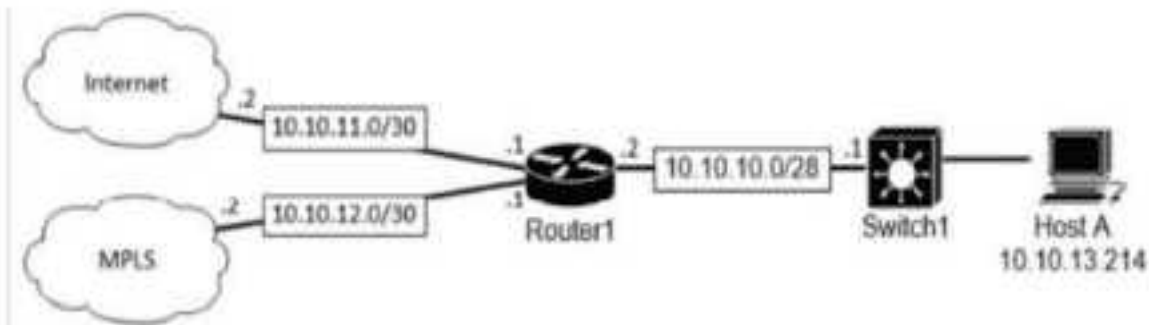
Explanation:

---

**Question: 1050**

---

Refer to the exhibit.



```

Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0

    209.165.200.0/27 is subnetted, 1 subnets
B       209.165.200.224 [20/0] via 10.10.12.2, 03:22:14
    209.165.201.0/27 is subnetted, 1 subnets
B       209.165.201.0 [20/0] via 10.10.12.2, 02:26:33
    209.165.202.0/27 is subnetted, 1 subnets
B       209.165.202.128 [20/0] via 10.10.12.2, 02:26:03
    10.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
C       10.10.10.0/28 is directly connected, GigabitEthernet0/0
C       10.10.11.0/30 is directly connected, FastEthernet2/0
C       10.10.12.0/30 is directly connected, GigabitEthernet0/1
O       10.10.13.0/25 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O       10.10.13.128/28 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O       10.10.13.144/28 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O       10.10.13.160/29 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O       10.10.13.208/29 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
S*     0.0.0.0/0 [1/0] via 10.10.11.2
  
```

What is the prefix length for the route that Router1 will use to reach Host A?

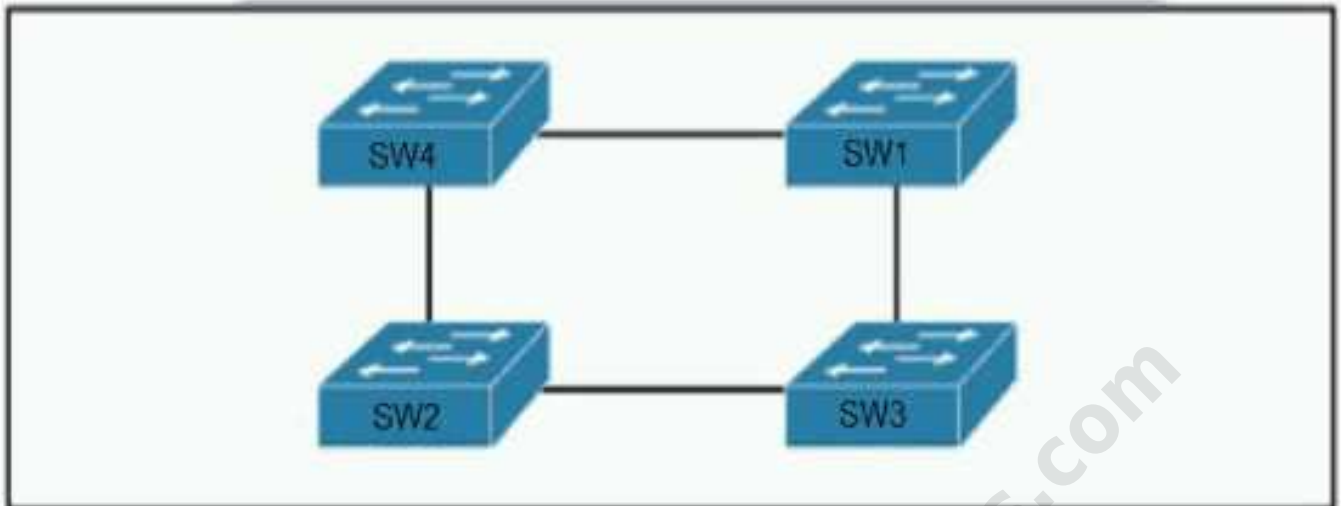
- A. /25
- B. /27
- C. /28
- D. /29

**Answer: D**

Explanation:

**Question: 1051**

Refer to the exhibit.



Given the following switch MAC addresses, which switch will be elected as the root bridge?

SW1: 0C:5A:05:53:27:17

SW2: 0C:0A:A8:1A:3C:9D

SW3: 0C:0A:18:81:83:19

SW4: 0C:0A:05:22:05:97

A. SW1

B. SW2

C. SW3

D. SW4

---

**Answer: C**

---

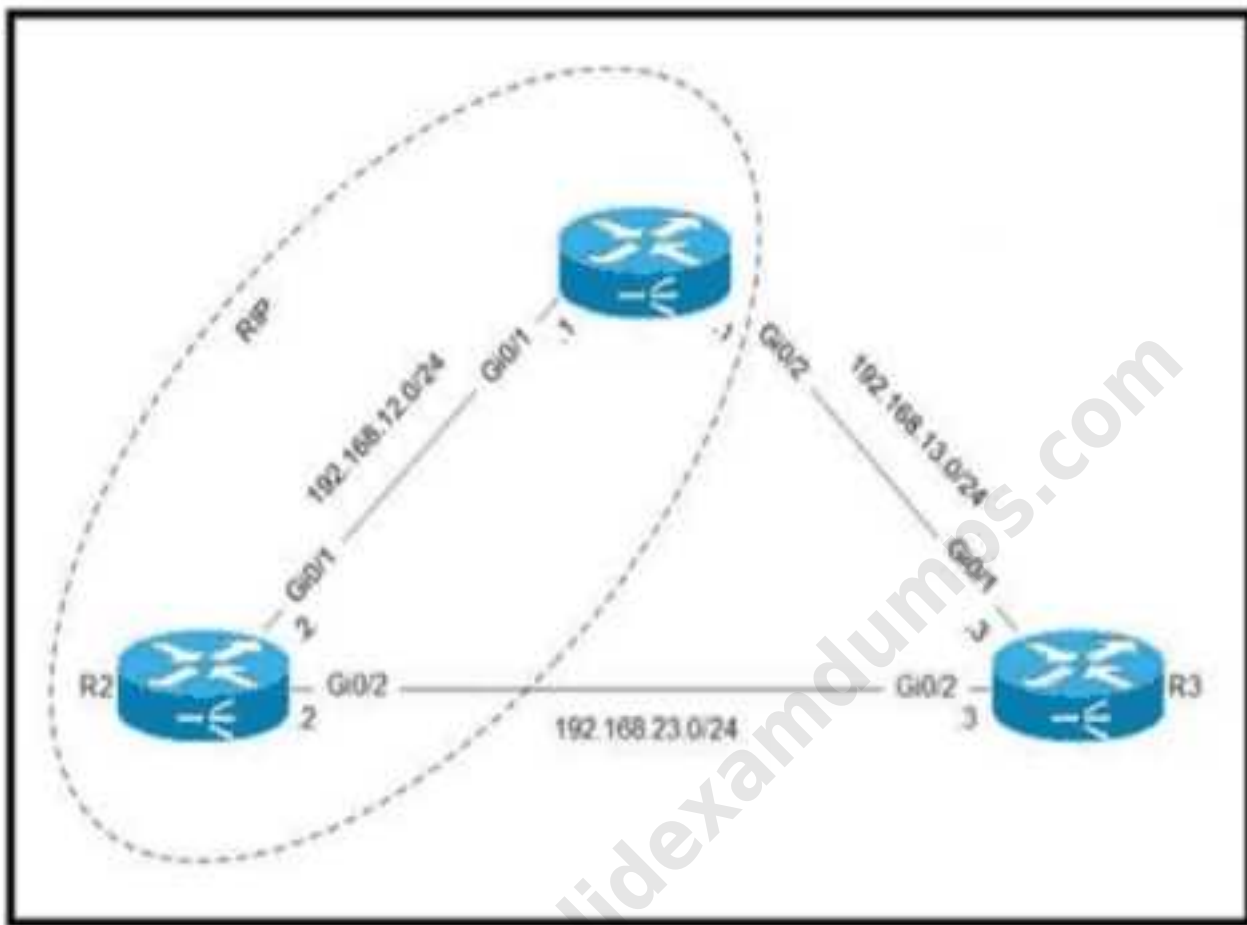
Explanation:

---

**Question: 1052**

---

Refer to the exhibit.



Routers R1 and R2 are configured with RIP as the dynamic routing protocol. A network engineer must configure R1 with a floating static route to serve as a backup route to network 192.168.23.0. Which command must the engineer configure on R1?

- A. ip route 192.168.23.0 255.255.255.255 192.168.13.3 121
- B. ip route 192.168.23.0 255.255.255.0 192.168.13.3 100
- C. ip route 192.168.23.0 255.255.255.0 192.168.13.3 121
- D. ip route 192.168.23.0 255.255.255.0 192.168.13.3

---

**Answer: C**

---

Explanation:

---

**Question: 1053**

---

R1 has learned route 192.168.12.0/24 via IS-IS, OSPF, RIP, and Internal EIGRP. Under normal operating conditions, which routing protocol's route will be installed in the routing table?

- A. IS-IS
- B. RIP
- C. OSPF
- D. Internal EIGRP

---

**Answer: D**

---

Explanation:

---

**Question: 1054**

---

Which set of actions satisfies the requirement for multifactor authentication?

- A. The user enters a user name and password, and then clicks a notification in an authentication app on a mobile device.
- B. The user swipes a key fob, then clicks through an email link.
- C. The user enters a PIN into an RSA token, and then enters the displayed RSA key on a login screen.
- D. The user enters a user name and password, and then re-enters the credentials on a second screen.

---

**Answer: A**

---

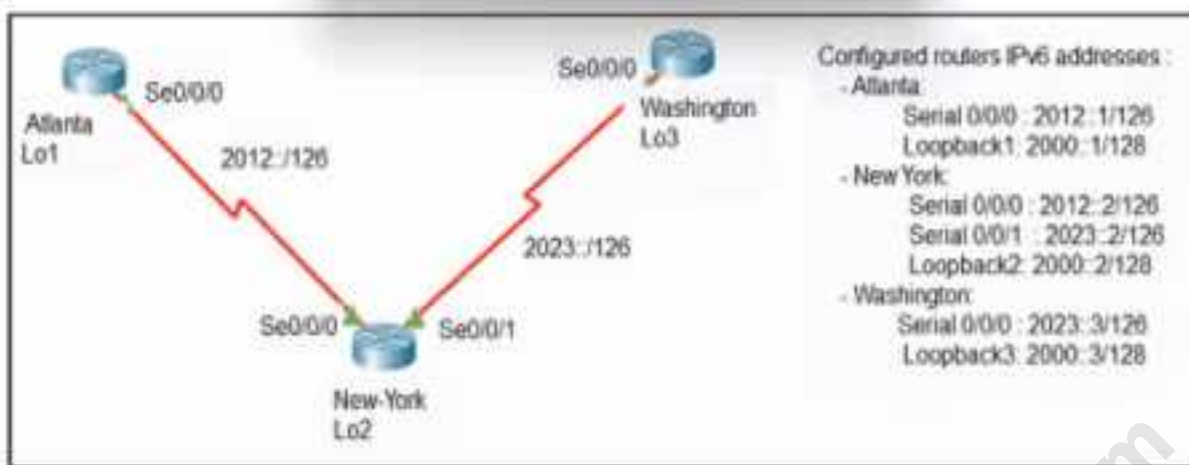
Explanation:

---

**Question: 1055**

---

Refer to the exhibit.



Refer to the exhibit. An engineer configured the New York router with static routes that point to the Atlanta and Washington sites. Which command must be configured on the Atlanta and Washington routers so that both sites are able to reach the Lo2 interface on the New York router?

- A. ipv6 route ::/0 Serial 0/0/0
- B. ipv6 route 0.0.0.0 0.0.0.0 Serial 0/0/0
- C. ipv6 route ::/0 2000::2
- D. ipv6 route ::/0 Serial 0/0/1

---

**Answer: A**

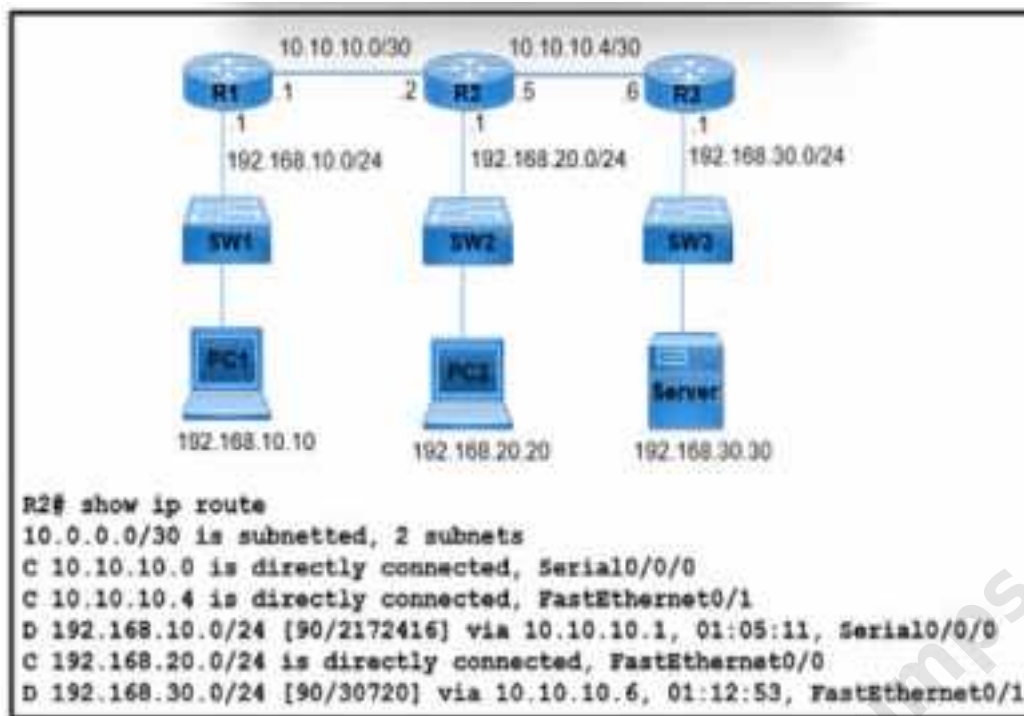
---

Explanation:

---

**Question: 1056**

Refer to the exhibit.



Refer to the exhibit. What is the next-hop IP address for R2 so that PC2 reaches the application server via EIGRP?

- A. 192.168.20.1
- B. 10.10.10.5
- C. 192.168.30.1
- D. 10.10.10.6

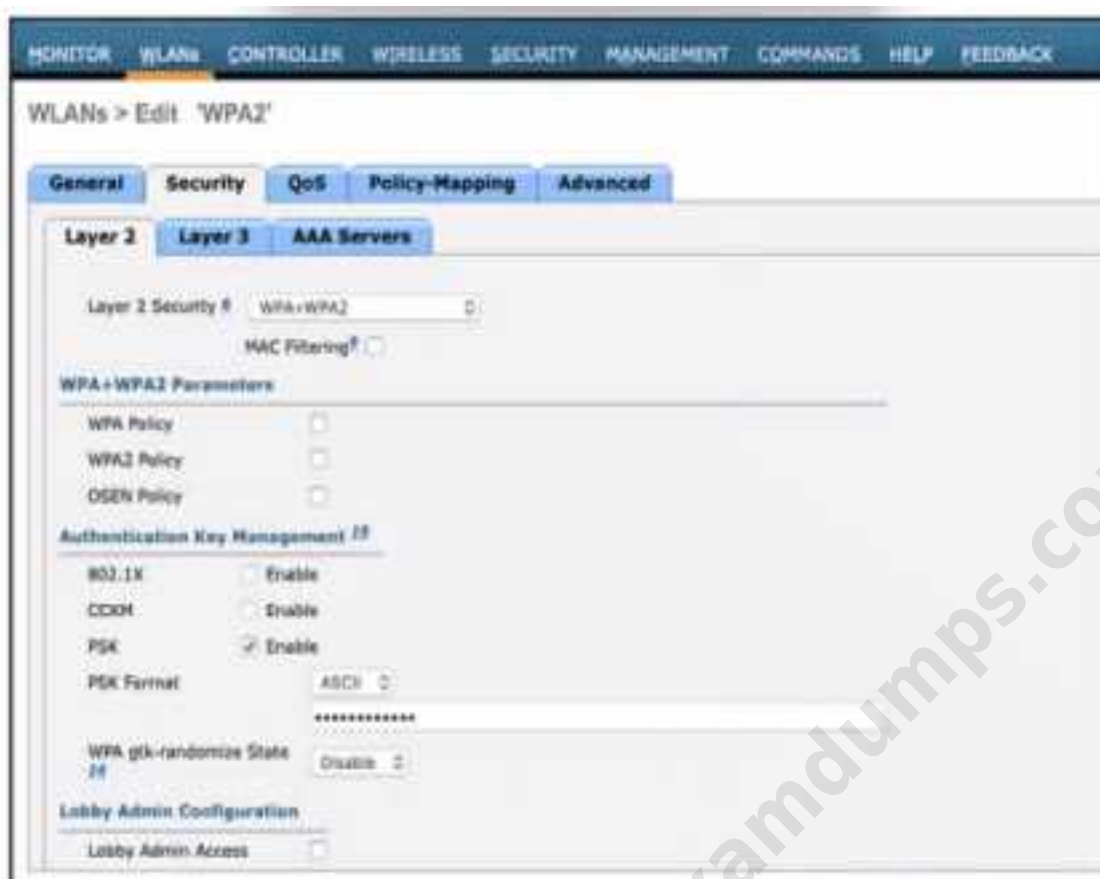
**Answer: D**

Explanation:

### Question: 1057

Refer to the exhibit.





Refer to the exhibit. A network engineer is configuring a WLAN to use a WPA2 PSK and allow only specific clients to join. Which two actions must be taken to complete the process? (Choose two.)

- A. Enable the 802.1X option for Authentication Key Management
- B. Enable the WPA2 Policy option
- C. Enable the CCKM option for Authentication Key Management
- D. Enable the MAC Filtering option
- E. Enable the OSN Policy option

---

**Answer: B,D**

---

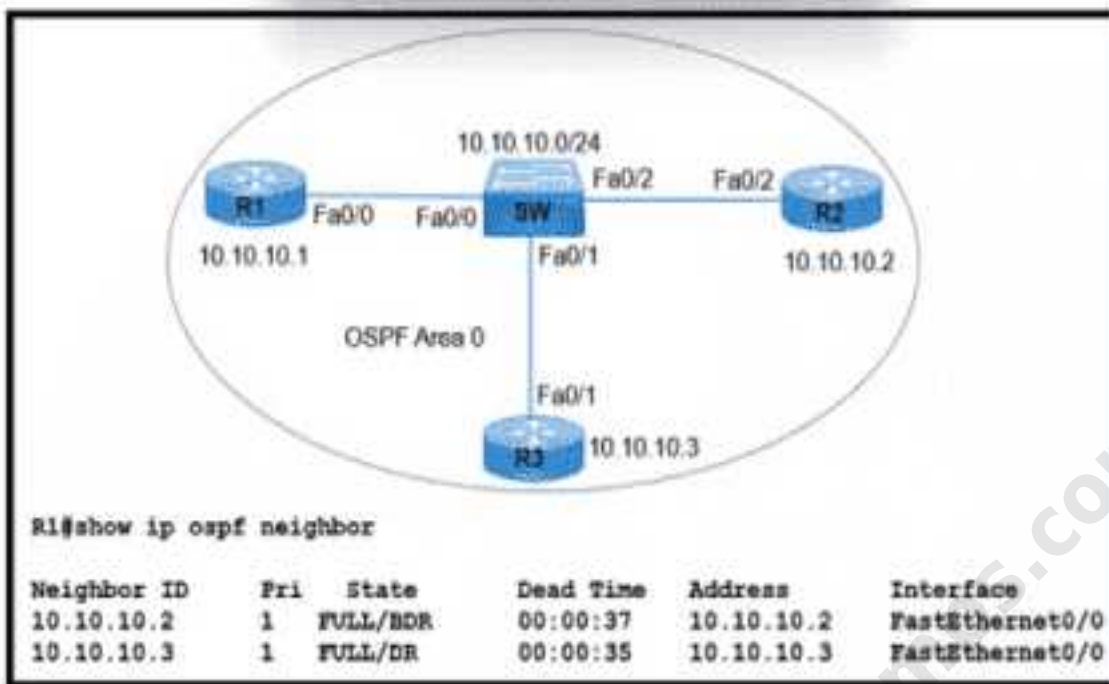
Explanation:

---

**Question: 1058**

---

Refer to the exhibit.



Refer to the exhibit. R1 has taken the DROTHER role in the OSPF DR/BDR election process. Which configuration must an engineer implement so that R1 is elected as the DR?

- A. R1(config)#interface FastEthernet 0/0R1(config-if)#ip ospf priority 1R1#clear ip ospf process
- B. R2(config)#interface FastEthernet 0/2R2(config-if)#ip ospf priority 1R2#clear ip ospf process
- C. R1(config)#interface FastEthernet 0/0R1(config-if)#ip ospf priority 200R1#clear ip ospf process
- D. R3(config)#interface FastEthernet 0/1R3(config-if)#ip ospf priority 200R3#clear ip ospf process

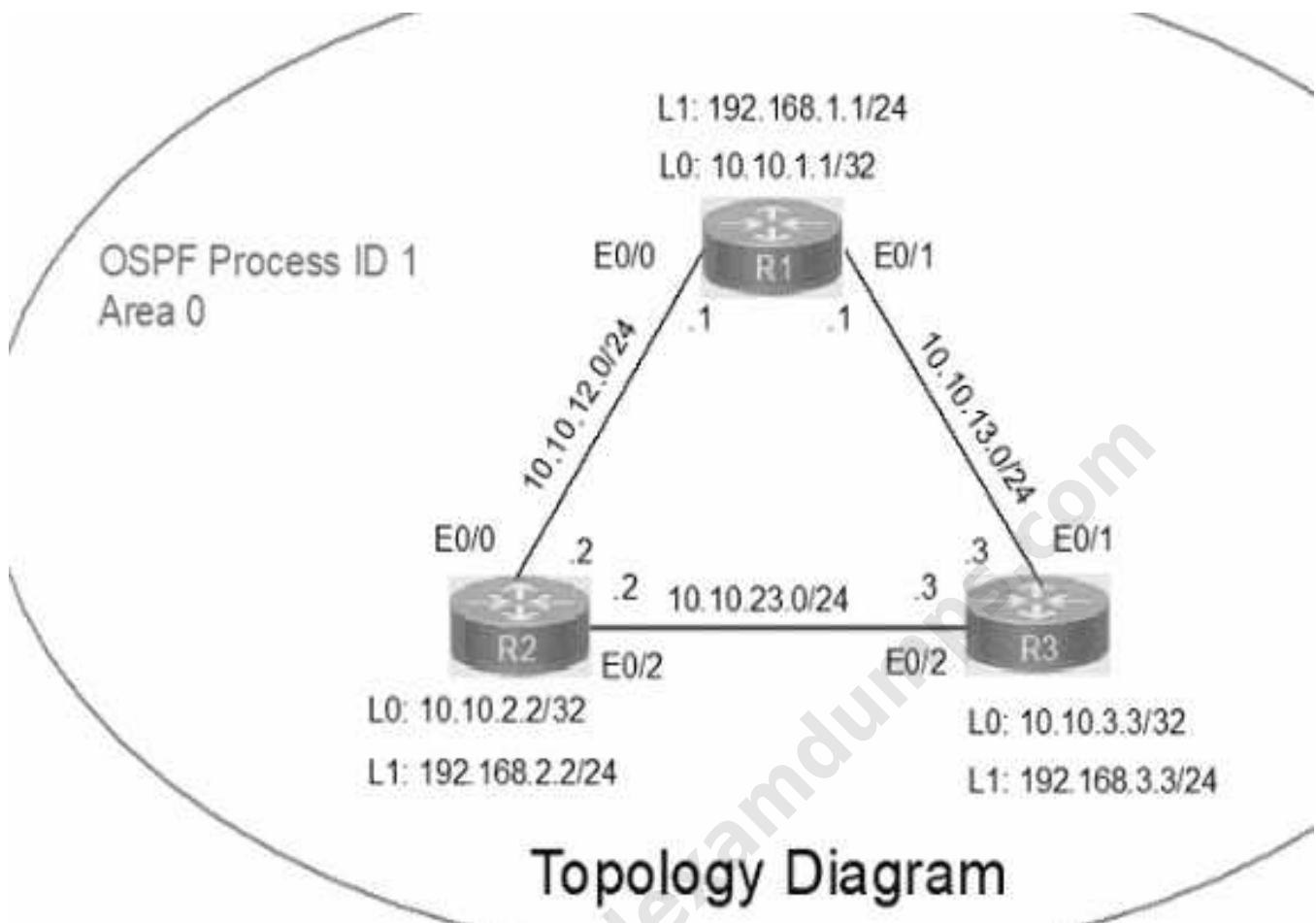
**Answer: C**

Explanation:

Topic 5, Simulations / Lab

**Question: 1059**

SIMULATION



## Guidelines

This is a lab item in which tasks will be performed on virtual devices.

- Refer to the **Tasks** tab to view the tasks for this lab item.
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- All necessary preconfigurations have been applied.
- Do not change the enable password or hostname for any device.
- **Save your configurations** to NVRAM before moving to the next item.
- Click **Next** at the bottom of the screen to submit this lab and move to the next question.
- When **Next** is clicked, the lab closes and cannot be reopened.

IP connectivity between the three routers is configured. OSPF adjacencies must be established.

1. Configure R1 and R2 Router IDs using the interface IP addresses from the link that is shared between them.
2. Configure the R2 links with a max value facing R1 and R3. R2 must become the DR. R1 and R3 links facing R2 must remain with the default OSPF configuration for DR election. Verify the configuration

after clearing the OSPF process.

3. Using a host wildcard mask, configure all three routers to advertise their respective Loopback1 networks.

4. Configure the link between R1 and R3 to disable their ability to add other OSPF routers.

---

**Answer: See the  
Explanation below.**

---

Explanation:

Answer as below configuration:

on R1

conf terminal

interface Loopback0

ip address 10.10.1.1 255.255.255.255

!

interface Loopback1

ip address 192.168.1.1 255.255.255.0

!

interface Ethernet0/0

no shut

ip address 10.10.12.1 255.255.255.0

ip ospf 1 area 0

duplex auto

!

interface Ethernet0/1

no shut

ip address 10.10.13.1 255.255.255.0

```
ip ospf 1 area 0

duplex auto

!

router ospf 1

router-id 10.10.12.1

network 10.10.1.1 0.0.0.0 area 0

network 192.168.1.0 0.0.0.255 area 0

!

copy run star
```

-----

On R2

```
conf terminal

interface Loopback0

ip address 10.10.2.2 255.255.255.255

!

interface Loopback1

ip address 192.168.2.2 255.255.255.0

!

interface Ethernet0/0

no shut

ip address 10.10.12.2 255.255.255.0

ip ospf priority 255

ip ospf 1 area 0

duplex auto
```

```
!  
interface Ethernet0/2  
no shut  
ip address 10.10.23.2 255.255.255.0  
ip ospf priority 255  
ip ospf 1 area 0  
duplex auto  
!  
router ospf 1  
network 10.10.2.2 0.0.0.0 area 0  
network 192.168.2.0 0.0.0.255 area 0  
!  
copy runs start
```

```
-----  
  
On R3  
conf ter  
interface Loopback0  
ip address 10.10.3.3 255.255.255.255  
!  
interface Loopback1  
ip address 192.168.3.3 255.255.255.0  
!  
interface Ethernet0/1  
no shut
```

```
ip address 10.10.13.3 255.255.255.0
```

```
ip ospf 1 area 0
```

```
duplex auto
```

```
!
```

```
interface Ethernet0/2
```

```
no shut
```

```
ip address 10.10.23.3 255.255.255.0
```

```
ip ospf 1 area 0
```

```
duplex auto
```

```
!
```

```
router ospf 1
```

```
network 10.10.3.3 0.0.0.0 area 0
```

```
network 192.168.3.0 0.0.0.255 area 0
```

```
!
```

```
copy run start
```

```
!
```

---

**Question: 1060**

---

**SIMULATION**

Physical connectivity is implemented between the two Layer 2 switches, and the network connectivity between them must be configured

1. Configure an LACP EtherChannel and number it as 1; configure it between switches SW1 and SVV2 using interfaces Ethernet0/0 and Ethernet0/1 on both sides. The LACP mode must match on both ends

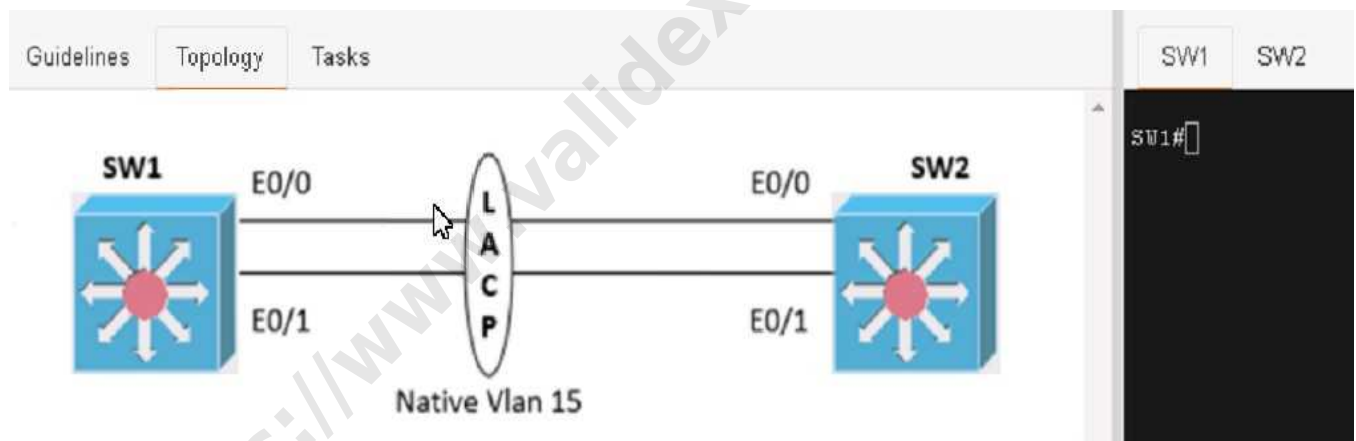
2 Configure the EtherChannel as a trunk link.

3. Configure the trunk link with 802.1 q tags.
4. Configure the native VLAN of the EtherChannel as VLAN 15.

## Guidelines

This is a lab item in which tasks will be performed on virtual devices.

- Refer to the **Tasks** tab to view the tasks for this lab item.
- Refer to the **Topology** tab to access the device console(s) and perform the tasks.
- Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
- All necessary preconfigurations have been applied.
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- **Save your configurations** to NVRAM before moving to the next item.
- Click **Next** at the bottom of the screen to submit this lab and move to the next question.
- When **Next** is clicked, the lab closes and cannot be reopened.



**Answer: See the  
Explanation below.**

Explanation:

Answer as below configuration:

On SW1:



```
conf terminal  
  
vlan 15  
  
exit  
  
interface range eth0/0 - 1  
  
channel-group 1 mode active  
  
exit  
  
interface port-channel 1  
  
switchport trunk encapsulation dot1q  
  
switchport mode trunk  
  
switchport trunk native vlan 15  
  
end  
  
copy run start
```

on SW2:

```
conf terminal  
  
vlan 15  
  
exit  
  
interface range eth0/0 - 1  
  
channel-group 1 mode active  
  
exit  
  
interface port-channel 1  
  
switchport trunk encapsulation dot1q  
  
switchport mode trunk  
  
switchport trunk native vlan 15  
  
end  
  
copy run start
```

---

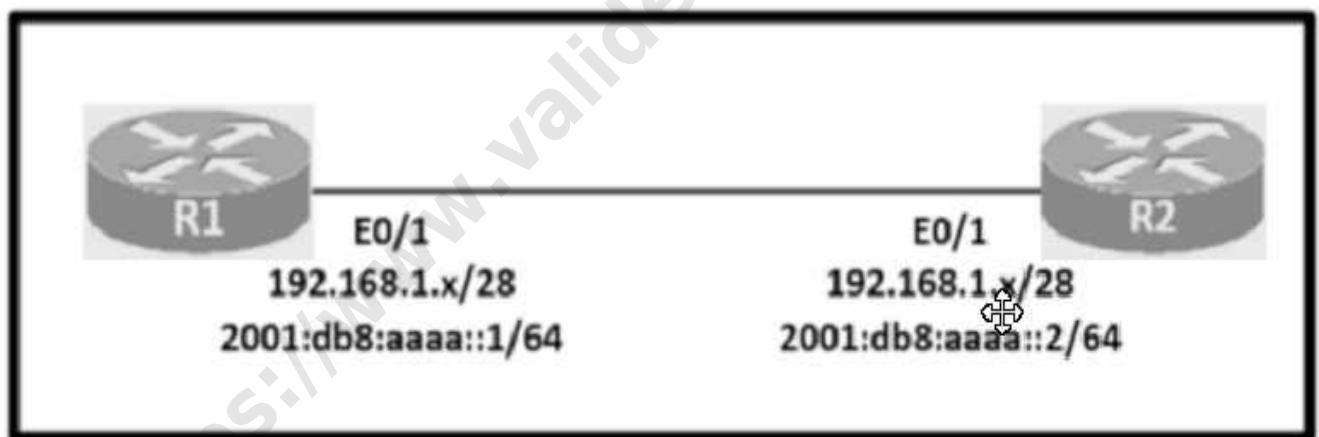
**Question: 1061**

---

**SIMULATION**

Configure IPv4 and IPv6 connectivity between two routers. For IPv4, use a /28 network from the 192.168.1.0/24 private range. For IPv6, use the first /64 subnet from the 2001:db8:aaaa::/48 subnet.

1. Using Ethernet0/1 on routers R1 and R2, configure the next usable/28 from the 192.168.1.0/24 range. The network 192.168.1.0/28 is unavailable.
2. For the IPv4 /28 subnet, router R1 must be configured with the first usable host address.
3. For the IPv4 /28 subnet, router R2 must be configured with the last usable host address.
4. For the IPv6 /64 subnet, configure the routers with the IP addressing provided from the topology.
5. A ping must work between the routers on the IPv4 and IPv6 address ranges.



The screenshot shows a network lab interface with three tabs: Guidelines, Topology, and Tasks. The Guidelines tab is active, displaying a list of instructions for the lab. On the right side, there are tabs for R1 and R2, with R1 currently selected. The R1 console window is visible, showing a command prompt.

**Guidelines**

This is a lab item in which tasks will be performed on virtual devices.

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- Do not change the enable password or hostname for any device.
- **Save your configurations** to NVRAM before moving to the next item.
- Click **Next** at the bottom of the screen to submit this lab and move to the next question.
- When **Next** is clicked, the lab closes and cannot be reopened.

**Answer: See the  
Explanation below.**

Explanation:

Answer as below configuration:

on R1

config terminal

ipv6 unicast-routing

inter eth0/1

ip addre 192.168.1.1 255.255.255.240

ipv6 addre 2001:db8:aaaa::1/64

not shut

end

copy running start

on R2

```
config terminal
ipv6 unicast-routing
inter eth0/1
ip address 192.168.1.14 255.255.255.240
ipv6 address 2001:db8:aaaa::2/64
not shut
end
copy running start
-----
for test from R1
ping ipv6 2001:db8:aaaa::1

for test from R2
ping ipv6 2001:db8:aaaa::2
```

---

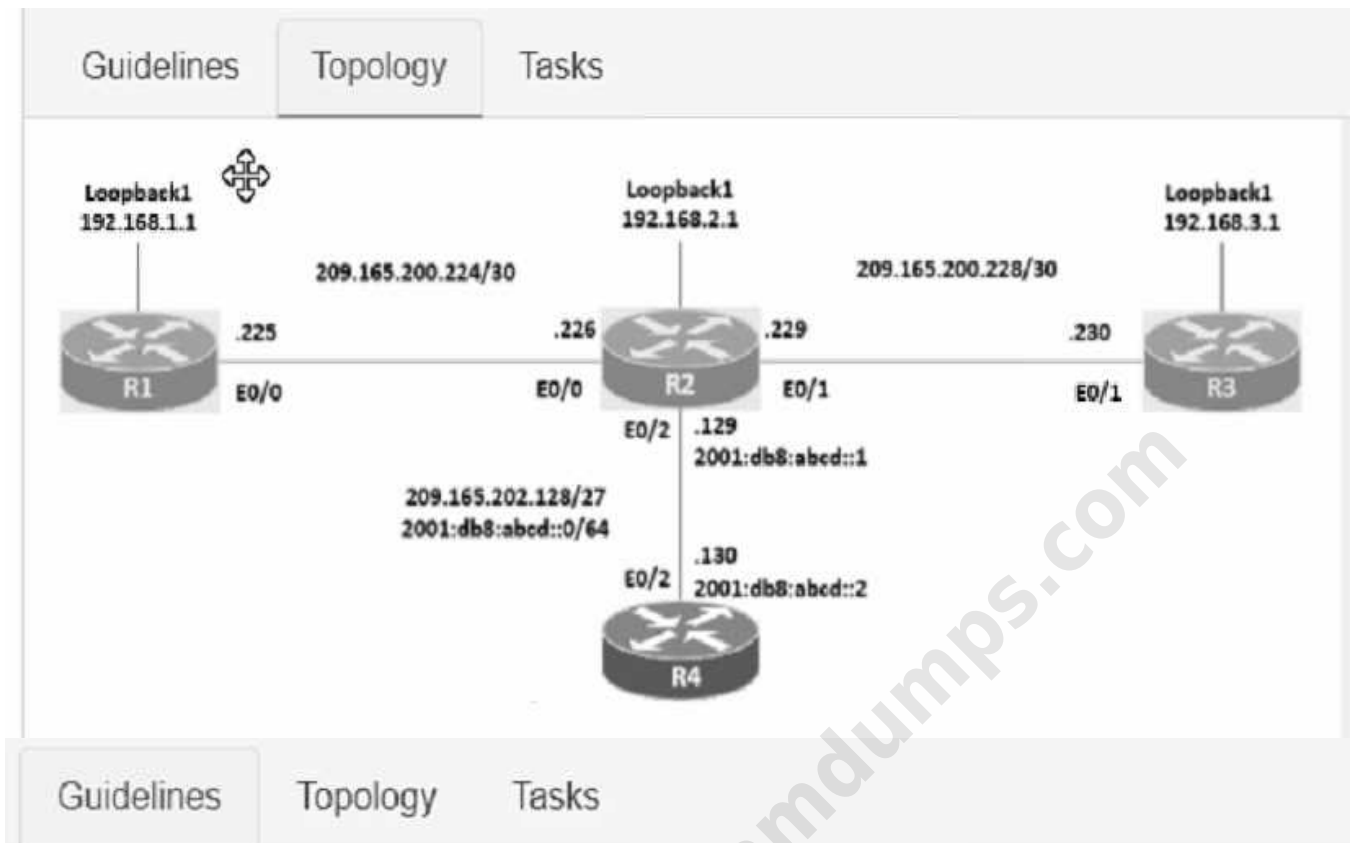
**Question: 1062**

---

**SIMULATION**

Connectivity between four routers has been established. IP connectivity must be configured in the order presented to complete the implementation. No dynamic routing protocols are included.

1. Configure static routing using host routes to establish connectivity from router R3 to the router R1 Loopback address using the source IP of 209.165.200.230.
2. Configure an IPv4 default route on router R2 destined for router R4.
3. Configure an IPv6 default router on router R2 destined for router R4.



## Guidelines

This is a lab item in which tasks will be performed on virtual devices.

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- Do not change the enable password or hostname for any device.
- **Save your configurations** to NVRAM before moving to the next item.
- Click **Next** at the bottom of the screen to submit this lab and move to the next question.
- When **Next** is clicked, the lab closes and cannot be reopened.

---

**Answer: See the  
Explanation below.**

---

Explanation:

Answer as below configuration:

1.- on R3

config terminal

ip route 192.168.1.1 255.255.255.255 209.165.200.229

end

copy running start

2.- on R2

config terminal

ip route 0.0.0.0 0.0.0.0 209.165.202.130

end

copy running start

3.- on R2

config terminal

ipv6 route ::/0 2001:db8:abcd::2

end

copy running start

---

**Question: 1063**

---

SIMULATION

## Guidelines

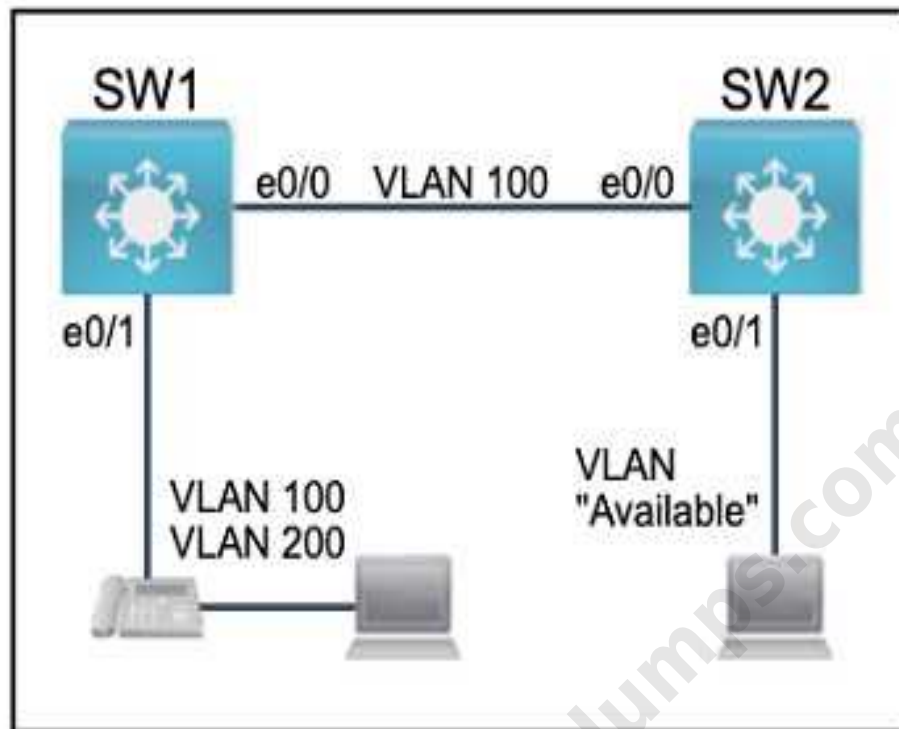
---

This is a lab item in which tasks will be performed on virtual devices.

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- Do not change the enable password or hostname for any device.
- **Save your configurations** to NVRAM before moving to the next item.
- Click **Next** at the bottom of the screen to submit this lab and move to the next question.
- When **Next** is clicked, the lab closes and cannot be reopened.

All physical cabling between the two switches is installed. Configure the network connectivity between the switches using the designated VLANs and interfaces.

1. Configure VLAN 100 named Compute and VLAN 200 named Telephony where required for each task.
2. Configure Ethernet0/1 on SW2 to use the existing VLAN named Available.
3. Configure the connection between the switches using access ports.
4. Configure Ethernet0/1 on SW1 using data and voice VLANs.
5. Configure Ethernet0/1 on SW2 so that the Cisco proprietary neighbor discovery protocol is turned off for the designated interface only.



**Answer: See the  
Explanation below.**

Explanation:

Answer as below configuration:

on sw1

enable

conf t

vlan 100

name Compute

vlan 200

name Telephony

int e0/1

switchport voice vlan 200



switchport access vlan 100

int e0/0

switchport mode access

do wr

on sw2

Vlan 99

Name Available

Int e0/1

Switchport access vlan 99

do wr

---

### Question: 1064

---

SIMULATION

## Guidelines

This is a lab item in which tasks will be performed on virtual devices.

- Refer to the **Tasks** tab to view the tasks for this lab item.
- Refer to the **Topology** tab to access the device console(s) and perform the tasks.
- Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
- All necessary preconfigurations have been applied.
- Do not change the enable password or hostname for any device.
- **Save your configurations** to NVRAM before moving to the next item.
- Click **Next** at the bottom of the screen to submit this lab and move to the next question.
- When **Next** is clicked, the lab closes and cannot be reopened.

Next

Three switches must be configured for Layer 2 connectivity. The company requires only the designated VLANs to be configured on their respective switches and permitted across any links

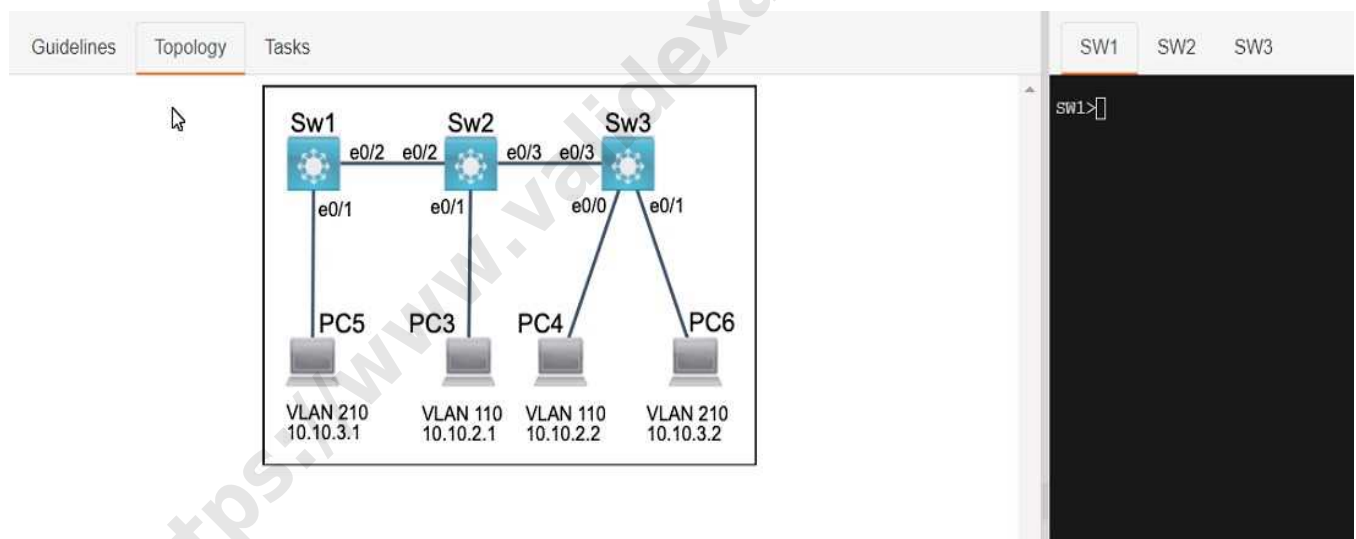
between switches for security purposes. Do not modify or delete VTP configurations.

The network needs two user-defined VLANs configured:

VLAN 110: MARKETING

VLAN 210: FINANCE

1. Configure the VLANs on the designated switches and assign them as access ports to the interfaces connected to the PCs.
2. Configure the e0/2 interfaces on Sw1 and Sw2 as 802.1q trunks with only the required VLANs permitted.
3. Configure the e0/3 interfaces on Sw2 and Sw3 as 802.1q trunks with only the required VLANs permitted.



**Answer: See the  
Explanation below.**

Explanation:

Answer as below configuration:

Sw1

enbale

config t

Vlan 210

Name FINANCE

Inter e0/1

Switchport access vlan 210

do wr

Sw2

Enable

config t

Vlan 110

Name MARKETING

Int e0/1

Switchport access vlan 110

do wr

Sw3

Enable

config t

Vlan 110

Name MARKETING

Vlan 210

Name FINANCE

Int e0/0

Switchport access vlan 110

Int e0/1

Switchport access vlan 210

Sw1

Int e0/1

Switchport allowed vlan 210

Sw2

Int e0/2

Switchport trunk allowed vlan 210

Sw3

Int e0/3

Switchport trunk allowed vlan 210

Switchport trunk allowed vlan 210,110

---

**Question: 1065**

---

SIMULATION

GuidelinesTopologyTasks

```
graph TD; R1((R1)) ---|E0/0| R2((R2)); R1 ---|E0/2| R3((R3)); R2 ---|E0/1| R3;
```

R1R2R3  
R1#

GuidelinesTopologyTasks

## Guidelines

This is a lab item in which tasks will be performed on virtual devices.

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- All necessary preconfigurations have been applied.
- Do not change the enable password or hostname for any device.
- **Save your configurations** to NVRAM before moving to the next item.
- Click **Next** at the bottom of the screen to submit this lab and move to the next question.
- When **Next** is clicked, the lab closes and cannot be reopened.

R1R2R3  
R1#

Connectivity between three routers has been established, and IP services must be configured in the order presented to complete the implementation. Tasks assigned include configuration of NAT, NTP, DHCP, and SSH services.

1. All traffic sent from R3 to the R1 Loopback address must be configured for NAT on R2. All source addresses must be translated from R3 to the IP address of Ethernet0/0 on R2, while using only a standard access list named NAT. To verify, a ping must be successful to the R1 Loopback address sourced from R3. Do not use NVI NAT configuration.

2. Configure R1 as an NTP server and R2 as a client, not as a peer, using the IP address of the R1 Ethernet0/2 interface. Set the clock on the NTP server for midnight on January 1, 2019.

3. Configure R1 as a DHCP server for the network 10.1.3.0/24 in a pool named TEST. Using a single command, exclude addresses 1-10 from the range. Interface Ethernet0/2 on R3 must be issued the IP address of 10.1.3.11 via DHCP.
4. Configure SSH connectivity from R1 to R3, while excluding access via other remote connection protocols. Access for user root and password Cisco must be set on router R3 using RSA and 1024 bits. Verify connectivity using an SSH session from router R1 using a destination address of 10.1.3.11. Do NOT modify console access or line numbers to accomplish this task.

---

**Answer: See the  
Explanation below.**

---

Explanation:

Answer as below configuration:

```
conf t
```

```
R1(config)#ntp master 1
```

```
R2(config)#ntp server 10.1.2.1
```

```
Exit
```

```
Router#clock set 00:00:00 jan 1 2019
```

```
ip dhcp pool TEST
```

```
network 10.1.3.0 255.255.255.0
```

```
ip dhcp excluded-address 10.1.3.1 10.1.3.10
```

```
R3(config)#int e0/3
```

```
R3(config)#int e0/2
```

```
ip address dhcp
```

```
no shut
```

```
crypto key generate RSA
```

```
1024
```

Copy run start

## Question: 1066

### SIMULATION

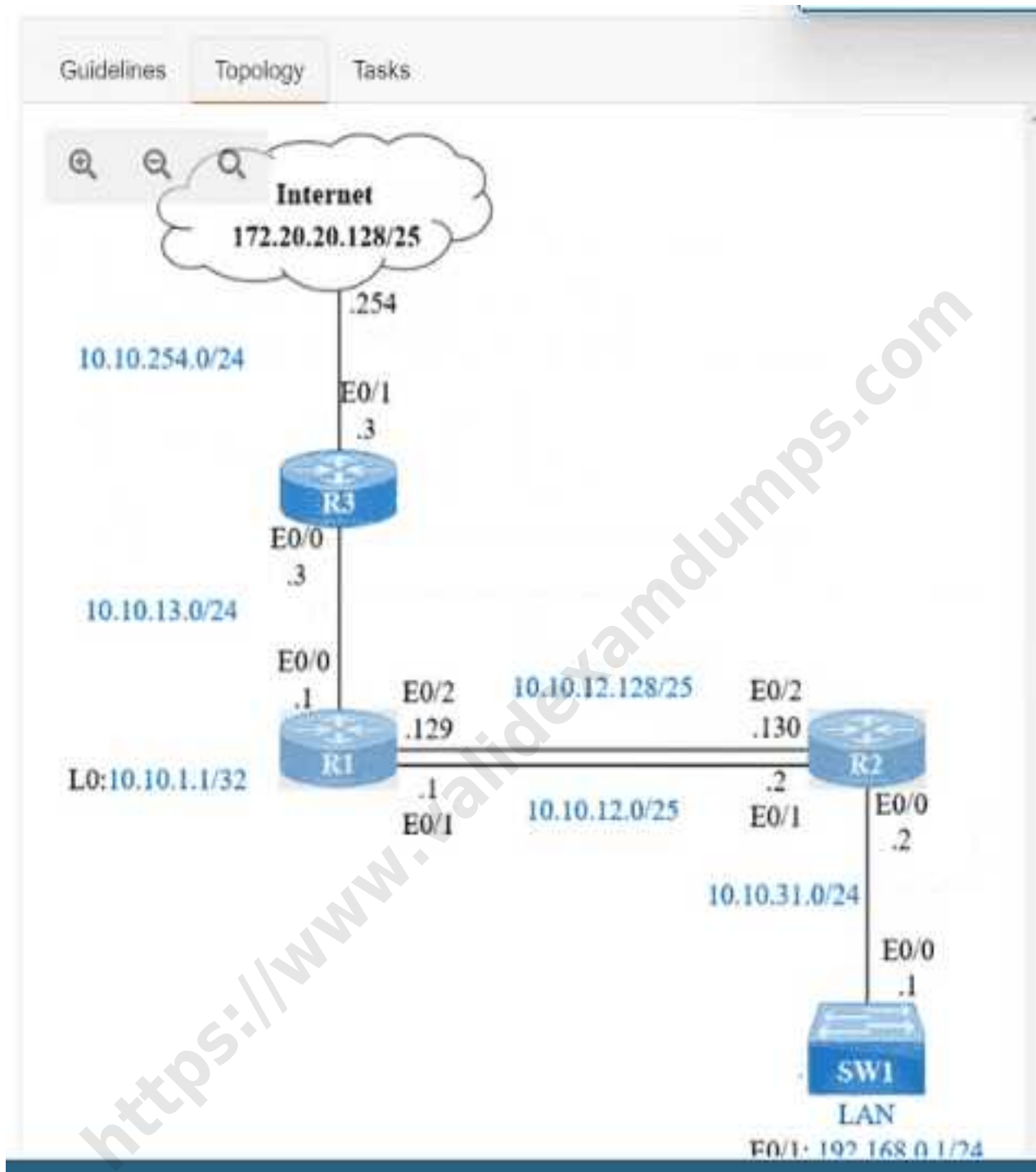
Guidelines   Topology   Tasks

## Guidelines

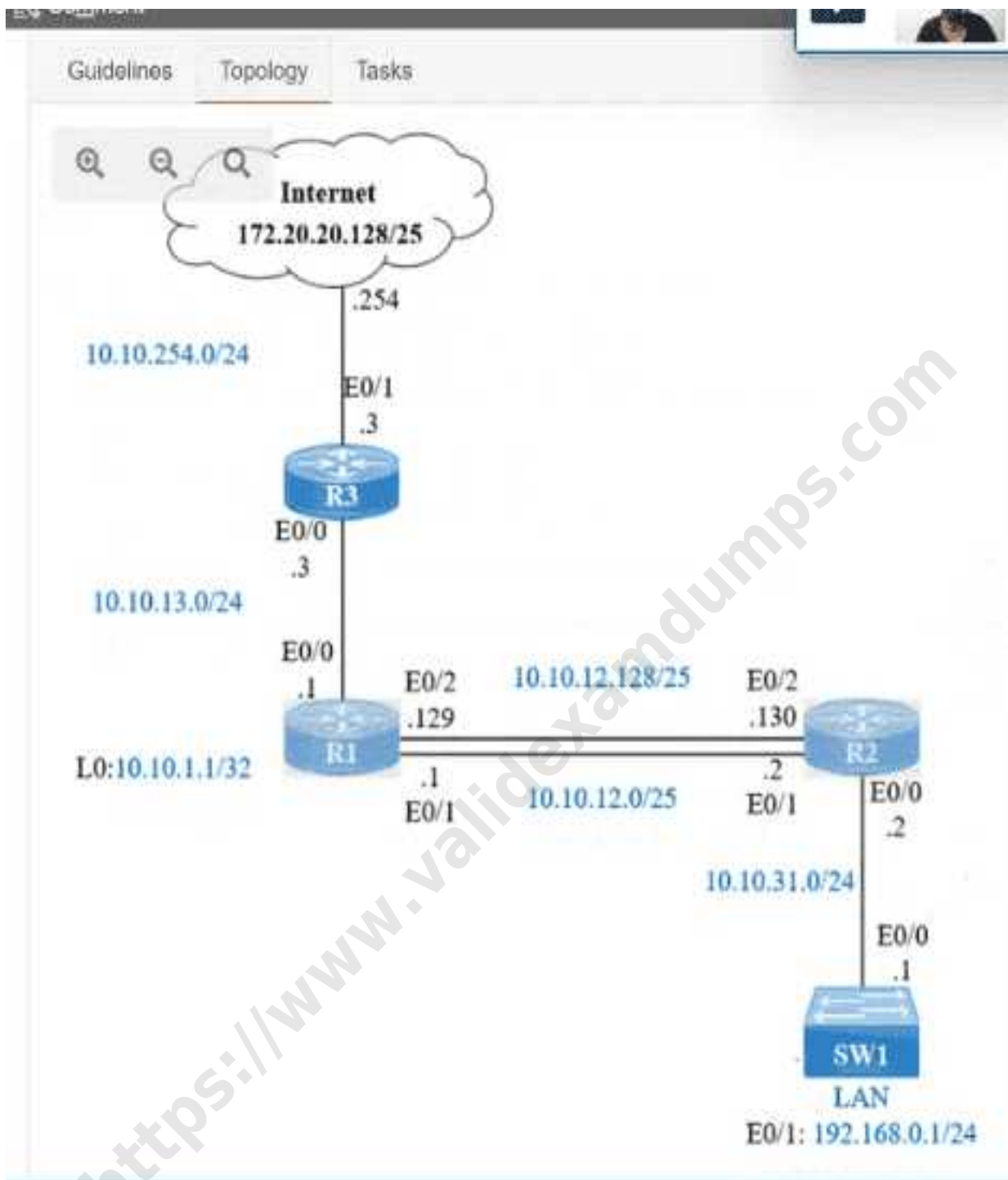
This is a lab item in which tasks will be performed on virtual devices.

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- Do not change the enable password or hostname for any device.
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- Click **Next** at the bottom of the screen to submit this lab and move to the next question.
- When **Next** is clicked, the lab closes and cannot be reopened.

The diagram illustrates a network topology with three devices: Router R1, Router R2, and Switch SW1. R1 and R2 are connected via their E0/2 interfaces with IP addresses 10.10.12.129 and 10.10.12.130 respectively. R2 is connected to SW1 via its E0/2 interface (10.10.31.2) and SW1's E0/0 interface (10.10.31.1). SW1 is connected to R1 via its E0/0 interface (10.10.1.1/32) and R1's E0/1 interface (10.10.12.0/25). The network is divided into three segments: 10.10.12.0/25, 10.10.31.0/24, and 10.10.1.1/32.







IP connectivity and OSPF are preconfigured on all devices where necessary. Do not make any changes to the IP addressing or OSPF. The company policy uses connected interfaces and next hops when configuring static routes except for load balancing or redundancy without floating static. Connectivity must be established between subnet 172.20.20.128/25 on the Internet and the LAN at 192.168.0.0/24 connected to SW1:

1. Configure reachability to the switch SW1 LAN subnet in router R2.

2. Configure default reachability to the Internet subnet in router R1.
3. Configure a single static route in router R2 to reach to the Internet subnet considering both redundant links between routers R1 and R2. A default route is NOT allowed in router R2.
4. Configure a static route in router R1 toward the switch SW1 LAN subnet where the primary link must be through Ethernet0/1. and the backup link must be through Ethernet0/2 using a floating route. Use the minimal administrative distance value when required.

---

**Answer: See the  
Explanation below.**

---

Explanation:

Answer as below configuration:

On R2:

Enable

Conf t

Ip route 192.168.1.0 255.255.255.0 10.10.31.1

On R1:

Enable

Conf t

Ip route 0.0.0.0 0.0.0.0 10.10.13.3

On R2

Ip route 172.20.20.128 255.255.255.128 e0/2

Ip route 172.20.20.128 255.255.255.128 e0/1

On R1

Ip route 192.168.0.0 255.255.255.0 e0/1

Ip route 192.168.0.0 255.255.255.0 10.10.12.2 3

Save all configurations after every router from anyone of these command

Do wr

Or

Copy run start

---

**Question: 1067**

---

#### SIMULATION

All physical cabling is in place. A company plans to deploy 32 new sites.

The sites will utilize both IPv4 and IPv6 networks.

1 . Subnet 172.25.0.0/16 to meet the subnet requirements and maximize

the number of hosts

Using the second subnet

- Assign the first usable IP address to e0/0 on Sw101
- Assign the last usable IP address to e0/0 on Sw102

2. Subnet to meet the subnet requirements and maximize

the number of hosts

c Using the second subnet

- Assign an IPv6 GUA using a unique 64-Bit interface identifier

on e0/0 on Sw101

- Assign an IPv6 GUA using a unique 64-Bit interface identifier

on e0/0 on swi02

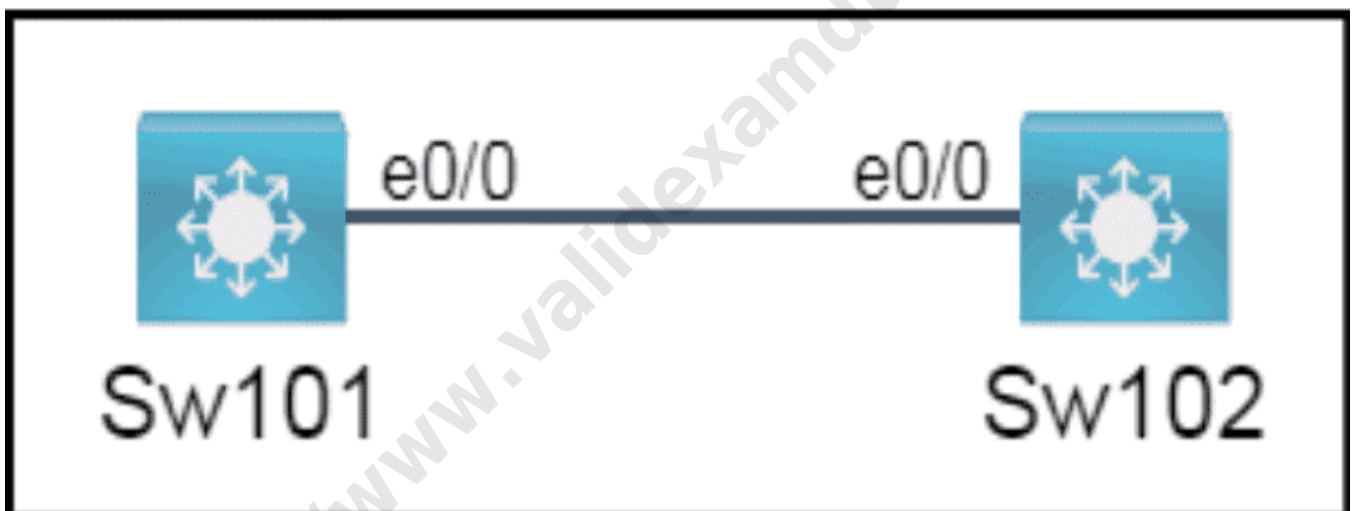
#### Guidelines

This is a lab item in which tasks will be performed on virtual devices.

- Refer to the Tasks tab to view the tasks for this lab item.
- Refer to the Topology tab to access the device console(s) and perform the tasks.
- Console access is available for all required devices by clicking the device icon or using

the tab(s) above the console window.

- All necessary preconfigurations have been applied.
- Do not change the enable password or hostname for any device.
- Save your configurations to NVRAM before moving to the next item.
- Click Next at the bottom of the screen to submit this lab and move to the next question.
- When Next is clicked, the lab closes and cannot be reopened.



**Answer: See the  
Explanation for the  
solution.**

Explanation:

To subnet 172.25.0.0/16 to meet the subnet requirements and maximize the number of hosts, you need to determine how many bits you need to borrow from the host portion of the address to create enough subnets for 32 sites. Since 32 is  $2^5$ , you need to borrow 5 bits, which means your new subnet mask will be /21 or 255.255.248.0. To find the second subnet, you need to add the value of the fifth bit (32) to the third octet of the network address (0), which gives you 172.25.32.0/21 as the second subnet. The first usable IP address in this subnet is 172.25.32.1, and the last usable IP address

is 172.25.39.254.

To assign the first usable IP address to e0/0 on Sw101, you need to enter the following commands on the device console:

```
Sw101#configure terminal Sw101(config)#interface e0/0 Sw101(config-if)#ip address 172.25.32.1 255.255.248.0 Sw101(config-if)#no shutdown Sw101(config-if)#end
```

To assign the last usable IP address to e0/0 on Sw102, you need to enter the following commands on the device console:

```
Sw102#configure terminal Sw102(config)#interface e0/0 Sw102(config-if)#ip address 172.25.39.254 255.255.248.0 Sw102(config-if)#no shutdown Sw102(config-if)#end
```

To subnet an IPv6 GUA to meet the subnet requirements and maximize the number of hosts, you need to determine how many bits you need to borrow from the interface identifier portion of the address to create enough subnets for 32 sites. Since 32 is  $2^5$ , you need to borrow 5 bits, which means your new prefix length will be /69 or ffff:fff:fff:fff8::/69 (assuming that your IPv6 GUA has a /64 prefix by default). To find the second subnet, you need to add the value of the fifth bit (32) to the fourth hextet of the network address (0000), which gives you xxxx:xxxx:xxxx:0020::/69 as the second subnet (where xxxx:xxxx:xxxx is your IPv6 GUA prefix). The first and last IPv6 addresses in this subnet are xxxx:xxxx:xxxx:0020::1 and xxxx:xxxx:xxxx:0027:ffff:ffff:ffff:ffe respectively.

To assign an IPv6 GUA using a unique 64-bit interface identifier on e0/0 on Sw101, you need to enter the following commands on the device console (assuming that your IPv6 GUA prefix is 2001:db8::/64):

```
Sw101#configure terminal Sw101(config)#interface e0/0 Sw101(config-if)#ipv6 address 2001:db8::20::1/69 Sw101(config-if)#no shutdown Sw101(config-if)#end
```

To assign an IPv6 GUA using a unique 64-bit interface identifier on e0/0 on Sw102, you need to enter the following commands on the device console (assuming that your IPv6 GUA prefix is 2001:db8::/64):

```
Sw102#configure terminal Sw102(config)#interface e0/0 Sw102(config-if)#ipv6 address 2001:db8::27::ffe/69 Sw102(config-if)#no shutdown Sw102(config-if)#end
```

---

## Question: 1068

---

### SIMULATION

All physical cabling is in place. Router R4 and PCI are fully configured and

inaccessible. R4's WAN interfaces use .4 in the last octet for each subnet.

Configurations should ensure that connectivity is established end-to-end.

1 . Configure static routing to ensure R1 prefers the path through R2 to

reach only PCI on R4's LAN

2. Configure static routing that ensures traffic sourced from R1 will take

an alternate path through R3 to PCI in the event of an outage along

the primary path

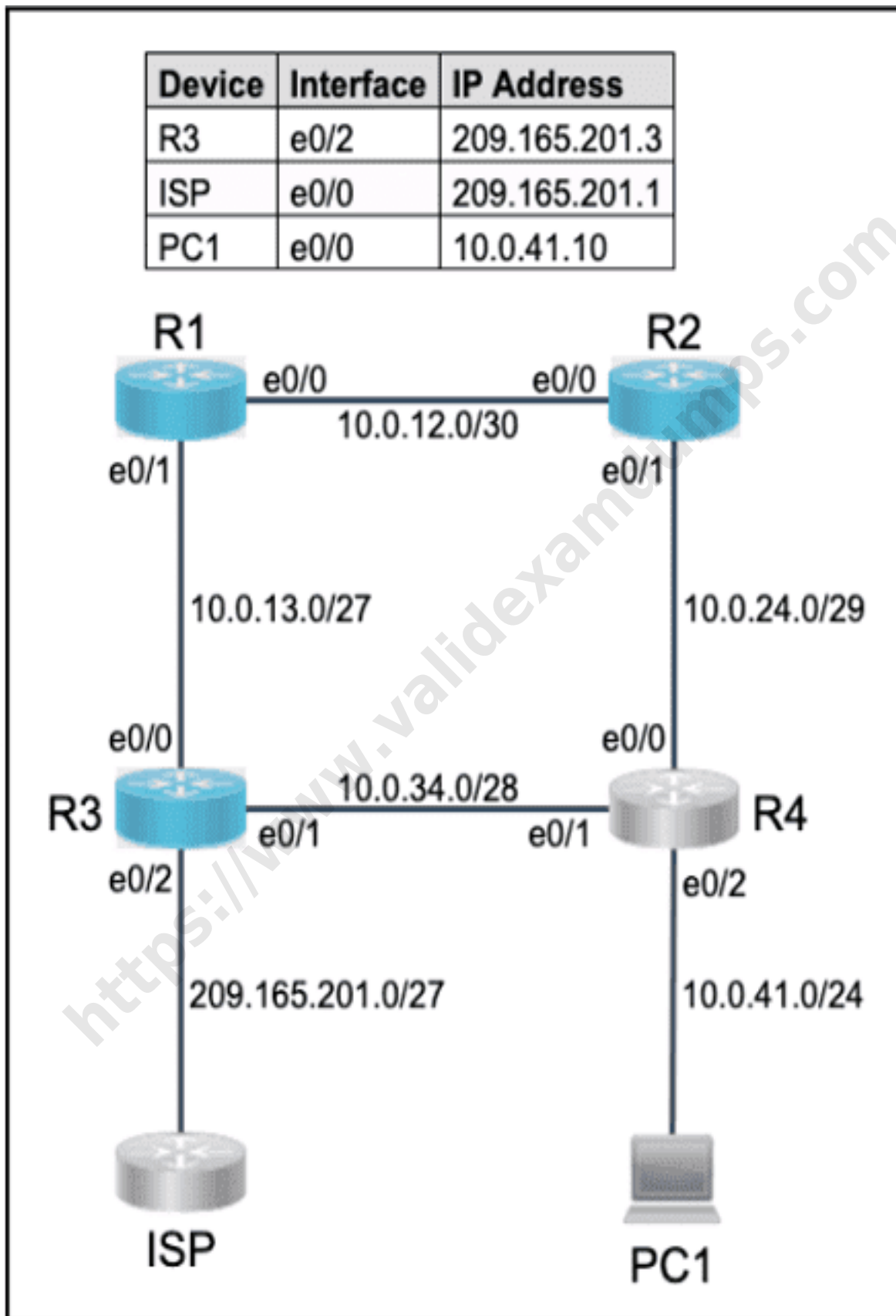
3. Configure default routes on R1 and R3 to the Internet using the least number of hops

#### Guidelines

This is a lab item in which tasks will be performed on virtual devices.

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- Save your configurations to NVRAM before moving to the next item.
- Click Next at the bottom of the screen to submit this lab and move to the next question.

- When Next is clicked, the lab closes and cannot be reopened.





---

**Answer: See the solution below in Explanation.**

---

Explanation:

To configure static routing on R1 to ensure that it prefers the path through R2 to reach only PC1 on R4's LAN, you need to create a static route for the host 10.0.0.100/8 with a next-hop address of 20.0.0.2, which is the IP address of R2's interface connected to R1. You also need to assign a lower administrative distance (AD) to this route than the default AD of 1 for static routes, so that it has a higher preference over other possible routes. For example, you can use an AD of 10 for this route. To create this static route, you need to enter the following commands on R1's console:

```
R1#configure terminal R1(config)#ip route 10.0.0.100 255.0.0.0 20.0.0.2 10 R1(config)#end
```

To configure static routing on R1 that ensures that traffic sourced from R1 will take an alternate path through R3 to PC1 in the event of an outage along the primary path, you need to create another static route for the host 10.0.0.100/8 with a next-hop address of 40.0.0.2, which is the IP address of R3's interface connected to R1. You also need to assign a higher AD to this route than the AD of the primary route, so that it has a lower preference and acts as a backup route. For example, you can use an AD of 20 for this route. This type of static route is also known as a floating static route. To create this static route, you need to enter the following commands on R1's console:

```
R1#configure terminal R1(config)#ip route 10.0.0.100 255.0.0.0 40.0.0.2 20 R1(config)#end
```

To configure default routes on R1 and R3 to the Internet using the least number of hops, you need to create a static route for the network 0.0.0.0/0 with a next-hop address of the ISP's interface connected to each router respectively. A default route is a special type of static route that matches any destination address and is used when no other specific route is available. The ISP's interface connected to R1 has an IP address of 10.0.0.4, and the ISP's interface connected to R3 has an IP address of 50.0.0.4. To create these default routes, you need to enter the following commands on each router's console:

On R1: R1#configure terminal R1(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.4 R1(config)#end

On R3: R3#configure terminal R3(config)#ip route 0.0.0.0 0.0.0.0 50.0.0.4 R3(config)#end

---

### Question: 1069

---

SIMULATION

Physical connectivity is implemented between the two Layer 2 switches, and the network connectivity between them must be configured.

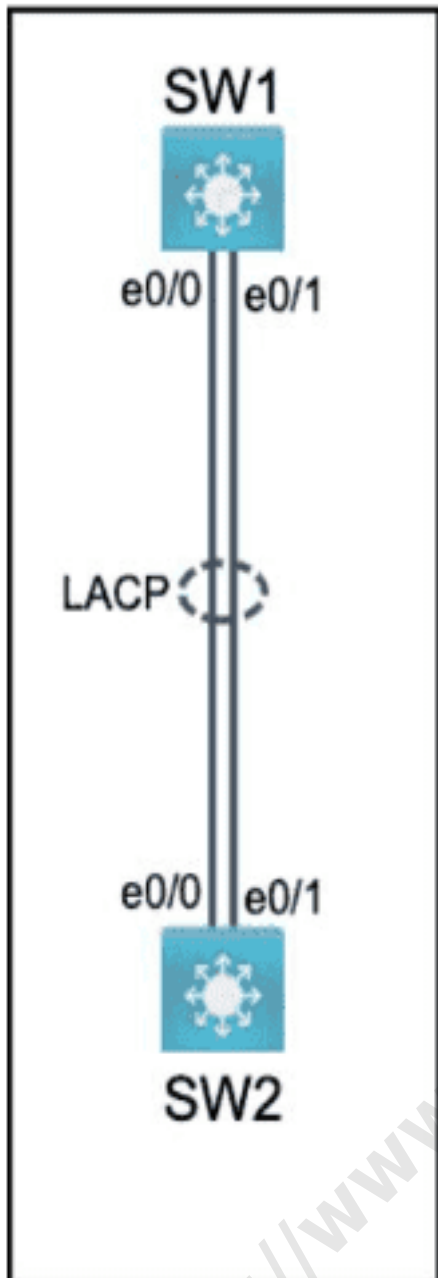
1. Configure an LACP EtherChannel and number it as 44; configure it between switches SW1 and SW2 using interfaces Ethernet0/0 and Ethernet0/1 on both sides. The LACP mode must match on both ends.
2. Configure the EtherChannel as a trunk link.
3. Configure the trunk link with 802.1q tags.
4. Configure VLAN 'MONITORING' as the untagged VLAN of the EtherChannel.

=====

#### Guidelines

This is a lab item in which tasks will be performed on virtual devices.

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- Save your configurations to NVRAM before moving to the next item.
- Click Next at the bottom of the screen to submit this lab and move to the next question.
- When Next is clicked, the lab closes and cannot be reopened.



**Answer: Solution is given below explanation.**

Explanation:

To configure an LACP EtherChannel and number it as 44, configure it between switches SW1 and SW2 using interfaces Ethernet0/0 and Ethernet0/1 on both sides, configure the EtherChannel as a trunk link, configure the trunk link with 802.1q tags, and configure VLAN 'MONITORING' as the untagged VLAN of the EtherChannel, you need to follow these steps:

On both SW1 and SW2, enter the global configuration mode by using the configure terminal command.

On both SW1 and SW2, select the two interfaces that will form the EtherChannel by using the interface range ethernet 0/0 - 1 command. This will enter the interface range configuration mode.

On both SW1 and SW2, set the protocol to LACP by using the channel-protocol lacp command.

On both SW1 and SW2, assign the interfaces to an EtherChannel group number 44 by using the channel-group 44 mode active command. This will create a logical interface named Port-channel44 and set the LACP mode to active on both ends. The LACP mode must match on both ends for the EtherChannel to form.

On both SW1 and SW2, exit the interface range configuration mode by using the exit command.

On both SW1 and SW2, enter the Port-channel interface configuration mode by using the interface port-channel 44 command.

On both SW1 and SW2, configure the Port-channel interface as a trunk link by using the switchport mode trunk command.

On both SW1 and SW2, configure the Port-channel interface to use 802.1q tags for VLAN identification by using the switchport trunk encapsulation dot1q command.

On both SW1 and SW2, configure VLAN 'MONITORING' as the untagged VLAN of the Port-channel interface by using the switchport trunk native vlan MONITORING command.

On both SW1 and SW2, exit the Port-channel interface configuration mode by using the exit command.

On both SW1 and SW2, save the configuration to NVRAM by using the copy running-config startup-config command.

---

**Question: 1070**

---

Which feature, when used on a WLC, allows it to bundle its distribution system ports into one 802.3ad group?

- A. QinQ
- B. ISL
- C. PAgP
- D. LAG

---

**Answer: D**

---

Explanation:

---

**Question: 1071**

---

What is the maximum length of characters used in an SSID?

- A. 16
- B. 32
- C. 48
- D. 64

---

**Answer: B**

---

Explanation:

---

**Question: 1072**

---

An engineer must update the configuration on two PCs in two different subnets to communicate locally with each other. One PC is configured with IP address 192.168.25.128/25 and the other with 192.168.25.100/25. Which network mask must the engineer configure on both PCs to enable the communication?

- A. 255.255.255.224
- B. 255.255.255.248
- C. 255.255.255.0
- D. 255.255.255.252

---

**Answer: C**

---

Explanation:

**Question: 1073**

Which two QoS tools provide congestion management? (Choose two.)

- A. PBR
- B. FRTS
- C. PQ
- D. CBWFQ
- E. CAR

**Answer: C**

Explanation:

**Question: 1074**

Refer to the exhibit.

The screenshot displays a network configuration interface with tabs for General, Security, QoS, Policy-Mapping, and Advanced. The QoS tab is selected. On the left, under the QoS section, the 'P2P Blocking Action' is set to 'Disabled'. On the right, under the 'DHCP' section, the 'DHCP Server' is set to 'Override'. Under the 'Management Frame Protection (MFP)' section, 'MFP Client Protection' is set to 'Optional'. Under the 'DTIM Period (in beacon intervals)' section, the values for 802.11a/n (1 - 255) and 802.11b/g/n (1 - 255) are both set to 1. Under the 'NAC' section, 'NAC State' is set to 'None'. Under the 'Load Balancing and Band Select' section, 'Client Load Balancing' and 'Client Band Select' are both set to 'Optional'. Under the 'Passive Client' section, there are no visible settings.

Refer to the exhibit. An administrator is configuring a new WLAN for a wireless network that has these requirements:

Dual-band clients that connect to the WLAN must be directed to the 5-GHz spectrum.

Wireless clients on this WLAN must be able to apply VLAN settings from RADIUS attributes.

Which two actions meet these requirements? (Choose two.)

- A. Enable the Aironet IE option.
- B. Enable the Coverage Hole Detection option.
- C. Set the MFP Client Protection option to Required
- D. Enable the client band select option.
- E. Enable the allow AAA Override option

---

**Answer: D,E**

---

Explanation:

---

**Question: 1075**

---

Which statement describes virtualization on containers?

- A. It is a type of operating system virtualization that allows the host operating system to control the different CPU memory processes.
- B. It emulates a physical computer and enables multiple machines to run with many operating systems on a physical machine.
- C. It separates virtual machines from each other and allocates memory, processors, and storage to compute.
- D. It contains a guest operating system and virtual partition of hardware for OS and requires application libraries.

---

**Answer: A**

---

Explanation:

---

**Question: 1076**

---

An engineer requires a switch interface to actively attempt to establish a trunk link with a neighbor

switch. What command must be configured?

- A. switchport mode dynamic desirable
- B. switchport mode trunk
- C. switchport nonegotiate
- D. switchport mode dynamic auto

---

**Answer: A**

---

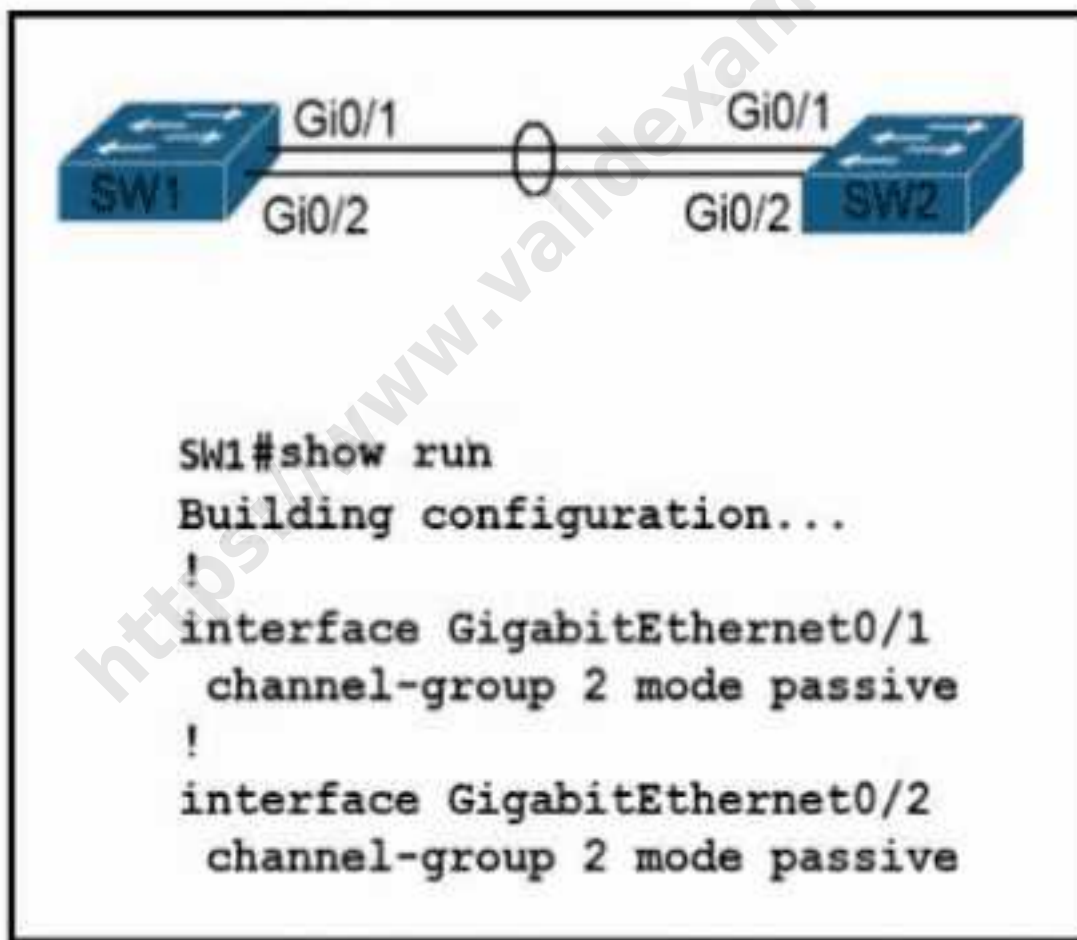
Explanation:

---

**Question: 1077**

---

Refer to the exhibit.



Refer to the exhibit. Which configuration enables SW2 to establish an LACP EtherChannel?

- A.



- B. B.
- C. C.
- D. D.

---

**Answer: C**

---

Explanation:

---

**Question: 1078**

---

What is represented by the word "switch" within this JSON schema?

```
1 [
2  {"IDS": "IPS22", "port": "1e3/46"},
3  {"load balancer": "LB12", "port": "1e6/38"},
4  {"switch": "SW18", "port": "ge2/41"},
5 ]
```

- A. array
- B. value
- C. key
- D. object

---

**Answer: C**

---

Explanation:

---

**Question: 1079**

---

Why would a network administrator choose to implement RFC 1918 address space?

- A. to route traffic on the internet

- B. to provide flexibility in the IP network design
- C. to provide overlapping address space with another network
- D. to limit the number of hosts on the network

---

**Answer: B**

---

Explanation:

---

**Question: 1080**

---

Refer to the exhibit.

```
AA#show ip route
      10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
C       10.10.0.0/30 is directly connected, GigabitEthernet0/1
L       10.10.0.1/32 is directly connected, GigabitEthernet0/1
O       10.20.0.0/30 [110/2] via 10.0.0.2, 00:00:40, GigabitEthernet0/0
O       10.30.0.0/30 [110/2] via 10.0.0.2, 00:00:40, GigabitEthernet0/0
      172.16.0.0/24 is subnetted, 1 subnets
S       172.16.10.0 [1/0] via 10.0.0.2
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/2
L       192.168.10.1/32 is directly connected, GigabitEthernet0/2
S       192.168.20.0/24 [1/0] via 192.168.10.2
```

Refer to the exhibit. What is the administrative distance for the advertised prefix that includes the host IP address 10.30.0.1?

- A. 10.0.0.2
- B. 110
- C. 30
- D. 2

---

**Answer: B**

---

Explanation:

**Question: 1081**

DRAG DROP

Drag and drop the characteristic from the left onto the cable type on the right.

|                                                  |                  |
|--------------------------------------------------|------------------|
| transmits data in the form of electronic signals | copper           |
| used for high-throughput over short distances    |                  |
| supplies conduit for PoE implementations         |                  |
| transmits signals using pulses of light          | multi-mode fiber |
|                                                  |                  |

**Answer:**

Explanation:

|                                                  |
|--------------------------------------------------|
| copper                                           |
| transmits data in the form of electronic signals |
| supplies conduit for PoE implementations         |
| multi-mode fiber                                 |
| used for high-throughput over short distances    |
| transmits signals using pulses of light          |

**Question: 1082**

What is the main purpose of SSH management access?

- A. To support DES 56-bit and 3DES (168-bit) ciphers
- B. To enable secured access to the inbound management interface
- C. To validate management access with username and domain name only
- D. To allow passwords protected with HTTPS encryption to be sent

---

**Answer: B**

---

Explanation:

---

**Question: 1083**

---

An organization developed new security policies and decided to print the policies and distribute them to all personnel so that employees review and apply the policies. Which element of a security program is the organization implementing?

- A. Asset identification
- B. User training
- C. Physical access control
- D. Vulnerability control

---

**Answer: B**

---

Explanation:

---

**Question: 1084**

---

How does automation leverage data models to reduce the operational complexity of a managed network?

- A. Reduces the response time for specific requests to devices with many interfaces
- B. Categorizes traffic and provides insights

- C. Allows the controller to be vendor-agnostic
- D. Streamlines monitoring using SNMP and other polling tools

---

**Answer: C**

---

Explanation:

---

**Question: 1085**

---

What is the total number of users permitted to simultaneously browse the controller management pages when using the AireOS GUI?

- A. 2
- B. 5
- C. 8
- D. 9

---

**Answer: C**

---

Explanation:

---

**Question: 1086**

---

How does MAC learning function?

- A. Enabled by default on all VLANs and interfaces
- B. Forwards frames to a neighbor port using CDP
- C. Overwrites the known source MAC address in the address table
- D. Protects against denial of service attacks

---

**Answer: A**

---

Explanation:

---

**Question: 1087**

---

Which header must be included in a REST request from an application that requires JSON-formatted content?

- A. Content-Type: application/json
- B. Accept-Encoding: application/json
- C. Accept: application/json
- D. Accept-Language: application/json

---

**Answer: A**

---

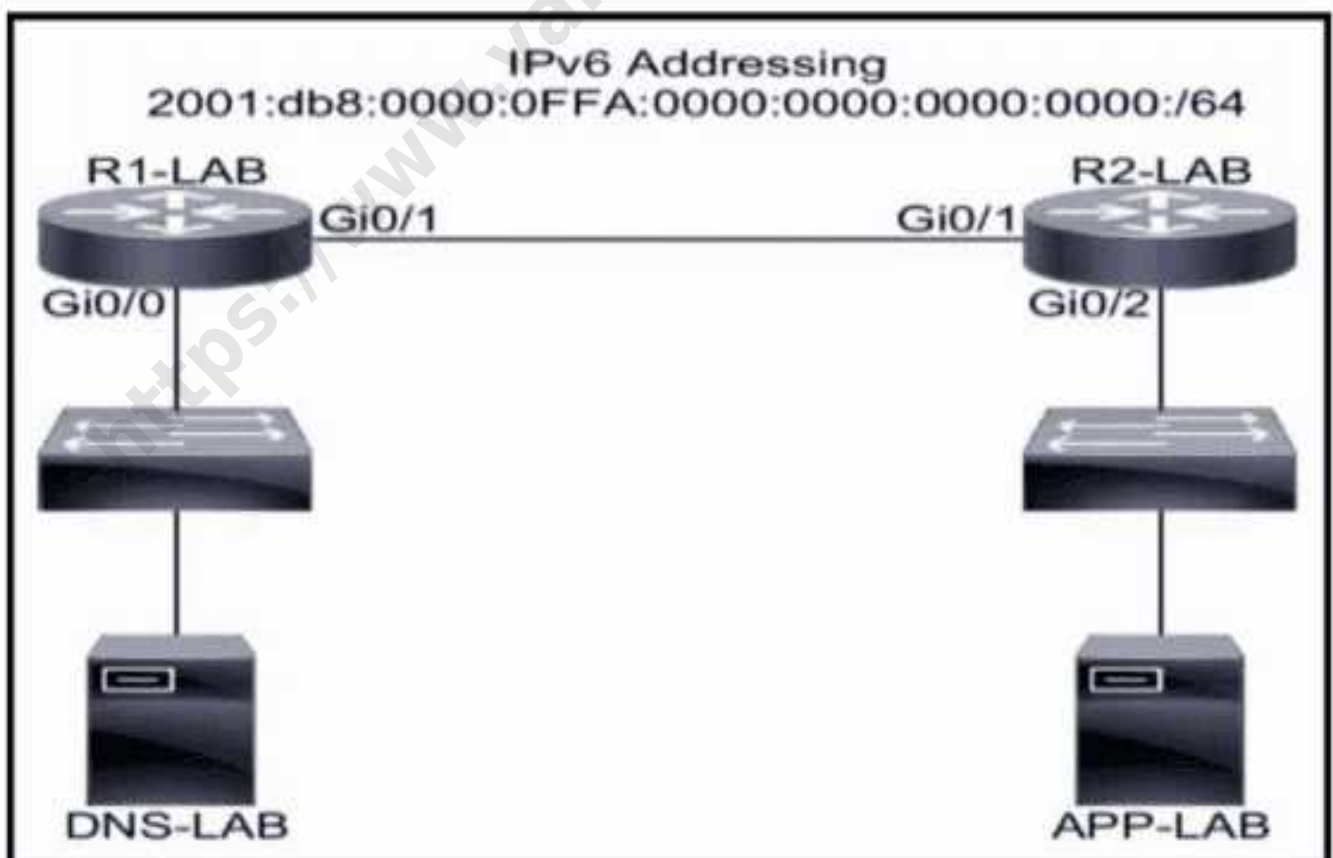
Explanation:

---

**Question: 1088**

---

Refer to the exhibit.



Refer to the exhibit. The routers R1-LAB and R2-LAB are configured with link-local addresses. What command must be applied to interface Gi0/0 on R1-LAB for an automated address self-assignment on the IPv6 network?

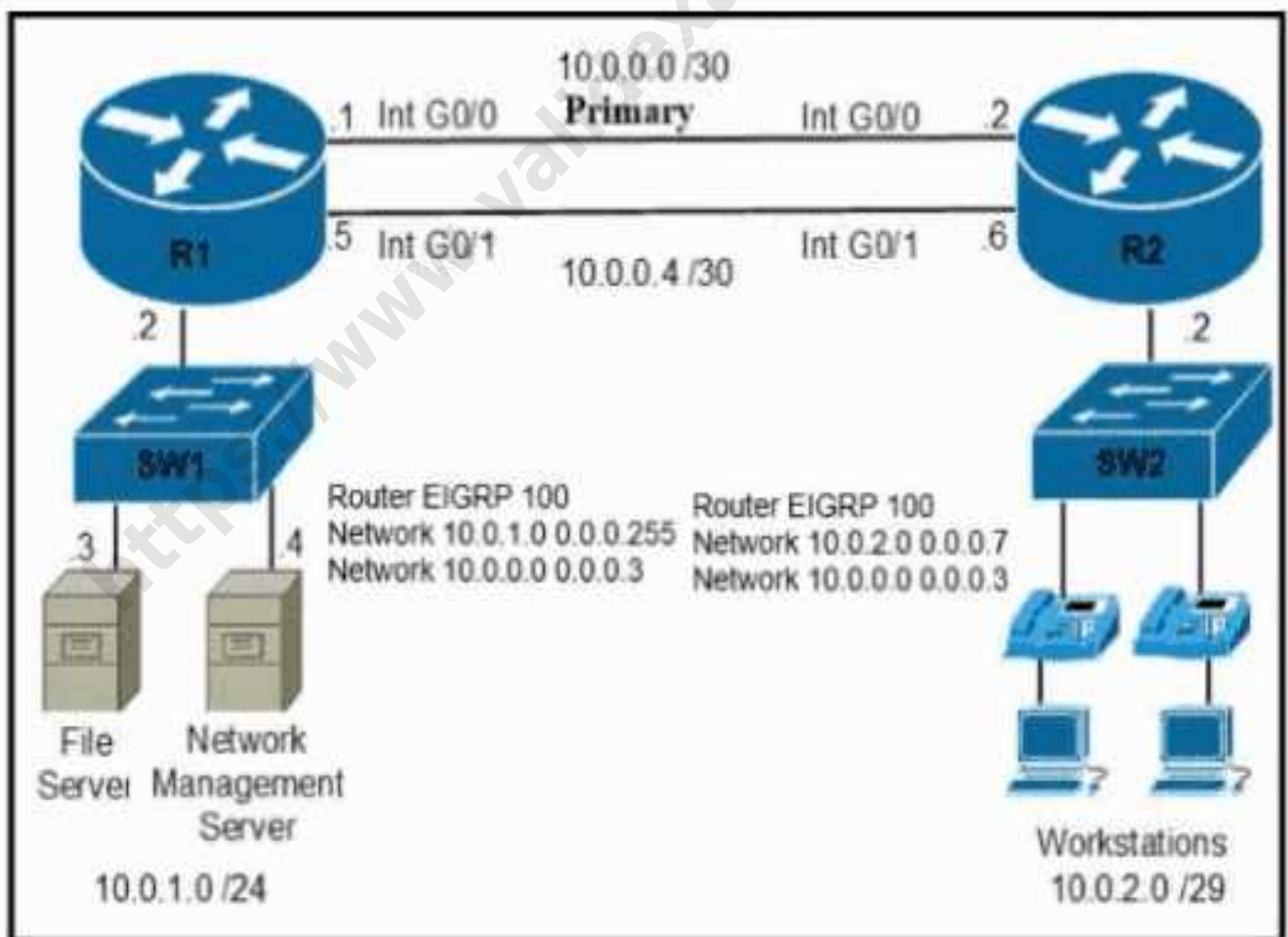
- A. ipv6 address 2001:db8:1:0FFA:0::/64
- B. ipv6 address 2001:db8:0:0FFA::1/64
- C. ipv6 address 2001:db8:0:0FFA::/64 eui-64
- D. ipv6 address 2001:db8:0:0FFA::/64 anycast

**Answer: C**

Explanation:

### Question: 1089

Refer to the exhibit.



Refer to the exhibit. A secondary route is required on router R1 to pass traffic to the LAN network on

R2 if the primary link fails. Which command must be entered to configure the router?

- A. ip route 10.0.2.0 255.255.255.240 10.0.0.7 92
- B. ip route 10.0.2.0 255.255.255.248 10.0.0.6 91
- C. ip route 10.0.2.0 256.255.255.240 10.0.0.6 91
- D. ip route 10.0.2.0 255.255.255.248 null0 93

---

**Answer: B**

---

Explanation:

---

**Question: 1090**

---

What is the difference between the TCP and UDP protocols?

- A. TCP ensures ordered, reliable data delivery, and UDP offers low latency and high throughput.
- B. TCP is used for transmitting data over the internet, and UDP is used for transmitting data over a local network.
- C. TCP manages multicast and broadcast data transfers, and UDP only handles unicast communications.
- D. TCP is used to ensure data integrity in a file transfer, and UDP is used to broadcast a message to multiple recipients.

---

**Answer: A**

---

Explanation:

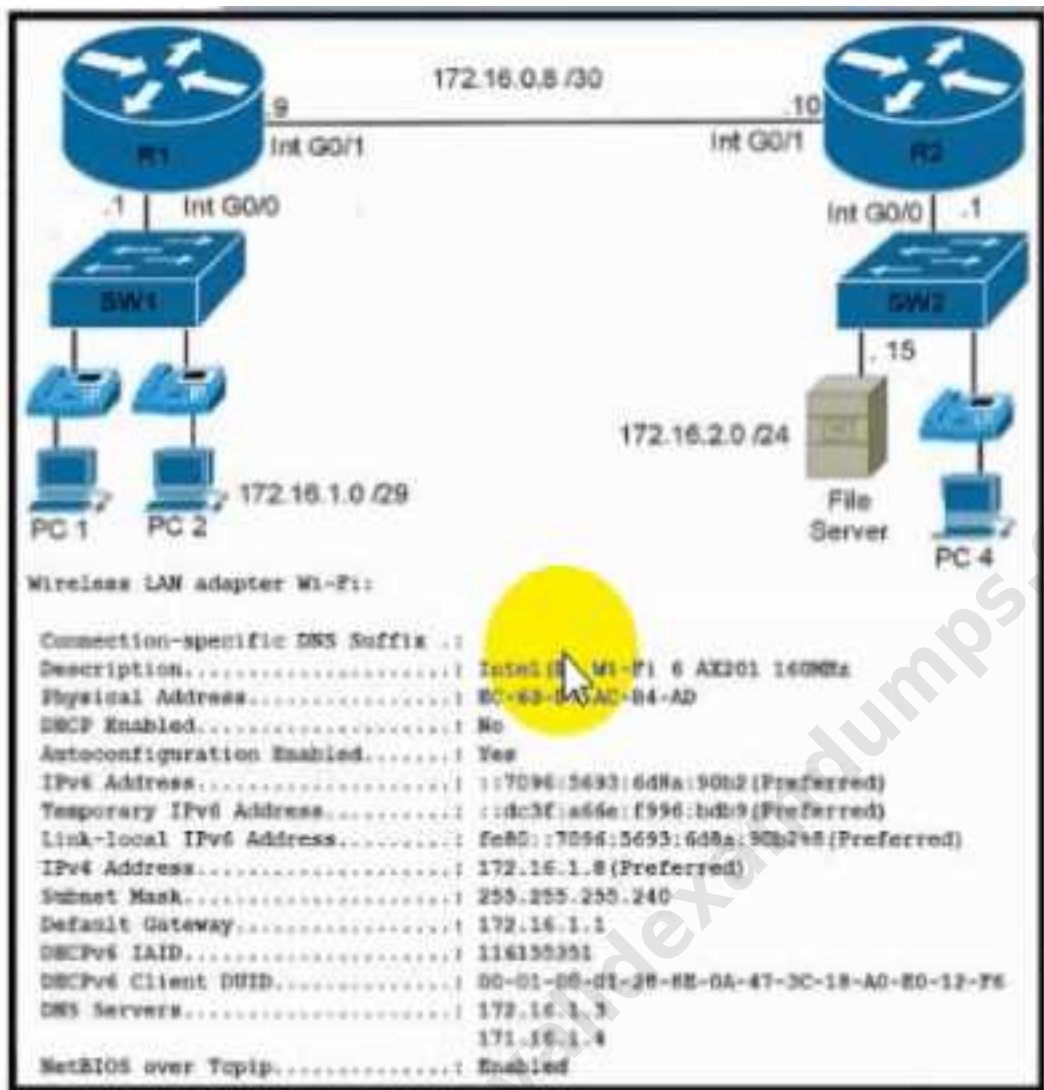
---

**Question: 1091**

---

Refer to the exhibit.





Refer to the exhibit. During initial configuration testing, the Windows workstation PC1 cannot connect with the 172.16.2.0/24 network.

Which set of actions corrects the configuration?

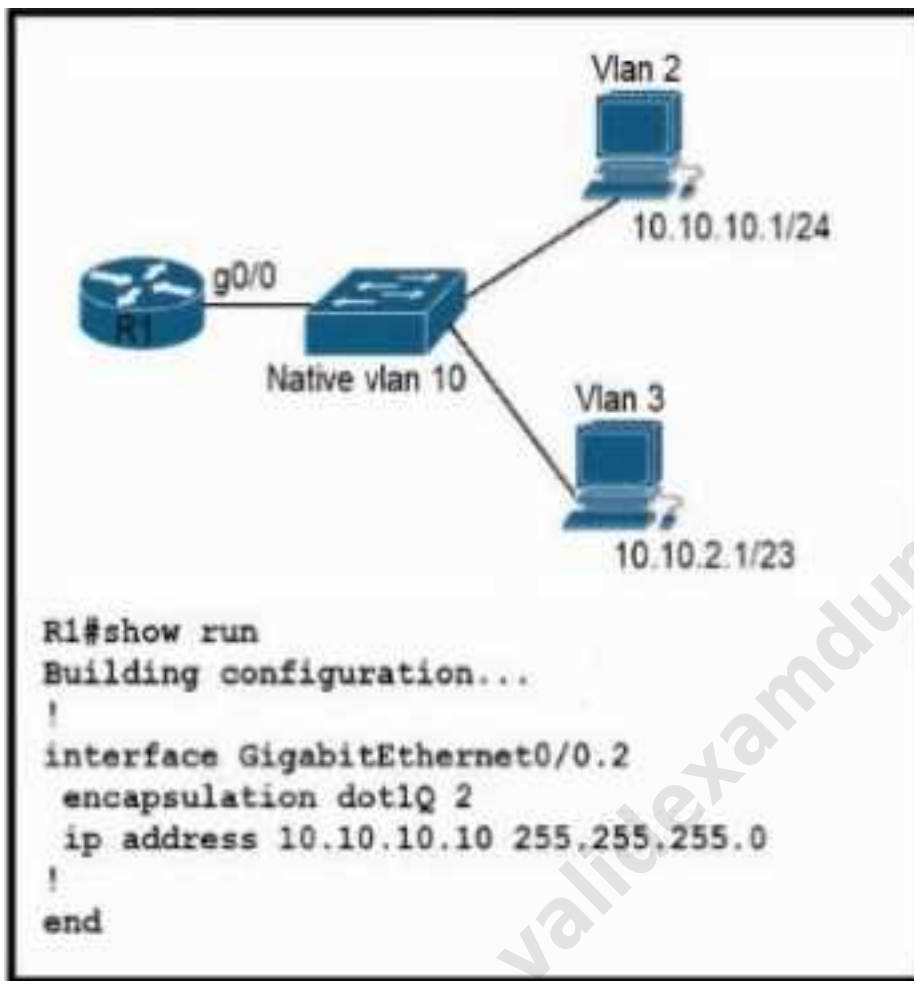
- A. Change the IP address to 172.16.1.6 and change the subnet mask to 255.255.255.248.
- B. Change the IP address to 172.16.1.6 and change the DNS servers to 172.16.1.12 and 172.16.1.13.
- C. Change the IP address to 172.16.1.9 and change the default gateway to 172.16.1.7.
- D. Change the IP address to 172.16.1.9 and change the DNS server to 172.16.1.12 only.

**Answer: A**

Explanation:

**Question: 1092**

Refer to the exhibit.



Refer to the exhibit. Configurations for the switch and PCs are complete.

Which configuration must be applied so that VLANs 2 and 3 communicate back and forth?

- A. interface GigabitEthernet0/0 ip address 10.10.2.10 255.255.252.0
- B. interface GigabitEthernet0/0.3 encapsulation dot1Q 3 native ip address 10.10.2.10 255.255.252.0
- C. interface GigabitEthernet0/0.10 encapsulation dot1Q 3
- D. interface GigabitEthernet0/0.3 encapsulation dot1Q 10 ip address 10.10.2.10 255.255.252.0

**Answer: B**

Explanation:

**Question: 1093**

Refer to the exhibit.

```
SW1
configure terminal
interface range GigabitEthernet 0/1-2
    switchport mode trunk
    channel-group 1 mode active

SW2
configure terminal
interface range GigabitEthernet 0/1-2
    switchport mode trunk
interface Port-channel1
    switchport mode trunk
```

Refer to the exhibit. An LACP EtherChannel between two directly connected switches is in the configuration process.

Which command must be configured on switch SW2's Gi0/1-2 interfaces to establish the channel to SW1?

- A. channel-group 1 mode desirable
- B. channel-group 1 mode on
- C. channel-group 1 mode auto
- D. channel-group 1 mode active

---

**Answer: D**

---

Explanation:

---

### Question: 1094

---

What is an advantage of using SDN versus traditional networking when it comes to security?

- A. SDN security is managed near the perimeter of the network with firewalls, VPNs, and IPS, and traditional networking security policies are created based on telemetry data.
- B. SDN exposes an API to configure locally per device for security policies, and traditional networking uses northbound API for network admin interface for configuring security policies.

C. SDN creates a unified control point making security policies consistent across all devices, and traditional networking must be configured device by device, leaving room for error.

D. SDN devices communicate with each other to establish a security policy, and in traditional networking, devices communicate upstream to a central location to establish a security policy.

---

**Answer: C**

---

Explanation:

---

**Question: 1095**

---

Refer to the exhibit.

```
SW1# show etherchannel summary
Flags: D - down          P - bundled in port-channel
      I - stand-alone s - suspended
      H - Hot-standby (LACP only)
      R - Layer3          S - Layer2
      U - in use          f - failed to allocate aggregator
      M - not in use, minimum links not met
      u - unsuitable for bundling
      w - waiting to be aggregated
      d - default port
      A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (RU)        LACP        Et0/0 (P)  Et0/1 (P)
```

Refer to the exhibit. A network engineer is adding another physical interface as a new member to the existing Port-Channel1 bundle.

Which command set must be configured on the new interface to complete the process?

- A. switchport mode trunk channel-group 1 mode active
- B. no switchport channel-group 1 mode active
- C. no switchport channel-group 1 mode on
- D. switchport switchport mode trunk

---

**Answer: B**

---

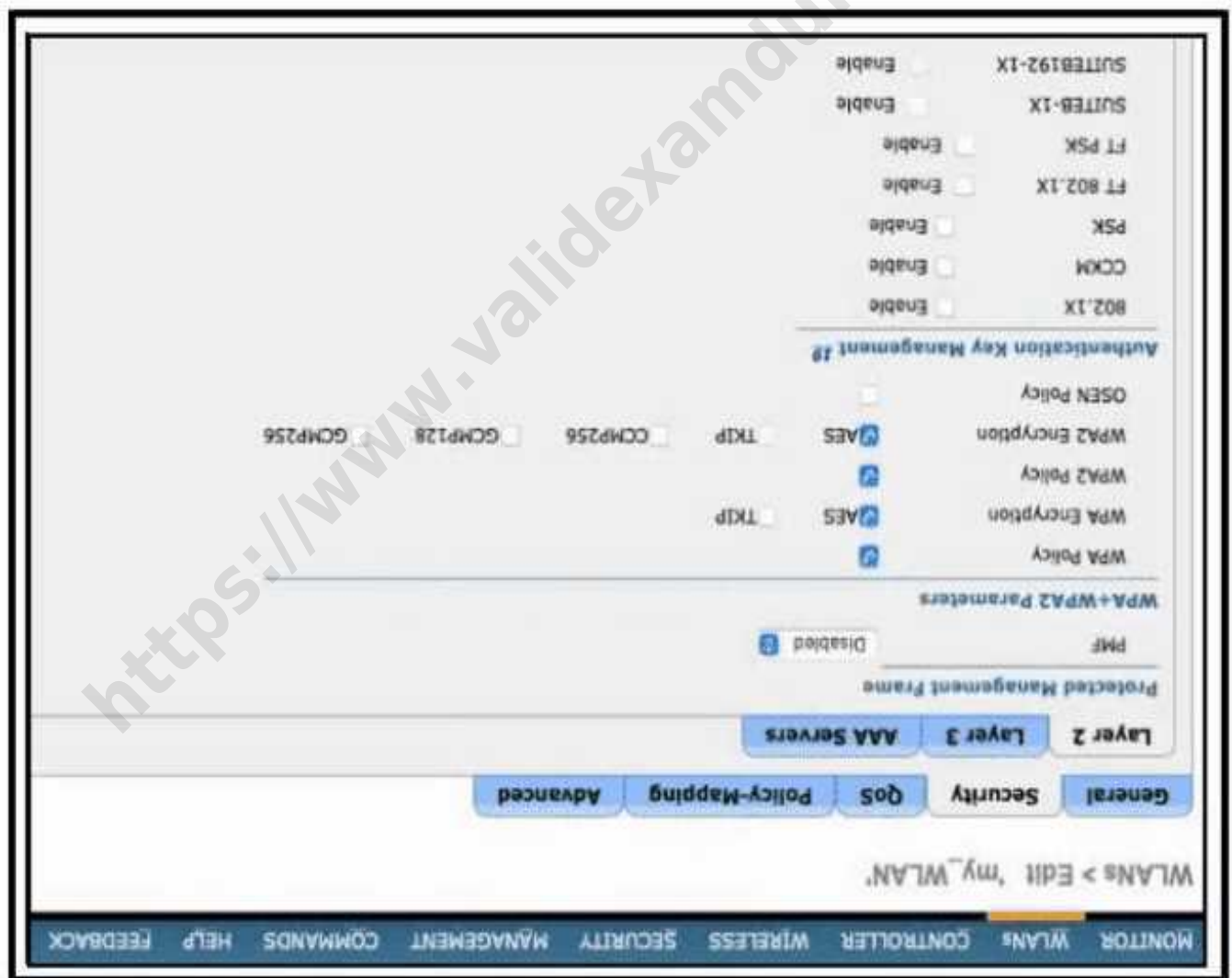
Explanation:

---

**Question: 1096**

---

Refer to the exhibit.



Refer to the exhibit.

Which configuration is needed to configure a WLAN with WPA2 only and with a password that is 63

characters long?

- A. Disable WPA Policy and WPA Encryption and then enable PSK using ASCII.
- B. Enable PSK and FT PSK and then disable WPA Policy.
- C. Disable WPA Encryption and then enable FT PSK.
- D. Enable PSK using Hex format and then disable WPA Policy.

---

**Answer: A**

---

Explanation:

---

**Question: 1097**

---

What is the function of generative AI in network operations?

- A. It disables unused services.
- B. It deploys network firmware updates.
- C. It creates synthetic network configurations.
- D. It computes optimal data storage solutions.

---

**Answer: C**

---

Explanation:

---

**Question: 1098**

---

DRAG DROP

Drag and drop the characteristic from the left onto the IPv6 address type on the right.

|                                                               |              |
|---------------------------------------------------------------|--------------|
| may be used by multiple organizations at the same time        | Unique Local |
| never used as a source address                                |              |
| sends packets to a group address rather than a single address |              |
| allows sites to be combined without address conflicts         | Multicast    |
|                                                               |              |
|                                                               |              |

---

**Answer:**

---

Explanation:

|                                                               |
|---------------------------------------------------------------|
| Unique Local                                                  |
| may be used by multiple organizations at the same time        |
| allows sites to be combined without address conflicts         |
| Multicast                                                     |
| never used as a source address                                |
| sends packets to a group address rather than a single address |

---

**Question: 1099**

---

Which key function is provided by the data plane?

- A. Making routing decisions
- B. Originating packets
- C. Forwarding traffic to the next hop
- D. Exchanging routing table data



---

**Answer: C**

---

Explanation:

---

**Question: 1100**

---

Refer to the exhibit.

```
R1#show ip route

Gateway of last resort is 10.0.0.2 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
C 10.0.0.0/24 is directly connected, FastEthernet0/0
L 10.0.0.1/32 is directly connected, FastEthernet0/0
C 10.0.1.0/24 is directly connected, FastEthernet0/1
L 10.0.1.1/32 is directly connected, FastEthernet0/1
C 10.0.2.0/24 is directly connected, FastEthernet1/0
L 10.0.2.1/32 is directly connected, FastEthernet1/0
C 10.0.3.0/24 is directly connected, FastEthernet1/1
L 10.0.3.1/32 is directly connected, GigabitEthernet1/1
O 10.0.4.0/29 [110/2] via 10.0.4.2 00:00:03, GigabitEthernet1/1
S 10.1.0.0/16 [1/0] via 10.0.3.2
S 10.1.3.0/24 [1/0] via 10.0.3.2
S* 0.0.0.0/0 [1/0] via 10.0.0.2
```

Refer to the exhibit.

How does router R1 forward packets destined to 10.0.4.10?

- A. via 10.0.4.2
- B. via 10.0.0.2
- C. via FastEthernet0/1
- D. via FastEthernet1/1

---

**Answer: B**

---



Explanation:

---

**Question: 1101**

---

What is a characteristic of a Layer 2 switch?

- A. Uses routers to create collision domains
- B. Responsible for sending data in a particular sequence
- C. Avoids MAC address storage for faster transmission
- D. Uses the data link layer for communications

---

**Answer: D**

---

Explanation:

---

**Question: 1102**

---

Which interface is used to send traffic to the destination network?

- 10.90.207.87/26 [110/1912] via F0/7
- 10.90.207.87/26 [110/28968] via F0/6
- 10.90.207.87/26 [120/14] via F0/4
- 10.90.207.87/26 [120/11] via F0/5
- A. F0/7
- B. F0/5
- C. F0/4
- D. F0/6

---

**Answer: A**

---

Explanation:

---

**Question: 1103**

---

Where are the real-time control functions processed in a split MAC architecture?

- A. Centralized cloud management platform
- B. Central WLC
- C. Individual AP
- D. Client device

---

**Answer: C**

---

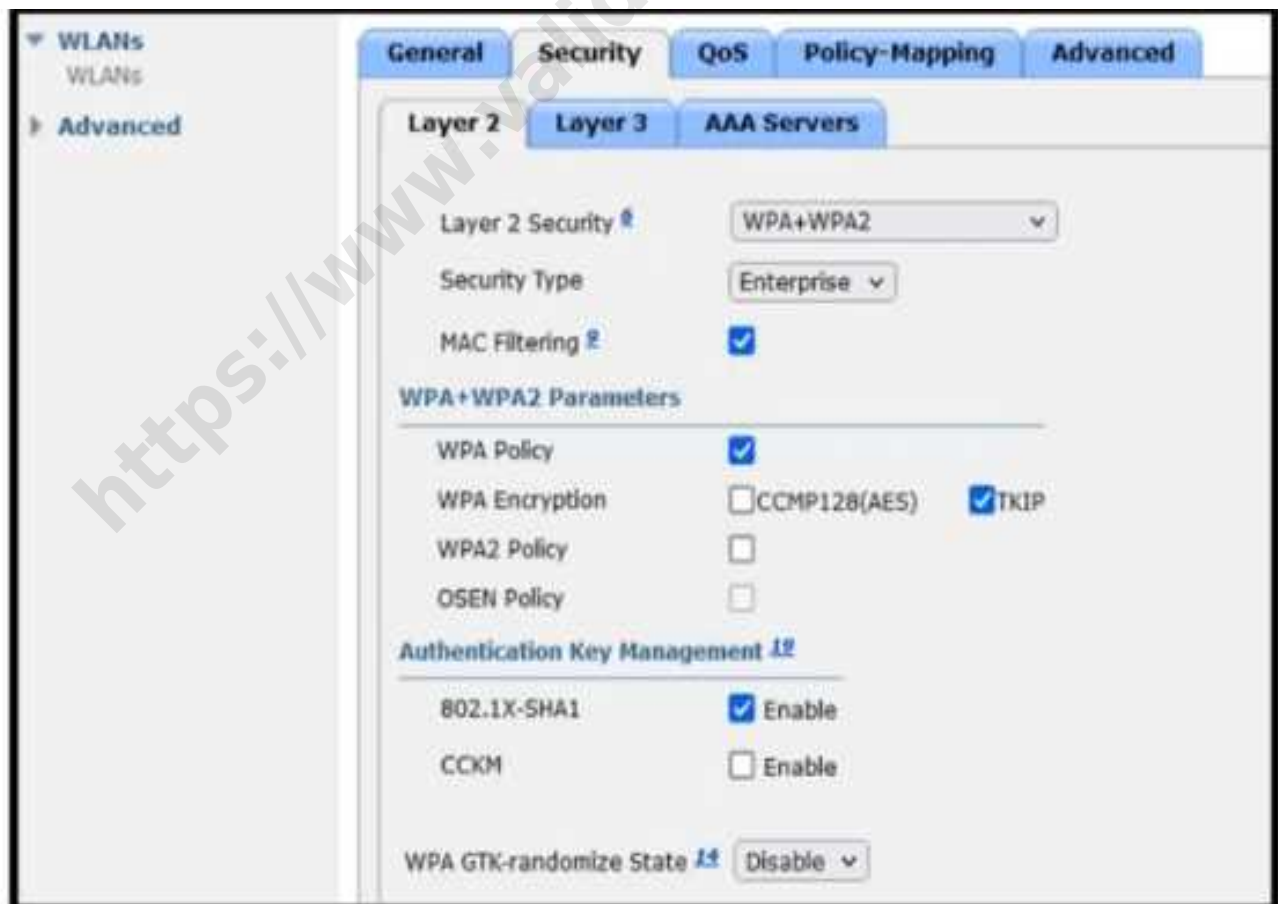
Explanation:

---

**Question: 1104**

---

Refer to the exhibit.



Refer to the exhibit. A guest WLAN must be created that prompts the client for a username and password on the local web page of the WLC. Which two actions must be performed on the Layer 2 tab before enabling the Authentication option on the Layer 3 tab? (Choose two.)

- A. Uncheck the WPA Policy option check box, and check the WPA2 Policy option check box.
- B. Uncheck the MAC Filtering option check box.
- C. Change the WPA Encryption option from TKIP to CCMP(128AES).
- D. Set the Security Type option to Personal.
- E. Set the Layer 2 Security option to None.

---

**Answer: A,D**

---

Explanation:

---

**Question: 1105**

---

What differentiates the TCP and UDP protocols?

- A. TCP tracks segments being transmitted or received by assigning segment numbers, and UDP adjusts data flow according to network conditions.
- B. TCP establishes a connection with the device on the other end before transferring, and UDP transfers without establishing a connection.
- C. TCP sends data at a constant rate with error checking on upper protocol layers, and UDP provides error-checking and sequencing.
- D. TCP immediately transmits data without waiting for a handshake, and UDP awaits a response from the receiver before sending additional data.

---

**Answer: B**

---

Explanation:

---

**Question: 1106**

---

Refer to the exhibit.

```
ip route 172.0.0.0 255.0.0.0 10.65.65.65
ip route 172.16.0.0 255.255.0.0 10.56.65.56
ip route 122.16.153.0 255.255.255.0 10.65.65.56
ip route 172.16.153.153 255.255.255.255 10.65.56.65
ip route 0.0.0.0 0.0.0.0 10.56.56.56
```

Refer to the exhibit. The static routes were implemented on the border router. What is the next hop IP address for a ping sent to 172.16.153.154 from the border router?

- A. 10.56.65.56
- B. 10.56.65.65
- C. 10.65.56.56
- D. 10.65.65.65

**Answer: A**

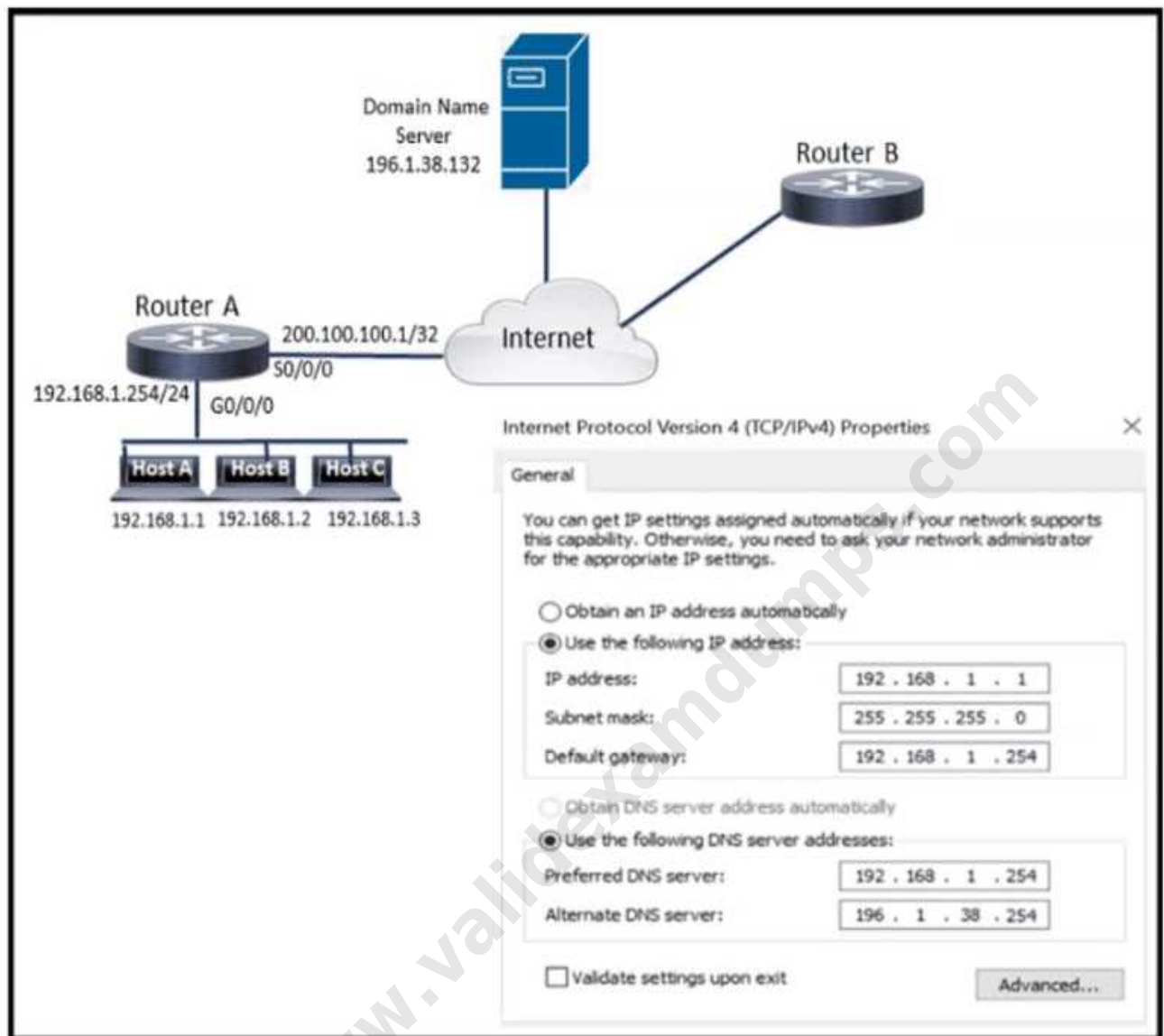
Explanation:

---

**Question: 1107**

---

Refer to the exhibit.



Refer to the exhibit. What is preventing host A from reaching the internet?

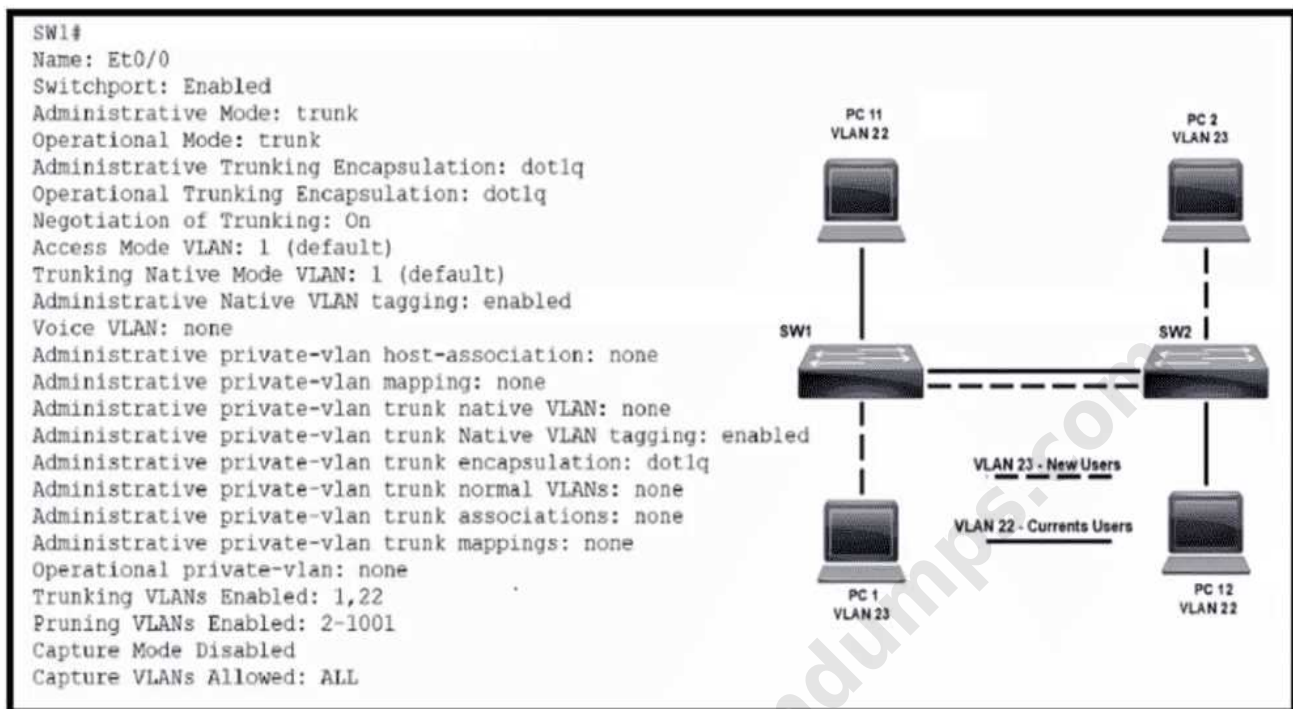
- A. The domain name server is unreachable.
- B. LAN and WAN network segments are different.
- C. IP address assignment is incorrect.
- D. The default gateway should be the first usable IP address.

**Answer: C**

Explanation:

**Question: 1108**

Refer to the exhibit.



Refer to the exhibit. VLAN 23 is being implemented between SW1 and SW2. The command show interface ethernet0/0 switchport has been issued on SW1. Ethernet0/0 on SW1 is the uplink to SW2. Which command when entered on the uplink interface allows PC 1 and PC 2 to communicate without impact to the communication between PC 11 and PC 12?

- A. switchport trunk allowed vlan 2-1001
- B. switchport trunk allowed vlan add 23
- C. switchport trunk allowed vlan 23
- D. switchport trunk allowed vlan 22-23

**Answer: B**

Explanation:

### Question: 1109

Which default condition must be considered when an encrypted mobility tunnel is used between two Cisco WLCs?

- A. TCP port 443 and UDP 21 are used.

- B. Control and data traffic encryption are enabled.
- C. The tunnel uses the IPsec protocol for encapsulation.
- D. The tunnel uses the EoIP protocol to transmit data traffic.

---

**Answer: A**

---

Explanation:

---

**Question: 1110**

---

Which interface is used to send traffic to the destination network?

- O 10.18.75.113/27 [110/6906] via G0/6
- O 10.18.75.113/27 [110/23018] via G0/3
- R 10.18.75.113/27 [120/16] via G0/16
- R 10.18.75.113/27 [120/14] via G0/23
- A. G0/23
- B. G0/3
- C. G0/16
- D. G0/6

---

**Answer: D**

---

Explanation:

---

**Question: 1111**

---

Which type of protocol is VRRP?

- A. uses dynamic IP address assignment
- B. allows two or more routers to act as a default gateway

- C. uses a destination IP address 224.0.0.102 for router-to-router communication
- D. uses Cisco-proprietary First Hop Redundancy Protocol

---

**Answer: D**

---

Explanation:

---

**Question: 1112**

---

Refer to the exhibit.

```
Gateway of last resort is 172.16.2.2 to network 0.0.0.0

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.10.10.0/24 is directly connected, GigabitEthernet0/0/0
L    10.10.10.3/32 is directly connected, GigabitEthernet0/0/0
  172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
S    172.16.1.33/32 is directly connected, GigabitEthernet0/0/1
C    172.16.2.0/23 is directly connected, GigabitEthernet0/0/1
L    172.16.2.1/32 is directly connected, GigabitEthernet0/0/1
S*   0.0.0.0/0 [1/0] via 172.16.2.2
```

Refer to the exhibit. A packet sourced from 10.10.10.1 is destined for 172.16.3.254. What is the subnet mask of the destination route?

- A. 0.0.0.0
- B. 255.255.254.0
- C. 255.255.255.0
- D. 255.255.255.255

---

**Answer: A**

---

Explanation:

---

**Question: 1113**

---



How does MAC learning function?

- A. inserts MAC addresses dynamically into the CAM table
- B. restricts ports to a maximum of 10 dynamically-learned addresses
- C. protects against denial of service attacks
- D. rewrites the source and destination MAC address

---

**Answer: A**

---

Explanation:

---

**Question: 1114**

---

What is a similarity between global and unique local IPv6 addresses?

- A. They are allocated by the same organization.
- B. They are routable on the global internet.
- C. They use the same process for subnetting.
- D. They are part of the multicast IPv6 group type.

---

**Answer: C**

---

Explanation:

---

**Question: 1115**

---

What is the purpose of the service-set identifier?

- A. It identifies the wired network to which a network device is connected.
- B. It identifies a wireless network for a mobile device to connect.
- C. It identifies the wireless network to which an application must connect.

D. It identifies the wired network to which a user device is connected.

**Answer: B**

Explanation:

### Question: 1116

Why are API keys used to enforce rate limiting?

- A. to uniquely identify clients to monitor their usage patterns
- B. to encrypt data to prevent excessive usage
- C. to contain embedded permissions that automatically expire
- D. to track the geographical location of each request

**Answer: A**

Explanation:

### Question: 1117

Refer to the exhibit.

The screenshot displays the Cisco Meraki dashboard interface. At the top, there is a navigation bar with the Cisco logo and several tabs: MONITOR, WLANs, CONTROLLER, WIRELESS, SECURITY, MANAGEMENT, COMMANDS (which is currently selected), HELP, and FEEDBACK. Below the navigation bar, the 'Commands' section is active, showing a list of commands on the left: Download File, Upload File, Reboot, Config Boot, Scheduled Reboot, Reset to Factory Default, Set Time, and Login Banner. The 'Download File' command is selected, leading to the 'Download file to Controller' configuration page. This page includes a 'File Type' dropdown menu, a 'Configuration File Encryption' checkbox, and a 'Transfer Mode' dropdown. Below these, the 'Server Details' section contains input fields for 'IP Address', 'Maximum retries' (set to 10), 'Timeout (seconds)' (set to 6), 'File Path' (set to /), and 'File Name'.

Refer to the exhibit. Which tasks must be performed on the Download File tab to install new software

using TCP port 22?)

- A. Set the File Type to Code, set the Transfer Mode to SFTP, and specify the IP address of the WLC.
- B. Set the File Type to Configuration, set the Transfer Mode to FTP, and specify the IP address of the file server.
- C. Set the File Type to Code, set the Transfer Mode to SFTP, and specify the IP address of the file server.
- D. Set the File Type to Configuration, set the Transfer Mode to SFTP, and specify the IP address of the WLC.

---

**Answer: C**

---

Explanation:

---

**Question: 1118**

---

What are two behaviors of a point-to-point WAN topology? (Choose two.)

- A. It uses a single router to route traffic between sites.
- B. It leverages a dedicated connection.
- C. It connects remote networks through a single line.
- D. It delivers redundancy between the central office and branch offices.
- E. It provides direct connections between each router in the topology.

---

**Answer: B,C**

---

Explanation:

---

**Question: 1119**

---

How does automation affect network management processes?

- A. It interoperates with ISE to define and manage patch and update schedules.

- B. It performs configuration updates based on user profiles.
- C. It improves the efficiency of system lifecycle management.
- D. It provides a reactive support model.

---

**Answer: C**

---

Explanation:

---

**Question: 1120**

---

Which interface condition is occurring in this output?

```
R9# show interface fa0/0
FastEthernet0/0 is up, line protocol is up
Hardware is DEC21140, address is ca02.7788.0000 (bia ca02.7788.0000)
Description: atlanta_subnet
Internet address is 10.32.102.2/30
MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (60 sec)
Full-duplex, 100 Mb/s, 100BaseTX/FX
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:01, output 00:00:00, output hang never
Last clearing of "show interface" counters 00:00:18
Input queue: 175/300/0/0 (size/max/drops/flushes); Total output drops: 100
Queueing strategy: fifo
Output queue: 50/300 (size/max)
30 second input rate 0 bits/sec, 0 packets/sec
30 second output rate 0 bits/sec, 0 packets/sec
7331 packets input, 7101162 bytes
Received 267 broadcasts (0 IP multicasts)
0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 watchdog
0 input packets with dribble condition detected
3927 packets output, 1440403 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
```

- A. duplex mismatch
- B. queueing
- C. bad NIC
- D. broadcast storm

---

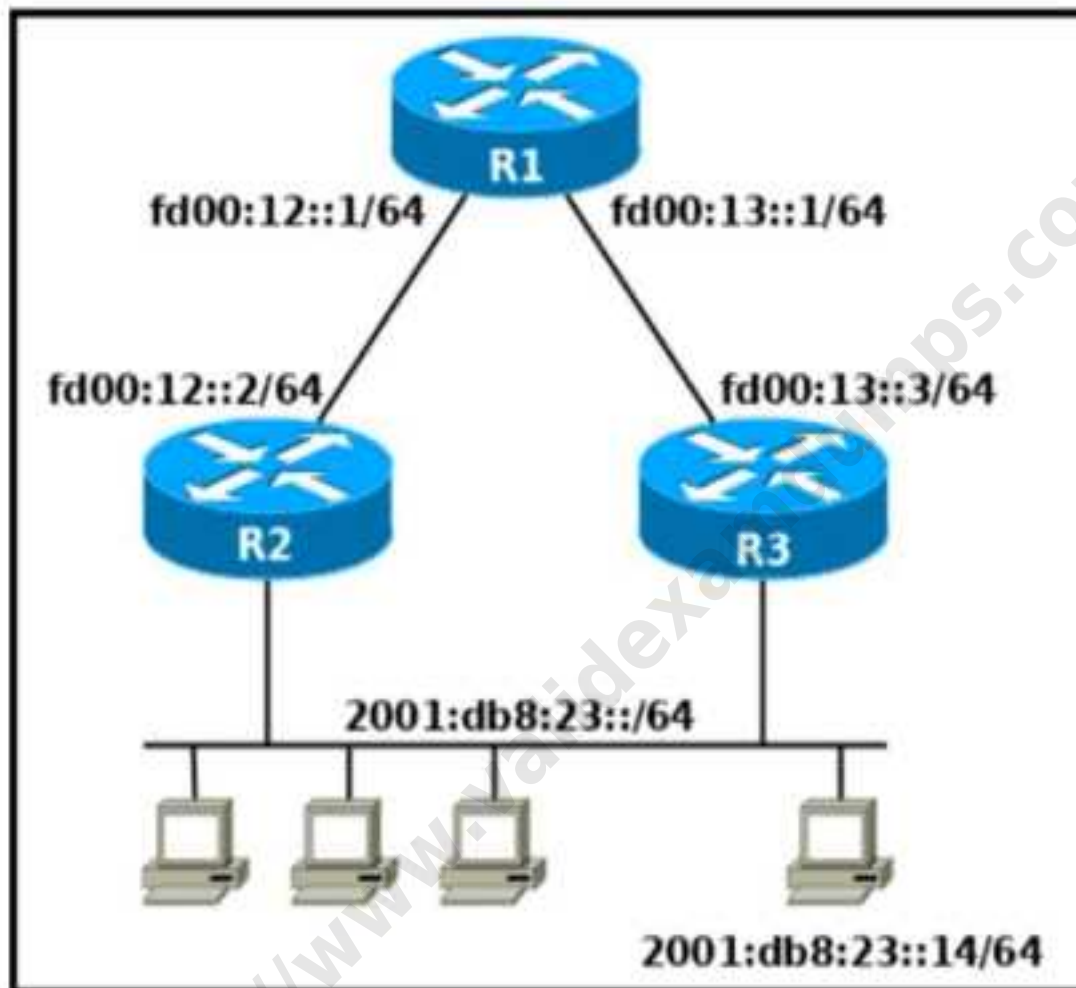
**Answer: A**

---

Explanation:

**Question: 1121**

Refer to the exhibit.



Refer to the exhibit. Which two commands, when configured on router R1, fulfill these requirements? (Choose two.) ' Packets toward the entire network 2001:db8:23: :/64 must be forwarded through router R2. ' Packets toward host 2001: db8:23::14 preferably must be forwarded through R3.

- A. ipv6 route 2001:db8:23: :/128 fd00:12::2
- B. ipv6 route 2001:db8:23::14/128 fd00:13::3
- C. ipv6 route 2001:db8:23::14/64 fd00:12::2 200
- D. ipv6 route 2001:db8:23: l4/64 fd00:12::2
- E. ipv6 route 2001:db8:23: :/64 fd00:12::2

---

**Answer: B,E**

---

Explanation:

---

**Question: 1122**

---

How does a network administrator securely manage an AP in lightweight mode?

- A. using the CLI via an out-of-band connection
- B. using the WLC GUI via HTTPS
- C. using the AP GUI via an in-band SSH connection
- D. using the CLI via a virtual interface with SSH

---

**Answer: B**

---

Explanation:

---

**Question: 1123**

---

Which IP address is used when an administrator must open a web-based management session with a lightweight AP?

- A. WLCIP
- B. gateway IP
- C. autonomous AP IP
- D. ACS IP

---

**Answer: A**

---

Explanation:

---

**Question: 1124**

---

How is AI used to identify issues within network traffic?

- A. It exclusively predicts device malfunctions.
- B. It enhances data packet delivery speeds.
- C. It simplifies traffic route mapping.
- D. It analyzes patterns for anomaly detection.

---

**Answer: D**

---

Explanation:

---

**Question: 1125**

---

Why does an administrator choose to implement a remote access IPsec VPN?

- A. to establish an encrypted tunnel between a remote user and a private network over the internet
- B. to allow access to an enterprise network using any internet-enabled location via a web browser using SSL
- C. to provide a secure link between an HTTPS server, authentication subsystem, and an end-user
- D. to use cryptography for authentication between a device and user over a negotiated VPN gateway

---

**Answer: A**

---

Explanation:

---

**Question: 1126**

---

What are two benefits of private IPv4 addressing? (Choose two.)

- A. provides external internet network connectivity
- B. provides unlimited address ranges



- C. propagates routing information to WAN links
- D. reuses addresses at multiple sites
- E. conserves globally unique address space

---

**Answer: D,E**

---

Explanation:

---

**Question: 1127**

---

Which interface on the WLC is used exclusively as a DHCP relay?

- A. distribution
- B. service
- C. AP-manager
- D. virtual

---

**Answer: B**

---

Explanation:

---

**Question: 1128**

---

Which role do predictive AI models play in network load balancing?

- A. They anticipate future traffic spikes.
- B. They assign IP addresses to devices.
- C. They select correct cabling types for deployment.
- D. They solely monitor historical traffic volumes.

---

**Answer: A**

---

Explanation:

---

**Question: 1129**

---

Refer to the exhibit.

```
%AMDP2 FE-5-COLL: AMDP2/FE 0/0/[DEC], Excessive collisions, TDR=[DEC], TRC=[DEC]
%DEC21140-5-COLL: [chars] excessive collisions
%ILACC-5-COLL: Unit [DEC], excessive collisions. TDR=[DEC]
%LANCE-5-COLL: Unit [DEC], excessive collisions. TDR=[DEC]
%PQUICC-5-COLL: Unit [DEC], excessive collisions. Retry limit [DEC] exceeded
%PQUICC_ETHER-5-COLL: Unit [DEC], excessive collisions. Retry limit [DEC] exceeded
```

What is occurring on this switch?

- A. A high number of frames smaller than 64 bytes are received.
- B. Frames are dropped after 16 failed transmission attempts.
- C. The internal transmit buffer is overloaded.
- D. An excessive number of frames greater than 1518 bytes are received.

---

**Answer: B**

---

Explanation:

---

**Question: 1130**

---

Refer to the exhibit.

```
AA#show ip route
      10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
C       10.10.0.0/30 is directly connected, GigabitEthernet0/1
L       10.10.0.1/32 is directly connected, GigabitEthernet0/1
O       10.20.0.0/30 [110/2] via 10.0.0.2, 00:00:40, GigabitEthernet0/0
O       10.30.0.0/30 [110/2] via 10.0.0.2, 00:00:40, GigabitEthernet0/0
      172.16.0.0/24 is subnetted, 1 subnets
S       172.16.10.0 [1/0] via 10.0.0.2
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/2
L       192.168.10.1/32 is directly connected, GigabitEthernet0/2
S       192.168.20.0/24 [1/0] via 192.168.10.2
```

Refer to the exhibit. What is the administrative distance for the advertised prefix that includes the host IP address 192.168.20.1?

- A. 0
- B. 192.168.10.2
- C. 24
- D. 1

---

**Answer: D**

---

Explanation:

---

**Question: 1131**

---

Which fact must the engineer consider when implementing syslog on a new network?

- A. Syslog defines the software or hardware component that triggered the message.
- B. There are 16 different logging levels (0-15).
- C. By default, all message levels are sent to the syslog server.
- D. The logging level defines the severity of a particular message.

---

**Answer: D**

---

Explanation:

---

**Question: 1132**

---

Which technology allows for logical Layer 3 separation on physical network equipment?

- A. Virtual Route Forwarding
- B. Virtual Switch System
- C. IPsec Transport Mode
- D. Time Division Multiplexer

---

**Answer: A**

---

Explanation:

---

**Question: 1133**

---

Which two features are provided by Ansible in network automation? (Choose two.)

- A. supplying network credentials
- B. role-based access control
- C. agentless deployment
- D. manual playbook runs
- E. launching job templates using version control

---

**Answer: A,C**

---

Explanation:

---

**Question: 1134**

---

Refer to the exhibit.

The screenshot shows the Cisco IOS configuration interface for a wireless network. The 'Security' tab is active. The configuration is as follows:

- Security Type:** Personal (dropdown)
- MAC Filtering:** ☐ (disabled)
- AutoConfig IPSK:** ☐ Enable (disabled)
- WPA2+WPA3 Parameters:**
  - Policy:** ☒ WPA2 ☐ WPA3
  - Encryption Cipher:** ☐ CCMP128(AES)
- Fast Transition:**
  - Fast Transition:** Disable (dropdown)
- Protected Management Frame:**
  - PMF:** Disabled (dropdown)
- Authentication Key Management:**
  - PSK Format:** ASCII (dropdown)
  - PSK:** ☐ Enable (disabled)
  - PSK-SHA2:** ☐ Enable (disabled)

Users will be using a preconfigured secret key and SSID and must have a secured key hashing algorithm configured. The AAA server must not be used for the user authentication method. Which action completes the task?

- A. Enable AutoConfig iPSK.
- B. SetCCMP128(AES).
- C. Configure PSK Format HEX with key string.
- D. Configure PSK-SHA2.

**Answer: D**

Explanation:

---

**Question: 1135**

---

What is represented by the word "LB13" within this JSON schema?

```
1 [
2  {"load balancer": "LB13", "port": "fe9/2"},
3  {"firewall": "FW20", "port": "e2/28"},
4  {"router": "R41", "port": "te7/27"},
5 ]
```

- A. value
- B. object
- C. array
- D. key

---

**Answer: A**

---

Explanation:

---

**Question: 1136**

---

Aswitch receives a frame with the destination MAC address 3C:5D: 7E:9F: 1A:2B.

Switch# show ethernet-frame-and-mac-address-table

**Switch# show ethernet-frame-and-mac-address-table**

Ethernet Frame:

```
+-----+
| Destination MAC Address: 3C:5D:7E:9F:1A:2B |
| Source MAC Address: B0:D1:14:24:34:31      |
| EtherType/Length: 0x0800 (IPv4)           |
| Payload: [Data]                           |
| Frame Check Sequence: [CRC]               |
+-----+
```

MAC Address Table:

| VLAN | MAC Address       | Type    | Ports   |
|------|-------------------|---------|---------|
| ---- | -----             | ----    | -----   |
| 9    | B0:D1:14:24:34:31 | Dynamic | Gi7/0/1 |
| 13   | A5:D5:10:54:66:67 | Static  | Gi7/0/2 |
| 21   | 7D:8E:9F:B0:D1:33 | Secure  | Gi7/0/3 |
| 28   | 06:17:B0:39:50:34 | Dynamic | Gi7/0/4 |
| 41   | 6C:7D:8E:9F:B0:35 | Static  | Gi7/0/5 |

How does the switch handle the frame?

- A. It ages out the frame until the MAC address becomes known.
- B. It drops the frame to avoid unnecessary network congestion.
- C. It switches the frame to a predetermined port based on settings.
- D. It floods the frame to all ports except the incoming port.

---

**Answer: D**

---

Explanation:

---

**Question: 1137**

---

Which interface is used to send traffic to the destination network?

- ☐ 10.76.170 161/26 | 110/102] via FO/17
- ☐ 10.76.170 161/26[110/27e31] via FO/20

R 10.76.170.161/26 [120/15] via FO/8

R 10.76.170.161/26 [120/10] via FO/12

- A. F0/8
- B. FO/20
- C. FO/12
- D. FO/17

---

**Answer: B**

---

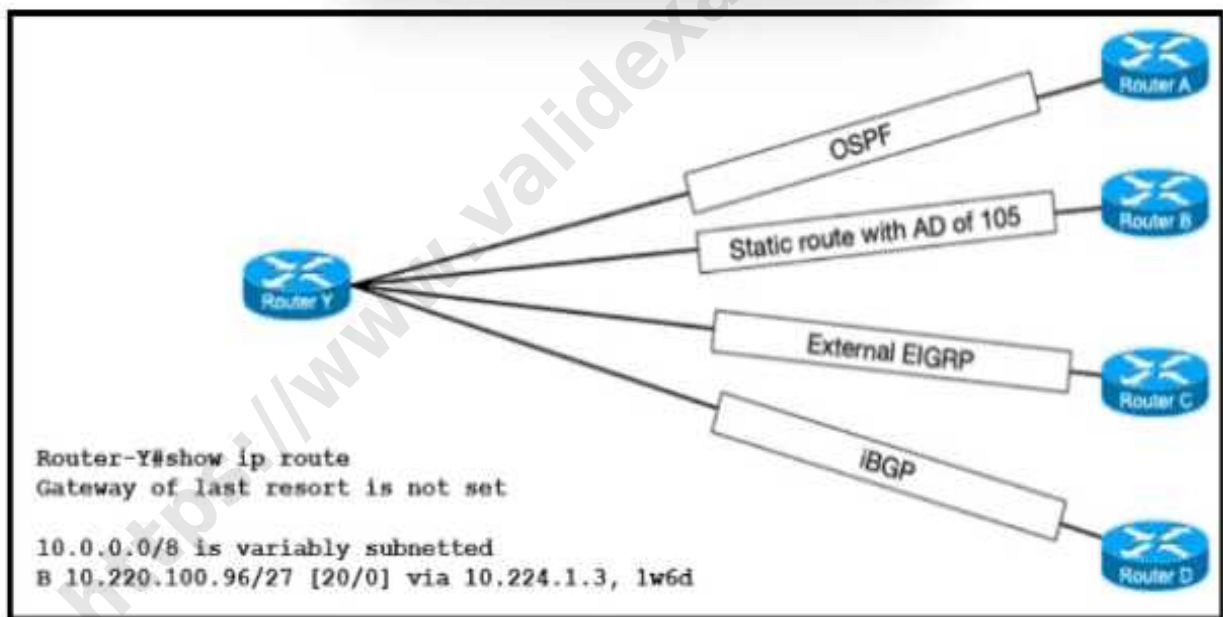
Explanation:

---

**Question: 1138**

---

Refer to the exhibit.



Refer to the exhibit. The route for 10.220.100.96/27 has been very unstable. The same route has four backups to routers A, B, C, and D via the respective methods. The routing protocol defaults for router Y have not been changed. When the current route for 10.220.100.96/27 becomes unavailable, which router will router Y use to route traffic to 10.220.100.96/27?

- A. router D
- B. router B
- C. router C



D. router A

---

**Answer: D**

---

Explanation:

---

**Question: 1139**

---

A new DHCP server has been deployed in a corporate environment with lease time set to eight hours. Which CMD command on a Windows-based device allows the engineer to verify the DHCP lease expiration?

- A. ipconfig /renew
- B. ipconfig
- C. ipconfig /all
- D. ipconfig /displaydns

---

**Answer: C**

---

Explanation:

---

**Question: 1140**

---

Which technology allows multiple operating systems to run a single physical server?

- A. cloud computing
- B. virtualization
- C. application hosting
- D. containers

---

**Answer: B**

---

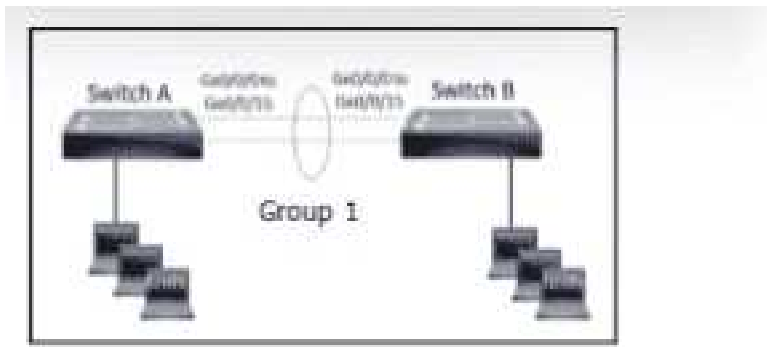
Explanation:

---

**Question: 1141**

---

Refer to the exhibit.



The LACP EtherChannel is configured, and the last change is to modify the interfaces on SwitchA to respond to packets received, but not to initiate negotiation. The interface range `gigabitethernet0/0-15` command is entered. What must be configured next?

- A. `SwitchA(config-if-range) #channel-group 1 mode desirable`
- B. `SwitchA(config-if-range) #channel-group 1 mode auto`
- C. `SwitchA(config-if-range) #channel-group 1 mode active`
- D. `SwitchA(config-if-range) #channel-group 1 mode passive`

---

**Answer: D**

---

Explanation:

---

**Question: 1142**

---

Which two principles must be considered when using per-hop behavior in QoS? (Choose two.)

- A. Policing is not supported on subinterfaces.
- B. Shaping and rate limiting have the same effect.
- C. Shaping drops excessive traffic without adding traffic delay.
- D. Shaping levels out traffic bursts by delaying excess traffic.

E. Policing is performed in the inbound and outbound directions.

---

**Answer: D,E**

---

Explanation:

---

**Question: 1143**

---

What is a function of a northbound API in an SDN environment?

- A. It supports distributed processing for configuration.
- B. It relies on global provisioning and configuration.
- C. It upgrades software and restores files.
- D. It provides orchestration and network automation services.

---

**Answer: D**

---

Explanation:

---

**Question: 1144**

---

Which security element uses a combination of one-time passwords, a login name, and a personal smartphone?

- A. software-defined segmentation
- B. multifactor authentication
- C. attribute-based access control
- D. rule-based access control

---

**Answer: B**

---

Explanation:

**Question: 1145**

DRAG DROP

Drag and drop the IPv6 address from the left onto the type on the right.

|                                       |                    |
|---------------------------------------|--------------------|
| fe80:1e60:1f59:01b8:ece8:e962:4364:7  | Global Unicast     |
| ff00:45fc:709a:d439:62cb:cdad:0a15:12 | Link-Local Unicast |
| fc00:302d:1b26:db7:ce8b:17ae:be9f:3   | Unique Local       |
| 2000:6a4f:74f8:f7b6:cb1a:934c:36ee:1  | Multicast          |

**Answer:**

Explanation:

|                                       |
|---------------------------------------|
| 2000:6a4f:74f8:f7b6:cb1a:934c:36ee:1  |
| fe80:1e60:1f59:01b8:ece8:e962:4364:7  |
| fc00:302d:1b26:db7:ce8b:17ae:be9f:3   |
| ff00:45fc:709a:d439:62cb:cdad:0a15:12 |

**Question: 1146**

Which AP mode wirelessly connects two separate network segments each set up within a different campus building?

- A. mesh
- B. local
- C. bridge
- D. point-to-point

---

**Answer: C**

---

Explanation:

---

**Question: 1147**

---

What is a characteristic of private IPv4 addressing?

- A. alleviates the shortage of IPv4 addresses
- B. reduces the forwarding table on network routers
- C. enables secure connectivity over the internet
- D. used as the NAT outside global IP address

---

**Answer: C**

---

Explanation:

---

**Question: 1148**

---

Which action prevents debug messages from being sent via syslog while allowing other messages when an abnormally high number of syslog messages are generated by a device with the debug process turned on?

- A. Use an access list to filter out the syslog messages.
- B. Turn off the logging monitor in global configuration mode.
- C. Disable logging to the console.
- D. Set the logging trap severity level to informational.

---

**Answer: D**

---

Explanation:

---

**Question: 1149**

---

In which way does generative AI aid network simulations?

- A. It deploys network firmware updates.
- B. It creates synthetic network configurations.
- C. It computes optimal data storage solutions.
- D. It produces a greenfield network design.

---

**Answer: B**

---

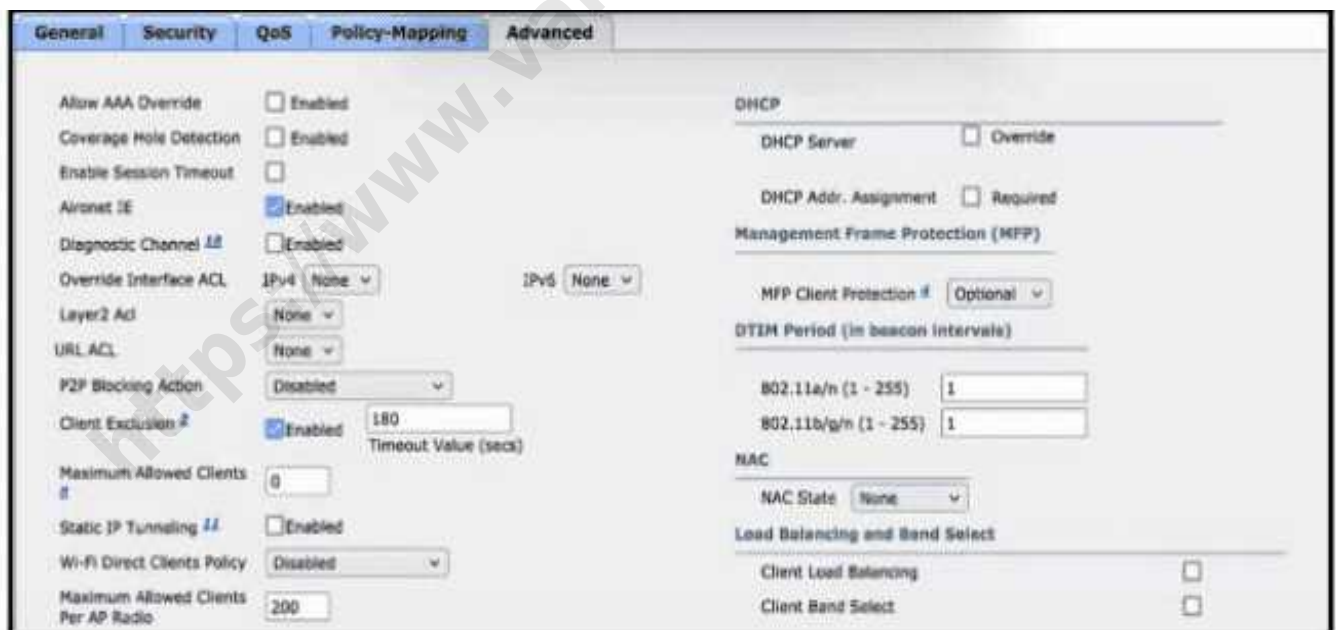
Explanation:

---

**Question: 1150**

---

Refer to the exhibit.



Refer to the exhibit. After a recent internal security audit, the network administrator decided to block all P2P-capable devices from the selected SSID. Which configuration setting must the administrator apply?

- A. Set the P2P Block Action to Drop.

- B. Select a correctly configured Layer 2 ACL.
- C. Set the Wi-Fi Direct Client Policy to Not-Allow.
- D. Set the MFP Client Protection to Required.

---

**Answer: A**

---

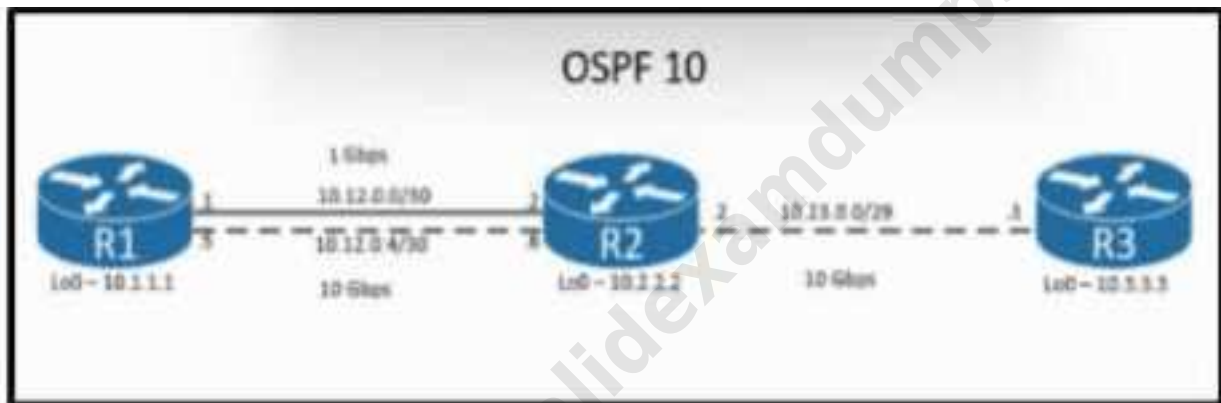
Explanation:

---

**Question: 1151**

---

Refer to the exhibit.



Refer to the exhibit. The loopback IP of R3 has been learned via the two interfaces on R1. R1 is configured with a reference bandwidth of 10 Gbps. Based on the metric calculations, which next hop IP would be used for outgoing routing?

- A. 10.12.0.5
- B. 10.12.0.2
- C. 10.12.0.1
- D. 10.12.0.6

---

**Answer: D**

---

Explanation:

---

**Question: 1152**

---

Refer to the exhibit.

| Prefix          | Interface | Next-hop  |
|-----------------|-----------|-----------|
| 152.168.32.0/22 | f0/0      | 10.10.1.2 |
| 152.168.32.0/24 | f1/0      | 10.10.2.2 |
| 152.168.32.0/26 | f2/0      | 10.10.3.2 |
| 152.168.32.0/23 | f3/0      | 10.10.4.2 |

Refer to the exhibit. Which next hop is used to route packets to the application server at 152.168.32.85?

- A. 10.10.1.2
- B. 10.10.2.2
- C. 10.10.3.2
- D. 10.10.4.2

**Answer: B**

Explanation:

---

**Question: 1153**

Refer to the exhibit.

| Current Neighbor Relationship |     |         |           |             |                    |
|-------------------------------|-----|---------|-----------|-------------|--------------------|
| Neighbor ID                   | Pri | State   | Dead Time | Address     | Interface          |
| 192.168.1.1                   | 1   | FULL/DR | 00:00:33  | 192.168.1.1 | GigabitEthernet0/0 |
| Desired Neighbor Relationship |     |         |           |             |                    |
| Neighbor ID                   | Pri | State   | Dead Time | Address     | Interface          |
| 192.168.1.1                   | 0   | FULL/ - | 00:00:31  | 192.168.1.1 | GigabitEthernet0/0 |

Refer to the exhibit. How must OSPF be configured on the GigabitEthernet0/0 interface of the neighbor device to achieve the desired neighbor relationship?



- A. Router(config)#interface GigabitEthernet 0/0Router(config-if)#ip ospf network point-to-point
- B. Router(config)#interface GigabitEthernet 0/0Router(config-if)#ip ospf cost 5
- C. Router(config)#interface GigabitEthernet 0/0Router(config-if)#ip ospf 1 area 2
- D. Router(config)#interface GigabitEthernet 0/0Router(config-if)#ip ospf priority 1

---

**Answer: A**

---

Explanation:

---

**Question: 1154**

---

A network security team noticed that an increasing number of employees are becoming victims of phishing attacks. Which security program should be implemented to mitigate the problem?

- A. Physical access control
- B. Software firewall enabled on all PCs
- C. Email system patches
- D. User awareness training

---

**Answer: D**

---

Explanation:

---

**Question: 1155**

---

DRAG DROP

Drag and drop the IPv6 address from the left onto the type on the right.

|                                        |                    |
|----------------------------------------|--------------------|
| fe80::a0f:3a8c:f04e:96ae:e7d5:1f2e:7   | Global Unicast     |
| ff00::0a21:c015:4a5b:48ff:0a17:a2fc:12 | Link-Local Unicast |
| fc00::98b2:3fc8:227:ca58:11f8:2663:3   | Unique Local       |
| 2000::d17d:ba4a:09c4:ec2d:64c2:edba0:1 | Multicast          |

---

**Answer:**

---

Explanation:

|                                        |
|----------------------------------------|
| 2000::d17d:ba4a:09c4:ec2d:64c2:edba0:1 |
| fe80::a0f:3a8c:f04e:96ae:e7d5:1f2e:7   |
| fc00::98b2:3fc8:227:ca58:11f8:2663:3   |
| ff00::0a21:c015:4a5b:48ff:0a17:a2fc:12 |

---

**Question: 1156**

---

Which QoS feature drops traffic that exceeds the committed access rate?

- A. Shaping
- B. Policing
- C. Weighted fair queuing
- D. FIFO

---

**Answer: B**

---

Explanation:

---

**Question: 1157**

---

Which statement describes virtual machines?

- A. They include a guest OS and the service.
- B. They enable the network to become agile and hardware-centric.
- C. They use a supervisor to provide management for services.
- D. They facilitate local management of infrastructure devices.

---

**Answer: A**

---

Explanation:

---

**Question: 1158**

---

Which feature of Digest Authentication prevents credentials from being sent in plaintext?

- A. SSL/TLS encryption
- B. Challenge-response mechanism
- C. Token-based authorization
- D. Public key infrastructure

---

**Answer: B**

---

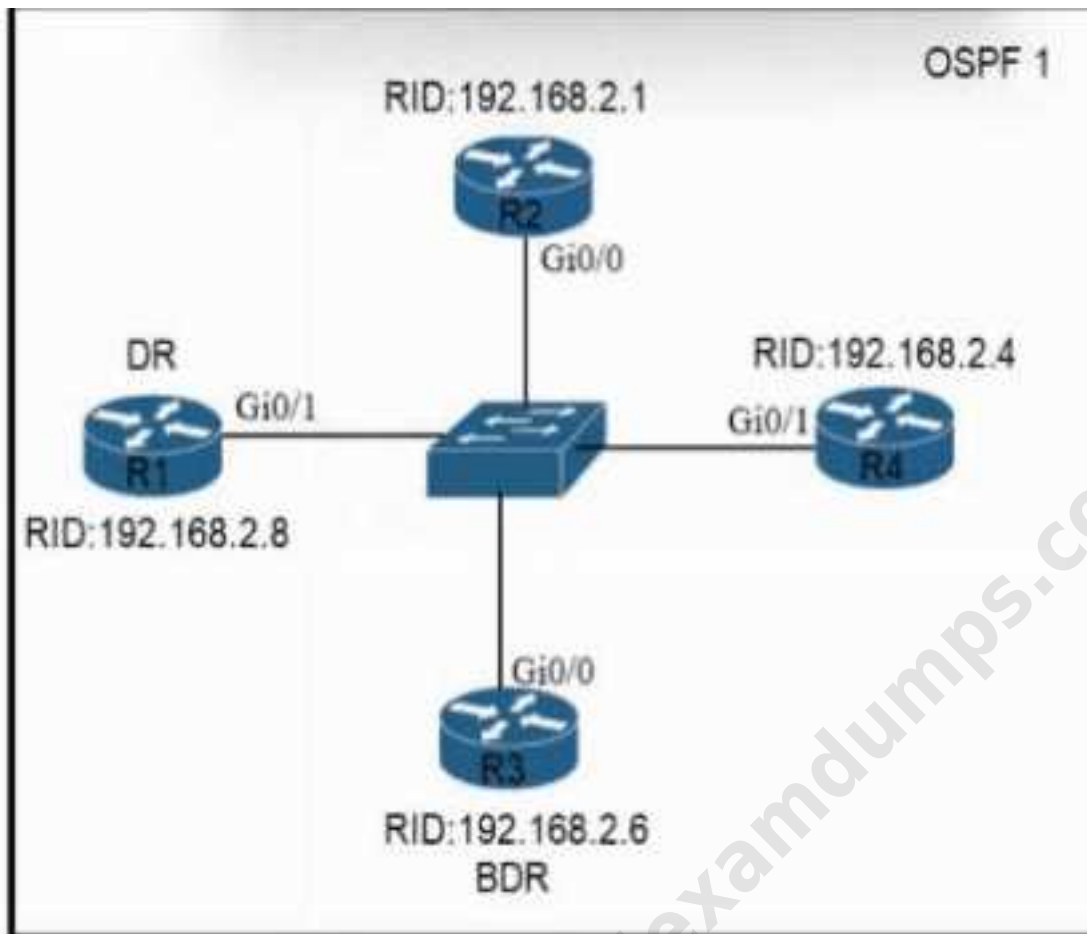
Explanation:

---

**Question: 1159**

---

Refer to the exhibit.



Refer to the exhibit. All routers in the network are configured. R2 must be the DR. After the engineer connected the devices, R1 was elected as the DR. Which command sequence must be configured on R2 to be elected as the DR in the network?

- ☐ R2(config)#interface gi0/0  
R2(config-if)#ip ospf priority 1
- ☐ R2(config)#router ospf 1  
R2(config-router)#router-id 192.168.2.7
- ☐ R2(config)#interface gi0/0  
R2(config-if)#ip ospf priority 100
- ☐ R2(config)#router ospf 1  
R2(config-router)#router-id 10.100.100.10

- A. Option A
- B. Option B
- C. Option C
- D. Option D

---

**Answer: C**

---

Explanation:

---

**Question: 1160**

---

What is the main capability of multifactor authentication?

- A. Identifying permissions for end users using three authentication factors
- B. Authenticating and authorizing end users using two authentication factors
- C. Confirming end-user identity using two or more authentication factors
- D. Verifying end-user access permissions using two authentication factors

---

**Answer: C**

---

Explanation:

---

**Question: 1161**

---

What is a characteristic of private IPv4 addressing?

- A. Reduces the forwarding table on network routers
- B. Used on the external interface of a firewall
- C. Used by ISPs when only one IP is needed to connect to the internet
- D. Address space which is isolated from the internet

---

**Answer: D**

---

Explanation:

---

**Question: 1162**

---

Refer to the exhibit.

```
Router#show ip route

Gateway of last resort is 172.17.0.2 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 172.17.0.2
10.0.0.0/8 is variably subnetted, 412 subnets, 10 masks
O E2 10.0.0.0/16 [110/10] via 10.2.24.1, 7w0d, Vlan82
O 10.2.17.0/24 [110/6] via 10.2.24.1, 6w4d, vlan82
O 10.2.17.0/24 [110/6] via 10.2.24.1, 6w4d, vlan82
O 10.2.23.0/24 [110/6] via 10.2.24.1, 7w0d, vlan82
<output suppressed>
C 10.173.5.0/24 is directly connected, vlan283
L 10.173.5.2/32 is directly connected, vlan283
```

Refer to the exhibit. What is the value of the administrative distance for the default gateway?

- A. 10
- B. 0
- C. 1
- D. 110

---

**Answer: C**

---

Explanation:

---

**Question: 1163**

---

What is a characteristic of encryption in wireless networks?

- A. uses a unidirectional handshake for authentication
- B. Intercepts data threats before they attack a network
- C. uses integrity checks to identify forgery attacks
- D. prevents intercepted data from being easily read

---

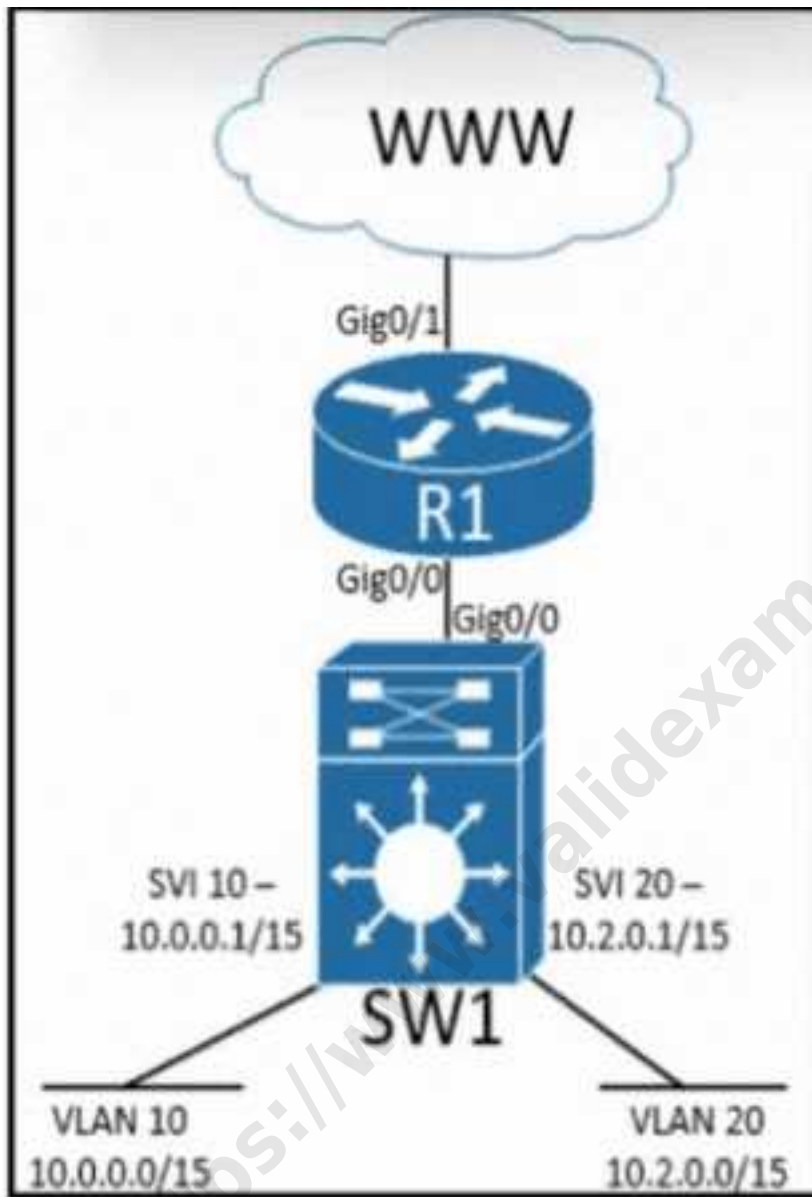
**Answer: D**

---

Explanation:

**Question: 1164**

Refer to the exhibit.



Refer to the exhibit. Inter-VLAN routing is configured on SW1. Client A is running Linux as an OS in VLAN 10 with a default gateway IP 10.0.0.1 but cannot ping client B in VLAN 20 running Windows. What action must be taken to verify that client A has the correct IP settings?

- A. Run the ipconfig command on client A and ensure that the IP address is within the host range of 10.0.0.1 - 10.0.255.254.
- B. Run the ifconfig command on client A to confirm that its IP and subnet mask fall within 255.254.0.0.
- C. Run the ipconfig command on client A to confirm that the correct 10.0.0.1 default gateway is used.

D. Run the ifconfig command on client A to confirm that the subnet mask is set to 255.255.128.0.

---

**Answer: B**

---

Explanation:

---

**Question: 1165**

---

Which two northbound APIs are found in a software-defined network? (Choose two.)

- A. OpenFlow
- B. REST
- C. OpFlex
- D. SOAP
- E. NETCONF

---

**Answer: B,D**

---

Explanation:

---

**Question: 1166**

---

Refer to the exhibit.



```

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP,
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area,
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2,
E1 - OSPF external type 1, E2 - OSPF external type 2, I - IS-IS,
su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2,
ia - IS-IS inter area, * - candidate default, I - per-user static route,
o - ODR, P - periodic download static route, H - NHRP, I - IISP, a -
application route, + - replicated route, % - next hop override,
p - overrides from Pfr

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C 10.10.10.0/24 is directly connected, GigabitEthernet0/0
L 10.10.10.1/32 is directly connected, GigabitEthernet0/0
C 10.10.10.0/24 is directly connected, GigabitEthernet1/0
L 10.10.10.1/32 is directly connected, GigabitEthernet1/0
C 10.10.10.0/24 is directly connected, GigabitEthernet2/0
L 10.10.10.1/32 is directly connected, GigabitEthernet2/0

192.168.18.0/24 is variable subnetted, 3 subnets, 3 masks
D 192.168.18.0/24 [90/3072] via 10.10.10.18, 00:13:10, GigabitEthernet0/0
R 192.168.18.0/27 [120/1] via 10.10.10.18, 00:00:13, GigabitEthernet1/0
O 192.168.18.0/28 [110/2] via 10.10.10.18, 00:28:56, GigabitEthernet2/0

```

Refer to the exhibit. Which interface does a packet take to reach the host address of 192.168.18.16?

- A. GigabitEthernet1/0
- B. GigabitEthernet0/0
- C. GigabitEthernet2/0
- D. Null0

---

**Answer: A**

---

Explanation:

---

### Question: 1167

---

What is a characteristic of a Layer 2 switch?

- A. Maintains stateful transaction information
- B. Responsible for sending data in a particular sequence

- C. Makes forwarding decisions based on MAC addresses
- D. Filters based on a transport layer protocol

---

**Answer: C**

---

Explanation:

---

**Question: 1168**

---

What is the default interface for in-band wireless network management on a WLC?

- A. Redundant port
- B. Out-of-band
- C. Service port
- D. Wireless management

---

**Answer: D**

---

Explanation:

---

**Question: 1169**

---

What is a function of Layer 3 switches?

- A. They route traffic between devices in different VLANs.
- B. They forward Ethernet frames between VLANs using only MAC addresses.
- C. They move frames between endpoints limited to IP addresses.
- D. They transmit broadcast traffic when operating in Layer 3 mode exclusively.

---

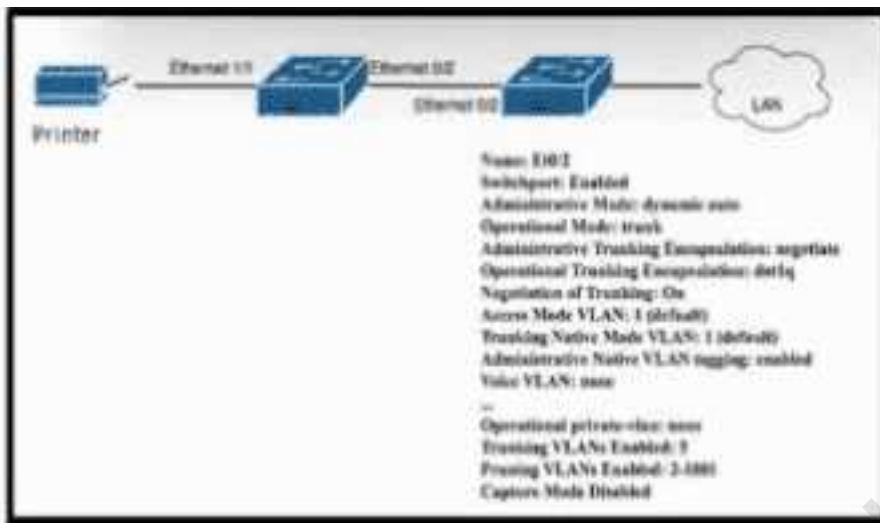
**Answer: C**

---

Explanation:

**Question: 1170**

Refer to the exhibit.



Refer to the exhibit. An administrator must connect SW\_1 and the printer to the network. SW\_2 requires DTP to be used for the connection to SW\_1. The printer is configured as an access port with VLAN 5. Which set of commands completes the connectivity?

- A. switchport mode dynamic autoswitchport private-vlan association host 5
- B. switchport mode dynamic autoswitchport trunk encapsulation negotiate
- C. switchport mode dynamic desirableswitchport trunk allowed vlan add 5
- D. switchport mode trunkswitchport trunk pruning vlan add 5

**Answer: C**

Explanation:

**Question: 1171**

Which interface is used to send traffic to the destination network?

- 10.174.208.210/28 [110/2708] via G0/21
- 10.174.208.210/28 [110/25445] via G0/20
- 10.174.208.210/28 [120/1] via G0/12

10.174.208.210/28 [120/4] via G0/16

- A. G0/12
- B. G0/20
- C. G0/21
- D. G0/16

---

**Answer: C**

---

Explanation:

---

**Question: 1172**

---

Which role does AI play in monitoring network data flow?

- A. It analyzes patterns for anomaly detection.
- B. It exclusively predicts device malfunctions.
- C. It simplifies traffic route mapping.
- D. It enhances data packet delivery speeds.

---

**Answer: A**

---

Explanation:

---

**Question: 1173**

---

How does MAC learning function?

- A. Sends frames with unknown destinations to a multicast group.
- B. Increases security on the management VLAN.
- C. Rewrites the source and destination MAC address.
- D. Associates the MAC address with the port on which it is received.

---

**Answer: D**

---

Explanation:

---

**Question: 1174**

---

Which interface is used to send traffic to the destination network?

10.249.210.56/25 [90/6144] via G0/15

10.249.210.56/25 [90/45053] via G0/13

10.249.210.56/25 [110/3693] via G0/16

10.249.210.56/25 [110/360] via G0/12

A. G0/16

B. G0/15

C. G0/13

D. G0/12

---

**Answer: B**

---

Explanation:

---

**Question: 1175**

---

Which switch port configuration must be configured when connected to an AP running in FlexConnect mode, and the WLANs use flex local switching?

A. access port with one VLAN

B. trunk port with pruned VLANs

C. Layer 3 port with an IP address

D. tagged port with MAC Filtering enabled

---

**Answer: B**

---

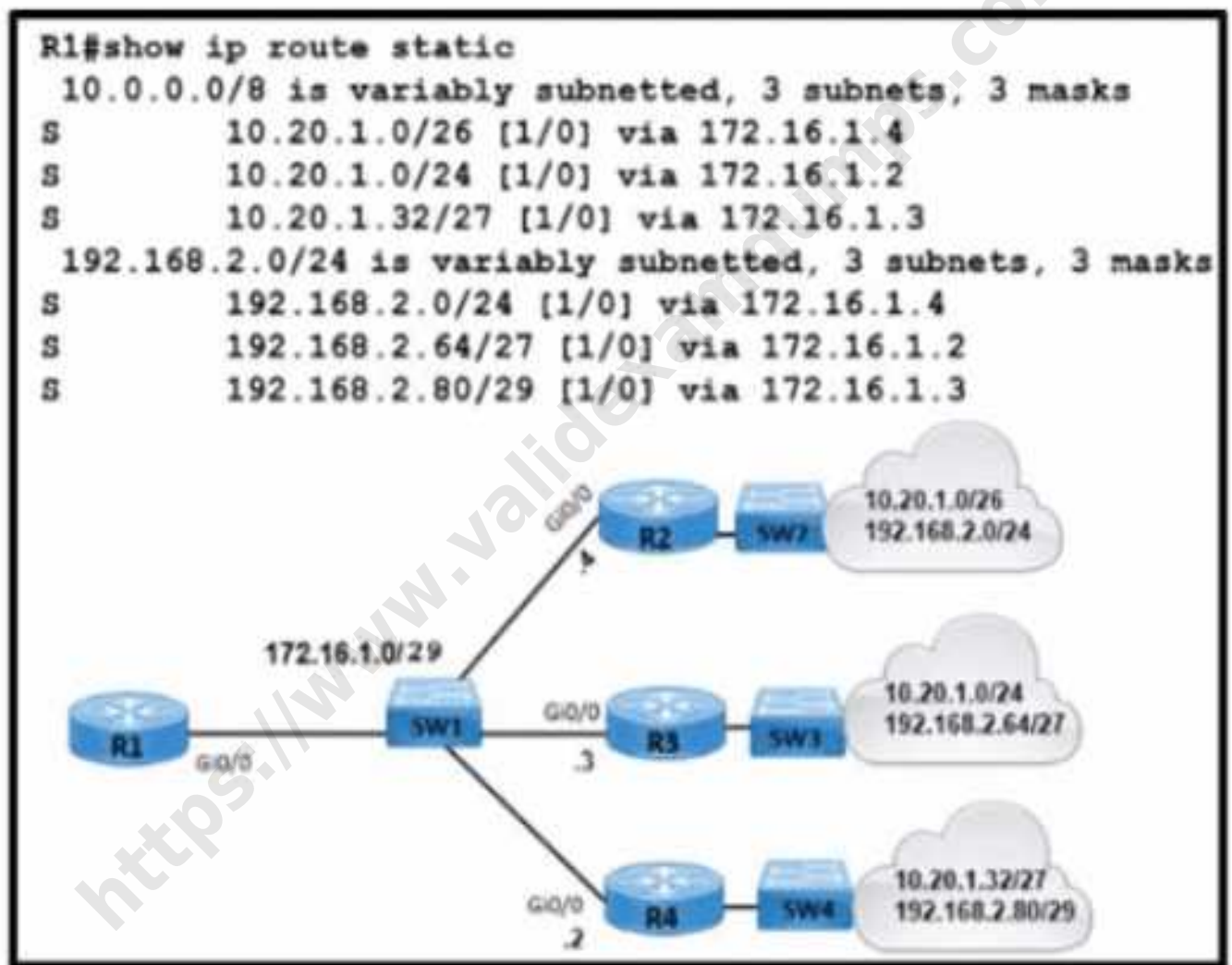
Explanation:

---

**Question: 1176**

---

Refer to the exhibit.



What is the correct next hop for router R1 to reach IP addresses 192.168.2.86 and 10.20.1.50?

- A. 172.16.1.4
- B. 172.16.1.1
- C. 172.16.1.2
- D. 172.16.1.3

---

**Answer: D**

---

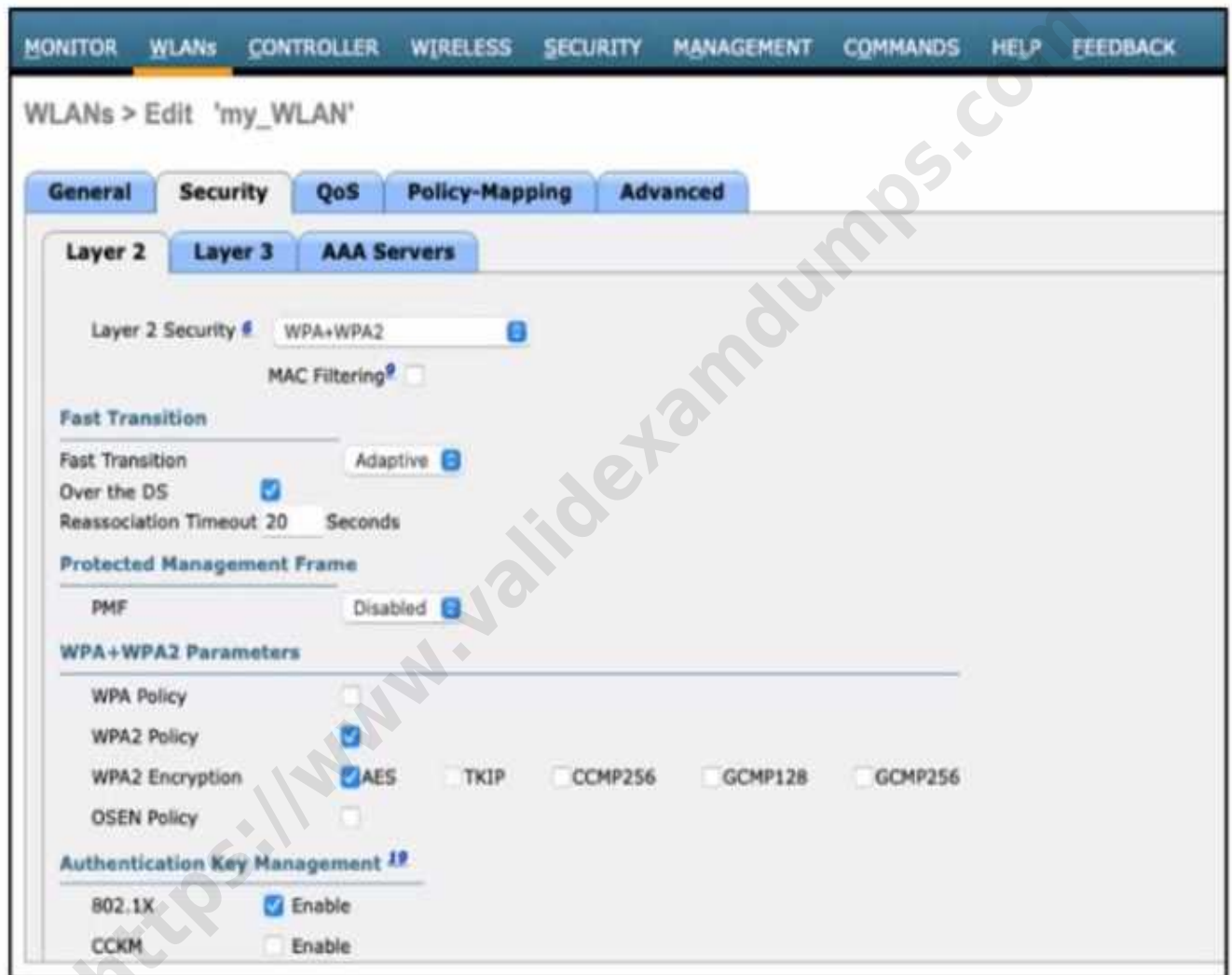
Explanation:

---

**Question: 1177**

---

Refer to the exhibit.



Refer to the exhibit. A network engineer is configuring a wireless LAN with Web Passthrough Layer 3 Web Policy. Which action must the engineer take to complete the configuration?

- A. Set the Layer 2 Security to 802.1X.
- B. Enable TKIP and CCMP256 WPA2 Encryption.
- C. Set the Layer 2 Security to None.
- D. Enable the WPA Policy.

---

**Answer: C**

---

Explanation:

---

**Question: 1178**

---

Refer to the exhibit.

The screenshot shows a network configuration interface with tabs for General, Security, QoS, Policy-Mapping, and Advanced. The Security tab is active, and the Layer 2 sub-tab is selected. The configuration for Layer 2 Security is as follows:

- Layer 2 Security: WPA2+WPA3
- Security Type: Enterprise
- MAC Filtering: ☒
- WPA2+WPA3 Parameters:
  - Policy: ☒ WPA2, ☒ WPA3
  - Encryption Cipher: ☐ COMP128(AES), ☐ CCMP256, ☐ GCMP128, ☒ GCMP256
- Fast Transition:
  - Fast Transition: Disable
- Protected Management Frame:
  - PMF: Optional
  - Comeback timer(1-10sec): 1
  - SA Query Timeout(100-500msec): 200

An engineer must document all Wi-Fi services on a new wireless LAN controller. The Wi-Fi SSID "Office\_WLan" has Layer 2 Security. What is determined by this configuration?

- A. There is a strong mutual authentication used between NAC and the network devices using x.509 standard.
- B. There is an extra layer of security that ensures only authorized devices with known MAC addresses



connect to the network.

C. There is a robust security mechanism configured to protect against various Layer 2 and Layer 3 attacks.

D. There is Galois cache algorithm configured that provides strong encryption and authentication.

---

**Answer: C**

---

Explanation:

---

**Question: 1179**

---

Refer to the exhibit.

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
Gateway of last resort is 192.168.30.10 to network 0.0.0.0
 192.168.30.0/29 is subnetted, 2 subnets
C      192.168.30.0 is directly connected, FastEthernet0/0
C      192.168.30.8 is directly connected, Serial0/0.1
 192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
O IA   192.168.10.32/28 [110/193] via 192.168.30.10, 00:18:49, Serial0/0.1
O IA   192.168.10.0/27 [110/192] via 192.168.30.10, 00:18:49, Serial0/0.1
 192.168.20.0/30 is subnetted, 1 subnets
O IA   192.168.20.0 [110/128] via 192.168.30.10, 00:18:49, Serial0/0.1
 192.168.50.0/32 is subnetted, 1 subnets
C      192.168.50.1 is directly connected, Loopback0
O*IA 0.0.0.0/0 [110/84] via 192.168.30.10, 00:10:36, Serial0/0.1
```

What is the metric for the route to the 192.168.10.33 host?

A. 84

B. 110

C. 192

D. 193

---

**Answer: B**

---

Explanation:

---

**Question: 1180**

---

What is a characteristic of private IPv4 addressing?

- A. traverses the Internet when an outbound ACL is applied
- B. used without allocation from a regional Internet authority
- C. shared address space between an ISP and subscriber
- D. reduces the need for isolated networks

---

**Answer: B**

---

Explanation:

---

**Question: 1181**

---

What is a reason to implement HSRP on a network?

- A. to identify the closest hop to the default gateway in a LAN network
- B. to provide load balancing over multiple gateways in a LAN network
- C. to optimally route traffic based on the forwarding capacity of the edge routing devices in the LAN network
- D. to ensure that user traffic in a LAN rapidly recovers from the failure of an edge routing device

---

**Answer: D**

---

Explanation:

---

**Question: 1182**

---

How does MAC learning function?

- A. rewrites the source and destination MAC address
- B. adds unknown source MAC addresses to the CAM table
- C. sends the frame back to the source to verify availability
- D. drops received MAC addresses not listed in the address table

---

**Answer: B**

---

Explanation:

---

**Question: 1183**

---

A default route must be configured to temporarily use a local broadband modem with the IP address of 192.168.1.1 until a WAN circuit is installed. The WAN circuit will use an external routing protocol that exchanges network prefixes between two separate autonomous systems on the Internet. Only a default route will be received by the ISP Which configuration must be applied to allow the new WAN circuit to take precedence when It Is In place?

- A. Ip route 0.0.0.0 0.0.0.0 192.168.1.1
- B. Ip route 0.0.0.0 0.0.0.0 192.168.1.1 25
- C. Ip route 0.0.0.0 0.0.0.0 192.168.1.1 track 1
- D. Ip route 0.0.0.0 0.0.0.0 192.168.1.1 20

---

**Answer: B**

---

Explanation:

---

**Question: 1184**

---

Which syslog message logging level displays interface line protocol up/down events?

- A. Informational
- B. debugging
- C. alerts
- D. notifications

---

**Answer: D**

---

Explanation:

---

**Question: 1185**

---

Refer to the exhibit.

```
R4#show ip route
Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 14 subnets, 2 masks
O       10.0.12.0/24 [100/2] via 10.0.24.2, 00:02:27, GigabitEthernet0/1
D       10.0.13.0/24 [90/3072] via 10.0.34.3, 00:00:19, GigabitEthernet0/0
C       10.0.24.0/24 is directly connected, GigabitEthernet0/1
L       10.0.24.4/32 is directly connected, GigabitEthernet0/1
C       10.0.34.0/24 is directly connected, GigabitEthernet0/0
L       10.0.34.4/32 is directly connected, GigabitEthernet0/0
C       10.0.45.0/24 is directly connected, GigabitEthernet0/2
L       10.0.45.4/32 is directly connected, GigabitEthernet0/2
R       10.2.0.0/24 [120/1] via 10.0.45.5, 00:00:08, GigabitEthernet0/2
O       10.255.1.1/32 [100/3] via 10.0.34.3, 00:02:27, GigabitEthernet0/0
        [100/3] via 10.0.24.2, 00:02:27, GigabitEthernet0/1
D       10.255.2.2/32 [90/130816] via 10.0.24.2, 00:14:46, GigabitEthernet0/1
O       10.255.3.3/32 [100/2] via 10.0.34.3, 00:02:27, GigabitEthernet0/0
C       10.255.4.4/32 is directly connected, Loopback0
R       10.255.5.5/32 [120/1] via 10.0.45.5, 00:00:08, GigabitEthernet0/2
```

Of the routes learned with dynamic routing protocols, which has the least preferred metric?

- A. Local
- B. EIGRP
- C. OSPF
- D. RIP

---

**Answer: D**

---

Explanation:

---

**Question: 1186**

---

What is represented in line 3 within this JSON schema?

```
1 [
2  {"switch": "SW_dallas", "port": "ge16"},
3  {"load balancer": "LB_munich", "port": "te0/26"},
4  {"VPN concentrator": "VPN_toronto", "port": "ge8/15"},
5 ]
```

- A. object
- B. key
- C. array
- D. value

---

**Answer: D**

---

Explanation:

---

**Question: 1187**

---

What is a characteristic of an SSID in wireless networks?

- A. provides protection against spyware
- B. eliminates network piggybacking
- C. associates a name to a wireless network
- D. allows easy file sharing between endpoints

---

**Answer: C**

---

Explanation:

---

**Question: 1188**

---

Which feature is mandatory when configuring a new SSID for a wireless network running WPA3-Personal mode?

- A. Opportunistic Wireless Encryption
- B. Protected Management Frame
- C. Enhanced Open
- D. Fast Transition

---

**Answer: B**

---

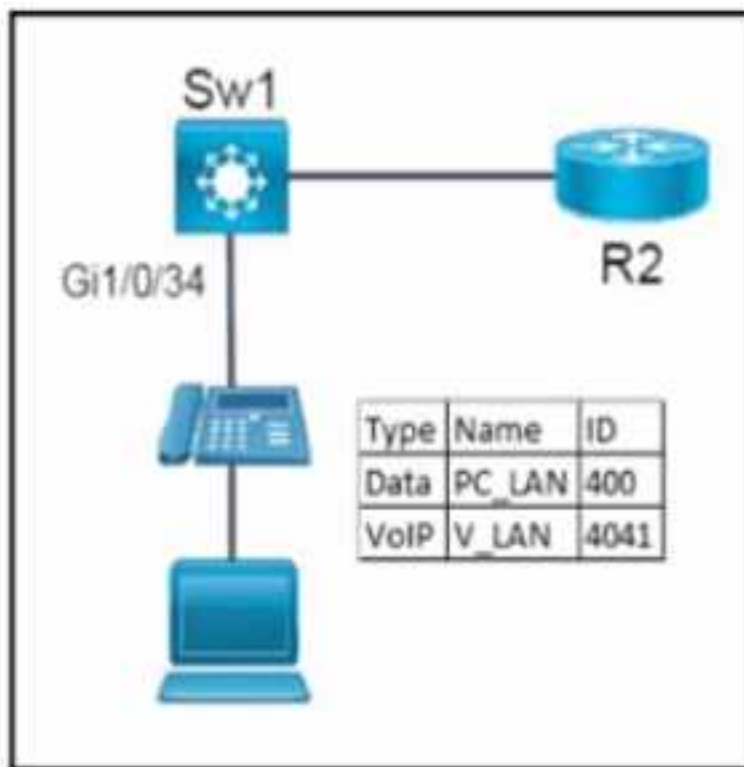
Explanation:

---

**Question: 1189**

---

Refer to the exhibit.



Network services must be enabled on interface Gi1/0/34. Which configuration meets the needs for this implementation?

A)

```
interface Gi1/0/34
switchport mode trunk
switchport
trunk allowed vlan 400, 4041
switchport voice vlan 4041
```

B)

```
interface Gi1/0/34
switchport mode trunk
switchport
trunk allowed native vlan 400
switchport
voice vlan 4041
```

C)

```
interface Gi1/0/34
switchport mode access
switchport
access vlan 4041
switchport voice vlan 400
```

D)

```
interface Gi1/0/34
switchport mode access
switchport
access vlan 400
switchport voice vlan 4041
```

A. Option

B. Option

C. Option

D. Option

---

**Answer: D**

---

Explanation:

---

**Question: 1190**

---

What is a characteristic of a Layer 2 switch?

A. tracks the number of active TCP connections

B. transmits exclusively at half duplex

C. offers link bundling to servers

D. uses routers to create collision domains

---

**Answer: C**

---

Explanation:



---

**Question: 1191**

---

Which combination of methods satisfies the minimum security requirements when a new multifactor authentication solution is deployed?

- A. password of 8 to 15 characters and personal 12-digit PIN
- B. authorized USB dongle and mobile phone
- C. fingerprint scanning and facial recognition
- D. complex password and time-based one-time password

---

**Answer: D**

---

Explanation:

---

**Question: 1192**

---

What is a characteristic of a Layer 2 switch?

- A. Provides a single broadcast domain for all connected devices
- B. Performs packet fragmentation and reassembly
- C. Tracks the number of active TCP connections
- D. Uses the data link layer for communications

---

**Answer: D**

---

Explanation:

---

**Question: 1193**

---

Which type of wired port is required when an AP offers one unique SSID, passes client data and management traffic, and is in autonomous mode?

- A. Default
- B. Access
- C. Trunk
- D. LAG

---

**Answer: B**

---

Explanation:

---

**Question: 1194**

---

Refer to the exhibit.

```
R1# show ip route
S   192.168.64.0/18 [1/0] via 10.1.1.1
O   192.168.64.0/18 [110/236855] via 10.1.1.2
O   192.168.64.0/18 [110/229840] via 10.1.1.3
S   192.168.64.0/19 [1/0] via Null0
```

Refer to the exhibit. How will router R1 handle packets destined to 192.168.64.22?

- A. It will use the static route to 10.1.1.1.
- B. It will use the route with the highest AD and highest destination IP.
- C. It will route the packets to 10.1.1.2.
- D. It will drop the packets.

---

**Answer: D**

---

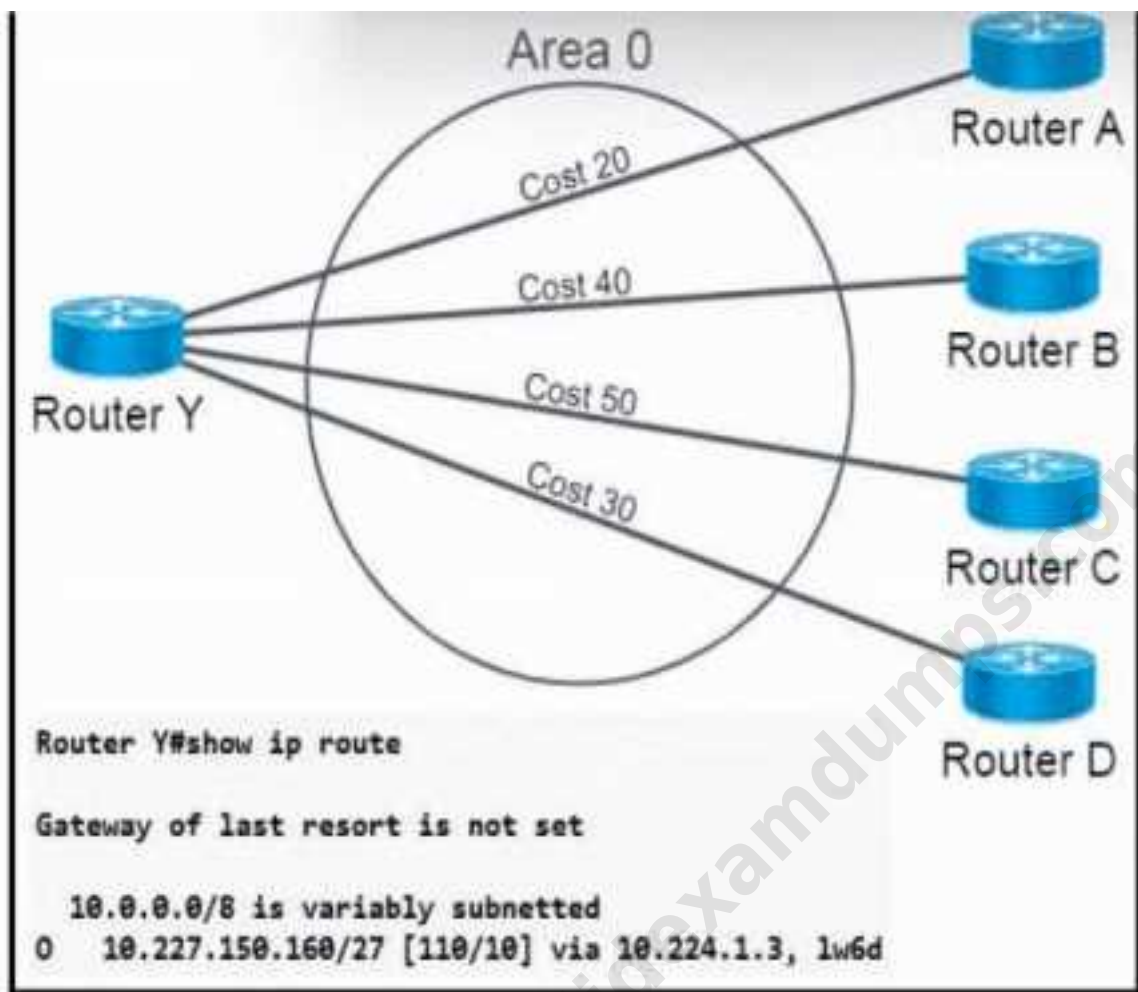
Explanation:

---

**Question: 1195**

---

Refer to the exhibit.



Refer to the exhibit. OSPF neighbors routers A, B, C, and D are sending a route for 10.227.150.160/27. When the current route for 10.227.150.160/27 becomes unavailable, which cost will router Y use to route traffic to 10.227.150.160/27?

- A. Cost 20
- B. Cost 30
- C. Cost 40
- D. Cost 50

---

**Answer: A**

---

Explanation:

---

**Question: 1196**

---

Which two transport layer protocols carry syslog messages? (Choose two.)

- A. UDP
- B. RTP
- C. TCP
- D. IP
- E. ARP

---

**Answer: A,C**

---

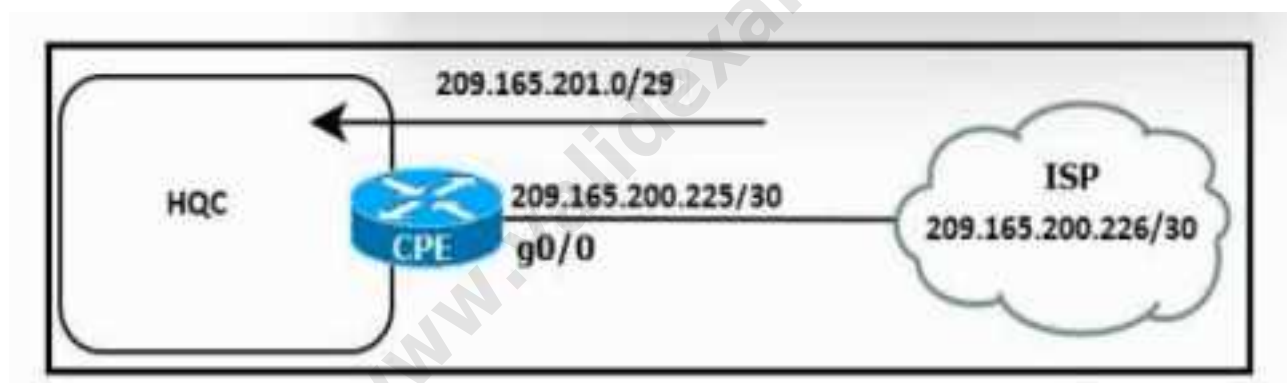
Explanation:

---

**Question: 1197**

---

Refer to the exhibit.



Refer to the exhibit. HQC needs to use a configuration that:

Handles up to 150,000 concurrent connections

Minimizes consumption of public IP addresses

Which configuration meets the requirements?

- A. ip nat pool NATPOOL 209.165.201.1 209.165.201.5 netmask 255.255.255.248ip nat inside source list HQC interface gigabitEthernet0/0 overload
- B. ip nat pool NATPOOL 209.165.200.225 209.165.200.226 netmask 255.255.255.252ip nat outside source list HQC pool NATPOOL overload
- C. ip nat pool NATPOOL 209.165.201.1 209.165.201.3 netmask 255.255.255.248ip nat inside source list HQC pool NATPOOL overload

D. ip nat pool NATPOOL 209.165.201.1 209.165.201.248 netmask 255.255.255.248 ip nat outside source list HQC pool NATPOOL overload

---

**Answer: A**

---

Explanation:

---

**Question: 1198**

---

How are API keys used to enforce rate limiting?

- A. To define the network path the API request should take
- B. To uniquely identify each client application
- C. To specify the type of data format the client prefers to receive
- D. To encrypt data sent in the API request

---

**Answer: B**

---

Explanation:

---

**Question: 1199**

---

Refer to the exhibit.

```
AccessSw1> show ntp associations
address      ref clock    st  when  poll reach  delay  offset  disp
*-2001:DB8:12::1  127.127.1.1  3   39   64   377   23.903 -5.581  2.077
* sys.peer, # selected, + candidate, - outliyer, x falseticker, - configured
```

Refer to the exhibit. A network engineer must replicate the AccessSw1 NTP configuration on a new switch. The engineer could not access privileged mode on AccessSw1 to view its configuration.

Which command must be applied to the new switch to replicate the configuration?

- A. ntp master
- B. ntp master 3

- C. ntp server 2001:db8:12::1
- D. ntp server 127.127.1.1

---

**Answer: C**

---

Explanation:

---

**Question: 1200**

---

Refer to the exhibit.

```
10.0.0.0/8 is variably subnetted, 15 subnets, 2 masks
O    10.0.12.0/24 [110/2] via 10.0.14.1, 00:17:46, GigabitEthernet0/2
C    10.0.14.0/24 is directly connected, GigabitEthernet0/2
L    10.0.14.4/32 is directly connected, GigabitEthernet0/2
O    10.0.23.0/24 [110/2] via 10.0.34.3, 00:17:46, GigabitEthernet0/0
C    10.0.34.0/24 is directly connected, GigabitEthernet0/0
L    10.0.34.4/32 is directly connected, GigabitEthernet0/0
C    10.0.45.0/24 is directly connected, GigabitEthernet0/1
L    10.0.45.4/32 is directly connected, GigabitEthernet0/1
O    10.1.0.1/32 [110/2] via 10.0.14.1, 00:17:46, GigabitEthernet0/2
R    10.2.0.0/24 [90/1] via 10.0.45.5, 00:00:13, GigabitEthernet0/1
O    10.255.1.1/32 [110/2] via 10.0.14.1, 00:17:46, GigabitEthernet0/2
O    10.255.2.2/32 [110/3] via 10.0.34.3, 00:17:46, GigabitEthernet0/0
    [110/3] via 10.0.14.1, 00:17:46, GigabitEthernet0/2
O    10.255.3.3/32 [110/2] via 10.0.34.3, 00:17:46, GigabitEthernet0/0
C    10.255.4.4/32 is directly connected, Loopback0
R    10.255.5.5/32 [90/1] via 10.0.45.5, 00:00:13, GigabitEthernet0/1
```

Refer to the exhibit. Which routes are configured with their default administrative distances?

- A. EIGRP
- B. OSPF
- C. RIP
- D. Local

---

**Answer: B**

---

Explanation:

---

**Question: 1201**

---

Which technology allows multiple operating systems to run a single physical server?

- A. Containers
- B. Cloud computing
- C. Application hosting
- D. Virtualization

---

**Answer: D**

---

Explanation:

---

**Question: 1202**

---

Which command must be configured to enable port security with a manually assigned MAC address of aabb.cc00.1234 for a VoIP handset on VLAN 4?

- A. switchport port-security mac-address aabb.cc00.1234
- B. switchport port-security mac-address aabb.cc00.1234 vlan 4
- C. mac-address-table static aabb.cc00.1234 vlan 4 interface fa0/1
- D. switchport port-security mac-address sticky

---

**Answer: B**

---

Explanation:

---

**Question: 1203**

---

Refer to the exhibit.

```
10.0.0.0/8 is variably subnetted, 15 subnets, 2 masks
O    10.0.12.0/24 [110/2] via 10.0.14.1, 00:17:46, GigabitEthernet0/2
C    10.0.14.0/24 is directly connected, GigabitEthernet0/2
L    10.0.14.4/32 is directly connected, GigabitEthernet0/2
O    10.0.23.0/24 [110/2] via 10.0.34.3, 00:17:46, GigabitEthernet0/0
C    10.0.34.0/24 is directly connected, GigabitEthernet0/0
L    10.0.34.4/32 is directly connected, GigabitEthernet0/0
C    10.0.45.0/24 is directly connected, GigabitEthernet0/1
L    10.0.45.4/32 is directly connected, GigabitEthernet0/1
O    10.1.0.1/32 [110/2] via 10.0.14.1, 00:17:46, GigabitEthernet0/2
R    10.2.0.0/24 [90/1] via 10.0.45.5, 00:00:13, GigabitEthernet0/1
O    10.255.1.1/32 [110/2] via 10.0.14.1, 00:17:46, GigabitEthernet0/2
O    10.255.2.2/32 [110/3] via 10.0.34.3, 00:17:46, GigabitEthernet0/0
    [110/3] via 10.0.14.1, 00:17:46, GigabitEthernet0/2
O    10.255.3.3/32 [110/2] via 10.0.34.3, 00:17:46, GigabitEthernet0/0
C    10.255.4.4/32 is directly connected, Loopback0
R    10.255.5.5/32 [90/1] via 10.0.45.5, 00:00:13, GigabitEthernet0/1
```

Refer to the exhibit. Which routes are configured with their default administrative distances?

- A. Local
- B. EIGRP
- C. RIP
- D. OSPF

---

**Answer: D**

---

Explanation:

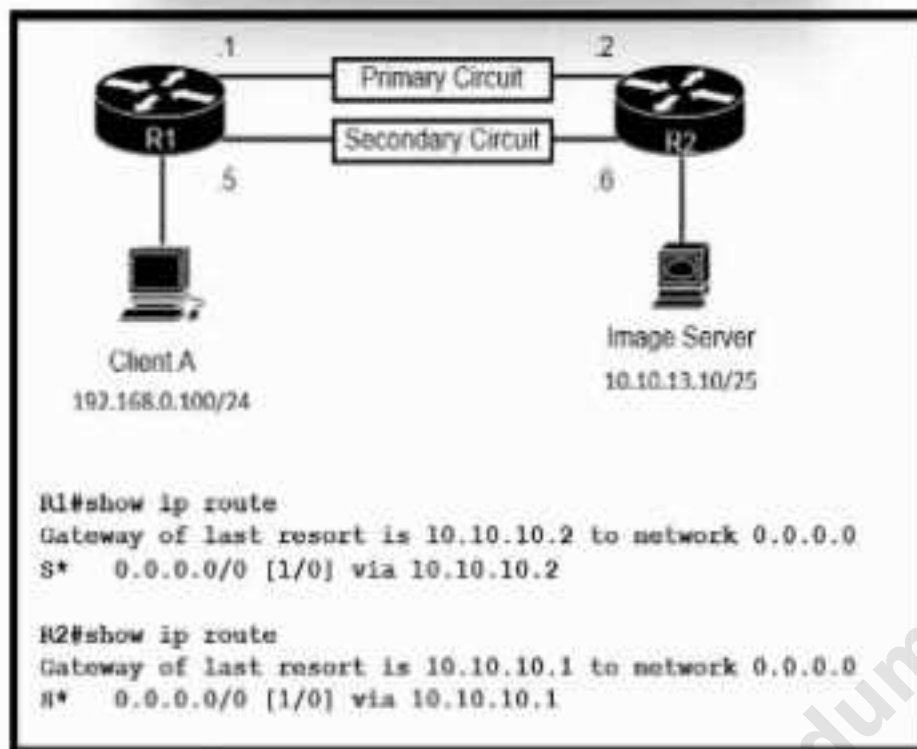
---

**Question: 1204**

---

Refer to the exhibit.





Refer to the exhibit. Routers R1 and R2 have been configured with their respective LAN interfaces. The two circuits are operational and reachable across the WAN. Which command set establishes failover redundancy if the primary circuit goes down?

- A. R1(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.6 R2(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.5
- B. R1(config)#ip route 10.10.13.10 255.255.255.255 10.10.10.6 R2(config)#ip route 192.168.0.100 255.255.255.255 10.10.10.5
- C. R1(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.6 R2(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.5
- D. R1(config)#ip route 10.10.13.10 255.255.255.255 10.10.10.2 R2(config)#ip route 192.168.0.100 255.255.255.255 10.10.10.1

---

**Answer: A**

---

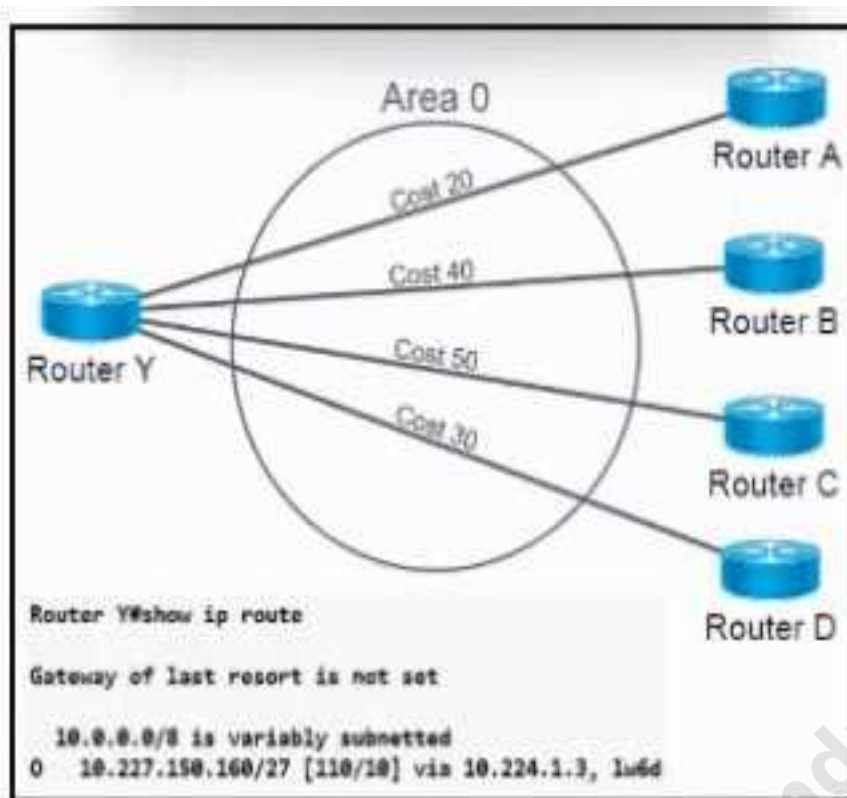
Explanation:

---

### Question: 1205

---

Refer to the exhibit.



Refer to the exhibit. OSPF neighbors routers A, B, C, and D are sending a route for 10.227.150.160/27. When the current route for 10.227.150.160/27 becomes unavailable, which cost will router Y use to route traffic to 10.227.150.160/27?

- A. cost 20
- B. cost 30
- C. cost 40
- D. cost 50

---

**Answer: C**

---

Explanation:

---

### Question: 1206

---

Which protocol should be used to transfer large files on a company intranet that allows UDP 69 through the firewall?

- A. FTP

- B. TFTP
- C. REST API
- D. SMTP

---

**Answer: A**

---

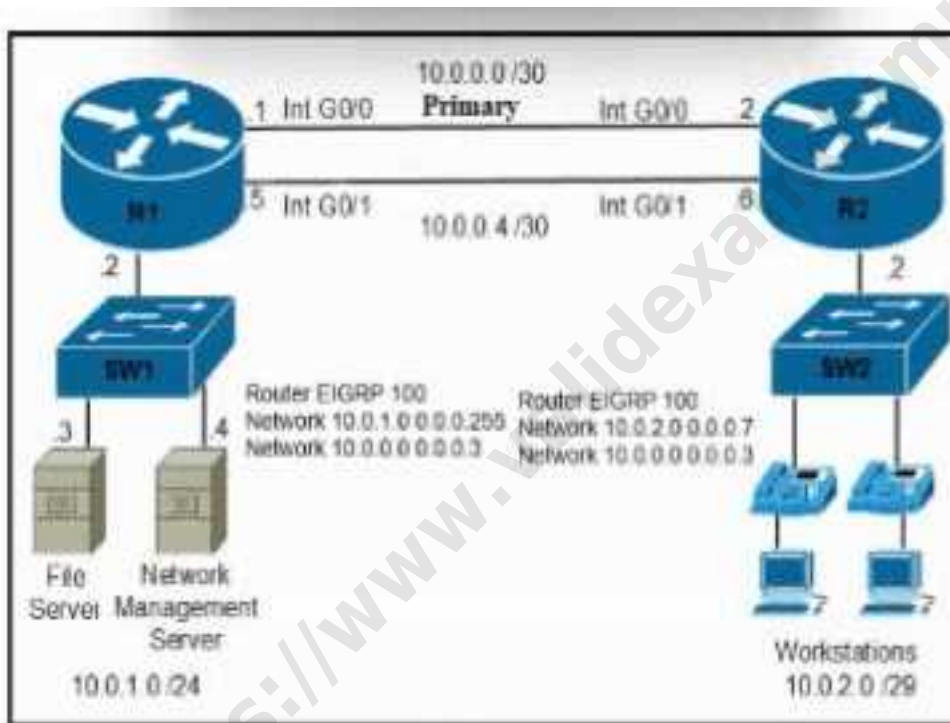
Explanation:

---

**Question: 1207**

---

Refer to the exhibit.



Refer to the exhibit. A secondary route is required on router R1 to pass traffic to the LAN network on R2 if the primary link fails. Which command must be entered to configure the router?

- A. ip route 10.0.2.0 255.255.255.240 10.0.0.6 91
- B. ip route 10.0.2.0 255.255.255.248 10.0.0.6 91
- C. ip route 10.0.2.0 255.255.255.248 null0 93
- D. ip route 10.0.2.0 255.255.255.240 10.0.0.7 92

---

**Answer: B**

---

Explanation:

---

**Question: 1208**

---

What is a characteristic of a Layer 2 switch?

- A. performs packet fragmentation and reassembly
- B. tracks the number of active TCP connections
- C. provides a single broadcast domain for all connected devices
- D. uses the data link layer for communications

---

**Answer: D**

---

Explanation:

---

**Question: 1209**

---

Which interface condition is occurring in this output?

```
R18# show interface fa0/0
FastEthernet0/0 is up, line protocol is up
Hardware is DEC21140, address is ca02.7788.c000 (bte ca02.7788.c000)
Description: dallas_subnet
Internet address is 10.32.102.2/30
MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
reliability 255/255, txload 255/255, rxload 255/255
Encapsulation ARPA, loopback not set
Keepalive set (50 sec)
Full-duplex, 100 Mbps, 100BaseTX/FX
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:01, output 00:00:00, output hang never
Last clearing of "show interface" counters 00:00:18
Input queue: 0/300/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/300 (size/max)
30 second input rate: 230000000 bits/sec, 40 packets/sec
30 second output rate: 200000000 bits/sec, 40 packets/sec
7331 packets input, 7101162 bytes
Received 267 broadcasts (0 IP multicasts)
0 runs, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 watchdog
0 input packets with dribble condition detected
3927 packets output, 1440403 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
```

- A. duplex mismatch
- B. bad NIC
- C. high throughput
- D. broadcast storm

---

**Answer: C**

---

Explanation:

---

**Question: 1210**

---

Refer to the exhibit.

WLAN

WLAN > Edit

General Security QoS Advanced

Layer2 Layer3 AAA Server

Layer 2 Security

MAC Filtering

**WPA+WPA2 Parameters**

WPA Policy ☐

WPA2 Policy ☐

WPA2 Encryption ☐ AES ☐ TKIP

Auth Key Mgmt

PSK Format

Refer to the exhibit. An engineer is using the Cisco WLC GUI to configure a WLAN for WPA2 encryption with AES and preshared key Cisc0123456. After the engineer selects the WPA + WPA2 option from the Layer 2 Security drop-down, which two tasks must they perform to complete the process? (Choose two)

- A. Select the WPA2 Policy and AES check boxes
- B. Select ASCII from the PSK Format drop-down list, enter the key, and leave the Auth Key Mgmt setting blank
- C. Select the WPA2 Policy, AES, and TKIP check boxes
- D. Select CCKM for the Auth Key Mgmt drop-down list, set the PSK Format to Hex, and enter the key
- E. Select PSK from the Auth Key Mgmt drop-down list, set the PSK Format to ASCII, and enter the key

---

**Answer: A,E**

---

Explanation:

---

**Question: 1211**

---

Which type of wired port is required when an AP offers one unique SSID, passes client data and management traffic, and is in autonomous mode?

- A. LAG
- B. default
- C. trunk
- D. Bopess

---

**Answer: B**

---

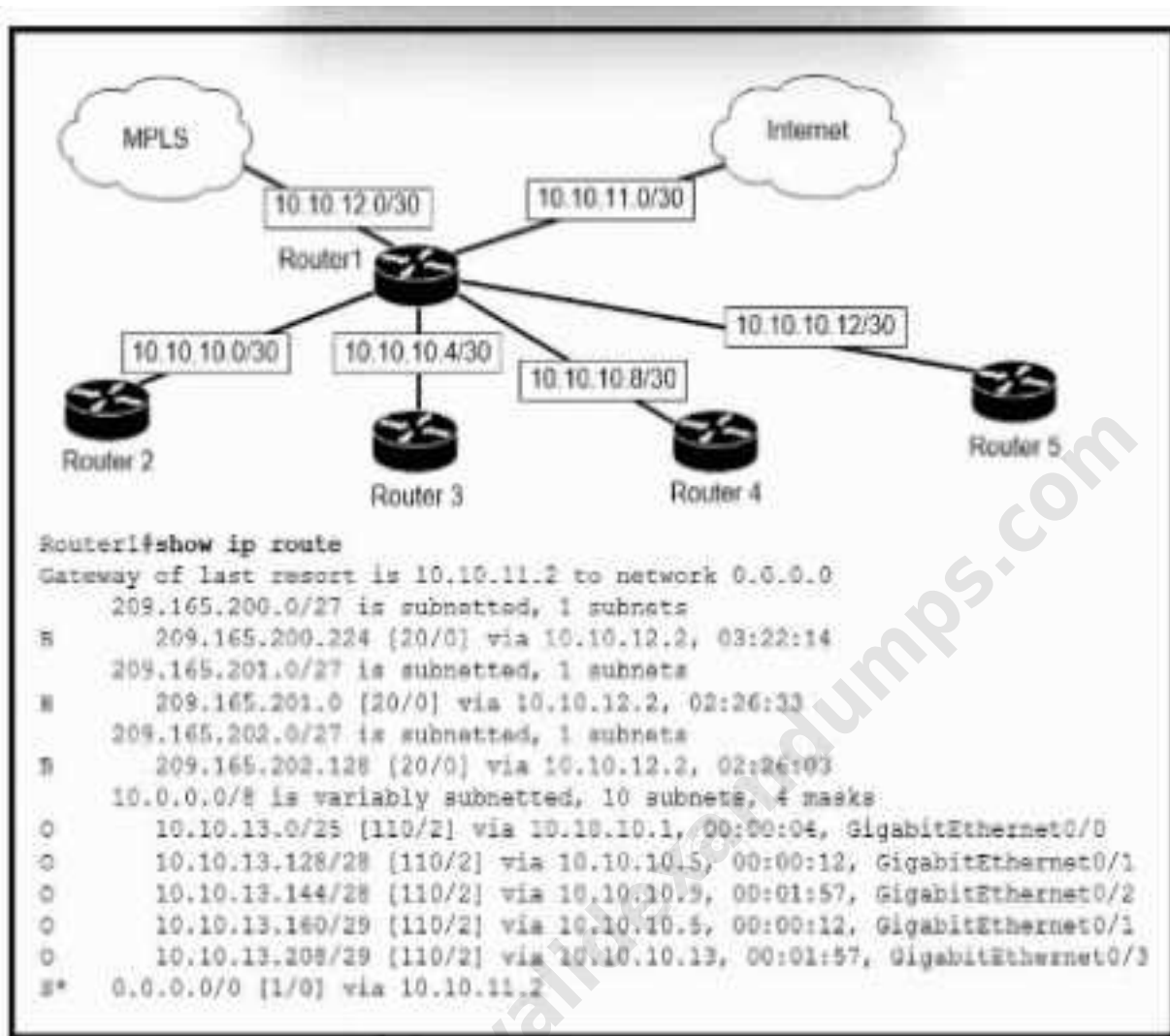
Explanation:

---

**Question: 1212**

---

Refer to the exhibit.



Refer to the exhibit. To which device does Router1 send packets that are destined to host 10.10.13.165?

- A. Router2
- B. Router3
- C. Router4
- D. Router5

**Answer: B**

Explanation:

### Question: 1213

Refer to the exhibit.





Refer to the exhibit. Which configuration for RTR-1 denies SSH access from PC-1 to any RTR-1 interface and allows all other traffic?

- A. access-list 100 deny tcp host 172.16.1.33 any eq 23  
access-list 100 permit ip any any  
interface GigabitEthernet0/0  
ip access-group 100 in
- B. access-list 100 deny tcp host 172.16.1.33 any eq 22  
access-list 100 permit ip any any  
interface GigabitEthernet0/0  
ip access-group 100 in
- C. access-list 100 deny tcp host 172.16.1.33 any eq 22  
access-list 100 permit ip any any  
line vty 0 15  
access-class 100 in
- D. access-list 100 deny tcp host 172.16.1.33 any eq 23  
access-list 100 permit ip any any  
line vty 0 15  
access-class 100 in

---

**Answer: C**

---

Explanation:

---

**Question: 1214**

---

In which way does a spine-and-leaf architecture allow for scalability in a network when additional access ports are required?

- A. A spine switch and a leaf switch are added with redundant connections between them.
- B. A spine switch is added with at least 40 GB uplinks.
- C. A leaf switch is added with a single connection to a core spine switch.
- D. A leaf switch is added with connections to every spine switch.

---

**Answer: D**

---

Explanation:

---

**Question: 1215**

---

What is the purpose of classifying network traffic in QoS?

- A. Writes the class identifier of a packet to a dedicated field in the packet header
- B. Services traffic according to its class
- C. Configures traffic-matching rules on network devices
- D. Identifies the type of traffic that will receive a particular treatment

---

**Answer: D**

---

Explanation:

---

**Question: 1216**

---

How is AI used to identify issues within network traffic?

- A. It enhances data packet delivery speeds
- B. It analyzes patterns for anomaly detection
- C. It exclusively predicts device malfunctions
- D. It guarantees zero packet loss in the network

---

**Answer: B**

---

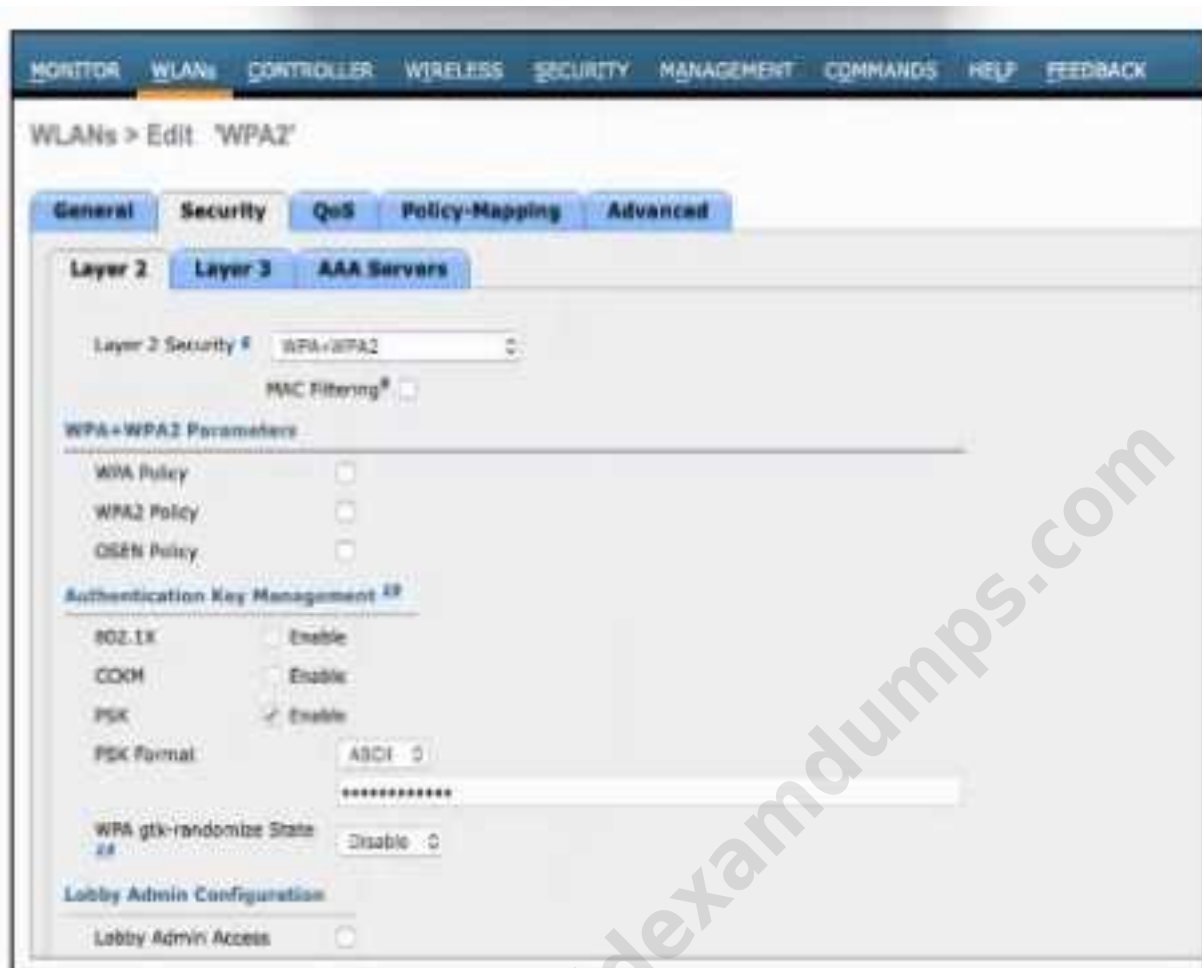
Explanation:

---

**Question: 1217**

---

Refer to the exhibit.



Refer to the exhibit. A network engineer is configuring a WLAN to use a WPA2 PSK and allow only specific clients to join. Which two actions must be taken to complete the process? (Choose two)

- A. Enable the MAC Filtering option
- B. Enable the OSEN Policy option
- C. Enable the CCKM option for Authentication Key Management
- D. Enable the WPA2 Policy option
- E. Enable the 802.1X option for Authentication Key Management

---

**Answer: A,D**

---

Explanation:

---

**Question: 1218**

---

What is a function of a Next-Generation IPS?

- A. Integrates with a RADIUS server to enforce Layer 2 device authentication rules
- B. Serves as a controller within a controller-based network
- C. Makes forwarding decisions based on learned MAC addresses
- D. Correlates user activity with network events

---

**Answer: D**

---

Explanation:

---

**Question: 1219**

---

Refer to the exhibit.



Refer to the exhibit. A static route must be configured on R14 to forward traffic for the 172.21.34.0/25 network that resides on R86. Which command must be used to fulfill the request?

- A. ip route 172.21.34.0 255.255.255.192 10.73.65.65
- B. ip route 172.21.34.0 255.255.255.0 10.73.65.65
- C. ip route 172.21.34.0 255.255.255.128 10.73.65.66
- D. ip route 172.21.34.0 255.255.128.0 10.73.65.64

---

**Answer: C**

---

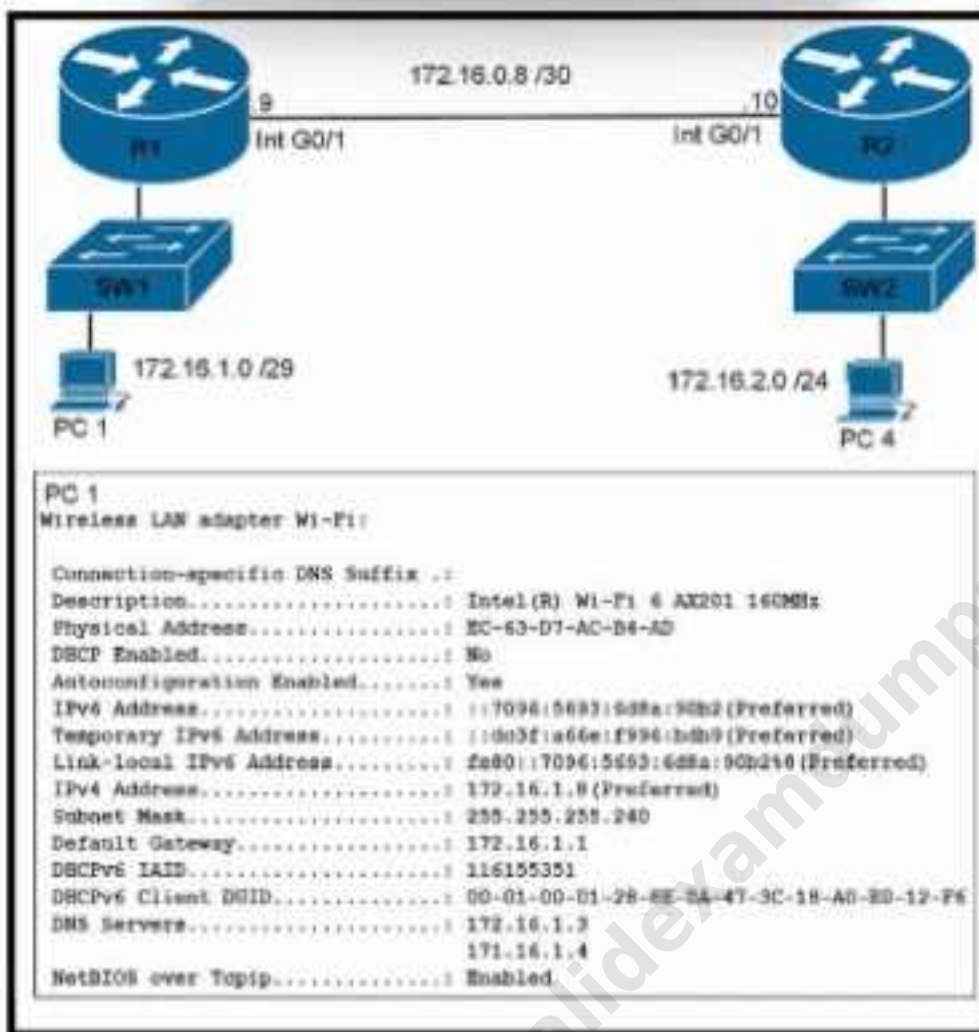
Explanation:

---

**Question: 1220**

---

Refer to the exhibit.



Refer to the exhibit. During initial configuration testing, the Windows workstation PC1 cannot connect with the 172.16.2.0/24 network. Which set of actions corrects the configuration?

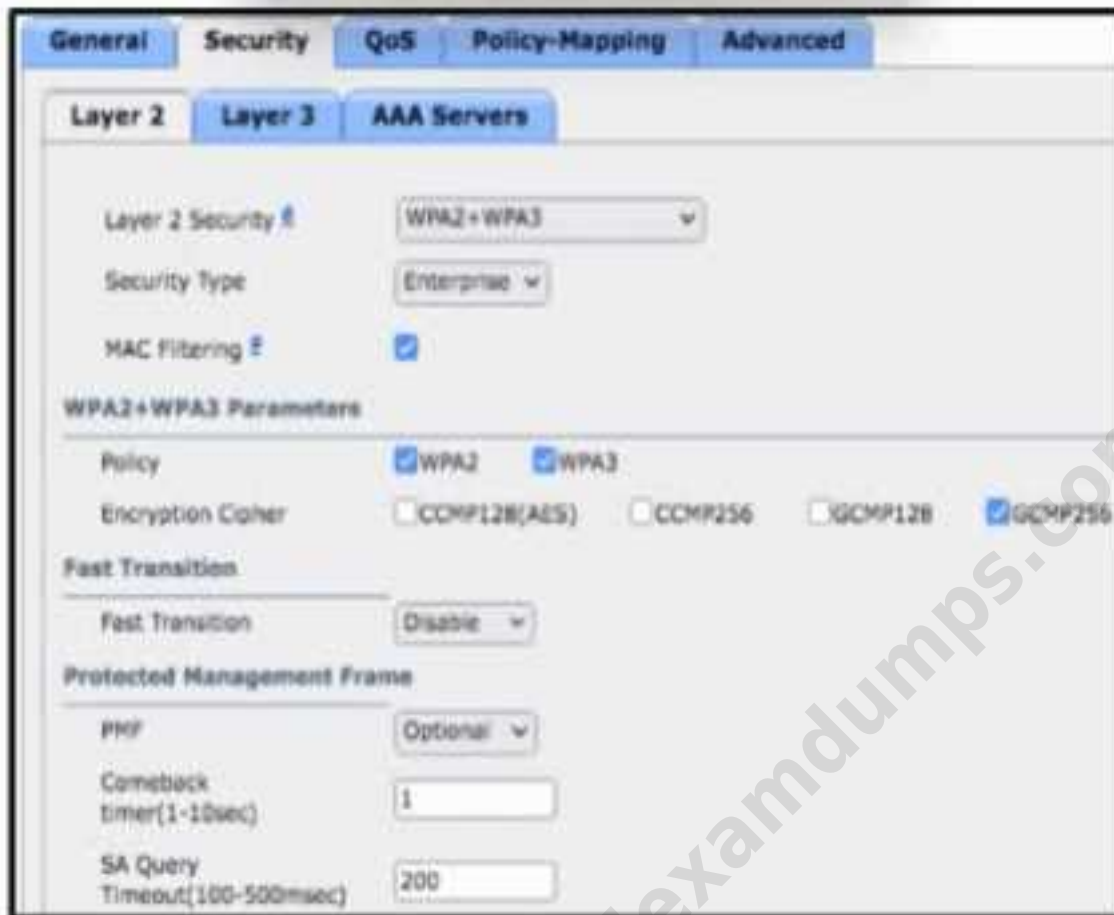
- A. Change the IP address to 172.16.1.6 and change the subnet mask to 255.255.255.248.
- B. Change the IP address to 172.16.1.9 and change the default gateway to 172.16.1.7.
- C. Change the IP address to 172.16.1.9 and change the DNS server to 172.16.1.12 only.
- D. Change the IP address to 172.16.1.6 and change the DNS servers to 172.18.1.12 and 172.16.1.13.

**Answer: B**

Explanation:

### Question: 1221

Refer to the exhibit.



Refer to the exhibit. The Wi-Fi SSID "Office\_WLAN" has Layer 2 Security configured with MAC filtering enabled. What additional security is provided by this specific feature?

- A. There is an extra layer of security that ensures only authorized devices with known MAC addresses connect to the network
- B. There is strong mutual authentication used between NAC and the network devices using X.509 standard
- C. All data frames exchanged between the client and the access point are encrypted
- D. There is a Galois cipher algorithm configured that provides strong encryption and authentication

---

**Answer: B**

---

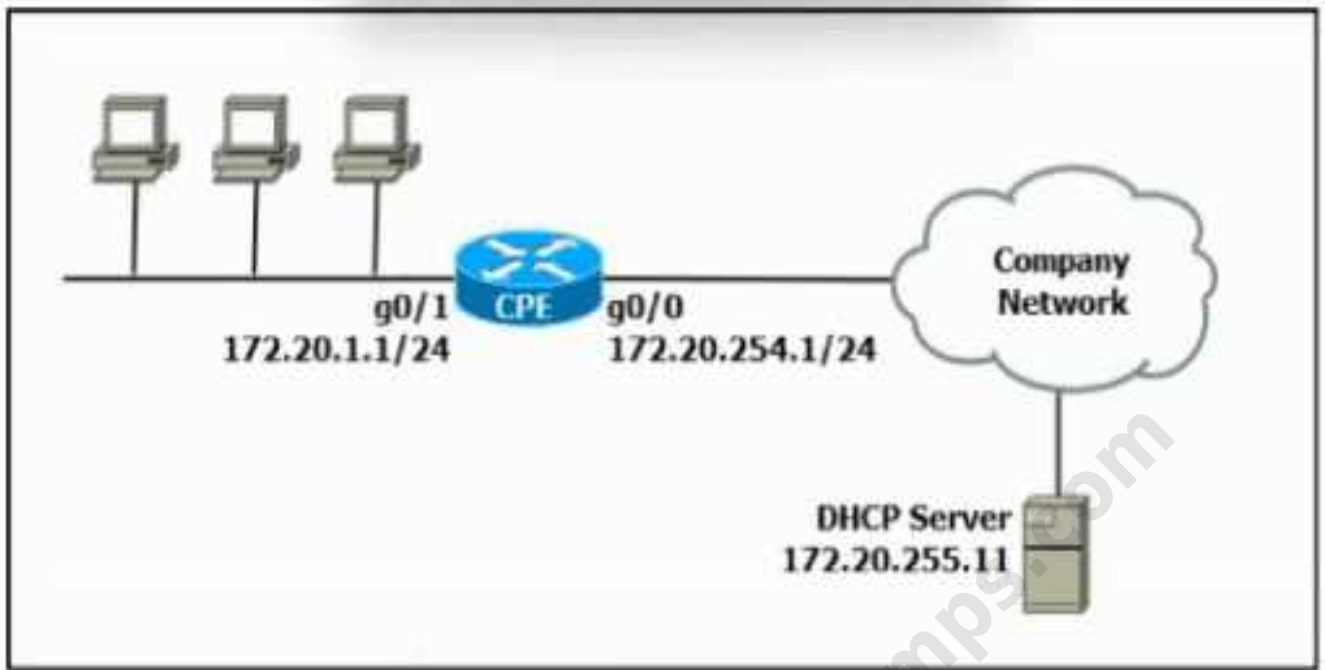
Explanation:

---

**Question: 1222**

---

Refer to the exhibit.



Refer to the exhibit. A network engineer must configure the CPE router to allow computers in the 172.20.1.0/24 network to obtain their IP configurations from the central DHCP server. Which configuration must the engineer apply to the CPE?

- A. interface GigabitEthernet0/1ip helper-address 172.20.254.1
- B. interface GigabitEthernet0/0ip helper-address 172.20.1.1
- C. interface GigabitEthernet0/0ip helper-address 172.20.255.1
- D. interface GigabitEthernet0/ip helper-address 172.20.255.11

---

**Answer: A**

---

Explanation:

---

### Question: 1223

---

DRAG DROP

An engineer is configuring an encrypted password for the enable command on a router where the local user database has already been configured. Drag and drop the configuration commands from the left into the correct sequence on the right. Not all commands are used.

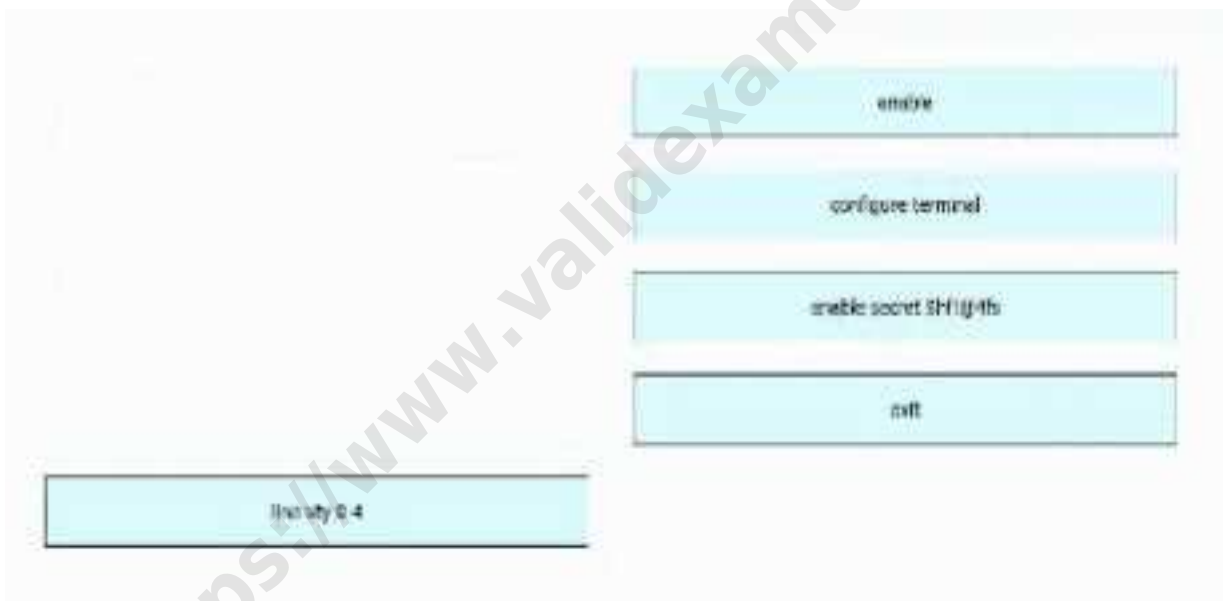


---

**Answer:**

---

Explanation:



---

**Question: 1224**

---

How are API keys used to enforce rate limiting?

- A. to specify the type of data format the client prefers to receive
- B. to define the network path the API request should take
- C. to encrypt data sent in the API request



D. to uniquely identify each client application

---

**Answer: D**

---

Explanation:

---

**Question: 1225**

---

Which action implements physical access control as part of the security program of an organization?

- A. configuring a password for the console port
- B. configuring enable passwords on network devices
- C. backing up syslogs at a remote location
- D. setting up IP cameras to monitor key infrastructure

---

**Answer: D**

---

Explanation:

---

**Question: 1226**

---

What is an advantage of using SDN versus traditional networking when it comes to security?

- A. Security is managed near the perimeter of the network with firewalls, VPNs, and IPS.
- B. Devices communicate with each other to establish a security policy.
- C. It creates a unified control point making security policies consistent across all devices.
- D. It exposes an API to configure locally per device for security policies.

---

**Answer: C**

---

Explanation:

**Question: 1227**

Refer to the exhibit.

```
Router#show ip route
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 14 subnets, 2 masks
R    10.0.12.0/24 [100/3] via 10.0.24.2, 00:02:27, GigabitEthernet0/1
D    10.0.13.0/24 [90/3072] via 10.0.34.3, 00:00:19, GigabitEthernet0/0
C    10.0.24.0/24 is directly connected, GigabitEthernet0/1
L    10.0.34.4/32 is directly connected, GigabitEthernet0/1
C    10.0.34.0/24 is directly connected, GigabitEthernet0/0
L    10.0.34.4/32 is directly connected, GigabitEthernet0/0
C    10.0.45.0/24 is directly connected, GigabitEthernet0/2
L    10.0.45.4/32 is directly connected, GigabitEthernet0/2
R    10.2.0.0/24 [120/1] via 10.0.45.5, 00:00:09, GigabitEthernet0/2
D    10.255.1.1/32 [100/1] via 10.0.34.3, 00:02:27, GigabitEthernet0/0
    [100/1] via 10.0.24.2, 00:02:27, GigabitEthernet0/1
D    10.255.2.2/32 [90/130816] via 10.0.24.2, 00:14:46, GigabitEthernet0/1
D    10.255.3.3/32 [100/2] via 10.0.34.3, 00:02:27, GigabitEthernet0/0
C    10.255.4.4/32 is directly connected, Loopback0
D    10.255.5.5/32 [120/1] via 10.0.45.5, 00:00:09, GigabitEthernet0/2
```

Refer to the exhibit. Of the routes learned with dynamic routing protocols, which has the least preferred metric?

- A. EIGRP
- B. OSPF
- C. Local
- D. RIP

---

**Answer: D**

---

Explanation:

**Question: 1228**

Refer to the exhibit.

```
AA#show ip route
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.0.0.0/30 is directly connected, GigabitEthernet0/0
L    10.0.0.1/32 is directly connected, GigabitEthernet0/0
C    10.10.0.0/30 is directly connected, GigabitEthernet0/1
L    10.10.0.1/32 is directly connected, GigabitEthernet0/1
O    10.20.0.0/30 [110/2] via 10.0.0.2, 00:00:40, GigabitEthernet0/0
O    10.30.0.0/30 [110/2] via 10.0.0.2, 00:00:40, GigabitEthernet0/0
172.16.0.0/24 is subnetted, 1 subnets
S    172.16.10.0 [1/0] via 10.0.0.2
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/2
L    192.168.10.1/32 is directly connected, GigabitEthernet0/2
S    192.168.20.0/24 [1/0] via 192.168.10.2
```

Refer to the exhibit. What is the administrative distance for the advertised prefix that includes the host IP address 192.168.20.1?

- A. 1
- B. 24
- C. 192.168.10.2
- D. 0

---

**Answer: A**

---

Explanation:

---

### Question: 1229

---

A network engineer starts to implement a new wireless LAN by configuring the authentication server and creating the dynamic interface. What must be performed next to complete the basic configuration?

- A. Create the new WLAN and bind the dynamic interface to it.
- B. Install the management interface and add the management IP.
- C. Enable Telnet and RADIUS access on the management interface.
- D. Configure high availability and redundancy for the access points.

---

**Answer: A**

---

Explanation:

---

**Question: 1230**

---

DRAG DROP

Drag and drop the steps in a standard DNS lookup operation from the left into the order on the right.

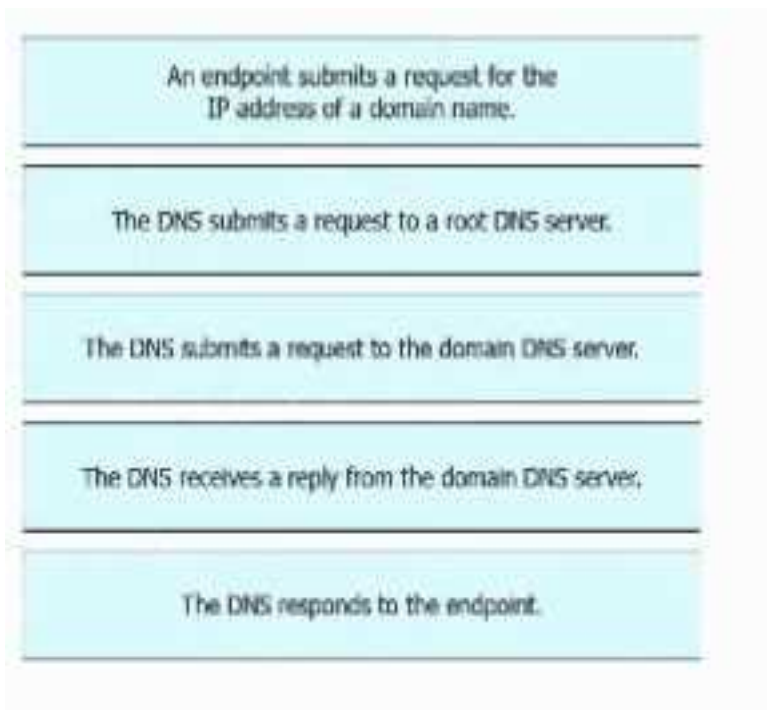
|                                                                    |   |
|--------------------------------------------------------------------|---|
| An endpoint submits a request for the IP address of a domain name. | 1 |
| The DNS receives a reply from the domain DNS server.               | 2 |
| The DNS responds to the endpoint.                                  | 3 |
| The DNS submits a request to a root DNS server.                    | 4 |
| The DNS submits a request to the domain DNS server.                | 5 |

---

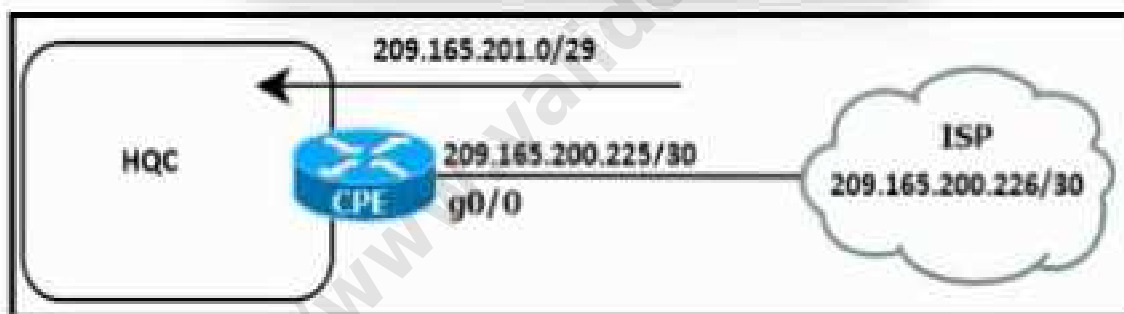
**Answer:**

---

Explanation:

**Question: 1231**

Refer to the exhibit.



Refer to the exhibit. HQC needs to use a configuration that:

handles up to 150,000 concurrent connections

minimizes consumption of public IP addresses

A. ip nat pool NATPOOL 209.165.201.1 209.165.201.3 netmask 255.255.255.248 ip nat inside source list HQC pool NATPOOL overload

B. ip nat pool NATPOOL 209.165.201.1 209.165.201.248 netmask 255.255.255.248 ip nat outside source list HQC pool NATPOOL overload

C. ip nat pool NATPOOL 209.165.200.225 209.165.200.226 netmask 255.255.255.252 ip nat outside source list HQC pool NATPOOL overload

D. ip nat pool NATPOOL 209.165.201.1 209.165.201.5 netmask 255.255.255.248 ip nat inside source list HQC interface gigabitEthernet0/0 overload

---

**Answer: D**

---

Explanation:

---

**Question: 1232**

---

Which cloud-provided service allows an organization to install its own operating system on a virtual machine?

- A. platform-as-a-service
- B. software-as-a-service
- C. infrastructure-as-a-service
- D. network-as-a-service

---

**Answer: C**

---

Explanation:

---

**Question: 1233**

---

When deploying a new network that includes both Cisco and third-party network devices, which redundancy protocol avoids the interruption of network traffic if the default gateway router fails?

- A. FHRP
- B. HSRP
- C. GLBP
- D. VRRP

---

**Answer: D**

---

Explanation:

**Question: 1234**

Refer to the exhibit.

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile B -
BGP, D - EIGRP EX - EIGRP external, O - OSPF, IA - OSPF inter area,
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2,
E1 - OSPF external type 1, E2 - OSPF external type 2, I - IS-IS,
su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area, * - candidate default, I - per-user static route, o - ODR,
P - periodic download static route, H - NHRP, I - LISP,
a - application route, + - replicated route, % - next hop override,
p - overrides from PBR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
C 10.10.10.16/28 is directly connected, FastEthernet 0/0
L 10.10.10.17/32 is directly connected, FastEthernet 0/0
C 10.10.10.8/29 is directly connected, FastEthernet 0/1
L 10.10.10.9/32 is directly connected, FastEthernet 0/1
C 10.10.10.4/30 is directly connected, FastEthernet 0/2
C 10.10.10.5/32 is directly connected, FastEthernet 0/2
C 10.10.10.0/30 is directly connected, Serial 0/0
C 10.10.10.1/32 is directly connected, Serial 0/0
```

Refer to the exhibit. Which interface does a packet take to reach the destination address of 10.10.10.147?

- A. Serial 0/0
- B. FastEthernet 0/2
- C. FastEthernet 0/0
- D. FastEthernet 0/1

**Answer: C**

Explanation:

**Question: 1235**

Refer to the exhibit.

```
SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#enable password test!2E
SW1#(config)# line con 0
SW1#(config-line)# password Labtest32!
SW1#(config-line)# exit
SW1#(config)#
```

Refer to the exhibit. A VTY password has been set to Labtest32! for remote access. Which commands are required to allow only SSH access and to hide passwords in the running configuration?

- A. SW1#(config-line)#login local SW1#(config-line)#exit SW1(config)#crypto key generate rsa
- B. SW1#(config-line)#login local SW1#(config-line)#exit SW1(config)#enable secret test!2E
- C. SW1#(config-line)#transport input ssh SW1#(config-line)#exit SW1(config)#service password-encryption
- D. SW1#(config-line)#exit SW1(config)#aaa new-model

---

**Answer: C**

---

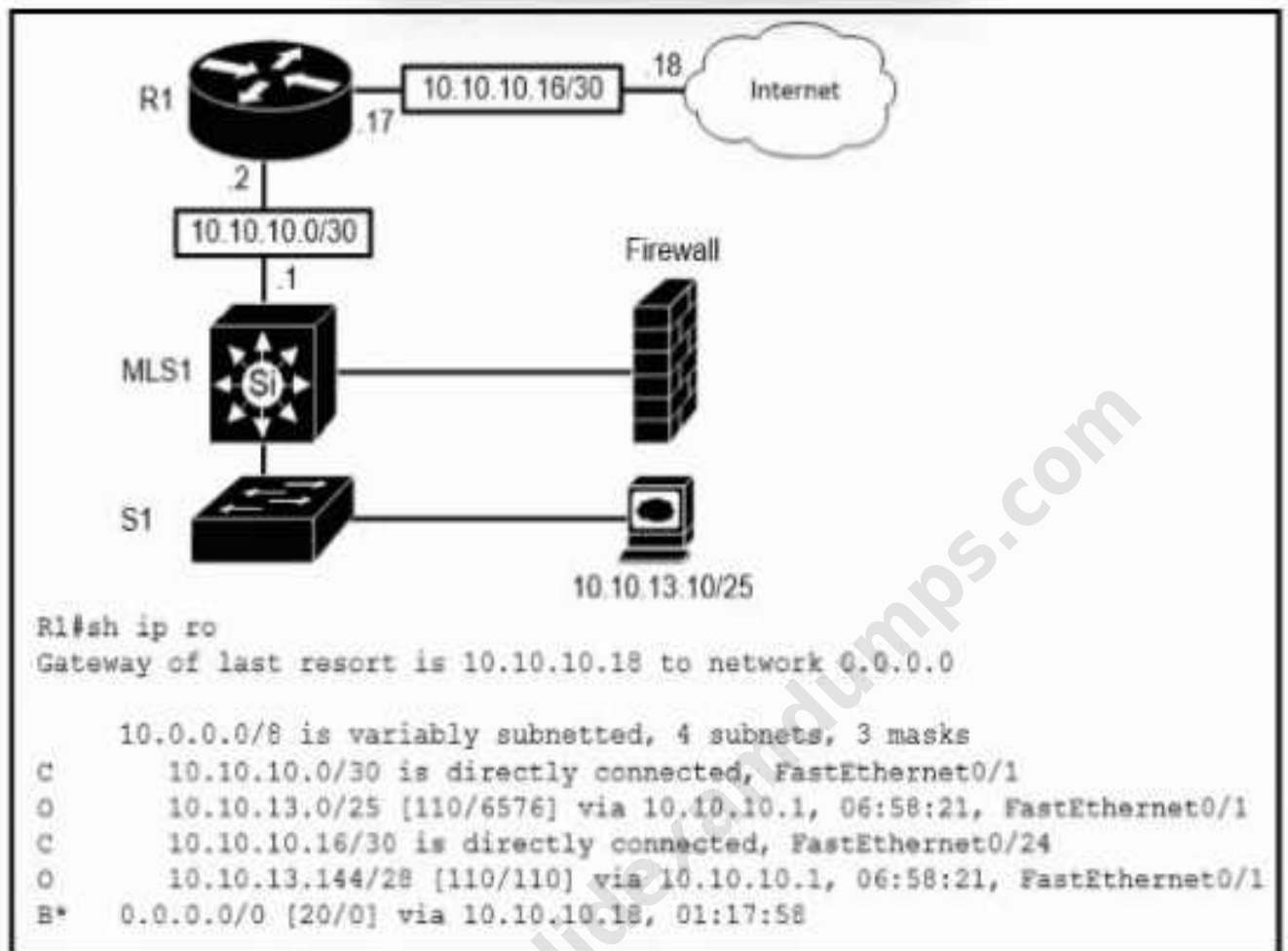
Explanation:

---

**Question: 1236**

Refer to the exhibit.





Refer to the exhibit. Which route type is configured to reach the Internet?

- A. host route
- B. floating static route
- C. default route
- D. network route

**Answer: C**

Explanation:

### Question: 1237

Refer to the exhibit.

```
1  [  
2  {"switch": "SW_admin", "interface": "ge7/41"},  
3  {"VPN concentrator": "VPN_finance", "interface": "fe9/5"},  
4  {"IDS": "IPS_admin", "interface": "te2/10"},  
5  ]
```

Refer to the exhibit. What is represented in line 2 within this JSON schema?

- A. key
- B. value
- C. object
- D. array

---

**Answer: A**

---

Explanation:

---

### Question: 1238

An engineer is configuring data and voice services to pass through the same port. The designated switch interface fastethernet0/1 must transmit packets using the same priority for data when they are received from the access port of the IP phone. Which configuration must be used?

A)

```
interface fastethernet0/1  
switchport priority extend cos 7
```

B)

```
interface fastethernet0/1  
switchport voice vlan untagged
```

C)

```
interface fastethernet0/1  
switchport voice vlan dot1p
```

D)

```
interface fastethernet0/1  
switchport priority extend trust
```

A. Option A

B. Option B

C. Option C

D. Option D

---

**Answer: A**

---

Explanation:

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